

MATH1131 Mathematics 1A 1231-41 1B MATH1141 Higher Mathematics 1A

Please read carefully.

These Notes form the basis for the algebra strand of MATH1131 and MATH1141. However, not all of the material in these Notes is included in the MATH1131 or MATH1141 algebra syllabuses. A detailed syllabus will be uploaded to Moodle.

In using these Notes, you should remember the following points:

1. Most courses at university present new material at a faster pace than you will have been accustomed to in high school, so it is essential that you start working right from the beginning of the session and continue to work steadily throughout the session. Make every effort to keep up with the lectures and to do problems relevant to the current lectures.
2. These Notes are **not** intended to be a substitute for attending lectures or tutorials. The lectures will expand on the material in the notes and help you to understand it.
3. These Notes may seem to contain a lot of material but not all of this material is equally important. One aim of the lectures will be to give you a clearer idea of the relative importance of the topics covered in the Notes.
4. Use the tutorials for the purpose for which they are intended, that is, to ask questions about both the theory and the problems being covered in the current lectures.
5. Some of the material in these Notes is more difficult than the rest. This extra material is marked with the symbol [H]. Material marked with an [X] is intended for students in MATH1141.
6. Problems marked with [V] have a video solution available from Moodle.
7. It is **essential** for you to do **problems** which are given at the end of each chapter. If you find that you do not have time to attempt all of the problems, you should at least attempt a representative selection of them.
8. You will be expected to use the computer algebra package Maple in tests and understand Maple syntax and output for the end of semester examination.
8. You will probably find some of the ideas in Chapters 6 and 7 quite difficult at first because they are expressed in a general and abstract manner. However, as you work through the examples in the chapters and the problems at the ends of the chapters you should find that the ideas become much clearer to you.

PROBLEM SETS

At the end of each chapter there is a set of problems. Some of the problems are very easy, some are less easy but still routine and some are quite hard. To help you decide which problems to try first, each problem is marked with an **[R]**, an **[H]** or an **[X]**. The problems marked **[R]** form a basic set of problems which you should try first. Problems marked **[H]** are harder and can be left until you have done the problems marked **[R]**. You *do* need to make an attempt at the **[H]** problems because problems of this type will occur on tests and in the exam. If you have difficulty with the **[H]** problems, ask for help in your tutorial.

There are a number of questions marked **[M]**, indicating that Maple is required in the solution of the problem. Questions marked with a **[V]** have a video solution available from the Moodle course page.

The computer package Maple will be used in the algebra course. An introduction to Maple is included in the booklet titled *First Year Maple Notes*.

ALGEBRA PROBLEM SCHEDULE

Solving problems and writing mathematics clearly are two separate skills that need to be developed through practice. We recommend that you keep a workbook to practice *writing* solutions to mathematical problems. The range of questions suitable for each week will be provided on Moodle along with a suggestion of specific recommended problems to do before your classroom tutorials.

The Online Tutorials will develop your problem solving skills, and give you examples of mathematical writing. Online Tutorials help build your understanding from lectures towards solving problems on your own.

THEORY IN THE ALGEBRA COURSE

The theory is regarded as an essential part of this course and it will be examined both in class tests and in the end of year examination.

You should make sure that you can give DEFINITIONS of the following ideas:

Chapter 6. Subspace of a vector space, linear combination of a set of vectors, span of a set of vectors, linear independence of a set of vectors, spanning set for a vector space, basis for a vector space, dimension of a vector space.

Chapter 7. Linear function, kernel and nullity of a linear function, image and rank of a linear function.

Chapter 8. Eigenvalue and eigenvector, diagonalizable matrix.

Chapter 9. Probability, statistical independence, conditional probability, discrete random variable, expected value (mean) of a random variable, variance of a random variable, binomial distribution, geometric distribution.

You should be able to give STATEMENTS of the following theorems and propositions.

Chapter 6. Theorem 1 of §6.3, Propositions 1 and 3 and Theorem 2 of §6.4, Proposition 1 and Theorems 2, 3, 4, 5 and 6 of §6.5, Theorems 1, 2, 3, 4, 5, 6 and 7 of §6.6.

Chapter 7. Theorems 2, 3 and 4 of §7.1, Theorem 1 and 2 of §7.2, Proposition 7 and Theorems 1, 5, 8, 9 and 10 of §7.4.

Chapter 8. Theorems 1, 2 and 3 of §8.1, Theorem 1 and 2 of §8.2.

You should be able to give PROOFS of the following theorems and propositions.

Chapter 6. Theorem 2 of §6.4, Theorems 2 and 3 of §6.5, Theorem 2 of §6.6.

Chapter 7. Theorem 2 of §7.1, Theorem 1 of §7.2, Theorems 1, 5 and 8 of §7.4.

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Chapter 1 INTRODUCTION TO VECTORS

*“You see, the earth takes twenty-four hours to
 turn round on its axis —”*
“Talking of axes,” said the Duchess, “chop off her head.”
 Lewis Carroll, Alice in Wonderland.

The aims of this chapter are to introduce the idea of “vector” and in a relatively informal and intuitive manner, and to illustrate applications of these ideas to the geometry of lines and planes.

Until quite recently, the main applications of vectors had been in the physical and engineering sciences. However the study of vectors has now become an important branch of modern pure and applied mathematics, and vectors are now being used in such diverse fields as economics and management science, psychology and the social sciences, chemistry and chemical engineering, mechanical and electrical engineering, computer science, numerical analysis and computational mathematics.

The definition of vectors used in mathematics courses is essentially algebraic in nature whereas the one use by physicists is geometric. We shall begin with the geometric approach then we shall introduce the algebraic definition and show how they relate to one another. As we shall see however, the algebraic definition of a vector is not limited to describing quantities that arise in physics and engineering.

1.1 Vector quantities

Vector quantities, as opposed to scalar quantities, are very important in an understanding of the laws of physics and engineering.

A **scalar** quantity is anything that can be specified by a single number. Examples of scalar quantities are temperature, distance, mass and speed. For example, specifying the speed of a car as 60 km per hour only involves the single real number 60.

A **vector** quantity is one which is specified by both a magnitude and a direction. Examples of vector quantities are displacement, velocity, force and electric field. For example, specifying the velocity of a car as 60 km per hour northeast involves a **vector** of magnitude 60 and direction northeast.

The usual notational convention in books is to differentiate vector quantities from scalar ones by denoting them by boldface symbols such as **a**, and we shall do this in these notes. In handwriting, one usually signifies that a quantity is a vector by using a tilde sign under the letter (as in $\underline{a}$), or by

writing an arrow above the letter (as in $\vec{a}$). Because the properties of scalar and vector quantities are quite different, it is vital that you distinguish them, especially in solutions to problems.

The magnitude of the vector $\mathbf{a}$ is usually denoted by $|\mathbf{a}|$. Note that this is always a non-negative real number, and that $|\mathbf{a}| = 0$ only when $\mathbf{a}$ is the zero vector, usually denoted by $\mathbf{0}$.

Definition 1. *The **zero vector** is the vector $\mathbf{0}$ of magnitude zero, and undefined direction.*

1.1.1 Geometric vectors

To represent a vector on a diagram we draw an arrow (i.e. a directed line segment) where the length of the arrow is the magnitude of the vector, and the direction of the arrow is the direction of the vector. An arrow can be specified by its *initial point* (the tail) and its *terminal point* (the head). In figure 1, a vector $\mathbf{a}$ is represented by an arrow with initial point P and terminal point Q . We denote this arrow by $\overrightarrow{PQ}$.

Two vectors are said to be equal if they have the same magnitude and direction. As the arrows $\overrightarrow{AB}$ and $\overrightarrow{EF}$ have the same length and direction as $\overrightarrow{PQ}$, these arrows all represent the same vector. We can write

$$\overrightarrow{PQ} = \overrightarrow{AB} = \overrightarrow{EF}.$$

Each vector may be represented by many arrows, but each arrow only represents one vector. Nonetheless, we shall sometimes find it convenient to blur the distinction and write expressions like $\mathbf{a} = \overrightarrow{PQ}$, when we really mean that $\overrightarrow{PQ}$ represents $\mathbf{a}$.

NOTE. Since we use the intuitive notion of direction in our physical world, apparently we can only have two dimensional or three dimensional geometric vectors. We shall introduce the algebraic definition of vectors including the higher dimensional ones in the next section. Though we can still talk about higher dimensional geometric vectors, the notions of length and direction will depend on the algebraic nature of the vectors.

There are two equivalent ways to add two vectors together. The first addition rule is often known as the triangle law for vector addition.

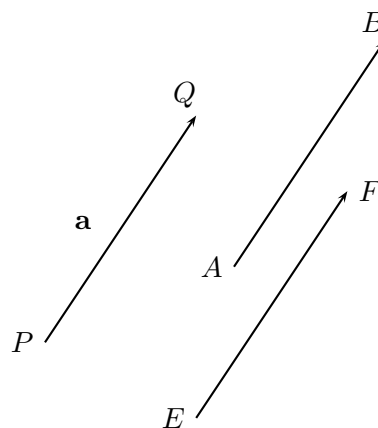


Figure 1.

Definition 2. (The addition of vectors). *On a diagram drawn to scale, draw an arrow representing the vector $\mathbf{a}$. Now draw an arrow representing $\mathbf{b}$ whose initial point lies at the terminal point of the arrow representing $\mathbf{a}$. The arrow which goes from the initial point of $\mathbf{a}$ to the terminal point of $\mathbf{b}$ represents the vector $\mathbf{c} = \mathbf{a} + \mathbf{b}$.*

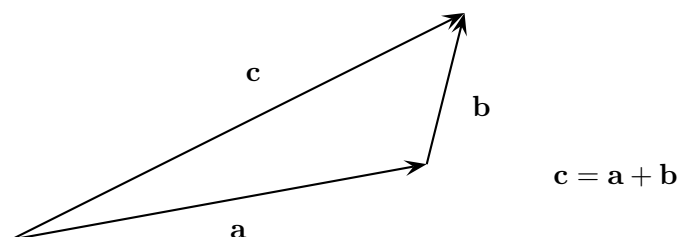


Figure 2: Addition of Vectors.

The second addition rule is known as the parallelogram law for vector addition.

Definition 3. (Alternate definition of vector addition). *On a diagram drawn to scale, draw an arrow representing the vector $\mathbf{a}$. Now draw an arrow representing $\mathbf{b}$ whose initial point lies at the initial point of the arrow representing $\mathbf{a}$. Draw the parallelogram with the two arrows as adjacent sides. The initial point and the two terminal points are vertices of the parallelogram. The arrow which goes from the initial point of $\mathbf{a}$ to the fourth vertex of the parallelogram represents the vector $\mathbf{c} = \mathbf{a} + \mathbf{b}$.*

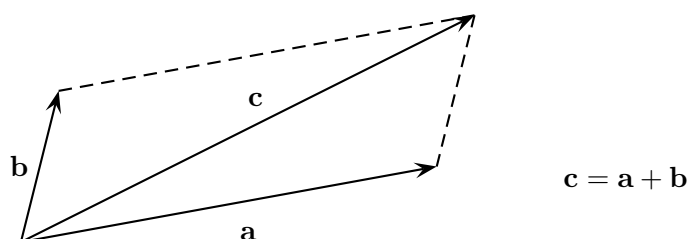


Figure 3: Addition of Vectors.

Obviously, these two definitions are equivalent.

Vector addition is of course quite different from usual addition of numbers, but they do share some important properties. In particular, for any vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$,

$$\begin{array}{ll} \mathbf{a} + \mathbf{b} = \mathbf{b} + \mathbf{a}, & \text{(Commutative law of vector addition)} \\ \text{and } (\mathbf{a} + \mathbf{b}) + \mathbf{c} = \mathbf{a} + (\mathbf{b} + \mathbf{c}). & \text{(Associative law of vector addition)} \end{array}$$

These laws, which follow from basic geometry (see Figures 4 and 5), assert that it makes no difference in what order, or in what grouping we add vectors.

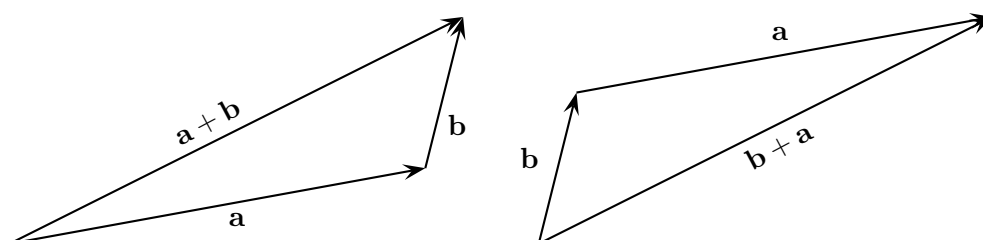


Figure 4: Commutative Law of Vector Addition.

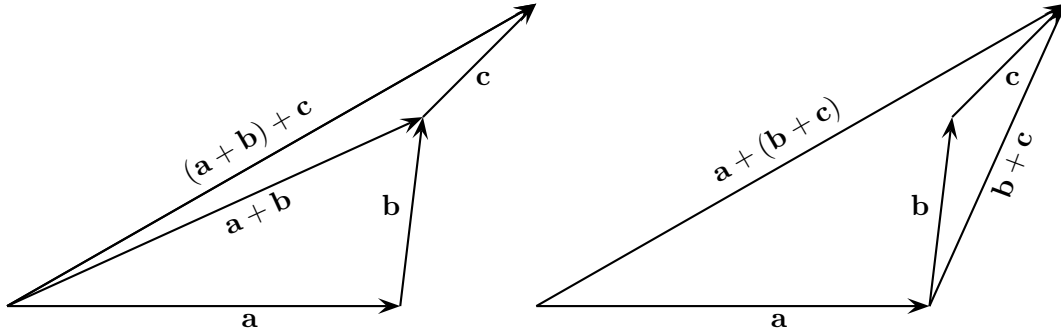


Figure 5: Associative Law of Vector Addition.

The law of addition can easily be extended to cover the zero vector by the natural condition that $\mathbf{a} + \mathbf{0} = \mathbf{0} + \mathbf{a} = \mathbf{a}$ for all vectors $\mathbf{a}$. We then can introduce the negative of a vector and the subtraction of vectors.

Definition 4. (Negative of a vector and subtraction). The *negative* of $\mathbf{a}$, written $-\mathbf{a}$ is a vector such that

$$\mathbf{a} + (-\mathbf{a}) = -\mathbf{a} + \mathbf{a} = \mathbf{0}.$$

If $\mathbf{a}$ and $\mathbf{b}$ are vectors, we define the subtraction by

$$\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b}).$$

From the definition, the vector $-\mathbf{a}$ is the unique vector which has the same magnitude as $\mathbf{a}$, but the direction is opposite to that of $\mathbf{a}$. As shown in Figure 6, the arrow representing $\mathbf{a} - \mathbf{b}$ can be drawn by first reversing the arrow representing $\mathbf{b}$, and then adding the vectors $\mathbf{a}$ and $-\mathbf{b}$.

On the other hand if we represent the vectors $\mathbf{a}$ and $\mathbf{b}$ by arrows with the same initial point O . Let P and Q be the terminal points of the two vectors, respectively. Suppose that $OPRQ$ is a parallelogram. From definition, we have

$$\begin{aligned} \overrightarrow{OQ} + \overrightarrow{QP} &= \overrightarrow{OP} \\ \mathbf{b} + \overrightarrow{QP} &= \mathbf{a} \\ \overrightarrow{QP} &= \mathbf{a} - \mathbf{b} \end{aligned}$$

Hence the diagonal $\overrightarrow{QP}$ represents the difference $\mathbf{a} - \mathbf{b}$. Moreover, $\mathbf{a} - \mathbf{b}$ is the vector which can be represented by the arrow from the terminal point of $\mathbf{b}$ to the terminal point of $\mathbf{a}$.

We now define the operation of multiplying a vector by a real number. Roughly speaking, to multiply a vector $\mathbf{a}$ by a real number λ , all we do is stretch the vector by a factor of λ , whilst keeping its direction unchanged. We need to be careful if λ is not positive.

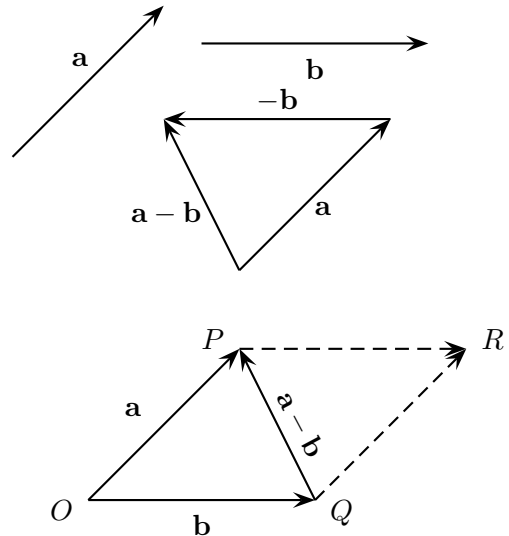


Figure 6: Subtraction of Vectors.

Definition 5. (Multiplication of a vector by a scalar). Let $\mathbf{a}$ and $\mathbf{b}$ be vectors and let $\lambda \in \mathbb{R}$.

1. If $\lambda > 0$, then $\lambda\mathbf{a}$ is the vector whose magnitude is $\lambda|\mathbf{a}|$ and whose direction is the same as that of $\mathbf{a}$.
2. If $\lambda = 0$, then $\lambda\mathbf{a} = \mathbf{0}$.
3. If $\lambda < 0$, then $\lambda\mathbf{a}$ is the vector whose length is $|\lambda||\mathbf{a}|$ and whose direction is the opposite of the direction of $\mathbf{a}$.

From the above, the negative of $\mathbf{a}$ and the product of the scalar -1 and the vector $\mathbf{a}$ are both the vector which has the same magnitude as $\mathbf{a}$, but the direction is the opposite of that of $\mathbf{a}$. Hence we have

$$-\mathbf{a} = (-1)\mathbf{a}.$$

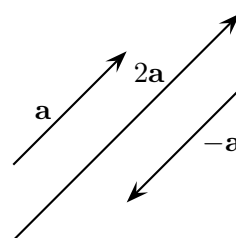


Figure 7: Scalar Multiplication.

We have already seen the commutative and associative laws of vector addition. There are some other important properties of scalar multiplication and vector addition. Let $\mathbf{a}$ and $\mathbf{b}$ be vectors, λ and μ be real numbers, then:

$\lambda(\mu\mathbf{a}) = (\lambda\mu)\mathbf{a},$	(Associative law of multiplication by a scalar)
$(\lambda + \mu)\mathbf{a} = \lambda\mathbf{a} + \mu\mathbf{a},$	(Scalar distributive law)
$\lambda(\mathbf{a} + \mathbf{b}) = \lambda\mathbf{a} + \lambda\mathbf{b}.$	(Vector distributive law)

The vector distributive law for the case that $\lambda > 0$ follows from properties of similar triangles and Figure 8. The proof for the other cases and the proofs of the other two laws are left as exercises for the 1141 students.

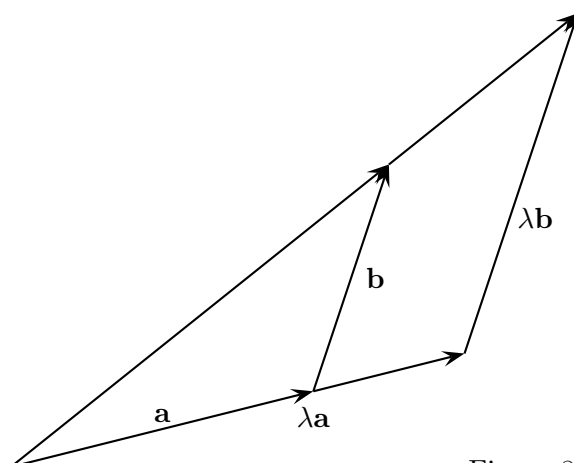


Figure 8: Vector Distributive Law.

We can use these rules to simplify vector expressions.

Example 1. Simplify $3(2\mathbf{a} - \mathbf{b}) + (\mathbf{a} - 2\mathbf{b})$.

SOLUTION. In this example, we shall quote all the rules that we used.

$$\begin{aligned}
 & 3(2\mathbf{a} - \mathbf{b}) + (\mathbf{a} - 2\mathbf{b}) \\
 = & 3[2\mathbf{a} + (-1)\mathbf{b}] + [\mathbf{a} + (-1)(2\mathbf{b})] && \text{(Definition of subtraction; } -\mathbf{a} = (-1)\mathbf{a}) \\
 = & [3(2\mathbf{a}) + 3((-1)\mathbf{b})] + [\mathbf{a} + (-1)(2\mathbf{b})] && \text{(Vector distributive law)} \\
 = & [6\mathbf{a} + (-3)\mathbf{b}] + [\mathbf{a} + (-2)\mathbf{b}] && \text{(Associative law of multiplication by a scalar)} \\
 = & (6\mathbf{a} + \mathbf{a}) + [(-3)\mathbf{b} + (-2)\mathbf{b}] && \text{(Associative law and commutative law)} \\
 = & (6 + 1)\mathbf{a} + [(-3) + (-2)]\mathbf{b} && \text{(Scalar distributive law)} \\
 = & 7\mathbf{a} - 5\mathbf{b} && \text{(Definition of subtraction; } -\mathbf{a} = (-1)\mathbf{a})
 \end{aligned}$$

In practice, we simply write

$$3(2\mathbf{a} - \mathbf{b}) + (\mathbf{a} - 2\mathbf{b}) = 6\mathbf{a} - 3\mathbf{b} + \mathbf{a} - 2\mathbf{b} = 7\mathbf{a} - 5\mathbf{b}.$$

◇

Example 2. Simplify $2\overrightarrow{AC} - \overrightarrow{OC} + \overrightarrow{OA}$.

SOLUTION. Let $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OC} = \mathbf{c}$.

$$2\overrightarrow{AC} - \overrightarrow{OC} + \overrightarrow{OA} = 2(\mathbf{c} - \mathbf{a}) - \mathbf{c} + \mathbf{a} = 2\mathbf{c} - 2\mathbf{a} - \mathbf{c} + \mathbf{a} = \mathbf{c} - \mathbf{a} = \overrightarrow{AC}.$$

◇

1.1.2 Two dimensional vector quantities

At first, we can apply geometric vectors to prove some geometry theorems. In some aspects, vectors do have some advantages. For instance, we can prove a quadrilateral $OABC$ to be a parallelogram simply by proving that $\overrightarrow{OA} = \overrightarrow{BC}$. To prove two lines PQ and RS are parallel, we need to show that there exists a real number λ such that $\overrightarrow{PQ} = \lambda\overrightarrow{RS}$.

Example 3. In a triangle OAB , take D and E such that $OD : DA = OE : EB = 1 : 2$. Prove that DE is parallel to AB and the length of DE is $\frac{1}{3}$ times that of that of AB .

Proof. Since D divides OA in the ratio $1 : 2$, the vector $\overrightarrow{OD}$ has the same direction as $\overrightarrow{OA}$ and its length is $\frac{1}{3}$ times that of $\overrightarrow{OA}$. Hence $\overrightarrow{OD} = \frac{1}{3}\overrightarrow{OA}$. Similarly, we also have $\overrightarrow{OE} = \frac{1}{3}\overrightarrow{OB}$. Thus

$$\begin{aligned}
 \overrightarrow{DE} &= \overrightarrow{OE} - \overrightarrow{OD} = \frac{1}{3}\overrightarrow{OB} - \frac{1}{3}\overrightarrow{OA} \\
 &= \frac{1}{3}(\overrightarrow{OB} - \overrightarrow{OA}) = \frac{1}{3}\overrightarrow{AB}.
 \end{aligned}$$

Hence DE is parallel to AB and the length of DE is $\frac{1}{3}$ times that of that of AB . $\square$

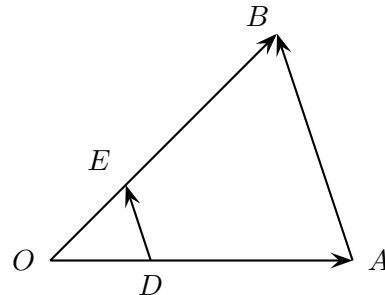


Figure 9.

[X] **Example 4.** Let $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. Prove that P is a point on the line AB between A and B if and only if $\overrightarrow{OP} = (1 - \lambda)\mathbf{a} + \lambda\mathbf{b}$ for some real number $0 < \lambda < 1$.

Proof. If P is a point on AB between A and B , there exists a real number λ between 0 and 1 such that $\overrightarrow{AP} = \lambda\overrightarrow{AB}$.

$$\begin{aligned}\text{Hence, } \overrightarrow{OP} &= \overrightarrow{OA} + \overrightarrow{AP} = \overrightarrow{OA} + \lambda\overrightarrow{AB} \\ &= \mathbf{a} + \lambda(\mathbf{b} - \mathbf{a}) = (1 - \lambda)\mathbf{a} + \lambda\mathbf{b}.\end{aligned}$$

Conversely, if $\overrightarrow{OP} = (1 - \lambda)\mathbf{a} + \lambda\mathbf{b}$ and $0 < \lambda < 1$, we have

$$\overrightarrow{AP} = [(1 - \lambda)\mathbf{a} + \lambda\mathbf{b}] - \mathbf{a} = \lambda(\mathbf{b} - \mathbf{a}) = \lambda\overrightarrow{AB}.$$

Since $0 < \lambda < 1$, the point P lies on AB between A and B . $\square$

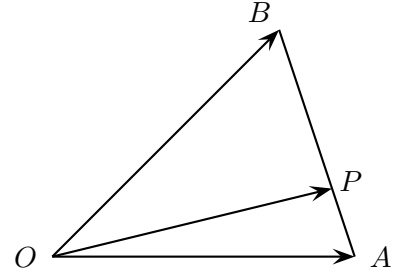


Figure 10.

[X] **Example 5.** Prove that the three medians of a triangle are concurrent.

Proof. Name the vertices of a triangle by O , A and B . Let D, E, F be the midpoints of OB, OA, AB respectively. Suppose that AD and BE intersect at G .

Let $\overrightarrow{OA} = \mathbf{a}$ and $\overrightarrow{OB} = \mathbf{b}$. Hence

$$\overrightarrow{OE} = \frac{1}{2}\mathbf{a} \quad \text{and} \quad \overrightarrow{OD} = \frac{1}{2}\mathbf{b}.$$

Since G lies on both AD and BE and inside the triangle, from Example 4 there exist real numbers λ and μ such that

$$\overrightarrow{OG} = (1 - \lambda)\mathbf{a} + \lambda\left(\frac{1}{2}\mathbf{b}\right) = (1 - \mu)\mathbf{b} + \mu\left(\frac{1}{2}\mathbf{a}\right).$$

By rearranging terms, we get

$$(1 - \lambda)\mathbf{a} - \frac{1}{2}\mu\mathbf{a} = (1 - \mu)\mathbf{b} - \frac{1}{2}\lambda\mathbf{b}.$$

Since $\mathbf{a}$ cannot be a non-zero scalar multiple of $\mathbf{b}$, we have

$$(1 - \lambda) - \frac{1}{2}\mu = 0 \quad \text{and} \quad (1 - \mu) - \frac{1}{2}\lambda = 0.$$

By solving the above simultaneous equations, we have $\lambda = \frac{2}{3}$ and $\mu = \frac{2}{3}$. So we have $\overrightarrow{OG} = \frac{1}{3}(\mathbf{a} + \mathbf{b})$.

Since F is the midpoint of AB , so $\overrightarrow{OF} = \frac{1}{2}(\mathbf{a} + \mathbf{b})$. Thus $\overrightarrow{OG}$ and $\overrightarrow{OF}$ are in the same direction. Hence G lies on OF and therefore the three medians of the triangle OAB are concurrent. $\square$

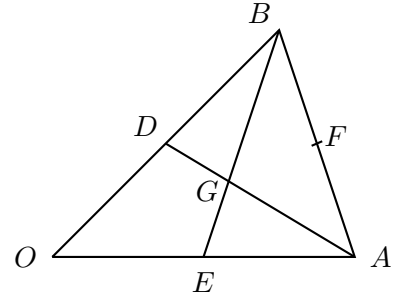


Figure 11.

Many calculations with vector quantities in the plane can be done geometrically using scale diagrams.

Example 6. A yacht sails from a pier in the direction N 60° E for 15 km, then turns to N 45° W for 10 km. What are the distance and the bearing of the yacht from the pier?

SOLUTION. As shown in Figure 12, after the first leg, the yacht is at P which is $15 \cos 30^\circ$ km east and $15 \sin 30^\circ$ km north of the pier, O . The yacht then moves, $10 \cos 45^\circ$ km west and $10 \sin 45^\circ$ km north, to Q . The yacht is then

$$15 \cos 30^\circ - 10 \cos 45^\circ \approx 5.919 \text{ km east of } O$$

and $15 \sin 30^\circ + 10 \sin 45^\circ \approx 14.571 \text{ km north of } O.$

$$\text{Hence } OQ \approx \sqrt{(5.919)^2 + (14.571)^2} \approx 15.73,$$

$$\theta \approx \tan^{-1} \left(\frac{14.571}{5.919} \right) \approx 67^\circ 54'.$$

Here, θ is angle of OQ measured from the east. The distance and the bearing of the yacht from the pier are then 15.73 km and N $22^\circ 6'$ E. $\diamond$

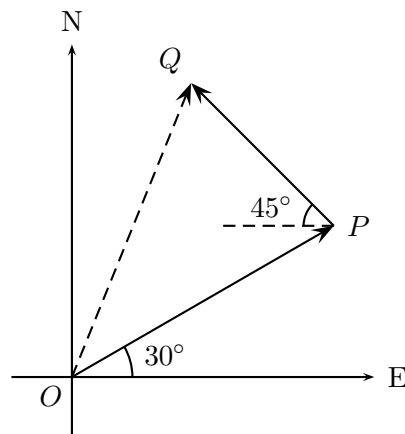


Figure 12.

In adding the two displacement vectors $\overrightarrow{OP}$ and $\overrightarrow{PQ}$, we add and subtract displacements in the east-west and the north-south direction. We then specified the sum $\overrightarrow{OQ}$ by its length and direction. We shall see how this relates to the algebraic definition of vector in the next section.

NOTE. For many of the problems in physical world, the vector quantities are in three dimensions rather than two. At least in theory, the same methods in this section could be used to solve such problems. Each vector could be represented as an arrow in space, and then we can add two vectors and multiply a vector by a scalar.

1.2 Vector quantities and $\mathbb{R}^n$

In Example 6 of Section 1.1.2, if we denote the vector of length 1 km towards the east by $\mathbf{i}$ and the vector of length 1 km towards the north by $\mathbf{j}$, by the definition of geometric vectors we can write

$$\begin{aligned} \overrightarrow{OP} &= 15 \cos 30^\circ \mathbf{i} + 15 \sin 30^\circ \mathbf{j}, \\ \overrightarrow{PQ} &= -10 \cos 45^\circ \mathbf{i} + 10 \sin 45^\circ \mathbf{j}, \quad \text{and} \end{aligned}$$

By the associative, commutative and distributive laws of geometric vectors, we have

$$\begin{aligned} \overrightarrow{OQ} &= \overrightarrow{OP} + \overrightarrow{PQ} \\ &= (15 \cos 30^\circ - 10 \cos 45^\circ)\mathbf{i} + (15 \sin 30^\circ + 10 \sin 45^\circ)\mathbf{j}. \end{aligned}$$

In a more convenient way, we can write

$$\begin{aligned}\overrightarrow{OP} &= \begin{pmatrix} 15 \cos 30^\circ \\ 15 \sin 30^\circ \end{pmatrix}, \\ \overrightarrow{PQ} &= \begin{pmatrix} -10 \cos 45^\circ \\ 10 \sin 45^\circ \end{pmatrix}, \\ \overrightarrow{OQ} &= \begin{pmatrix} 15 \cos 30^\circ - 10 \cos 45^\circ \\ 15 \sin 30^\circ + 10 \sin 45^\circ \end{pmatrix}.\end{aligned}$$

There is no reason why we cannot generalise this to all two dimensional vectors, three dimensional vectors and beyond.

1.2.1 Vectors in $\mathbb{R}^2$

We first choose two vectors, conventionally denoted by $\mathbf{i}$ and $\mathbf{j}$, of unit length, and at right angles to each other so that $\mathbf{j}$ is pointing at an angle of $\frac{\pi}{2}$ anticlockwise from $\mathbf{i}$. The vectors $\mathbf{i}$ and $\mathbf{j}$ are known as the standard **basis vectors** for $\mathbb{R}^2$. See Figure 13.

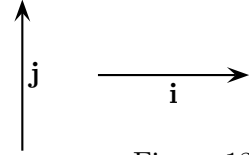


Figure 13.

As shown in Figure 14, every vector $\mathbf{a}$ can be ‘resolved’ (in a unique way) into the sum of a scalar multiple of $\mathbf{i}$ plus a scalar multiple of $\mathbf{j}$. That is there are unique real numbers a_1 and a_2 such that $\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j}$. If the direction θ of $\mathbf{a}$ is measured from the direction of $\mathbf{i}$, then these scalars can be easily found using the formulae

$$a_1 = |\mathbf{a}| \cos \theta, \quad \text{and} \quad a_2 = |\mathbf{a}| \sin \theta.$$

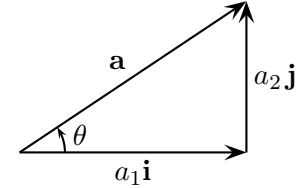


Figure 14.

We call a_1, a_2 the **components** of $\mathbf{a}$.

Now, every vector $\mathbf{a}$ in the plane can be specified by these two unique real numbers. We can write $\mathbf{a}$ in form of a **column vector** or a **2-vector** $\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$. We call the numbers a_1, a_2 the **components** of $\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$. In this case, $\mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

The column vector $\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ is also called the **coordinate vector** with respect to the basis vectors $\left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$ and a_1, a_2 are also called the **coordinates** of the vector.

Theorem 1. Let $\mathbf{a}$ and $\mathbf{b}$ be (geometric) vectors, and let $\lambda \in \mathbb{R}$. Suppose that $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$. Then

1. the coordinate vector for $\mathbf{a} + \mathbf{b}$ is $\begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \end{pmatrix}$;

2. the coordinate vector for $\lambda \mathbf{a}$ is $\begin{pmatrix} \lambda a_1 \\ \lambda a_2 \end{pmatrix}$.

Proof. The basis vectors are $\mathbf{i}$ and $\mathbf{j}$. We first give a detailed proof of the second part.

$$\begin{aligned} \lambda \mathbf{a} &= \lambda(a_1 \mathbf{i} + a_2 \mathbf{j}) && \text{(by definition of coordinate vectors)} \\ &= \lambda a_1 \mathbf{i} + \lambda a_2 \mathbf{j} && \text{(by vector distributive law)} \end{aligned}$$

Hence the coordinate vector for $\lambda \mathbf{a}$ is $\begin{pmatrix} \lambda a_1 \\ \lambda a_2 \end{pmatrix}$.

For the first part, by associative, commutative and distributive laws, we have

$$\begin{aligned} \mathbf{a} + \mathbf{b} &= (a_1 \mathbf{i} + a_2 \mathbf{j}) + (b_1 \mathbf{i} + b_2 \mathbf{j}) \\ &= (a_1 \mathbf{i} + b_1 \mathbf{i}) + (a_2 \mathbf{j} + b_2 \mathbf{j}) \\ &= (a_1 + b_1) \mathbf{i} + (a_2 + b_2) \mathbf{j} \end{aligned}$$

Hence the coordinate vector for $\mathbf{a} + \mathbf{b}$ is $\begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \end{pmatrix}$. □

We then can define the mathematics structure $\mathbb{R}^2$, which is the set of 2-vectors, by

$$\mathbb{R}^2 = \left\{ \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} : a_1, a_2 \in \mathbb{R} \right\},$$

with addition and multiplication by a scalar defined by — for any $\begin{pmatrix} a_1 \\ a_2 \end{pmatrix}, \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} \in \mathbb{R}^2$ and $\lambda \in \mathbb{R}$,

$$\begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \end{pmatrix} \quad \text{and} \quad \lambda \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} = \begin{pmatrix} \lambda a_1 \\ \lambda a_2 \end{pmatrix}.$$

The elements in $\mathbb{R}^2$ are called **vectors** and sometimes **column vectors**. It is obvious that the set $\mathbb{R}^2$ is closed under addition and scalar multiplication. Like geometric vectors, the vectors in $\mathbb{R}^2$ also obey the commutative, associative, and distributive laws. There is a zero vector $\mathbf{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$ and a negative $\begin{pmatrix} -a_1 \\ -a_2 \end{pmatrix}$ for any vector $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$.

1.2.2 Vectors in $\mathbb{R}^n$

The concept of components can easily be generalised from two dimensions to any number of dimensions.

Definition 1. Let n be a positive integer. The set $\mathbb{R}^n$ is defined by

$$\mathbb{R}^n = \left\{ \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix} : a_1, a_2, \dots, a_n \in \mathbb{R} \right\}.$$

An element $\begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$ in $\mathbb{R}^n$ is called an n -vector or simply a vector; and $a_1, a_2, \dots, a_n$ are called the components of the vector.

NOTE. We say that two vectors in $\mathbb{R}^n$ are equal if the corresponding components are equal. In

other words $\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$ if and only if $a_1 = b_1, \dots, a_n = b_n$.

Example 1. 1. Clearly $\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \in \mathbb{R}^3$; $\begin{pmatrix} 0.5 \\ 1.4 \\ 2.3 \\ -4.1 \end{pmatrix} \in \mathbb{R}^4$.

2. $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$ are different elements of $\mathbb{R}^3$. ◇

As in $\mathbb{R}^2$, we can define addition of two vectors in $\mathbb{R}^n$ and multiplication of a vector in $\mathbb{R}^n$ by a scalar in $\mathbb{R}$.

Definition 2. Let $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$ be vectors in $\mathbb{R}^n$ and λ be a real number.

We define the sum of $\mathbf{a}$ and $\mathbf{b}$ by $\mathbf{a} + \mathbf{b} = \begin{pmatrix} a_1 + b_1 \\ a_2 + b_2 \\ \vdots \\ a_n + b_n \end{pmatrix}$.

We define the scalar multiplication of $\mathbf{a}$ by λ by $\lambda \mathbf{a} = \begin{pmatrix} \lambda a_1 \\ \lambda a_2 \\ \vdots \\ \lambda a_n \end{pmatrix}$.

NOTE. The addition rule tells us how to add two vectors with the same number of components. The sum of vectors with different numbers of components is not defined.

Example 2. 1. $\begin{pmatrix} 4 \\ -3 \\ 5 \end{pmatrix} + \begin{pmatrix} 2 \\ 4 \\ -2 \end{pmatrix} = \begin{pmatrix} 4+2 \\ -3+4 \\ 5+(-2) \end{pmatrix} = \begin{pmatrix} 6 \\ 1 \\ 3 \end{pmatrix}.$

2. $\begin{pmatrix} 1.3 \\ 2 \\ 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 3.2 \\ -2 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4.5 \\ 0 \\ 2 \\ 2 \end{pmatrix}.$

3. $3 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix}.$

4. $\pi \begin{pmatrix} -1 \\ 2 \\ 0 \\ 5 \end{pmatrix} = \begin{pmatrix} -\pi \\ 2\pi \\ 0 \\ 5\pi \end{pmatrix}.$

5. $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 2 \\ 0 \\ 4 \\ -1 \end{pmatrix}$ is *not defined!* $\diamond$

Proposition 2. Let $\mathbf{u}$, $\mathbf{v}$ and $\mathbf{w}$ be vectors in $\mathbb{R}^n$.

1. $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}.$ (Commutative Law of Addition)

2. $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w}).$ (Associative Law of Addition)

[X] *Proof.* We prove the commutative law, while proving the associative law will be left as an exercise.

Let $\mathbf{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}.$

$$\begin{aligned}
\mathbf{u} + \mathbf{v} &= \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} + \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} \\
&= \begin{pmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{pmatrix} && \text{(definition of addition in } \mathbb{R}^n \text{)} \\
&= \begin{pmatrix} v_1 + u_1 \\ \vdots \\ v_n + u_n \end{pmatrix} && \text{(commutative law of real numbers)} \\
&= \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} + \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} && \text{(definition of addition in } \mathbb{R}^n \text{)} \\
&= \mathbf{v} + \mathbf{u}
\end{aligned}$$

□

Definition 3. Let n be a positive integer.

1. The **zero vector** in $\mathbb{R}^n$, denoted by $\mathbf{0}$, is the vector with all n components 0.

2. Let $\mathbf{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in \mathbb{R}^n$. The **negative** of $\mathbf{a}$, denoted by $-\mathbf{a}$ is the vector $\begin{pmatrix} -a_1 \\ \vdots \\ -a_n \end{pmatrix}$.

3. Let $\mathbf{a}, \mathbf{b}$ be vectors in $\mathbb{R}^n$. We define the difference, $\mathbf{a} - \mathbf{b}$, by

$$\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b}).$$

NOTE. Let $\mathbf{a}, \mathbf{b}$ be vectors in $\mathbb{R}^n$ and $\mathbf{0}$ be the zero vector in $\mathbb{R}^n$.

1. $\mathbf{a} + \mathbf{0} = \mathbf{0} + \mathbf{a} = \mathbf{a}$.

2. $\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} - \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix} = \begin{pmatrix} a_1 - b_1 \\ \vdots \\ a_n - b_n \end{pmatrix}.$

3. $\mathbf{a} + (-\mathbf{a}) = (-\mathbf{a}) + \mathbf{a} = \mathbf{0}$.

Example 3. 1. $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ is the zero vector in $\mathbb{R}^3$, while $\begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$ is the zero vector in $\mathbb{R}^4$

$$2. \begin{pmatrix} 4 \\ -3 \\ 5 \end{pmatrix} - \begin{pmatrix} 2 \\ 4 \\ -2 \end{pmatrix} = \begin{pmatrix} 4-2 \\ -3-4 \\ 5-(-2) \end{pmatrix} = \begin{pmatrix} 2 \\ -7 \\ 7 \end{pmatrix}.$$

Proposition 3. Let λ, μ be scalars and $\mathbf{u}, \mathbf{v}$ be vectors in $\mathbb{R}^n$.

1. $\lambda(\mu\mathbf{v}) = (\lambda\mu)\mathbf{v}$ (Associative Law of Scalar Multiplication)
2. $(\lambda + \mu)\mathbf{v} = \lambda\mathbf{v} + \mu\mathbf{v}$ (Scalar Distributive Law)
3. $\lambda(\mathbf{u} + \mathbf{v}) = \lambda\mathbf{u} + \lambda\mathbf{v}$ (Vector distributive Law)

1.3 $\mathbb{R}^n$ and analytic geometry

1.3.1 Two dimensions

From an algebraic point of view, $\mathbb{R}^2$ behaves very much the same as $\mathbb{R}^3$ — the rules for addition and scalar multiplication are more or less the same. However, the different sets $\mathbb{R}^1, \mathbb{R}^2, \mathbb{R}^3, \mathbb{R}^4, \dots$ tend to appear in different types of applications.

The sets $\mathbb{R}^2$ and $\mathbb{R}^3$ are particularly important for solving problems which lie in a plane or in “space”. You are probably quite familiar with working with the coordinates of points in a plane. In this section we shall look more closely at the relationship between points in the plane and their coordinates.

Let’s suppose that we have a plane with several objects in it, as in Figure 15.

To solve problems such as “what is the distance between A and B?”, it is often convenient to work with the coordinates of these points. In real life, points don’t come with (x, y) coordinates attached.

We have to

- (i) specify an origin point O somewhere in the plane;
- (ii) specify a unit of length;
- (iii) specify a direction (usually called the x -direction); and
- (iv) specify a y -direction (usually at angle $\frac{\pi}{2}$ anti-clockwise to the x -direction).

Having done this we can then find the **coordinates** (x_1, y_1) of A (with respect to our chosen coordinate system) by requiring that A is x_1 units from the origin in the x -direction and y_1 units from the origin in the y -direction. In this case we represent A by the **2-tuple**, or ordered pair, (x_1, y_1) .



Figure 15.

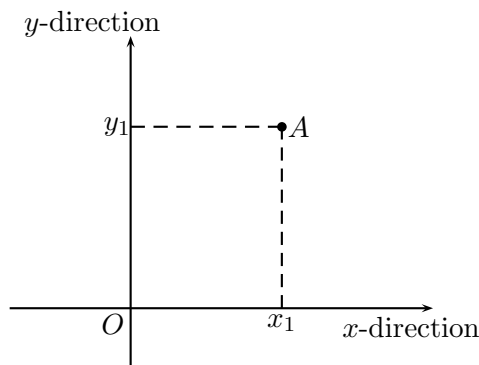


Figure 16.

The line through O in the x -direction is called the x -axis. The y -axis is similarly defined.

With the choice of the origin O , the point A is naturally associated with a vector $\overrightarrow{OA}$ which is called the **position vector** of A with respect to the origin. Then we choose $\mathbf{i}$ to be the unit vector in the x -direction and $\mathbf{j}$ to be the unit vector in the y -direction. The scalars x_1 and y_1 are often known as the **components** of $\mathbf{a} = \overrightarrow{OA}$ in the x and y -directions respectively. As in section 2.2.1, the position vector of A is represented by the coordinate vector $\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ with respect to the basis vectors $\{\mathbf{i}, \mathbf{j}\}$.

When a coordinate system is specified, each point in the plane corresponds to a unique 2-tuple (x, y) , and equivalently a unique column vector $\begin{pmatrix} x \\ y \end{pmatrix}$. Conversely, once the coordinate system has been fixed, each 2-tuple (x, y) or equivalently each $\begin{pmatrix} x \\ y \end{pmatrix} \in \mathbb{R}^2$ corresponds to a unique point in the plane. We shall often denote the position vector of A by $\mathbf{a}$.

It is a consequence of Pythagoras' Theorem that if the points A and B in the plane have position vectors $\begin{pmatrix} x_1 \\ y_1 \end{pmatrix}$ and $\begin{pmatrix} x_2 \\ y_2 \end{pmatrix}$ respectively, then the distance between A and B is given by

$$\text{dist}(A, B) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}.$$

Now, we can naturally define the length of a 2-vector $\mathbf{a}$. Choose the point A such that $\overrightarrow{OA}$ represents $\mathbf{a}$. The length of $\mathbf{a}$ is defined to be $\text{dist}(O, A)$. It is not difficult to see that as displacements in a plane, two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^2$ is parallel if and only if there is a non-zero real number λ such that $\mathbf{a} = \lambda\mathbf{b}$.

NOTE. The use of coordinates to solve problems in geometry is a relatively recent introduction. Although geometry has been studied for thousands of years, this “analytic geometry” was only introduced by R. Descartes (from whose name the term ‘Cartesian’ is derived) in the early 17th century.

1.3.2 Three dimensions

You can give coordinates to points in space just as you give coordinates to points in the plane. Again however, we need to fix our coordinate system. Now, let us

- (i) specify an origin point O somewhere in space;
- (ii) specify a unit of length;
- (iii) specify three directions, usually called the x , y and z directions, each at right angles to the others.

We choose unit vectors $\mathbf{i}, \mathbf{j}, \mathbf{k}$, respectively, in the x -direction, y -direction, z -direction. The **coordinates** of a point A with respect to the basis $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ are given by a **3-tuple** (x_1, y_1, z_1) such that

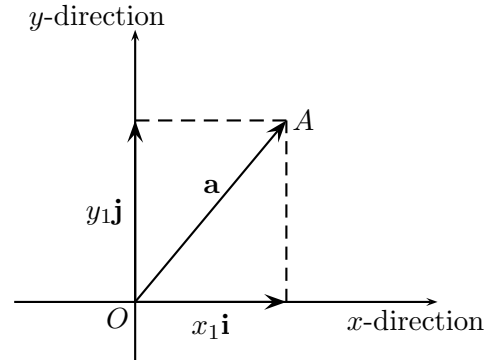


Figure 17.

A is x_1 units from the origin in the x -direction, y_1 units from the origin in the y -direction, and z_1 units from the origin in the z -direction. The **position vector** of A , $\overrightarrow{OA}$, is $\mathbf{a} = x_1\mathbf{i} + y_1\mathbf{j} + z_1\mathbf{k}$, which can also be represented by $\begin{pmatrix} x_1 \\ y_1 \\ z_1 \end{pmatrix} \in \mathbb{R}^3$.

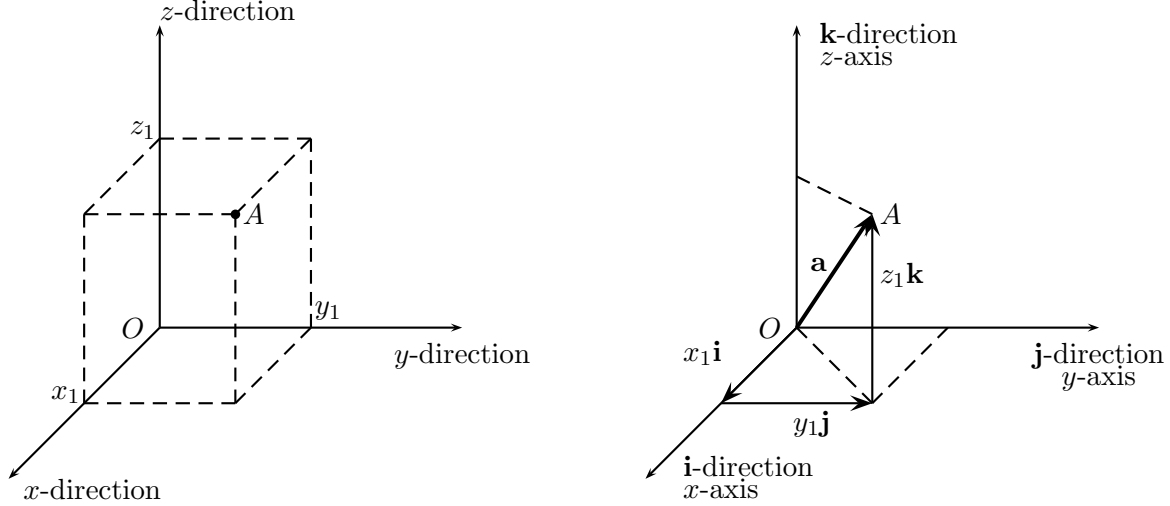


Figure 18: Right-handed Coordinate System.

The coordinate system in Figure 18 is what is called **right-handed Cartesian coordinate system**. Using your right hand, if you point your index finger in the x -direction, your middle finger in the y -direction, then your extended thumb will point in the z -direction. This is the orientation that is conventionally used in mathematics and physics.

Again once we have set up our coordinate system, each point in space corresponds uniquely to a 3-tuple and uniquely to a 3-vector (and vice-versa). It is rather hard to draw pictures of objects in three dimensions (on two dimensional paper!). Nevertheless, this identification will provide us with an important way of visualising relationships between vectors in $\mathbb{R}^3$.

Since the basis vectors chosen are of unit length and are perpendicular to each other, by Pythagoras' theorem the length of $\overrightarrow{OA}$ is $\sqrt{x_1^2 + y_1^2 + z_1^2}$. Let B be another point with position vector $\mathbf{b} = \begin{pmatrix} x_2 \\ y_2 \\ z_2 \end{pmatrix}$, the distance between A and B is the same as the length of the vector $\mathbf{b} - \mathbf{a} = \overrightarrow{AB}$, i.e.

$$\text{dist}(A, B) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}.$$

Similar to $\mathbb{R}^2$, if $\mathbf{a} = \overrightarrow{OA}$, the length of $\mathbf{a}$ is $\text{dist}(O, A)$. Two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^3$ are parallel if and only if there is a non-zero real number λ such that $\mathbf{a} = \lambda\mathbf{b}$.

1.3.3 n -dimensions

We have used elements in $\mathbb{R}^2$ and $\mathbb{R}^3$ to represent the position vectors of points in the plane and in the three-dimensional space, respectively. How do we interpret n -vectors in $\mathbb{R}^n$ as position vectors of points in an n -dimensional space which generalises the 2-dimensional and the 3-dimensional spaces? To study the geometry in the n -dimensional space, we need the notions of length and direction.

In $\mathbb{R}^2$ or $\mathbb{R}^3$, two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ are parallel if and only if there exists a non-zero real number λ such that $\mathbf{b} = \lambda\mathbf{a}$. We use this idea as definition.

Definition 1. Two non-zero vectors $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^n$ are said to be **parallel** if $\mathbf{b} = \lambda\mathbf{a}$ for some non-zero real number λ .
They are said to be in the same **direction** if $\lambda > 0$.

Example 1. The vectors $\begin{pmatrix} 3 \\ -1 \\ 2 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 6 \\ -2 \\ 4 \\ 8 \end{pmatrix}$ are parallel because $\begin{pmatrix} 6 \\ -2 \\ 4 \\ 8 \end{pmatrix} = 2 \begin{pmatrix} 3 \\ -1 \\ 2 \\ 4 \end{pmatrix}$.

The vectors $\begin{pmatrix} 3 \\ 0 \\ 2 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 6 \\ 1 \\ 4 \\ 2 \end{pmatrix}$ are not parallel because we cannot find a scalar λ such that $\begin{pmatrix} 6 \\ 1 \\ 4 \\ 2 \end{pmatrix} = \lambda \begin{pmatrix} 3 \\ 0 \\ 2 \\ 1 \end{pmatrix}$. ◇

Definition 2. Let $\mathbf{e}_j$ be the vector in $\mathbb{R}^n$ with a 1 in the j th component and 0 for all the other components. Then the n vectors $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ are called the **standard basis vectors** for $\mathbb{R}^n$.

Example 2. The standard basis vectors for $\mathbb{R}^3$ are

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = a_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + a_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

The standard basis vectors for $\mathbb{R}^4$ are $\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}$. ◇

We imagine that we can choose a point in the n -dimensional space as the origin O and n axes in the directions of these n standard basis vectors respectively. Then we can define the **coordinates** of a point A with respect to this coordinate system in the same way as in 3-dimensional space. For instance, if A is a_i units from O in the direction of $\mathbf{e}_i$ for each $1 \leq i \leq n$, the coordinates of A are given by an n -**tuple** $(a_1, \dots, a_n)$. Thus the **coordinate vector**, or the **position vector**, of the

point A is $\mathbf{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in \mathbb{R}^n$. We can also discuss n -dimensional geometric vectors. For example,

the vector $\overrightarrow{OA}$, which has the initial point O and the terminal point A , represents $\mathbf{a}$.

Example 3. i) The coordinate vector for the displacement from the point A , with coordinates $(2, 3)$ to the point B with coordinates $(-1, 2)$ is

$$\overrightarrow{AB} = \overrightarrow{OB} - \overrightarrow{OA} = \begin{pmatrix} -1 \\ 2 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -3 \\ -1 \end{pmatrix}.$$

ii) The coordinate vector for the line segment joining the point A with coordinates $(2, 4, -5, 3)$ to the point B with coordinates $(-3, 4, 0, 4)$ is

$$\overrightarrow{AB} = \begin{pmatrix} -3 \\ 4 \\ 0 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ 4 \\ -5 \\ 3 \end{pmatrix} = \begin{pmatrix} -5 \\ 0 \\ 5 \\ 1 \end{pmatrix}.$$

iii) If the position vector of the point A is $\begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix}$ and the displacement from A to B has coordinate vector $\begin{pmatrix} 4 \\ 2 \\ -3 \end{pmatrix}$, then the position vector for B is

$$\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{AB} = \begin{pmatrix} -1 \\ 2 \\ 2 \end{pmatrix} + \begin{pmatrix} 4 \\ 2 \\ -3 \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ -1 \end{pmatrix}.$$

Thus B has coordinates $(3, 4, -1)$. ◇

We can also define the length of a vector and the distance between two points which have coordinate vectors in $\mathbb{R}^n$.

Definition 3. The **length** of a vector $\mathbf{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in \mathbb{R}^n$ is defined by

$$|\mathbf{a}| = \sqrt{a_1^2 + \cdots + a_n^2}.$$

Remark 1. The quantity $|\mathbf{a}|$ is sometimes called the **magnitude** or **norm** of the vector $\mathbf{a}$.

Example 4. Find the lengths of $\mathbf{a} = \begin{pmatrix} 2 \\ -4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ -3 \\ 2 \\ -6 \\ 4 \end{pmatrix}$.

SOLUTION. $|\mathbf{a}| = \sqrt{4 + 16} = 2\sqrt{5}$; $|\mathbf{b}| = \sqrt{1 + 9 + 4 + 36 + 16} = \sqrt{66}$. ◇

Definition 4. The **distance** between two points A and B with position vectors in $\mathbb{R}^n$ is the length of the vector $\overrightarrow{AB}$.

Then, if $\mathbf{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$ is the coordinate vector of A and $\mathbf{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$ is the coordinate vector of B , we have

$$\overrightarrow{AB} = \mathbf{b} - \mathbf{a} = \begin{pmatrix} b_1 - a_1 \\ \vdots \\ b_n - a_n \end{pmatrix},$$

so the distance between A and B is

$$|\overrightarrow{AB}| = |\mathbf{b} - \mathbf{a}| = \sqrt{(b_1 - a_1)^2 + \dots + (b_n - a_n)^2}.$$

Example 5. Find the distance between the point A with coordinates $(3, -2, 5, 1)$ and the point B with coordinates $(-1, 2, -8, 4)$.

SOLUTION. As $\overrightarrow{AB} = \begin{pmatrix} -4 \\ 4 \\ -13 \\ 3 \end{pmatrix}$, the distance is $|\overrightarrow{AB}| = \sqrt{16 + 16 + 169 + 9} = \sqrt{210}$. $\diamond$

NOTE. Until now, we distinguish an n -dimensional space from $\mathbb{R}^n$ which contains all the position vectors of the points in the space. From now on, we shall blur the difference. So we often refer to $\mathbb{R}^2$ as the plane and the $\mathbb{R}^3$ as the three dimensional space. In our algebra notes, a point in an n -space is usually referred as an n -tuple of coordinates $(a_1, \dots, a_n)$ and the corresponding position

vector is denoted by an n -vector $\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$.

1.4 Lines

Suppose now that we wish to express a line in the plane in terms of coordinate vectors. One form for the equation of a line is $y = mx + b$, where m is the slope of the line and b is the y -intercept.

Since a line is really just a set of points, each point on the line will have a corresponding coordinate vector in $\mathbb{R}^2$. There is a simple way to describe the coordinate vector of each point on a line in the plane.

Let us look firstly at a line through the origin, $y = mx$. One way of reading this equation is to say that the line is the set of all points whose coordinates are of the form (x, mx) . In other words, the set

$$S = \left\{ \mathbf{x} \in \mathbb{R}^2 : \mathbf{x} = x \begin{pmatrix} 1 \\ m \end{pmatrix}, \text{ for some } x \in \mathbb{R} \right\}$$

is the set of all coordinate vectors for points on the line.

There is another way of thinking about this. Let $\mathbf{v}$ be some non-zero displacement in the plane. Let Q be a point in the plane such that $\overrightarrow{OQ} = \mathbf{v}$. Consider the set S of all points P in the plane whose displacement from the origin is $\lambda \mathbf{v}$ for some real number λ . It is not difficult to see that S is the set of all points on the line through O parallel to $\mathbf{v}$ or equivalently S is the line through O and Q . See Figure 19.

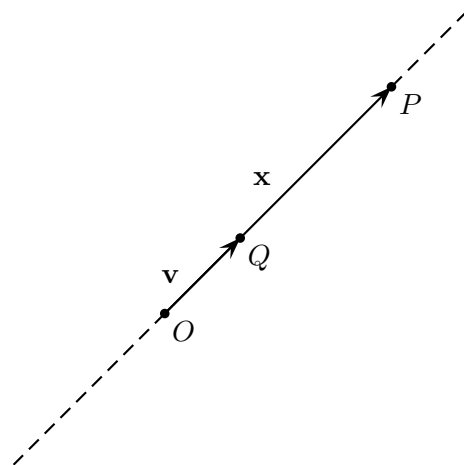


Figure 19: A Line Through O .

For each value of the ‘parameter’ λ , we obtain a point, in the form of a coordinate vector, on the line. It is for this reason that such a description is often called a **parametric vector form** for the line.

This is how we are going to define a ‘line’ spanned by a vector in $\mathbb{R}^n$.

Definition 1. Let $\mathbf{v}$ be any non-zero vector in $\mathbb{R}^n$. The set

$$S = \{ \mathbf{x} \in \mathbb{R}^n : \mathbf{x} = \lambda \mathbf{v}, \text{ for some } \lambda \in \mathbb{R} \}.$$

is the **line** in $\mathbb{R}^n$ **spanned** by $\mathbf{v}$, and we call the expression $\mathbf{x} = \lambda \mathbf{v}$, a **parametric vector form** for this line.

Example 1. Draw a diagram of the line in $\mathbb{R}^2$ spanned by $(1, 2)$, and find a parametric vector form for this line.

SOLUTION. This is the set of all points of the form $\lambda(1, 2) = (\lambda, 2\lambda)$ for some $\lambda \in \mathbb{R}$. In the plane, it is the line going through the origin and the point with coordinates $(1, 2)$. (See Figure 20.)

A parametric vector form is

$$\mathbf{x} = \lambda \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \lambda \in \mathbb{R}.$$

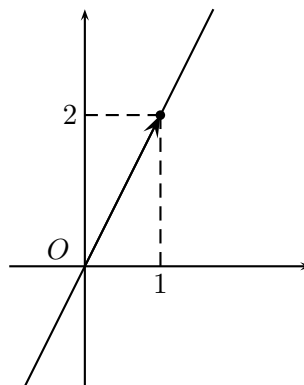


Figure 20.

◇

Example 2. Write a parametric vector form for the line in $\mathbb{R}^4$ spanned by the vector $\begin{pmatrix} 3 \\ 5 \\ -6 \\ 8 \end{pmatrix}$.

A solution is $\mathbf{x} = \lambda \begin{pmatrix} 3 \\ 5 \\ -6 \\ 8 \end{pmatrix}$ for $\lambda \in \mathbb{R}$.

◇

NOTE. A given line does not have a unique parametric vector form. The following are also parametric vector forms of the same line.

$$\mathbf{x} = \lambda \begin{pmatrix} -3 \\ -5 \\ 6 \\ -8 \end{pmatrix} \quad \text{and} \quad \mathbf{x} = \lambda \begin{pmatrix} 9 \\ 15 \\ -18 \\ 24 \end{pmatrix}.$$

On the other hand, the name of the parameter in the expression is not important. For example the forms

$$\mathbf{x} = \mu \begin{pmatrix} 3 \\ 5 \\ -6 \\ 8 \end{pmatrix} \quad \text{for } \mu \in \mathbb{R} \quad \text{and} \quad \mathbf{x} = s \begin{pmatrix} 3 \\ 5 \\ -6 \\ 8 \end{pmatrix} \quad \text{for } s \in \mathbb{R}$$

also represent the same line.

Most lines in the plane, or in three dimensional space do not go through the origin of the coordinate system being used. Consider the line in the plane going through the points A and B . To find the coordinate vector of some point P on this line, we shall calculate the displacement from O to P .

Clearly

$$\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP}.$$

Because A , P and B are collinear, thus $\overrightarrow{AP}$ is parallel to $\overrightarrow{AB}$. That is, $\overrightarrow{AP} = \lambda \overrightarrow{AB}$, for some real number λ . Let $\mathbf{a}$ denote the position vector of A and $\mathbf{x}$ the position vector of P and let $\mathbf{v}$ denote the displacement from A to B . In terms of these vectors, what we have just said is

$$\mathbf{x} = \mathbf{a} + \lambda \mathbf{v}, \quad \text{for some } \lambda \in \mathbb{R}.$$

Again, each value of the parameter λ gives the position vector of a point on the line, and each point on the line can be found by choosing an appropriate value of λ .

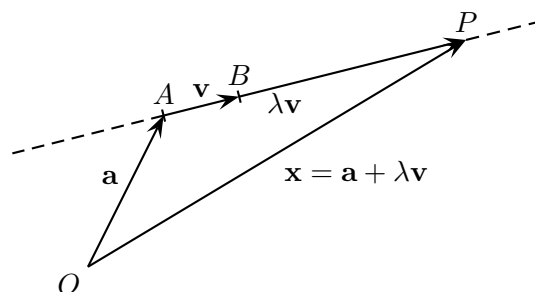


Figure 21.

Definition 2. A **line** in $\mathbb{R}^n$ is any set of vectors of the form

$$S = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} = \mathbf{a} + \lambda \mathbf{v}, \text{ for some } \lambda \in \mathbb{R}\},$$

where $\mathbf{a}$ and $\mathbf{v} \neq \mathbf{0}$ are fixed vectors in $\mathbb{R}^n$. The expression

$$\mathbf{x} = \mathbf{a} + \lambda \mathbf{v}, \text{ for some } \lambda \in \mathbb{R},$$

is a **parametric vector form** of the line through $\mathbf{a}$ parallel to $\mathbf{v}$.

Example 3. Find a parametric vector form of the line through the point $(2, -3, 1, 6)$ parallel to

the vector $\mathbf{v} = \begin{pmatrix} 1 \\ 4 \\ -6 \\ 2 \end{pmatrix}$.

SOLUTION. Each point on the line has coordinate vector of the form

$$\mathbf{x} = \begin{pmatrix} 2 \\ -3 \\ 1 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 4 \\ -6 \\ 2 \end{pmatrix}, \quad \text{for some } \lambda \in \mathbb{R}. \quad (*)$$

◇

NOTE. Once again, a line may have different parametric vector forms. For instance, the following line

$$\mathbf{x} = \begin{pmatrix} 3 \\ 1 \\ -5 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 4 \\ -6 \\ 2 \end{pmatrix}, \quad \text{for some } \lambda \in \mathbb{R}. \quad (**)$$

is one which passes through $(3, 1, -5, 8)$ parallel to $\mathbf{v}$. Note that $\begin{pmatrix} 2 \\ -3 \\ 1 \\ 6 \end{pmatrix} + \begin{pmatrix} 1 \\ 4 \\ -6 \\ 2 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ -5 \\ 8 \end{pmatrix}$, so

$(3, 1, -5, 8)$ is another point on the line with equation $(*)$. Thus both $(*)$ and $(**)$ are parametric vector forms of the same line.

1.4.1 Lines in $\mathbb{R}^2$

We can write the equation of a line in Cartesian form $y = mx + d$ and also in parametric vector form $\mathbf{x} = \mathbf{a} + \lambda \mathbf{v}$. It is often necessary to convert between these two forms.

Conversion from Cartesian form to parametric vector form.

The equation $y = mx + d$ can be converted to parametric vector form as follows:

Suppose $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$ is the position vector of a point on the line and set $x = \lambda$. Then, since $y = m\lambda + d$, we obtain

$$\mathbf{x} = \begin{pmatrix} \lambda \\ m\lambda + d \end{pmatrix} = \begin{pmatrix} 0 \\ d \end{pmatrix} + \begin{pmatrix} \lambda \\ m\lambda \end{pmatrix} = \begin{pmatrix} 0 \\ d \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ m \end{pmatrix}.$$

Thus a parametric vector form for the line is

$$\mathbf{x} = \begin{pmatrix} 0 \\ d \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ m \end{pmatrix}, \quad \text{for } \lambda \in \mathbb{R}.$$

You should think of this as saying that the equation $y = mx + d$ represents a line through the point $(0, d)$ parallel to the vector $\begin{pmatrix} 1 \\ m \end{pmatrix}$.

Example 4. The equation $y = 3x + 2$ has a parametric vector form

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \quad \text{for } \lambda \in \mathbb{R}.$$

The line passes through $(0, 2)$ and is parallel to $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$. ◇

The equation $y = mx + d$ is a special case of the more general linear equation in two variables x and y , which is given by

$$ax + by = c,$$

where a , b , and c are fixed numbers. If $b \neq 0$, this linear equation can easily be converted to the $y = mx + d$ form by dividing by b . Alternatively, the equation $ax + by = c$ can be converted directly to parametric vector form by setting either x or y as a parameter. The following example illustrates this technique.

Example 5. Find parametric vector forms for the lines in $\mathbb{R}^2$ given by

$$\text{i) } 2x - 4y = 6, \quad \text{ii) } 2x = 8, \quad \text{and} \quad \text{iii) } 4y = 8.$$

SOLUTION. i) On setting $y = \lambda$, we have $2x - 4\lambda = 6$, and hence $x = 3 + 2\lambda$. Hence a parametric vector form for the line is

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 3 + 2\lambda \\ \lambda \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \quad \text{for } \lambda \in \mathbb{R}.$$

ii) Note that $2x = 8$ means $x = 4$. So the x has a fixed value but the y value varies. We need to set $y = \lambda$. Hence a parametric vector form for the line is

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 4 \\ \lambda \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad \text{for } \lambda \in \mathbb{R}.$$

iii) In this case the equation fixes $y = 2$, whereas x can have any real value. We therefore set $x = \lambda$ as the parameter and obtain

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \lambda \\ 2 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \text{for } \lambda \in \mathbb{R}.$$

as a parametric vector form. ◇

Conversion from parametric vector form to Cartesian form.

Let us start from a line in parametric vector form

$$\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} + \lambda \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}.$$

Obviously, we can find two different points (x_1, y_1) and (x_2, y_2) on the line by choosing two different values for the parameter, then we can find the Cartesian equation of the line by the two-point form $\frac{y - y_1}{x - x_1} = \frac{y_2 - y_1}{x_2 - x_1}$. However, we would like to use a method which can be easily generalised to the higher dimension cases.

By comparing the components of the vectors on the left side and right side of the parametric vector form, we can express the line as a pair of parametric equations:

$$\begin{cases} x = a_1 + \lambda v_1, \\ y = a_2 + \lambda v_2. \end{cases}$$

Then we can eliminate the parameter λ to get the line in Cartesian form.

Example 6. Find the Cartesian form of each of the following lines which are given in parametric vector form.

$$\text{i) } \mathbf{x} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \text{ for } \lambda \in \mathbb{R}, \quad \text{ii) } \mathbf{x} = \begin{pmatrix} 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \text{ for } \lambda \in \mathbb{R}.$$

SOLUTION. i) By comparing the components, we get the parametric equations $x = 3 + 2\lambda$ and $y = 1 + \lambda$. If we eliminate the parameter λ , we find

$$\lambda = \frac{x - 3}{2} = y - 1,$$

which on rearranging gives

$$y = \frac{x - 1}{2}, \quad \text{or} \quad x - 2y = 1.$$

ii) For the second line, the parametric equations are $x = 3 + 2\lambda$ and $y = 1$. Since y has a fixed value 1 and x can be any real number, the Cartesian form of the line is $y = 1$. $\diamond$

1.4.2 Lines in $\mathbb{R}^3$

A line in $\mathbb{R}^3$ is still of the form $\mathbf{x} = \mathbf{a} + \lambda \mathbf{v}$, but now the vectors $\mathbf{x}, \mathbf{a}$ and $\mathbf{v}$ each have three components or three coordinates.

An alternative Cartesian (or symmetric) form for the equation of a line is sometimes used in engineering. This form can be obtained as follows:

Let $\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$, $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ be vectors in $\mathbb{R}^3$. Then the parametric vector

equation of the line through (a_1, a_2, a_3) parallel to $\begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}$ is

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} + \lambda \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}.$$

Thus the parametric equations of the line are

$$x = a_1 + \lambda v_1, \quad y = a_2 + \lambda v_2, \quad z = a_3 + \lambda v_3.$$

Eliminating the parameter λ yields (if all $v_i \neq 0$), the **Cartesian form**

$$\frac{x - a_1}{v_1} = \frac{y - a_2}{v_2} = \frac{z - a_3}{v_3} (= \lambda).$$

If v_1, v_2 or v_3 is 0, then x, y or z will, respectively, be constant.

Example 7. Find the Cartesian form for the lines

$$\text{i) } \mathbf{x} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 5 \\ 6 \end{pmatrix}, \text{ for } \lambda \in \mathbb{R}, \quad \text{ii) } \mathbf{x} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 0 \\ 6 \end{pmatrix}, \text{ for } \lambda \in \mathbb{R}.$$

SOLUTION. i) Here $x = 2 + 3\lambda$, $y = -3 + 5\lambda$, $z = 1 + 6\lambda$. Thus, eliminating λ gives

$$\frac{x - 2}{3} = \frac{y + 3}{5} = \frac{z - 1}{6}.$$

ii) In this case, $x = 2 + 3\lambda$, $y = -3$, $z = 1 + 6\lambda$. By eliminating λ , the Cartesian form of the line is

$$\frac{x - 2}{3} = \frac{z - 1}{6} \quad \text{and} \quad y = -3.$$

◇

We have seen how to convert a parametric vector form of a line in $\mathbb{R}^3$ to the Cartesian form. To convert a line from Cartesian to the parametric vector form, we find a point on the line and a vector parallel to the line. Alternatively, we can introduce a parameter.

Example 8. Find a parametric vector form for the line

$$\frac{x - 4}{7} = \frac{2y + 3}{2} = -\frac{z}{6}.$$

SOLUTION. METHOD 1. Besides forming parametric equations and then eliminating the parameter, we can rewrite the Cartesian equations as

$$\frac{x - 4}{7} = \frac{y + \frac{3}{2}}{1} = \frac{z}{-6}.$$

Obviously, this is a line through $(4, -\frac{3}{2}, 0)$ parallel to $\begin{pmatrix} 7 \\ 1 \\ -6 \end{pmatrix}$. Hence a parametric vector form of the line is

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ -\frac{3}{2} \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ 1 \\ -6 \end{pmatrix}, \quad \text{for } \lambda \in \mathbb{R}.$$

METHOD 2. Set each term to λ . That is

$$\frac{x-4}{7} = \frac{2y+3}{2} = -\frac{z}{6} = \lambda.$$

By rearranging terms, we get the parametric equations

$$x = 7\lambda + 4, \quad y = \lambda - \frac{3}{2}, \quad z = -6\lambda.$$

In vector form, we have

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 7\lambda + 4 \\ \lambda - \frac{3}{2} \\ -6\lambda \end{pmatrix} = \begin{pmatrix} 4 \\ -\frac{3}{2} \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 7 \\ 1 \\ -6 \end{pmatrix}.$$

Then we shall get the same parametric vector form as the one we have obtained by Method 1. $\diamond$

Example 9. Find a parametric vector form of the line defined by

$$x - 2y = 4 \quad \text{and} \quad z = 1.$$

SOLUTION. Set y to be a parameter λ . Then $x = 4 + 2\lambda$. In vector form, we have

$$\mathbf{x} = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 + 2\lambda \\ \lambda \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}.$$

Hence a parametric vector form of the line is:

$$\mathbf{x} = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} \quad \text{for } \lambda \in \mathbb{R}.$$

$\diamond$

Example 10. Find a parametric vector form of the line through the points $A(1, 3, -4)$ and $B(2, 0, 6)$.

SOLUTION. The line passes through the point $A(1, 3, -4)$ and it is parallel to the vector

$$\overrightarrow{AB} = \begin{pmatrix} 2 \\ 0 \\ 6 \end{pmatrix} - \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix} = \begin{pmatrix} 1 \\ -3 \\ 10 \end{pmatrix}.$$

Hence a parametric vector form of the line is

$$\mathbf{x} = \begin{pmatrix} 1 \\ 3 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -3 \\ 10 \end{pmatrix}, \quad \text{for } \lambda \in \mathbb{R}.$$

$\diamond$

1.4.3 Lines through two given points (in $\mathbb{R}^n$)

The methods used in handling lines in $\mathbb{R}^3$ can be generalised to deal with lines in $\mathbb{R}^n$. We have seen the definition of a parametric vector form of a line in $\mathbb{R}^n$ (Definition 2 on page 22) — $\mathbf{x} = \mathbf{a} + \lambda \mathbf{v}$. We can also use the symmetric form of a line in $\mathbb{R}^n$. In particular, if none of the components of $\mathbf{v}$ is

0, the **symmetric form**, or the **Cartesian form** of a line through $(a_1, \dots, a_n)$ parallel $\mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$ is

$$\frac{x_1 - a_1}{v_1} = \frac{x_2 - a_2}{v_2} = \dots = \frac{x_n - a_n}{v_n}, \quad \text{where } \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}.$$

Example 11. Find the equation of the line in $\mathbb{R}^4$ through the points $A(1, 0, 3, -2)$ and $B(2, 1, 0, -1)$ in parametric vector form and Cartesian form.

SOLUTION. The line is parallel to the vector $\begin{pmatrix} 2 \\ 1 \\ 0 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ 0 \\ 3 \\ -2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ -3 \\ 1 \end{pmatrix}$. Thus the line in parametric vector form is

$$\mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 3 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ -3 \\ 1 \end{pmatrix}, \quad \text{for } \lambda \in \mathbb{R}.$$

The Cartesian form of the line is

$$\frac{x_1 - 1}{1} = \frac{x_2}{1} = \frac{x_3 - 3}{-3} = \frac{x_4 + 2}{1}.$$

◇

More generally, a vector equation of the line joining A to B , with position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, is

$$\mathbf{x} = \mathbf{a} + \lambda(\mathbf{b} - \mathbf{a}) = (1 - \lambda)\mathbf{a} + \lambda\mathbf{b}, \quad \lambda \in \mathbb{R}.$$

Different values of λ correspond to different points on this line. For example, the value $\lambda = \frac{1}{5}$, gives the position vector

$$\mathbf{x} = \frac{4}{5}\mathbf{a} + \frac{1}{5}\mathbf{b} = \mathbf{a} + \frac{1}{5}(\mathbf{b} - \mathbf{a}).$$

This vector can be written as

$$\mathbf{x} = \overrightarrow{OA} + \frac{1}{5} \overrightarrow{AB}.$$

In $\mathbb{R}^2$ or $\mathbb{R}^3$, it is the position vector of the point P which divides AB in the ratio $1 : 4$ (see Figure 22).

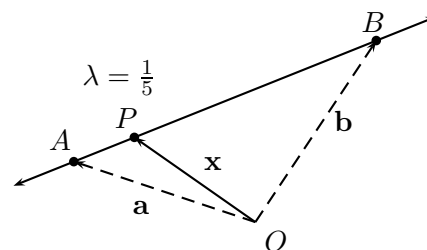


Figure 22.

Generally, in $\mathbb{R}^n$, if P lies between A and B , then $AP : PB = \lambda : 1 - \lambda$, for some λ such that $0 < \lambda < 1$. Therefore, the position vector of P is

$$(1 - \lambda)\mathbf{a} + \lambda\mathbf{b},$$

and the set all the points on the line segment AB in $\mathbb{R}^n$ is

$$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} = (1 - \lambda)\mathbf{a} + \lambda\mathbf{b}, \text{ for } 0 \leq \lambda \leq 1\}.$$

Example 12. Find the medians of the triangle with vertices A $(1, 2, 3)$, B $(-2, 1, -4)$ and C $(4, -1, 2)$.

SOLUTION. A median of a triangle is a line through a vertex and the midpoint of the opposite side. The median through A is therefore the line which passes through the point A and the midpoint P of the line segment BC , as shown in Figure 23.

Let $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ be the position vectors of A , B , C , respectively.

Since P is the midpoint of BC , the coordinate vector of P is

$$\frac{1}{2}(\mathbf{b} + \mathbf{c}) = \frac{1}{2} \left[\begin{pmatrix} -2 \\ 1 \\ -4 \end{pmatrix} + \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} \right] = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

and hence

$$\overrightarrow{AP} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \\ -4 \end{pmatrix}.$$

Since A lies on the median and the vector $\overrightarrow{AP}$ is parallel to the median, we can write

$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 0 \\ -2 \\ -4 \end{pmatrix}, \quad \lambda_1 \in \mathbb{R}$$

as a parametric vector form for the median through A .

Similarly, we can obtain the following parametric vector forms for the medians through B and C :

$$\mathbf{x} = \begin{pmatrix} -2 \\ 1 \\ -4 \end{pmatrix} + \lambda_2 \left[\begin{pmatrix} \frac{5}{2} \\ \frac{1}{2} \\ \frac{5}{2} \end{pmatrix} - \begin{pmatrix} -2 \\ 1 \\ -4 \end{pmatrix} \right] = \begin{pmatrix} -2 \\ 1 \\ -4 \end{pmatrix} + \lambda_2 \begin{pmatrix} \frac{9}{2} \\ -\frac{1}{2} \\ \frac{13}{2} \end{pmatrix}, \quad \lambda_2 \in \mathbb{R};$$

and

$$\mathbf{x} = \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} + \lambda_3 \begin{pmatrix} -\frac{9}{2} \\ \frac{5}{2} \\ -\frac{5}{2} \end{pmatrix}, \quad \lambda_3 \in \mathbb{R}.$$

◇

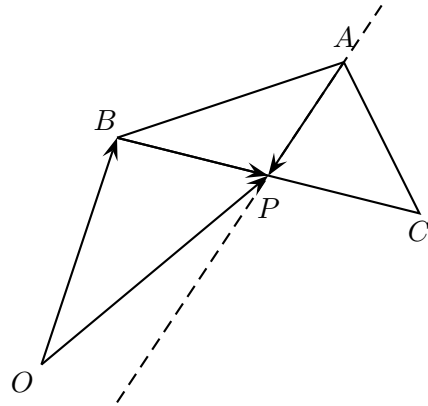


Figure 23.

Example 13. Are the lines $\mathbf{x} = \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} + \lambda_1 \begin{pmatrix} -2 \\ -3 \\ 4 \end{pmatrix}$ and $\mathbf{x} = \begin{pmatrix} 4 \\ 0 \\ 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ \frac{9}{2} \\ -6 \end{pmatrix}$ parallel?

SOLUTION. Since two vectors are parallel if and only if one is a non-zero scalar multiple of the other, and $\begin{pmatrix} 3 \\ \frac{9}{2} \\ -6 \end{pmatrix} = -\frac{3}{2} \begin{pmatrix} -2 \\ -3 \\ 4 \end{pmatrix}$, we can conclude that the two lines are parallel. $\diamond$

1.5 Planes

We have seen that the set of all vectors $\mathbf{x} \in \mathbb{R}^n$ which are scalar multiples of a fixed non-zero vector $\mathbf{v} \in \mathbb{R}^n$, i.e. the set $\{\mathbf{x} \in \mathbb{R}^n: \mathbf{x} = \lambda \mathbf{v}, \text{ for } \lambda \in \mathbb{R}\}$, has a simple geometric interpretation as a line through the origin. In this section we shall show that the set of all sums of scalar multiples of two non-zero non-parallel vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ has a geometric interpretation as a plane through the origin. Furthermore, we shall look at how to find equations for planes.

1.5.1 Linear combination and span

The set of all sums of scalar multiples of some collection of vectors is very important in linear algebra, and therefore has a special name. In the following definitions the word vector is used to mean either a geometric vector or an element of $\mathbb{R}^n$.

Definition 1. A **linear combination** of two vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ is a sum of scalar multiples of $\mathbf{v}_1$ and $\mathbf{v}_2$. That is, it is a vector of the form

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2,$$

where λ_1 and λ_2 are scalars.

Example 1. i) The vector $\begin{pmatrix} -3 \\ 2 \\ 6 \end{pmatrix}$ is a linear combination of the vectors $\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix}$ since

$$\begin{pmatrix} -3 \\ 2 \\ 6 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + (-2) \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix}.$$

ii) The vector $\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$ is not a linear combination of $\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix}$ since there are no real numbers λ_1 and λ_2 such that

$$\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} = \lambda_1 \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ -1 \\ 0 \end{pmatrix}.$$

(Can you prove this?) $\diamond$

Definition 2. The **span** of two vectors $\mathbf{v}_1$ and $\mathbf{v}_2$, written $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$, is the set of all linear combinations of $\mathbf{v}_1$ and $\mathbf{v}_2$. That is, it is the set

$$\text{span}(\mathbf{v}_1, \mathbf{v}_2) = \{\mathbf{x} : \mathbf{x} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2, \text{ for some } \lambda_1, \lambda_2 \in \mathbb{R}\}.$$

Example 2. Given that $\mathbf{v}_1$ and $\mathbf{v}_2$ are non-zero, non-parallel vectors in $\mathbb{R}^3$. What does $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$ look like?

Take $\mathbf{x}$ in $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$. So $\mathbf{x} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2$, for some $\lambda_1, \lambda_2 \in \mathbb{R}$. By the parallelogram law of vector addition, $\mathbf{x}$ lies in the plane passing through O and contains $\mathbf{v}_1$ and $\mathbf{v}_2$.

Conversely, suppose that X is a point with position vector $\mathbf{x}$ on this plane and OX is neither parallel to $\mathbf{v}_1$ nor parallel to $\mathbf{v}_2$. Since $\mathbf{v}_1$ and $\mathbf{v}_2$ are non-zero and non-parallel, there exists a parallelogram $OAXB$, such that OA and OB are parallel to $\mathbf{v}_1$ and $\mathbf{v}_2$ respectively. Note also that, in this case, $\overrightarrow{OA} = \lambda_1 \mathbf{v}_1$ and $\overrightarrow{OB} = \lambda_2 \mathbf{v}_2$, for some real numbers λ_1, λ_2 . By the parallelogram law, we have $\mathbf{x} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2$ which is in $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$.

When OX is parallel to $\mathbf{v}_1$, we have $\mathbf{x} = \lambda_1 \mathbf{v}_1$ which is also in the span. Similarly, $\mathbf{x}$ is also in the span when OX is parallel to $\mathbf{v}_2$.

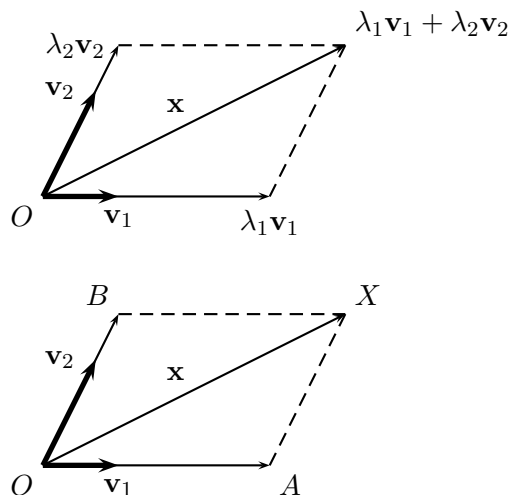


Figure 24.

Hence, $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$ is the plane through the origin and parallel to $\mathbf{v}_1$ and $\mathbf{v}_2$ with equation

$$\mathbf{x} = \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2, \quad \text{for some } \lambda_1, \lambda_2 \in \mathbb{R}.$$

◇

Note that The above construction does not work if $\mathbf{v}_1$ and $\mathbf{v}_2$ are parallel or one of them is $\mathbf{0}$.

Theorem 1. In $\mathbb{R}^3$, the span of any two non-zero non-parallel vectors is a plane through the origin.

We can extend this to $\mathbb{R}^n$ using:

Definition 3. A **plane through the origin** is the span of any two (non-zero) non-parallel vectors.

Example 3. Describe geometrically $\text{span}\left(\begin{pmatrix} 3 \\ -4 \\ 5 \end{pmatrix}, \begin{pmatrix} 4 \\ 2 \\ 7 \end{pmatrix}\right)$, and decide if it is a plane or a line.

SOLUTION. By definition, the span is the set of all vectors $\mathbf{x} \in \mathbb{R}^3$ such that

$$\mathbf{x} = \lambda_1 \begin{pmatrix} 3 \\ -4 \\ 5 \end{pmatrix} + \lambda_2 \begin{pmatrix} 4 \\ 2 \\ 7 \end{pmatrix} \quad \text{for some } \lambda_1, \lambda_2 \in \mathbb{R}.$$

As $\begin{pmatrix} 3 \\ -4 \\ 5 \end{pmatrix}$ is not a scalar multiple of $\begin{pmatrix} 4 \\ 2 \\ 7 \end{pmatrix}$, the span is a plane through the origin parallel to the vectors $\begin{pmatrix} 3 \\ -4 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 2 \\ 7 \end{pmatrix}$. $\diamond$

Example 4. Describe geometrically $\text{span} \left(\begin{pmatrix} 3 \\ 5 \\ 2 \\ 7 \\ 8 \end{pmatrix}, \begin{pmatrix} 6 \\ 10 \\ 4 \\ 14 \\ 16 \end{pmatrix} \right)$ and decide whether it is a plane or a line.

SOLUTION. By definition, the span is the set of all vectors $\mathbf{x} \in \mathbb{R}^5$ satisfying

$$\mathbf{x} = \lambda_1 \begin{pmatrix} 3 \\ 5 \\ 2 \\ 7 \\ 8 \end{pmatrix} + \lambda_2 \begin{pmatrix} 6 \\ 10 \\ 4 \\ 14 \\ 16 \end{pmatrix} \quad \text{for some } \lambda_1, \lambda_2 \in \mathbb{R}.$$

As $\begin{pmatrix} 6 \\ 10 \\ 4 \\ 14 \\ 16 \end{pmatrix} = 2 \begin{pmatrix} 3 \\ 5 \\ 2 \\ 7 \\ 8 \end{pmatrix}$, the span is the line $\mathbf{x} = \lambda \begin{pmatrix} 3 \\ 5 \\ 2 \\ 7 \\ 8 \end{pmatrix}$ for some $\lambda \in \mathbb{R}$. $\diamond$

1.5.2 Parametric vector form of a plane

The span of two non-zero non-parallel vectors is a plane through the origin. For planes that do not pass through the origin we use a similar approach to that we use for lines.

Consider first three vectors $\mathbf{a}$, $\mathbf{v}_1$, $\mathbf{v}_2 \in \mathbb{R}^3$. We shall assume that $\mathbf{v}_1$ and $\mathbf{v}_2$ are non-zero non-parallel. Let

$$S = \{\mathbf{x} \in \mathbb{R}^3 : \mathbf{x} = \mathbf{a} + \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 \text{ for } \lambda_1, \lambda_2 \in \mathbb{R}\}.$$

This set consists of all the vectors in $\mathbb{R}^3$ for which $\mathbf{x} - \mathbf{a} \in \text{span}(\mathbf{v}_1, \mathbf{v}_2)$. That is, S contains all the points $\mathbf{x}$ for which $\mathbf{x} - \mathbf{a}$ lies in the plane through the origin spanned by $\mathbf{v}_1$ and $\mathbf{v}_2$. Thus S is the plane in $\mathbb{R}^3$ which passes through the point with position vector $\mathbf{a}$, and is parallel to $\mathbf{v}_1$ and $\mathbf{v}_2$. The picture is shown in Figure 25.

So in $\mathbb{R}^3$,

$$\mathbf{x} = \mathbf{a} + \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2, \quad \text{for } \lambda_1, \lambda_2 \in \mathbb{R}$$

is an equation of the plane passing through $\mathbf{a}$ and parallel to the non-zero non-parallel vectors $\mathbf{v}_1, \mathbf{v}_2$. This is called a parametric vector form of the plane.

As before we shall use this as a way of defining what we mean by a plane in $\mathbb{R}^n$.

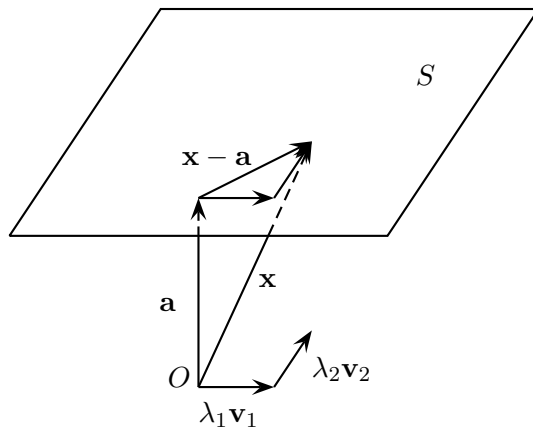


Figure 25: $\mathbf{x} = \mathbf{a} + \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2$.

Definition 4. Let $\mathbf{a}$, $\mathbf{v}_1$ and $\mathbf{v}_2$ be fixed vectors in $\mathbb{R}^n$, and suppose that $\mathbf{v}_1$ and $\mathbf{v}_2$ are not parallel. Then the set

$$S = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} = \mathbf{a} + \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2, \text{ for some } \lambda_1, \lambda_2 \in \mathbb{R}\},$$

is the **plane** through the point with position vector $\mathbf{a}$, parallel to the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$. The expression

$$\mathbf{x} = \mathbf{a} + \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2, \text{ for } \lambda_1, \lambda_2 \in \mathbb{R},$$

is called a **parametric vector form** of the plane.

NOTE. For a given plane, there is not a unique parametric vector form. For example, a plane parallel to $\mathbf{v}_1$ and $\mathbf{v}_2$ is also parallel to $\mathbf{v}_1 + \mathbf{v}_2$ and $\mathbf{v}_1 - \mathbf{v}_2$.

From the definition, a point P with position vector $\mathbf{p} \in \mathbb{R}^n$ is said to lie on the plane through $\mathbf{a}$ parallel to $\mathbf{v}_1, \mathbf{v}_2$, if there exist real numbers λ_1, λ_2 such that $\mathbf{p} = \mathbf{a} + \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2$.

Example 5. Find a parametric vector form for the plane through the point $(1, 3, 0)$, parallel to the vectors $\begin{pmatrix} 4 \\ 2 \\ 9 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 3 \\ 4 \end{pmatrix}$.

SOLUTION. One answer (there are many possible!) is

$$\mathbf{x} = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} + \lambda_1 \begin{pmatrix} 4 \\ 2 \\ 9 \end{pmatrix} + \lambda_2 \begin{pmatrix} -2 \\ 3 \\ 4 \end{pmatrix}, \quad \text{for } \lambda_1, \lambda_2 \in \mathbb{R}.$$

◇

Example 6. Describe geometrically the set

$$\left\{ \mathbf{x} \in \mathbb{R}^4 : \mathbf{x} = \begin{pmatrix} 1 \\ 4 \\ 2 \\ 1 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} -3 \\ 2 \\ -4 \\ 1 \end{pmatrix}, \quad \text{for } \lambda_1, \lambda_2 \in \mathbb{R} \right\}.$$

SOLUTION. Since $\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} -3 \\ 2 \\ -4 \\ 1 \end{pmatrix}$ are not parallel (i.e. they are not scalar multiples of each

other), this set is the plane in $\mathbb{R}^4$ through the point $(1, 4, 2, 1)$ parallel to the vectors $\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}$ and

$$\begin{pmatrix} -3 \\ 2 \\ -4 \\ 1 \end{pmatrix}.$$

◇

Example 7. Let A , B and C be three points in $\mathbb{R}^n$ whose coordinate vectors with respect to some coordinate system are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively. Find a parametric vector form for the plane which passes through A , B and C .

SOLUTION. Consider the plane through A parallel to $\overrightarrow{AB}$ and $\overrightarrow{AC}$. This has a parametric vector form:

$$\mathbf{x} = \mathbf{a} + \lambda_1 \overrightarrow{AB} + \lambda_2 \overrightarrow{AC} = \mathbf{a} + \lambda_1(\mathbf{b} - \mathbf{a}) + \lambda_2(\mathbf{c} - \mathbf{a}), \quad \text{for } \lambda_1, \lambda_2 \in \mathbb{R}.$$

When $\lambda_1 = \lambda_2 = 0$ we have $\mathbf{x} = \mathbf{a}$. When $\lambda_1 = 1$, $\lambda_2 = 0$ we have $\mathbf{x} = \mathbf{b}$, and when $\lambda_1 = 0$, $\lambda_2 = 1$ we have $\mathbf{x} = \mathbf{c}$. Hence it is the plane through A , B and C .

◇

Example 8. Find a parametric vector form for the plane through the three points $A(2, -1, 3)$, $B(-1, 4, 4)$ and $C(3, -1, 2)$.

SOLUTION. The plane is parallel to $\overrightarrow{AB} = \begin{pmatrix} -1 \\ 4 \\ 4 \end{pmatrix} - \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} -3 \\ 5 \\ 1 \end{pmatrix}$ and $\overrightarrow{BC} = \begin{pmatrix} 4 \\ -5 \\ -2 \end{pmatrix}$, and passes through the point $A(2, -1, 3)$. Hence a parametric vector form for the plane is

$$\mathbf{x} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} -3 \\ 5 \\ 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 4 \\ -5 \\ -2 \end{pmatrix}, \quad \text{for } \lambda_1, \lambda_2 \in \mathbb{R}.$$

◇

1.5.3 Cartesian form of a plane in $\mathbb{R}^3$

In $\mathbb{R}^3$, any equation of the form

$$ax_1 + bx_2 + cx_3 = d$$

represents plane when not all a, b, c are zero. In $\mathbb{R}^n$, $n > 3$, if not all $a_1, a_2, \dots, a_n$ are zero, an equation of the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = d$$

is called a hyperplane.

Example 9. Find a parametric vector form for the plane $x_1 - 3x_2 + 4x_3 = 4$, and hence find a point on the plane and two vectors parallel to the plane.

SOLUTION. We set $x_2 = \lambda_1$ and $x_3 = \lambda_2$. Then solving for x_1 gives $x_1 = 4 + 3\lambda_1 - 4\lambda_2$. Hence

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 4 + 3\lambda_1 - 4\lambda_2 \\ \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 0 \end{pmatrix} + \lambda_1 \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} -4 \\ 0 \\ 1 \end{pmatrix},$$

for $\lambda_1, \lambda_2 \in \mathbb{R}$. Thus the plane passes through the point $(4, 0, 0)$ and is parallel to $\begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ 0 \\ 1 \end{pmatrix}$. ◇

As we did for lines, we can find a Cartesian equation for a plane by eliminating the parameters λ_1, λ_2 from the parametric vector form.

Example 10. Find the Cartesian form of the plane $\mathbf{x} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$ for $\lambda, \mu \in \mathbb{R}$.

SOLUTION. Let $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$. By comparing the components, we get the following parametric equations

$$x_1 = 1 + \lambda - \mu, \quad x_2 = 1 + \lambda \quad \text{and} \quad x_3 = 2 + \lambda + 2\mu.$$

Hence from the second equation, we get $\lambda = x_2 - 1$. Substituting this into the first equation, we have $x_1 = 1 + (x_2 - 1) - \mu$, so $\mu = x_2 - x_1$. Now substitute these values of λ and μ into the third parametric equation. The Cartesian form of the plane is, therefore,

$$x_3 = 2 + (x_2 - 1) + 2(x_2 - x_1) \quad \text{or} \quad 2x_1 - 3x_2 + x_3 = 1.$$

◇

1.6 Vectors and Maple

This section shows how to handle vectors in Maple. The capacity to plot simple curves in the plane is illustrated. The 3-dimensional plot facility needs an x-terminal.

The following instruction loads the linear algebra package:

```
with(LinearAlgebra);
```

Before attempting anything involving vectors you may wish to plot a piece of a parabola parametrically with:

```
plot([2*t, t^2, t=-4..4]);
```

To enter the vectors $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 0 \\ 9 \end{pmatrix}$, we type:

```
a:=<1,2,3>;
```

```
b:=<2,6,9>;
```

The command

```
a[2];
```

selects the second component of $\mathbf{a}$.

Since $\mathbf{a}$ and $\mathbf{b}$ have the same number of components, they may be added with:

```
a+b;
```

Multiplication by a scalar is handled analogously:

```
4*a;
```

Now start another example with

```
a:=<2,3>; b:=<1,4>;
```

If $\mathbf{a}$ and $\mathbf{b}$ are two vectors, a parametric vector form for the line through $\mathbf{a}$ and $\mathbf{b}$ is given by:

```
v:=b-a; x:=a+t*v;
```

To plot the line segment from $t = -2$ to $t = 3$, use:

```
plot([x[1],x[2],t=-2..3]);
```

Now let $\mathbf{a}$, $\mathbf{v}_1$ and $\mathbf{v}_2$ be 3-vectors. The plane $\mathbf{x} = \mathbf{a} + s\mathbf{v}_1 + t\mathbf{v}_2$ can be entered as:

```
x:=a+s*v1+t*v2;
```

To find out about the three-dimensional plot facility, just type:

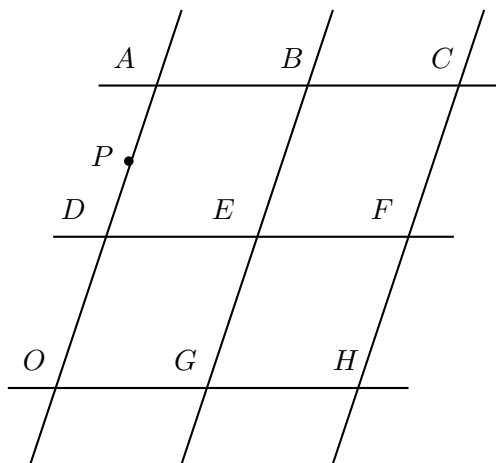
```
?plot3d
```

Problems for Chapter 1

Questions marked with [R] are routine, [H] harder, [M] Maple and [X] are for MATH1141 only. You should make sure that you can do the easier questions before you tackle the more difficult questions. Questions marked with a [V] have video solutions available on Moodle.

Problems 1.1 : Vector quantities

1. [R][V] Given that ABC , DEF , and OGH are equally spaced parallel lines, as are ADO , BEG and CFH . P is the mid point of AD .



If $\overrightarrow{OH} = \mathbf{h}$ and $\overrightarrow{OA} = \mathbf{a}$, express the following in terms of $\mathbf{a}$ and $\mathbf{h}$.

- a) $\overrightarrow{OC}$, b) $\overrightarrow{HA}$, c) $\overrightarrow{GC}$, d) $\overrightarrow{OP}$, e) $\overrightarrow{GP}$.

2. [R] Simplify

- a) $\overrightarrow{AB} - \overrightarrow{OB} + \overrightarrow{OA}$, b) $\overrightarrow{AB} - \overrightarrow{CB} + 3\overrightarrow{DA} + 3\overrightarrow{CD}$.

3. [R] Express each of the following in terms of $\mathbf{a}$ and $\mathbf{b}$.

- a) $3(2\mathbf{a} + \mathbf{b}) - 2(5\mathbf{a} - \mathbf{b})$,
 b) $2(p\mathbf{a} + q\mathbf{b}) + 3(r\mathbf{a} - s\mathbf{b})$ where $p, q, r, s \in \mathbb{R}$.

4. [R] Let ABC be a triangle with $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$, $\overrightarrow{OC} = \mathbf{c}$ where O is the origin.

- a) If M is the midpoint of the line segment AB and P is the midpoint of the line segment CB express the vectors $\overrightarrow{OM}$ and $\overrightarrow{OP}$ in terms of $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$.
 b) Show that $\overrightarrow{MP}$ is parallel to $\overrightarrow{AC}$ and has half its length.

5. [H][V] Given a convex quadrilateral $ABCD$, prove, using vectors, that the quadrilateral formed by joining the midpoints of AB , BC , CD , and DA is a parallelogram.

6. [R] Use geometric vectors to solve the following problems. In each case, draw a careful picture and then use trigonometry to find the answer. If your picture is accurate, you may wish to use a ruler and protractor to confirm your result.
- An ant crawls 10 cm due east in a straight line and then crawls 5 cm northeast in a straight line. What is the ant's final displacement from its starting point?
 - An ant is standing at the western edge of a moving walkway which is moving at 12 cm per sec in the direction due South. The ant starts to walk at 5 cm/sec across the walkway in the direction perpendicular to its edge. If the walkway is 40 cm wide, find the displacement of the ant from its starting point just as it steps off the walkway.
 - An observer on a wharf sees a yacht sailing at 15 km per hour southeast. A sailor on the yacht is watching a container ship and sees it sailing at 25 km per hour due north. What is the velocity of the container ship as seen by the observer on the wharf?
 - A rower is rowing across a river. His rowing speed is 2 km per hour and there is a current flowing in the river at 1 km per hour. Find the direction that the rower must row to go directly across the river. If the river is 300 metres wide, how long will it take him to cross the river?
7. [X] Town B is 18 km N 18° W from Town A . Town C is 25 km N 36° E from B . Town D is 20 km S 72° E from C . Town F is 15 km S 25° W from D . What are the distance and bearing of Town F from Town A ?
8. [X] Let O, A, B, C be points in a plane. Suppose that X is the midpoint of BC , Y is the point on AC with $AY : YC = 3 : 1$, and Z is the point on AB with $AZ : ZB = 3 : 1$. Let $\overrightarrow{OA} = \mathbf{a}$, $\overrightarrow{OB} = \mathbf{b}$, and $\overrightarrow{OC} = \mathbf{c}$.
- Write down the vectors $\overrightarrow{OX}$, $\overrightarrow{OY}$ and $\overrightarrow{OZ}$ in terms of $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$.
 - Let T be the point on AX with $AT : TX = 6 : 1$. Write $\overrightarrow{OT}$ in terms of $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$.
 - Show that T lies on both CZ and YB .

Problems 1.2 : Vector quantities and $\mathbb{R}^n$

9. [R][V] Find $\mathbf{u} + 2\mathbf{v} - 3\mathbf{w}$ (if possible) given that

- $\mathbf{u} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$, $\mathbf{w} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$;
- $\mathbf{u} = \begin{pmatrix} 2 \\ 3 \\ 3 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} 7 \\ 6 \\ -1 \end{pmatrix}$, $\mathbf{w} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$;
- $\mathbf{u} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} -1 \\ 2 \\ 1 \\ 0 \end{pmatrix}$, $\mathbf{w} = \begin{pmatrix} 2 \\ 1 \\ 3 \\ 1 \end{pmatrix}$;

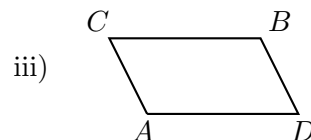
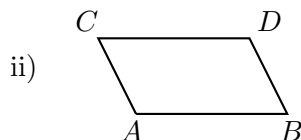
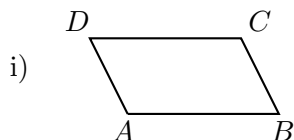
d) $\mathbf{u} = \begin{pmatrix} -1 \\ 3 \\ 5 \end{pmatrix}, \mathbf{v} = \begin{pmatrix} 10 \\ 2 \\ -3 \end{pmatrix}, \mathbf{w} = \begin{pmatrix} 0 \\ 0 \end{pmatrix};$

e) $\mathbf{u} = 2\mathbf{i} + 3\mathbf{j} - 2\mathbf{k}, \mathbf{v} = \mathbf{i} - 2\mathbf{j} + \mathbf{k}, \mathbf{w} = -\mathbf{i} + \mathbf{j} - \mathbf{k}.$

10. [R] A car travels 3km due North then 5km Northeast. Use coordinate vectors to find the distance and direction from the starting point.
11. [R] Solve Problems 6 (a) and (b) using coordinate vectors.
12. [R] Suppose that $\mathbf{v} = \begin{pmatrix} a_1 \\ b_1 \\ c_1 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} a_2 \\ b_2 \\ c_2 \end{pmatrix}$ are vectors in $\mathbb{R}^3$; λ and μ are real numbers. Prove the scalar distributive law $(\lambda + \mu)\mathbf{v} = \lambda\mathbf{v} + \mu\mathbf{v}$ and the vector distributive law $\lambda(\mathbf{v} + \mathbf{w}) = \lambda\mathbf{v} + \lambda\mathbf{w}$.
13. [X] Prove the associative law of vector addition in $\mathbb{R}^n$. (Proposition 2 on page 12).
14. [X] Prove Proposition 3 on page 14.

Problems 1.3 : $\mathbb{R}^n$ and analytic geometry

15. [R][V] Let $\mathbf{v} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$. Draw coordinate axes and mark in the points whose coordinate vectors are $\mathbf{v}$, $-\mathbf{v}$, $\mathbf{w}$, $\mathbf{v} + \mathbf{w}$, $2\mathbf{v}$ and $\mathbf{v} - \mathbf{w}$.
16. [R] Given the following points A , B , C and D , are the vectors $\overrightarrow{AB}$ and $\overrightarrow{CD}$ parallel?
- a) $A = (1, 2, 3)$, $B = (-2, 3, 4)$, $C = (-3, -4, 7)$, $D = (4, -6, -9)$;
 b) $A = (3, 2, 5)$, $B = (5, -3, -6)$, $C = (-2, 3, 7)$, $D = (0, -2, -4)$;
 c) $A = (12, -4, 6)$, $B = (2, 6, -4)$, $C = (5, -2, 9)$, $D = (0, 3, 4)$.
- Do any of these sets of 4 points form a parallelogram?
17. [R] a) Prove that $A(1, 2, 1)$, $B(4, 7, 8)$, $C(6, 4, 12)$ and $D(3, -1, 5)$ are the vertices of a parallelogram.
 b) Which of the following rough sketches correctly represents the parallelogram from part (a)?



18. [R] Show that the points $A(1, 2, 3)$, $B(3, 8, 1)$, $C(7, 20, -3)$ are collinear.

19. **[R][V]** Show that the points $A(-1, 2, 1)$, $B(4, 6, 3)$, $C(-1, 2, -1)$ are not collinear.

20. **[R]** Show that the points A, B, C in $\mathbb{R}^3$ with coordinate vectors

$$\mathbf{a} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 0 \\ 1 \\ 4 \end{pmatrix} \quad \text{and} \quad \mathbf{c} = \begin{pmatrix} 6 \\ -5 \\ -2 \end{pmatrix}$$

are collinear.

21. **[H]** If $A(-1, 3, 4)$, $B(4, 6, 3)$, $C(-1, 2, 1)$ and D are the vertices of a parallelogram, find all the possible coordinates for the point D .

22. **[H]** Consider three non-collinear points D, E, F in $\mathbb{R}^3$ with coordinate vectors $\mathbf{d}$, $\mathbf{e}$ and $\mathbf{f}$. There are exactly 3 points in $\mathbb{R}^3$ which, taken one at a time with D, E and F , form a parallelogram. Calculate vector expressions for the three points.

23. **[R][V]** Let $A = (2, 3, -1)$ and $B = (4, -5, 7)$. Find the midpoint of A and B . Find the point Q on the line through A and B such that B lies between A and Q and BQ is three times as long as AB .

24. **[R]** The coordinate vectors, relative to the origin O , of the points A and B are respectively $\mathbf{a}$ and $\mathbf{b}$. State, in terms of $\mathbf{a}$ and $\mathbf{b}$, the position vector of the point T which lies on AB and is such that $\overrightarrow{AT} = 2\overrightarrow{TB}$.

25. **[R]** List the standard basis vectors for $\mathbb{R}^5$.

26. **[R]** For each of the following vectors, find its length and find a vector of length one (“unit” vectors) parallel to it.

$$\mathbf{a} = \begin{pmatrix} 4 \\ -4 \\ 2 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 2 \\ 1 \\ 0 \\ 3 \end{pmatrix}, \quad \mathbf{c} = \begin{pmatrix} 4 \\ 0 \\ 1 \\ -2 \\ 0 \end{pmatrix}.$$

27. **[R][V]** Find the distances between each of the following pairs of points with coordinate vectors:

$$\text{a) } \begin{pmatrix} 8 \\ -4 \\ 2 \end{pmatrix}, \begin{pmatrix} -6 \\ 1 \\ 0 \end{pmatrix}; \quad \text{b) } \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 5 \\ -7 \\ -7 \end{pmatrix}; \quad \text{c) } \begin{pmatrix} 3 \\ 0 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} -2 \\ 6 \\ 1 \\ 3 \end{pmatrix}.$$

28. **[R]** A triangle has vertices A, B and C which have coordinate vectors $\begin{pmatrix} 4 \\ 1 \\ 7 \end{pmatrix}$, $\begin{pmatrix} 7 \\ -4 \\ 6 \end{pmatrix}$ and $\begin{pmatrix} 6 \\ 2 \\ 8 \end{pmatrix}$

respectively. Find the lengths of the sides of the triangle and deduce that the triangle is right-angled.

29. [H] Construct a cube in $\mathbb{R}^3$ with the length of each edge 1. Show that the face diagonal has length $\sqrt{2}$ and the long diagonal $\sqrt{3}$. Try to generalise this idea to $\mathbb{R}^4$ and show that there are now diagonals of length $\sqrt{2}$, $\sqrt{3}$ and 2. How many vertices does a 4-cube have?
30. [X] Find 10 vectors in $\mathbb{R}^{10}$, each pair of which is $5\sqrt{2}$ apart. Can you now find an 11th such vector?

Problems 1.4 : Lines

31. [R][V] Find the coordinate vector for the displacement vector $\overrightarrow{AB}$ and parametric vector forms for the lines through the points A and B with coordinates
- a) $A(1, 2), B(2, 7)$; b) $A(1, 2, -1), B(-1, -1, 5)$;
 c) $A(1, 2, 1), B(7, 2, 3)$; d) $A(1, 2, -1, 3), B(-1, 3, 1, 1)$.
32. [R] Does the point $(3, 5, 7)$ lie on the line $\mathbf{x} = \begin{pmatrix} -1 \\ 3 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 2 \\ 1 \end{pmatrix}$?
33. [R] Find parametric vector forms for the following lines in $\mathbb{R}^2$:
- a) $y = 3x + 4$; b) $3x + 2y = 6$; c) $y = -7x$;
 d) $y = 4$; e) $x = -2$.
- In each case indicate the direction of the line and a point through which the line passes.
34. [R] Find a parametric vector form and a Cartesian form for each of the following lines
- a) through the points $(-4, 1, 3)$ and $(2, 2, 3)$;
 b) through $(1, 2, -3)$ parallel to the vector $\begin{pmatrix} 4 \\ -5 \\ 6 \end{pmatrix}$;
 c) through $(1, -1, 1)$ parallel to the line joining the points $(2, 2, 1)$ and $(7, 1, 3)$;
 d) through $(1, 0, 0)$ parallel to the line joining the points $(3, 2, -1)$ and $(3, 5, 2)$.
35. [R] Let A, B, P be points in $\mathbb{R}^3$ with position vectors

$$\mathbf{a} = \begin{pmatrix} 7 \\ -2 \\ 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ -5 \\ 0 \end{pmatrix} \quad \text{and} \quad \mathbf{p} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$

Let Q be the point on AB such that $AQ = \frac{2}{3}AB$.

- a) Find $\mathbf{q}$, the position vector of Q .
 b) Find the parametric vector equation of the line that passes through P and Q .

36. [R] Decide whether each of the following statements is true or false.

- a) The lines $y = 3x - 4$ and $\mathbf{x} = \begin{pmatrix} 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 12 \end{pmatrix}$ are parallel.
- b) The lines $\mathbf{x} = \begin{pmatrix} 3 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ 4 \end{pmatrix}$ and $2x + 3y = 8$ are parallel.
- c) The lines $\mathbf{x} = \begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 10 \\ 2 \\ 8 \end{pmatrix}$ and $\frac{x+10}{5} = y-7 = \frac{z+3}{4}$ are parallel.
- d) The line $\mathbf{x} = \begin{pmatrix} 3 \\ -2 \\ 7 \end{pmatrix} + \lambda \begin{pmatrix} 10 \\ 0 \\ -4 \end{pmatrix}$ and the line

$$\frac{x+10}{5} = \frac{z+3}{-2} \quad \text{and} \quad y = -5$$

are parallel.

37. [X] Suppose A and B are points with coordinate vectors $\mathbf{a}$ and $\mathbf{b}$, respectively. Write down a parametric vector form for

- a) the line segment AB .
- b) the ray from B through A .
- c) all points P which lie on the line through A and B such that A is between P and B .
- d) all points Q which lie on the line through A and B and are closer to B than A .

38. [X] Give a geometric interpretation of the following sets. In each set, $\lambda \in \mathbb{R}$.

- a) $S = \left\{ \mathbf{x} : \mathbf{x} = \begin{pmatrix} 1 \\ 3 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} -3 \\ 1 \\ 7 \end{pmatrix} \text{ for } 0 \leq \lambda \leq 1 \right\}.$
- b) $S = \left\{ \mathbf{x} : \mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 4 \\ 0 \\ -7 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 5 \\ 9 \\ 3 \\ 6 \end{pmatrix} \text{ for } -1 \leq \lambda \leq 5 \right\}.$
- c) $S = \left\{ \mathbf{x} : \mathbf{x} = \lambda \begin{pmatrix} 6 \\ -2 \\ 7 \\ 2 \\ -1 \\ 5 \end{pmatrix} + (1-\lambda) \begin{pmatrix} 0 \\ 4 \\ 8 \\ 3 \\ -5 \\ 4 \end{pmatrix} \text{ for } 0 \leq \lambda \leq 1 \right\}.$
- d) $S = \left\{ \mathbf{x} : \mathbf{x} = \begin{pmatrix} 1 \\ 4 \\ -6 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 0 \\ -1 \\ 5 \end{pmatrix} \text{ for } \lambda \geq 0 \right\}.$

$$\text{e) } S = \left\{ \mathbf{x} : \mathbf{x} = \begin{pmatrix} 3 \\ 1 \\ -4 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ -2 \\ 7 \end{pmatrix} \text{ for } |\lambda| \geq 2 \right\}.$$

Problems 1.5 : Planes

39. [R] Find a parametric vector form for the planes passing through the points

$$\text{a) } (0, 0, 0), (3, -1, 2), (1, 4, -6); \quad \text{b) } (1, 4, -2), (2, 6, 4), (1, -10, 3).$$

40. [R] For each of the following sets of vectors, decide if the set is a line or a plane, give a point on the line or plane, and give vectors parallel to the line or plane, i.e., geometrically describe the sets.

$$\text{a) } S = \left\{ \mathbf{x} : \mathbf{x} = \lambda_1 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda_2 \begin{pmatrix} -2 \\ 3 \\ 4 \end{pmatrix} \text{ for } \lambda_1, \lambda_2 \in \mathbb{R} \right\}.$$

$$\text{b) } S = \left\{ \mathbf{x} : \mathbf{x} = \begin{pmatrix} 3 \\ 1 \\ 2 \\ 4 \end{pmatrix} + \lambda_1 \begin{pmatrix} -2 \\ 1 \\ 3 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 4 \\ -2 \\ -6 \\ -4 \end{pmatrix} \text{ for } \lambda_1, \lambda_2 \in \mathbb{R} \right\}.$$

$$\text{c) } \text{span} \left(\begin{pmatrix} 3 \\ 2 \\ 1 \\ 2 \end{pmatrix}, \begin{pmatrix} -9 \\ -6 \\ -3 \\ -6 \end{pmatrix} \right).$$

$$\text{d) } S = \left\{ \mathbf{x} : \mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \mathbf{y} \text{ for } \mathbf{y} \in \text{span} \left(\begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 8 \\ 2 \\ 4 \end{pmatrix} \right) \right\}.$$

41. [R][V] Find parametric vector forms for the planes

$$\text{a) } \text{through the point } (1, 2, 3) \text{ parallel to } \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \text{ and } \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix};$$

$$\text{b) } \text{through the points } (3, 1, 4), (-1, 2, 4), (6, 7, -2);$$

$$\text{c) } \text{through the points } (-2, 4, 1, 6), (3, 2, 6, -1), (1, 4, 0, 0);$$

$$\text{d) } 4x_1 - 3x_2 + 6x_3 = 12, \text{ where } \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3;$$

$$\text{e) } 5x_2 - 6x_3 = 5, \text{ where } \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3;$$

f) through the point $(1, 2, 3, 4)$ parallel to the lines $\mathbf{x} = \begin{pmatrix} -3 \\ 1 \\ 2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 0 \\ -4 \\ 5 \end{pmatrix}$ and

$$\frac{x_1 - 5}{7} = \frac{x_2 + 6}{2} = \frac{x_3 - 2}{-3} = \frac{x_4 + 1}{-5}.$$

42. [R] Find parametric vector forms to describe the following planes in $\mathbb{R}^3$.

- a) $x_1 + x_2 + x_3 = 0$. b) $3x_1 - x_2 + 4x_3 = 12$.
 c) $x_2 + 6x_3 = -1$. d) $x_3 = 2$.

43. [H] Show that the line $\mathbf{x} = t \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$

- a) lies on the plane $4x - 5y - z = 0$, and b) is parallel to the plane $3x - 3y - z = 2$.

44. [H] a) Find the intersection of the line $\mathbf{x} = \begin{pmatrix} 2+t \\ 3-t \\ 4t \end{pmatrix}$ and the plane $2x + 3y + z = 16$.

b) Find the intersection of the line $\mathbf{x} = \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix}$ and the plane $9x + 4y - z = 0$.

45. [H] a) Write the plane $\mathbf{x} = \begin{pmatrix} -3 \\ 2 \\ 6 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 4 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 0 \\ 3 \end{pmatrix}$ in Cartesian form.

b) Write the plane $\mathbf{x} = \begin{pmatrix} 6 \\ -1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 6 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}$ in Cartesian form.

46. [H] Consider the line $\frac{x-3}{-2} = \frac{y+2}{3} = z-1$ and the plane $2x + y + 3z = 23$ in $\mathbb{R}^3$.

- a) Find a parametric vector form for the line.
 b) Hence find where the line meets the plane.

47. [H] Let ℓ be the line $\frac{x-6}{5} = \frac{y-4}{2} = \frac{z-1}{-2}$ in $\mathbb{R}^3$.

- a) Express the line ℓ in parametric vector form.
 b) Find the coordinates of the point where ℓ meets the plane $2x + y - z = 1$.

48. [X] The following sets of points represent simple geometric figures in a plane. λ_1 and λ_2 are real numbers. For each problem draw a sketch in the (λ_1, λ_2) plane and a second sketch in $\mathbb{R}^2, \mathbb{R}^3$ or $\mathbb{R}^4$ (!) as appropriate. For each problem identify the geometric shape.

- a) $S = \left\{ \mathbf{x} : \mathbf{x} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad \text{for } 0 \leq \lambda_1 \leq 1, \ 0 \leq \lambda_2 \leq 1 \right\}.$
- b) $S = \left\{ \mathbf{x} : \mathbf{x} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 2 \\ 3 \end{pmatrix} \quad \text{for } 0 \leq \lambda_1 \leq 1, \ 0 \leq \lambda_2 \leq \lambda_1 \right\}.$
- c) $S = \left\{ \mathbf{x} : \mathbf{x} = \lambda_1 \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix} \quad \text{for } 0 \leq \lambda_1 \leq 6, \ 0 \leq \lambda_2 \leq 8 \right\}.$
- d) $S = \left\{ \mathbf{x} : \mathbf{x} = \lambda_1 \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 4 \\ -2 \\ 3 \end{pmatrix} \quad \text{for } 0 \leq \lambda_1 \leq 6, \ 0 \leq \lambda_2 \leq \lambda_1 \right\}.$
- e) $S = \left\{ \mathbf{x} : \mathbf{x} = \lambda_1 \begin{pmatrix} 2 \\ 1 \\ -2 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 4 \\ -2 \\ 3 \\ -1 \end{pmatrix} \quad \text{for } 0 \leq \lambda_1 \leq 6, \ 0 \leq \lambda_1 \leq \lambda_2 \right\}.$

49. [X] Write down the sets of points corresponding to the following:

- a) A “parallelogram” with the three vertices $A(1, 3, 4, 2)$, $B(-2, 1, 0, 5)$ and $C(-4, 0, 6, 8)$.
Hint: Look at Question 48 a), and assume B and C are adjacent to A .
- b) The triangle with the three vertices given in part a) of this question.
Hint: Look at Question 48 b).
- c) All three parallelograms which have the three vertices given in part a).

50. [X] Given two planes in $\mathbb{R}^n$, $n \geq 3$:

$$\mathbf{x} = \mathbf{a} + s_1 \mathbf{u}_1 + s_2 \mathbf{u}_2 \quad \text{and} \quad \mathbf{x} = \mathbf{b} + t_1 \mathbf{v}_1 + t_2 \mathbf{v}_2,$$

for $s_1, s_2, t_1, t_2 \in \mathbb{R}$. These two planes are said to be parallel if $\text{span}(\mathbf{u}_1, \mathbf{u}_2) = \text{span}(\mathbf{v}_1, \mathbf{v}_2)$.

Consider the pair of planes in $\mathbb{R}^4$ with equations

$$\mathbf{x} = s_1 \mathbf{e}_1 + s_2 \mathbf{e}_2 \quad \text{and} \quad \mathbf{x} = \mathbf{e}_4 + t_1 \mathbf{e}_2 + t_2 \mathbf{e}_3$$

for $s_1, s_2, t_1, t_2 \in \mathbb{R}$, where $\{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3, \mathbf{e}_4\}$ is the standard basis of $\mathbb{R}^4$.

Show that these form a pair of skew planes; that is, they are non-parallel and non-intersecting.

Chapter 2

VECTOR GEOMETRY

“Why,” said the Gryphon, “you first form into a line along the sea-shore—”
“Two lines!” cried the Mock Turtle.
Lewis Carroll, Alice in Wonderland.

In Chapters 1 and 4 we have shown how vectors can be used to solve geometric problems involving points, lines and planes.

Our aim in this chapter is to show how vectors can be used to solve geometric problems involving lengths, distances, areas, angles and volumes. For simplicity, and because of the fundamental importance of two and three dimensions in the physical sciences and engineering, we will concentrate on problems in two and three dimensions. However, we shall see that many of the two and three dimensional results can be easily generalised to $\mathbb{R}^n$. The key idea is to use **theorems** in $\mathbb{R}^2$ and $\mathbb{R}^3$ to motivate **definitions** in $\mathbb{R}^n$ for $n > 3$.

2.1 Lengths

We have defined lengths of vectors in $\mathbb{R}^n$ and distance between two points on page 18. We expect that lengths and distances in $\mathbb{R}^n$ should have the same essential properties as those in two and three dimensional spaces. For example, we expect them to be real non-negative numbers. Some of these essential properties for $\mathbb{R}^n$ are proved in the following proposition.

Proposition 1. For all $\mathbf{a} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$,

1. $|\mathbf{a}|$ is a real number,
2. $|\mathbf{a}| \geq 0$,
3. $|\mathbf{a}| = 0$ if and only if $\mathbf{a} = \mathbf{0}$,
4. $|\lambda\mathbf{a}| = |\lambda| |\mathbf{a}|$.

Proof. From Definition 3 on page 18

$$|\mathbf{a}| = \sqrt{a_1^2 + \cdots + a_n^2}.$$

As $\mathbf{a} \in \mathbb{R}^n$, all a_k^2 are real and non-negative, and hence $a_1^2 + \cdots + a_n^2 \geq 0$. Thus, properties 1 and 2 hold.

Property 3 holds since a sum of non-negative numbers is zero if and only if every term in the sum is zero.

The proof of Property 4 is as follows. From the definitions of $\lambda \mathbf{a}$ and the length of a vector, we have

$$|\lambda \mathbf{a}| = \sqrt{(\lambda a_1)^2 + \cdots + (\lambda a_n)^2} = |\lambda| \sqrt{a_1^2 + \cdots + a_n^2} = |\lambda| |\mathbf{a}|.$$

□

Example 1. $|-6\mathbf{a}| = 6|\mathbf{a}|$.

◇

2.2 The dot product

Most introductory courses in trigonometry include a statement of the cosine rule for triangles. This rule can be stated in vector form as follows.

Proposition 1. Cosine Rule for Triangles. If the sides of a triangle in $\mathbb{R}^2$ or $\mathbb{R}^3$ are given by the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$, then

$$|\mathbf{c}|^2 = |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2|\mathbf{a}| |\mathbf{b}| \cos \theta,$$

where θ is the interior angle between $\mathbf{a}$ and $\mathbf{b}$.

Proof. From Figure 1, since $\triangle BPA$ is a right-angled triangle, we have

$$|\mathbf{c}|^2 = |\overrightarrow{BA}|^2 = |\overrightarrow{BP}|^2 + |\overrightarrow{PA}|^2.$$

$$\text{But, } |\overrightarrow{BP}| = |\overrightarrow{OP}| - |\overrightarrow{OB}| = |\mathbf{a}| \cos \theta - |\mathbf{b}|$$

$$\text{and } |\overrightarrow{PA}| = |\mathbf{a}| \sin \theta,$$

and hence

$$\begin{aligned} |\mathbf{c}|^2 &= (|\mathbf{a}| \cos \theta - |\mathbf{b}|)^2 + |\mathbf{a}|^2 \sin^2 \theta \\ &= (|\mathbf{a}| \cos \theta - |\mathbf{b}|)^2 + |\mathbf{a}|^2 \sin^2 \theta \\ &= |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2|\mathbf{a}| |\mathbf{b}| \cos \theta. \end{aligned}$$

□

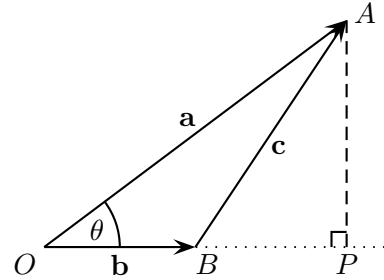


Figure 1: The Cosine Rule.

If we write $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ and use the formula for the length of $\mathbf{c} = \mathbf{a} - \mathbf{b}$, we have

$$\begin{aligned} |\mathbf{c}|^2 &= (a_1 - b_1)^2 + (a_2 - b_2)^2 + (a_3 - b_3)^2 \\ &= (a_1^2 + a_2^2 + a_3^2) + (b_1^2 + b_2^2 + b_3^2) - 2(a_1b_1 + a_2b_2 + a_3b_3) \\ &= |\mathbf{a}|^2 + |\mathbf{b}|^2 - 2(a_1b_1 + a_2b_2 + a_3b_3). \end{aligned}$$

On comparing this expression with the cosine rule, we find

$$a_1b_1 + a_2b_2 + a_3b_3 = |\mathbf{a}| |\mathbf{b}| \cos \theta.$$

This expression also applies immediately to vectors in the plane, where $a_3 = b_3 = 0$.

Let us now make the following definition.

Definition 1. The **dot product** of two vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ is

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + \cdots + a_n b_n = \sum_{k=1}^n a_k b_k.$$

Remark 1. You may have already encountered this product as the **scalar product** of two vectors. The name “scalar product” is helpful for reminding us that this operation gives us a scalar quantity (see also: Example 4 of Section 5.2).

For the special case of $\mathbb{R}^2$ or $\mathbb{R}^3$, we then have

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta.$$

Students of physics and engineering should note that it is this geometric result which is usually taken as the definition of the dot product in physics and engineering courses.

Example 1. Find the dot product of $\begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$, and hence find the cosine of the angle between the vectors.

SOLUTION. The dot product is

$$\begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} = -2 + 3 + 8 = 9,$$

and the lengths are $\sqrt{21}$ and $\sqrt{14}$, and hence

$$\cos \theta = \frac{9}{\sqrt{(21)(14)}} = \frac{9}{7\sqrt{6}}.$$

◇

Notice that the value of $\cos \theta$ does not uniquely define the value of the angle θ . It is conventional to define the angle between two vectors as an angle θ in the interval $[0, \pi]$, so that the value of $\cos \theta$ does uniquely define the value of θ .

Example 2. On choosing the angle in the interval $[0, \pi]$, the angle between the vectors $\begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$ of Example 1 is

$$\theta = \cos^{-1} \frac{9}{7\sqrt{6}} = 1.018 \dots \text{ radians.}$$

◇

2.2.1 Arithmetic properties of the dot product

Some of the main properties of the dot product are summarised in the following proposition.

Proposition 2. For all vectors $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^n$ and scalars $\lambda \in \mathbb{R}$,

1. $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$, and hence $|\mathbf{a}| = \sqrt{\mathbf{a} \cdot \mathbf{a}}$;
2. $\mathbf{a} \cdot \mathbf{b}$ is a scalar, i.e., is a real number;
3. **Commutative Law:** $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$;
4. $\mathbf{a} \cdot (\lambda \mathbf{b}) = \lambda(\mathbf{a} \cdot \mathbf{b})$;
5. **Distributive Law:** $\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}$;

Proof. Properties 1 and 2 follow immediately from the definitions of length and dot product.

The proof of the commutative law is as follows.

$$\begin{aligned}\mathbf{a} \cdot \mathbf{b} &= a_1 b_1 + \cdots + a_n b_n, \quad \text{and} \\ \mathbf{b} \cdot \mathbf{a} &= b_1 a_1 + \cdots + b_n a_n.\end{aligned}$$

But, $a_k b_k = b_k a_k$ (all a_k and b_k are real numbers), and hence the two expressions are equal.

The proof of Property 4 follows immediately on expanding the dot products on each side and using the properties of real numbers.

Finally, to prove the distributive law we note that

$$\begin{aligned}\mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) &= a_1(b_1 + c_1) + \cdots + a_n(b_n + c_n) \\ &= (a_1 b_1 + \cdots + a_n b_n) + (a_1 c_1 + \cdots + a_n c_n) \\ &= \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}\end{aligned}$$

□

2.2.2 Geometric interpretation of the dot product in $\mathbb{R}^n$

We have seen above that the dot product between two vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$ has a geometric interpretation in terms of lengths and angles as

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 = |\mathbf{a}| |\mathbf{b}| \cos \theta,$$

where $\theta \in [0, \pi]$ is the angle between $\mathbf{a}$ and $\mathbf{b}$. Now, in $\mathbb{R}^n$, we have defined lengths of vectors (Definition 3 on page 18) and the dot product (Definition 1 of Section 2.2), but the idea of an angle has not been defined. It is reasonable to try to **define** an angle in $\mathbb{R}^n$ so that the geometric interpretation of the dot product in $\mathbb{R}^n$ is the same as the geometric interpretation of it in $\mathbb{R}^2$ or $\mathbb{R}^3$. We therefore define an angle in $\mathbb{R}^n$ as follows.

Definition 2. If $\mathbf{a}, \mathbf{b}$ are non-zero vectors in $\mathbb{R}^n$, then the **angle** θ between $\mathbf{a}$ and $\mathbf{b}$ is given by

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}, \quad \text{where } \theta \in [0, \pi].$$

Example 3. Find the angle between $\mathbf{a} = \begin{pmatrix} -1 \\ 2 \\ -3 \\ -1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ -4 \\ 0 \\ -1 \end{pmatrix}$.

SOLUTION. Using the definition of length, dot product and angle in $\mathbb{R}^n$, we have

$$|\mathbf{a}| = \sqrt{15}, \quad |\mathbf{b}| = \sqrt{21}, \quad \mathbf{a} \cdot \mathbf{b} = -2 - 8 + 0 + 1 = -9,$$

$$\cos \theta = -\frac{9}{\sqrt{(15)(21)}} = -\frac{3}{\sqrt{35}},$$

and hence, on choosing the angle between 0 and π , we have

$$\theta = \cos^{-1} \frac{-3}{\sqrt{35}} = 2.103 \dots \text{ radians.}$$

◇

Now, in Definition 2 of an angle in $\mathbb{R}^n$, we have assumed that the definition makes sense, i.e., that the equation

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|}$$

can always be solved to obtain a real number as the value for the angle θ . But, since $-1 \leq \cos \theta \leq 1$ for real numbers, a real solution for θ is possible if and only if

$$-1 \leq \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}| |\mathbf{b}|} \leq 1.$$

A proof that this inequality is true for all non-zero vectors in $\mathbb{R}^n$ is given in the following important theorem.

Theorem 3 (The Cauchy-Schwarz Inequality). *If $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$, then $-|\mathbf{a}| |\mathbf{b}| \leq \mathbf{a} \cdot \mathbf{b} \leq |\mathbf{a}| |\mathbf{b}|$.*

[X] *Proof.* Note first that the inequality is clearly true if either $\mathbf{a}$ or $\mathbf{b}$ is a zero vector. For $\mathbf{b} \neq \mathbf{0}$, consider

$$q(\lambda) = |\mathbf{a} - \lambda \mathbf{b}|^2 \text{ for } \lambda \in \mathbb{R}.$$

Then, from the properties of lengths and dot products in $\mathbb{R}^n$, we have that $q \geq 0$ for all $\lambda \in \mathbb{R}$ and hence that

$$0 \leq q = (\mathbf{a} - \lambda \mathbf{b}) \cdot (\mathbf{a} - \lambda \mathbf{b}) = |\mathbf{a}|^2 - 2\lambda \mathbf{a} \cdot \mathbf{b} + \lambda^2 |\mathbf{b}|^2.$$

This $q(\lambda)$ is a quadratic function of λ which has a minimum at

$$\lambda = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2}.$$

The minimum value is

$$q_0 = q\left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2}\right) = |\mathbf{a}|^2 - \frac{(\mathbf{a} \cdot \mathbf{b})^2}{|\mathbf{b}|^2}.$$

Now, as $q(\lambda) \geq 0$ for all λ , we have

$$0 \leq q_0 = |\mathbf{a}|^2 - \frac{(\mathbf{a} \cdot \mathbf{b})^2}{|\mathbf{b}|^2}.$$

Thus,

$$(\mathbf{a} \cdot \mathbf{b})^2 \leq |\mathbf{a}|^2 |\mathbf{b}|^2,$$

and therefore

$$-|\mathbf{a}| |\mathbf{b}| \leq \mathbf{a} \cdot \mathbf{b} \leq |\mathbf{a}| |\mathbf{b}|,$$

and the proof is complete. $\square$

Another useful inequality which follows immediately from the Cauchy-Schwarz inequality is the following inequality for lengths of vectors.

Theorem 4 (Minkowski's Inequality (or the Triangle Inequality)). *For $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$,*

$$|\mathbf{a} + \mathbf{b}| \leq |\mathbf{a}| + |\mathbf{b}|.$$

Proof.

$$|\mathbf{a} + \mathbf{b}|^2 = (\mathbf{a} + \mathbf{b}) \cdot (\mathbf{a} + \mathbf{b}) = |\mathbf{a}|^2 + 2\mathbf{a} \cdot \mathbf{b} + |\mathbf{b}|^2.$$

But, from the Cauchy-Schwarz inequality, $\mathbf{a} \cdot \mathbf{b} \leq |\mathbf{a}| |\mathbf{b}|$, and hence

$$|\mathbf{a} + \mathbf{b}|^2 \leq |\mathbf{a}|^2 + 2|\mathbf{a}| |\mathbf{b}| + |\mathbf{b}|^2 = (|\mathbf{a}| + |\mathbf{b}|)^2.$$

On taking positive square roots of both sides of this inequality, we then obtain the result to be proved. $\square$

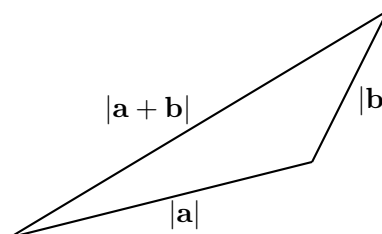


Figure 2: The Triangle Inequality.

As illustrated in Figure 2, the geometric interpretation of the triangle inequality is that the sum of two sides of a triangle is greater than or equal to the third side.

2.3 Applications: orthogonality and projection

The dot product has many applications. Two important applications are for testing if two vectors are at right angles and for projecting one vector on another.

2.3.1 Orthogonality of vectors

The dot product provides a simple test for vectors being at right angles to each other. We begin with the following definition.

Definition 1. Two vectors $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^n$ are said to be **orthogonal** if $\mathbf{a} \cdot \mathbf{b} = 0$.

Note that the zero vector is orthogonal to every vector, including itself.

On using the formula for the dot product in terms of angles, we see that two vectors are orthogonal if

$$\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta = 0,$$

i.e., if either $|\mathbf{a}|$ or $|\mathbf{b}|$ or $\cos \theta$ is zero. As a vector is zero if its length is zero, and $\cos \theta = 0$ only when θ is a right angle, we have the result that vectors $\mathbf{a}$ and $\mathbf{b}$ are orthogonal if either vector is the zero vector or if the vectors are at right angles to each other.

NOTE. Two non-zero vectors at right angles to each other are also said to be **perpendicular** to each other or to be **normal** to each other.

By the definitions of length and orthogonality, the set of standard basis vectors $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is a set of vectors of unit length at right angles to each other. Sets of vectors with this property are of great practical importance, and they have been given a special name.

Definition 2. An *orthonormal set of vectors* in $\mathbb{R}^n$ is a set of vectors which are of unit length and mutually orthogonal.

The connection between the dot product and lengths and angles provides a simple test for a set of vectors to be an orthonormal set.

Example 1. The three standard basis vectors $\mathbf{e}_1$, $\mathbf{e}_2$, and $\mathbf{e}_3$ of $\mathbb{R}^3$ form an orthonormal set.

Proof. The three vectors are of unit length since

$$|\mathbf{e}_1|^2 = \mathbf{e}_1 \cdot \mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 1,$$

and similarly, $|\mathbf{e}_2|^2 = 1$ and $|\mathbf{e}_3|^2 = 1$.

The vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ are orthogonal since

$$\mathbf{e}_1 \cdot \mathbf{e}_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0.$$

Similarly, $\mathbf{e}_1$ and $\mathbf{e}_3$ are orthogonal since $\mathbf{e}_1 \cdot \mathbf{e}_3 = 0$. Finally, $\mathbf{e}_2$ and $\mathbf{e}_3$ are orthogonal since $\mathbf{e}_2 \cdot \mathbf{e}_3 = 0$.

Thus, the three vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ are each of unit length and they are mutually orthogonal, and hence they form an orthonormal set. $\square$

Note that a compact form of writing the conditions for an orthonormal set in $\mathbb{R}^n$ are that, for $1 \leq i \leq n$ and $1 \leq j \leq n$,

$$\mathbf{e}_i \cdot \mathbf{e}_j = \begin{cases} 0 & \text{for } i \neq j; \\ 1 & \text{for } i = j. \end{cases}$$

Example 2. Show that the three vectors $\mathbf{u}_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$, $\mathbf{u}_2 = \begin{pmatrix} -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} \end{pmatrix}$, $\mathbf{u}_3 = \begin{pmatrix} \frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{pmatrix}$ form an orthonormal set. Find scalars $\lambda_1, \lambda_2, \lambda_3$ such that $\mathbf{e}_1 = \lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \lambda_3 \mathbf{u}_3$.

SOLUTION. The three vectors are of unit length, since

$$\mathbf{u}_1 \cdot \mathbf{u}_1 = \left(\frac{1}{\sqrt{2}}\right)^2 + 0^2 + \left(-\frac{1}{\sqrt{2}}\right)^2 = 1,$$

and similarly $\mathbf{u}_2 \cdot \mathbf{u}_2 = 1$, $\mathbf{u}_3 \cdot \mathbf{u}_3 = 1$. They are mutually orthogonal since

$$\mathbf{u}_1 \cdot \mathbf{u}_2 = \left(\frac{1}{\sqrt{2}}\right)\left(-\frac{1}{\sqrt{3}}\right) + 0 \times \frac{1}{\sqrt{3}} + \left(-\frac{1}{\sqrt{2}}\right)\left(-\frac{1}{\sqrt{3}}\right) = 0,$$

and similarly $\mathbf{u}_1 \cdot \mathbf{u}_3 = 0$, $\mathbf{u}_2 \cdot \mathbf{u}_3 = 0$. The three vectors therefore form an orthonormal set.

For writing $\mathbf{e}_1$ as a linear combination of the three vectors, again we use the dot product.

$$\begin{aligned}\mathbf{u}_1 \cdot \mathbf{e}_1 &= \mathbf{u}_1 \cdot (\lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \lambda_3 \mathbf{u}_3) \\ \frac{1}{\sqrt{2}} &= \lambda_1(\mathbf{u}_1 \cdot \mathbf{u}_1) + \lambda_2(\mathbf{u}_1 \cdot \mathbf{u}_2) + \lambda_3(\mathbf{u}_1 \cdot \mathbf{u}_3),\end{aligned}$$

we have $\lambda_1 = \frac{1}{\sqrt{2}}$. Similarly,

$$\lambda_2 = \mathbf{u}_2 \cdot \mathbf{e}_1 = -\frac{1}{\sqrt{3}} \quad \text{and} \quad \lambda_3 = \mathbf{u}_3 \cdot \mathbf{e}_1 = \frac{1}{\sqrt{6}}.$$

In other words, we can write $\mathbf{e}_1$ as a linear combination of the three vectors:

$$\mathbf{e}_1 = \frac{1}{\sqrt{2}}\mathbf{u}_1 - \frac{1}{\sqrt{3}}\mathbf{u}_2 + \frac{1}{\sqrt{6}}\mathbf{u}_3.$$

◇

The method used in the above example can generally be used to write any vector in the span of an orthonormal set as a linear combination of vectors in this set.

Here is an application of orthogonality to the geometry of triangles.

Example 3. Show that the three altitudes of a triangle are concurrent, i.e., they intersect at a point.

SOLUTION. As shown in Figure 3, let the three vertices be A , B and C (coordinate vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$), and let D and E be the points at which the altitudes from A and B intersect the opposite sides of the triangle. Finally, let P (coordinate vector $\mathbf{p}$) be the point of intersection of the altitudes AD and BE .

Then, we can prove the result by showing that P is also on the altitude from C to AB .

Now, as P is on the altitude from A to the opposite side BC , we have that $\overrightarrow{AP}$ is perpendicular to $\overrightarrow{BC}$, and hence

$$\begin{aligned} (\mathbf{p} - \mathbf{a}) \cdot (\mathbf{c} - \mathbf{b}) &= 0, \\ \text{i.e., } \mathbf{p} \cdot \mathbf{c} - \mathbf{p} \cdot \mathbf{b} - \mathbf{a} \cdot \mathbf{c} + \mathbf{a} \cdot \mathbf{b} &= 0. \end{aligned} \quad (1)$$

Similarly, as P is on the altitude from B to the opposite side CA , we have

$$\begin{aligned} (\mathbf{p} - \mathbf{b}) \cdot (\mathbf{a} - \mathbf{c}) &= 0, \\ \text{i.e., } \mathbf{p} \cdot \mathbf{a} - \mathbf{p} \cdot \mathbf{c} - \mathbf{b} \cdot \mathbf{a} + \mathbf{b} \cdot \mathbf{c} &= 0. \end{aligned} \quad (2)$$

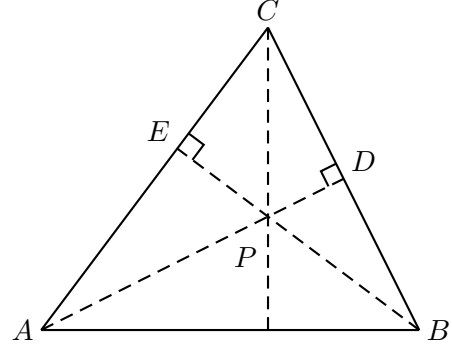


Figure 3: Altitudes of a Triangle.

Then, on adding (1) and (2) and grouping terms, we obtain

$$(\mathbf{p} - \mathbf{c}) \cdot (\mathbf{a} - \mathbf{b}) = 0,$$

i.e., $\overrightarrow{CP} \cdot \overrightarrow{BA} = 0$. Thus, $\overrightarrow{CP}$ is perpendicular to $\overrightarrow{BA}$ and hence P is on the altitude from C to the opposite side BA . The result is proved. $\diamond$

2.3.2 Projections

The geometric idea of a projection of a vector $\mathbf{a}$ on a non-zero vector $\mathbf{b}$ in $\mathbb{R}^2$ or $\mathbb{R}^3$ is shown in Figure 4, where $\overrightarrow{OP}$ is the projection.

Note that $\overrightarrow{OP}$ is parallel to $\mathbf{b}$ and $\overrightarrow{PA} = \mathbf{a} - \overrightarrow{OP}$ is perpendicular to $\mathbf{b}$. Note also that

$$|\overrightarrow{OP}| = |\mathbf{a}| \cos \theta = \frac{|\mathbf{a} \cdot \mathbf{b}|}{|\mathbf{b}|},$$

because $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$. Since $\frac{\mathbf{b}}{|\mathbf{b}|}$ is a unit vector,

$$\begin{aligned} \overrightarrow{OP} &= |\overrightarrow{OP}| \frac{\mathbf{b}}{|\mathbf{b}|} \quad \text{when } \theta \text{ is acute;} \\ \overrightarrow{OP} &= -|\overrightarrow{OP}| \frac{\mathbf{b}}{|\mathbf{b}|} \quad \text{when } \theta \text{ is obtuse.} \end{aligned}$$

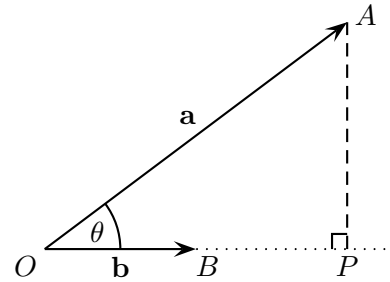


Figure 4: Projection of $\mathbf{a}$ on $\mathbf{b}$.

In both cases, $\overrightarrow{OP} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2} \right) \mathbf{b}$. Thus we can formally define a projection in terms of the dot product as follows.

Definition 3. For $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ and $\mathbf{b} \neq \mathbf{0}$, the **projection of $\mathbf{a}$ on $\mathbf{b}$** is

$$\text{proj}_{\mathbf{b}} \mathbf{a} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2} \right) \mathbf{b}.$$

The geometric properties in $\mathbb{R}^2$ used to motivate this definition can be proved to be true for projections in $\mathbb{R}^n$ also. We have

Proposition 1. $\text{proj}_{\mathbf{b}} \mathbf{a}$ is the unique vector $\lambda \mathbf{b}$ parallel to the non-zero vector $\mathbf{b}$ such that

$$(\mathbf{a} - \lambda \mathbf{b}) \cdot \mathbf{b} = 0. \quad (\#)$$

Proof. From $(\#)$ we have

$$0 = (\mathbf{a} - \lambda \mathbf{b}) \cdot \mathbf{b} = \mathbf{a} \cdot \mathbf{b} - \lambda |\mathbf{b}|^2.$$

For $\mathbf{b} \neq \mathbf{0}$, this equation can be solved for λ to obtain the unique solution

$$\lambda = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2},$$

and hence the unique solution of $(\#)$ is

$$\lambda \mathbf{b} = \left(\frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2} \right) \mathbf{b} = \text{proj}_{\mathbf{b}} \mathbf{a}.$$

The proof is complete. □

Alternative forms of writing the formula for a projection are sometimes useful. These are

$$\text{proj}_{\mathbf{b}} \mathbf{a} = (\mathbf{a} \cdot \hat{\mathbf{b}}) \hat{\mathbf{b}} = |\mathbf{a}| \cos \theta \hat{\mathbf{b}},$$

where $\hat{\mathbf{b}} = \frac{\mathbf{b}}{|\mathbf{b}|}$ is the unit vector in the direction of $\mathbf{b}$ and where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$.

Example 4. Find the projections of a vector $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ on the three standard basis vectors $\mathbf{e}_1$, $\mathbf{e}_2$, $\mathbf{e}_3$.

SOLUTION. $\mathbf{a} \cdot \mathbf{e}_1 = a_1$, $|\mathbf{e}_1|^2 = \mathbf{e}_1 \cdot \mathbf{e}_1 = 1$, and hence

$$\text{proj}_{\mathbf{e}_1} \mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{e}_1}{|\mathbf{e}_1|^2} \mathbf{e}_1 = a_1 \mathbf{e}_1.$$

Similarly, $a_2 \mathbf{e}_2$ is the projection of $\mathbf{a}$ on $\mathbf{e}_2$ and $a_3 \mathbf{e}_3$ is the projection of $\mathbf{a}$ on $\mathbf{e}_3$. ◇

Example 5. Find the projection of $\mathbf{a} = \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}$ on $\mathbf{b} = \begin{pmatrix} -4 \\ 1 \\ 5 \end{pmatrix}$.

SOLUTION. $\mathbf{a} \cdot \mathbf{b} = 3$, $|\mathbf{b}|^2 = 42$, and hence $\text{proj}_{\mathbf{b}} \mathbf{a} = \frac{1}{14} \begin{pmatrix} -4 \\ 1 \\ 5 \end{pmatrix}$. $\diamond$

Note that a simple formula for the length of the projection of $\mathbf{a}$ on $\mathbf{b}$ is

$$|\text{proj}_{\mathbf{b}} \mathbf{a}| = |\mathbf{a} \cdot \hat{\mathbf{b}}| = \frac{|\mathbf{a} \cdot \mathbf{b}|}{|\mathbf{b}|}.$$

Example 6. Find the length of the projection of $\begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix}$ on $\begin{pmatrix} -3 \\ 1 \\ -4 \end{pmatrix}$.

SOLUTION. As $\left| \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 1 \\ -4 \end{pmatrix} \right| = 8$, and as $\left| \begin{pmatrix} -3 \\ 1 \\ -4 \end{pmatrix} \right| = \sqrt{26}$, the length of the projection is $\frac{8}{\sqrt{26}}$. $\diamond$

2.3.3 Distance between a point and a line in $\mathbb{R}^3$

In $\mathbb{R}^2$ and $\mathbb{R}^3$, the distance between a point B and a line $\mathbf{x} = \mathbf{a} + \lambda \mathbf{d}$ is the shortest distance between the point and the line. In the diagram, the distance is $|\overrightarrow{PB}|$, where P is the point on the line such that $\angle APB$ is a right angle (for a general proof of this result, see Question 13 in Problems for Chapter 2). We can easily find $|\overrightarrow{AB}|$ from the coordinates of A and B , and we can find $|\overrightarrow{AP}|$ as it is the length of the projection of $\overrightarrow{AB}$ on the direction $\mathbf{d}$. Thus,

$$|\overrightarrow{AB}| = |\mathbf{b} - \mathbf{a}|, \quad |\overrightarrow{AP}| = \frac{|\overrightarrow{AB} \cdot \mathbf{d}|}{|\mathbf{d}|},$$

$$\text{and} \quad |\overrightarrow{PB}| = \sqrt{|\overrightarrow{AB}|^2 - |\overrightarrow{AP}|^2}.$$

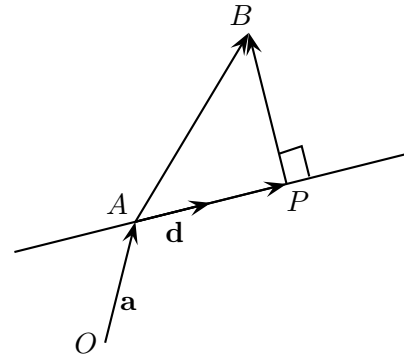


Figure 5: Shortest Distance between Point and Line.

Note that $\overrightarrow{AB}$ is the line segment joining some point A on the line to the given point B , while $\overrightarrow{AP}$ is the projection of this line segment on the direction of the line.

An alternative method of solving this problem is to use the fact that

$$\overrightarrow{PB} = \overrightarrow{AB} - \overrightarrow{AP} = \mathbf{b} - \mathbf{a} - \text{proj}_{\mathbf{d}}(\mathbf{b} - \mathbf{a}),$$

and then the shortest distance is $|\overrightarrow{PB}|$.

Example 7. Find the distance from the point $(2, -1, 3)$ to the line through the points $(0, 1, 4)$ and $(4, 2, 9)$.

SOLUTION. Let A , B and C be the points $(0, 1, 4)$, $(2, -1, 3)$ and $(4, 2, 9)$, respectively. Suppose that P is the foot of the perpendicular from B to the line AC . Referring to Figure 5, the length of the projection of $\overrightarrow{AB} = \begin{pmatrix} 2 \\ -2 \\ -1 \end{pmatrix}$ on the direction $\mathbf{d} = \overrightarrow{AC} = \begin{pmatrix} 4 \\ 1 \\ 5 \end{pmatrix}$ is

$$|\overrightarrow{AP}| = \frac{|\overrightarrow{AB} \cdot \mathbf{d}|}{|\mathbf{d}|} = \frac{|8 - 2 - 5|}{\sqrt{16 + 1 + 25}} = \frac{1}{\sqrt{42}}.$$

Together with $|\overrightarrow{AB}| = 3$, the distance from the point to the line is

$$|\overrightarrow{BP}| = \sqrt{3^2 - \frac{1}{42}} = \frac{\sqrt{377}}{\sqrt{42}}.$$

◇

2.4 The cross product

We now define the cross product of two vectors in three dimensions. One motivation for a cross product is to find a formula which gives a vector which is perpendicular to two other vectors.

Now, a vector $\mathbf{x}$ will be perpendicular to two vectors $\mathbf{a}$ and $\mathbf{b}$ if and only if the dot products $\mathbf{a} \cdot \mathbf{x}$ and $\mathbf{b} \cdot \mathbf{x}$ are both zero. Using coordinates, we can write

$$\begin{aligned} a_1x_1 + a_2x_2 + a_3x_3 &= 0 \\ b_1x_1 + b_2x_2 + b_3x_3 &= 0 \end{aligned}$$

This pair of equations can be easily solved in the usual way to obtain a solution which can be written in the form

$$\mathbf{x} = \lambda \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix},$$

where λ is a real parameter. This expression for $\mathbf{x}$ looks like some kind of “product” of $\mathbf{a}$ and $\mathbf{b}$.

Definition 1. The **cross product** of two vectors $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ in $\mathbb{R}^3$ is

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}.$$

Note that the cross product of two vectors is a **vector**. For this reason the cross product is often called the **vector product** of two vectors, in contrast to the dot product which is a scalar and is often called the **scalar product** of two vectors. Note also that the cross product $\mathbf{a} \times \mathbf{b}$ has

the important property that it is perpendicular to the two vectors $\mathbf{a}$ and $\mathbf{b}$. As an exercise you might like to check directly that $\mathbf{a} \times \mathbf{b}$ is orthogonal to $\mathbf{a}$ by checking that $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = 0$.

There are several tricks available for remembering the formula for a cross product. The most common trick is to use determinant notation. The more general theory of determinants will be covered in a later chapter. We define here a 2×2 determinant by

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc.$$

To find the cross product of two vectors $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and $\begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ we write these as rows in a 3×3 determinant:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}.$$

To calculate this, we use the following procedure. Firstly, take the vector $\mathbf{e}_1$ and multiply it by the 2×2 determinant obtained by deleting the row and column in which $\mathbf{e}_1$ is contained. That is, we write

$$\mathbf{e}_1 \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix}.$$

Then take $-\mathbf{e}_2$ and repeat the process, followed by $\mathbf{e}_3$. Each of the 2×2 determinants can be found using the definition above.

We write

$$\begin{aligned} \mathbf{a} \times \mathbf{b} &= \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \\ &= \mathbf{e}_1 \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \mathbf{e}_2 \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} + \mathbf{e}_3 \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \\ &= \mathbf{e}_1(a_2b_3 - a_3b_2) - \mathbf{e}_2(a_1b_3 - a_3b_1) + \mathbf{e}_3(a_1b_2 - a_2b_1) \\ &= \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} (a_2b_3 - a_3b_2) - \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} (a_1b_3 - a_3b_1) + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} (a_1b_2 - a_2b_1) \\ &= \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix}. \end{aligned}$$

The determinant is expanded along the first row and as usual $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ are the standard basis vectors of $\mathbb{R}^3$. While this may appear complicated at first, with practice it is much easier than using the formula for the cross product.

Example 1. Find the cross product of $\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ -5 \\ 6 \end{pmatrix}$.

SOLUTION. Using the determinant formula, we have

$$\begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \times \begin{pmatrix} 4 \\ -5 \\ 6 \end{pmatrix} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ 1 & -2 & 3 \\ 4 & -5 & 6 \end{vmatrix} = \mathbf{e}_1(-12 + 15) - \mathbf{e}_2(6 - 12) + \mathbf{e}_3(-5 + 8) = \begin{pmatrix} 3 \\ 6 \\ 3 \end{pmatrix}.$$

◇

2.4.1 Arithmetic properties of the cross product

Some of the basic properties of the cross product are summarised in the following proposition.

Proposition 1. For all $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$ and $\lambda \in \mathbb{R}$,

1. $\mathbf{a} \times \mathbf{a} = \mathbf{0}$, i.e., the cross product of a vector with itself is the zero vector.
2. $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$. The cross product is **not commutative**. If the order of vectors in the cross product is reversed, then the sign of the product is also reversed.
3. $\mathbf{a} \times (\lambda \mathbf{b}) = \lambda(\mathbf{a} \times \mathbf{b})$ and $(\lambda \mathbf{a}) \times \mathbf{b} = \lambda(\mathbf{a} \times \mathbf{b})$.
4. $\mathbf{a} \times (\lambda \mathbf{a}) = \mathbf{0}$, i.e., the cross product of parallel vectors is zero.
5. **Distributive Laws.** $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$ and $(\mathbf{a} + \mathbf{b}) \times \mathbf{c} = \mathbf{a} \times \mathbf{c} + \mathbf{b} \times \mathbf{c}$.

Proof. Each of the properties listed in Proposition 1 can be proved by expanding each side of the properties using the definition of cross product given in Definition 1. For example, for Property 2, we have on expansion that

$$\mathbf{a} \times \mathbf{b} = \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix} = - \begin{pmatrix} b_2a_3 - b_3a_2 \\ b_3a_1 - b_1a_3 \\ b_1a_2 - b_2a_1 \end{pmatrix} = -\mathbf{b} \times \mathbf{a}.$$

□

There are several useful relations for the cross products of the standard basis vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ in $\mathbb{R}^3$.

Proposition 2. The three standard basis vectors in $\mathbb{R}^3$ satisfy the relations

1. $\mathbf{e}_1 \times \mathbf{e}_1 = \mathbf{e}_2 \times \mathbf{e}_2 = \mathbf{e}_3 \times \mathbf{e}_3 = \mathbf{0}$,
2. $\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3$, $\mathbf{e}_2 \times \mathbf{e}_3 = \mathbf{e}_1$, $\mathbf{e}_3 \times \mathbf{e}_1 = \mathbf{e}_2$.

Proof. The proof of these relations follows immediately from Property 1 of Proposition 1 and from Definition 1. For example,

$$\mathbf{e}_1 \times \mathbf{e}_2 = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{vmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \mathbf{e}_3.$$

□

NOTE. The cross product is **not** associative. In fact,

$$\begin{aligned} (\mathbf{e}_1 \times \mathbf{e}_2) \times \mathbf{e}_2 &= -\mathbf{e}_1, \\ \text{but } \mathbf{e}_1 \times (\mathbf{e}_2 \times \mathbf{e}_2) &= \mathbf{0}. \end{aligned}$$

Proposition 3. Suppose A, B are points in $\mathbb{R}^3$ that have coordinate vectors $\mathbf{a}$ and $\mathbf{b}$, and $\angle AOB = \theta$ then $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}| |\mathbf{b}| \sin \theta$.

Proof. If $\mathbf{a} = \mathbf{0}$ or $\mathbf{b} = \mathbf{0}$ then both sides are 0 although θ is not defined. Also, since $0 \leq \theta \leq \pi$, $\sin \theta \geq 0$ we need only prove

$$|\mathbf{a} \times \mathbf{b}|^2 = |\mathbf{a}|^2 |\mathbf{b}|^2 \sin^2 \theta$$

$$\text{or } |\mathbf{a} \times \mathbf{b}|^2 = |\mathbf{a}|^2 |\mathbf{b}|^2 (1 - \cos^2 \theta) = |\mathbf{a}|^2 |\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2 \quad \text{by Section 2.2.}$$

To prove this identity we expand both sides

$$\begin{aligned} \text{L.H.S.} &= \left| \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix} \right|^2 \\ &= a_2^2 b_3^2 + a_3^2 b_2^2 - 2a_2 a_3 b_2 b_3 + a_3^2 b_1^2 + a_1^2 b_3^2 - 2a_3 a_1 b_3 b_1 + a_1^2 b_2^2 + a_2^2 b_1^2 - 2a_1 a_2 b_1 b_2 \end{aligned}$$

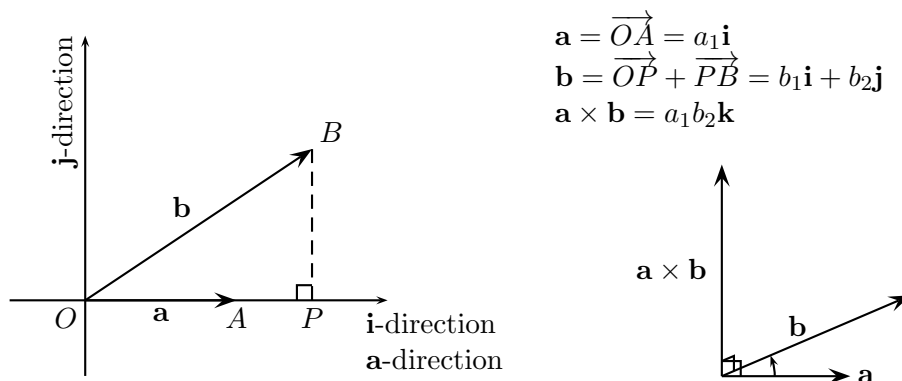
$$\begin{aligned} \text{R.H.S.} &= (a_1^2 + a_2^2 + a_3^2)(b_1^2 + b_2^2 + b_3^2) - (a_1 b_1 + a_2 b_2 + a_3 b_3)^2 \\ &= a_1^2 b_1^2 + a_1^2 b_2^2 + a_1^2 b_3^2 + a_2^2 b_1^2 + a_2^2 b_2^2 + a_2^2 b_3^2 + a_3^2 b_1^2 + a_3^2 b_2^2 + a_3^2 b_3^2 \\ &\quad - a_1^2 b_1^2 - a_2^2 b_2^2 - a_3^2 b_3^2 - 2a_1 a_2 b_1 b_2 - 2a_1 a_3 b_1 b_3 - 2a_2 a_3 b_2 b_3 \\ &= a_1^2 b_2^2 + a_1^2 b_3^2 + a_2^2 b_1^2 + a_2^2 b_3^2 + a_3^2 b_1^2 + a_3^2 b_2^2 - 2a_2 a_3 b_2 b_3 - 2a_3 a_1 b_3 b_1 - 2a_1 a_2 b_1 b_2 \end{aligned}$$

as required. □

2.4.2 A geometric interpretation of the cross product

We have constructed a cross product to be perpendicular to two given vectors. In this subsection we explore the geometric properties further.

Choose an orthonormal set of basis vectors so that $\mathbf{i}$ is in the direction of $\mathbf{a}$, and so that $\mathbf{i}$ and $\mathbf{j}$ are in the plane of $\mathbf{a}$ and $\mathbf{b}$, with $\mathbf{b} = b_1 \mathbf{i} + b_2 \mathbf{j}$, $b_1 > 0$. Finally, $\mathbf{k}$ is taken at right angles to the plane of $\mathbf{a}$ and $\mathbf{b}$ with its direction determined by a “right-hand rule”: using your right hand, point to $\mathbf{a}$ with your index finger and to $\mathbf{b}$ with your middle finger, then your thumb can be extended in the direction of $\mathbf{k}$. If we take the direction of $\mathbf{k}$ as normal to the page and pointing outwards, then the picture for the $\mathbf{i}, \mathbf{j}$ plane is as shown in Figure 6.

Figure 6: Geometry of $\mathbf{a} \times \mathbf{b}$.

Hence, we have

$$\mathbf{a} = \begin{pmatrix} a_1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ 0 \end{pmatrix} \quad \text{and} \quad \mathbf{a} \times \mathbf{b} = \begin{pmatrix} 0 \\ 0 \\ a_1 b_2 \end{pmatrix},$$

and we also have that the coordinates a_1 and b_2 are positive.

Now, since a_1 and b_2 are positive, the cross product is in the direction of $\mathbf{k}$, i.e., it is in a direction perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ as given by **the right-hand rule**.

Let θ denote the angle between $\mathbf{a}$ and $\mathbf{b}$. Note first that the angle θ is always chosen to be in the interval $[0, \pi]$ so that $\sin \theta \geq 0$. Also, the length of the vector $\mathbf{a} \times \mathbf{b}$ is

$$|\mathbf{a} \times \mathbf{b}| = a_1 b_2.$$

Now $a_1 = |\mathbf{a}|$, and from trigonometry, $b_2 = |\mathbf{b}| \sin \theta$, and hence $|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}| |\mathbf{b}| \sin \theta$, in agreement with Proposition 3.

We have therefore shown that:

$\mathbf{a} \times \mathbf{b}$ is a vector of length $|\mathbf{a}| |\mathbf{b}| \sin \theta$ in the direction perpendicular to both $\mathbf{a}$ and $\mathbf{b}$ as given by the right-hand rule.

This statement is usually taken as the definition of the cross product in physics and engineering courses.

NOTE. The above proof is valid since if $\mathbf{a}, \mathbf{b}$ are arbitrary in $\mathbb{R}^3$ then we can apply a rotation to move $\mathbf{a}$ to $a_1 \mathbf{i}$. We then rotate about the x -axis to move $\mathbf{b}$ to $\beta \mathbf{i} + \gamma \mathbf{j}$ with $\gamma > 0$. Since rotations preserve lengths and angle sizes the results

$$|\mathbf{a} \times \mathbf{b}| = |\mathbf{a}| |\mathbf{b}| \sin \theta$$

$$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} = 0, \quad (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{b} = 0 \quad \text{and}$$

$$(\mathbf{a}, \mathbf{b}, \mathbf{a} \times \mathbf{b}) \quad \text{form a right-hand triple}$$

holds for arbitrary vectors $\mathbf{a}, \mathbf{b}$ in $\mathbb{R}^3$.

2.4.3 Areas

The length of the cross product $\mathbf{a} \times \mathbf{b}$ also has an alternative geometric interpretation in terms of the area of a parallelogram with sides $\mathbf{a}$ and $\mathbf{b}$. Consider the picture of Figure 7.

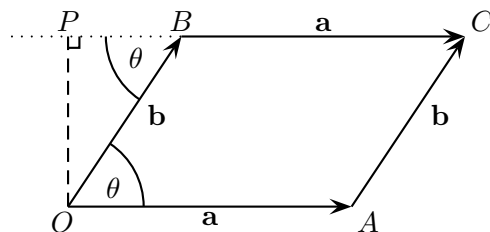


Figure 7: The Cross Product and the Area of a Parallelogram.

The area of the parallelogram $OACB$ in Figure 7 is “base times perpendicular height”.

$$\text{Area} = |\vec{OA}| |\vec{OB}| \sin \theta = |\mathbf{a}| |\mathbf{b}| \sin \theta = |\mathbf{a} \times \mathbf{b}|.$$

The vector $\mathbf{a} \times \mathbf{b}$ is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$, so it is normal to the plane of the parallelogram.

Example 2. Find the area of a parallelogram with vertices at points $A (1, 0, 1)$, $B (-2, 1, 3)$, and $C (3, 1, 4)$.

SOLUTION. One parallelogram can be formed with sides $\vec{AB} = \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix}$ and $\vec{AC} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$. The cross product is $\vec{AB} \times \vec{AC} = \begin{pmatrix} 1 \\ 13 \\ -5 \end{pmatrix}$, and hence the area is $\left| \begin{pmatrix} 1 \\ 13 \\ -5 \end{pmatrix} \right| = \sqrt{195}$. $\diamond$

NOTE. There are three parallelograms which can be formed from three vertices. The areas of all three are equal. As an exercise you might like to find the other two parallelograms with the same three vertices and check that they have the same area.

[X] 2.4.4 Shortest distance between lines in $\mathbb{R}^3$

It is necessary to consider the case of parallel lines and the case of skew (non-parallel) lines separately. Be careful, non-parallel lines in $\mathbb{R}^3$ may not intersect.

Parallel Lines. The distance between two parallel lines is the same as the distance between a point on one of the lines and the other line. We only need to use the method in 2.3.3 to find the distance between a point on one line and the other line.

Skew Lines. The distance, i.e. the shortest distance, between two skew lines is obtained by drawing a perpendicular to both lines. The direction of the perpendicular is in the direction of the cross product of the directions of the lines. The shortest distance is the length of the projection on this perpendicular of a line segment joining any point on one line to any point on the other.

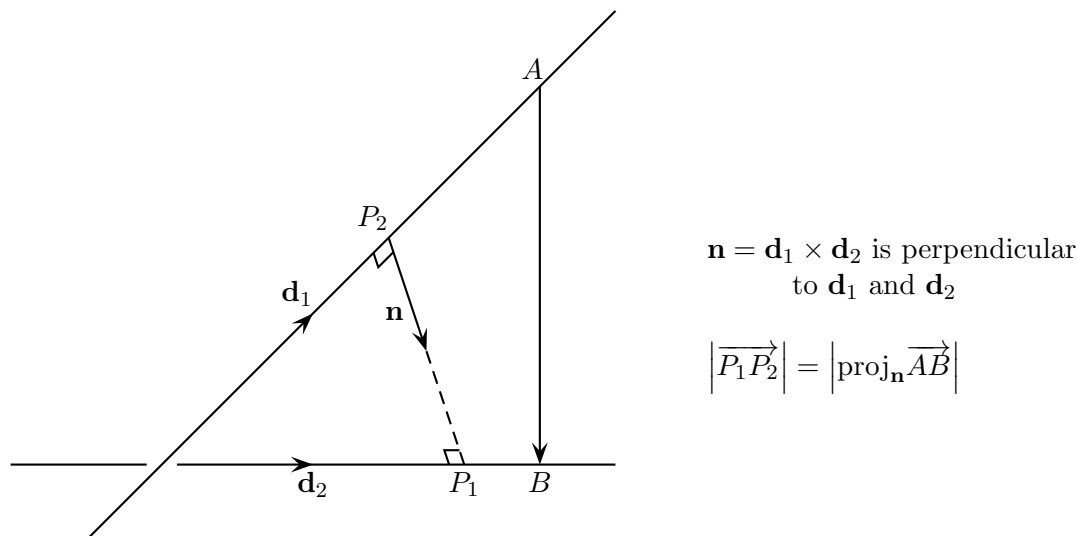


Figure 8: Shortest Distance between Skew Lines.

Example 3. Find the shortest distance between the lines

$$\mathbf{x} = \begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} \quad \text{and} \quad \mathbf{x} = \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix}.$$

SOLUTION. The cross product of the direction vectors of the lines is $\mathbf{n} = \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} \times \begin{pmatrix} 2 \\ 4 \\ 1 \end{pmatrix} = \begin{pmatrix} -7 \\ 1 \\ 10 \end{pmatrix}$,

and hence $\mathbf{n}$ is perpendicular to both lines. A line segment from a point on one line to a point on

the other is $\begin{pmatrix} 1 \\ 4 \\ 2 \end{pmatrix} - \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

The shortest distance is the length of the projection of the line segment $\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ on the perpendicular

$$\begin{pmatrix} -7 \\ 1 \\ 10 \end{pmatrix} \text{ and is given by, } \frac{\left| \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -7 \\ 1 \\ 10 \end{pmatrix} \right|}{\left| \begin{pmatrix} -7 \\ 1 \\ 10 \end{pmatrix} \right|} = \frac{4}{\sqrt{150}}. \quad \diamond$$

2.5 Scalar triple product and volume

There are two products that can be constructed from three vectors in three dimensions. They are

Definition 1. For $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$, the **scalar triple product** of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ is $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$.

Definition 2. For $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$, the **vector triple product** of $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ is $\mathbf{a} \times (\mathbf{b} \times \mathbf{c})$.

In this section we shall briefly examine the properties of the scalar triple product and give it a geometric interpretation as a volume. Although the vector triple product is useful in physics and engineering, we shall not consider it any further in this mathematics course.

It is important to notice that in evaluating the scalar triple product, the **cross product must be calculated before the dot product**. The expression $(\mathbf{a} \cdot \mathbf{b}) \times \mathbf{c}$ has no meaning, because $\mathbf{a} \cdot \mathbf{b}$ is a scalar and the cross product of a scalar and a vector has no meaning.

Some properties of the scalar triple product are listed in the following proposition.

Proposition 1. For $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$,

1. $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$, that is, the dot and cross can be interchanged.
2. $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = -\mathbf{a} \cdot (\mathbf{c} \times \mathbf{b})$, that is, the sign is reversed if the order of two vectors is reversed.
3. $\mathbf{a} \cdot (\mathbf{a} \times \mathbf{b}) = (\mathbf{a} \times \mathbf{a}) \cdot \mathbf{b} = 0$, that is, the scalar triple product is zero if any two vectors are the same.
4. The scalar triple product can be written using the determinant notation.

$$\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

This means that we replace $\mathbf{i}$ by a_1 , $\mathbf{j}$ by a_2 and $\mathbf{k}$ by a_3 in the determinant form of the cross product.

Proof. The proof of Property 1 follows immediately on using the definitions of the dot and cross product to expand the two expressions. We have

$$\begin{aligned} \mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) &= \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \cdot \begin{pmatrix} b_2c_3 - b_3c_2 \\ b_3c_1 - b_1c_3 \\ b_1c_2 - c_1b_2 \end{pmatrix} \\ &= a_1b_2c_3 - a_1b_3c_2 + a_2b_3c_1 - a_2b_1c_3 + a_3b_1c_2 - a_3c_1b_2. \\ (\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c} &= \begin{pmatrix} a_2b_3 - a_3b_2 \\ a_3b_1 - a_1b_3 \\ a_1b_2 - a_2b_1 \end{pmatrix} \cdot \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} \\ &= a_2b_3c_1 - a_3b_2c_1 + a_3b_1c_2 - a_1b_3c_2 + a_1b_2c_3 - a_2b_1c_3. \end{aligned}$$

The two fully expanded expressions are equal, and hence the result is proved.

Property 2 is an immediate consequence of the fact that $\mathbf{b} \times \mathbf{c} = -\mathbf{c} \times \mathbf{b}$.

Property 3 follows immediately from Property 1 and from the fact that $\mathbf{a} \times \mathbf{a} = \mathbf{0}$ for all $\mathbf{a}$.

Property 4 can be proved by expanding both the scalar triple product and the determinant and noting that the expansions of the two are equal. $\square$

NOTE. Property 3 gives one more proof that the cross product $\mathbf{a} \times \mathbf{b}$ is perpendicular to $\mathbf{a}$. Clearly, $\mathbf{b} \cdot (\mathbf{a} \times \mathbf{b}) = 0$, and hence $\mathbf{b}$ is also orthogonal to $\mathbf{a} \times \mathbf{b}$.

2.5.1 Volumes of parallelepipeds

A parallelepiped is a three-dimensional analogue of a parallelogram. Given three vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ we can form a parallelepiped as shown in Figure 9.

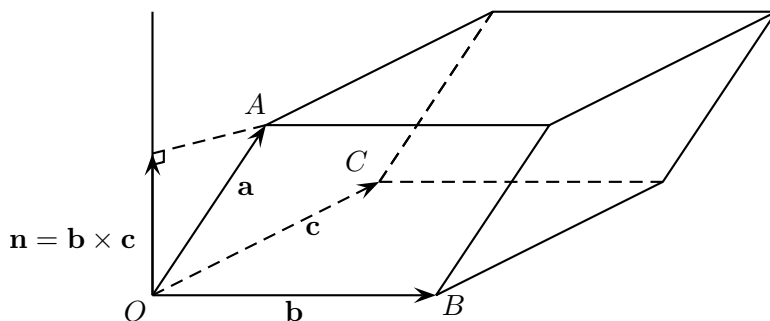


Figure 9: The Scalar Triple Product and the Volume of a Parallelepiped.

The parallelepiped formed from the vectors $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ is called the **parallelepiped spanned by $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$** .

From geometry, the volume of a parallelepiped is “area of base times perpendicular height”. The base is the parallelogram whose sides are the vectors $\mathbf{b}$ and $\mathbf{c}$, and hence from the results of Section 2.4.2, the area of the base is $|\mathbf{b} \times \mathbf{c}|$ and the direction of the perpendicular to the base is the direction of $\mathbf{n} = \mathbf{b} \times \mathbf{c}$. From the results of Section 2.3.2, the length of the projection of the vector $\mathbf{a}$ on the perpendicular $\mathbf{n}$ is the perpendicular height, and hence

$$\text{Perpendicular height} = \frac{|\mathbf{a} \cdot \mathbf{n}|}{|\mathbf{n}|} = \frac{|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|}{|\mathbf{b} \times \mathbf{c}|} = \frac{|\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|}{\text{Area of base}}.$$

Hence, the volume of the parallelepiped is given by the formula

$$\text{Volume} = |\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})|.$$

Example 1. Find the volume of the parallelepiped spanned by $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\begin{pmatrix} -2 \\ 4 \\ -1 \end{pmatrix}$, and $\begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix}$.

SOLUTION. Using the determinant formula for the scalar triple product, we have

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \cdot \left(\begin{pmatrix} -2 \\ 4 \\ -1 \end{pmatrix} \times \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix} \right) = \begin{vmatrix} 1 & 2 & 3 \\ -2 & 4 & -1 \\ 3 & 5 & 1 \end{vmatrix} = 1(4 + 5) - 2(-2 + 3) + 3(-10 - 12) = -59,$$

and hence the volume is $|-59| = 59$. $\diamond$

Notice that if the volume of the parallelepiped spanned by three vectors $\mathbf{a}$, $\mathbf{b}$, $\mathbf{c}$ is zero, i.e., if $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c}) = 0$, then the three vectors are in the same plane (that is, they are coplanar).

2.6 Planes in $\mathbb{R}^3$

Another useful application of vectors and dot and cross products is to the geometry of planes in three dimensions. There are three common forms for the equation of a plane in $\mathbb{R}^3$. These are “parametric vector form”, “Cartesian form”, and “point-normal form”. Each of these equations has a direct geometric interpretation. In this section we shall discuss the geometric interpretation of each of these forms and we shall show how to convert one form to another. We shall also show how to find the distance between a point and a plane in $\mathbb{R}^3$.

2.6.1 Equations of planes in $\mathbb{R}^3$

Parametric Vector Form. In Section 1.5, we showed that the equation of a plane through a given point with position vector $\mathbf{c}$ and parallel to two given non-parallel vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ could be written in a parametric vector form

$$\mathbf{x} = \mathbf{c} + \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 \text{ for } \lambda_1, \lambda_2 \in \mathbb{R}.$$

Cartesian Form. In Section 1.5, we have shown that the linear equation in three unknowns,

$$a_1x_1 + a_2x_2 + a_3x_3 = b,$$

represents a plane in $\mathbb{R}^3$. This linear equation is also often called the **Cartesian form** of the equation of a plane in $\mathbb{R}^3$. We have already shown that this Cartesian form can be converted to a parametric vector form by solving the linear equation.

It is important to note that a single linear equation $a_1x_1 + \cdots + a_nx_n = b$ is the equation of a plane only if $n = 3$. For $n = 2$, the equation is $a_1x_1 + a_2x_2 = b$ which represents a line. In general, solving a single linear equation with n unknowns yields a parametric vector form of solution containing $n - 1$ parameters, whereas a parametric vector equation of a plane must contain 2 parameters.

The Cartesian form can be given a geometric interpretation in terms of intercepts on the three coordinate axes.

We first divide the Cartesian form by the right hand side and rearrange it as

$$\frac{x_1}{d_1} + \frac{x_2}{d_2} + \frac{x_3}{d_3} = 1,$$

$$\text{where } d_1 = \frac{b}{a_1}, \quad d_2 = \frac{b}{a_2}, \quad d_3 = \frac{b}{a_3}.$$

Then, to obtain the intercept on the x_1 axis, we set $x_2 = x_3 = 0$ in the equation, and find $x_1 = d_1$. By a similar argument, the intercept on the x_2 axis is d_2 and the intercept on the x_3 axis is d_3 .

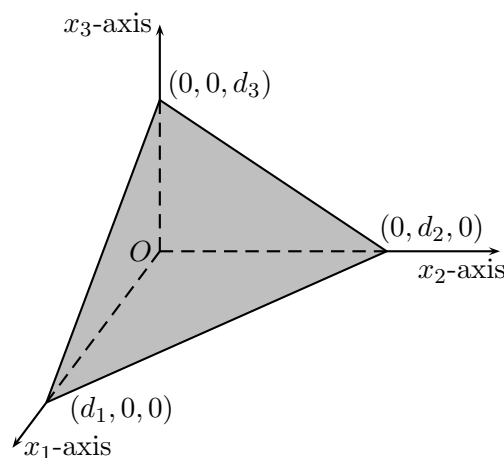


Figure 10: $a_1x_1 + a_2x_2 + a_3x_3 = b$.

The rule for obtaining intercepts is therefore to rewrite the equation with 1 on the right, and then the intercepts are the reciprocals of the coefficients of the variables. Notice that if the coefficient of x_1 is zero, then the plane is parallel to the x_1 axis. A similar result applies if the coefficient of x_2 or x_3 is zero.

Example 1. Write down the equation of a plane which has the intercepts 4, 7, -2 on the three coordinate axes.

SOLUTION. The coefficients of the variables are the reciprocals of the intercepts, and hence the equation is

$$\frac{x_1}{4} + \frac{x_2}{7} + \frac{x_3}{-2} = 1, \quad \text{or} \quad 14x_1 + 8x_2 - 28x_3 = 56.$$

◇

Point-Normal Form. A point-normal form is an equation of the form $\mathbf{n} \cdot (\mathbf{x} - \mathbf{c}) = 0$, where $\mathbf{n}$ and $\mathbf{c}$ are fixed vectors and $\mathbf{x}$ is the position vector of a point in space. If we rewrite this equation as $\mathbf{n} \cdot \mathbf{x} = \mathbf{n} \cdot \mathbf{c}$ and then expand in terms of coordinates, we obtain

$$n_1x_1 + n_2x_2 + n_3x_3 = n_1c_1 + n_2c_2 + n_3c_3 = b,$$

which is a Cartesian equation of a plane. Thus, the equation $\mathbf{n} \cdot (\mathbf{x} - \mathbf{c}) = 0$ is the equation of a plane. As for a Cartesian form, a point-normal equation is an equation of a plane for three-dimensional vectors only.

The name “point-normal form” for the equation $\mathbf{n} \cdot (\mathbf{x} - \mathbf{c}) = 0$ is based on the following geometric interpretation (see Figure 11). Clearly, $\mathbf{x} = \mathbf{c}$ is a solution of the equation, and hence $\mathbf{c}$ is the position vector of a point on the plane. If $\mathbf{x}$ is any other point in the plane, then the line segment $\mathbf{x} - \mathbf{c}$ lies in the plane. The equation then says that the vector $\mathbf{n}$ is normal to all line segments lying in the plane. The vector $\mathbf{n}$ is called a normal to the plane. A point-normal form is therefore the most convenient form of equation to use when a point on the plane and a normal to the plane are known.

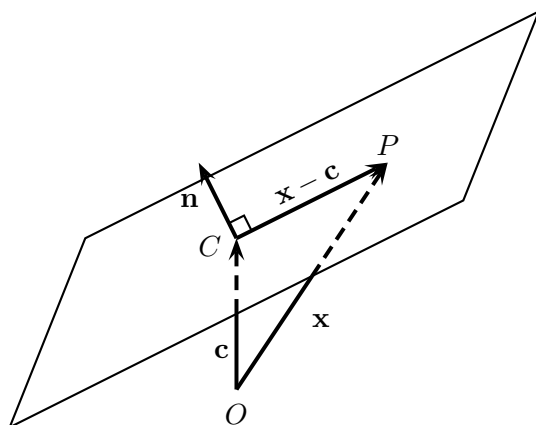


Figure 11: Point-Normal Form $\mathbf{n} \cdot (\mathbf{x} - \mathbf{c}) = 0$.

Example 2. Find a point-normal form for the plane which passes through the point $(1, -2, 3)$ and has $\begin{pmatrix} -3 \\ 4 \\ 5 \end{pmatrix}$ as a normal vector.

SOLUTION. $\begin{pmatrix} -3 \\ 4 \\ 5 \end{pmatrix} \cdot \left(\mathbf{x} - \begin{pmatrix} 1 \\ -2 \\ 3 \end{pmatrix} \right) = 0.$ $\diamond$

We shall now show how to convert from one form of equation to another by giving examples of the methods.

Example 3 (Conversion from point-normal to Cartesian form). Find an equation, in Cartesian form, of the plane which passes through the point $(2, 0, 4)$ and whose normal is $\begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix}$.

SOLUTION. A point-normal form is $\begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} \cdot \left(\mathbf{x} - \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix} \right) = 0$, and hence a corresponding Cartesian form is $2x_1 - 3x_2 + 5x_3 = \begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix} = 24.$ $\diamond$

Example 4 (Conversion from Cartesian to point-normal form). Find a point-normal form of for the plane described by the Cartesian equation

$$3x_1 - 7x_2 + 5x_3 = 21.$$

SOLUTION. A comparison of Cartesian and point-normal forms shows that the coefficients of x_1, x_2, x_3 are just the coordinates of a normal to the plane. Thus, a normal to the plane is $\mathbf{n} = \begin{pmatrix} 3 \\ -7 \\ 5 \end{pmatrix}$. To

find some point on the plane, we let $x_2 = x_3 = 0$, and then from the equation we find $x_1 = \frac{21}{3} = 7$. Hence $(7, 0, 0)$ is a point on the plane.

A point-normal form is therefore $\begin{pmatrix} 3 \\ -7 \\ 5 \end{pmatrix} \cdot \left(\mathbf{x} - \begin{pmatrix} 7 \\ 0 \\ 0 \end{pmatrix} \right) = 0.$ $\diamond$

Example 5 (Conversion from parametric vector to point-normal form). Find a point-normal form for the plane described by the parametric vector equation

$$\mathbf{x} = \begin{pmatrix} 3 \\ -5 \\ 1 \end{pmatrix} + \lambda_1 \begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix} + \lambda_2 \begin{pmatrix} -2 \\ 6 \\ 1 \end{pmatrix}.$$

SOLUTION. The plane is parallel to $\begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 6 \\ 1 \end{pmatrix}$. The cross product of these two vectors is therefore normal to the plane. Thus, a normal to the plane is $\mathbf{n} = \begin{pmatrix} 0 \\ 3 \\ 4 \end{pmatrix} \times \begin{pmatrix} -2 \\ 6 \\ 1 \end{pmatrix} = \begin{pmatrix} -21 \\ -8 \\ 6 \end{pmatrix}$. As $(3, -5, 1)$ is a point on the plane, a point-normal form is $\begin{pmatrix} -21 \\ -8 \\ 6 \end{pmatrix} \cdot \left(\mathbf{x} - \begin{pmatrix} 3 \\ -5 \\ 1 \end{pmatrix} \right) = 0$. $\diamond$

Example 6 (Conversion from parametric vector to Cartesian form). Find an equation, in Cartesian form, for the plane through the point $(2, -1, 3)$ parallel to the vectors $\begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix}$.

SOLUTION. The simplest way to carry out this conversion is to first find a point-normal form as in Example 5, and then to find a Cartesian form as in Example 3.

A normal to the plane is $\begin{pmatrix} 3 \\ -1 \\ 4 \end{pmatrix} \times \begin{pmatrix} 2 \\ 1 \\ 5 \end{pmatrix} = \begin{pmatrix} -9 \\ -7 \\ 5 \end{pmatrix}$.

A point-normal form is $\begin{pmatrix} -9 \\ -7 \\ 5 \end{pmatrix} \cdot \left(\mathbf{x} - \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} \right) = 0$.

Hence a Cartesian form is $-9x_1 - 7x_2 + 5x_3 = 4$. $\diamond$

Example 7 (Conversion from point-normal to parametric vector form). Find a parametric vector form for the plane which is normal to $\begin{pmatrix} -2 \\ 0 \\ 4 \end{pmatrix}$ and passes through the point $(0, 2, 1)$.

SOLUTION. One way to carry out this conversion is to first convert the point-normal we were given into a Cartesian form as in Example 3, and then convert the Cartesian form we found into a parametric vector form as in section 1.5.3.

A point-normal form is $\begin{pmatrix} -2 \\ 0 \\ 4 \end{pmatrix} \cdot \left(\mathbf{x} - \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} \right) = 0$. Hence, a Cartesian form is

$$-2x_1 + 4x_3 = \begin{pmatrix} -2 \\ 0 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} = 4.$$

The variable x_2 does not appear in the equation, but don't forget it. As in section 1.5.3, we set $x_2 = \lambda_1$ to be a real parameter. Also, x_3 can have any value in the equation, and hence we set $x_3 = \lambda_2$ to be a second real parameter. Then, we obtain $x_1 = -2 + 2x_3 = -2 + 2\lambda_2$. Then, on

rewriting the solution in vector form, we obtain a parametric vector form of the equation of the plane to be

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} -2 + 2\lambda_2 \\ \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} -2 \\ 0 \\ 0 \end{pmatrix} + \lambda_1 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}.$$

◇

Example 8. Find a point-normal form for the plane that passes through the three points $A(3, 1, 2)$, $B(0, -2, 1)$, and $C(1, 2, 3)$.

SOLUTION. The plane is parallel to the two line segments

$$\overrightarrow{AB} = \begin{pmatrix} -3 \\ -3 \\ -1 \end{pmatrix} \quad \text{and} \quad \overrightarrow{AC} = \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}.$$

Hence a normal to the plane is

$$\mathbf{n} = \overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} -3 \\ -3 \\ -1 \end{pmatrix} \times \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 5 \\ -9 \end{pmatrix}.$$

The point $A(3, 1, 2)$ is a point on the plane, and hence a point-normal form is

$$\begin{pmatrix} -2 \\ 5 \\ -9 \end{pmatrix} \cdot \left(\mathbf{x} - \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} \right) = 0.$$

◇

NOTE. Although we have not stated it explicitly, most of the solutions given to the examples in this section are not unique. For example, any multiple of a normal to a plane is still a normal, any multiple of a vector parallel to a plane is still a vector parallel to the plane, there are an infinite number of points on a plane etc. However, the intercepts of a plane on the coordinate axes are unique.

2.6.2 Distance between a point and a plane in $\mathbb{R}^3$

The distance from the point B to the plane is $|\overrightarrow{PB}|$, where $\overrightarrow{PB}$ is normal to the plane. If A is any point on the plane, $\overrightarrow{PB}$ is the projection of $\overrightarrow{AB}$ on any vector normal to the plane. The shortest distance is the “perpendicular distance” to the plane, and it is the length of the projection on a normal to the plane of a line segment from any point on the plane to the given point.

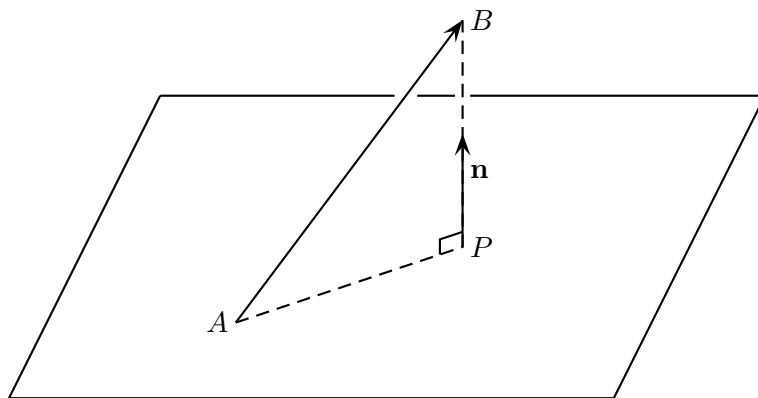


Figure 12: Shortest Distance between Point and Plane.

Example 9. Find the distance between the point $(2, -1, 3)$ and the plane

$$\mathbf{x} = \begin{pmatrix} 0 \\ 4 \\ 2 \end{pmatrix} + \lambda_1 \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} -3 \\ 1 \\ 5 \end{pmatrix}.$$

SOLUTION. $(0, 4, 2)$ is a point on the plane and $\mathbf{n} = \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} \times \begin{pmatrix} -3 \\ 1 \\ 5 \end{pmatrix} = \begin{pmatrix} 3 \\ -26 \\ 7 \end{pmatrix}$ is a vector normal to the plane. A vector from the point $(0, 4, 2)$ on the plane to the given point $(2, -1, 3)$ is $\begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} - \begin{pmatrix} 0 \\ 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix}$.

The shortest distance is the length of the projection of $\begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix}$ on $\mathbf{n}$, which is

$$\frac{\left| \begin{pmatrix} 2 \\ -5 \\ 1 \end{pmatrix} \cdot \mathbf{n} \right|}{|\mathbf{n}|} = \frac{143}{\sqrt{734}}.$$

◇

2.7 Geometry and Maple

Rather than list all the commands that you might need in this chapter, we suggest that you have a look through the commands available in the **LinearAlgebra** package using the on-line manual. The command

```
?LinearAlgebra;
```

opens a help page with a list of the commands options available in the linear algebra package. If you think that **CrossProduct** looks promising, you should click on its hyperlink, or type in:

```
?CrossProduct
```

in the Maple worksheet. The on-line help usually gives a few examples. Try these first then you can experiment with a few problems of your own.

The package `geom3d` contains many useful procedures for solving problems in three-dimensional geometry. In fact, every one of the problems for Section 5.7 can be solved with procedures in `geom3d`. The following is an outline of some of the things which you can do with `geom3d`.

The method of assigning a name to a point, line or plane is not quite what you might expect. To say that A is the point $(1, 2, 3)$, you do NOT enter `A:=[1,2,3];`, you enter `point(A,[1,2,3]);`. The command `line` is used to assign a name to a line. The line may be specified by giving two points on it or a point on it and a direction parallel to it. The command `plane` is used to assign a name to a plane. The plane may be specified by giving a Cartesian equation for it or three (non-collinear) points on it or a point on it and a normal direction or a point on it and two lines parallel to it. The command `sphere` is used to assign a name to a sphere. A sphere may be specified by giving its Cartesian equation or four points on it or the end-points of a diameter or its center and its radius. To display the specifications of one of these things you need to use `detail` (see the example below). For a plane, the detail includes a Cartesian equation for the plane. (If you want to find a normal to a plane p use

`NormalVector(p);`)

Be warned that if you specify a plane or sphere by means of an equation then Maple will want you to specify the names of the variables which are associated with the three axes. You can do this by listing them as a third argument to the `plane` or `sphere` command, as in

`plane(P,x+y+z=1,[x,y,z]);`

If you leave out the `[x,y,z]` then Maple will, rather strangely, prompt you to **enter the name of the x-axis**, to which you reply `x;`, and similarly for the other two axes.

When you have set up objects of these types you can, for example, use the command `distance` to find the distance between two of them or the command `intersection` to find the intersection of two of them (except the intersection of a line and a sphere) or the command `FindAngle` to find the angle between two of them. You can use the Maple `help` to find out more about any of these commands and to find out about the many other commands available in `geom3d`.

In the following example, we first label the points $A(0, 1, 2)$ and $B(2, 3, 1)$ and the line AB through A and B . Applying `detail` to AB shows that the direction of the line is $\begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ and the line can be expressed in parametric vector form as

$$\mathbf{x} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix} + t \begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}, \quad t \in \mathbb{R}.$$

Then we assign the label P to the plane through $C(4, 5, 6)$ with normal $(1, 1, 1)$ and use `detail` to find that P can be described by the Cartesian equation

$$x + y + z = 15.$$

Then we assign the label X to the point of intersection of the line AB and the plane P and find that the coordinates of X are $(8, 9, -2)$. Finally, we find that the plane ABC through the three points A, B, C can be described by the Cartesian equation

$$12x - 12y + 12 = 0.$$

Notice that we use a colon : to suppress the output of most of the commands because the output would just be an echo of assigned names.

```

with(geom3d):
point(A,[0,1,2]),point(B,[2,3,1]):
line(AB,[A,B]):
detail(AB);
Warning, assume that the name of the parameter in the parametric equations is _t
Warning, assuming that the names of the axes are _x, _y, and _z
name of the object: AB
form of the object: line3d
equation of the line: [_x = 2*_t, _y = 1+2*_t, _z = 2-_t]
point(C,[4,5,6]):
plane(P,[C,[1,1,1]]):
detail(P);
Warning, assuming that the names of the axes are _x, _y and _z
name of the object: P
form of the object: plane3d
equation of the plane: -15+_x+_y+_z = 0
intersection(X,AB,P):
detail(X);
name of the object: X
form of the object: point3d
coordinates of the point: [8, 9, -2]
plane(ABC,[A,B,C]):
detail(ABC);
Warning, assuming that the names of the axes are _x, _y and _z
name of the object: ABC
form of the object: plane3d
equation of the plane: 12+12*_x-12*_y = 0

```

Problems for Chapter 2

Questions marked with [R] are routine, [H] harder, [M] Maple and [X] are for MATH1141 only. You should make sure that you can do the easier questions before you tackle the more difficult questions. Questions marks with a [V] have video solutions available on Moodle.

Problems 2.2: The dot product

1. [R][V] Find the angles between the following pairs of vectors:

$$\text{a) } \begin{pmatrix} -2 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 0 \end{pmatrix}; \quad \text{b) } \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} -2 \\ 5 \\ 1 \end{pmatrix}; \quad \text{c) } \begin{pmatrix} 7 \\ 1 \\ -2 \end{pmatrix}, \begin{pmatrix} 3 \\ -11 \\ 5 \end{pmatrix}; \quad \text{d) } \begin{pmatrix} 3 \\ 0 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} -2 \\ 6 \\ 1 \\ 3 \end{pmatrix}.$$

2. [H] Find the cosines of the internal angles of the triangles whose vertices have the following coordinate vectors:

$$\begin{aligned} \text{a) } & A \begin{pmatrix} 4 \\ 0 \\ 2 \end{pmatrix}, B \begin{pmatrix} 6 \\ 2 \\ 1 \end{pmatrix} \text{ and } C \begin{pmatrix} 5 \\ 1 \\ 6 \end{pmatrix}; & \text{b) } & A \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}, B \begin{pmatrix} -1 \\ 3 \\ 0 \end{pmatrix} \text{ and } C \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix}; \\ \text{c) } & A \begin{pmatrix} 1 \\ -2 \\ 0 \\ 3 \end{pmatrix}, B \begin{pmatrix} 0 \\ 4 \\ -2 \\ 5 \end{pmatrix} \text{ and } C \begin{pmatrix} -2 \\ 1 \\ 0 \\ 3 \end{pmatrix}. \end{aligned}$$

3. [R] A cube has vertices at the 8 points $O(0, 0, 0)$, $A(1, 0, 0)$, $B(1, 1, 0)$, $C(0, 1, 0)$, $D(0, 0, 1)$, $E(1, 0, 1)$, $F(1, 1, 1)$, $G(0, 1, 1)$. Sketch the cube, and then find the angle between the diagonals $\overrightarrow{OF}$ and $\overrightarrow{AG}$.

4. [H][V] Prove the following properties of dot products for vectors $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$. :

$$\text{a) } \mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}, \quad \text{b) } \mathbf{a} \cdot (\lambda \mathbf{b}) = \lambda(\mathbf{a} \cdot \mathbf{b}), \quad \text{c) } \mathbf{a} \cdot (\mathbf{b} + \mathbf{c}) = \mathbf{a} \cdot \mathbf{b} + \mathbf{a} \cdot \mathbf{c}.$$

5. [X] Prove that $||\mathbf{a}| - |\mathbf{b}|| \leq |\mathbf{a} - \mathbf{b}|$ for all $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$.

HINT. See the proof of Minkowski's inequality in Section 2.2.2.

6. [H] Use the dot product to prove that the diagonals of a square intersect at right angles.

Problems 2.3 : Applications: orthogonality and projection

7. [R] Let $\mathbf{u}_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{pmatrix}$, $\mathbf{u}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$, $\mathbf{u}_3 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}$ and $\mathbf{a} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$. Show that the set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthonormal set. Find scalars $\lambda_1, \lambda_2, \lambda_3$ such that $\mathbf{a} = \lambda_1 \mathbf{u}_1 +$

$$\lambda_2 \mathbf{u}_2 + \lambda_3 \mathbf{u}_3.$$

HINT. See Examples 2 of Section 2.3.

8. [H] Consider the triangle ABC in $\mathbb{R}^3$ formed by the points $A(3, 2, 1)$, $B(4, 4, 2)$ and $C(6, 1, 0)$.
- Find the coordinates of the midpoint M of the side BC .
 - Find the angle BAC .
 - Find the area of the triangle ABC .
 - Find the coordinates of the point D on BC such that AD is perpendicular to BC .
9. [R][V] Find the following projections:
- the projection of $\begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}$ on $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$,
 - the projection of $\begin{pmatrix} 2 \\ -1 \\ 2 \\ 4 \end{pmatrix}$ on $\begin{pmatrix} -1 \\ 3 \\ 0 \\ 2 \end{pmatrix}$,
 - the projection of $\begin{pmatrix} -2 \\ 2 \\ 7 \end{pmatrix}$ on the direction of the line $\mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$.
10. [R] Find the shortest distances between
- the point $(-2, 1, 5)$ and the line $\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ -5 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ 3 \\ -4 \end{pmatrix}$;
 - the point $(0, 3, 8)$ and the line $\frac{x_1 - 1}{1} = \frac{x_2 - 2}{-1} = \frac{x_3 - 3}{4}$;
 - [X] the point $(11, 2, -1)$ and the line of intersection of the planes
- $$\mathbf{x} \cdot \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} = 0 \quad \text{and} \quad \mathbf{x} = \lambda_1 \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ 1 \\ -3 \end{pmatrix}.$$
11. [X] A point P in $\mathbb{R}^n$ has coordinate vector $\mathbf{p}$. Find the coordinate vector of the point Q which is the reflection of P in the line ℓ which passes through the point $\mathbf{a}$ parallel to the direction $\mathbf{d}$.
- NOTE. Define Q to be the point which lies in the same plane as P and ℓ with ℓ bisecting the interval PQ .
12. [X] Fix $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$ with $\mathbf{b} \neq \mathbf{0}$. Let $q(\lambda) = |\mathbf{a} - \lambda \mathbf{b}|^2$.
- Show $q(\lambda)$ is a minimum when $\lambda = \lambda_0 = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2}$.

- b) Determine $q(\lambda_0)$ and hence show that $-|\mathbf{a}| |\mathbf{b}| \leq \mathbf{a} \cdot \mathbf{b} \leq |\mathbf{a}| |\mathbf{b}|$.
13. [X] Let B be a point in $\mathbb{R}^n$ with coordinate vector $\mathbf{b}$. Let $\mathbf{x} = \mathbf{a} + \lambda \mathbf{d}$, $\lambda \in \mathbb{R}$ be the equation of a line. Do the following:
- a) Show that the square of the distance from B to an arbitrary point $\mathbf{x}$ on the line is given by

$$q(\lambda) = |\mathbf{b} - \mathbf{a}|^2 - 2\lambda(\mathbf{b} - \mathbf{a}) \cdot \mathbf{d} + \lambda^2 |\mathbf{d}|^2.$$

- b) Find the shortest distance between the point B and the line by minimising $q(\lambda)$.
- c) If P is the point on the line closest to B , show that

$$\overrightarrow{PB} = \mathbf{b} - \mathbf{a} - \text{proj}_{\mathbf{d}}(\mathbf{b} - \mathbf{a}),$$

and show that $\overrightarrow{PB}$ is orthogonal to the direction $\mathbf{d}$ of the line.

NOTE. This problem proves that the shortest distance between a point and a line is obtained by “dropping a perpendicular from the point to the line”.

Problems 2.4 : The cross product

14. [R] Find the cross product $\mathbf{a} \times \mathbf{b}$ of the following pairs of vectors:
- a) $\mathbf{a} = \begin{pmatrix} 0 \\ 2 \\ -4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$, b) $\mathbf{a} = \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} -2 \\ 6 \\ 1 \end{pmatrix}$,
- c) $\mathbf{a} = \begin{pmatrix} 1 \\ 9 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 0 \\ -5 \end{pmatrix}$.
15. [R][V] Find a vector which is perpendicular to $\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 0 \\ 4 \end{pmatrix}$.
16. [H] Prove the following properties of cross products for vectors $\mathbf{a}, \mathbf{b}, \mathbf{c} \in \mathbb{R}^3$:
- a) $\mathbf{a} \times \mathbf{a} = \mathbf{0}$; b) $\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$;
c) $\mathbf{a} \times (\lambda \mathbf{b}) = \lambda(\mathbf{a} \times \mathbf{b})$; d) $\mathbf{a} \times (\mathbf{b} + \mathbf{c}) = \mathbf{a} \times \mathbf{b} + \mathbf{a} \times \mathbf{c}$.
17. [R] Find the areas of, and the normals to the planes of, the following parallelograms:
- a) the parallelogram spanned by $\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix}$;
- b) a parallelogram which has vertices at the three points $A(0, 2, 1)$, $B(-1, 3, 0)$ and $C(3, 1, 2)$ and sides $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

18. [R][V] Find the areas of the triangles with the following vertices:

- a) $A(0, 2, 1)$, $B(-1, 3, 0)$ and $C(3, 1, 2)$;
- b) $A(2, 2, 0)$, $B(-1, 0, 2)$ and $C(0, 4, 3)$.

19. [R] Let D, E, F be the points with coordinate vectors

$$\mathbf{d} = \begin{pmatrix} 5 \\ 6 \\ 7 \end{pmatrix}, \mathbf{e} = \begin{pmatrix} 6 \\ 7 \\ 8 \end{pmatrix}, \mathbf{f} = \begin{pmatrix} 7 \\ 8 \\ 10 \end{pmatrix}$$

- a) Calculate $\cos(\angle DEF)$ as a surd.
- b) Calculate the area of $\triangle DEF$ as a surd.

20. [X] Find the shortest distances between

- a) the line through $(1, 2, 3)$ parallel to $\begin{pmatrix} 3 \\ 0 \\ 1 \end{pmatrix}$ and the line through $(0, 2, 5)$ parallel to $\begin{pmatrix} 3 \\ -2 \\ 2 \end{pmatrix}$;
- b) the line through the points $(1, 3, 1)$ and $(1, 5, -1)$ and the line through the points $(0, 2, 1)$ and $(1, 2, -3)$;
- c) the lines $\mathbf{x} = \begin{pmatrix} 2 \\ 7 \\ 8 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ 3 \\ -5 \end{pmatrix}$ and $\frac{x_1 - 1}{-10} = \frac{x_2 - 2}{1} = \frac{x_3 - 3}{4}$.

21. [X] Let $\mathbf{a}, \mathbf{b}, \mathbf{c}$ be three vectors in $\mathbb{R}^3$ which satisfy the relations $\mathbf{b} = \mathbf{c} \times \mathbf{a}$ and $\mathbf{c} = \mathbf{a} \times \mathbf{b}$.

- a) Show that $\mathbf{a}, \mathbf{b}$ and $\mathbf{c}$ are a set of mutually orthogonal vectors.
- b) Show that $\mathbf{b}$ and $\mathbf{c}$ are of equal length and that if $\mathbf{b} \neq \mathbf{0}$, then $\mathbf{a}$ is a unit vector (i.e. a vector of length 1).

22. [X] A tetrahedron has vertices A, B, C and D with coordinate vectors for the points being

$$\mathbf{a} = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} -1 \\ 4 \\ 1 \end{pmatrix}, \mathbf{c} = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix} \text{ and } \mathbf{d} = \begin{pmatrix} -3 \\ 1 \\ 2 \end{pmatrix}.$$

Find parametric vector equations for the two altitudes of the tetrahedron which pass through the vertices A and B , and determine whether the two altitudes intersect or not.

NOTE. An altitude of a tetrahedron through a vertex is a line through the vertex and perpendicular to the opposite face.

23. [X] Points A, B, C and D have coordinate vectors $\begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 0 \\ -1 \end{pmatrix}$, respectively.

- a) Find a parametric vector equations of the line through A and B and the line through C and D .
- b) Find the shortest distance between the lines AB and CD .
- c) Find the point P on AB and point Q on CD such that PQ is the shortest distance between the lines AB and CD .

Problems 2.5 : Scalar triple product and volume

24. [R][V] Show that $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$ can be written in the form

$$\begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix},$$

where the vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ are replaced by the scalars a_1, a_2, a_3 .

25. [R] Find the volumes of the following parallelepipeds:

- a) the parallelepiped spanned by $\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$;
- b) a parallelepiped which has vertices at the four points $A(2, 1, 3)$, $B(-2, 1, 4)$, $C(0, 4, 1)$ and $D(3, -1, 0)$, with sides $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{AD}$.

26. [R] Show that the four points A, B, C, O with coordinate vectors $\begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$, $\begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix}$, $\begin{pmatrix} 6 \\ 1 \\ 1 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ are coplanar.

Problems 2.6 : Planes in $\mathbb{R}^3$

27. [R][V] Find parametric vector, point-normal, and Cartesian forms for the following planes:

- a) the plane through $(1, 2, -2)$ perpendicular to $\begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$;
- b) the plane through $(1, 2, -2)$ parallel to $\begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$;
- c) the plane through the three points $(1, 2, -2)$, $(-1, 1, 2)$ and $(2, 3, 1)$;
- d) the plane with intercepts $-1, 2$ and -4 on the x_1, x_2 and x_3 axes;

- e) [X] the plane through $(1, 2, -2)$ which is parallel to $\begin{pmatrix} -1 \\ 2 \\ -2 \end{pmatrix}$ and the line of intersection of the planes

$$\mathbf{x} \cdot \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = 0 \quad \text{and} \quad \mathbf{x} = \lambda_1 \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}.$$

28. [R] Consider four points O, A, B, C in $\mathbb{R}^3$ with coordinate vectors

$$\mathbf{0} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \mathbf{a} = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \mathbf{c} = \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}.$$

Let Π be the plane through A and parallel to the lines OB and OC .

- Find a parametric vector form for Π .
 - Find a vector $\mathbf{n}$ normal to Π .
 - Use the point normal form to find a Cartesian equation for Π .
29. [R] Find the following projections:
- the projection of $\begin{pmatrix} 2 \\ 3 \\ 8 \end{pmatrix}$ on the normal to the plane $2x_1 + 2x_2 + x_3 = 4$;
 - [X] the projection of $\begin{pmatrix} 2 \\ 2 \\ -1 \end{pmatrix}$ on the line of intersection of the planes

$$\mathbf{x} \cdot \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} = 0 \quad \text{and} \quad \mathbf{x} = \lambda_1 \begin{pmatrix} 2 \\ 1 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ 1 \\ -3 \end{pmatrix}.$$

30. [R][V] Find the shortest distances between

- the point $(2, 6, -5)$ and the plane $\left(\mathbf{x} - \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right) \cdot \begin{pmatrix} -2 \\ 4 \\ 4 \end{pmatrix} = 0$;
- the point $(1, 4, 1)$ and the plane $2x_1 - x_2 + x_3 = 5$;
- the point $(1, 2, 1)$ and the plane with intercepts at $3, -1, 2$ on the three axes;
- the origin and the plane through the three points $(2, 1, 3)$, $(5, 3, 1)$ and $(5, 1, 2)$.

31. [R] Let P be the plane in $\mathbb{R}^3$ through the points $A = (1, 2, 0)$, $B = (0, 1, 2)$, and $C = (-1, 3, 1)$.
- Find a parametric vector form for the plane P .
 - Find a vector $\mathbf{n}$ normal to the plane P .

- c) Find a point normal form for the plane P .
- d) Find the shortest distance from the point $Q = (2, 4, 5)$ to the plane P .
32. [X] a) Let $\mathbf{a}$ and $\mathbf{v}$ be two non-zero vectors in $\mathbb{R}^3$. Show how to write $\mathbf{v}$ as $\mathbf{c} + \mathbf{d}$ where $\mathbf{c}$ is parallel to $\mathbf{a}$ and $\mathbf{d}$ is perpendicular to $\mathbf{a}$.
- b) Consider the plane

$$\Pi = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \lambda_1, \lambda_2 \in \mathbb{R} \right\}$$

and the vector $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$. By using a) (or otherwise) express $\mathbf{v}$ as $\mathbf{c} + \mathbf{d}$ where $\mathbf{d}$ is parallel to Π and $\mathbf{c}$ is perpendicular to Π . (We call $\mathbf{d}$ the projection of $\mathbf{v}$ onto Π).

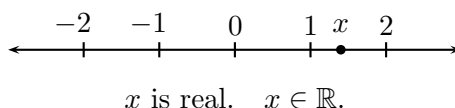
Chapter 3

COMPLEX NUMBERS

*“Ignorance of Axioms”, the Lecturer continued,
“is a great drawback in life. It wastes so much time
to have to say them over and over again.”*

Lewis Carroll, Sylvie and Bruno Concluded.

The main purpose of this chapter is to introduce the system of complex numbers. In the calculus part of this subject we concentrate on the set $\mathbb{R}$ of real numbers which we think of as corresponding to the points on the number line.



If one wants to solve all quadratic equations then the real numbers do not suffice. In particular, if $b^2 - 4ac < 0$ then it is not possible to find real solutions to $ax^2 + bx + c = 0$. In this chapter, we will construct a larger set of numbers in which such an equation can be solved. Indeed, in this larger set, every polynomial equation has at least one solution. Much of mathematics becomes simpler (not more complicated) when complex numbers are used and applications to areas such as physics, chemistry, electrical and mechanical engineering, oceanography, economics and the theory of dynamical systems are made simpler by using complex numbers.

3.1 A review of number systems

Before going into the details of the complex number system, it is worth while reviewing some of the basic properties of the number systems which you have already met in primary and high school mathematics. This review is not intended as a rigorous development of the theory of number systems – such a rigorous development is quite difficult and is sometimes given in more advanced courses.

The first system of numbers that you will have met consists of the set of **natural numbers** (or counting numbers)

$$\mathbb{N} = \{0, 1, 2, \dots\},$$

together with rules of addition and multiplication. This set of numbers has the property that addition or multiplication of natural numbers always produces another natural number, whereas subtraction or division may not. Thus $3 - 5$ and $\frac{3}{5}$ are not natural numbers. We say that the set of natural numbers is **closed** under the operations of addition and multiplication, whereas the set is not closed under the operations of subtraction or division.

A very limited class of equations have solutions in $\mathbb{N}$. For example, in $\mathbb{N}$, $x+3=7$ and $5x+2=17$ can be solved, but $x+7=3$ and $5x+2=18$ can not! Thus, to solve all linear equations, a larger set of numbers is required.

A set of numbers that is closed under subtraction (i.e., for which subtraction is always possible) can be obtained by **extending** the natural number system by introducing a new number (-1) . Then, after using the usual rules for addition and multiplication, the set of **integers**

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

is obtained.

However, the set $\mathbb{Z}$ is not closed under division as, for example, $\frac{3}{5}$ is not an integer. At this stage $x+7=3$ has a unique solution but $5x+2=18$ still has no solution. To solve such an equation, we need fractions. Thus, we extend the system of integers to the set of **rational numbers**, $\mathbb{Q}$, defined by

$$\mathbb{Q} = \left\{ \frac{p}{q} : p, q \in \mathbb{Z} \text{ for } q \neq 0 \right\}.$$

This system is then closed under the four standard operations of arithmetic of addition, subtraction, multiplication and division (division by zero excluded).

Now that the rationals are in the set that we are focusing on, all equations in one variable with rational coefficients, such as $5x+2=18$ or $\frac{3}{2}x+\frac{1}{4}=1$ can be solved. All solutions will be rational although sometimes a solution may happen to be an integer. Indeed, if we now consider the general equation in one variable $ax+b=c$ with $a, b, c \in \mathbb{Q}$ then this has a unique solution $x=(c-b)/a$ unless $a=0$.

The rationals are the first and primary example of a mathematical concept called a **field**. (See definition 1.) A field is a set (of numbers) which satisfy “twelve number laws”. These laws, or axioms as they are called, form a minimal list of properties that one needs in order to be able to add, subtract, multiply and divide (by non-zero numbers).

Much elementary mathematics can be carried out using rational numbers, as is done by your calculator. The set of real numbers, $\mathbb{R}$, contains all the rationals, along with numbers such as $\sqrt{2}, \sqrt{3}, \pi, e$, etc. which are **not** rational. The proofs that $\sqrt{2}, \sqrt{3}, \sqrt{5}, \dots$ are irrational are very straightforward. The proofs that π and e are irrational are harder. You will see a proof of the irrationality of e later in the year. The last question in the HSC Extension 2 2003 paper outlines a proof of the irrationality of π .

Using real numbers, we can now solve some quadratic and higher degree equations. For example, $x^2-3=0$ has 2 real solutions $x=\pm\sqrt{3}$ and $x^3-x^2-3x+3=0$ has 3 real solutions $x=1, \pm\sqrt{3}$. The set of real numbers also satisfies all twelve number laws and thus forms a field. Hence all equations $ax+b=c$ with $a, b, c \in \mathbb{R}$, $a \neq 0$, have a unique solution.

Note that the general motivation behind the development of number systems sketched above is that some set of numbers is extended to a new set by introducing new “numbers” so that some operation (e.g., subtraction, division, or finding lengths of sides of squares) is always possible in the extended set.

We conclude this section by introducing the definition of a field.

Definition 1. Let $\mathbb{F}$ be a non-empty set of elements for which a rule of addition (+) and a rule of multiplication are defined. Then the system is a **field** if the following twelve axioms (or fundamental number laws) are satisfied.

1. **Closure under Addition.** If $x, y \in \mathbb{F}$ then $x + y \in \mathbb{F}$.
2. **Associative Law of Addition.** $(x + y) + z = x + (y + z)$ for all $x, y, z \in \mathbb{F}$.
3. **Commutative Law of Addition.** $x + y = y + x$ for all $x, y \in \mathbb{F}$.
4. **Existence of a Zero.** There exists an element of $\mathbb{F}$ (usually written as 0) such that $0 + x = x + 0 = x$ for all $x \in \mathbb{F}$.
5. **Existence of a Negative.** For each $x \in \mathbb{F}$, there exists an element $w \in \mathbb{F}$ (usually written as $-x$) such that $x + w = w + x = 0$.
6. **Closure under Multiplication.** If $x, y \in \mathbb{F}$ then $xy \in \mathbb{F}$.
7. **Associative Law of Multiplication.** $x(yz) = (xy)z$ for all $x, y, z \in \mathbb{F}$.
8. **Commutative Law of Multiplication.** $xy = yx$ for all $x, y \in \mathbb{F}$.
9. **Existence of a One.** There exists a non-zero element of $\mathbb{F}$ (usually written as 1) such that $x1 = 1x = x$ for all $x \in \mathbb{F}$.
10. **Existence of an Inverse for Multiplication.** For each non-zero $x \in \mathbb{F}$, there exists an element w of $\mathbb{F}$ (usually written as $1/x$ or x^{-1}) such that $xw = wx = 1$.
11. **Distributive Law.** $x(y + z) = xy + xz$ for all $x, y, z \in \mathbb{F}$.
12. **Distributive Law.** $(x + y)z = xz + yz$, for all $x, y, z \in \mathbb{F}$.

3.2 Introduction to complex numbers

We cannot find a real number x such that $x^2 + 1 = 0$, since the square of real number is never negative. In the sixteenth century, mathematicians wrote down formal solutions to quadratic equations such as $x^2 + 1 = 0$ in the form $x = \pm\sqrt{-1}$. If this object $\sqrt{-1}$ were treated just like an ordinary number, but with the property that $\sqrt{-1}\sqrt{-1} = -1$, then indeed these values of x satisfy the above equation. The problem was that no-one really had any idea what writing down $\sqrt{-1}$ meant! No real-life problems that were being treated at the time ever had meaningful solutions that involved these “imaginary” numbers.

In the eighteenth and nineteenth centuries however, these imaginary numbers became more and more useful. Leonhard Euler introduced the now standard notation i for $\sqrt{-1}$ and showed that the use of these numbers allows one to express deep relationships between the trigonometric functions

and the exponential function.

Let us begin with the general quadratic equation $ax^2 + bx + c = 0$, $a \neq 0$, then by multiplying both sides of the equation by $4a$ and completing the square, we have

$$\begin{aligned} 4a^2x^2 + 4abx + 4ac &= 0 \\ (2ax + b)^2 &= b^2 - 4ac \end{aligned}$$

Hence, if $\Delta = b^2 - 4ac \geq 0$

$$\begin{aligned} 2ax + b &= \pm\sqrt{\Delta} \\ \text{or } x &= \frac{-b \pm \sqrt{\Delta}}{2a} \end{aligned}$$

and the equation has been solved.

If $\Delta = b^2 - 4ac < 0$, we can express the solutions using the complex number i . For example, if $x^2 = -4$, then we can write

$$x^2 = \pm\sqrt{-4} = \pm 2\sqrt{-1} = \pm 2i.$$

So, returning to our quadratic equation $ax^2 + bx + c = 0$ with $\Delta = b^2 - 4ac < 0$, we write $\Delta = -(4ac - b^2)$

$$\begin{aligned} \text{thus } \sqrt{\Delta} &= \pm i\sqrt{4ac - b^2} \quad \text{and} \\ x &= \frac{-b \pm i\sqrt{4ac - b^2}}{2a} \quad \text{are our 2 solutions.} \end{aligned}$$

We now define the set $\mathbb{C}$ of **complex numbers** by:

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}.$$

A complex number written in the form $a + bi$, where $a, b \in \mathbb{R}$, is said to be in **Cartesian form**. The real number a is called the *real part* of $a + bi$, and b is called the *imaginary part*. The set $\mathbb{C}$ contains all the real numbers (when $b = 0$). Numbers of the form bi , with b real ($b \neq 0$), are called **purely imaginary numbers**. The set of complex numbers also satisfies the twelve number laws, and so it also forms a field.

Example 1. Some examples of complex numbers in Cartesian form are

$$3 + 4i, \quad 2 - i = 2 + (-1)i, \quad -5i = 0 + (-5)i, \quad 6 = 6 + 0i \quad \text{and} \quad \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}.$$

◇

NOTE. The symbol ◇ indicates the end of an example.

3.3 The rules of arithmetic for complex numbers

Let $z = a + bi$ and $w = c + di$, where $a, b, c, d \in \mathbb{R}$.

Addition and subtraction. We define the sum, $z + w$, by

$$z + w = (a + c) + (b + d)i.$$

and the difference, $z - w$, by

$$z - w = (a - c) + (b - d)i.$$

That is, we add or subtract the real parts and the imaginary parts separately.

Example 1. If $z = 6 + 5i$ and $w = -3 + 4i$ then $z + w = 3 + 9i$ and $z - w = 9 + i$. $\diamond$

Multiplication. Expanding out, we have

$$(a + bi)(c + di) = ac + bci + adi + (bi)(di) = ac + (bc + ad)i + (bd)i^2.$$

Since $i^2 = -1$, this can be simplified to $(ac - bd) + (bc + ad)i$. Hence we define the product, zw , by

$$zw = (ac - bd) + (bc + ad)i.$$

Example 2. If $z = 6 + 5i$ and $w = -3 + 4i$ then

$$zw = (6 + 5i)(-3 + 4i) = -18 - 20 - 15i + 24i = -38 + 9i.$$

$\diamond$

(It is wise to multiply out the terms producing real numbers first and then purely imaginary numbers second.)

Division. To divide two complex numbers we use a similar process to that used in “rationalising the denominator”. For example,

$$\frac{1}{2 + \sqrt{3}} = \frac{(2 - \sqrt{3})}{(2 + \sqrt{3})(2 - \sqrt{3})} = \frac{2 - \sqrt{3}}{4 - 3} = 2 - \sqrt{3},$$

rationalises the denominator by multiplying both numerator and denominator by the denominator with the sign of $\sqrt{3}$ changed.

A similar process can be applied when dividing complex numbers:

$$\frac{z}{w} = \frac{a + bi}{c + di} = \frac{(a + bi)(c - di)}{(c + di)(c - di)} = \frac{ac + bd + (bc - ad)i}{c^2 + d^2} = \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2} i.$$

Thus we define the quotient $\frac{z}{w}$, ($w \neq 0$) by

$$\frac{z}{w} = \frac{ac + bd}{c^2 + d^2} + \frac{bc - ad}{c^2 + d^2} i.$$

Note that the quotient of two complex numbers is also a complex number.

Example 3. If $z = 3 - 4i$ and $w = 1 + 2i$ then

$$\frac{z}{w} = \frac{3 - 4i}{1 + 2i} = \frac{(3 - 4i)(1 - 2i)}{(1 + 2i)(1 - 2i)} = \frac{-5 - 10i}{5} = -1 + 2i.$$

$\diamond$

NOTE. The formula for division by $w = c + id$ given above fails if and only if $c^2 + d^2 = 0$. But, since c and d are real, $c^2 + d^2 = 0$ if and only if $c = 0$ and $d = 0$, that is, if and only if $w = 0$. Thus the formula for complex division fails if and only if the denominator is 0.

Before concluding this section on complex number arithmetic, we should point out that we now have three examples of fields; namely, the rational numbers $\mathbb{Q}$, the real numbers $\mathbb{R}$ and the complex numbers $\mathbb{C}$.

Proposition 1. [X] *The following properties hold for addition of complex numbers:*

1. **Uniqueness of Zero.** There is one and only one zero in $\mathbb{C}$.
2. **Cancellation Property.** If $z, v, w \in \mathbb{C}$ satisfy $z + v = z + w$, then $v = w$.

Proposition 2. [X] *The following properties hold for multiplication of complex numbers:*

1. $0z = 0$ for all complex numbers z .
2. $(-1)z = -z$ for all complex numbers z .
3. **Cancellation Property.** If $z, v, w \in \mathbb{C}$ satisfy $zv = zw$ and $z \neq 0$, then $v = w$.
4. If $z, w \in \mathbb{C}$ satisfy $zw = 0$, then either $z = 0$ or $w = 0$ or both.

One final point should be made. So far we have stressed the similarities between real number arithmetic and complex number arithmetic. However, there are also many important differences. One important difference is that while it makes sense to say that a real number is positive or that one real number is greater than (or less than) another, it does **not** make sense to say that a complex number is positive or that one complex number is greater than (or less than) another. That is, complex numbers cannot be *ordered*.

3.4 Real parts, imaginary parts and complex conjugates

Definition 1. The **real part** of $z = a + bi$ (written $\operatorname{Re}(z)$), where $a, b \in \mathbb{R}$, is given by

$$\operatorname{Re}(z) = a.$$

Definition 2. The **imaginary part** of $z = a + bi$ (written $\operatorname{Im}(z)$), where $a, b \in \mathbb{R}$, is given by

$$\operatorname{Im}(z) = b.$$

Example 1. If $z = 6 - 5i$ then $\operatorname{Re}(z) = 6$ and $\operatorname{Im}(z) = -5$.

◇

NOTE.

1. The imaginary part of a complex number is a real number.
2. Two complex numbers are equal if and only if their real parts are equal and their imaginary parts are equal. That is

$$a + bi = c + di \quad \text{if and only if} \quad a = c \text{ and } b = d,$$

where $a, b, c, d \in \mathbb{R}$.

Example 2. Find real numbers a, b such that $(2a + b) + (3a - 2b)i = 4 - i$.

SOLUTION. By comparing the real parts and imaginary parts, we have

$$2a + b = 4 \quad \text{and} \quad 3a - 2b = -1.$$

Hence $a = 1, b = 2$. ◇

The **complex conjugate** of a complex number is defined as follows.

Definition 3. If $z = a + bi$, where $a, b \in \mathbb{R}$, then the **complex conjugate** of z is $\bar{z} = a - bi$.

Example 3. $\overline{3 - 4i} = 3 + 4i$, $\bar{6} = 6$, $\overline{-10i} = 10i$. ◇

Example 4. Let $z = 2 + i$, $w = 1 + 2i$. We have

$$\overline{z + w} = \overline{3 + 3i} = 3 - 3i.$$

Note that $\bar{z} + \bar{w} = (2 - i) + (1 - 2i) = 3 - 3i$. So $\overline{z + w} = \bar{z} + \bar{w}$. ◇

Properties of the Complex Conjugate.

1. $\overline{\bar{z}} = z$.
2. $\overline{z + w} = \bar{z} + \bar{w}$ and $\overline{z - w} = \bar{z} - \bar{w}$.
3. $\overline{zw} = \bar{z}\bar{w}$ and $\overline{\left(\frac{z}{w}\right)} = \frac{\bar{z}}{\bar{w}}$.
4. $\operatorname{Re}(z) = \frac{1}{2}(z + \bar{z})$ and $\operatorname{Im}(z) = \frac{1}{2i}(z - \bar{z})$.
5. If $z = a + bi$, then $z\bar{z} = a^2 + b^2$, so $z\bar{z} \in \mathbb{R}$ and $z\bar{z} \geq 0$.

The proofs of most of these properties are straightforward. For example, the proof of property 4 is as follows.

If $z = a + bi$ then $\bar{z} = a - bi$ and

$$\begin{aligned} z + \bar{z} &= (a + bi) + (a - bi) = 2a = 2\operatorname{Re}(z) \\ z - \bar{z} &= (a + bi) - (a - bi) = 2bi = 2i\operatorname{Im}(z). \end{aligned}$$

The proofs of the remaining properties are left as exercises.

So far, we handle complex numbers by writing them in $a + bi$ form. With the above properties, we can write the real and imaginary parts of a complex number in terms of the complex number and its conjugate. We can also prove some general results in a simpler way.

Example 5. Let $z, w \in \mathbb{C}$ such that $z\bar{z} = w\bar{w}$. Prove that $\frac{z+w}{z-w}$ is purely imaginary.

Proof. To show that $\frac{z+w}{z-w}$ is purely imaginary, we only need to show that $\operatorname{Re}\left(\frac{z+w}{z-w}\right)$ is 0. Using $\alpha + \bar{\alpha} = 2\operatorname{Re}(\alpha)$,

$$\begin{aligned} \operatorname{Re}\left(\frac{z+w}{z-w}\right) &= \frac{1}{2} \left(\frac{z+w}{z-w} + \overline{\left(\frac{z+w}{z-w}\right)} \right) \\ &= \frac{1}{2} \left(\frac{z+w}{z-w} + \frac{\bar{z}+\bar{w}}{\bar{z}-\bar{w}} \right) \\ &= \frac{(z+w)(\bar{z}-\bar{w}) + (z-w)(\bar{z}+\bar{w})}{2(z-w)(\bar{z}-\bar{w})} \\ &= \frac{z\bar{z} - z\bar{w} + w\bar{z} - w\bar{w} + z\bar{z} + z\bar{w} - w\bar{z} - w\bar{w}}{2(z-w)(\bar{z}-\bar{w})} \\ &= \frac{2(z\bar{z} - w\bar{w})}{2(z-w)(\bar{z}-\bar{w})} \end{aligned}$$

which is 0 since $z\bar{z} = w\bar{w}$. The result follows. $\square$

NOTE. The symbol $\square$ indicates the end of a proof.

3.5 The Argand diagram

In the rest of this chapter, unless otherwise stated, we shall assume $a, b, x, y \in \mathbb{R}$ when we say $z = a + bi$ or $z = x + yi$ is a complex number.

There is a simple and extremely useful geometric picture of complex numbers which is obtained by identifying a complex number $z = a + bi$ with the point in the xy -plane whose coordinates are (a, b) . For example, $4 + 3i$ is represented by $(4, 3)$, 4 by $(4, 0)$ and $3i$ by $(0, 3)$. The coordinate plane with complex numbers plotted in this way is called an **Argand diagram**, as shown in Figure 1.

Note that a number $z = a + bi$ is plotted with $\operatorname{Re}(z) = a$ in the x -direction and $\operatorname{Im}(z) = b$ in the y -direction. For this reason, the x -axis is usually called the **real axis** and the y -axis is usually called the **imaginary axis**.

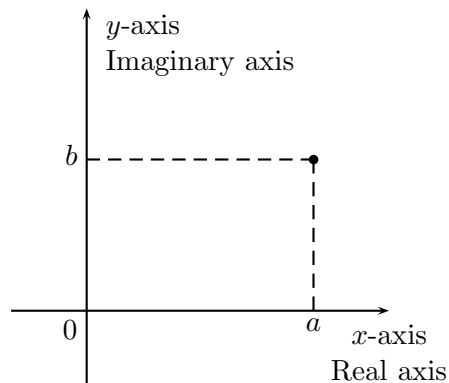


Figure 1: The Argand Diagram.

Example 1. Plot the numbers 4 , -4 , i , $-2i$, $-2 + 3i$, $3 - 4i$, $-3 - 4i$ on an Argand diagram. The solution is shown in Figure 2. $\diamond$

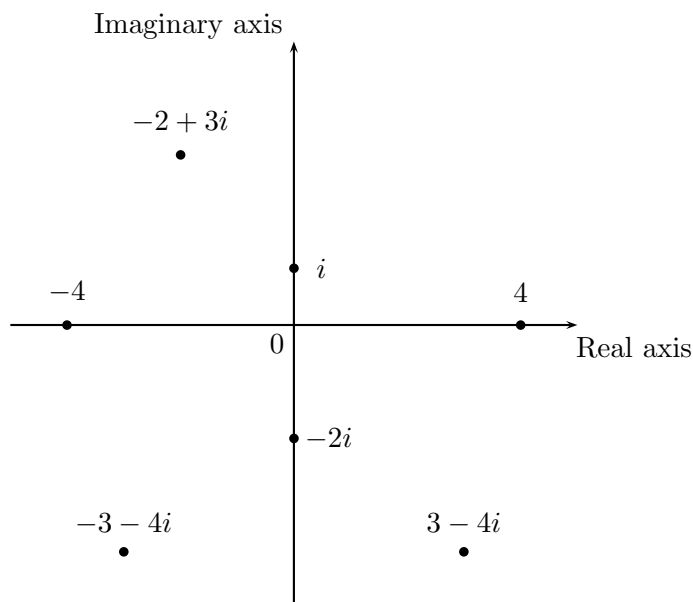


Figure 2: Plots of 4 , -4 , i , $-2i$, $-2 + 3i$, $3 - 4i$, $-3 - 4i$, on the Argand Diagram.

Note that we can write either i and $-2i$ on the y -axis above, or simply mark 1 and -2 , remembering that this is the imaginary axis.

Plotting real and imaginary parts leads to a simple geometric picture of addition and subtraction of complex numbers. Addition and subtraction is done by adding or subtracting x -coordinates and y -coordinates separately, as shown in Figure 3.

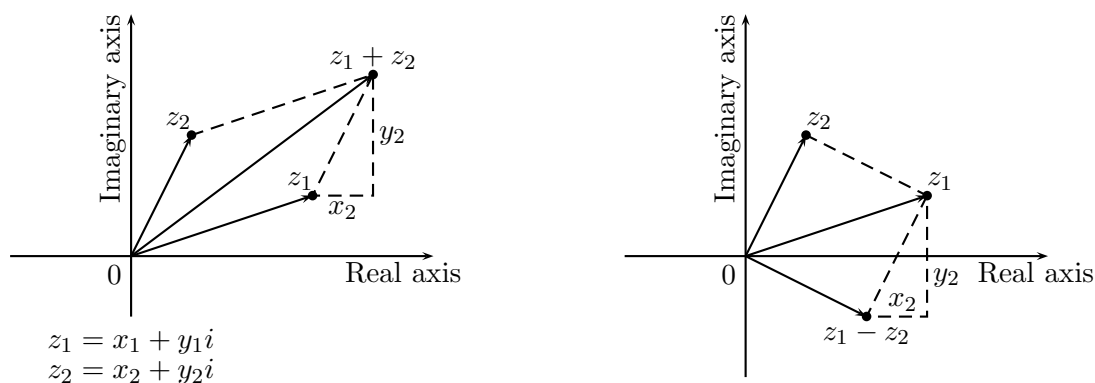


Figure 3: Addition and Subtraction of Complex Numbers.

3.6 Polar form, modulus and argument

An alternative representation for complex numbers, which proves to be very useful, is obtained by using plane polar coordinates r and θ instead of the Cartesian coordinates x and y . The coordinate r is the distance of a point from the origin, and θ is an angle measured from the positive x -axis, as shown in Figure 4.

Take a complex number $z \neq 0$, then from Figure 4, Pythagoras' theorem gives

$$r = \sqrt{x^2 + y^2} \quad \text{with} \quad r > 0.$$

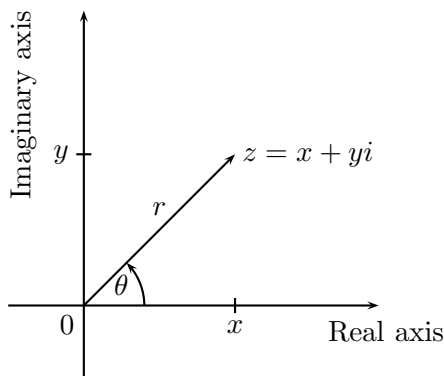


Figure 4: Polar Coordinates of a Complex Number.

By trigonometry,

$$\cos \theta = \frac{x}{\sqrt{x^2 + y^2}} = \frac{x}{r} \quad \text{and} \quad \sin \theta = \frac{y}{\sqrt{x^2 + y^2}} = \frac{y}{r}.$$

Thus the relations between the real and imaginary parts of $z = x + yi$ and the polar coordinates r and θ are

$$\operatorname{Re}(z) = x = r \cos \theta \quad \text{and} \quad \operatorname{Im}(z) = y = r \sin \theta,$$

and hence a complex number $z \neq 0$ can be written using the polar coordinates r and θ as:

$$z = r(\cos \theta + i \sin \theta).$$

It is important to note here that the angle θ for a given complex number $z = x + yi$ is not uniquely defined; since adding or subtracting 2π produces exactly the same values for x and y and hence the same complex number z . This result is summarised in the following proposition.

Proposition 1 (Equality of Complex Numbers). *Two complex numbers*

$$z_1 = r_1(\cos \theta_1 + i \sin \theta_1) \quad \text{and} \quad z_2 = r_2(\cos \theta_2 + i \sin \theta_2), \quad z_1, z_2 \neq 0$$

are equal if and only if $r_1 = r_2$ and $\theta_1 = \theta_2 + 2k\pi$, where k is any integer.

[X] *Proof.* Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$.

If $r_1 = r_2$ and $\theta_1 = \theta_2 + 2k\pi$, then $\cos \theta_1 = \cos \theta_2$, $\sin \theta_1 = \sin \theta_2$,

$$x_1 = r_1 \cos \theta_1 = r_2 \cos \theta_2 = x_2$$

$$y_1 = r_1 \sin \theta_1 = r_2 \sin \theta_2 = y_2$$

and $z_1 = z_2$.

Conversely, if $z_1 = z_2$, then we have $x_1 = x_2$ and $y_1 = y_2$. Hence,

$$r_1 = \sqrt{x_1^2 + y_1^2} = \sqrt{x_2^2 + y_2^2} = r_2,$$

and since $r_1, r_2 \neq 0$,

$$\cos \theta_1 = \frac{x_1}{r_1} = \frac{x_2}{r_2} = \cos \theta_2 \quad \text{and} \quad \sin \theta_1 = \frac{y_1}{r_1} = \frac{y_2}{r_2} = \sin \theta_2,$$

and hence, $\cos(\theta_1 - \theta_2) = \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2 = \cos^2 \theta_1 + \sin^2 \theta_1 = 1$
so $\theta_1 - \theta_2 = 2k\pi$, $\theta_1 = \theta_2 + 2k\pi$ for $k \in \mathbb{Z}$. □

The polar coordinate r that we have associated with a complex number is often called the **modulus** of the complex number. The formal definition is:

Definition 1. For $z = x + yi$, where $x, y \in \mathbb{R}$, we define the **modulus** of z to be

$$|z| = \sqrt{x^2 + y^2}.$$

The quantity $|z|$ is also called the **magnitude** of z or the **absolute value** of z . Note that it has a geometric interpretation as the distance $r = |z|$ of the point z from the origin in an Argand diagram. Note also that $z\bar{z} = |z|^2$.

Example 1. $|-4| = 4$, $|2i| = 2$, and $|3 - 4i| = \sqrt{3^2 + (-4)^2} = 5$. ◇

Argument and Principal Argument.

The polar coordinate θ that we have associated with a complex number is often called an **argument** of the complex number and is written as $\arg(z)$. As mentioned above, such an angle can be increased or decreased by 2π without changing the corresponding complex number. It is desirable to define a particular argument that is unique for a given complex number. We do this by choosing a value θ of the argument so that $-\pi < \theta \leq \pi$. This is called the **principal argument** of z and is written as $\text{Arg}(z)$.

The following diagrams illustrate the possible positions of a complex number z in each of the four quadrants.

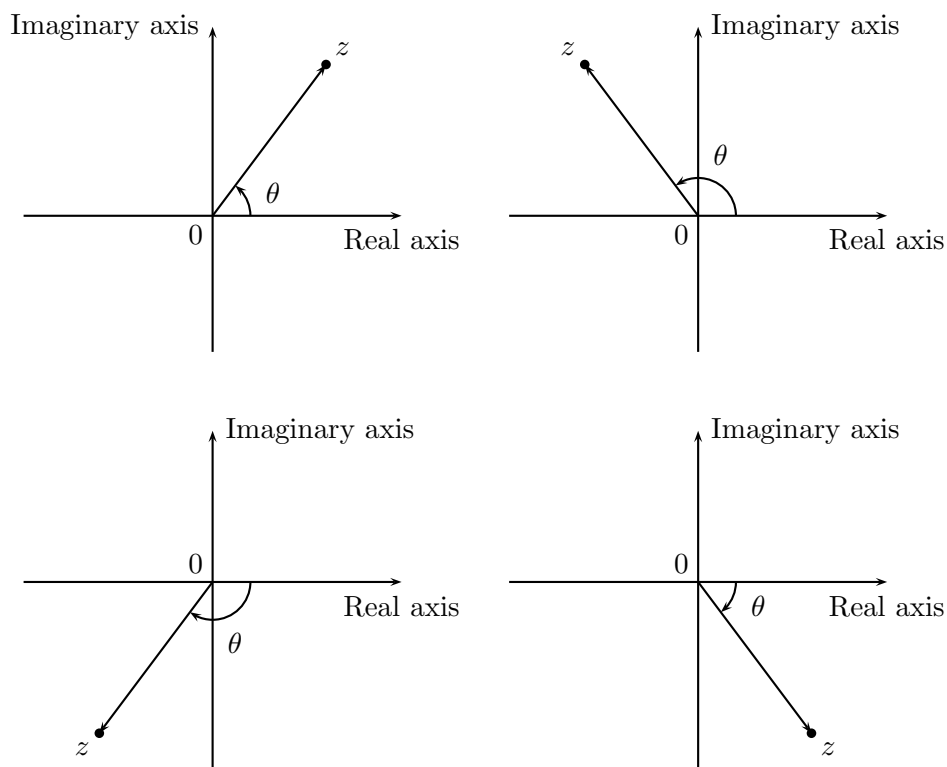


Figure 5: Principal Argument.

We can see from the diagrams, that if z lies in the first or second quadrant, we measure the principal argument θ anticlockwise. For z in the 3rd or the 4th quadrant, we measure θ , as a negative angle, clockwise. Note that we leave $\text{Arg}(0)$ undefined.

One useful strategy is always to draw a diagram, then use the tangent ratio to find the *acute* angle α formed by the corresponding triangle and use this to find the principal argument.

Example 2. Find the arguments of $1 + i$, $-1 + \sqrt{3}i$, $-\sqrt{3} - i$ and $1 - \sqrt{3}i$.

SOLUTION.

We first plot $z = 1 + i$ on an Argand diagram as in Figure 6. The complex number lies in the 1st quadrant, and so $0 < \text{Arg}(z) < \pi/2$. As shown in the figure,

$$\tan \theta = 1, \quad \text{and} \\ \text{Arg}(1 + i) = \theta = \frac{\pi}{4}.$$

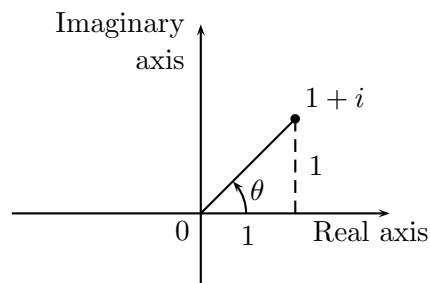


Figure 6.

From Figure 7, the complex number $-1 + \sqrt{3}i$ lies in the 2nd quadrant. Also,

$$\tan \alpha = \sqrt{3}, \quad \text{so} \quad \alpha = \frac{\pi}{3}.$$

Hence

$$\text{Arg}(-1 + \sqrt{3}i) = \theta = \pi - \alpha = \frac{2\pi}{3}.$$

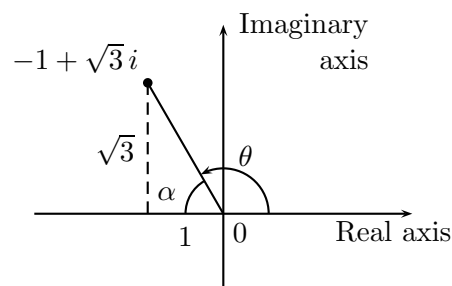


Figure 7.

From Figure 8, the complex number $-\sqrt{3} - i$ lies in the 3rd quadrant. Also,

$$\tan \alpha = \frac{1}{\sqrt{3}}, \quad \text{so} \quad \alpha = \frac{\pi}{6}.$$

Hence

$$\text{Arg}(-\sqrt{3} - i) = -\pi + \alpha = -\frac{5\pi}{6}.$$

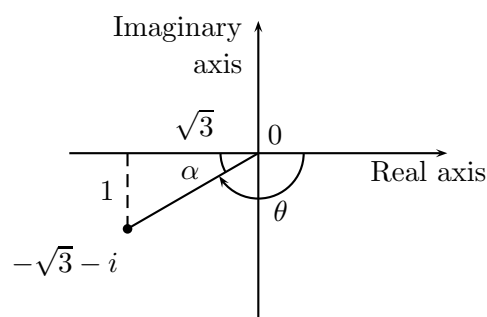


Figure 8.

From Figure 9, the complex number $1 - \sqrt{3}i$ lies in the 4th quadrant. Also

$$\tan \alpha = \sqrt{3}, \quad \text{so} \quad \alpha = \frac{\pi}{3}.$$

Hence,

$$\text{Arg}(1 - \sqrt{3}i) = -\alpha = -\frac{\pi}{3}.$$

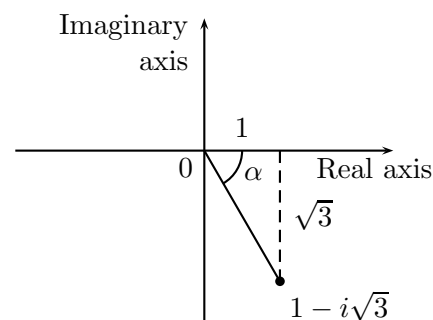


Figure 9.

◇

In the special case that a given complex number is real or purely imaginary, we can easily read the principal argument from an Argand diagram.

Example 3. Find the principal argument for each of the numbers 4, -4 , i and $-2i$.

SOLUTION. From Figure 2 in Section 3.5, we can easily read the arguments as

$$\text{Arg}(4) = 0, \quad \text{Arg}(-4) = \pi, \quad \text{Arg}(i) = \frac{\pi}{2}, \quad \text{Arg}(-2i) = -\frac{\pi}{2}.$$

◇

Example 4. Find the “ $a + ib$ ” form of the complex number with modulus 4 and argument $-\frac{\pi}{6}$.

SOLUTION.

$$z = 4 \left(\cos \left(-\frac{\pi}{6} \right) + i \sin \left(-\frac{\pi}{6} \right) \right) = 4 \left(\frac{\sqrt{3}}{2} - \frac{1}{2}i \right) = 2\sqrt{3} - 2i.$$

◇

3.7 Properties and applications of the polar form

We begin by proving a useful lemma.

Lemma 1. For any real numbers θ_1 and θ_2

$$(\cos \theta_1 + i \sin \theta_1)(\cos \theta_2 + i \sin \theta_2) = \cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2).$$

Proof. Expanding the left hand side, we obtain

$$(\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\cos \theta_1 \sin \theta_2 + \sin \theta_1 \cos \theta_2).$$

Then, using the standard trigonometric formulae

$$\cos(\theta_1 + \theta_2) = \cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2,$$

and

$$\sin(\theta_1 + \theta_2) = \cos \theta_1 \sin \theta_2 + \sin \theta_1 \cos \theta_2,$$

the result follows. □

Lemma 1 can be used to derive a very important and useful theorem for integer powers of complex numbers in polar forms. This theorem is called De Moivre’s Theorem.

Theorem 2 (De Moivre’s Theorem). For any real number θ and integer n

$$(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta. \quad (\#)$$

Proof. We shall prove this theorem by proving that the condition (#) holds for the four separate cases of $n > 0$, $n = 0$, $n = -1$ and $n < -1$, and hence that it holds for all $n \in \mathbb{Z}$.

CASE 1. $n > 0$. The proof is by induction (see Section 3.11).

We first note that (#) is obviously true for $n = 1$.

We now show that, if (#) is true for some value of $n > 0$, then it is also true for $n + 1$. Assuming (#) is true for some value of $n > 0$, we have

$$\begin{aligned} (\cos \theta + i \sin \theta)^{n+1} &= (\cos \theta + i \sin \theta)(\cos \theta + i \sin \theta)^n \\ &= (\cos \theta + i \sin \theta)(\cos n\theta + i \sin n\theta) \\ &= \cos(n+1)\theta + i \sin(n+1)\theta \quad (\text{from lemma 1}), \end{aligned}$$

and hence (#) is true for $n + 1$.

Now, we have already seen that (#) is true for $n = 1$, and hence, from the first principle of induction, it is also true for all integers $n \geq 1$.

CASE 2. $n = 0$. The condition (#) is true for this case, provided we use the convention that $z^0 = 1$ for any complex number z .

CASE 3. $n = -1$. By definition, $z^{-1} = \frac{1}{z}$. Then, applying the division rule for complex numbers to $z = \cos \theta + i \sin \theta$, we have

$$\begin{aligned} (\cos \theta + i \sin \theta)^{-1} &= \frac{1}{\cos \theta + i \sin \theta} \\ &= \frac{1}{\cos \theta + i \sin \theta} \times \frac{\cos \theta - i \sin \theta}{\cos \theta - i \sin \theta} \\ &= \frac{\cos \theta - i \sin \theta}{\cos^2 \theta + \sin^2 \theta} \\ &= \cos(-\theta) + i \sin(-\theta), \end{aligned}$$

where we have used the trigonometric identities

$$\cos \theta = \cos(-\theta), \quad \sin \theta = -\sin(-\theta) \quad \text{and} \quad \cos^2 \theta + \sin^2 \theta = 1.$$

The condition (#) is therefore true for $n = -1$ also.

CASE 4. $n < -1$. Let $m = -n$ so that $m > 1$. Note that

$$\begin{aligned} (\cos \theta + i \sin \theta)^n &= (\cos \theta + i \sin \theta)^{-m} \\ &= ((\cos \theta + i \sin \theta)^{-1})^m \\ &= (\cos(-\theta) + i \sin(-\theta))^m && \text{(from case 3)} \\ &= \cos(-m\theta) + i \sin(-m\theta) && \text{(from case 1)} \\ &= \cos(n\theta) + i \sin(n\theta). \end{aligned}$$

□

De Moivre's Theorem provides a simple formula for integer powers of complex numbers. However, it can also be used to suggest a meaning for complex powers of complex numbers. To make this extension to complex powers it is actually sufficient just to give a meaning to the exponential function for imaginary exponents. We first make the following definition.

Definition 1. (Euler's Formula). For real θ , we define

$$e^{i\theta} = \cos \theta + i \sin \theta.$$

This definition may appear somewhat arbitrary at first, but there are several reasons why it is reasonable. First recall that for real constants a , θ and ϕ , we have for integer n ,

$$(e^{a\theta})^n = e^{an\theta}, \quad (1)$$

$$e^{a\theta} e^{a\phi} = e^{a(\theta+\phi)}, \quad (2)$$

$$e^0 = 1, \quad (3)$$

$$\frac{d}{d\theta} (e^{a\theta}) = ae^{a\theta}. \quad (4)$$

In fact, properties (3) and (4) are often taken as a definition of the exponential function for real numbers.

The next point to notice is that if a is replaced by i and if $e^{i\theta}$ is replaced by $(\cos \theta + i \sin \theta)$ in these four formulae, then all four formulae are still satisfied. In fact, equation (1) would be De Moivre's Theorem, equation (2) would be lemma 1, equation (3) would obviously be true, and equation (4) would be

$$\frac{d}{d\theta} (\cos \theta + i \sin \theta) = i(\cos \theta + i \sin \theta) = -\sin \theta + i \cos \theta,$$

which is also true (provided we assume that differentiation of expressions containing the symbol i can be carried out in the same way as if i were a real constant). Thus, $\cos \theta + i \sin \theta$ has exactly the same properties that we would like $e^{i\theta}$ to have, and so the definition

$$e^{i\theta} = \cos \theta + i \sin \theta$$

is consistent with our experience with other exponential functions.

Note also that, since cosine is an even function and sine is an odd function, so

$$e^{-i\theta} = \cos(-\theta) + i \sin(-\theta) = \cos \theta - i \sin \theta,$$

which is the conjugate of $e^{i\theta}$.

We should also say that in more sophisticated treatments of complex functions, $e^{i\theta}$ will be defined by other means (usually a "power series") and Euler's formula then becomes a theorem.

De Moivre's Theorem and Euler's formula have a wide variety of uses, ranging from calculation of powers of complex numbers to calculation of roots of complex numbers to derivation of trigonometric formulae etc. We shall examine some of these applications in the remainder of this chapter.

3.7.1 The arithmetic of polar forms

Using Euler's formula, we can rewrite the complex number $z = r(\cos \theta + i \sin \theta)$ in an alternative and more usual form. We will call this form, the **polar form** of the complex number.

Definition 2. The **polar form** for a non-zero complex number z is

$$z = re^{i\theta}$$

where $r = |z|$ and $\theta = \text{Arg}(z)$.

Four important special cases are:

$$1 = e^0, \quad i = e^{i\pi/2}, \quad -1 = e^{i\pi}, \quad -i = e^{-i\pi/2}.$$

Some further examples of polar forms are as follows.

Example 1. $-4 = 4e^{i\pi}$, $-4i = 4e^{-i\pi/2}$, $1 - i\sqrt{3} = 2e^{-i\pi/3}$, $-1 - i\sqrt{3} = 2e^{-2i\pi/3}$. $\diamond$

(Note: At High School you may have taken $r(\cos \theta + i \sin \theta)$ as the polar form and abbreviated it to $r \text{cis}(\theta)$. This notation is not used in this course.)

Since $e^{i\theta} = e^{i(\theta+2k\pi)}$ for all k an integer, it is sometimes convenient to express the polar form using a general argument, that is,

$$z = e^{i\theta}, \quad \text{where } \theta = \text{Arg}(z) + 2k\pi, \quad k \in \mathbb{Z}.$$

Using Euler's formula, we can rewrite the equality proposition for polar forms (Proposition 1 of Section 3.6) as

$$z_1 = r_1 e^{i\theta_1} = r_2 e^{i\theta_2} = z_2$$

if and only if $r_1 = r_2$ and $\theta_1 = \theta_2 + 2k\pi$ for $k \in \mathbb{Z}$.

The polar form is very useful for multiplication and division of complex numbers. The formulae for multiplication and division of polar forms are:

$$z_1 z_2 = r_1 e^{i\theta_1} r_2 e^{i\theta_2} = r_1 r_2 e^{i(\theta_1 + \theta_2)},$$

$$\frac{z_1}{z_2} = \frac{r_1 e^{i\theta_1}}{r_2 e^{i\theta_2}} = \frac{r_1}{r_2} e^{i(\theta_1 - \theta_2)}.$$

It is sometimes useful to express these results in terms of the modulus and argument. In this case, we have for multiplication,

$$|z_1 z_2| = r_1 r_2 = |z_1| |z_2| \quad \text{and} \quad \text{Arg}(z_1 z_2) = \text{Arg}(z_1) + \text{Arg}(z_2) + 2k\pi,$$

where k is an integer, chosen so that $-\pi < \text{Arg}(z_1 z_2) \leq \pi$. That is, the rule is to *multiply* the moduli and *add* the arguments. For division, we have

$$\left| \frac{z_1}{z_2} \right| = \frac{r_1}{r_2} = \frac{|z_1|}{|z_2|} \quad \text{and} \quad \text{Arg}\left(\frac{z_1}{z_2}\right) = \text{Arg}(z_1) - \text{Arg}(z_2) + 2k\pi,$$

where k is an integer chosen so that $-\pi < \text{Arg}(z_1/z_2) \leq \pi$. That is, the rule is to *divide* the moduli and *subtract* the arguments.

Example 2. Use polar forms to find the modulus and argument of $(-1 - i)(1 - i\sqrt{3})$ and $(-1 - i)/(1 - i\sqrt{3})$.

SOLUTION. The modulus and argument of $-1 - i$ and $1 - i\sqrt{3}$ are

$$\begin{aligned} |-1 - i| &= \sqrt{2} & \text{and} & & \text{Arg}(-1 - i) &= -\frac{3\pi}{4}, \\ |1 - i\sqrt{3}| &= 2 & \text{and} & & \text{Arg}(1 - i\sqrt{3}) &= -\frac{\pi}{3}. \end{aligned}$$

Hence, for multiplication, we have

$$\left| (-1 - i)(1 - i\sqrt{3}) \right| = |-1 - i| |1 - i\sqrt{3}| = 2\sqrt{2},$$

and, for some integer k ,

$$\text{Arg}\left((-1 - i)(1 - i\sqrt{3})\right) = -\frac{3\pi}{4} + \left(-\frac{\pi}{3}\right) + 2k\pi = -\frac{13\pi}{12} + 2k\pi.$$

Then, on choosing $k = 1$, to obtain a principal argument in the interval $(-\pi, \pi]$, we have

$$\text{Arg}\left((-1 - i)(1 - i\sqrt{3})\right) = -\frac{13\pi}{12} + 2\pi = \frac{11\pi}{12}.$$

For division, we have

$$\left| \frac{-1 - i}{1 - i\sqrt{3}} \right| = \frac{|-1 - i|}{|1 - i\sqrt{3}|} = \frac{\sqrt{2}}{2}$$

and, for some integer k ,

$$\text{Arg}\left(\frac{-1 - i}{1 - i\sqrt{3}}\right) = -\frac{3\pi}{4} - \left(-\frac{\pi}{3}\right) + 2k\pi = -\frac{5\pi}{12},$$

where $k = 0$ has been chosen to obtain a principal argument in the interval $(-\pi, \pi]$. $\diamond$

The next example shows a reasonably simple method for finding the square roots of a complex number in Cartesian form. It rests on the observation that if $z \in \mathbb{C}$ and $z = a + bi$ with $a, b \in \mathbb{R}$, then

$$|z^2| = |z \cdot z| = |z| |z| = |z|^2 \quad \text{or} \quad |(a + bi)^2| = a^2 + b^2,$$

and also $\text{Re}(z^2) = a^2 - b^2$, $\text{Im}(z^2) = 2ab$.

Example 3. Find the square roots of $-5 - 12i$.

SOLUTION. We want to find all complex solutions to $z^2 = -5 - 12i$. Writing $z = a + bi$, this is equivalent to finding all real solutions (a, b) to the equation $(a + bi)^2 = -5 - 12i$. Expanding the left-hand side and equating real and imaginary parts, we have

$$\begin{aligned} a^2 - b^2 &= -5 \\ 2ab &= -12. \end{aligned}$$

Also,

$$a^2 + b^2 = |(a + bi)^2| = |-5 - 12i| = \sqrt{(-5)^2 + (-12)^2} = 13.$$

So we want to solve the system of equations

$$\begin{aligned}a^2 + b^2 &= 13 \\a^2 - b^2 &= -5 \\2ab &= -12.\end{aligned}$$

Solving the first pair in this system, we get $a^2 = 4$ and $b^2 = 9$. This means that $a = \pm 2$ and $b = \pm 3$. The third equation is satisfied precisely when we choose opposite signs for a and b . Therefore the two square roots of $-5 - 12i$ are

$$z = 2 - 3i \quad \text{and} \quad -2 + 3i,$$

or more compactly,

$$z = \pm(2 - 3i).$$

◇

Example 4. Use the quadratic formula to solve the equation

$$z^2 - (4 + i)z + (5 + 5i) = 0.$$

SOLUTION. We solve this using the quadratic formula. The proof of the quadratic formula uses only field axioms and the fact of the existence of square roots and thus carries over directly to quadratic polynomials with complex coefficients. The details are left as an exercise.

Proceeding, the solutions to $z^2 - (4 + i)z + (5 + 5i) = 0$ are

$$z = \frac{4 + i \pm r}{2}$$

where r is a square root of $(4 + i)^2 - 4(5 + 5i) = -5 - 12i$.

From the previous example, the square roots of $-5 - 12i$ are $\pm(2 - 3i)$. Thereby the roots to the quadratic are

$$z = \frac{4 + i + (2 - 3i)}{2} = 3 - i \quad \text{and} \quad z = \frac{4 + i - (2 - 3i)}{2} = 1 + 2i$$

We can easily check that $z = 3 - i$ and $z = 1 + 2i$ are roots to the equation by substitution. ◇

Example 5. Show that the set of numbers of unit modulus, that is, the set

$$S = \{z \in \mathbb{C} : |z| = 1\},$$

is closed under multiplication and division.

SOLUTION. For closure under multiplication we must prove that the product $z_1 z_2 \in S$ for all $z_1, z_2 \in S$. Now, if $z_1, z_2 \in S$, then $|z_1| = 1$ and $|z_2| = 1$. Then

$$|z_1 z_2| = |z_1| |z_2| = 1$$

also. Thus, $z_1 z_2 \in S$, and hence S is closed under multiplication.

For closure under division we must prove that $z_1/z_2 \in S$ for all $z_1, z_2 \in S$. For z_1/z_2 , we have

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|} = 1,$$

as $|z_1| = 1$ and $|z_2| = 1$. Thus $z_1/z_2 \in S$, and hence S is closed under division. ◇

Example 6. [X] Suppose a and b are real numbers (not both zero) and $w = (az + b)/(bz + a)$. Show that, if $|z| = 1$, then $|w| = 1$.

SOLUTION. We have

$$|w| = \left| \frac{az + b}{bz + a} \right| = \frac{|az + b|}{|bz + a|}.$$

Now, $|z| = 1$, so $z = \cos \theta + i \sin \theta$, θ real.

$$\begin{aligned} |az + b| &= |a(\cos \theta + i \sin \theta) + b| = \sqrt{(a \cos \theta + b)^2 + (a \sin \theta)^2} \\ &= \sqrt{a^2 + 2ab \cos \theta + b^2} \end{aligned}$$

$$\text{and } |bz + a| = |b(\cos \theta + i \sin \theta) + a| = \sqrt{a^2 + 2ab \cos \theta + b^2}.$$

So $|w| = 1$. ◇

3.7.2 Powers of complex numbers

With the exception of very simple cases such as squares and cubes, the simplest method of calculating powers of a complex number z is to use the polar form. Thus, if

$$z = re^{i\theta},$$

then the properties of exponentials give

$$z^n = r^n e^{in\theta}.$$

Example 7. Calculate $(1 + i\sqrt{3})^{10}$.

SOLUTION. For $z = 1 + i\sqrt{3}$, we have $|z| = 2$, $\text{Arg}(z) = \frac{\pi}{3}$ and so $z = 2e^{i\pi/3}$. Hence,

$$(1 + i\sqrt{3})^{10} = (2e^{i\pi/3})^{10} = 2^{10} e^{10i\pi/3} = 2^{10} e^{-2i\pi/3} = 2^{10} \left(\cos \frac{2\pi}{3} - i \sin \frac{2\pi}{3} \right) = 2^9 (-1 - i\sqrt{3}).$$

◇

3.7.3 Roots of complex numbers

The polar form can also be used to find roots of complex numbers, where a root of a complex number is defined as follows.

Definition 3. A complex number z is an **n th root** of a number z_0 if z_0 is the n th power of z , that is, z is the n th root of z_0 if $z^n = z_0$.

If $z_0 \neq 0$ the n th roots of z_0 can be found by equating the polar forms of z^n and z_0 . Thus, if $z = re^{i\theta}$ and $z_0 = r_0e^{i\theta_0}$, we have

$$z^n = r^n e^{in\theta} = r_0 e^{i\theta_0},$$

and hence, on equating the moduli and arguments, we have

$$r^n = r_0 \quad \text{and} \quad n\theta = \theta_0 + 2k\pi \quad \text{for } k \in \mathbb{Z}.$$

Thus

$$r = r_0^{1/n} \quad \text{and} \quad \theta = \frac{\theta_0 + 2k\pi}{n} = \frac{\theta_0}{n} + \frac{2k\pi}{n} \quad \text{for } k \in \mathbb{Z},$$

and

$$z = re^{i\theta} = r_0^{\frac{1}{n}} e^{i\left(\frac{\theta_0 + 2k\pi}{n}\right)}. \quad (*)$$

As k ranges over the integers, z takes precisely n different values. These n values can be found by letting k take any n consecutive values. Thus z_0 has precisely n distinct n th roots.

Example 8. Find all fifth roots of unity.

SOLUTION. The modulus and argument of $z_0 = 1$ are $r_0 = |1| = 1$ and $\theta_0 = \text{Arg}(1) = 0$.

Hence, the fifth roots of unity are complex numbers, z , given by

$$z^5 = e^{2k\pi i} \quad \text{for } k \in \mathbb{Z},$$

By (*) above, these roots are

$$z = e^{i\left(\frac{0+2k\pi}{5}\right)},$$

where the five consecutive integers for k are chosen to be $-2, -1, 0, 1, 2$ so that each argument $\theta = \frac{2k\pi}{5}$ lies in $(-\pi, \pi]$.

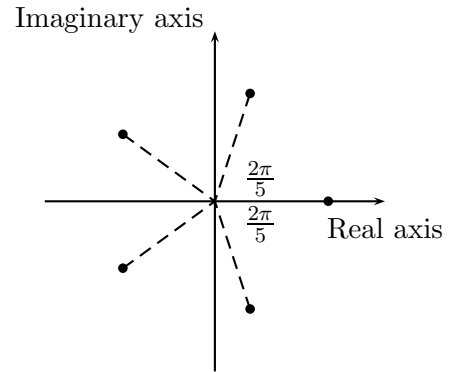


Figure 10: The Five Fifth Roots of Unity.

Polar forms for the five fifth roots of unity are therefore

$$e^{-4\pi i/5}, e^{-2\pi i/5}, 1, e^{2\pi i/5}, e^{4\pi i/5}.$$

◇

Note that these solutions occur in conjugate pairs, eg. $e^{2\pi i/5}$ and $e^{-2\pi i/5}$ and so on.

As shown in Figure 10, these fifth roots are equally spaced out on a circle of radius 1 with angles $\frac{2\pi}{5}$ between them.

Example 9. Find all sixth roots of -2 .

SOLUTION. The modulus and argument of -2 are $|-2| = 2$ and $\text{Arg}(-2) = \pi$.

Hence, the sixth roots of -2 are complex numbers z , such that

$$z^6 = 2e^{(\pi+2k\pi)i} \quad \text{for } k \in \mathbb{Z}.$$

So, the sixth roots are

$$z = 2^{\frac{1}{6}} e^{i\frac{(2k+1)\pi}{6}}, \quad \text{for } k = -3, -2, -1, 0, 1, 2.$$

Writing $\alpha = 2^{\frac{1}{6}}$, the six roots are: $\alpha e^{-i\frac{5\pi}{6}}$, $\alpha e^{-i\frac{\pi}{2}}$, $\alpha e^{-i\frac{\pi}{6}}$, $\alpha e^{i\frac{\pi}{6}}$, $\alpha e^{i\frac{\pi}{2}}$, and $\alpha e^{i\frac{5\pi}{6}}$.

In “ $a + ib$ ” form, these sixth roots are

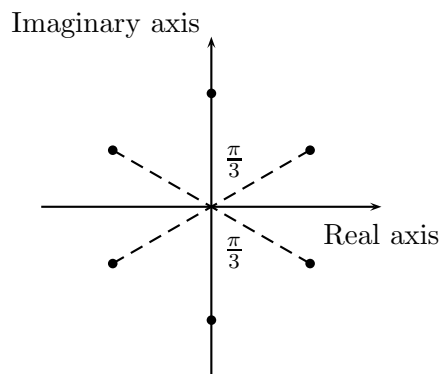


Figure 11: The Six Sixth Roots of -2 .

$$-2^{\frac{1}{6}} \left(\frac{\sqrt{3}+i}{2} \right), -2^{\frac{1}{6}}i, 2^{\frac{1}{6}} \left(\frac{\sqrt{3}-i}{2} \right), 2^{\frac{1}{6}} \left(\frac{\sqrt{3}+i}{2} \right), 2^{\frac{1}{6}}i, -2^{\frac{1}{6}} \left(\frac{\sqrt{3}-i}{2} \right)$$

◇

There is again a simple geometric picture of this result. From figure 11, the 6 roots lie on a circle of radius $2^{\frac{1}{6}}$ with angles $\frac{2\pi}{6} = \frac{\pi}{3}$ between them.

In summary, there are always exactly n n th roots of a non-zero complex number. On the Argand diagram, these n th roots of z_0 lie on a circle of radius $|z_0|^{\frac{1}{n}}$ with angles $\frac{2\pi}{n}$ between them.

3.8 Trigonometric applications of complex numbers

A large variety of trigonometric formulae can be obtained by using complex numbers and the Binomial Theorem. This theorem is stated below, and a proof is given in Section 3.12.

Theorem 1 (Binomial Theorem). *If $a, b \in \mathbb{C}$ and $n \in \mathbb{N}$, then*

$$\begin{aligned} (a+b)^n &= a^n + na^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 + \frac{n(n-1)(n-2)}{3!}a^{n-3}b^3 + \cdots + nab^{n-1} + b^n \\ &= \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k, \end{aligned}$$

where the numbers $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ are the binomial coefficients.

For small values of n the binomial coefficients may be easily calculated using **Pascal's triangle**.

n	BINOMIAL COEFFICIENTS													
0					1									
1					1		1							
2				1		2		1						
3				1		3		3		1				
4				1		4		6		4		1		
5				1		5		10		10		5		1

Figure 12: Pascal's Triangle.

Note that each coefficient (except the 1's at the end) is obtained by adding the two coefficients immediately above it.

Example 1. For $n = 4$, the coefficients are 1, 4, 6, 4, 1, and hence

$$(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4.$$

◇

Sine and cosine of multiples of θ .

It is now easy to express $\cos n\theta$ or $\sin n\theta$ in terms of powers of $\cos \theta$ or $\sin \theta$ by using De Moivre's Theorem.

Example 2. Find a formula for $\cos 4\theta$ in terms of powers of $\cos \theta$ and $\sin \theta$.

SOLUTION. Using De Moivre's Theorem and the Binomial Theorem gives

$$\begin{aligned}\cos 4\theta + i \sin 4\theta &= (\cos \theta + i \sin \theta)^4 \\ &= \cos^4 \theta + 4 \cos^3 \theta (i \sin \theta) + 6 \cos^2 \theta (i \sin \theta)^2 + 4 \cos \theta (i \sin \theta)^3 + (i \sin \theta)^4.\end{aligned}$$

Then, on using $i^2 = -1$, $i^3 = -i$, $i^4 = 1$, and separately equating real and imaginary parts on the left and right hand sides gives

$$\cos 4\theta = \cos^4 \theta - 6 \cos^2 \theta \sin^2 \theta + \sin^4 \theta$$

and

$$\sin 4\theta = 4 (\cos^3 \theta \sin \theta - \cos \theta \sin^3 \theta).$$

Note that the formula for $\cos 4\theta$ can be rewritten in term of powers of $\cos \theta$ only by using $\sin^2 \theta = 1 - \cos^2 \theta$ to replace the sine terms by cosines. ◇

Powers of sine and cosine.

Euler's formula shows that the trigonometric functions, sine and cosine, are closely related to the exponential function with an imaginary exponent.

Using Euler's formula, we note that

$$\begin{aligned}e^{in\theta} &= \cos n\theta + i \sin n\theta, \\ e^{-in\theta} &= \cos(-n\theta) + i \sin(-n\theta) = \cos n\theta - i \sin n\theta.\end{aligned}$$

On first adding and then subtracting these formulae, we obtain the important formulae

$$\cos n\theta = \frac{1}{2} (e^{in\theta} + e^{-in\theta}), \quad \sin n\theta = \frac{1}{2i} (e^{in\theta} - e^{-in\theta}).$$

In particular, we have

$$\cos \theta = \frac{1}{2} (e^{i\theta} + e^{-i\theta}), \quad \sin \theta = \frac{1}{2i} (e^{i\theta} - e^{-i\theta}).$$

We can apply the above formulae to derive trigonometric formulae which relate powers of $\sin \theta$ or $\cos \theta$ to sines or cosines of multiples of θ .

Example 3. Find a formula for $\sin^3 \theta$ in terms of sines of multiples of θ .

SOLUTION. Using the formula $\sin n\theta = \frac{1}{2i} (e^{in\theta} - e^{-in\theta})$, for $n = 1$ and $n = 3$, and the Binomial Theorem gives

$$\begin{aligned} \sin^3 \theta &= \left(\frac{1}{2i} (e^{i\theta} - e^{-i\theta}) \right)^3 = -\frac{1}{8i} (e^{i\theta} - e^{-i\theta})^3 \\ &= -\frac{1}{8i} (e^{i3\theta} - 3e^{i\theta} + 3e^{-i\theta} - e^{-i3\theta}) \\ &= -\frac{1}{4} \left(\frac{1}{2i} (e^{i3\theta} - e^{-i3\theta}) - 3 \cdot \frac{1}{2i} (e^{i\theta} - e^{-i\theta}) \right) \\ &= \frac{3}{4} \sin \theta - \frac{1}{4} \sin 3\theta. \end{aligned}$$

◇

Example 4. Find a formula for $\cos^5 \theta$ in terms of cosines of multiples of θ .

SOLUTION. Using the formula $\cos n\theta = \frac{1}{2} (e^{in\theta} + e^{-in\theta})$, for $n = 1, 3, 5$, and the Binomial Theorem gives

$$\begin{aligned} \cos^5 \theta &= \left(\frac{1}{2} (e^{i\theta} + e^{-i\theta}) \right)^5 = \frac{1}{32} (e^{i\theta} + e^{-i\theta})^5 \\ &= \frac{1}{32} (e^{i5\theta} + 5e^{i3\theta} + 10e^{i\theta} + 10e^{-i\theta} + 5e^{-i3\theta} + e^{-i5\theta}) \\ &= \frac{1}{16} \left(\frac{1}{2} (e^{i5\theta} + e^{-i5\theta}) + \frac{5}{2} (e^{i3\theta} + e^{-i3\theta}) + 5 (e^{i\theta} + e^{-i\theta}) \right) \\ &= \frac{1}{16} (\cos 5\theta + 5 \cos 3\theta + 10 \cos \theta). \end{aligned}$$

◇

Formulae of this type are very useful in integration. For example, using the above identity,

$$\int \cos^5 \theta \, d\theta = \frac{1}{16} \left(\frac{1}{5} \sin 5\theta + \frac{5}{3} \sin 3\theta + 10 \sin \theta \right) + C.$$

[X] **Example 5.** Find the sum of

$$\cos \theta + \cos(2\theta) + \cdots + \cos(n\theta).$$

SOLUTION. We use the fact that $\cos \theta$ is the real part of $e^{i\theta}$. The required sum is then the real part of the sum

$$S_n = e^{i\theta} + e^{2i\theta} + \cdots + e^{ni\theta}.$$

Since $e^{ki\theta} = (e^{i\theta})^k$, the sum S_n is a geometric progression with ratio of successive terms given by $R = e^{i\theta}$. Then, on using the formula for the sum of a geometric progression (see Section 3.11), we have

$$S_n = R + R^2 + \cdots + R^n = R(1 + R + \cdots + R^{n-1}) = R \frac{1 - R^n}{1 - R} = e^{i\theta} \frac{1 - e^{in\theta}}{1 - e^{i\theta}}.$$

We require the real part of S_n . The simplest way of finding the real part is to use the following trick. Note that

$$1 - e^{i\theta} = e^{i\theta/2} (e^{-i\theta/2} - e^{i\theta/2}) = -2ie^{i\theta/2} \sin\left(\frac{\theta}{2}\right),$$

and hence that

$$S_n = e^{i\theta} \frac{e^{in\theta/2} \sin\left(\frac{n\theta}{2}\right)}{e^{i\theta/2} \sin\left(\frac{\theta}{2}\right)} = e^{i(n+1)\theta/2} \frac{\sin\left(\frac{n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}.$$

The required sum of the cosine terms is therefore

$$\sum_{k=1}^n \cos(k\theta) = \operatorname{Re}(S_n) = \cos\left(\frac{(n+1)\theta}{2}\right) \frac{\sin\left(\frac{n\theta}{2}\right)}{\sin\left(\frac{\theta}{2}\right)}.$$

◇

Sums of this type are used in X-ray diffraction, solid-state physics, chemistry, signal processing in electrical engineering, and tomography, as well as in many other areas where periodic functions and waves must be analysed.

3.9 Geometric applications of complex numbers

We have seen that the Argand diagram can be used to represent every complex number as a point in a plane. This plane is frequently called the **complex plane**. Relations between complex numbers can therefore be given a geometric interpretation in this complex plane, and, conversely, the geometry of a plane can be represented by algebraic relations between complex numbers. This connection between geometry and complex numbers has proved to be a very useful and powerful tool in mathematics and in areas such as physics, electrical engineering, fluid dynamics, oceanography, aerodynamics and mechanical engineering, among others.

In this section, we shall look at some simple geometrical examples. From Figure 3 in Section 3.5, we can see that the complex numbers 0, z_1 , z_2 and $z_1 + z_2$ form a parallelogram. For the geometric interpretation of the difference of two complex numbers, we shall use a different approach.

Example 1. Give a geometric interpretation of $|z - w|$ and $\operatorname{Arg}(z - w)$.

SOLUTION. Let $z = x + iy$, $w = a + ib$. Note that $z - w = (x - a) + i(y - b)$. Now

$$|z - w| = \sqrt{(x - a)^2 + (y - b)^2}$$

which is the distance between the points representing z and w .

To understand the geometric interpretation of $\text{Arg}(z - w)$, we represent the complex number $z - w$ by the directed line segment (the vector) from w to z . We plot z and w on the Argand diagram for each of the four cases as shown in Figure 13. The angle α satisfies

$$\sin \alpha = \frac{y - b}{|z - w|} \quad \text{and} \quad \cos \alpha = \frac{x - a}{|z - w|},$$

and hence $\alpha = \text{Arg}(z - w)$.

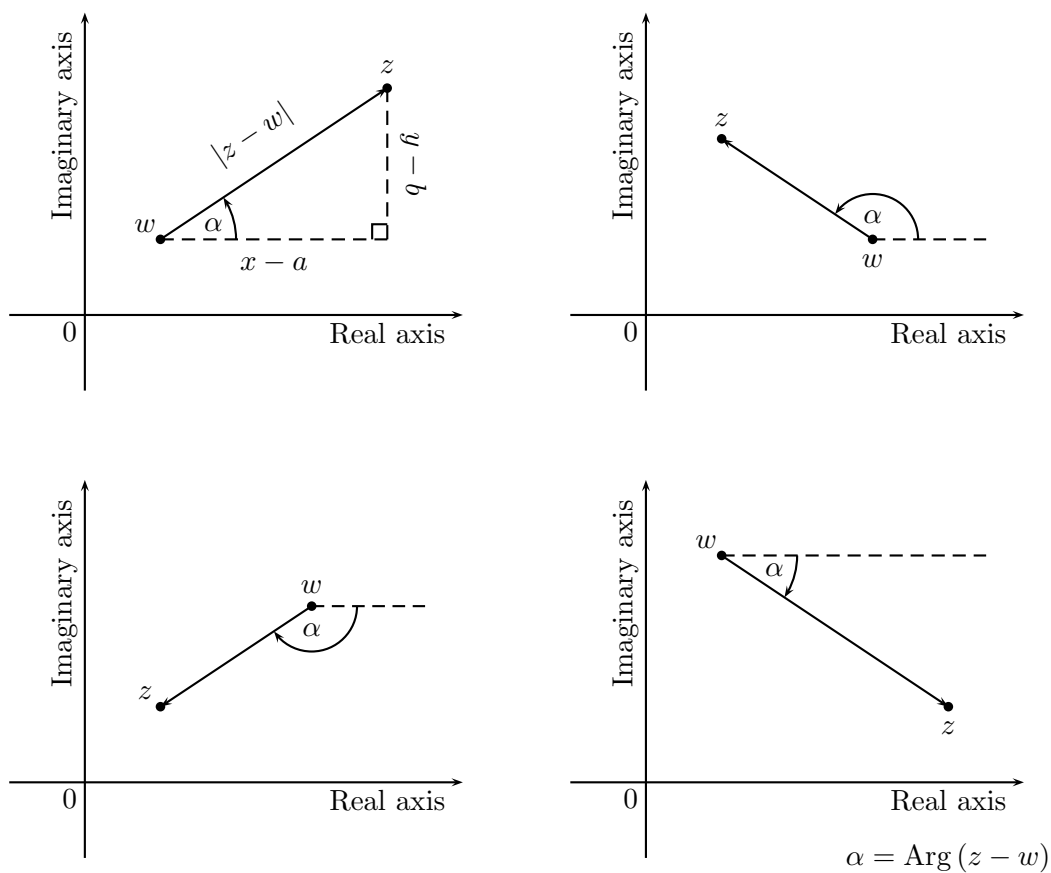


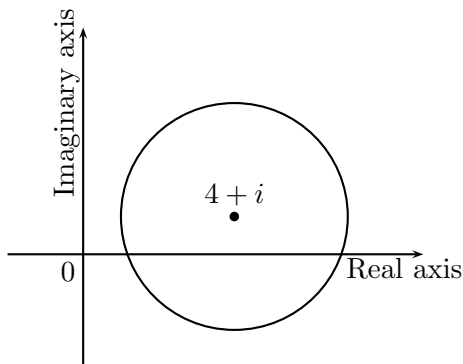
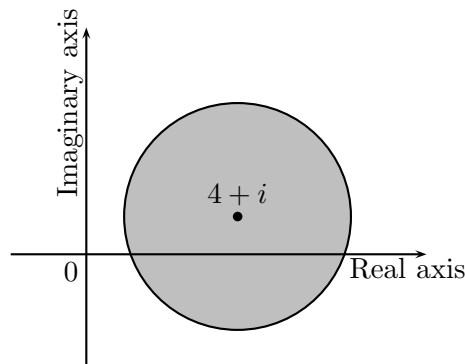
Figure 13: Geometric Interpretation of $\text{Arg}(z - w)$.

The fact that $|z - w|$ is the distance between the points z and w and that $\text{Arg}(z - w)$ is the angle between the arrow from w to z and a line in the direction of the positive real axis are important and should be remembered. $\diamond$

Example 2. Give a geometric interpretation of each of the following sets of points

- a) $|z - 4 - i| = 3$, and b) $|z - 4 - i| \leq 3$.

SOLUTION. $|z - 4 - i|$ is the distance of a point z from the point $4 + i$. Therefore $|z - 4 - i| = 3$ says z is at a distance of 3 from the point $4 + i$. Thus, the equation describes a circle of radius 3 with centre at the point $4 + i$. Similarly, $|z - 4 - i| \leq 3$ is the region in which all points are at a distance at most 3 from the point $4 + i$, and hence the inequality describes the disc of radius 3 with centre at $4 + i$ (with the boundary circle included). The plots are shown in Figures 14 and 15.

Figure 14: $\{z \in \mathbb{C} : |z - 4 - i| = 3\}$.Figure 15: $\{z \in \mathbb{C} : |z - 4 - i| \leq 3\}$.

An alternative way of arriving at a geometric interpretation of a set is to use the “ $x + yi$ ” (or Cartesian) form of a complex number. For example, using $z = x + yi$, we can square the equation $|z - 4 - i| = 3$ and rewrite it to obtain

$$|z - 4 - i|^2 = |(x - 4) + (y - 1)i|^2 = (x - 4)^2 + (y - 1)^2 = 9,$$

which corresponds to the equation of a circle with centre at $x = 4$ and $y = 1$ and of radius 3. $\diamond$

Example 3. Sketch the set $\{z \in \mathbb{C} : -\frac{\pi}{3} \leq \text{Arg}(z - 4 - i) \leq \frac{\pi}{4}\}$.

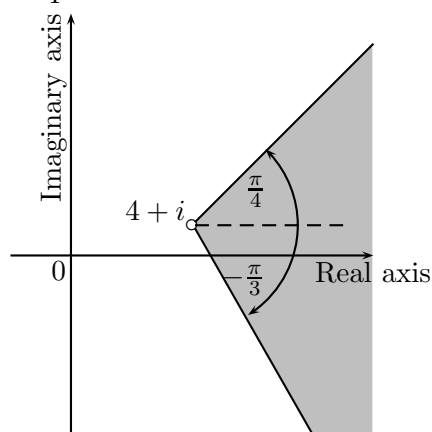
SOLUTION. The above set is the set of all complex numbers z such that

$$-\frac{\pi}{3} \leq \text{Arg}(z - 4 - i) \leq \frac{\pi}{4}.$$

Now, as shown in Example 1, $\text{Arg}(z - 4 - i)$ represents the angle between the line segment from $4 + i$ to z and a line parallel to the real axis, and hence the required sketch is as shown in Figure 16. Notice that the point $z = 4 + i$ is omitted from the set. This is because the function $\text{Arg}(z - 4 - i)$ is undefined there. The boundary line segments are

$$\left\{z \in \mathbb{C} : \text{Arg}(z - 4 - i) = -\frac{\pi}{3}\right\},$$

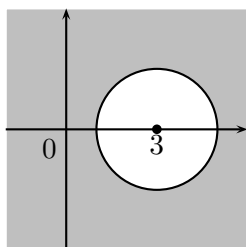
$$\left\{z \in \mathbb{C} : \text{Arg}(z - 4 - i) = \frac{\pi}{4}\right\}.$$

Figure 16: $\{z : -\frac{\pi}{3} \leq \text{Arg}(z - 4 - i) \leq \frac{\pi}{4}\}$.

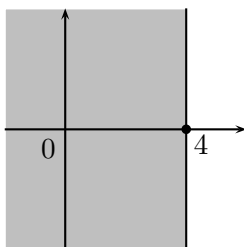
$\diamond$

Example 4. Sketch the set $\{z \in \mathbb{C} : |z - 3| \geq 2 \text{ and } \operatorname{Re}(z) \leq 4\}$.

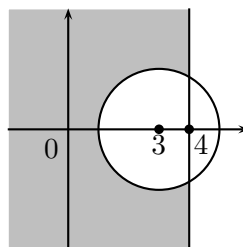
SOLUTION. The given set is the set of all complex numbers z which satisfy both $|z - 3| \geq 2$ and $\operatorname{Re}(z) \leq 4$. The inequality $|z - 3| \geq 2$ describes the region on and outside the circle of radius 2 centred at 3, while $\operatorname{Re}(z) = x \leq 4$ corresponds to the half-plane on and to the left of the line $x = 4$. The required set is the intersection of these two regions as both inequalities must be satisfied. The plots are shown in Figure 17. In practise, only the last diagram need to be shown.



$$\{z \in \mathbb{C} : |z - 3| \geq 2\}$$



$$\{z \in \mathbb{C} : \operatorname{Re}(z) \leq 4\}$$



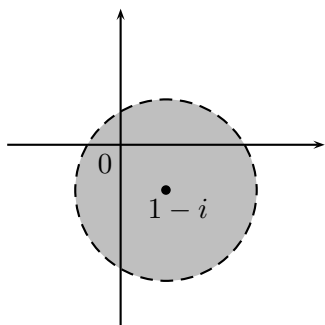
$$\{z \in \mathbb{C} : |z - 3| \geq 2 \text{ and } \operatorname{Re}(z) \leq 4\}$$

Figure 17: Argand Diagrams for Example 4.

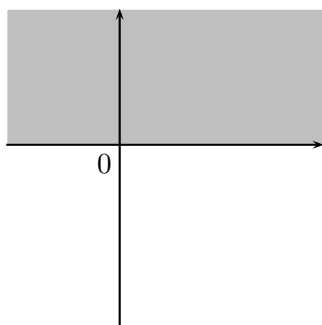
◇

Example 5. Sketch the set $\{z \in \mathbb{C} : |z - 1 + i| < 2 \text{ or } \operatorname{Im}(z) \geq 0\}$.

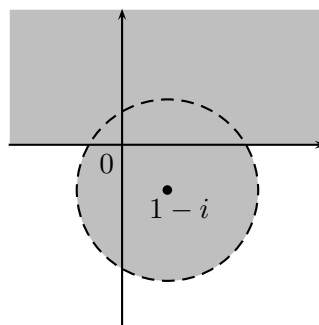
SOLUTION. The inequality $|z - 1 + i| < 2$ represents the open disc of radius 2 with centre at $1 - i$.



$$\{z \in \mathbb{C} : |z - 1 + i| < 2\}$$



$$\{z \in \mathbb{C} : \operatorname{Im} z \geq 0\}$$



$$\{z \in \mathbb{C} : |z - 1 + i| < 2 \text{ or } \operatorname{Im}(z) \geq 0\}$$

Figure 18: Argand Diagrams for Example 5.

To indicate that the circle $|z - 1 + i| = 2$ is not included in the set, we draw it with dashes. Also, $\operatorname{Im}(z) = y \geq 0$ describes the half-plane on and above the real axis. The required set is the union of these two regions (as **either** inequality being satisfied puts the point in the set). ◇

3.10 Complex polynomials

We saw at the beginning of this chapter that there are polynomials which have no roots in the real numbers. Instead we defined the number i to be the solution for $x^2 + 1 = 0$. We shall now investigate in more detail the properties of polynomials and their roots when we work over the complex numbers.

Definition 1. Suppose n is a natural number and $a_0, a_1, \dots, a_n$ are complex numbers with $a_n \neq 0$. Then the function $p : \mathbb{C} \rightarrow \mathbb{C}$ such that

$$p(z) = a_0 + a_1z + a_2z^2 + \dots + a_nz^n$$

is called a polynomial of degree n .

The zero polynomial is defined to be the function $p(z) = 0$ and we do not define its degree.

NOTE. If $a_0, a_1, \dots, a_n$ are real and z takes only real values, then we say that the polynomial is defined over $\mathbb{R}$.

Example 1. $p_1(z) = 1 + 3z^2$, $p_2(z) = 1 + i - 4iz^3$ are examples of polynomials.

Note that $p_1(z)$ has real coefficients. If it takes complex values, we say it is a polynomial defined over $\mathbb{C}$. However, if it only takes real values, we say it is a polynomial defined over $\mathbb{R}$.

3.10.1 Roots and factors of polynomials

An important mathematical and practical problem concerning polynomials is that of factorising them into simpler polynomials. This factorisation problem is closely related to the problem of finding the roots (or zeroes) of polynomials.

Definition 2. A number α is a **root** (or **zero**) of a polynomial p if $p(\alpha) = 0$.

Definition 3. Let p be a polynomial. Then, if there exist polynomials p_1 and p_2 such that $p(z) = p_1(z)p_2(z)$ for all complex z , then p_1 and p_2 are called **factors** of p .

Theorem 1 (Remainder Theorem). The remainder r which results when $p(z)$ is divided by $z - \alpha$ is given by $r = p(\alpha)$.

Theorem 2 (Factor Theorem). A number α is a root of p if and only if $z - \alpha$ is a factor of $p(z)$.

The major difference between polynomials over the complex numbers and polynomials over the real numbers is contained in the following theorem.

Theorem 3 (The Fundamental Theorem of Algebra). A polynomial of degree $n \geq 1$ has at least one root in the complex numbers.

We shall not try to prove the fundamental theorem here. The proof is usually given in courses on functions of a complex variable.

There are several important points to note about the fundamental theorem.

The theorem is not true in general for polynomials over $\mathbb{R}$. For example, the real quadratic q defined by $q(z) = 1 + z^2$ does not have any real roots. In contrast, it has the two complex roots $\pm i$.

We can combine the Fundamental Theorem of Algebra and the Factor Theorem to prove the following extremely important theorem.

Theorem 4 (Factorisation Theorem). *Every polynomial of degree $n \geq 1$ has a factorisation into n linear factors of the form*

$$p(z) = a(z - \alpha_1)(z - \alpha_2) \dots (z - \alpha_n), \quad (\#)$$

where the n complex numbers $\alpha_1, \alpha_2, \dots, \alpha_n$ are roots of p and where a is the coefficient of z^n .

[X] *Proof.* To prove the theorem we use induction on the degree of p .

Condition $(\#)$ is obviously true for $\deg(p) = 1$. We now prove that, for $n \geq 1$, $(\#)$ is true for polynomials of degree $n + 1$ whenever it is true for polynomials of degree n .

Let p be a polynomial of degree $n + 1$, where $n \geq 1$. Then, by the Fundamental Theorem there exists a root $\alpha_1 \in \mathbb{C}$ of p , and by the Factor Theorem $z - \alpha_1$ is a factor of p . Hence $p(z) = (z - \alpha_1)p_1(z)$, where p_1 is a polynomial of degree n . Now, if $(\#)$ is true for polynomials of degree n , then there are n complex numbers $\alpha_2, \dots, \alpha_{n+1}$ and a complex number a such that

$$p_1(z) = a(z - \alpha_2) \dots (z - \alpha_{n+1}).$$

Thus,

$$p(z) = (z - \alpha_1)p_1(z) = a(z - \alpha_1)(z - \alpha_2) \dots (z - \alpha_{n+1}).$$

Hence, $(\#)$ is true for polynomials of degree $n + 1$ whenever it is true for polynomials of degree n , and as $(\#)$ is true for $n = 1$, it is true for all $n \geq 1$, by induction.

Finally, note that in the factorisation shown in $(\#)$ a is the coefficient of the z^n term in p . $\square$

Example 2. Find the roots and factors of the quadratics

$$p_1(z) = z^2 + 2z + 1, \quad p_2(z) = 2z^2 - 9z + 4 \quad \text{and} \quad p_3(z) = z^2 + z + 1.$$

SOLUTION. It is straightforward to factorise $p_1(z)$:

$$p_1(z) = (z + 1)^2.$$

So the polynomial has a repeated root -1 , with multiplicity 2.

Likewise, we have

$$p_2(z) = (2z - 1)(z - 4).$$

The roots of this polynomial are $\frac{1}{2}$ and 4.

The polynomial $p_3(z)$ does not easily factor. Using the quadratic formula, the roots are $z = \frac{-1 \pm i\sqrt{3}}{2}$, so

$$p_3(z) = \left(z - \frac{1}{2}(-1 + i\sqrt{3})\right) \left(z - \frac{1}{2}(-1 - i\sqrt{3})\right).$$

◇

The Factorisation Theorem guarantees that a polynomial of degree n always has n roots, but it does not tell us how to actually find these roots. For polynomials of degree $n \leq 4$, exact formulae for the roots in terms of the coefficients have been found. For quadratics, the exact formula for the roots is well known and very easy to use. For cubic polynomials (degree 3) the exact formula is called Cardano's formula and it is also reasonably easy to use. For quartic polynomials (degree 4) there is also an exact formula for the roots but the formula is very complicated. However, for polynomials of degree $n > 4$, there are no formulae for the roots in terms of square roots, cube roots, etc. A proof of this fact is given in courses on Galois Theory.

In general, it is either difficult or impossible to find exact roots, and hence an exact factorisation, for higher degree polynomials. It is usually necessary to resort to approximate numerical methods to find the roots of a polynomial. At the present time the best general-purpose numerical method is based on finding the "eigenvalues" of a "companion matrix" for the polynomial. These numerical methods are discussed in advanced courses on numerical matrix algebra.

There are, however, some very simple types of higher degree polynomials for which exact roots, and hence an exact factorisation, can be found. One such case is for polynomials of the form $p(z) = z^n - a$, since in this case the solutions of

$$z^n - a = 0 \quad \text{or equivalently of} \quad z^n = a,$$

are the n th roots of the number a . This problem has been discussed in Section 3.7.3.

Example 3. Factorise $z^6 + 1$.

SOLUTION. We first solve $z^6 + 1 = 0$ or $z^6 = -1$ by finding the sixth roots of -1 . Since the modulus and argument of -1 are $|-1| = 1$ and $\text{Arg}(-1) = \pi$, we have

$$z^6 = e^{(\pi+2k\pi)i} \quad \text{for } k \in \mathbb{Z}.$$

So, the sixth roots in polar forms are

$$z = e^{i\frac{(2k+1)\pi}{6}}, \text{ for } k = -3, -2, -1, 0, 1, 2.$$

That is, $e^{-i\frac{5\pi}{6}}$, $e^{-i\frac{\pi}{2}}$, $e^{-i\frac{\pi}{6}}$, $e^{i\frac{\pi}{6}}$, $e^{i\frac{\pi}{2}}$, and $e^{i\frac{5\pi}{6}}$.

By the Factor Theorem, we have

$$z^6 + 1 = (z - e^{i\pi/2}) (z - e^{-i\pi/2}) (z - e^{i\pi/6}) (z - e^{-i\pi/6}) (z - e^{5i\pi/6}) (z - e^{-5i\pi/6}).$$

An alternative form can be obtained by putting the roots into the " $a + ib$ " form. You should do this as an exercise, and you should obtain the result

$$z^6 + 1 = (z - i)(z + i) \left(z - \frac{1}{2}(\sqrt{3} + i)\right) \left(z - \frac{1}{2}(\sqrt{3} - i)\right) \left(z - \frac{1}{2}(-\sqrt{3} + i)\right) \left(z - \frac{1}{2}(-\sqrt{3} - i)\right).$$

◇

3.10.2 Factorisation of polynomials with real coefficients

This is an important special case of the general theory. In the examples of factorisation we have given above in Examples 2 and 3, all of the polynomials have real coefficients. For the quadratics in Example 2, we showed one case of two equal real roots, one case of two distinct real roots, and one case of two complex roots. In Example 3, we obtained six complex roots of $z^6 + 1$. However, if you examine the complex roots in Examples 2 and 3, you will see that the roots occur in conjugate pairs. Thus, in the quadratic example the roots $(-1 + i\sqrt{3})/2$ and $(-1 - i\sqrt{3})/2$ are a conjugate pair, while in the sixth-root example there are three conjugate pairs $\{e^{-i\pi/2}, e^{i\pi/2}\}$, $\{e^{-i\pi/6}, e^{i\pi/6}\}$, and $\{e^{-5i\pi/6}, e^{5i\pi/6}\}$. These results are examples of a useful proposition which is:

Proposition 5. If α is a root of a polynomial p with **real** coefficients, then the complex conjugate $\bar{\alpha}$ is also a root of p .

Proof. Let p be a polynomial with real coefficients given by

$$p(z) = a_0 + a_1z + a_2z^2 + \cdots + a_nz^n \quad \text{for all } z \in \mathbb{C},$$

where $a_0, a_1, \dots, a_n$ are real numbers. Then, since α is a root of p ,

$$p(\alpha) = a_0 + a_1\alpha + a_2\alpha^2 + \cdots + a_n\alpha^n = 0.$$

On taking the complex conjugate of this equation, and using the facts that if a_k is real and b is any complex number, then

$$\overline{a_k b} = \overline{a_k} \bar{b} = a_k \bar{b}$$

and that the complex conjugate of α^k is $(\bar{\alpha})^k$, we have

$$0 = \overline{p(\alpha)} = a_0 + a_1\bar{\alpha} + a_2\bar{\alpha}^2 + \cdots + a_n\bar{\alpha}^n = p(\bar{\alpha}).$$

Hence, $\bar{\alpha}$ is also a root of p . □

An immediate consequence of this proposition is that the roots of a complex polynomial with real coefficients are either real or occur in conjugate pairs. This fact can be used to obtain a factorisation of a polynomial with real coefficients into linear or quadratic factors with real coefficients.

Proposition 6. If p is a polynomial with real coefficients, then p can be factored into linear and quadratic factors all of which have real coefficients.

Proof. For a polynomial with real coefficients, if α is a root which is not real then so is $\bar{\alpha}$, and hence both $z - \alpha$ and $z - \bar{\alpha}$ are factors of p . On multiplying these factors, we obtain

$$(z - \alpha)(z - \bar{\alpha}) = z^2 - (\alpha + \bar{\alpha})z + \alpha\bar{\alpha}.$$

Then, from section 3.3, we have that

$$\alpha + \bar{\alpha} = 2\operatorname{Re}(\alpha), \quad \text{and} \quad \alpha\bar{\alpha} = |\alpha|^2,$$

where $\operatorname{Re}(\alpha)$ and $|\alpha|^2$ are both real numbers. Hence,

$$q(z) = (z - \alpha)(z - \bar{\alpha}) = z^2 - 2\operatorname{Re}(\alpha)z + |\alpha|^2$$

is a quadratic with real coefficients. Thus, all complex factors can be replaced in pairs by quadratic factors with real coefficients, and the proof is complete. □

Example 4. Factorise $z^5 + 32$ into linear and quadratic factors with real coefficients.

SOLUTION. We first solve $z^5 = -32$ to find the fifth roots of -32 . Using the usual procedure, we obtain the five solutions

$$2e^{i\pi/5}, \quad 2e^{i3\pi/5}, \quad 2e^{-i\pi/5}, \quad 2e^{i\pi} = -2, \quad 2e^{-3i\pi/5},$$

which consist of one real root and two conjugate pairs.

The factors corresponding to a conjugate pair can be replaced by a quadratic factor with real coefficients. For example,

$$\begin{aligned} (z - 2e^{i\pi/5})(z - 2e^{-i\pi/5}) &= z^2 - 2(e^{i\pi/5} + e^{-i\pi/5})z + 4 \\ &= z^2 - 4z \cos \frac{\pi}{5} + 4. \end{aligned}$$

The required factorisation is therefore

$$\begin{aligned} z^5 + 32 &= (z + 2)(z - 2e^{i\pi/5})(z - 2e^{-i\pi/5})(z - 2e^{3i\pi/5})(z - 2e^{-3i\pi/5}) \\ &= (z + 2)(z^2 - 4z \cos \frac{\pi}{5} + 4)(z^2 - 4z \cos \frac{3\pi}{5} + 4). \end{aligned}$$

Note that this factorisation is certainly not obvious. If you multiply out the right hand side and compare coefficients you will find some very surprising relations between $\cos \frac{\pi}{5}$ and $\cos \frac{3\pi}{5}$. $\diamond$

3.11 Appendix: A note on proof by induction

In mathematics and logic a proposition is a meaningful statement which can only be either true or false. Proof by induction is a method of proof which is often used to prove that some proposition $P(n)$ is true for all integers $n \geq n_0$, where n_0 is a fixed integer.

Example 1. For each integer $n \geq 1$, let $P(n)$ be the proposition

$$1 + 2 + 3 + \cdots + n = \frac{1}{2}n(n + 1).$$

In this example, $P(n)$ gives the formula for the sum of the first n positive integers. $\diamond$

Example 2. For each integer $n \geq 0$, let $P(n)$ be the proposition

$$1 + r + r^2 + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r},$$

where $r \neq 1$ is a fixed number. In this example, $P(n)$ gives the formula for the sum of $n + 1$ terms of a geometric progression with ratio r . $\diamond$

Example 3. For each integer $n \geq 2$, let $P(n)$ be the proposition that “ n can be completely factored into prime numbers”. $\diamond$

As we shall see, all the propositions $P(n)$ in examples 1, 2 and 3 are true.

There are several versions of “proof by induction”, but the two most commonly used are based on what are sometimes called the first and second principles of induction.

First Principle of Induction. Let $n_0 \in \mathbb{Z}$, and let $P(n)$, for $n \geq n_0$, be propositions. Then, if

1. $P(n_0)$ is true, and
2. for each $n \geq n_0$, $P(n+1)$ is true whenever $P(n)$ is true,

then $P(n)$ is true for all $n \geq n_0$.

Let us return to the previous examples.

Example 1. We prove the formula for the sum of the first n integers, i.e., that:

For $n \geq 1$,

$$1 + 2 + 3 + \cdots + n = \frac{1}{2}n(n+1).$$

Proof. We first note that the proposition $P(1)$ states that $1 = \frac{1}{2} \times 1 \times 2$, which is clearly true.

Now, if $P(n)$ is true for some fixed $n \geq 1$, then the sum of $n+1$ integers is

$$\begin{aligned} 1 + 2 + \cdots + n + (n+1) &= \frac{1}{2}n(n+1) + (n+1) \\ &= \frac{1}{2}(n+1)(n+2) = \frac{1}{2}(n+1)((n+1)+1), \end{aligned}$$

and hence $P(n+1)$ is also true. Thus, for all $n \geq 1$, $P(n+1)$ is true whenever $P(n)$ is true.

Therefore, from the first principle of induction, $P(n)$ is true for all integers $n \geq 1$. □

Example 2. We prove the formula for the sum of a geometric progression, i.e., that:

For $n \in \mathbb{N}$ and $r \neq 1$,

$$1 + r + \cdots + r^n = \frac{1 - r^{n+1}}{1 - r}.$$

Proof. We first note that the proposition $P(0)$ states that

$$1 = \frac{1 - r}{1 - r},$$

which is clearly true for $r \neq 1$.

Now, if $P(n)$ is true for some fixed integer $n \geq 0$, then the sum of $n+2$ terms is

$$1 + r + \cdots + r^n + r^{n+1} = \frac{1 - r^{n+1}}{1 - r} + r^{n+1} = \frac{1 - r^{n+2}}{1 - r} = \frac{1 - r^{(n+1)+1}}{1 - r},$$

and hence $P(n+1)$ is also true. Thus, for any integer $n \geq 0$, $P(n+1)$ is true whenever $P(n)$ is true.

Therefore, from the first principle of induction, $P(n)$ is true for all $n \geq 0$. □

We now briefly describe the second principle of induction.

Second Principle of Induction. Let $n_0 \in \mathbb{Z}$, and let $P(n)$, for $n \geq n_0$, be propositions. Then, if

1. $P(n_0)$ is true, and
2. for each $n \geq n_0$, $P(n+1)$ is true whenever all propositions $P(m)$ are true for $m = n_0, \dots, n$ (i.e. $n_0 \leq m \leq n$),

then $P(n)$ is true for all $n \geq n_0$.

Example 3. Every integer $n \geq 2$ can be factored into primes.

Proof. The proposition $P(n)$ is that n can be factored into primes. $P(2)$ then asserts that 2 can be factored into primes. This result is clearly true, since 2 is itself a prime.

We now show that, if $P(m)$ is true for all integers m with $2 \leq m \leq n$, then $P(n+1)$ is also true.

Now, the integer $n+1$ must either be prime or not prime.

Case 1. $n+1$ is prime. Then $n+1$ is already factored and hence $P(n+1)$ is true in this case.

Case 2. $n+1$ is not prime. Then, there are two integers m_1 and m_2 less than $n+1$ such that $n+1 = m_1 m_2$. But, as $P(m)$ is true for $2 \leq m \leq n$, m_1 and m_2 can both be factored into primes, and hence $n+1$ can also be factored into primes. Thus, $P(n+1)$ is true in this case also.

We have shown that $P(2)$ is true, and that $P(n+1)$ is true whenever $P(m)$ is true for all m with $2 \leq m \leq n$. Hence, from the second principle of induction, $P(n)$ is true for all integers $n \geq 2$. $\square$

In fact the two apparently different types of induction are the same, as we can easily show. Simply let the proposition $Q(n)$ be “ $P(m)$ is true for $m = n_0, \dots, n$ ”. Then use the first type of proof.

3.12 Appendix: The Binomial Theorem

Theorem 1. If $a, b \in \mathbb{C}$ and $n \in \mathbb{N}$, then

$$\begin{aligned} (a+b)^n &= a^n + na^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 + \dots + nab^{n-1} + b^n \\ &= \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k, \end{aligned}$$

where the numbers

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

are the binomial coefficients.

NOTE. We are using the convention that $a^0 = 1$ and $0! = 1$.

Proof. The proof of the theorem is based on the first principle of induction. In this case, the proposition $P(n)$ to be proved true is that the formula given above for $(a+b)^n$ is correct.

For $n = 1$, $P(1)$ asserts that $(a+b)^1 = a^1 + b^1$, which is clearly true.

Now, if the formula is correct for some integer $n \geq 1$, then we have

$$\begin{aligned}(a+b)^{n+1} &= (a+b)(a+b)^n \\ &= (a+b) \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k.\end{aligned}$$

On multiplying out, we find that the coefficient of a^{n+1} and b^{n+1} are 1 and that the coefficient of $a^{n+1-k}b^k$ is

$$\binom{n}{k} + \binom{n}{k-1}.$$

Now,

$$\begin{aligned}\binom{n}{k} + \binom{n}{k-1} &= \frac{n!}{k!(n-k)!} + \frac{n!}{(k-1)!(n-k+1)!} \\ &= \frac{n!}{k!(n-k+1)!} (n-k+1+k) \\ &= \frac{(n+1)!}{k!(n-k+1)!} = \binom{n+1}{k}.\end{aligned}$$

Hence,

$$\begin{aligned}(a+b)^{n+1} &= a^{n+1} + \sum_{k=1}^n \binom{n+1}{k} a^{n+1-k} b^k + b^{n+1} \\ &= \sum_{k=0}^{n+1} \binom{n+1}{k} a^{n+1-k} b^k.\end{aligned}$$

and hence $P(n+1)$ is true.

We have therefore shown that $P(1)$ is true, and that, for all $n \geq 1$, $P(n+1)$ is true whenever $P(n)$ is true. Thus, $P(n)$ is true for all $n \geq 1$ by induction. $\square$

NOTE. The relation

$$\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$$

shows why Pascal's triangle works.

3.13 Complex numbers and Maple

Maple is a Symbolic Computing Package which enables computers to do algebra and calculus. Information about computing and Maple is available from the School of Mathematics web site, my eLearning Vista, the MATH1131/1141 information booklet and the Computing Notes for the First Year Mathematics Courses.

This section gives a few hints on how it can be used for complex numbers. The symbol I is used for the complex number i . Multiplication is denoted by $*$ and exponentiation by the 'circumflex'

or ‘uparrow’ $\wedge$. Complex number arithmetic is normally done automatically. However, in some situations the command `evalc` is required to get a complex expression in $a + ib$ from, where a, b are real. Some examples are:

```
(3+5*I)^3;
```

gives the correct value of $(3 + 5i)^3 = -198 + 10i$ on the screen.

```
Re(%);
```

gives the real part of this last expression. If you now try:

```
conjugate(%);
```

the complex conjugate of the expression before last will appear. The command

```
evalc(polar(2,Pi));
```

can be used to enter complex numbers (here $2e^{\pi i}$) in polar form and this command gives the correct value -2 .

Instructions on how to convert complex numbers into polar form are available ‘on line’ by means of

```
?convert[polar]
```

Alternatively, you can use Maple Help menu. Various other conversion operations are also available.

Problems for Chapter 3

“Why,” said the Dodo, “the best way to explain it is to do it.”
- Lewis Carroll, Alice in Wonderland.

Questions marked with [R] are routine, [H] harder, [M] Maple and [X] are for MATH1141 only. You should make sure that you can do the easier questions before you tackle the more difficult questions. Questions marks with a [V] have video solutions available on Moodle.

Problems 3.1 : A review of number systems

- [R] Solve (if possible) the following equations for $x \in \mathbb{N}$, $x \in \mathbb{Z}$, $x \in \mathbb{Q}$ and $x \in \mathbb{R}$.
 - $x + 25 = 0$, $3x - 9 = 0$, $3x + 9 = 0$, $3x + 10 = 0$.
 - $x^2 + 4x - 5 = 0$, $2x^2 - 13x + 15 = 0$, $x^2 - x - 1 = 0$, $x^2 + 3x + 4 = 0$.
 - $\sin(\pi x/3) = 0$, $\sin(x/3) = 0$.
- [R] Is the set $\{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ closed under addition? Prove your answer.
- [H] Can any finite set of integers be closed under addition? Prove your answer.
- [R] Is the set $\{-1, 1\}$ closed under multiplication and division?

Problems 3.3 : The rules of arithmetic for complex numbers

- [R][V] Let $z = 2 + 3i$, $w = -1 + 2i$. Calculate $3z$, z^2 , $z + 2w$, $z(w + 3)$, $\frac{z}{w}$, $\frac{w}{z}$.
- [R] Write the following expressions in $a + ib$ or “Cartesian” form:
 - $\frac{1+i}{1+2i}$, b) $\frac{2-i}{3+i} - \frac{3-i}{2+i}$.
- [R] If $z = a + ib$, express the following in “Cartesian” form:
 - z^2 , b) $\frac{1}{z}$ c) $\frac{z+1}{z-1}$
- [R][V] Use the quadratic formula to find all complex roots of the following polynomials.
 - $z^2 + z + 1$, b) $z^2 + 2z + 3$, c) $z^2 - 6z + 10$,
 - $-2z^2 + 6z - 3$, e) $z^4 + 5z^2 + 4$.
- [H] Show that $[(\sqrt{3} + 1) + (\sqrt{3} - 1)i]^3 = 16(1 + i)$.
- [R] Simplify $(r + s)^2$ where r is the square root of $3 + 4i$ with non negative real part and s is the square root of $3 - 4i$ with non negative real part.

11. [H][V] Simplify $\left(\frac{a+bi}{a-bi}\right)^2 - \left(\frac{a-bi}{a+bi}\right)^2$ where a and b are real numbers not both zero.

Problems 3.4 : Real parts, imaginary parts and complex conjugates

12. [R] Find $\operatorname{Re}(z)$, $\operatorname{Im}(z)$ and $\bar{z}$ for $z = -1 + i$, $2 + 3i$, $2 - 3i$, $\frac{2-i}{1+i}$, $\frac{1}{(1+i)^2}$.
13. [R] Let $z = 1 + 2i$ and $w = 3 - 4i$. Calculate z^2 and $\frac{\bar{z}}{w}$, expressing the answers in Cartesian form.
14. [R][V] Given that $2z + 3w = 1 + 12i$ and $\bar{z} - \bar{w} = 3 - i$, find z and w .
15. [R] By evaluating each side of the equations, check that $\overline{zw} = \bar{z}\bar{w}$, and $\overline{\left(\frac{z}{w}\right)} = \frac{\bar{z}}{\bar{w}}$ are satisfied by the complex numbers $z = 2 + 3i$, $w = -1 + 2i$.
16. [R] Prove that for any two complex numbers z and w
- $\operatorname{Im}(z) = \frac{1}{2i}(z - \bar{z})$
 - $2\operatorname{Re}(z) = z + \bar{z}$
 - $\overline{(z - w)} = \bar{z} - \bar{w}$
 - $\overline{\left(\frac{1}{z}\right)} = \frac{1}{\bar{z}}$,
 - $\overline{zw} = \bar{z}\bar{w}$
 - $\overline{\left(\frac{z}{w}\right)} = \frac{\bar{z}}{\bar{w}}$.
17. [H]
- Use the properties of the complex conjugate to show that if the complex number α is a root of a quadratic equation $ax^2 + bx + c = 0$ with a, b, c being **real** coefficients, then so is $\bar{\alpha}$.
 - Write down the monic quadratic polynomial with real coefficients which has $3 - 2i$ as one of its roots.
 - Does the result of a) generalise to higher degree polynomials?

Problems 3.5 : The Argand diagram and 3.6 : Polar form, modulus and argument

18. [R][V] Find the modulus, principal argument and polar form of each of the following numbers and plot them on an Argand diagram:
- $6 + 6i$,
 - -4 ,
 - $\sqrt{3} - i$,
 - $\frac{-1}{\sqrt{2}} - \frac{i}{\sqrt{2}}$,
 - $-7 + 3i$.
19. [R] If $z = 4 + 3i$ and $w = 2 + i$ find $|3z - 3iw|$, $\operatorname{Im}((1 - i)z - 3|w|)$.
20. [H] If $z = 1 + i$, calculate the powers z^j for $j = 1, 2, \dots, 10$ and plot them on an Argand diagram. Is there a pattern? What is the smallest positive integer n such that z^n is a real number?

21. [R] Find the “ $a + ib$ ” form of the complex numbers whose moduli and principal arguments are
- a) $|z| = 3$, $\text{Arg}(z) = \frac{\pi}{3}$; b) $|z| = 3$, $\text{Arg}(z) = \frac{5\pi}{6}$;
 c) $|z| = 3$, $\text{Arg}(z) = -\frac{2\pi}{3}$; d) $|z| = 3$, $\text{Arg}(z) = -\frac{\pi}{6}$;
 e) [H] $|z| = 3$, $\text{Arg}(z) = \frac{\pi}{8}$.
22. [R][V] a) Show that $z\bar{z} = |z|^2$. Hence, or otherwise, show that if $|z| = 1$, then $\bar{z} = z^{-1}$.
 b) Show that $|z| = |\bar{z}|$ for all $z \in \mathbb{C}$.
 c) If $z = r(\cos \theta + i \sin \theta)$, show that a polar form for the complex conjugate is $\bar{z} = r(\cos(-\theta) + i \sin(-\theta))$.
23. [H] Show that $\text{Re}\left(\frac{1-z}{1+z}\right) = 0$ for any complex z with $|z| = 1$.
24. [H] Use $z\bar{z} = |z|^2$ to prove the identity $|z_1 + z_2|^2 + |z_1 - z_2|^2 = 2(|z_1|^2 + |z_2|^2)$.
25. [H] Use $z\bar{z} = |z|^2$ to show that
- $$|1 - \bar{z}w|^2 - |z - w|^2 = (1 - |z|^2)(1 - |w|^2)$$
- and deduce that $|1 - \bar{z}w|^2 = |z - w|^2$ if either z or w lies on the unit circle.
26. [R] Plot the following complex numbers on an Argand diagram:
- a) $2e^{\frac{i\pi}{4}}$, b) $3e^{\frac{5i\pi}{6}}$, c) $e^{-\frac{2i\pi}{3}}$, d) $2e^{-\frac{i\pi}{2}}$, e) $4e^{i\pi}$.
27. [R] Let $z = (1 - i)$ and $w = 2e^{i\pi/3}$. Calculate w^6 , $z - w$ and $\frac{w}{\bar{z}}$ and express your answers in Cartesian form.
28. [R] For $z = 3e^{-5\pi i/6}$ and $w = 1 + i$, find $\text{Re}(iw + \bar{z}^2)$.
29. [R] Solve $|e^{i\theta} - 1| = 2$ for $-\pi < \theta \leq \pi$.
30. [R] Find $\text{Arg}(-1 + i)$ and $\text{Arg}(-\sqrt{3} + i)$ and hence find the principal arguments of the complex numbers $(-1 + i)(-\sqrt{3} + i)$ and $\frac{-1 + i}{-\sqrt{3} + i}$.
31. [H][V] Let $z = (1 + \sqrt{3}i)$ and $w = (1 + i)$. Find $\text{Arg } z$ and $\text{Arg } w$ and hence $\text{Arg}(zw)$. Evaluate zw and hence show that $\cos \frac{7\pi}{12} = \frac{1 - \sqrt{3}}{2\sqrt{2}}$. Find a similar expression for $\sin \frac{7\pi}{12}$.
32. [R] Find polar forms for $z = 1 + i\sqrt{3}$ and $w = 1 - i$, and hence find first the polar forms and then the “ $a + ib$ ” forms of zw , z^9 , and $\left(\frac{z}{\bar{w}}\right)^{12}$.

Problems 3.7 : Properties and applications of the polar form

33. [R] Find the polar, and hence also the Cartesian form for:

$$\text{a) } (\sqrt{3} + i)^5, \quad \text{b) } \left(\frac{-1 + i}{\sqrt{2}} \right)^{1002}, \quad \text{c) } \left(\frac{1 + \sqrt{3}i}{2} \right)^{-8}.$$

34. [H] Find the square roots (in Cartesian Form) of

$$\text{a) } 21 - 20i, \quad \text{b) } -16 + 30i, \quad \text{c) } 24 + 70i.$$

35. [H] a) Explain why multiplying a complex number z by $e^{i\theta}$ rotates the point represented by z anticlockwise about the origin, through an angle θ .

b) The point represented by the complex number $1 + i$ is rotated anticlockwise about the origin through an angle of $\frac{\pi}{6}$. Find its image in polar and Cartesian form.

c) Find the complex number (in Cartesian form) obtained by rotating $6 - 7i$ anticlockwise about the origin through an angle $\frac{3\pi}{4}$.

36. [H] If $z = re^{i\theta}$, $0 \leq \theta \leq \frac{\pi}{2}$, show that

$$\begin{aligned} \text{a) } |(1 - i)z^2| &= \sqrt{2}r^2 & \text{b) } \text{Arg}((1 - i)z^2) &= 2\theta - \frac{\pi}{4}, \\ \text{c) } \left| \frac{1 + i\sqrt{3}}{z} \right| &= \frac{2}{r}, & \text{d) } \text{Arg}\left(\frac{1 + i\sqrt{3}}{z}\right) &= \frac{\pi}{3} - \theta. \end{aligned}$$

37. [H][V] Find the roots (in Cartesian Form) of

$$\begin{aligned} \text{a) } z^2 - 3z + (3 - i) &= 0, & \text{b) } z^2 - (7 - i)z + (14 - 5i) &= 0, \\ \text{c) } z^2 + (4 - i)z + (1 + 13i) &= 0. \end{aligned}$$

38. [R] Find the seventh roots of -1 and plot the roots on an Argand diagram.

39. [R] Find the sixth roots of i and plot the roots on an Argand diagram.

40. [H] Find the fifth roots of $16 - 16i\sqrt{3}$ and plot the roots on an Argand diagram.

41. [H] Find all $z \in \mathbb{C}$ satisfying $(z - 6 + i)^3 = -27$.

42. [H][V] Show that if ω is an n th root of unity ($\omega \neq 1$ and $n > 1$) then

$$\omega + \omega^2 + \cdots + \omega^n = 0.$$

Hint: Sum the geometric progression.

43. [X] Show that the set $\{z \in \mathbb{C} : |z| \leq 1\}$ is closed under multiplication. Is the set closed under division (zero excluded)? Is the set closed under addition or subtraction?

44. [X] Use the properties of complex conjugates to show that if $a, b \in \mathbb{R}$ and $|z| = 1$, then $|a + bz| = |az + b|$. Hint: You might find the results of Question 22 useful.

45. [X] Suppose a and b are real numbers (not both zero) and $w = \frac{az + bz^{-1}}{bz + az^{-1}}$. Show that if $|z| = 1$, then $|w| = 1$.
Hint: You might find the results of Question 22 useful.

46. [X] Let z, w be complex numbers.

- a) Using polar forms, show that $|\operatorname{Re}(z\bar{w})| \leq |z||w|$.
- b) Use the result in (a) to show that $|z + w| \leq |z| + |w|$, and interpret the result geometrically. Hint: Write $|z + w|^2 = (z + w)(\bar{z} + \bar{w})$ and expand.

47. [X] Let

$$z = \frac{i(1 + is)}{1 - is} \quad \text{where } s \in \mathbb{R}.$$

a) Show that

$$\operatorname{Arg}(z) = \begin{cases} \frac{\pi}{2} + 2 \tan^{-1} s & \text{for } s \leq 1 \\ -\frac{3\pi}{2} + 2 \tan^{-1} s & \text{for } s > 1. \end{cases}$$

b) Describe geometrically what happens to z as s increases from $-\infty$ to ∞ .

48. [H] Suppose $\theta, \phi \neq \frac{\pi}{2}(2k + 1)$ where k is an integer. Use the fact that, for $z \neq -1$,

$$z = \frac{1 + z}{1 + z^{-1}}$$

a) to find the real and imaginary parts of

$$\frac{1 + \cos 2\theta + i \sin 2\theta}{1 + \cos 2\theta - i \sin 2\theta};$$

b) to show that if n is a positive integer then

$$\left(\frac{1 + \sin \phi + i \cos \phi}{1 + \sin \phi - i \cos \phi} \right)^n = \cos n \left(\frac{\pi}{2} - \phi \right) + i \sin n \left(\frac{\pi}{2} - \phi \right).$$

49. [X] For $n > 1$, let $\omega_1, \omega_2, \dots, \omega_n$ be the n distinct n th roots of 1 and let A_k be the point on the Argand diagram which represents ω_k . Let P represent any point z on the unit circle, and let PA_k denote the distance from P to A_k .

a) Prove that $(PA_k)^2 = (z - \omega_k)(\bar{z} - \bar{\omega}_k)$.

- b) Deduce that $\sum_{k=1}^n (PA_k)^2 = 2n$.
- c) Now let P represent the point x on the real axis, $-1 < x < 1$, prove that

$$\prod_{k=1}^n PA_k = 1 - x^n.$$

Problems 3.8 : Trigonometric applications of complex numbers

50. [R][V] Using De Moivre's theorem and the binomial theorem, prove the identity
- $$\cos 3\theta = 4\cos^3 \theta - 3\cos \theta.$$
51. [R] a) Use De Moivre's Theorem to express $\cos 6\theta$ and $\sin 6\theta$ in terms $\cos \theta$ and $\sin \theta$.
b) Write $\cos 6\theta$ in terms of $\cos \theta$ only
52. [H] Express $\cos 7\theta$ and $\sin 7\theta$ in terms of powers of $\cos \theta$ and $\sin \theta$.
53. [R] a) Derive a formula for $\cos \theta$ in terms of $e^{i\theta}$ and $e^{-i\theta}$.
b) Deduce a formula for $\cos^6 \theta$ in terms of $\cos k\theta$, $1 \leq k \leq 6$.
c) Show that $\int_0^{\frac{\pi}{2}} \cos^6 \theta \, d\theta = \frac{5\pi}{32}$.
54. [R][V] Express $\sin^5 \theta$ and $\cos^4 \theta$ in terms of sines or cosines of multiples of θ , and hence find their integrals.
55. [X] a) Use De Moivre's Theorem to express $\cos 5\theta$ as a polynomial $p(x)$ in $x = \cos \theta$.
b) Put $\theta = 36^\circ = \frac{\pi}{5}$ and show that $x = \cos \frac{\pi}{5}$ is a root of $P(x) = 16x^5 - 20x^3 + 5x + 1$.
c) Check that $P(x) = (x+1)(4x^2 - 2x - 1)^2$.
d) What are the 5 roots of $P(x)$? Give full reasons for your answer.
e) Deduce that $\cos \frac{\pi}{5} + \cos \frac{3\pi}{5} + \cos \frac{7\pi}{5} + \cos \frac{9\pi}{5} = 1$ and $\cos \frac{\pi}{5} \cos \frac{3\pi}{5} \cos \frac{7\pi}{5} \cos \frac{9\pi}{5} = \frac{1}{16}$.
56. [X] Let $\omega_1, \omega_2, \dots, \omega_n$ be the n distinct n th roots of unity ($n \geq 1$). Show that if k is an integer then

$$\omega_1^k + \omega_2^k + \dots + \omega_n^k$$

equals 0 or n . Find the values of k for which the sum is n .

Hint: Write the roots in polar form and sum the resulting geometric progression. See Example 5 of Section 3.8.

57. [X] Show that if θ is not a multiple of 2π , then the imaginary part of

$$\frac{1 - e^{i(n+1)\theta}}{1 - e^{i\theta}} \quad \text{is} \quad \frac{\sin\left(\frac{1}{2}(n+1)\theta\right) \sin\left(\frac{1}{2}n\theta\right)}{\sin \frac{1}{2}\theta}.$$

Hint. See Example 5 of Section 3.8.

58. [X] Find the sum of

$$\sin \theta + \sin 2\theta + \cdots + \sin n\theta.$$

Hint. See previous exercise.

59. [X] a) Calculate the sum of the series

$$S = e^{i\theta} - \frac{e^{3i\theta}}{3^2} + \frac{e^{5i\theta}}{3^4} - \frac{e^{7i\theta}}{3^6} + \cdots$$

- b) Hence show that

$$\sin \theta - \frac{\sin 3\theta}{3^2} + \frac{\sin 5\theta}{3^4} - \frac{\sin 7\theta}{3^6} + \cdots = \frac{72 \sin \theta}{82 + 18 \cos 2\theta}.$$

Problems 3.9 : Geometric applications of complex numbers

60. [R][V] Sketch the set of points on the complex plane corresponding to each of the following:

- | | |
|---|--|
| a) $ z - i \leq 2$, | b) $ z - i \leq 2$ or $-\frac{\pi}{3} \leq \text{Arg}(z - i) \leq \frac{2\pi}{3}$, |
| c) $ z \geq 2$ and $ \text{Im}(z) \leq 3$, | d) $\text{Re}(z) \geq \text{Im}(z)$, |
| e) $ z - i = z + i $, | f) $ z - 1 - i < 1$ and $-\frac{\pi}{4} < \text{Arg}(z - 1 - i) \leq \frac{\pi}{2}$, |
| g) $ z - i = 2 z + i $, | h) [X] $ z - i + z + i = 6$. |

61. [R] Sketch the following on two carefully labelled Argand diagrams.

- | |
|---|
| a) $S_1 = \{z : \text{Re}(z) \geq 3 \text{Im}(z) \text{ and } z - (3 + i) > 2\}$, |
| b) $S_2 = \left\{z : z - i < z + i \text{ and } -\frac{\pi}{6} \leq \text{Arg}(z - i) \leq \frac{\pi}{6}\right\}$. |

62. [R] Let $S = \{z \in \mathbb{C} : \text{Im}(z) > -4 \text{ and } |z - 1 - i| \geq 3\}$.

- | |
|---|
| a) Sketch S on a carefully labelled Argand diagram. |
| b) Does $2 + 4i$ belong to S ? |

63. [R] Let z be a complex number. Prove that $|z - \text{Re}(z)| \leq |z - x|$ for all real numbers x . Draw a sketch to illustrate the result.

64. [H] Let z, w be complex numbers.

- | |
|--|
| a) Sketch the subset of the complex plane defined by $w = e^{i\alpha}$ for $-\pi < \alpha \leq \pi$. |
| b) Given that $\text{Arg}(z) = \theta$, prove that $ z - e^{i\theta} \leq z - e^{i\alpha} $ for all $\alpha \in \mathbb{R}$. |
| c) Give a geometric interpretation of the result in part b). |

Problems 3.10 : Complex polynomials

65. [R] Use the remainder theorem to find the following remainders when.
- $2 + 3z - z^2 + 6z^3$ is divided by $z - 5$,
 - $1 - 6z + 5z^2 - 8z^3 + 2z^4$ is divided by $z + 2$,
 - $3z + 2z^2 + z^3$ is divided by $z - 1 - i$.
66. [R][V] Use the remainder theorem and the factor theorem to show that $z - 2$ is a factor of $p(z) = 30 - 17z - 3z^2 + 2z^3$. Then divide p by $z - 2$ and hence find all linear factors of p .
67. [R] Use the method of the previous question to show that $z - 1$ and $z + 2$ are factors of $p(z) = -8 - 6z + 7z^2 + 6z^3 + z^4$. Then find all linear factors of p .
68. [R] Find all linear factors of
- $z^5 + i$,
 - $z^6 + 8$.
69. [H]
 - Factorise $x^8 - 1$ into real linear and real quadratic factors.
 - Repeat for $x^6 + 8$.
70. [R][V] Factorise $z^4 + 4$ over the rational numbers.
71. [R] Factorise the polynomial $z^4 + i$ into complex linear factors.
72. [R][V]
 - Solve the equation $z^6 = -1$ where $z \in \mathbb{C}$.
 - Plot your solutions from part a) as points in the Argand diagram.
 - Write $z^6 + 1$ as a product of complex linear factors.
 - Write $z^6 + 1$ as a product of real quadratic factors.
73. [H] Let $p(z) = z^6 + z^4 + z^2 + 1$.
- By using the identity, $(z^2 - 1)p(z) = z^8 - 1$, find all 6 complex roots of $p(z)$ in polar form.
 - Hence factorise $p(z)$ into complex linear factors.
 - Factorise $p(z)$ into a product of 3 real irreducible quadratic polynomials.
74. [H] Let $p(z) = 1 + z + z^2 + z^3 + z^4$.
- Solve $z^5 - 1 = 0$ and hence factorise $p(z)$ into linear factors.
 - Find all linear and quadratic factors with real coefficients for $p(z)$.
 - Divide the equation $p(z) = 0$ by z^2 . Let $x = z + \frac{1}{z}$ and deduce that $x^2 + x - 1 = 0$.
 - Deduce that

$$\cos \frac{2\pi}{5} = \frac{-1 + \sqrt{5}}{4} \quad \text{and} \quad \cos \frac{4\pi}{5} = \frac{-1 - \sqrt{5}}{4}$$

75. [H] Consider $f(t) = t^6 + t^5 - t^4 - 5t^3 - 6t^2 - 6t - 4$. Given that $-1 + i$ is a root of f and that f also has two real integer roots,

- a) factorise f into complex linear factors,
- b) factorise f into linear and quadratic factors with real coefficients.

76. [H][V] Let $f(z) = z^5 - 2z^4 + 2z^3 - 5z^2 + 10z - 10$. Given that $1 + i$ is a root, find all solutions to $f(z) = 0$.

77. [X] a) By considering $z^9 - 1$ as a difference of two cubes, write

$$1 + z + z^2 + z^3 + z^4 + z^5 + z^6 + z^7 + z^8$$

as a product of two real factors one of which is a quadratic.

- b) Solve $z^9 - 1 = 0$ and hence write down the six solutions of $z^6 + z^3 + 1 = 0$.
- c) By letting $y = z + \frac{1}{z}$ and dividing $z^6 + z^3 + 1 = 0$ by z^3 , deduce that

$$\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9} = 0.$$

78. [X] Let $p(z) = 3z - z^3 + 5z^4 + z^6$. You are told that the six roots of p , say $\alpha_1, \dots, \alpha_6$, are distinct.

- a) Prove that at least two of these roots are real.
- b) Show that

$$\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4 + \alpha_5 + \alpha_6 = 0.$$

- c) Hence or otherwise, show that there is at least one root with positive real part, and at least one root with negative real part.
- d) Show that if $|z| > 3$, then

$$|3z - z^3 + 5z^4| < |z^6|.$$

- e) Hence or otherwise show that for $j = 1, \dots, 6$, $|\alpha_j| \leq 3$.

79. [X] Cardan (approximately 1545) gave a formula for the roots of the cubic equation

$$x^3 + ax + b = 0$$

in the form $x = u - v$, where

$$u = \left\{ -\frac{b}{2} + \sqrt{\left(\frac{b}{2}\right)^2 + \frac{a^3}{27}} \right\}^{\frac{1}{3}} \quad v = \left\{ \frac{b}{2} + \sqrt{\left(\frac{b}{2}\right)^2 + \frac{a^3}{27}} \right\}^{\frac{1}{3}}$$

and where the cube roots u and v must be selected to satisfy $uv = \frac{a}{3}$. It can be shown that there are three pairs of values of u and v which satisfy the above conditions.

If $a \neq 0$ a simpler way of writing Cardan's formula is that the three roots of the cubic are of the form

$$x = u - \frac{a}{3u},$$

where u satisfies

$$u^3 = -\frac{b}{2} + \sqrt{\left(\frac{b}{2}\right)^2 + \frac{a^3}{27}}.$$

- a) Use the simpler version of Cardan's formula to find all three roots of $x^3 - 6x + 4 = 0$. (Note that complex numbers are used in the calculation even though all three roots are real).
- b) Use the fact that one root of the cubic $x^3 - 6x + 4$ is 2 to factor the cubic as $(x-2)q(x)$ where $q(x)$ is a quadratic. Hence find all roots of the cubic. Hence deduce that

$$\cos \frac{5\pi}{12} = \frac{-1 + \sqrt{3}}{2\sqrt{2}} \quad \text{and} \quad \cos \frac{11\pi}{12} = -\frac{1 + \sqrt{3}}{2\sqrt{2}}.$$

80. [X] a) Show that $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$.
- b) Make the substitution $x = k \cos \theta$ in the equation $x^3 - ax - b = 0$ and show that the left hand side will become $\cos 3\theta$ if we choose k such that $k^2 = \frac{4}{3}a$ (assume $a > 0$!).
- c) With the above choice of k show that the cubic equation can then be written as

$$\cos 3\theta = \frac{4b}{k^3} \quad \text{provided} \quad -1 \leq \frac{4b}{k^3} \leq 1.$$

- d) Use the method outlined above to find the three real roots of $x^3 - 6x - 4 = 0$.

81. [X] Consider the polynomial

$$p(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where the all coefficients $a_0, a_1, \dots, a_n$ are integers. Show that, if r/s is a rational root of p for which the integers r and s have no common factors, then r is a divisor of a_0 and s is a divisor of a_n . Hence find all rational roots of

- a) $-5 + 3x - x^2 + 3x^3$, b) $5 - 22x + x^2 + 28x^3$, c) $4 - x - 100x^2 + 25x^3$.

82. [X] Let M, N be positive integers. If $x^M(1-x)^N$ is divided by $(1+x^2)$, and the remainder is $ax + b$, show that $a = (\sqrt{2})^N \sin \frac{(2M-N)\pi}{4}$ and $b = (\sqrt{2})^N \cos \frac{(2M-N)\pi}{4}$.

Problems 3.11 : Appendix: A note on proof by induction

83. [R][V] Prove by mathematical induction that for all positive integers n ,

$$1 \cdot 2 + 2 \cdot 3 + \cdots + n(n+1) = \frac{1}{3} n(n+1)(n+2).$$

84. [R] Prove that, for all integers $n \geq 1$,

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{1}{6} n(n+1)(2n+1).$$

85. [R] Prove that, for all integers $n \geq 1$,

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \frac{1}{4} n^2(n+1)^2.$$

86. [H] Prove that, for all integers $n \geq 1$,

$$1^4 + 2^4 + 3^4 + \cdots + n^4 = \frac{1}{30} n(n+1)(6n^3 + 9n^2 + n - 1).$$

87. [X] Prove, by induction, that the sum to k terms of

$$1^2 - 3^2 + 5^2 - 7^2 + 9^2 - 11^2 + \cdots$$

is $-8n^2$ if $k = 2n$ and $8n^2 + 8n + 1$ if $k = 2n + 1$.

88. [X] A sequence of real numbers $\{a_n\}_{n=1}^\infty$ is defined recursively by

$$a_1 = 1, \quad a_{n+1} = \sum_{j=1}^n \frac{a_j}{2^j} \quad \text{for } n \geq 1.$$

Use the second Principle of Induction to prove that $a_n \leq 1$ for all $n \geq 1$.

89. [X] Suppose we draw n lines in the plane with no three lines concurrent and no two lines parallel. Let s_n denote the number of regions into which these lines divide the plane. For example, $s_1 = 2, s_2 = 4, s_3 = 7, \dots$. Prove that $s_{n+1} = s_n + (n+1)$. Deduce by induction that $s_n = \frac{1}{2}n(n+1) + 1$.

Problems 3.13 : Complex numbers and Maple

90. [M] Write a Maple command (or commands) to evaluate the complex number $(\sqrt{2} + 7i)^{13}$ in Cartesian or “ $a + ib$ ” form.
91. [M] Use Maple to evaluate $(5 + i)^4(239 - i)$. Then use de Moivre’s Theorem to show that $\frac{\pi}{4} = 4 \cot^{-1} 5 - \cot^{-1} 239$.

*“I can do Addition,” she said, “if you give me time
— but I can’t do Subtraction under ANY circumstances!”*
Lewis Carroll, Through the Looking Glass.

Chapter 4

LINEAR EQUATIONS AND MATRICES

*One glass lemonade (Why ca'n't you drink water, like me?)
three sandwiches (They never put in half mustard enough.
I told the young woman so, to her face ...)
and seven biscuits. Total one-and-twopence.
Lewis Carroll, A Tangled Tale.*

Linear equations and matrices are very important because they are used as mathematical models in virtually all areas in which mathematics is applied in the modern world and because they also appear at the heart of computational algorithms for solving a vast array of quite diverse mathematical problems.

The processes described in this chapter are appropriate for the general theory of handling systems of linear equations. They work well for most small-scale examples.

An understanding of linear equations and matrices is essential for much of the later work in algebra in this book so you must master the material in this chapter.

4.1 Introduction to linear equations

You are already familiar with solving linear equations in one variable such as $2x = 6$, and system of two simultaneous linear equations in two variables such as $\begin{cases} 2x + y = 3 \\ 3x - y = 7 \end{cases}$.

A linear equation in two variables such as $x - 2y = 1$ has infinitely many solutions, since once y is specified, then x is determined and conversely. It is convenient to write the solutions to such an equation parametrically, by introducing a parameter λ .

Example 1. Express solutions of $x - 2y = 1$ parametrically.

SOLUTION. We put $y = \lambda$ then $x = 2\lambda + 1$, so we can write

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 2\lambda + 1 \\ \lambda \end{pmatrix}, \quad \lambda \in \mathbb{R},$$

as the parametric solution to $x - 2y = 1$. Notice that this is not unique, since we could also have put $x = \lambda$ and solved for y . $\diamond$

Two linear simultaneous equations in two unknowns may have:

(1) No solution, for example $\begin{cases} x + y = 3 \\ x + y = 5 \end{cases}$.

(2) Unique solution, for example $\begin{cases} 2x + y = 3 \\ 3x - y = 7 \end{cases}$.

(3) Infinitely many solutions, for example $\begin{cases} x - 2y = 1 \\ 2x - 4y = 2 \end{cases}$.

Since these equations correspond to straight lines in the plane, case (1) represents two distinct parallel lines; case (2), two non-parallel lines and in case (3), the two lines are the same.

Now consider some examples of two equations in *three* unknowns. Assume that none of the equations is of the form $0x_1 + 0x_2 + 0x_3 = b$. The solutions of such simultaneous equations can be interpreted as the points of intersection (if any) of two planes in $\mathbb{R}^3$. From geometry, we expect that two planes either

1. intersect along a line, or
2. are parallel and do not intersect, or
3. are parallel with all points in common, (that is, they are the same plane).

We convert each system to another *equivalent system*, i.e. a system which has the same solution set, until we get one which gives us a simple form of the solution set.

Example 2. Find the solution of the system of equations

$$x_1 + x_2 + x_3 = 5 \tag{1}$$

$$3x_1 + 4x_2 + 7x_3 = 20 \tag{2}$$

SOLUTION. To solve this pair, we first eliminate x_1 from equation (2) by subtracting 3 times equation (1) from equation (2). The new equation (2) is then

$$x_2 + 4x_3 = 5 \tag{2'}$$

Therefore the original system is equivalent to

$$x_1 + x_2 + x_3 = 5 \tag{1}$$

$$x_2 + 4x_3 = 5 \tag{2'}$$

Equation (2') has infinitely many solutions which can be represented using a parameter. Put $x_3 = \lambda$, then

$$x_2 = 5 - 4\lambda.$$

Substituting this back to equation (1), we have

$$x_1 + (5 - 4\lambda) + \lambda = 5,$$

so $x_1 = 3\lambda.$

We write the solutions as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3\lambda \\ 5 - 4\lambda \\ \lambda \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}, \quad \lambda \in \mathbb{R}.$$

The geometric interpretation of this example is that equations (1) and (2) each represents a plane, and the solutions of system represent the points of intersection of the two planes. In this particular example, the intersection of the two planes is a line. This line passes through the point $(0, 5, 0)$ and is parallel to the vector $\begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}$. $\diamond$

Example 3. Find the solution of the system

$$2x_1 - 3x_2 + x_3 = 20 \tag{1}$$

$$4x_1 - 6x_2 + 2x_3 = 34. \tag{2}$$

SOLUTION. As usual, we eliminate x_1 from equation (2). On subtracting $2 \times$ equation (1) from equation (2), we obtain the new equation (2') as

$$0x_1 + 0x_2 + 0x_3 = -6, \tag{2'}$$

which clearly has no solution. Thus the system has no solution. The geometric interpretation is that the planes represented by equations (1) and (2) are parallel and do not intersect. $\diamond$

Example 4. Find the solutions of the system

$$x_1 - 3x_2 + x_3 = 20 \tag{1}$$

$$2x_1 - 6x_2 + 2x_3 = 40. \tag{2}$$

SOLUTION. Eliminating x_1 from equation (2) gives the new equation (2') as

$$0x_1 + 0x_2 + 0x_3 = 0. \tag{2'}$$

The given system is thus equivalent to

$$x_1 - 3x_2 + x_3 = 20 \tag{1}$$

$$0x_1 + 0x_2 + 0x_3 = 0 \tag{2'}$$

Thus the solutions to (1) and (2) are just the solutions to (1) and conversely. There are infinitely many solutions, but here two of the unknowns need to be specified in order to obtain the third. Thus **two** parameters λ_1, λ_2 need to be introduced. Put $x_2 = \lambda_1$ and $x_3 = \lambda_2$, then

$$x_1 = 20 + 3\lambda_1 - \lambda_2,$$

and we write the solutions as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 20 + 3\lambda_1 - \lambda_2 \\ \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} 20 \\ 0 \\ 0 \end{pmatrix} + \lambda_1 \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \quad \text{for } \lambda_1, \lambda_2 \in \mathbb{R}.$$

Hence the system has an infinite number of solutions. The geometric interpretation of this example is that equations (1) and (2) represent the same plane and the ‘solution’ is just a parametric vector form of expression for that plane. $\diamond$

Now, what about a system of more than two equations in two or three variables? Let us exclude the cases that one or more of the equations in the system is of the form $0x_1 + 0x_2 = b$ or $0x_1 + 0x_2 + 0x_3 = b$. In such cases, the system is either has no solutions or is equivalent to a system of two or less equations.

For a system of three equations in two variables, we can interpret the three equations as three lines in a plane. Geometrically, we have three cases.

1. The three lines are concurrent, so the system has a unique solution.
2. The three lines do not have a point in common, so the system has no solution. (See Figure 1 for cases in which the three lines are distinct.)
3. The three equations represent the same line. The system has infinitely many solutions.

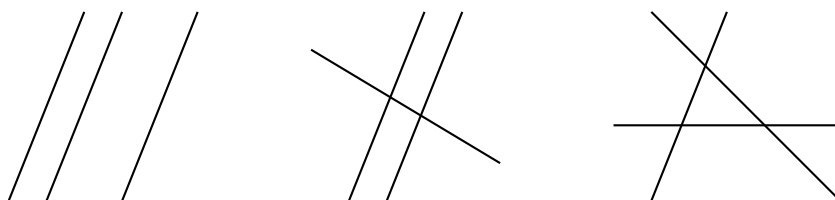


Figure 1: Three Lines with no Point in Common.

For a system of three equations in three variables, the three equations represent three planes in three dimensional space. Geometrically, we have four cases.

1. The three planes intersect at one point, so the system has a unique solution.
2. The three planes do not have a point in common, so the system has no solution. (See Figure 2 for cases in which the three planes are distinct.)
3. The three planes intersect in a line, so there are infinitely many solutions.
4. The three equations represent the same plane. Again, the system has infinitely many solutions.

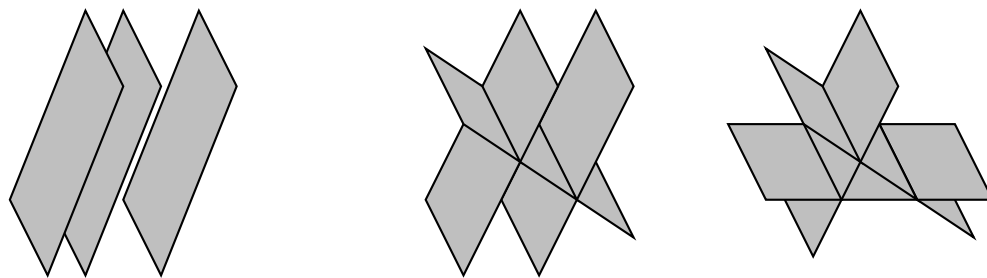


Figure 2: Three Planes with no Point in Common.

4.2 Systems of linear equations and matrix notation

We will now consider the general system of simultaneous linear equations and develop a systematic method of solution.

Definition 1. A **system** of m **linear equations** in n variables is a set of m linear equations in n variables which must be simultaneously satisfied. Such a system is of the form

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = b_1 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = b_2 \\ \vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots \qquad \qquad \qquad \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = b_m \end{cases} \quad (*)$$

Note carefully the position of the subscripts in (*): a_{ij} is the coefficient of the variable x_j in the i th equation, that is, the **equation index is first** and the **variable index is second**.

A **solution** to a system of equations is the set of values of the variables which simultaneously satisfy all the equations. We normally write a solution in form of a column vector in $\mathbb{R}^n$. For

instance, if $x_1 = \alpha_1, \dots, x_n = \alpha_n$ satisfy the equations simultaneously, the vector $\begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix}$ is a

solution. A system of equations is said to be **consistent** if it has at least one solution. Otherwise, the system is said to be **inconsistent**.

In particular, the system is called homogeneous when all the b_i 's are zero.

Definition 2. The system (*) is said to be **homogeneous** if $b_1 = 0, \dots, b_m = 0$.

Example 1.

$$\begin{cases} 2x_1 + 3x_2 - x_3 + x_4 = 6 \\ x_1 - 4x_2 + 5x_3 = 2 \\ x_1 - 6x_3 = 5 \\ x_1 + 2x_2 - x_4 = 7 \end{cases}$$

is a system of 4 linear equations in 4 unknowns. ◇

The variables $x_1, x_2, \dots, x_n$ in the system (*) are really just place marker for the coefficients. Hence it is convenient to adopt a shorthand notation for (*) which records only the coefficients. This is called the **augmented matrix** for the system (*):

$$\left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right).$$

Example 2. The system in Example 1 has augmented matrix:

$$\left(\begin{array}{cccc|c} 2 & 3 & -1 & 1 & 6 \\ 1 & -4 & 5 & 0 & 2 \\ 1 & 0 & -6 & 0 & 5 \\ 1 & 2 & 0 & -1 & 7 \end{array} \right).$$

NOTE.

1. A rectangular array of numbers is called a **matrix**. So the above matrix form of the system of linear equations is called the corresponding augmented matrix. The vertical line is not part of the matrix. It is used to separate the right hand side of the system from the coefficients on the left hand side.
2. We call $(a_{i1} \ a_{i2} \ \cdots \ a_{in} \mid b_i)$ the i th *row* of the augmented matrix which is denoted by R_i . The entries in R_i of the matrix correspond to the coefficients and the right hand side of the i -th equation.

3. We call $\begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{pmatrix}$ the j th *column* which is denoted by C_j . The entries in C_j correspond to the coefficients of x_j when $1 \leq j \leq n$. The last column C_{n+1} is the right hand side of the system.

We denote the matrix of coefficients by A , the column of the right hand side as the vector $\mathbf{b}$ and the column vector of the variables as $\mathbf{x}$. That is

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}, \quad \text{and} \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

Using such notation, the augmented matrix can be abbreviated by $(A|\mathbf{b})^1$.

The drawback of using augmented matrix is the omission of the variables. We introduce the notion of the **matrix equation** form. The system of equations is represented by the matrix equation $A\mathbf{x} = \mathbf{b}$. The matrix A is called the **coefficient matrix** and $\mathbf{x}$ is called the **unknown vector**.

When we write $A\mathbf{x}$, we mean the vector

$$\begin{pmatrix} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n \end{pmatrix}.$$

¹Formally speaking, $(A|\mathbf{b})$ should be $\left(\begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \mid \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} \right)$, but for simplicity we denote the augmented matrix $\left(\begin{array}{ccc|c} a_{11} & \cdots & a_{1n} & b_1 \\ \vdots & \ddots & \vdots & \vdots \\ a_{m1} & \cdots & a_{mn} & b_m \end{array} \right)$ by $(A|\mathbf{b})$.

This is our first meeting with “matrix multiplication”, which we shall investigate in more detail in Chapter 5.

Finally we can also write the system of linear equations in **vector equation** or **vector form**.

$$x_1 \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix} + x_2 \begin{pmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{pmatrix} + \cdots + x_n \begin{pmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}.$$

This vector equation can be written more concisely as

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n = \mathbf{b},$$

where

$$\mathbf{a}_j = \begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{mj} \end{pmatrix} \quad \text{for } 1 \leq j \leq n \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}.$$

Notice that the vector $\mathbf{a}_j$ contains the coefficients of the variable x_j in the system of linear equations. The vector $\mathbf{b}$ is called the right-hand-side vector.

This vector form of the system of equations shows that solving the system corresponds to expressing the right-hand-side vector $\mathbf{b}$ as a linear combination of the vectors $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$.

Example 3. The system of linear equations

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 1 \\ 4x_1 + 5x_2 + 6x_3 = -1 \\ 7x_1 - 5x_2 - 9x_3 = 0 \end{cases}$$

may be written in vector form as

$$x_1 \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix} + x_2 \begin{pmatrix} 2 \\ 5 \\ -5 \end{pmatrix} + x_3 \begin{pmatrix} 3 \\ 6 \\ -9 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix};$$

or represented by the matrix equation $A\mathbf{x} = \mathbf{b}$ with

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & -5 & -9 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix};$$

or represented by the augmented matrix

$$(A|\mathbf{b}) = \left(\begin{array}{ccc|c} 1 & 2 & 3 & 1 \\ 4 & 5 & 6 & -1 \\ 7 & -5 & -9 & 0 \end{array} \right).$$

◇

Example 4. Write the system of linear equations corresponding to the augmented matrix

$$\left(\begin{array}{cccc|c} 1 & 3 & 6 & 7 & -2 \\ -2 & 0 & 5 & -4 & 3 \\ 7 & 0 & 0 & -5 & -10 \end{array} \right).$$

SOLUTION. Because there are 4 columns before the “|”, the system has 4 variables, which we will call x_1, x_2, x_3 and x_4 (although other names are acceptable). Because $(A|\mathbf{b})$ has 3 rows, the system has 3 equations. Reading off the coefficients of row 1, row 2, and row 3 in turn, we obtain the system

$$\begin{cases} x_1 + 3x_2 - 6x_3 + 7x_4 = -2 \\ -2x_1 + 5x_3 - 4x_4 = 3 \\ 7x_1 - 5x_4 = -10 \end{cases}$$

Although it is conventional to omit variables with zero coefficients when writing a system of linear equations, it is essential in matrix notation that every row contains exactly the same number of entries and that every column also contains exactly the same number of entries so 0s must *not* be omitted from matrices or vectors. $\diamond$

Example 5. Write the system of linear equations corresponding to the matrix equation $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{pmatrix} 3 & -2 & 6 & 7 & -8 \\ 5 & 3 & -2 & -7 & 4 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 7 \\ -3 \end{pmatrix}.$$

SOLUTION. The matrix A has 5 columns, so there are 5 variables, which we call x_1, x_2, x_3, x_4, x_5 . Also, A has 2 rows, so there are 2 equations.

Reading off the coefficients from the two rows, we obtain the system

$$\begin{cases} 3x_1 - 2x_2 + 6x_3 + 7x_4 - 8x_5 = 7 \\ 5x_1 + 3x_2 - 2x_3 - 7x_4 + 4x_5 = -3 \end{cases}$$

$\diamond$

Example 6. Write down the $A\mathbf{x} = \mathbf{b}$ form of the system of linear equations corresponding to the vector equation

$$x_1 \begin{pmatrix} 2 \\ -3 \\ 4 \end{pmatrix} + x_2 \begin{pmatrix} 5 \\ 0 \\ -1 \end{pmatrix} + x_3 \begin{pmatrix} -4 \\ 8 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 6 \\ 7 \\ -2 \end{pmatrix} = \begin{pmatrix} 0 \\ -6 \\ 2 \end{pmatrix}.$$

SOLUTION. The vector associated with x_j is the j th column of the coefficient matrix so as a matrix equation:

$$\begin{pmatrix} 2 & 5 & -4 & 6 \\ -3 & 0 & 8 & 7 \\ 4 & -1 & 0 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 0 \\ -6 \\ 2 \end{pmatrix}.$$

$\diamond$

4.3 Elementary row operations

When you solve a pair a simultaneous linear equations, you add (or subtract) a multiple of one equation from the other equation. Each equation in a system of linear equations is represented by one row of the corresponding augmented matrix. This means that the operation of adding a multiple of one equation from another equation can be carried out on the augmented matrix by adding a multiple of one row from another row. This is an example of what we call a **row operation**.

In this section we introduce the three **elementary row operations**. These operations are applied to the augmented matrix in order to find the solution to a system of m linear equations in n variables. Each of these operations gives us a new augmented matrix which is **equivalent** to the old augmented matrix in the following sense: the system represented by the new matrix has exactly the same solution set as the system represented by the old matrix.

We also show how to record row operations to make it easier to check your calculations.

4.3.1 Interchange of equations

If two equations are interchanged, then the set of solutions of the new system is clearly the same as that of the original system. The corresponding operation for an augmented matrix is to interchange two complete rows.

Example 1. Consider the system

$$\begin{array}{lcl} (1) & 2x_2 + 3x_3 & = 5 \\ (2) & -x_1 + 3x_2 + x_3 & = 6 \\ (3) & 2x_1 + 4x_2 + 7x_3 & = 8 \end{array} \quad \text{or} \quad \begin{array}{l} R_1 \\ R_2 \\ R_3 \end{array} \left(\begin{array}{ccc|c} 0 & 2 & 3 & 5 \\ -1 & 3 & 1 & 6 \\ 2 & 4 & 7 & 8 \end{array} \right).$$

Interchanging equations (1) and (2), or rows R_1 and R_2 , yields

$$\begin{array}{lcl} (1') = (2) & -x_1 + 3x_2 + x_3 & = 6 \\ (2') = (1) & 2x_2 + 3x_3 & = 5 \\ (3) & 2x_1 + 4x_2 + 7x_3 & = 8 \end{array} \quad \begin{array}{l} \text{New } R_1 = \text{old } R_2 \\ \text{New } R_2 = \text{old } R_1 \\ R_3 \end{array} \left(\begin{array}{ccc|c} -1 & 3 & 1 & 6 \\ 0 & 2 & 3 & 5 \\ 2 & 4 & 7 & 8 \end{array} \right).$$

This should be recorded as

$$\left(\begin{array}{ccc|c} 0 & 2 & 3 & 5 \\ -1 & 3 & 1 & 6 \\ 2 & 4 & 7 & 8 \end{array} \right) \xrightarrow{R_2 \leftrightarrow R_1} \left(\begin{array}{ccc|c} -1 & 3 & 1 & 6 \\ 0 & 2 & 3 & 5 \\ 2 & 4 & 7 & 8 \end{array} \right).$$

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4.3.2 Adding a multiple of one equation to another

The operation of adding a multiple of one equation to another does not change the solution set of the system of equations. For example, consider the system of equations

$$\begin{cases} a_1x_1 + \cdots + a_nx_n = b \\ \alpha_1x_1 + \cdots + \alpha_nx_n = \beta \end{cases} \quad (1)$$

If we add λ times equation (1) to equation (2), we obtain the system

$$\begin{cases} a_1x_1 + \cdots + a_nx_n = b \\ (\alpha_1 + \lambda a_1)x_1 + \cdots + (\alpha_n + \lambda a_n)x_n = \beta + \lambda b \end{cases} \quad (2)$$

Proposition 1. The solution of the system (1) is the same as the solution of the system (2).

Proof. Let $\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ be a solution of system (1). Then we have

$$a_1x_1 + \cdots + a_nx_n = b$$

$$\text{and } (\alpha_1 + \lambda a_1)x_1 + \cdots + (\alpha_n + \lambda a_n)x_n = (\alpha_1x_1 + \cdots + \alpha_nx_n) + \lambda(a_1x_1 + \cdots + a_nx_n) \\ = \beta + \lambda b,$$

and so system (2) is satisfied.

Conversely, let $\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$ be a solution of system (2). Then, we have

$$a_1x_1 + \cdots + a_nx_n = b$$

$$\text{and } \alpha_1x_1 + \cdots + \alpha_nx_n = (\alpha_1 + \lambda a_1)x_1 + \cdots + (\alpha_n + \lambda a_n)x_n - \lambda(a_1x_1 + \cdots + a_nx_n) \\ = (\beta + \lambda b) - \lambda b = \beta,$$

and so system (1) is satisfied. $\square$

In the augmented-matrix notation, the addition of a scalar multiple of one equation to another is equivalent to addition of a scalar multiple of one complete row to another row.

Example 2. Consider the system

$$\begin{array}{lcl} (1) & x_1 & + \quad 2x_2 & + \quad 3x_3 & = & 5 \\ (2) & -x_1 & - \quad 4x_2 & + \quad x_3 & = & 6 \\ (3) & 2x_1 & + \quad 10x_2 & + \quad 7x_3 & = & 8 \end{array} \quad \text{or} \quad \begin{array}{l} R_1 \left(\begin{array}{ccc|c} 1 & 2 & 3 & 5 \end{array} \right) \\ R_2 \left(\begin{array}{ccc|c} -1 & -4 & 1 & 6 \end{array} \right) \\ R_3 \left(\begin{array}{ccc|c} 2 & 10 & 7 & 8 \end{array} \right) \end{array}$$

On adding equation (1) to equation (2) in the above system, we obtain

$$\begin{array}{lcl} (1) & x_1 & + \quad 2x_2 & + \quad 3x_3 & = & 5 \\ (2') & & - \quad 2x_2 & + \quad 4x_3 & = & 11 \\ (3) & 2x_1 & + \quad 10x_2 & + \quad 7x_3 & = & 8 \end{array} \quad \text{or} \quad \begin{array}{l} R_1 \left(\begin{array}{ccc|c} 1 & 2 & 3 & 5 \end{array} \right) \\ R_2 \left(\begin{array}{ccc|c} 0 & -2 & 4 & 11 \end{array} \right) \\ R_3 \left(\begin{array}{ccc|c} 2 & 10 & 7 & 8 \end{array} \right) \end{array},$$

where the new $R_2 = \text{old } R_2 + R_1$.

This should be recorded as

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ -1 & -4 & 1 & 6 \\ 2 & 10 & 7 & 8 \end{array} \right) \xrightarrow{R_2 = R_2 + R_1} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & -2 & 4 & 11 \\ 2 & 10 & 7 & 8 \end{array} \right).$$

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4.3.3 Multiplying an equation by a non-zero number

We can multiply one equation by a non-zero constant without changing the solutions of the system. For example the systems of equations corresponding to the two matrices

$$\left(\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 8 \\ 0 & 0 & 5 & 6 & 9 \\ 0 & 0 & 0 & 7 & 3 \end{array} \right) \quad \text{and} \quad \left(\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 8 \\ 0 & 0 & 1 & \frac{6}{5} & \frac{9}{5} \\ 0 & 0 & 0 & 1 & \frac{3}{7} \end{array} \right)$$

are equivalent.

This should be recorded as

$$\left(\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 8 \\ 0 & 0 & 5 & 6 & 9 \\ 0 & 0 & 0 & 7 & 3 \end{array} \right) \xrightarrow[\begin{array}{l} R_2 = \frac{1}{5}R_2 \\ R_3 = \frac{1}{7}R_3 \end{array}]{\begin{array}{l} R_2 = \frac{1}{5}R_2 \\ R_3 = \frac{1}{7}R_3 \end{array}} \left(\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 8 \\ 0 & 0 & 1 & \frac{6}{5} & \frac{9}{5} \\ 0 & 0 & 0 & 1 & \frac{3}{7} \end{array} \right).$$

Row operations should be done one after the other. Although in the above we have done two row operations at once, there is no difference from performing the operations one after the other. However, we should not make the following **mistake**.

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ -1 & -4 & 1 & 6 \\ 2 & 10 & 7 & 8 \end{array} \right) \xrightarrow[\begin{array}{l} R_2 = R_2 + R_1 \\ R_1 = R_1 + R_2 \end{array}]{\begin{array}{l} R_2 = R_2 + R_1 \\ R_1 = R_1 + R_2 \end{array}} \left(\begin{array}{ccc|c} 0 & -2 & 4 & 11 \\ 0 & -2 & 4 & 11 \\ 2 & 10 & 7 & 8 \end{array} \right)$$

The two systems are not equivalent. If we do want to perform the row operations $R_2 = R_2 + R_1$ then $R_1 = R_1 + R_2$, we should have

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ -1 & -4 & 1 & 6 \\ 2 & 10 & 7 & 8 \end{array} \right) \xrightarrow{R_2 = R_2 + R_1} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 5 \\ 0 & -2 & 4 & 11 \\ 2 & 10 & 7 & 8 \end{array} \right) \xrightarrow{R_1 = R_1 + R_2} \left(\begin{array}{ccc|c} 1 & 0 & 7 & 16 \\ 0 & -2 & 4 & 11 \\ 2 & 10 & 7 & 8 \end{array} \right).$$

4.4 Solving systems of equations

This section describes a process for solving systems of linear equations.

The process consists of two distinct stages.

1. In the first stage, known as **Gaussian elimination**, we use two types of row operation (interchange of rows and subtracting a multiple of one row from another row) to produce an equivalent system in a simpler form which is known as **row-echelon form**. From the row-echelon form we can tell many things about solutions of the system. In particular, we can tell whether the system has no solution, a unique solution or infinitely many solutions.
2. If the system does have solutions, the second stage is to find them. It can be carried out by either of two methods.
 - (a) We can use further row operations to obtain an even simpler form which is called **reduced row-echelon form**. From this form we can read off the solution(s).
 - (b) From the row echelon form, we can read off the value (possibly in terms of parameters) for at least one of the variables. We substitute this value into one of the other equations and get the value for another variable, and so on. This process, which is called **back-substitution**, will be described fully later.

4.4.1 Row-echelon form and reduced row-echelon form

We begin by defining some special forms of matrix.

Definition 1. *In any matrix*

1. a **leading row** is one which is not all zeros,
2. the **leading entry** in a leading row is the first (i.e. leftmost) non-zero entry,
3. a **leading column** is a column which contains the leading entry for some row.

For example, in the following matrix the 1st row is a leading row with leading entry 5 and the 2nd row is a non-leading row. The 2nd column is a leading column but the 1st and 3rd columns are non-leading columns.

$$\begin{pmatrix} 0 & \textcircled{5} & 7 \\ 0 & 0 & 0 \end{pmatrix}$$

Definition 2. *A matrix is said to be in **row-echelon form** if*

1. all leading rows are above all non-leading rows (so any all-zero rows are at the bottom of the matrix), and
2. in every leading row, the leading entry is further to the right than the leading entry in any row higher up in the matrix.

The following are examples of augmented matrices which are in row-echelon form. The circled entries are leading entries. We shall see later that the position of leading entries in a row-echelon form gives important information about solubility of the system.

$$\begin{aligned}
(1) \quad & \left(\begin{array}{ccc|c} \textcircled{2} & 3 & 4 & 11 \\ 0 & \textcircled{-3} & 2 & 7 \\ 0 & 0 & \textcircled{4} & 8 \end{array} \right), & (2) \quad & \left(\begin{array}{cccc|c} \textcircled{-5} & 0 & 4 & 1 & 0 \\ 0 & \textcircled{2} & 3 & 2 & 1 \\ 0 & 0 & 0 & 0 & \textcircled{2} \end{array} \right), \\
(3) \quad & \left(\begin{array}{ccc|c} \textcircled{6} & 2 & 3 & 4 \\ 0 & 0 & \textcircled{7} & 7 \end{array} \right), & (4) \quad & \left(\begin{array}{ccccc|c} \textcircled{-5} & 0 & 2 & 4 & 1 & 0 \\ 0 & 0 & \textcircled{2} & 3 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right), \\
(5) \quad & \left(\begin{array}{ccc|c} \textcircled{2} & 3 & 4 & 11 \\ 0 & \textcircled{-3} & 2 & 7 \\ 0 & 0 & \textcircled{4} & 8 \\ 0 & 0 & 0 & 0 \end{array} \right), & (6) \quad & \left(\begin{array}{cccc|c} \textcircled{-5} & 0 & 4 & 1 & 0 \\ 0 & \textcircled{2} & 3 & 2 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right).
\end{aligned}$$

The following are examples of augmented matrices which are NOT in row-echelon form.

$$\begin{aligned}
(7) \quad & \left(\begin{array}{ccc|c} 0 & \textcircled{6} & 3 & 4 \\ \textcircled{5} & 7 & 0 & 0 \end{array} \right), & (8) \quad & \left(\begin{array}{ccc|c} \textcircled{-5} & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & \textcircled{2} & 3 & 1 \end{array} \right), \\
(9) \quad & \left(\begin{array}{cccc|c} \textcircled{5} & 3 & 5 & -4 & 6 \\ 0 & 0 & \textcircled{6} & -6 & 7 \\ 0 & 0 & \textcircled{7} & 0 & -8 \end{array} \right)
\end{aligned}$$

Example (8) does not satisfy Condition 1 of Definition 2. Examples (7) and (9) do not satisfy Condition 2.

Warning. The definition of row-echelon form varies from one book to another and the terminology in Definition 1 is not universally adopted.

Definition 3. A matrix is said to be in **reduced row-echelon form** if it is in row-echelon form and in addition

3. every leading entry is 1, and
4. every leading entry is the only non-zero entry in its column.

The following are examples of augmented matrices which are in *reduced* row-echelon form.

$$(10) \left(\begin{array}{ccc|c} \textcircled{1} & 0 & 0 & 1 \\ 0 & \textcircled{1} & 0 & 2 \\ 0 & 0 & \textcircled{1} & 3 \end{array} \right),$$

$$(11) \left(\begin{array}{ccc|c} 0 & \textcircled{1} & 2 & 0 \\ 0 & 0 & 0 & \textcircled{1} \\ 0 & 0 & 0 & 0 \end{array} \right),$$

$$(12) \left(\begin{array}{ccccc|c} \textcircled{1} & -1 & 0 & 2 & 0 & 2 \\ 0 & 0 & \textcircled{1} & 1 & 0 & -1 \\ 0 & 0 & 0 & 0 & \textcircled{1} & 6 \end{array} \right).$$

Later in this section, we shall see that we can easily read the solutions of a system with its augmented matrix in reduced row-echelon form. The solution set of a system with its augmented matrix in row-echelon form can be found by a process called *back-substitution*.

4.4.2 Gaussian elimination

In this subsection we shall see how the operations of interchanging two rows and of adding a multiple of a row from another row can be used to take a system of equations $A\mathbf{x} = \mathbf{b}$ and transform it into an equivalent system $U\mathbf{x} = \mathbf{y}$ such that the augmented matrix $(U|\mathbf{y})$ is in row-echelon form. The method we shall describe is called the Gaussian-elimination algorithm.

We now describe the steps in the algorithm and illustrate each step by applying it to the following augmented matrices.

$$\text{a) } (A|\mathbf{b}) = \left(\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 2 & 3 & 1 & 3 \\ 1 & 3 & 3 & 2 \end{array} \right), \quad \text{b) } (A|\mathbf{b}) = \left(\begin{array}{ccccc|c} 0 & 0 & 0 & 2 & 3 & -1 \\ 0 & 3 & -3 & 3 & 3 & -6 \\ 0 & 0 & 0 & 1 & 3 & -2 \\ 0 & 6 & -6 & 6 & 3 & -9 \end{array} \right).$$

Step 1. Select a pivot element.

We shall use the following rule to choose what is called a **pivot element**: start at the left of the matrix and move to the right until you reach a column which is not all zeros, then go down this column and choose the first non-zero entry as the pivot entry. The column containing the pivot entry is called the **pivot column** and the row containing the pivot entry is called the **pivot row**.

NOTE. When solving linear equations on a computer, a more complicated pivot selection rule is generally used in order to minimise “rounding errors”.

In examples (a) and (b), our pivot selection rule selects the circled entries below as pivot entries for the first step of Gaussian elimination.

$$\text{a) } \left(\begin{array}{ccc|c} \textcircled{1} & 2 & 1 & 3 \\ 2 & 3 & 1 & 3 \\ 1 & 3 & 3 & 2 \end{array} \right), \quad \text{b) } \left(\begin{array}{ccccc|c} 0 & 0 & 0 & 2 & 3 & -1 \\ 0 & \textcircled{3} & -3 & 3 & 3 & -6 \\ 0 & 0 & 0 & 1 & 3 & -2 \\ 0 & 6 & -6 & 6 & 3 & -9 \end{array} \right).$$

In example (a), row 1 is the pivot row and column 1 is the pivot column. In example (b), row 2 is the pivot row and column 2 is the pivot column.

Step 2. By a row interchange, swap the pivot row and the top row if necessary.

The rule is that the first pivot row of the augmented matrix must finish as the first row of $(U|\mathbf{y})$. You can achieve this by interchanging row 1 and the pivot row.

In example (a), the pivot row is already row 1, so no row interchange is needed. In example (b), the pivot row is row 2, so rows 1 and 2 of the augmented matrix must be interchanged. This is shown as

$$\text{b) } \left(\begin{array}{ccccc|c} 0 & 0 & 0 & 2 & 3 & -1 \\ 0 & \textcircled{3} & -3 & 3 & 3 & -6 \\ 0 & 0 & 0 & 1 & 3 & -2 \\ 0 & 6 & -6 & 6 & 3 & -9 \end{array} \right) \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccccc|c} 0 & \textcircled{3} & -3 & 3 & 3 & -6 \\ 0 & 0 & 0 & 2 & 3 & -1 \\ 0 & 0 & 0 & 1 & 3 & -2 \\ 0 & 6 & -6 & 6 & 3 & -9 \end{array} \right).$$

Step 3. Eliminate (i.e., reduce to 0) all entries in the pivot column below the pivot element.

We can do this by adding suitable multiples of the pivot row to the *lower* rows. After this process the pivot column is in the correct form for the final row-echelon matrix ($U|\mathbf{y}$).

In example (a), we can use the row operations $R_2 = R_2 + (-2)R_1$ and $R_3 = R_3 + (-1)R_1$ to reduce the pivot column to the required form. In recording the row operations, we can simply write $R_2 = R_2 - 2R_1$ and $R_3 = R_3 - R_1$.

$$\text{a) } \left(\begin{array}{ccc|c} \textcircled{1} & 2 & 1 & 3 \\ 2 & 3 & 1 & 3 \\ 1 & 3 & 3 & 2 \end{array} \right) \xrightarrow{\substack{R_2 = R_2 - 2R_1 \\ R_3 = R_3 - R_1}} \left(\begin{array}{ccc|c} \textcircled{1} & 2 & 1 & 3 \\ 0 & -1 & -1 & -3 \\ 0 & 1 & 2 & -1 \end{array} \right).$$

In example (b), the row operation $R_4 = R_4 - 2R_1$ reduces the pivot column to the required form. This gives

$$\text{b) } \left(\begin{array}{ccccc|c} 0 & \textcircled{3} & -3 & 3 & 3 & -6 \\ 0 & 0 & 0 & 2 & 3 & -1 \\ 0 & 0 & 0 & 1 & 3 & -2 \\ 0 & 6 & -6 & 6 & 3 & -9 \end{array} \right) \xrightarrow{R_4 = R_4 - 2R_1} \left(\begin{array}{ccccc|c} 0 & \textcircled{3} & -3 & 3 & 3 & -6 \\ 0 & 0 & 0 & 2 & 3 & -1 \\ 0 & 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 & -3 & 3 \end{array} \right).$$

Step 4. Repeat steps 1 to 3 on the submatrix of rows and columns strictly to the right of and below the pivot element and stop when the augmented matrix is in row-echelon form.

Note that the top row in step 2 here should mean the top row of the submatrix. In example (a), the required operation is

$$\left(\begin{array}{ccc|c} \textcircled{1} & 2 & 1 & 3 \\ 0 & \textcircled{-1} & -1 & -3 \\ 0 & 1 & 2 & -1 \end{array} \right) \xrightarrow{R_3 = R_3 + R_2} \left(\begin{array}{ccc|c} \textcircled{1} & 2 & 1 & 3 \\ 0 & \textcircled{-1} & -1 & -3 \\ 0 & 0 & 1 & -4 \end{array} \right).$$

Then we have reduced the matrix to row-echelon from

$$\left(\begin{array}{ccc|c} \textcircled{1} & 2 & 1 & 3 \\ 0 & \textcircled{-1} & -1 & -3 \\ 0 & 0 & \textcircled{1} & -4 \end{array} \right).$$

In example (b), the required operations are

$$\begin{aligned} \left(\begin{array}{ccccc|c} 0 & \textcircled{3} & -3 & 3 & 3 & -6 \\ 0 & 0 & 0 & 2 & 3 & -1 \\ 0 & 0 & 0 & 1 & 3 & -2 \\ 0 & 0 & 0 & 0 & -3 & -3 \end{array} \right) &\xrightarrow{R_3 = R_3 - \frac{1}{2}R_2} \left(\begin{array}{ccccc|c} 0 & \textcircled{3} & -3 & 3 & 3 & -6 \\ 0 & 0 & 0 & \textcircled{2} & 3 & -1 \\ 0 & 0 & 0 & 0 & \frac{3}{2} & -\frac{3}{2} \\ 0 & 0 & 0 & 0 & -3 & -3 \end{array} \right) \\ &\xrightarrow{R_4 = R_4 + 2R_3} \left(\begin{array}{ccccc|c} 0 & \textcircled{3} & -3 & 3 & 3 & -6 \\ 0 & 0 & 0 & \textcircled{2} & 3 & -1 \\ 0 & 0 & 0 & 0 & \textcircled{\frac{3}{2}} & -\frac{3}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right) = (U|\mathbf{y}). \end{aligned}$$

In both examples we have now reached a matrix which is in row-echelon form, so we have completed the process of Gaussian elimination for these examples.

NOTE.

1. If you are doing Gaussian elimination without the aid of a computer, you don't have to stick rigidly to the pivot selection rule which we stated above. For example, the value 1 is very convenient as a pivot entry, so when the leftmost non-zero column has a 1 in it you may find it best to use this 1 as your pivot entry in preference to some other non-zero entry which is above the 1.

For example, if you have the augmented matrix

$$\left(\begin{array}{cc|c} 13 & 27 & 174 \\ 1 & 2 & 13 \end{array} \right)$$

you should do

$$\begin{aligned} \left(\begin{array}{cc|c} 13 & 27 & 174 \\ 1 & 2 & 13 \end{array} \right) &\xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{cc|c} 1 & 2 & 13 \\ 13 & 27 & 174 \end{array} \right) \\ &\xrightarrow{R_2 = R_2 - 13R_1} \left(\begin{array}{cc|c} 1 & 2 & 13 \\ 0 & 1 & 5 \end{array} \right) \xrightarrow{R_1 = R_1 - 2R_2} \left(\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 5 \end{array} \right) \end{aligned}$$

rather than applying the rule stated above, which would give

$$\left(\begin{array}{cc|c} 13 & 27 & 174 \\ 1 & 2 & 13 \end{array} \right) \xrightarrow{R_2 = R_2 - \frac{1}{13}R_1} \left(\begin{array}{cc|c} 13 & 27 & 174 \\ 0 & -\frac{1}{13} & -\frac{5}{13} \end{array} \right) \text{ etc.}$$

2. The legal row operations are just

$$\begin{aligned} R_i &\leftrightarrow R_j, \\ R_i &= \lambda R_i \quad \text{where } \lambda \neq 0, \\ R_i &= R_i + \lambda R_j, \end{aligned}$$

and they are supposed to be done one after the other. Be careful to avoid doing simultaneously two operations like

$$R_1 = R_1 - 2R_2 \quad \text{and} \quad R_2 = R_1 - 2R_2.$$

See page 139. If all the entries in R_i are integers and have a common factor λ , we should use $R_i = \frac{1}{\lambda}R_i$ to produce integers with smaller magnitude.

4.4.3 Transformation to reduced row-echelon form

Given a matrix in row-echelon form, we can transform it into *reduced* row-echelon form by using two types of elementary row operations — multiplication of a row by a constant and adding a multiple of one row to another row.

The procedure is as follows.

Start with the lowest row which is not all zeros. Multiply it by a suitable constant to make its leading entry 1. Then add multiples of this row to higher rows to get all zeros in the column above the leading entry of this row. Repeat this procedure with the second lowest non-zero row, and so on.

We shall now see how this procedure applies to examples 1 to 6 of Section 4.4.1 (all of which are already in row-echelon form) and how the solution (if any) of each system relates to the row-echelon form.

Example 1. The transformation to reduced row-echelon form goes like this.

$$\begin{aligned} \left(\begin{array}{ccc|c} 2 & 3 & 4 & 11 \\ 0 & -3 & 2 & 7 \\ 0 & 0 & 4 & 8 \end{array} \right) & \xrightarrow{R_3 = \frac{1}{4}R_3} \left(\begin{array}{ccc|c} 2 & 3 & 4 & 11 \\ 0 & -3 & 2 & 7 \\ 0 & 0 & 1 & 2 \end{array} \right) & \xrightarrow{\substack{R_1 = R_1 - 4R_3 \\ R_2 = R_2 - 2R_3}} \left(\begin{array}{ccc|c} 2 & 3 & 0 & 3 \\ 0 & -3 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right) \\ & \xrightarrow{R_2 = -\frac{1}{3}R_2} \left(\begin{array}{ccc|c} 2 & 3 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array} \right) & \xrightarrow{R_1 = R_1 - 3R_2} \left(\begin{array}{ccc|c} 2 & 0 & 0 & 6 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array} \right) \\ & \xrightarrow{R_1 = \frac{1}{2}R_1} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 2 \end{array} \right) \end{aligned}$$

The final augmented matrix represents the system

$$\begin{aligned} x_1 + 0x_2 + 0x_3 &= 3 \\ 0x_1 + x_2 + 0x_3 &= -1 \\ 0x_1 + 0x_2 + x_3 &= 2, \end{aligned}$$

so the solution is $x_1 = 3$, $x_2 = -1$, $x_3 = 2$. ◇

Notice that in the original row-echelon form, **every column of the coefficient matrix** (the part of the augmented matrix to the left of the vertical bar) **is a leading column** and that the system of equations has a **unique solution**.

Example 2. In this case we can get to reduced row-echelon form just by multiplying each row by a suitable constant: $R_1 = -\frac{1}{5}R_1$, $R_2 = \frac{1}{2}R_2$ and $R_3 = \frac{1}{2}R_3$. The reduced row-echelon form is then

$$\left(\begin{array}{cccc|c} 1 & 0 & -\frac{4}{5} & -\frac{1}{5} & 0 \\ 0 & 1 & \frac{3}{2} & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 1 \end{array} \right).$$

Since the third row of the reduced row-echelon form represents the equation

$$0x_1 + 0x_2 + 0x_3 + 0x_4 = 1,$$

the system has no solution. $\diamond$

In general, if an augmented matrix has a row of the form

$$(0 \ 0 \ 0 \cdots 0 \mid \alpha), \text{ where } \alpha \neq 0,$$

then the corresponding system of equations has no solution. Note that if the **right-hand-side column** is a **leading** column, then the system of equations has **no solution**, i.e. **inconsistent**.

Example 3. The transformation to reduced row-echelon form goes like this.

$$\left(\begin{array}{ccc|c} 6 & 2 & 3 & 4 \\ 0 & 0 & 1 & 4 \end{array} \right) \xrightarrow{R_1 = R_1 - 3R_2} \left(\begin{array}{ccc|c} 6 & 2 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right) \xrightarrow{R_1 = \frac{1}{6}R_1} \left(\begin{array}{ccc|c} \textcircled{1} & \frac{1}{3} & 0 & \frac{1}{6} \\ 0 & 0 & \textcircled{1} & 1 \end{array} \right).$$

The final augmented matrix represents the system

$$x_1 + \frac{1}{3}x_2 = \frac{1}{6}, \quad x_3 = 1.$$

The second equations tells us that x_3 must equal 1. For the first equation, we need to set x_2 as a parameter, say λ . This means that the system must have infinitely many solutions. In fact $\mathbf{x}$ will be a solution if and only if

$$x_1 = \frac{1}{6} - \frac{1}{3}\lambda, \quad x_2 = \lambda, \quad x_3 = 1.$$

This can be rewritten in vector form as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} \frac{1}{6} - \frac{1}{3}\lambda \\ \lambda \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{6} \\ 0 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -\frac{1}{3} \\ 1 \\ 0 \end{pmatrix} \quad \text{for } \lambda \in \mathbb{R}.$$

This is a parametric vector form of a line in $\mathbb{R}^3$ through the point $(\frac{1}{6}, 0, 1)$ parallel to the vector $\begin{pmatrix} -\frac{1}{3} \\ 1 \\ 0 \end{pmatrix}$. $\diamond$

A variable x_i is a *leading variable* if the i th column of the row-echelon matrix is a leading column. It is a *non-leading variable* if the i th column of the matrix is a non-leading column. Notice that the leading variables in the echelon form are x_1 and x_3 while x_2 is non-leading, so this example illustrates a general rule which says that **non-leading variables can be chosen arbitrarily** by setting parameter, and then the leading variables can be written in terms of the parameters.

Example 4. In this example, the third row corresponds to the equation

$$0x_1 + 0x_2 + 0x_3 + 0x_4 + 0x_5 = 0$$

which is satisfied by every $\mathbf{x}$, and so is superfluous. However, such rows should not be deleted as doing so may cause confusion in some applications that you will encounter in later courses.

The transformation to reduced echelon form gives

$$\left(\begin{array}{ccccc|c} \textcircled{1} & 0 & 0 & -\frac{1}{5} & \frac{1}{5} & \frac{1}{5} \\ 0 & 0 & \textcircled{1} & \frac{3}{2} & 1 & \frac{1}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right).$$

Columns 2, 4 and 5 are non-leading columns, so the variables x_2 , x_4 and x_5 are non-leading variables and in terms of parameters we have $x_2 = \lambda_1$, $x_4 = \lambda_2$ and $x_5 = \lambda_3$. We can then read off the values of the leading variables as

$$x_1 = \frac{1}{5} + \frac{1}{5}\lambda_2 - \frac{1}{5}\lambda_3, \quad x_3 = \frac{1}{2} - \frac{3}{2}\lambda_2 - \lambda_3.$$

This set of solutions can be expressed in vector form as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} \frac{1}{5} \\ 0 \\ \frac{1}{2} \\ 0 \\ 0 \end{pmatrix} + \lambda_1 \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} -\frac{1}{5} \\ 0 \\ -\frac{3}{2} \\ 1 \\ 0 \end{pmatrix} + \lambda_3 \begin{pmatrix} -\frac{1}{5} \\ 0 \\ -1 \\ 0 \\ 1 \end{pmatrix} \text{ for } \lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}.$$

◇

Examples 3 and 4 illustrate that **the parameters in the solution are the values of the non-leading variables**, so **the number of parameters in the solution equals the number of non-leading columns in the row-echelon form of the coefficient matrix**.

In general, if a system is consistent and there are non-leading columns in a row-echelon form of the system, then the system has *infinitely many solutions*.

Example 5. In this example, row 4 is all zeros and so is superfluous. The remaining rows are the same as in example 1, so the solution is identical to the solution in example 1. ◇

Example 6. Transformation to reduced row-echelon form gives

$$\left(\begin{array}{cccc|c} 1 & 0 & -\frac{4}{5} & -\frac{1}{5} & 0 \\ 0 & 1 & \frac{3}{2} & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

We let the values of the non-leading variables x_3 and x_4 be arbitrary parameters, say λ_1 and λ_2 . We can then read off the values for the leading variables x_1 and x_2 (in terms of λ_1 and λ_2) and express the set of solutions in vector form as

$$\mathbf{x} = \lambda_1 \begin{pmatrix} \frac{4}{5} \\ -\frac{3}{2} \\ 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} \frac{1}{5} \\ -1 \\ 0 \\ 1 \end{pmatrix} \text{ for } \lambda_1, \lambda_2 \in \mathbb{R}.$$

This is a parametric vector form of a plane in $\mathbb{R}^4$ which contains the origin and is parallel to the vectors $\begin{pmatrix} \frac{4}{5} \\ -\frac{3}{2} \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} \frac{1}{5} \\ -1 \\ 0 \\ 1 \end{pmatrix}$. $\diamond$

We end this subsection by summarising the rule for getting solutions from a reduced echelon form of the system:

Assign an arbitrary parameter as the value for each non-leading variable. Then read off expressions for the leading variables in terms of the arbitrary parameters which you have introduced.

4.4.4 Back-substitution

When solving small systems by hand, you may prefer to use this procedure as an alternative to the transformation of a row-echelon form into reduced row-echelon form. Instead of doing further row operations on the augmented matrix, we go to the equations represented by the row-echelon form and proceed as follows.

Assign an arbitrary parameter value to each non-leading variable. Then read off from the last non-trivial equation an expression for the last leading variable in terms of your arbitrary parameters. Substitute this expression back into the second last equation to get an expression for the second last leading variable, and so on.

Example 7. We will redo example 4 of the last subsection, using back-substitution instead of transformation to reduced echelon form. The row-echelon form is

$$\left(\begin{array}{ccccc|c} \textcircled{-5} & 0 & 2 & 4 & 1 & 0 \\ 0 & 0 & \textcircled{2} & 3 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right).$$

SOLUTION. The non-leading variables are x_2 , x_4 and x_5 so we let $x_2 = \lambda_1$, $x_4 = \lambda_2$ and $x_5 = \lambda_3$. Then the second row corresponds to the equation

$$2x_3 + 3\lambda_2 + 2\lambda_3 = 1$$

which gives

$$x_3 = \frac{1}{2} - \frac{3}{2}\lambda_2 - \lambda_3.$$

Substituting this back into the equation represented by the first row gives

$$-5x_1 + 2\left(\frac{1}{2} - \frac{3}{2}\lambda_2 - \lambda_3\right) + 4\lambda_2 + \lambda_3 = 0.$$

We solve this for x_1 and get

$$x_1 = \frac{1}{5} + \frac{1}{5}\lambda_2 - \frac{1}{5}\lambda_3.$$

Note that this is the same solution as we got by transforming to reduced row-echelon form. $\diamond$

4.5 Deducing solubility from row-echelon form

We can determine the number of solutions to a system by examining the **position of leading entries** in a row-echelon form for the augmented matrix of the system. The rules for doing this are summarised in the following proposition.

Proposition 1. If the augmented matrix for a system of linear equations can be transformed into an equivalent row-echelon form $(U|\mathbf{y})$ then:

1. The system has **no solution** if and only if the right hand column $\mathbf{y}$ is a **leading column**.
2. If the right hand column $\mathbf{y}$ is a NOT a leading column then the system has:
 - a) A **unique** solution if and only if **every variable is a leading variable**.
 - b) **Infinitely many** solutions if and only if there is **at least one non-leading variable**.
In this case, each non-leading variable becomes a parameter in the general expression for all solutions and the number of parameters in the solution equals the number of non-leading variables.

It often happens in practice that you want to consider the system $A\mathbf{x} = \mathbf{b}$ for one fixed coefficient matrix A but many different vectors $\mathbf{b}$. For this reason it is useful to know the answer to the following question: if we know a row-echelon form U for the *coefficient matrix* A (not the augmented matrix $(A|\mathbf{b})$), what can we say about solubility of the system $A\mathbf{x} = \mathbf{b}$ for an arbitrary right hand side $\mathbf{b}$?

If U has an all-zero row then it will obviously be possible to find at least one $\mathbf{b} \in \mathbb{R}^n$ such that the row-echelon form $(U|\mathbf{y})$ for $(A|\mathbf{b})$ has a row of the form

$$(0 \ 0 \ \cdots \ 0 \ | \ \alpha), \ \alpha \neq 0$$

and this represents an equation which has no solution. On the other hand, if U has no non-leading row (see page 140) then no $\mathbf{b}$ can give rise to an impossible row of the above type and we can always get a solution by back-substitution. In this case the number of solutions will be infinite if and only if there are non-leading columns in U .

The following proposition sums up what we can say about solubility for arbitrary $\mathbf{b}$.

Proposition 2. If A is an $m \times n$ matrix which can be transformed by elementary row operations into a row-echelon form U then the system $A\mathbf{x} = \mathbf{b}$ has

- a) **at least one** solution for each $\mathbf{b}$ in $\mathbb{R}^m$ if and only if U has **no non-leading rows**,
- b) **at most one** solution for each $\mathbf{b}$ in $\mathbb{R}^m$ if and only if U has **no non-leading columns**,
- c) **exactly one** solution for each $\mathbf{b}$ in $\mathbb{R}^m$ if and only if U has **no non-leading rows and no non-leading columns**.

When U has neither non-leading row nor non-leading column, $A\mathbf{x} = \mathbf{b}$ always has a unique solution. Otherwise, whether the matrix equation has solution depends on the vector $\mathbf{b}$. We shall study examples on finding a condition on $\mathbf{b}$ for a solution to exist and we shall discuss how to find a *specific formula* for a solution $\mathbf{x}$ expressed in terms of an arbitrary right hand side $\mathbf{b}$ in the next section.

4.6 Solving $A\mathbf{x} = \mathbf{b}$ for indeterminate $\mathbf{b}$

To solve the equation $A\mathbf{x} = \mathbf{b}$, we apply row operations to reduce the corresponding augmented matrix $(A|\mathbf{b})$.

Example 1. Solve the system of equations

$$\begin{cases} 2x_1 + 3x_2 = b_1 \\ 3x_1 + 4x_2 = b_2 \end{cases}$$

SOLUTION. We reduce the augmented matrix as follows.

$$\left(\begin{array}{cc|c} 2 & 3 & b_1 \\ 3 & 4 & b_2 \end{array} \right) \xrightarrow{R_2 = R_2 - \frac{3}{2}R_1} \left(\begin{array}{cc|c} 2 & 3 & b_1 \\ 0 & -\frac{1}{2} & -\frac{3}{2}b_1 + b_2 \end{array} \right).$$

From the second row of the row-echelon form, we have

$$-\frac{1}{2}x_2 = -\frac{3}{2}b_1 + b_2 \quad \text{or} \quad x_2 = 3b_1 - 2b_2.$$

Substitute it into the equation represented by the first row,

$$2x_1 + 3(3b_1 - 2b_2) = b_1 \quad \text{or} \quad x_1 = -4b_1 + 3b_2.$$

◇

NOTE. We could write the system of equations as

$$\begin{cases} 2x_1 + 3x_2 = 1b_1 + 0b_2 \\ 3x_1 + 4x_2 = 0b_1 + 1b_2 \end{cases}$$

Provided the interpretation of the columns is clearly understood the row reduction can be rewritten as

$$\left(\begin{array}{cc|cc} 2 & 3 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array} \right) \xrightarrow{R_2 = R_2 - \frac{3}{2}R_1} \left(\begin{array}{cc|cc} 2 & 3 & 1 & 0 \\ 0 & -\frac{1}{2} & -\frac{3}{2} & 1 \end{array} \right).$$

Then we can proceed to reduce the matrix to reduced row-echelon form or apply back-substitution as before.

In the above example, the system always has a unique solution for any $\mathbf{b}$. The system of equations in the next example does not always have solution. A condition on $\mathbf{b}$ has to be satisfied for a solution to exist.

Example 2. Find a condition for $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$ so that the system of equations

$$\begin{cases} 2x_1 + 3x_2 = b_1 \\ 4x_1 + 6x_2 = b_2 \end{cases}$$

has solutions. Find the solution set if the condition is satisfied.

SOLUTION. The reduction process yields

$$\left(\begin{array}{cc|c} 2 & 3 & b_1 \\ 4 & 6 & b_2 \end{array} \right) \xrightarrow{R_2 = R_2 - 2R_1} \left(\begin{array}{cc|c} 2 & 3 & b_1 \\ 0 & 0 & b_2 - 2b_1 \end{array} \right).$$

The second row represents the equation $0x_1 + 0x_2 = -2b_1 + b_2$. Thus for solutions to exist, the condition $-2b_1 + b_2 = 0$ has to be satisfied. If solutions exist, we have to assign the non-leading variable x_2 a parameter λ . Then the equation corresponding to the first row becomes

$$2x_1 + 3\lambda = b_1, \quad \text{that is} \quad x_1 = -\frac{3}{2}\lambda + \frac{1}{2}b_1.$$

The solution set is then

$$\left\{ \mathbf{x} \in \mathbb{R}^2 : \mathbf{x} = \begin{pmatrix} \frac{1}{2}b_1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -\frac{3}{2} \\ 1 \end{pmatrix}, \lambda \in \mathbb{R} \right\}.$$

◇

Sometimes there is more than one condition for solutions to exist.

Example 3. Find conditions on b_1, b_2, b_3, b_4 for the following system of equations to be consistent. Find the solutions when these conditions are satisfied.

$$\begin{cases} x_1 + 2x_2 = b_1 \\ 3x_1 + 5x_2 = b_2 \\ 5x_1 + 7x_2 = b_3 \\ 7x_1 + 9x_2 = b_4 \end{cases}$$

SOLUTION.

$$\begin{aligned} \left(\begin{array}{cc|c} 1 & 2 & b_1 \\ 3 & 5 & b_2 \\ 5 & 7 & b_3 \\ 7 & 9 & b_4 \end{array} \right) & \xrightarrow{\begin{array}{l} R_2 = R_2 - 3R_1 \\ R_3 = R_3 - 5R_1 \\ R_4 = R_4 - 7R_1 \end{array}} \left(\begin{array}{cc|c} 1 & 2 & b_1 \\ 0 & -1 & b_2 - 3b_1 \\ 0 & -3 & b_3 - 5b_1 \\ 0 & -5 & b_4 - 7b_1 \end{array} \right) \\ & \xrightarrow{\begin{array}{l} R_3 = R_3 - 3R_2 \\ R_4 = R_4 - 5R_2 \end{array}} \left(\begin{array}{cc|c} 1 & 2 & b_1 \\ 0 & -1 & -3b_1 + b_2 \\ 0 & 0 & 4b_1 - 3b_2 + b_3 \\ 0 & 0 & 8b_1 - 5b_2 + b_4 \end{array} \right) \end{aligned}$$

From rows 3 and 4, the original system of equations has a solution if and only if $b_3 = -4b_1 + 3b_2$ and $b_4 = -8b_1 + 5b_2$. If the condition is satisfied, we get $x_2 = 3b_1 - b_2$ from row 2. Then from row 1, we have $x_1 = b_1 - 2(3b_1 - b_2) = -5b_1 + 2b_2$. ◇

4.7 General properties of the solution of $A\mathbf{x} = \mathbf{b}$

We have seen that every system of linear equations can be written as $A\mathbf{x} = \mathbf{b}$ and every such system has either no solution, a unique solution or an infinite number of solutions. For all systems of linear equations that we have solved in this chapter which have infinitely many solutions, the solutions can be written in parametric vector form as

$$\mathbf{x} = \mathbf{x}_p + \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \cdots + \lambda_k \mathbf{v}_k,$$

where $\mathbf{v}_1, \dots, \mathbf{v}_k$ are non-zero vectors and where the parameters $\lambda_1, \dots, \lambda_k$ are scalars. It is not difficult to see that this is generally true for all systems of linear equations with infinitely many solutions.

Furthermore, we shall prove that the vector $\mathbf{x}_p$ is a solution of the system $A\mathbf{x} = \mathbf{b}$, and the vectors $\mathbf{v}_i$ which go with the parameters in the solution are solutions of the homogeneous equation $A\mathbf{x} = \mathbf{0}$. It does not depend on how we reduce the matrix to an echelon form.

We begin with some important propositions about solutions of $A\mathbf{x} = \mathbf{0}$.

Proposition 1. $A\mathbf{x} = \mathbf{0}$ has $\mathbf{x} = \mathbf{0}$ as a solution.

Proof. Since $A\mathbf{x} = \mathbf{0}$ represents the system of equations

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = 0 \end{cases},$$

so $x_1 = x_2 = \cdots = x_n = 0$ satisfies all the equations. $\square$

Proposition 2. If $\mathbf{v}$ and $\mathbf{w}$ are solutions of $A\mathbf{x} = \mathbf{0}$ then so are $\mathbf{v} + \mathbf{w}$ and $\lambda\mathbf{v}$ for any scalar λ .

Proof. Since $\mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix}$ are solutions, we have

$$\begin{cases} a_{11}v_1 + a_{12}v_2 + \cdots + a_{1n}v_n = 0 \\ a_{21}v_1 + a_{22}v_2 + \cdots + a_{2n}v_n = 0 \\ \vdots \\ a_{m1}v_1 + a_{m2}v_2 + \cdots + a_{mn}v_n = 0 \end{cases}, \quad (1)$$

$$\begin{cases} a_{11}w_1 + a_{12}w_2 + \cdots + a_{1n}w_n = 0 \\ a_{21}w_1 + a_{22}w_2 + \cdots + a_{2n}w_n = 0 \\ \vdots \\ a_{m1}w_1 + a_{m2}w_2 + \cdots + a_{mn}w_n = 0 \end{cases}. \quad (2)$$

If we add the i th equations of (1) and (2) for $1 \leq i \leq m$, we shall get

$$\begin{cases} a_{11}(v_1 + w_1) + a_{12}(v_2 + w_2) + \cdots + a_{1n}(v_n + w_n) = 0 \\ a_{21}(v_1 + w_1) + a_{22}(v_2 + w_2) + \cdots + a_{2n}(v_n + w_n) = 0 \\ \vdots \\ a_{m1}(v_1 + w_1) + a_{m2}(v_2 + w_2) + \cdots + a_{mn}(v_n + w_n) = 0 \end{cases}.$$

Hence $\mathbf{v} + \mathbf{w}$ is a solution of $A\mathbf{x} = \mathbf{0}$.

If we multiply both sides of every equation of (1) by λ , then

$$\begin{cases} a_{11}(\lambda v_1) + a_{12}(\lambda v_2) + \cdots + a_{1n}(\lambda v_n) = 0 \\ a_{21}(\lambda v_1) + a_{22}(\lambda v_2) + \cdots + a_{2n}(\lambda v_n) = 0 \\ \vdots \\ a_{m1}(\lambda v_1) + a_{m2}(\lambda v_2) + \cdots + a_{mn}(\lambda v_n) = 0 \end{cases}.$$

Hence $\lambda\mathbf{v}$ is also a solution. $\square$

A further important proposition about homogeneous systems can be obtained by induction from the last proposition.

Proposition 3. Let $\mathbf{v}_1, \dots, \mathbf{v}_k$ be solutions of $A\mathbf{x} = \mathbf{0}$ for $1 \leq j \leq k$. Then $\lambda_1 \mathbf{v}_1 + \dots + \lambda_k \mathbf{v}_k$ is also a solution of $A\mathbf{x} = \mathbf{0}$ for all $\lambda_1, \dots, \lambda_k \in \mathbb{R}$.

We now turn to the properties of the solutions of $A\mathbf{x} = \mathbf{b}$.

Proposition 4. If $\mathbf{v}$ and $\mathbf{w}$ are solutions of $A\mathbf{x} = \mathbf{b}$ then $\mathbf{v} - \mathbf{w}$ is a solution of $A\mathbf{x} = \mathbf{0}$.

Proof. Since $\mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$ and $\mathbf{w} = \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix}$ are solutions of $A\mathbf{x} = \mathbf{b}$, we have

$$\begin{cases} a_{11}v_1 + a_{12}v_2 + \dots + a_{1n}v_n = b_1 \\ a_{21}v_1 + a_{22}v_2 + \dots + a_{2n}v_n = b_2 \\ \vdots \\ a_{m1}v_1 + a_{m2}v_2 + \dots + a_{mn}v_n = b_n \end{cases}, \quad (1)$$

$$\begin{cases} a_{11}w_1 + a_{12}w_2 + \dots + a_{1n}w_n = b_1 \\ a_{21}w_1 + a_{22}w_2 + \dots + a_{2n}w_n = b_2 \\ \vdots \\ a_{m1}w_1 + a_{m2}w_2 + \dots + a_{mn}w_n = b_n \end{cases}. \quad (2)$$

If we subtract the i th equation of (2) from the i th equation of (1) for $1 \leq i \leq m$, we shall get

$$\begin{cases} a_{11}(v_1 - w_1) + a_{12}(v_2 - w_2) + \dots + a_{1n}(v_n - w_n) = 0 \\ a_{21}(v_1 - w_1) + a_{22}(v_2 - w_2) + \dots + a_{2n}(v_n - w_n) = 0 \\ \vdots \\ a_{m1}(v_1 - w_1) + a_{m2}(v_2 - w_2) + \dots + a_{mn}(v_n - w_n) = 0 \end{cases}.$$

Hence $\mathbf{v} - \mathbf{w}$ is a solution of $A\mathbf{x} = \mathbf{0}$. □

We can now prove the following result.

Proposition 5. Let $\mathbf{x}_p$ be a solution of $A\mathbf{x} = \mathbf{b}$.

1. If $\mathbf{x} = \mathbf{0}$ is the only solution of the homogeneous equation $A\mathbf{x} = \mathbf{0}$ then $\mathbf{x}_p$ is the unique solution of $A\mathbf{x} = \mathbf{b}$.
2. If the homogeneous equation has non-zero solutions $\mathbf{v}_1, \dots, \mathbf{v}_k$ then

$$\mathbf{x} = \mathbf{x}_p + \lambda_1 \mathbf{v}_1 + \dots + \lambda_k \mathbf{v}_k,$$

is also a solution of $A\mathbf{x} = \mathbf{b}$ for all $\lambda_1, \dots, \lambda_k \in \mathbb{R}$.

Proof. For part 1, let $\mathbf{v}$ be another solution of $A\mathbf{x} = \mathbf{b}$. By Proposition 4, $\mathbf{v} - \mathbf{x}_p$ is a solution of $A\mathbf{x} = \mathbf{0}$. Since $\mathbf{0}$ is the only solution of $A\mathbf{x} = \mathbf{0}$, we have $\mathbf{v} - \mathbf{x}_p = \mathbf{0}$ and that is $\mathbf{v} = \mathbf{x}_p$. Hence $\mathbf{x}_p$ is the unique solution of $A\mathbf{x} = \mathbf{b}$.

For part 2, let $\mathbf{v} = \lambda_1 \mathbf{v}_1 + \dots + \lambda_k \mathbf{v}_k$. By Proposition 3, we know $\mathbf{v}$ is also a solution of $A\mathbf{x} = \mathbf{0}$. With a similar method used in proving Proposition 2, we can show that $\mathbf{x}_p + \mathbf{v}$ is a solution of $A\mathbf{x} = \mathbf{b}$. □

NOTE. The form of solutions in part 2 of Proposition 5 raises the question of what is the minimum number of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ required to yield all the solutions of $A\mathbf{x} = \mathbf{b}$. We will return to this question in Chapter 6 when we discuss the ideas of “spanning sets” and “linear independence”. For the present, it is sufficient to note that the solution method we have used in this chapter does find **all** solutions of $A\mathbf{x} = \mathbf{b}$.

Example 1. In Section 4.4.2 we found a row-echelon form for the augmented matrix of $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{pmatrix} 0 & 0 & 0 & 2 & 3 \\ 0 & 3 & -3 & 3 & 3 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 6 & -6 & 6 & 3 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} -1 \\ -6 \\ -2 \\ -9 \end{pmatrix}$$

as

$$\left(\begin{array}{ccccc|c} 0 & 3 & -3 & 3 & 3 & -6 \\ 0 & 0 & 0 & 2 & 3 & -1 \\ 0 & 0 & 0 & 0 & \frac{3}{2} & -\frac{3}{2} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right).$$

By back-substitution, $x_5 = -1$, $x_4 = 1$. By setting $x_1 = \lambda_1$ and $x_3 = \lambda_2$, we also have $x_2 = \lambda_2 - 2$. Hence the solutions are

$$\mathbf{x} = \begin{pmatrix} \lambda_1 \\ \lambda_2 - 2 \\ \lambda_2 \\ 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \\ 0 \\ 1 \\ -1 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} \quad \text{for } \lambda_1, \lambda_2 \in \mathbb{R}.$$

The first vector is a particular solution (corresponding to $\lambda_1 = \lambda_2 = 0$) of $A\mathbf{x} = \mathbf{b}$ and the vectors which go with the parameters are solutions of the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

As an exercise, can you write down the two systems of 4 equations in 5 variables corresponding to $A\mathbf{x} = \mathbf{b}$ and $A\mathbf{x} = \mathbf{0}$ and verify that the first vector is a solution to the non-homogeneous system and the two other vectors are solutions to the homogeneous system? $\diamond$

4.8 Applications

Linear equations appear in many applications. In this section we first present some geometric applications and then some applications in some other fields.

4.8.1 Geometry

In Section 4.1, we have shown some simple geometric applications of linear equations to lines in $\mathbb{R}^2$ and planes in $\mathbb{R}^3$. In this section we show how some common problems involving lines and planes in $\mathbb{R}^n$ can be solved using linear equations.

Example 1. Does $\begin{pmatrix} 3 \\ 0 \\ 5 \\ 6 \end{pmatrix}$ belong to $\text{span} \left(\begin{pmatrix} 1 \\ -2 \\ 3 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 4 \\ 1 \\ 2 \end{pmatrix} \right)$?

SOLUTION. The vector $\begin{pmatrix} 3 \\ 0 \\ 5 \\ 6 \end{pmatrix} \in \text{span} \left(\begin{pmatrix} 1 \\ -2 \\ 3 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 4 \\ 1 \\ 2 \end{pmatrix} \right)$ if and only if

$$\begin{pmatrix} 3 \\ 0 \\ 5 \\ 6 \end{pmatrix} = \lambda_1 \begin{pmatrix} 1 \\ -2 \\ 3 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ 4 \\ 1 \\ 2 \end{pmatrix} \text{ for some } \lambda_1, \lambda_2 \in \mathbb{R}.$$

On equating corresponding coordinates on both sides of this vector equation, we have

$$\begin{cases} \lambda_1 & = & 3 \\ -2\lambda_1 + 4\lambda_2 & = & 0 \\ 3\lambda_1 + \lambda_2 & = & 5 \\ 2\lambda_1 + 2\lambda_2 & = & 6 \end{cases}$$

The augmented matrix is

$$(A|\mathbf{b}) = \left(\begin{array}{cc|c} 1 & 0 & 3 \\ -2 & 4 & 0 \\ 3 & 1 & 5 \\ 2 & 2 & 6 \end{array} \right)$$

This is equivalent to

$$(U|\mathbf{y}) = \left(\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 4 & 6 \\ 0 & 0 & -\frac{11}{2} \\ 0 & 0 & 0 \end{array} \right).$$

As the system has no solution, $\begin{pmatrix} 3 \\ 0 \\ 5 \\ 6 \end{pmatrix}$ is not in $\text{span} \left(\begin{pmatrix} 1 \\ 2 \\ -3 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 4 \\ 1 \\ 2 \end{pmatrix} \right)$. ◇

Example 2. Is $\mathbf{v} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$ parallel to the plane

$$\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}?$$

SOLUTION. The vector $\mathbf{v} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$ is parallel to the plane if and only if it is a linear combination of the two vectors $\begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}$. Hence, $\mathbf{v}$ is parallel to the plane if and only if there are real

numbers λ_1 and λ_2 such that

$$\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix} = \lambda_1 \begin{pmatrix} -1 \\ 3 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix}.$$

On equating corresponding components of the vectors on both sides, we obtain

$$\begin{cases} -\lambda_1 + 3\lambda_2 = 2 \\ 3\lambda_1 + \lambda_2 = 4 \\ 2\lambda_1 + 4\lambda_2 = 6 \end{cases}$$

The augmented matrix is

$$(A|\mathbf{b}) = \left(\begin{array}{cc|c} -1 & 3 & 2 \\ 3 & 1 & 4 \\ 2 & 4 & 6 \end{array} \right),$$

which becomes, by Gaussian elimination,

$$(U|\mathbf{y}) = \left(\begin{array}{cc|c} -1 & 3 & 2 \\ 0 & 10 & 10 \\ 0 & 0 & 0 \end{array} \right).$$

Since the right hand column is non-leading, this system has a solution for λ_1 and λ_2 , and hence

$\begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$ is parallel to the plane.

[Note that this question can also be solved by geometric considerations using dot and cross products.] $\diamond$

Example 3. Find the intersection of the line through $(3, 2, 1, 4)$ parallel to $\begin{pmatrix} 1 \\ 2 \\ -3 \\ 6 \end{pmatrix}$ and the plane

through $(3, 1, -4, 7)$ parallel to $\begin{pmatrix} 2 \\ 1 \\ 4 \\ 5 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ 3 \\ 0 \\ 6 \end{pmatrix}$.

SOLUTION. The equation of the line is

$$\mathbf{x} = \begin{pmatrix} 3 \\ 2 \\ 1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -3 \\ 6 \end{pmatrix} \text{ for } \lambda \in \mathbb{R},$$

and the equation of the plane is

$$\mathbf{x} = \begin{pmatrix} 3 \\ 1 \\ -4 \\ 7 \end{pmatrix} + \mu_1 \begin{pmatrix} 2 \\ 1 \\ 4 \\ 5 \end{pmatrix} + \mu_2 \begin{pmatrix} -1 \\ 3 \\ 0 \\ 6 \end{pmatrix} \text{ for } \mu_1, \mu_2 \in \mathbb{R}.$$

An intersection occurs for values of λ, μ_1, μ_2 for which

$$\mathbf{x} = \begin{pmatrix} 3 \\ 2 \\ 1 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ -3 \\ 6 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \\ -4 \\ 7 \end{pmatrix} + \mu_1 \begin{pmatrix} 2 \\ 1 \\ 4 \\ 5 \end{pmatrix} + \mu_2 \begin{pmatrix} -1 \\ 3 \\ 0 \\ 6 \end{pmatrix}.$$

On rearranging, we have

$$\lambda \begin{pmatrix} 1 \\ 2 \\ -3 \\ 6 \end{pmatrix} - \mu_1 \begin{pmatrix} 2 \\ 1 \\ 4 \\ 5 \end{pmatrix} - \mu_2 \begin{pmatrix} -1 \\ 3 \\ 0 \\ 6 \end{pmatrix} = \begin{pmatrix} 0 \\ -1 \\ -5 \\ 3 \end{pmatrix}.$$

On equating coordinates on both sides, we obtain a set of 4 equations in 3 unknowns

$$\begin{cases} \lambda - 2\mu_1 + \mu_2 = 0 \\ 2\lambda - \mu_1 - 3\mu_2 = -1 \\ -3\lambda - 4\mu_1 = -5 \\ 6\lambda - 5\mu_1 - 6\mu_2 = 3 \end{cases}$$

After forming an augmented matrix, and using Gaussian elimination, we obtain

$$(U|\mathbf{y}) = \left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 3 & -5 & -1 \\ 0 & 0 & -\frac{41}{3} & -\frac{25}{3} \\ 0 & 0 & 0 & \frac{681}{123} \end{array} \right).$$

The system of equations has no solution, so the given line and plane in $\mathbb{R}^4$ do not intersect. $\diamond$

4.8.2 Chemical engineering

A greatly simplified example which illustrates an application of systems of linear equations to oil refining is as follows.

Example 4. An oil company has three refineries at Sydney, Melbourne and Brisbane. Each refinery makes four products: super petrol, unleaded petrol, diesel fuel and aviation fuel. The amount of each product that each refinery can make per hour is given in the following table.

	Sydney	Melbourne	Brisbane
Super (litres/hour)	10000	20000	5000
Unleaded (litres/hour)	2000	5000	1000
Diesel (litres/hour)	500	800	200
Aviation fuel (litres/hour)	100	200	100

The oil company has to decide how many hours per day to run each refinery in order to produce a required amount of product each day.

Do the following:

- (a) Set up a linear equation model for the company.
- (b) On Monday, the oil company requires **exactly** 610,000 litres of super, 137,000 litres of unleaded, 26,400 litres of diesel and 7,200 litres of aviation fuel. Find the number of hours each refinery should be run or show that the desired production levels cannot be met.
- (c) On Tuesday, the amount of super, unleaded and diesel required is the same, but the amount of aviation fuel required is increased to 7,800 litres. Again find the number of hours each refinery should be run or show that the desired production levels cannot be met.

SOLUTION. Let

x_1 = number of hours for which Sydney refinery runs,

x_2 = number of hours for which Melbourne refinery runs,

x_3 = number of hours for which Brisbane refinery runs,

and let b_1 , b_2 , b_3 and b_4 be the amounts of super, unleaded, diesel and aviation fuel respectively required each day. Then the equations are:

$$\begin{array}{rclclcl} \text{Super} & 10000 x_1 & + & 20000 x_2 & + & 5000 x_3 & = & b_1 \\ \text{Unleaded} & 2000 x_1 & + & 5000 x_2 & + & 1000 x_3 & = & b_2 \\ \text{Diesel} & 500 x_1 & + & 800 x_2 & + & 200 x_3 & = & b_3 \\ \text{Aviation} & 100 x_1 & + & 200 x_2 & + & 100 x_3 & = & b_4 \end{array}$$

The coefficient matrix, unknowns vector and right-hand side are therefore

$$A = \begin{pmatrix} 10000 & 20000 & 5000 \\ 2000 & 5000 & 1000 \\ 500 & 800 & 200 \\ 100 & 200 & 100 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}.$$

On Monday

$$(A|\mathbf{b}) = \left(\begin{array}{ccc|c} 10000 & 20000 & 5000 & 610000 \\ 2000 & 5000 & 1000 & 137000 \\ 500 & 800 & 200 & 26400 \\ 100 & 200 & 100 & 7200 \end{array} \right)$$

is equivalent to

$$(U|\mathbf{y}) = \left(\begin{array}{ccc|c} 10000 & 20000 & 5000 & 610000 \\ 0 & 1000 & 0 & 15000 \\ 0 & 0 & -50 & -1100 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

which has the solution $x_3 = 22$, $x_2 = 15$, $x_1 = 20$.

On Tuesday

$$(A|\mathbf{b}) = \left(\begin{array}{ccc|c} 10000 & 20000 & 5000 & 410000 \\ 2000 & 5000 & 1000 & 137000 \\ 500 & 800 & 200 & 26400 \\ 100 & 200 & 100 & 7800 \end{array} \right)$$

is equivalent to

$$(U|\mathbf{y}) = \left(\begin{array}{ccc|c} 10000 & 2000 & 5000 & 610000 \\ 0 & 1000 & 0 & 15000 \\ 0 & 0 & -50 & -1100 \\ 0 & 0 & 0 & 600 \end{array} \right)$$

which has no solution, so production levels can not be met. $\diamond$

The model used here is too simple to reflect accurately what is done in practice. In practice the problem is generalised in two ways. First, the equations are replaced by inequalities; i.e., the problem is changed from one of having to produce **exactly** the required amount of product to one of having to produce **at least** the required amount of product. Secondly, the problem is changed to one of having to **minimise the cost of production** subject to the constraints of having to produce the required amount of product. This generalised problem is called a **linear programming problem**. Linear programming is very important in economic, financial and manufacturing applications and so on where problems involving thousands of variables and inequalities are routinely solved, and result in savings of millions of dollars.

4.8.3 Economics

A simplified example of an application of linear equations to economics is as follows.

Example 5. The island of Wotsit-Matta has a simple economy in which the only goods produced are wheat, iron and pigs. The Wotsit-Mattas have worked out that the following amounts of wheat, iron and pigs are required on 31 December of each year for each unit of wheat, iron and pigs produced in the following year:

Requirements	Wheat (per tonne)	Iron (per tonne)	Pigs (per hundred)
Wheat(tonnes)	.5333	4.286	2.0000
Iron(tonnes)	.02557	.2857	.0500
Pigs (hundreds)	.0400	.5714	.5000

That is, to produce 1 tonne of wheat in a particular year we need to have available on 31 December of the previous year 0.5333 tonnes of wheat, 0.02557 tonnes of iron and 0.0400 hundred pigs, and similarly for iron and pigs.

Assuming that **all** of the wheat, iron and pigs available at 31 December of one year are used to produce wheat, iron and pigs in the following year, find the amount of wheat, iron and pigs at 31 December, 1988 and then at 31 December, 1989 if

- the Wotsit-Mattas have 180 tonnes of wheat, 8.401 tonnes of iron and 2400 pigs available at 31 December, 1987, and
- the Wotsit-Mattas have 180 tonnes of wheat, 8 tonnes of iron and 2400 pigs available at 31 December, 1987.

SOLUTION. We first set up a linear equations model for the Wotsit-Matta economy. Note that the unknowns are the amounts of wheat, iron and pigs produced in a year, and the right-hand sides are the amounts available from the previous year.

We let

$$\begin{aligned}x_1 &= \text{number of tonnes of wheat produced in a year} \\x_2 &= \text{number of tonnes of iron produced in a year} \\x_3 &= \text{number of hundreds of pigs produced in a year}\end{aligned}$$

and let

$$\begin{aligned}b_1 &= \text{number of tonnes of wheat available from the previous year} \\b_2 &= \text{number of tonnes of iron available from the previous year} \\b_3 &= \text{number of hundreds of pigs available from the previous year}\end{aligned}$$

Then the linear-equation model for the economy is

$$\begin{array}{rclclcl} \text{Wheat used} & .5333x_1 & + & 4.286x_2 & + & 2x_3 & = & b_1 \\ \text{Iron used} & .02667x_1 & + & .2857x_2 & + & .05x_3 & = & b_2 \\ \text{Pigs used} & .04x_1 & + & .5714x_2 & + & .5x_3 & = & b_3 \end{array}$$

We can now solve for the production in years 1988 and 1989.

CASE 1.(a). On 31 December, 1987, $\mathbf{b} = \begin{pmatrix} 180 \\ 8.401 \\ 24 \end{pmatrix}$. On solving for the production in 1988 we obtain $\mathbf{x} = \begin{pmatrix} 180 \\ 8.403 \\ 24 \end{pmatrix}$, that is, 180 tonnes wheat, 8.403 tonnes iron and 2400 pigs.

This 1988 production is then used as the right-hand side $\mathbf{b}$ in the equations for the 1989 production. On solving $A\mathbf{x} = \begin{pmatrix} 180 \\ 8.403 \\ 24 \end{pmatrix}$ as before, we obtain the solution $\mathbf{x} = \begin{pmatrix} 179.9 \\ 8.425 \\ 23.98 \end{pmatrix}$, and so the 1989 production is 179.9 tonnes wheat, 8.425 tonnes iron and 2398 pigs.

CASE 2.(b). On 31 December, 1987, $\mathbf{b} = \begin{pmatrix} 180 \\ 8 \\ 24 \end{pmatrix}$. We find that the solution for 1988 production is 200.1 tonnes wheat, 4.658 tonnes iron and 2667 pigs.

On using this 1988 production as the right-hand side for the 1989 production, we obtain the 1989 production as 432.2 tonnes wheat, -34.37 tonnes iron and 5787 pigs.

◇

NOTE. The negative result for iron in case (b) means that the Wotsit-Matta economy is in deep trouble. Possible actions that the Wotsit-Mattas might take include importing iron from Australia (if they can find the foreign exchange) or alternatively, having a great party and eating sufficient wheat and pigs to stop the overheating in these sections of their economy. A complete mathematical explanation of the marked difference in behaviour between cases (a) and (b) requires a knowledge of eigenvalues and eigenvectors (which will be studied in MATH1231).

4.9 Matrix reduction and Maple

This section deals with the reduction of partitioned matrices to row echelon and reduced row echelon forms. It should be noted that Maple can handle symbolic entries, so we can solve systems that involve general right-hand-side vectors. The first instruction needed is:

```
with(LinearAlgebra):
```

which loads the linear algebra package. If you wish to experiment with a matrix with, say, 3 rows and 4 columns and with small integer entries, just try:

```
RandomMatrix(3,4);
```

though if you want to vary your random matrix you should try first:

```
randomize(301092);
```

Integers other than 301092 will give rise to other random matrices. To assign the name A to this, you use:

```
A:=RandomMatrix(3,4);
```

The 3×3 identity matrix may be obtained with the command:

```
Id:=IdentityMatrix(3);
```

An easy way to enter the values of a matrix is to command:

```
C:=< <1,4> | <2,5> | <3,6> >;
```

Note that we are inputting the entries of the matrix *columnwise* to give a 2×3 matrix called C . You may recover the second column with the command:

```
Column(%,2);
```

If you want to augment two matrices A_1 , A_2 with the same number (r) of rows to obtain what is denoted in the main text by the partitioned matrix $(A_1|A_2)$, the command

```
< A1 | A2 >;
```

will do the trick. However Maple is blind to the partitioning. The following example, using a given 3×4 matrix A , is worth trying:

```
b:=< b1 , b2 , b3 >;
```

```
Ab:=< A | b >;
```

Here Ab is the name we have given to the resulting 3×5 matrix.

The reduction to echelon form of the matrix A is achieved by:

```
GaussianElimination(A);
```

The reduced row-echelon form is obtained with the command:

```
ReducedRowEchelonForm(%);
```

either after using `GaussianElimination` or directly. Here the `%` refers to the previous expression. `GaussianElimination` does not always do what one would like. Try `GaussianElimination(A)` with

```
A:=< <1,2,3> | <1,2,3> | <1,2,3> | <a,b,c> >;
```

There is a facility for finding the general solution of a set of linear equations. For example, the pair $1x_1 + 2x_2 + 3x_3 + 7x_4 = b_1$, $2x_1 + 4x_2 + 5x_3 + 9x_4 = b_2$ can be handled by the commands:

```
A:=< <1,2> | <2,4> | <3,5> | <7,9> >;
```

```
b:=< b1,b2 >;
```

```
Ab:=< A | b >;
```

```
GaussianElimination(%);
```

```
BackwardSubstitute(%);
```

The same answer is obtained from:

```
GaussianElimination(Ab);
```

```
BackwardSubstitute(%);
```

Experimenting with this may help motivate the reduced row-echelon form. You should also experiment with:

```
BackwardSubstitute(Ab);
```

to see what can go wrong if you leave out the Gaussian elimination step.

Rows 1 and 3 of the matrix C may be swapped with the command:

```
RowOperation(C,[1,3]);
```

You may replace row 4 by row 4 plus $\frac{7}{2}$ times row 3 in the matrix A to row 4 with

```
RowOperation(A,[4,3],7/2);
```

You may multiply row 4 of the matrix A by $\frac{7}{2}$ by the command:

```
RowOperation(A,4,7/2);
```

You should now practise reducing a ‘random’ matrix A by hand and comparing your answer with that obtained from `GaussianElimination` and `ReducedRowEchelonForm`. Other useful commands may be found by typing in:

```
?Pivot
```

```
?LinearSolve
```

Problems for Chapter 4

Questions marked with [R] are routine, [H] harder, [M] Maple and [X] are for MATH1141 only. You should make sure that you can do the easier questions before you tackle the more difficult questions. Questions marks with a [V] have video solutions available on Moodle.

Problems 4.1 : Introduction to linear equations

1. [R] Find the solution set of each of the following linear equation.
 - a) $2x_1 - 5 = 0$ as an equation of one variable, then as an equation in two variables, and then three variables.
 - b) $x_1 + 2x_2 = 4$ as an equation of two variables, then three variables.
 - c) $2x_1 - 3x_2 + x_3 = 2$ as an equation of three variables.
2. [R][V] Determine algebraically whether the following systems of equations have a unique solution, no solution, or an infinite number of solutions. Draw graphs to illustrate your answers.

$$\begin{array}{ll} \text{a)} & \begin{array}{rcl} 3x_1 & + & 2x_2 = 6 \\ 9x_1 & + & 6x_2 = 36 \end{array} & \text{b)} & \begin{array}{rcl} 3x_1 & + & 2x_2 = 6 \\ 9x_1 & + & 4x_2 = 36 \end{array} \end{array}$$

$$\text{c)} \quad \begin{array}{rcl} x_1 & - & 5x_2 = 5 \\ 6x_1 & - & 30x_2 = 30 \end{array}$$

3. [H] Find conditions on the coefficients a_{11} , a_{12} , a_{21} , a_{22} , b_1 , b_2 so that the system of equations

$$\begin{array}{rcl} a_{11}x_1 & + & a_{12}x_2 = b_1 \\ a_{21}x_1 & + & a_{22}x_2 = b_2 \end{array}$$

has a) a unique solution, b) no solution, and c) an infinite number of solutions. For simplicity, assume $a_{11} \neq 0$.

4. [X] Repeat the previous question with no simplifying assumptions. That is, find general conditions which apply for all possible values of the coefficients.
5. [R] Find and geometrically describe the solutions for the following systems of linear equations.

$$\begin{array}{ll} \text{a)} & \begin{array}{rcl} x_1 & + & 2x_2 + 3x_3 = 5 \\ 2x_1 & + & 5x_2 + 8x_3 = 12 \end{array} & \text{b)} & \begin{array}{rcl} 4x_1 & + & 5x_2 - 2x_3 = 16 \\ 8x_1 & + & 10x_2 - 4x_3 = 20 \end{array} \end{array}$$

$$\text{c)} \quad \begin{array}{rcl} 4x_1 & + & 5x_2 - 2x_3 = 16 \\ 8x_1 & + & 10x_2 - 4x_3 = 32 \end{array}$$

6. [X] Prove algebraically that two distinct planes in $\mathbb{R}^3$ either intersect in a line or are parallel with no points in common. Use a linear equation in three unknowns to represent a plane in $\mathbb{R}^3$.

7. [R] Show that $x_1 = 2 - 2\lambda$, $x_2 = \lambda$, $x_3 = 3 + 2\lambda$, where λ is any real number, satisfy the system of equations

$$\begin{array}{rrcrcl} x_1 & + & 4x_2 & - & x_3 & = & -1 \\ 2x_1 & + & 4x_2 & & & = & 4 \\ & & 6x_2 & - & 3x_3 & = & -9 \end{array}$$

Problems 4.2 : Systems of linear equations and matrix notation

8. [R][V] Write each of the following systems of equations in vector form, as a matrix equation $A\mathbf{x} = \mathbf{b}$, and in the augmented matrix $(A|\mathbf{b})$ form.

a)
$$\begin{array}{rrcrcl} 3x_1 & - & 3x_2 & + & 4x_3 & = & 6 \\ 5x_1 & + & 2x_2 & - & 3x_3 & = & 7 \\ -x_1 & - & x_2 & + & 6x_3 & = & 8 \end{array}$$

b)
$$\begin{array}{rrcrcl} x_1 & + & 3x_2 & + & 7x_3 & + & 8x_4 & = & -2 \\ 3x_1 & + & 2x_2 & - & 5x_3 & - & x_4 & = & 7 \\ & & 3x_2 & + & 6x_3 & - & 6x_4 & = & 5 \end{array}$$

9. [R] Write the system of equations, the matrix equation and the augmented matrix form corresponding to the vector equation

$$x_1 \begin{pmatrix} 1 \\ 0 \\ -6 \\ 7 \end{pmatrix} + x_2 \begin{pmatrix} -3 \\ 6 \\ -1 \\ 9 \end{pmatrix} + x_3 \begin{pmatrix} 0 \\ 6 \\ -4 \\ 11 \end{pmatrix} = \begin{pmatrix} 10 \\ -2 \\ 0 \\ 5 \end{pmatrix}.$$

Problems 4.3 : Elementary row operations

10. [R] For each of the following matrices, find the appropriate elementary row operations to describe the transformation from one matrix to the next. Also continue the row reduction until the matrix is in row echelon form.

a)
$$\left(\begin{array}{ccc|c} 1 & 4 & 2 & 3 \\ 2 & 6 & 3 & 0 \\ 4 & -2 & 4 & 4 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & 4 & 2 & 3 \\ 0 & -2 & -1 & -6 \\ 0 & -18 & -4 & -8 \end{array} \right),$$

b)
$$\left(\begin{array}{ccc|c} 3 & 4 & 1 & 3 \\ 2 & 8 & 0 & 2 \\ 0 & 8 & 3 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccc|c} 1 & -4 & 1 & 1 \\ 1 & 4 & 0 & 1 \\ 0 & 8 & 3 & 0 \end{array} \right).$$

11. [M] Write down the output when the Maple command `RowOperation(A, [2,1], 3);` is applied to the matrix

$$A = \left(\begin{array}{ccc|c} 2 & 4 & 1 & 2 \\ 3 & 2 & 4 & 1 \\ 1 & 3 & 1 & 3 \end{array} \right).$$

Problems 4.4 : Solving systems of equations

12. [R] For each of the following augmented matrices do the following. Determine whether the matrix is in row-echelon form as defined in Section 4.4.1. If the matrix is in row-echelon form, identify the leading elements, leading rows, leading columns, and non-leading columns.

$$\text{a) } \left(\begin{array}{ccc|c} 3 & 2 & 1 & 10 \\ 0 & 4 & 2 & 8 \\ 0 & 0 & -7 & 14 \end{array} \right), \quad \text{b) } (\begin{array}{ccc|c} 3 & 2 & 1 & 10 \end{array}), \quad \text{c) } \left(\begin{array}{ccc|c} 3 & 2 & 1 & 10 \\ 4 & 0 & 2 & 8 \\ 0 & 0 & -7 & 14 \end{array} \right),$$

$$\text{d) } \left(\begin{array}{ccc|c} 3 & 2 & 1 & 10 \\ 0 & 4 & 2 & 8 \end{array} \right), \quad \text{e) } \left(\begin{array}{ccc|c} 0 & 3 & 1 & 6 \\ 0 & 0 & 1 & 5 \end{array} \right), \quad \text{f) } \left(\begin{array}{ccc|c} 3 & 2 & 1 & 10 \\ 0 & 4 & 2 & 8 \\ 0 & 0 & -7 & 14 \\ 0 & 0 & 0 & 0 \end{array} \right),$$

$$\text{g) } \left(\begin{array}{ccc|c} 3 & 2 & 1 & 10 \\ 0 & 4 & 2 & 8 \\ 0 & 0 & -7 & 14 \\ 0 & 0 & 0 & 6 \end{array} \right), \quad \text{h) } \left(\begin{array}{ccc|c} 3 & 2 & 1 & 10 \\ 0 & 4 & 2 & 8 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 6 \end{array} \right).$$

13. [R] Find the solutions to the following systems of equations. If possible give a geometric interpretation of the solution.

$$\begin{array}{rclcl} \text{a) } & 3x_1 & + & 2x_2 & + & x_3 & = & 10 \\ & & & 4x_2 & + & 2x_3 & = & 8 \\ & & & & - & 7x_3 & = & 14 \end{array}$$

$$\begin{array}{rclcl} \text{b) } & 3x_1 & + & 2x_2 & + & x_3 & + & x_4 & = & 10 \\ & & & 4x_2 & + & 2x_3 & - & 4x_4 & = & 8 \\ & & & & - & 7x_3 & + & 14x_4 & = & 14 \end{array}$$

14. [R][V] For each of the following systems of equations, do the following:

- i) Write down the corresponding augmented matrix.
- ii) Use Gaussian elimination to transform the augmented matrix into row-echelon form.
- iii) Solve each system of equations writing your answer in vector form.

- a)
$$\begin{array}{rcl} x_1 & - & 2x_2 = 5 \\ 3x_1 & + & x_2 = 8 \end{array}$$
- b)
$$\begin{array}{rcl} x_1 & - & 2x_2 & - & 3x_3 = 3 \\ 2x_1 & + & 4x_2 & + & 10x_3 = 14 \end{array}$$
- c)
$$\begin{array}{rcl} x_1 & - & 2x_2 & + & 3x_3 = 11 \\ 2x_1 & - & x_2 & + & 3x_3 = 10 \\ 4x_1 & + & x_2 & - & x_3 = 4 \end{array}$$
- d)
$$\begin{array}{rcl} 2x_1 & - & 2x_2 & + & 4x_3 = -3 \\ 3x_1 & - & 3x_2 & + & 6x_3 = -4 \\ & & 5x_2 & + & 2x_3 = 9 \end{array}$$
- e)
$$\begin{array}{rcl} x_1 & + & 2x_2 & + & 4x_3 = 10 \\ -3x_1 & + & 3x_2 & + & 15x_3 = 15 \\ -2x_1 & - & x_2 & + & x_3 = -5 \end{array}$$
- f)
$$\begin{array}{rcl} x_1 & - & 4x_2 & - & 5x_3 = -6 \\ 2x_1 & - & x_2 & - & x_3 = 2 \\ 3x_1 & + & 9x_2 & + & 12x_3 = 30 \end{array}$$
- g)
$$\begin{array}{rcl} x_1 & + & 2x_2 & - & x_3 & + & x_4 = 4 \\ & & x_2 & - & x_3 & + & x_4 = 1 \\ 3x_1 & + & 2x_2 & & & - & 2x_4 = 3 \\ 5x_1 & + & 3x_2 & & & - & x_4 = 9 \end{array}$$
- h)
$$\begin{array}{rcl} x_1 & + & 2x_2 & - & x_3 & + & x_4 = 4 \\ & & x_2 & - & x_3 & + & x_4 = 1 \\ 3x_1 & + & 2x_2 & & & - & 2x_4 = 3 \end{array}$$

15. [R] For each of the following augmented matrices, find a *reduced* row-echelon form. Then write down all solutions of the corresponding system of equations and try to give a geometric interpretation of the solutions.

a) $\left(\begin{array}{ccc|c} 2 & 4 & 1 & 4 \\ 0 & 1 & 2 & -2 \\ 0 & 0 & -1 & 2 \end{array} \right),$ b) $\left(\begin{array}{cccc|c} 1 & 2 & 3 & 4 & 1 \\ 0 & -1 & 5 & 6 & 2 \\ 0 & 0 & 1 & 7 & 3 \end{array} \right).$

Problems 4.5 : Deducing solubility from row-echelon form

16. [R][V] For each of the following augmented matrices, **without** solving, decide whether or not the corresponding system of linear equations has a unique solution, no solution or infinitely many solutions.

a) $\left(\begin{array}{ccc|c} 3 & 2 & 1 & 10 \\ 0 & 4 & 2 & 8 \\ 0 & 0 & -7 & 14 \end{array} \right),$ b) $\left(\begin{array}{ccc|c} 3 & 2 & 1 & 10 \\ 0 & 4 & 2 & 8 \\ 0 & 0 & -7 & 14 \\ 0 & 0 & 0 & 6 \end{array} \right),$ c) $(\begin{array}{ccc|c} 3 & 2 & 1 & 10 \end{array}),$

d) $\left(\begin{array}{ccc|c} 3 & 2 & 1 & 10 \\ 0 & 4 & 2 & 8 \end{array} \right),$ e) $\left(\begin{array}{ccc|c} 3 & 2 & 1 & 10 \\ 0 & 4 & 2 & 8 \\ 0 & 0 & -7 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right).$

17. [H][V] Determine which values of k , if any, will give a) a unique solution b) no solution

c) infinitely many solutions to the system of equations

$$\begin{aligned}x + y + kz &= 2 \\ 3x + 4y + 2z &= k \\ 2x + 3y - z &= 1.\end{aligned}$$

18. [H] For which values of λ do the equations

$$\begin{aligned}x + 2y + \lambda z &= 1 \\ -x + \lambda y - z &= 0 \\ \lambda x - 4y + \lambda z &= -1\end{aligned}$$

have a) no solutions, b) infinitely many solutions, c) a unique solution?

19. [H] Consider the equation

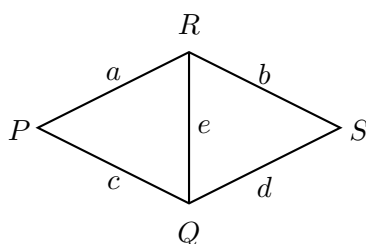
$$\begin{pmatrix} 1 & 2 & 3 & 0 \\ 0 & 2 & 2 & -1 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & a \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ a \\ a + 2b \end{pmatrix}.$$

For what values of a and b does the equation have

- a) a unique solution, b) no solution,
c) infinitely many solutions? d) In the case of (c), determine all solutions.
20. [H] You are an auditor for a company whose four executives make regular business trips on four routes and you suspect that at least one of the executives has been overstating her expenses. You don't know how much it costs to travel each route, but you know that it is the same for all the executives. You know the number of trips each executive made on each route in a certain period and you know the total expenses claimed by each executive for this period. If the numbers of trips are as shown in the table below, do you have sufficient information to be sure that someone is cheating? State your reasoning clearly.

	Route			
	1	2	3	4
Executive A	0	1	1	2
Executive B	1	2	0	1
Executive C	3	4	0	1
Executive D	2	1	3	3

21. [H][V] P, Q, R and S are four cities connected by highways which are labelled as shown in the diagram.



A hire car operator in P makes a note of the number of kilometres travelled by five customers who made trips starting and ending at P . He knows that the routes travelled by the five customers were as follows: abdc abdea cddc cdbec aedbec

Can he determine the length of each of the five highways? State your reasoning clearly.

Problems 4.6 : Solving $Ax = b$ for indeterminate b

22. [R] For each of the following systems of linear equations, find x_1 , x_2 and x_3 in terms of b_1 , b_2 and b_3 .

$$\begin{array}{lcl} \text{a)} & x_1 & - 2x_2 + 3x_3 = b_1 \\ & & x_2 - 3x_3 = b_2 \\ & -2x_1 & + 3x_2 - 2x_3 = b_3 \end{array} \qquad \begin{array}{lcl} \text{b)} & 2x_1 & - 4x_3 = b_1 \\ & 3x_1 + x_2 & - 2x_3 = b_2 \\ & -2x_1 - x_2 & - x_3 = b_3 \end{array}$$

23. [R] Show that the system of equations $x + y + 2z = a$, $x + z = b$ and $2x + y + 3z = c$ are consistent if and only if $c = a + b$.

24. [R] For the following systems, find conditions on the right-hand-side vector b which ensure that the system has a solution.

$$\begin{array}{lcl} \text{a)} & 2x_1 & - 4x_3 = b_1 \\ & 3x_1 + x_2 & - 2x_3 = b_2 \\ & -2x_1 - x_2 & = b_3 \end{array} \qquad \begin{array}{lcl} \text{b)} & x_1 + x_2 + 3x_3 & - x_4 = b_1 \\ & 2x_1 - x_2 & + 2x_4 = b_2 \\ & x_1 - 2x_2 - 3x_3 & + 3x_4 = b_3 \\ & 3x_2 + 6x_3 & - 4x_4 = b_4 \end{array}$$

Problems 4.7 : General properties of the solution of $Ax = b$

25. [R] Show that $x = \begin{pmatrix} 7 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$, $\lambda \in \mathbb{R}$ are the solutions of

$$\begin{array}{rcl} x_1 & - & 2x_2 + 2x_3 = 3 \\ 2x_1 & - & 6x_2 + 4x_3 = 2 \\ -2x_1 & + & 4x_2 - 4x_3 = -6 \end{array}$$

and that $x = \lambda \begin{pmatrix} -2 \\ 0 \\ 1 \end{pmatrix}$, $\lambda \in \mathbb{R}$ are the solutions of the corresponding homogeneous system

$$\begin{array}{rcl} x_1 & - & 2x_2 + 2x_3 = 0 \\ 2x_1 & - & 6x_2 + 4x_3 = 0 \\ -2x_1 & + & 4x_2 - 4x_3 = 0 \end{array}$$

Problems 4.8 : Applications

Some of these problems can be solved using more elementary geometric techniques. But try your best to also find solutions that involve solving systems of linear equations.

26. [R] Does the point $(-3, 3, 6)$ lie on the plane

$$\mathbf{x} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} + \lambda_1 \begin{pmatrix} -1 \\ 2 \\ 4 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}?$$

27. [R] Is the vector $\begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$ in $\text{span} \left(\begin{pmatrix} -1 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} \right)$?

28. [R] Is the vector $\begin{pmatrix} 1 \\ 1 \\ 4 \\ 12 \end{pmatrix}$ in $\text{span} \left(\begin{pmatrix} 3 \\ -1 \\ 4 \\ 6 \end{pmatrix}, \begin{pmatrix} 4 \\ -2 \\ 4 \\ 3 \end{pmatrix} \right)$?

29. [R] Can $\begin{pmatrix} 3 \\ 1 \\ -2 \\ 4 \end{pmatrix}$ be expressed as a linear combination of $\begin{pmatrix} 1 \\ 0 \\ -3 \\ 7 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -1 \\ 5 \\ 6 \end{pmatrix}$?

30. [R] Do the lines $\mathbf{x} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}$ and $\mathbf{x} = \begin{pmatrix} 12 \\ 15 \\ 7 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix}$ intersect?

31. [R] Is the vector $\begin{pmatrix} 5 \\ 7 \\ -1 \end{pmatrix}$ parallel to the plane $\mathbf{x} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix}$ for $\lambda_1, \lambda_2 \in \mathbb{R}$?

32. [H] Show that the line $\frac{x-1}{2} = \frac{y}{3} = \frac{z+1}{-1}$ is parallel to the plane

$$\mathbf{x} = \lambda_1 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}, \quad \lambda_1, \lambda_2 \in \mathbb{R}.$$

33. [H] Find the intersection (if any) of the line $\mathbf{x} = \begin{pmatrix} 0 \\ 18 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$ for $\mu \in \mathbb{R}$ and the plane

$$\mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix} \text{ for } \lambda_1, \lambda_2 \in \mathbb{R}.$$

34. [R] Find the intersection (if any) of the planes $8x_1 + 8x_2 + x_3 = 35$ and

$$\mathbf{x} = \begin{pmatrix} 6 \\ -2 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} -2 \\ 1 \\ 3 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \text{ for } \lambda_1, \lambda_2 \in \mathbb{R}.$$

35. [H] Are the planes

$$\mathbf{x} = \begin{pmatrix} 1 \\ -4 \\ 2 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 2 \\ 1 \\ -2 \\ 7 \end{pmatrix} + \lambda_2 \begin{pmatrix} -3 \\ 1 \\ 5 \\ 2 \end{pmatrix} \text{ for } \lambda_1, \lambda_2 \in \mathbb{R}$$

and

$$\mathbf{x} = \begin{pmatrix} 2 \\ -4 \\ 1 \\ 3 \end{pmatrix} + \mu_1 \begin{pmatrix} 3 \\ -1 \\ 2 \\ 4 \end{pmatrix} + \mu_2 \begin{pmatrix} -1 \\ 4 \\ 2 \\ 6 \end{pmatrix} \text{ for } \mu_1, \mu_2 \in \mathbb{R}$$

parallel?

36. [R] Show that the 3 planes with Cartesian equations

$$\begin{aligned} x + 3y + 2z &= 5 \\ 2x + y - z &= 2 \\ 7x + 11y + 4z &= 13 \end{aligned}$$

do not intersect at one point.

37. [H] Consider the following system of equations

$$\begin{aligned} x + y - z &= 1 \\ 2x - 4y + 2z &= 2 \\ 3x - 3y + z &= 3 \end{aligned}$$

- Use Gaussian elimination and back-substitution to find the solution(s), if any, of the above equations.
- Use your result in part a) to decide whether the three planes represented by the equations are parallel, intersecting in a straight line, intersecting at a point or have some other configuration.

38. [R] Find a polynomial $p(x)$ of degree 2 satisfying $p(1) = 5$, $p(2) = 7$, $p(3) = 13$.

39. [R] The total of the ages of my brother, my sister and myself is 140 years. I am seven times the difference between their ages (my sister is older than my brother) and in seven years I will be half their combined ages now. How old are we?

40. [R] In a trip to Asia a traveller spent \$90 a day for hotels in Bangkok, \$60 a day in Singapore and \$60 a day in Kuala Lumpur. For food the traveller spent \$60 a day in Bangkok, \$90 a day in Singapore, and \$60 a day in Kuala Lumpur. In addition the traveller spent \$30 a day in other expenses in each city. The traveller's diary shows that the total hotel bill was \$1020, total food bill was \$960, and total other expenses were \$420. Find the number of days the traveller spent in each city, or show that the diary must be wrong.
41. [R] A dietician is planning a meal consisting of three foods. A serving of the first food contains 5 units of protein, 2 units of carbohydrates and 3 units of iron. A serving of the second food contains 10 units of protein, 3 units of carbohydrates and 6 units of iron. A serving of the third food contains 15 units of protein, 2 units of carbohydrates and 1 unit of iron. How many servings of each food should be used to create a meal containing 55 units of protein, 13 units of carbohydrates and 17 units of iron?
42. [X] A farmer owns a 12-hectare farm on which he grows wheat, oats and barley. Each hectare of cereal crop planted has certain requirements for labour, fertiliser and irrigation water as shown in the following table.

Crop	Labour (hours per week)	Fertiliser (kilograms)	Irrigation Water kilolitres)
Wheat (per hectare)	6	150	72
Oats (per hectare)	6	100	48
Barley (per hectare)	2	70	36
Amount available	48	700	612

Answer the following questions

- Set up a linear equation model for the system.
 - Find the solution (if any) of the model.
 - Replace the equations by inequalities assuming that not all of the available land, labour time, fertiliser and irrigation water have to be used. Then introduce 4 new "slack" variables which represent the amounts of unused land, labour, fertiliser and irrigation water respectively.
 - Can you find any reasonable solutions for the systems of linear equations in (c), i.e. solutions in which the variables are non-negative?
43. [X] In this problem we shall calculate the area of a spherical triangle. Consider the surface of a sphere of unit radius of area 4π . A great circle on a sphere is the intersection of that sphere with a plane through the centre. If two great circles meet at antipodal points P, P' let the angle θ between them be the angle $0 < \theta < \pi$ between the tangents to the two circles at P . ($\pi - \theta$ is also the angle between the two great circles). Finally define a spherical triangle to be the region bounded by 3 great circles meeting A, B, C with angles α, β, γ .
- The areas bounded by 2 great circles are called lunes. Show their areas are 2θ , $2(\pi - \theta)$.

- b) Show the surface of the sphere is divided by a spherical triangle into 8 regions equal in area in pairs.
- c) Use parts a) and b) to set up a simple system of 4 linear equations in the 4 areas.
- d) Hence show $\text{area } ABC = \alpha + \beta + \gamma - \pi$.

Problems 4.9 : Matrix reduction and Maple

44. [M] The Maple session below (for which the package `LinearAlgebra` has been loaded) calculates the intersection of 3 planes Π_1 , Π_2 and Π_3 .

- a) Write down the cartesian equations for Π_1 , Π_2 and Π_3 .
- b) Give a full geometric description of the intersection of Π_1 , Π_2 and Π_3 .
- c) Express the intersection of Π_1 , Π_2 and Π_3 in cartesian form.

```
> A:=<<1,3,2>|<2,6,4>|<-1,-1,-1>>;
```

$$A := \begin{bmatrix} 1 & 2 & -1 \\ 3 & 6 & -1 \\ 2 & 4 & -1 \end{bmatrix}$$

```
> b:=<2,12,7>;
```

$$b := \begin{bmatrix} 2 \\ 12 \\ 7 \end{bmatrix}$$

```
> LinearSolve(A,b);
```

$$\begin{bmatrix} 5 - 2t_2 \\ t_2 \\ 3 \end{bmatrix}$$

45. [M] > with(LinearAlgebra):

```
> A:=<<1,3,4,7>|<-2,6,2,-8>|<1,8,7,6>|<a,b,c,d>>;
```

$$A := \begin{bmatrix} 1 & -2 & 1 & a \\ 3 & 6 & 8 & b \\ 4 & 2 & 7 & c \\ 7 & -8 & 6 & d \end{bmatrix}$$

```
> GaussianElimination(A);
```

$$\begin{bmatrix} 1 & -2 & 1 & a \\ 0 & 12 & 5 & b - 3a \\ 0 & 0 & -\frac{7}{6} & c - \frac{3a}{2} - \frac{5b}{6} \\ 0 & 0 & 0 & d - a + 2b - 3c \end{bmatrix}$$

```
>
```

- a) The above is a Maple session designed to calculate where 4 planes in $\mathbb{R}^3$ meet. What are the equations of the planes?
- b) What are the condition(s) on a, b, c, d for the planes to meet at a point?
- c) If $a = 1$, $b = 2$, $c = 3$ and the planes meet, where do they meet?

Chapter 5 MATRICES

*“It seems very pretty,” she said when she had finished it,
 “but it’s RATHER hard to understand! . . .
 . . . Somehow it seems to fill my head with ideas
 — only I don’t exactly know what they are!”
 Lewis Carroll, Through the Looking Glass.*

In Chapter 4, matrices are treated as devices to help in writing systems of linear equations. They are also important in many other areas of mathematics and their applications.

In this chapter, we shall define the addition of two matrices, the multiplication of two matrices and the multiplication of a matrix by a scalar. Under these operations of addition and scalar multiplication matrices behave much like vectors and indeed like real numbers. For instance, both matrices and real numbers obey associative laws and distributive laws. Yet, there are differences between matrices and real numbers. Multiplication of real numbers is commutative, but, as we shall see, matrix multiplication is not. Every non-zero real number has an multiplicative inverse which is the reciprocal of that number. However, not all matrices have inverses. We shall study the matrix arithmetic and algebra and an important function on the set of square matrices—the determinant function.

5.1 Matrix arithmetic and algebra

In this section we describe the arithmetic of matrices, including equality and addition of matrices, multiplication of a matrix by a scalar, and multiplication of matrices. In general, **division is not defined for matrices**, although an inverse can be defined for some matrices (see Section 5.3).

Definition 1. An $m \times n$ (read “ m by n ”) **matrix** is an array of m rows and n columns of numbers of the form

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}.$$

The number a_{ij} in the matrix A lies in row i and column j . It is called the ij th **entry** or ij th **element** of the matrix.

NOTE.

1. An $m \times 1$ matrix is called a column vector, while a $1 \times n$ matrix is called a row vector.
2. We use M_{mn} to stand for the set of all $m \times n$ matrices, i.e., the set of all matrices with m rows and n columns.

In general, we will assume that the entries in a matrix can be complex numbers. However, it is sometimes necessary to distinguish between real matrices, in which all entries a_{ij} are real numbers, and complex matrices, in which all a_{ij} are complex numbers. In this case, we use $M_{mn}(\mathbb{R})$ for the set of all real $m \times n$ matrices and $M_{mn}(\mathbb{C})$ for the set of all complex $m \times n$ matrices. Likewise, $M_{mn}(\mathbb{Q})$ is used for the set of all rational $m \times n$ matrices.

3. We say an $m \times n$ matrix is of **size** $m \times n$.
4. When we say “let $A = (a_{ij})$ ”, we are specifying a matrix of fixed size, in which, for each given i, j , the ij th entry is a_{ij} . On the other hand, for a given matrix A , we denote the entry in the i th row and j th column by $[A]_{ij}$.

Example 1. The matrix $A = \begin{pmatrix} 1 & 2 & -1 & 2 \\ 3 & -4 & 2 & 5 \\ -6 & -3 & 1 & 4 \end{pmatrix}$ is of size 3×4 and $[A]_{24} = 5$. ◇

When the number of rows or columns of a matrix is larger than 9, we need to use a comma to separate the row number and the column number. If B is a matrix of size 5×12 , we shall use $[B]_{2,4}$ and $[B]_{3,11}$ to denote the entry in the second row and fourth column and the entry in the third row and the eleventh column, respectively.

5.1.1 Equality, addition and multiplication by a scalar

The rules for equality of matrices, addition of matrices and multiplication of a matrix by a scalar are straightforward generalisations of the corresponding rules for vectors in $\mathbb{R}^n$.

Definition 2. Two matrices A and B are defined to be **equal** if

1. the number of rows of A equals the number of rows of B ,
2. the number of columns of A equals the number of columns of B ,
3. $[A]_{ij} = [B]_{ij}$ for all i and j .

To summarise, equal matrices have the same size and their corresponding entries are equal. Addition of matrices is defined as follows.

Definition 3. If A and B are $m \times n$ matrices, then the **sum** $C = A + B$ is the $m \times n$ matrix whose entries are

$$[C]_{ij} = [A]_{ij} + [B]_{ij} \quad \text{for all } i, j.$$

To understand the proofs of the properties of matrices, we need to know the notation $[A]_{ij}$ well. The row i column j entry of the matrix $A + B$ is denoted by $[A + B]_{ij}$. The definition simply says that this entry is the same as $[A]_{ij} + [B]_{ij}$, that is the sum of the corresponding entries of the matrices A and B .

Note that addition of matrices of different sizes is **not defined**.

Example 2.

$$\begin{pmatrix} 2 & 3 & 5 \\ 4 & -3 & 2 \end{pmatrix} + \begin{pmatrix} -1 & 1 & 4 \\ 2 & 4 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 4 & 9 \\ 6 & 1 & 1 \end{pmatrix}$$

but

$$\begin{pmatrix} -1 & 1 & 4 \\ 2 & 4 & -1 \end{pmatrix} + \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix}$$

is not defined as the sizes are different. ◇

Proposition 1. Let A , B and C be $m \times n$ matrices.

1. $A + B = B + A$. (Commutative Law of Addition)
2. $(A + B) + C = A + (B + C)$. (Associative Law of Addition)

Proof. We only prove the Associative law.

$$\begin{aligned} [(A + B) + C]_{ij} &= [A + B]_{ij} + [C]_{ij} && \text{(by definition, the sum of } (A + B) \text{ and } C) \\ &= ([A]_{ij} + [B]_{ij}) + [C]_{ij} && \text{(by definition, the sum of } A \text{ and } B) \\ &= [A]_{ij} + ([B]_{ij} + [C]_{ij}) && \text{(associative law of numbers)} \\ &= [A]_{ij} + [B + C]_{ij} && \text{(by definition, the sum of } B \text{ and } C) \\ &= [A + (B + C)]_{ij} && \text{(by definition, the sum of } A \text{ and } (B + C)) \end{aligned}$$

Hence $(A + B) + C = A + (B + C)$. □

We now define what is meant by a zero matrix.

Definition 4. A **zero matrix** (written $\mathbf{0}$) is a matrix in which every entry is zero.

Example 3.

$$\mathbf{0} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \quad \text{is the zero matrix in } M_{22}.$$

and

$$\mathbf{0} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \quad \text{is the zero matrix in } M_{23}.$$

◇

Note that when writing the zero matrix by hand, you can write $\underline{0}$ to distinguish it from the zero vector and the number 0.

Proposition 2. Let A be a matrix and $\mathbf{0}$ be the zero matrix, both in M_{mn} . Then

$$A + \mathbf{0} = \mathbf{0} + A = A.$$

Proof. Similar to the proof of the associative law of addition but skipping the detail explanation, we have

$$[A + \mathbf{0}]_{ij} = [A]_{ij} + 0 = [A]_{ij}.$$

Hence $A + \mathbf{0} = A$. We can also prove that $\mathbf{0} + A = A$ in a similar way. □

Definition 5. For any matrix $A \in M_{mn}$, the **negative** of A is the $m \times n$ matrix $-A$ with entries

$$[-A]_{ij} = -[A]_{ij} \quad \text{for all } 1 \leq i \leq m, 1 \leq j \leq n.$$

Using the above definition of the negative of a matrix, we can define subtraction by

$$A - B = A + (-B) \quad \text{for all } A, B \in M_{mn}.$$

As with addition, we cannot subtract one matrix from another one of different size.

Example 4. Let $A = \begin{pmatrix} 1 & 3 & -2 \\ 2 & -1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & -4 & 5 \\ 3 & 0 & -2 \end{pmatrix}$. We have

$$-B = \begin{pmatrix} -2 & 4 & -5 \\ -3 & 0 & 2 \end{pmatrix} \quad \text{and} \quad A - B = \begin{pmatrix} -1 & 7 & -7 \\ -1 & -1 & 5 \end{pmatrix}.$$

◇

Proposition 3. If A is an $m \times n$ matrix, then

$$A + (-A) = (-A) + A = \mathbf{0}.$$

We now define multiplication of a matrix by a scalar, that is, by a number (real or complex).

Definition 6. If A is an $m \times n$ matrix and λ is a scalar, then the **scalar multiple** $B = \lambda A$, of A is the $m \times n$ matrix whose entries are

$$[B]_{ij} = \lambda[A]_{ij} \text{ for all } 1 \leq i \leq m, 1 \leq j \leq n.$$

that is, each entry of the matrix is multiplied by the scalar λ .

Example 5.

$$3 \begin{pmatrix} 2 & 3 & 5 \\ 4 & -3 & 2 \end{pmatrix} = \begin{pmatrix} 6 & 9 & 15 \\ 12 & -9 & 6 \end{pmatrix}.$$

◇

Proposition 4. Let λ, μ be scalars and A, B be matrices in M_{mn} .

1. $\lambda(\mu A) = (\lambda\mu)A$ (Associative Law of Scalar Multiplication)
2. $(\lambda + \mu)A = \lambda A + \mu A$ (Scalar Multiplication is distributive over Scalar Addition)
3. $\lambda(A + B) = \lambda A + \lambda B$ (Scalar Multiplication is distributive over Matrix Addition)

The above rules for equality, matrix addition and multiplication by a scalar are essentially the same as the corresponding rules for vectors in $\mathbb{R}^n$. The matrix operations also obey the basic laws of arithmetic as do addition and multiplication by a scalar in $\mathbb{R}^n$. For instance, in both M_{mn} and $\mathbb{R}^n$, the commutative laws, associative laws and distributive laws hold.

5.1.2 Matrix multiplication

In Section 4.2, we represented the system of linear equations

$$\begin{array}{cccccc} a_{11}x_1 & + & a_{12}x_2 & + & \cdots & + & a_{1n}x_n & = & b_1 \\ a_{21}x_1 & + & a_{22}x_2 & + & \cdots & + & a_{2n}x_n & = & b_2 \\ \vdots & & \vdots & & & & \vdots & & \vdots \\ a_{m1}x_1 & + & a_{m2}x_2 & + & \cdots & + & a_{mn}x_n & = & b_m \end{array}$$

by the matrix equation $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}.$$

It makes sense to use the above relation between the matrix equation $A\mathbf{x} = \mathbf{b}$ and the system of linear equations as the basis for the definition of the product of two matrices. The column vector $\mathbf{x} \in \mathbb{R}^n$ may be regarded as an $n \times 1$ matrix and the column vector $\mathbf{b} = A\mathbf{x} \in \mathbb{R}^m$ may be regarded as an $m \times 1$ matrix.

Definition 7. If $A = (a_{ij})$ is an $m \times n$ matrix and $\mathbf{x}$ is an $n \times 1$ matrix with entries x_i , then the **product** $\mathbf{b} = A\mathbf{x}$ is the $m \times 1$ matrix whose entries are given by

$$b_i = a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n = \sum_{k=1}^n a_{ik}x_k \quad \text{for } 1 \leq i \leq m.$$

Example 6. If A is a 3×2 matrix and $\mathbf{x}$ is a 2×1 matrix given by

$$A = \begin{pmatrix} 2 & -3 \\ 5 & -1 \\ -7 & 4 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} 6 \\ 9 \end{pmatrix}$$

then the product $A\mathbf{x}$ is the 3×1 matrix

$$A\mathbf{x} = \begin{pmatrix} 2(6) + (-3)(9) \\ 5(6) + (-1)(9) \\ (-7)(6) + 4(9) \end{pmatrix} = \begin{pmatrix} -15 \\ 21 \\ -6 \end{pmatrix}.$$

◇

To obtain a suitable general definition of multiplication of two matrices, we replace the matrix $\mathbf{x}$ with 1 column by a matrix X with p columns. We can then think of the matrix A multiplying each of the columns of X using the above rule.

Definition 8. Let A be an $m \times n$ matrix and X be an $n \times p$ matrix and let $\mathbf{x}_j$ be the j th column of X . Then the **product** $B = AX$ is the $m \times p$ matrix whose j th column $\mathbf{b}_j$ is given by

$$\mathbf{b}_j = A\mathbf{x}_j \quad \text{for } 1 \leq j \leq p.$$

The matrix X is the **augmented matrix** of the p column vectors, i.e. $X = (\mathbf{x}_1 | \cdots | \mathbf{x}_p)$. So the definition can be written as

$$A(\mathbf{x}_1 | \cdots | \mathbf{x}_p) = (A\mathbf{x}_1 | \cdots | A\mathbf{x}_p).$$

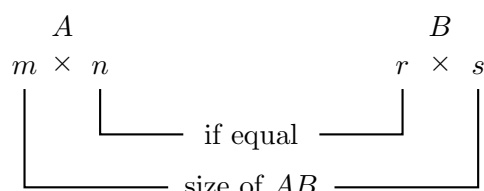
By combining Definitions 6 and 7, we can give an equivalent definition of a matrix product in terms of the entries of the matrices as:

Definition 9. (Alternative) If A is an $m \times n$ matrix and X is an $n \times p$ matrix, then the **product** AX is the $m \times p$ matrix whose entries are given by the formula

$$[AX]_{ij} = [A]_{i1}[X]_{1j} + \cdots + [A]_{in}[X]_{nj} = \sum_{k=1}^n [A]_{ik}[X]_{kj} \quad \text{for } 1 \leq i \leq m, 1 \leq j \leq p.$$

NOTE.

1. Let A be an $m \times n$ matrix and B be an $r \times s$ matrix. The product AB is defined **only** when the number of columns of A is the same as the number of rows of B , i.e. $n = r$. If $n = r$, the size of AB will be $m \times s$.



2. Matrix multiplication is a row-times-column process: we get the (row i , column j) entry of AX by going across the i th row of A and down the j th column of X multiplying and adding as we go.

Example 7. For

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & -1 \end{pmatrix} \quad \text{and} \quad X = \begin{pmatrix} 1 & -1 & 2 \\ 2 & -2 & 3 \\ 3 & -3 & 1 \end{pmatrix}$$

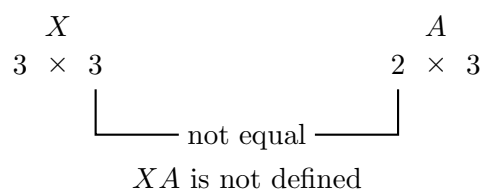
the product $AX = (b_{ij})$ can be easily obtained.

$$A \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 14 \\ -2 \end{pmatrix}, \quad A \begin{pmatrix} -1 \\ -2 \\ -3 \end{pmatrix} = \begin{pmatrix} -14 \\ 2 \end{pmatrix}, \quad A \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} = \begin{pmatrix} 11 \\ 1 \end{pmatrix}.$$

$$\text{Hence} \quad AX = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 0 & -1 \end{pmatrix} \begin{pmatrix} 1 & -1 & 2 \\ 2 & -2 & 3 \\ 3 & -3 & 1 \end{pmatrix} = \begin{pmatrix} 14 & -14 & 11 \\ -2 & 2 & 1 \end{pmatrix}.$$

◇

Warning. For some A and X , $AX \neq XA$, i.e., matrix multiplication does **not** satisfy the commutative law. This can happen when one product is defined and the other is not. It can happen even when both are defined. Thus, for the A and X just given, the product XA is not defined as the number of columns of X does not equal the number of rows of A .



Here is an example in which both AX and XA are defined but are different:

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \quad \text{and} \quad X = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix},$$

in which case

$$AX = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 4 & 8 \end{pmatrix},$$

whereas

$$XA = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 6 \\ 2 & 8 \end{pmatrix}.$$

A matrix is said to be **square** if it has the same number of rows as columns. For example, of the four matrices displayed below, the second and fourth are square:

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \quad \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

The **diagonal** of a square matrix consists of the positions on the line from the top left to the bottom right. More precisely, the diagonal entries of an $n \times n$ square matrix (a_{ij}) are $a_{11}, a_{22}, \dots, a_{nn}$. The matrix is said to be an **identity matrix** if its diagonal entries are all 1 and all other entries are 0. The following are identity matrices

$$(1) \quad \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

For each integer n there is one and only one $n \times n$ identity matrix. It is denoted by I_n , or just by I if there is no risk of ambiguity.

Definition 10. An **identity matrix** (written I) is a square matrix with 1's on the diagonal and 0's off the diagonal.

Example 8.

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{is the identity matrix in } M_{22}.$$

and

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad \text{is the identity matrix in } M_{33}.$$

◇

Although matrix multiplication does not satisfy a commutative law of multiplication, the operation does satisfy some laws.

Proposition 5 (Properties of Matrix Multiplication). Let A , B , C be matrices and λ be a scalar.

1. If the product AB exists, then $A(\lambda B) = \lambda(AB) = (\lambda A)B$.
2. **Associative Law of Matrix Multiplication.** If the products AB and BC exist, then $A(BC) = (AB)C$.
3. $AI = A$ and $IA = A$, where I represents identity matrices of the appropriate (possibly different) sizes.
4. **Right Distributive Law.** If $A + B$ and AC exist, then $(A + B)C = AC + BC$.
5. **Left Distributive Law.** If $B + C$ and AB exist, then $A(B + C) = AB + AC$.

Proof. We will prove the right distributive law, and leave the proof of the remainder as a problem at the end of the chapter. As $A + B$ exists, A and B must be the same size. Also, as AC exists, the number of columns of A must be equal to the number of rows of C . Therefore we let A, B be $m \times n$ matrices and C be an $n \times p$ matrix. Then BC also exists and the row i , column j entry of $(A + B)C$ is

$$\begin{aligned} [(A + B)C]_{ij} &= \sum_{k=1}^n [A + B]_{ik} [C]_{kj} \\ &= \sum_{k=1}^n ([A]_{ik} + [B]_{ik}) [C]_{kj} \\ &= \sum_{k=1}^n [A]_{ik} [C]_{kj} + \sum_{k=1}^n [B]_{ik} [C]_{kj} \\ &= [AC]_{ij} + [BC]_{ij} \\ &= [AC + BC]_{ij}. \end{aligned}$$

Hence $(A + B)C = AC + BC$ as claimed. □

Example 9. If $A \in M_{23}$ is given by

$$A = \begin{pmatrix} 1 & 2 & 4 \\ -3 & 5 & 7 \end{pmatrix},$$

then by direct multiplication

$$AI = \begin{pmatrix} 1 & 2 & 4 \\ -3 & 5 & 7 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = A,$$

and

$$IA = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 4 \\ -3 & 5 & 7 \end{pmatrix} = A.$$

Note that in this case the right and left I 's must be of different sizes for the matrix products to exist. ◇

Example 10. Here is an example to verify the associative law of multiplication. For

$$A = \begin{pmatrix} 2 & 1 & -2 \\ 3 & 4 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} -1 & 5 \\ 0 & -3 \\ 2 & 6 \end{pmatrix}, \quad C = \begin{pmatrix} -5 & 2 \\ -3 & -6 \end{pmatrix},$$

we have

$$A(BC) = \begin{pmatrix} 2 & 1 & -2 \\ 3 & 4 & 2 \end{pmatrix} \begin{pmatrix} -10 & -32 \\ 9 & 18 \\ -28 & -32 \end{pmatrix} = \begin{pmatrix} 45 & 18 \\ -50 & -88 \end{pmatrix}$$

and

$$(AB)C = \begin{pmatrix} -6 & -5 \\ 1 & 15 \end{pmatrix} \begin{pmatrix} -5 & 2 \\ -3 & -6 \end{pmatrix} = \begin{pmatrix} 45 & 18 \\ -50 & -88 \end{pmatrix}.$$

◇

Using the properties of matrix operations, we can simplify expressions in unknown matrices in almost the same way as simplifying expressions in algebra.

Example 11. By distributive laws and associative laws, we can write

$$A(A + 2B)A = A^3 + 2ABA.$$

Here A^3 obviously means AAA . Thanks to the associative law of matrix multiplication, we do not have to specify ABA to be $(AB)A$ or $A(BA)$ because $(AB)A = A(BA)$. Note that we cannot write the second term as $2A^2B$ because matrix multiplication is **not** commutative.

Example 12. Expand the expression $(A + I)^2$.

SOLUTION.

$$\begin{aligned} (A + I)^2 &= (A + I)(A + I) \\ &= (A + I)A + (A + I)I \\ &= (A^2 + IA) + (AI + I) \\ &= A^2 + A + A + I \\ &= A^2 + 2A + I \end{aligned}$$

Can you identify all the rules used in each step?

◇

Note that, in general, $(A + B)^2 \neq A^2 + 2AB + B^2$.

5.1.3 Matrix arithmetic and systems of linear equations

In Section 4.2 we used the matrix notation $A\mathbf{x} = \mathbf{b}$ as a shorthand notation for a system of linear equations. Then we used this as the motivation to define multiplication in Definition 7 in 5.1.2. That is, when we multiply the coefficient matrix A with the unknown column vector $\mathbf{x}$, we obtain a column vector $A\mathbf{x}$. If we equate the components of the column vector $A\mathbf{x}$ with those of $\mathbf{b}$, we shall recover the system of equations. Thus $A\mathbf{x} = \mathbf{b}$ is not simply a shorthand notation, it *is* a matrix equation. In other words, we have the following proposition.

Proposition 6. Let $A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}$, $\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} b_1 \\ \vdots \\ b_n \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$.

The vector $\mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}$ is a solution to the matrix equation $A\mathbf{x} = \mathbf{b}$, i.e. $A\mathbf{v} = \mathbf{b}$ if and only if $x_1 = v_1, \dots, x_n = v_n$ is a solution to the system of equations

$$\begin{cases} a_{11}x_1 + \cdots + a_{1n}x_n = b_1 \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n = b_m \end{cases}.$$

Using this we can rewrite the proof of the propositions in Section 4.7 in a much shorter way. As an example, let us rewrite the proof of the following.

Proposition 2 (Section 4.7). If $\mathbf{v}$ and $\mathbf{w}$ are solutions of $A\mathbf{x} = \mathbf{0}$ then so are $\mathbf{v} + \mathbf{w}$ and $\lambda\mathbf{v}$ for any scalar λ .

Proof. Since $\mathbf{v}$ and $\mathbf{w}$ are solutions of $A\mathbf{x} = \mathbf{0}$, we have

$$A(\mathbf{v} + \mathbf{w}) = A\mathbf{v} + A\mathbf{w} = \mathbf{0} + \mathbf{0} = \mathbf{0}.$$

Hence $\mathbf{v} + \mathbf{w}$ is also a solution.

On the other hand, we have

$$A(\lambda\mathbf{v}) = \lambda(A\mathbf{v}) = \lambda\mathbf{0} = \mathbf{0}.$$

Thus $\lambda\mathbf{v}$ is also a solution. □

5.2 The transpose of a matrix

Definition 1. The **transpose** of an $m \times n$ matrix A is the $n \times m$ matrix A^T (read ‘ A transpose’) with entries given by

$$[A^T]_{ij} = [A]_{ji}.$$

Example 1. The transpose of

$$A = \begin{pmatrix} 3 & 2 \\ -4 & 0 \\ 0 & 1 \\ 1 & -5 \end{pmatrix} \quad \text{is} \quad A^T = \begin{pmatrix} 3 & -4 & 0 & 1 \\ 2 & 0 & 1 & -5 \end{pmatrix}.$$

◇

Example 2. The transpose of

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} \in M_{23} \quad \text{is} \quad A^T = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{13} & a_{23} \end{pmatrix} \in M_{32}.$$

◇

NOTE. The columns of A^T are the rows of A and the rows of A^T are the columns of A .

5.2.1 Some uses of transposes

Vectors. As noted in Chapter 3, column vectors are often written as the transpose of ‘row vectors’.

Example 3. The transpose of the column vector

$$\mathbf{x} = \begin{pmatrix} 2 \\ 5 \\ -2 \\ 1 \end{pmatrix}$$

is the row vector $\mathbf{x}^T = (2 \ 5 \ -2 \ 1) = \mathbf{y}$; conversely, $\mathbf{x} = \mathbf{y}^T$.

◇

Products of Vectors. So far we have not defined the product of two vectors. However, the product of a row vector and a column vector makes sense as a special case of matrix multiplication.

Example 4. Let $\mathbf{a}$ and $\mathbf{b}$ be two vectors in $\mathbb{R}^n$. Then, using the usual rule for matrix multiplication, we have

$$\mathbf{a}^T \mathbf{b} = (a_1 \ a_2 \ \dots \ a_n) \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix} = a_1 b_1 + a_2 b_2 + \dots + a_n b_n.$$

This product is a 1×1 matrix¹, which we sometimes regard simply as a scalar, and so $\mathbf{a}^T \mathbf{b}$ is often called the **scalar product** of the two vectors $\mathbf{a}$ and $\mathbf{b}$ in $\mathbb{R}^n$. Note also that $\mathbf{b}^T \mathbf{a} = \mathbf{a}^T \mathbf{b}$. ◇

Example 5. If

$$\mathbf{a} = \begin{pmatrix} 1 \\ 4 \\ -2 \\ 3 \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} 2 \\ -3 \\ 0 \\ 6 \end{pmatrix}$$

then

$$\mathbf{a}^T \mathbf{b} = (1 \ 4 \ -2 \ 3) \begin{pmatrix} 2 \\ -3 \\ 0 \\ 6 \end{pmatrix} = 2 - 12 + 0 + 18 = 8 = (2 \ -3 \ 0 \ 6) \begin{pmatrix} 1 \\ 4 \\ -2 \\ 3 \end{pmatrix} = \mathbf{b}^T \mathbf{a}.$$

¹ Technically, we should write the matrix product $\mathbf{a}^T \mathbf{b}$ as a matrix with brackets. In practice we write the result as a single number without brackets. Thus if $\mathbf{a} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 2 \\ 3 \end{pmatrix}$, then we write $\mathbf{a}^T \mathbf{b} = 8$, not $\mathbf{a}^T \mathbf{b} = (8)$.

The order of multiplication is very important. If $\mathbf{a}$ and $\mathbf{b}$ are (column) vectors, then the product $\mathbf{ab}^T$ is not a scalar but a matrix. $\diamond$

Example 6. For $\mathbf{a} \in \mathbb{R}^m$ and $\mathbf{b} \in \mathbb{R}^n$

$$\mathbf{ab}^T = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_m \end{pmatrix} (b_1 \ b_2 \ \cdots \ b_n) = \begin{pmatrix} a_1 b_1 & a_1 b_2 & \cdots & a_1 b_n \\ a_2 b_1 & a_2 b_2 & \cdots & a_2 b_n \\ \vdots & \vdots & \ddots & \vdots \\ a_m b_1 & a_m b_2 & \cdots & a_m b_n \end{pmatrix}.$$

 $\diamond$

Example 7. If

$$\mathbf{a} = \begin{pmatrix} 1 \\ 4 \\ -2 \\ 3 \end{pmatrix} \quad \text{and} \quad \mathbf{b} = \begin{pmatrix} 2 \\ -3 \\ 0 \\ 6 \end{pmatrix}$$

then

$$\mathbf{ab}^T = \begin{pmatrix} 2 & -3 & 0 & 6 \\ 8 & -12 & 0 & 24 \\ -4 & 6 & 0 & -12 \\ 6 & -9 & 0 & 18 \end{pmatrix}.$$

 $\diamond$

Note that the expressions $\mathbf{ab}$, $\mathbf{a}^T \mathbf{b}^T$ have no meaning. The expression $\mathbf{ba}^T$ has a meaning as a matrix. However, $\mathbf{ba}^T \neq \mathbf{ab}^T$ in general.

Example 8. For the vectors $\mathbf{a}$ and $\mathbf{b}$ of Example 7

$$\mathbf{ba}^T = \begin{pmatrix} 2 & 8 & -4 & 6 \\ -3 & -12 & 6 & -9 \\ 0 & 0 & 0 & 0 \\ 6 & 24 & -12 & 18 \end{pmatrix}.$$

Note that the matrix in Example 8 is the transpose of the matrix in Example 7. $\diamond$

5.2.2 Some properties of transposes

Proposition 1. The transpose of a transpose is the original matrix, i.e., $(A^T)^T = A$.

Proof. A transpose is obtained by changing the order of the subscripts on each entry, and hence

$$[(A^T)^T]_{ij} = [A^T]_{ji} = [A]_{ij}.$$

Thus all matrix entries of $(A^T)^T$ are equal to corresponding entries of A , and hence the matrices are equal. $\square$

Example 9.

$$\left(\left(\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}^T \right)^T \right) = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}^T = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}.$$

◇

Proposition 2. If $A, B \in M_{mn}$ and $\lambda, \mu \in \mathbb{R}$, then $(\lambda A + \mu B)^T = \lambda A^T + \mu B^T$.

Proof. For $A, B \in M_{mn}$ and $\lambda, \mu \in \mathbb{R}$, the matrix $(\lambda A + \mu B)^T$ has matrix entries

$$\begin{aligned} [(\lambda A + \mu B)^T]_{ij} &= [\lambda A + \mu B]_{ji} \\ &= [\lambda A]_{ji} + [\mu B]_{ji} \\ &= \lambda a_{ji} + \mu b_{ji} \\ &= \lambda [A^T]_{ij} + \mu [B^T]_{ij} \\ &= [\lambda A^T + \mu B^T]_{ij}. \end{aligned}$$

Thus, as corresponding matrix entries are equal, we have $(\lambda A + \mu B)^T = \lambda A^T + \mu B^T$, and the result is proved. □

Proposition 3. If AB exists, then $(AB)^T = B^T A^T$.

That is, if a product exists, then the transpose of a product is the product of the transposes, but with the order of multiplication **reversed**.

Proof. Since AB exists, the number of columns of A must equal the number of rows of B . Suppose $A \in M_{mn}$ and $B \in M_{np}$. Then the matrix entries of $(AB)^T$ are

$$[(AB)^T]_{ij} = [AB]_{ji} = \sum_{k=1}^n [A]_{jk} [B]_{ki}.$$

Now, by definition of transpose, $B^T \in M_{pn}$ and $A^T \in M_{nm}$. Thus the product $B^T A^T$ exists. The matrix entries of this product are

$$[B^T A^T]_{ij} = \sum_{k=1}^n [B^T]_{ik} [A^T]_{kj} = \sum_{k=1}^n [B]_{ki} [A]_{jk} = \sum_{k=1}^n [A]_{jk} [B]_{ki} = [(AB)^T]_{ij}.$$

The result is proved. □

Example 10. Let

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 3 & -3 & 6 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} -2 & 5 & -1 \\ 4 & -1 & 5 \\ 1 & 3 & 2 \end{pmatrix}.$$

Then

$$(AB)^T = \begin{pmatrix} 5 & 0 & 7 \\ -12 & 36 & -6 \end{pmatrix}^T = \begin{pmatrix} 5 & -12 \\ 0 & 36 \\ 7 & -6 \end{pmatrix}$$

and

$$B^T A^T = \begin{pmatrix} -2 & 4 & 1 \\ 5 & -1 & 3 \\ -1 & 5 & 2 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 2 & -3 \\ -1 & 6 \end{pmatrix} = \begin{pmatrix} 5 & -12 \\ 0 & 36 \\ 7 & -6 \end{pmatrix}.$$

◇

Example 11. For column vectors, $\mathbf{a}, \mathbf{b}$, we have from Propositions 1 and 3 that

$$(\mathbf{a}\mathbf{b}^T)^T = (\mathbf{b}^T)^T \mathbf{a}^T = \mathbf{b}\mathbf{a}^T,$$

as noted for the special case of the vectors in Examples 7 and 8.

◇

Symmetric matrices are especially important in certain applications of matrices to geometry and physics,

Definition 2. A matrix is said to be a **symmetric** matrix if $A = A^T$.

Note that the entries of a symmetric matrix A satisfy $[A]_{ij} = [A^T]_{ji} = [A]_{ji}$ for all i, j , and hence a symmetric matrix must be square.

Example 12. The matrix

$$A = \begin{pmatrix} 1 & 3 & -4 & 6 \\ 3 & 7 & 10 & -11 \\ -4 & 10 & -2 & 5 \\ 6 & -11 & 5 & 4 \end{pmatrix}$$

is a symmetric matrix.

◇

Example 13. Let A and B be symmetric matrices. Prove that $A + B$ is symmetric but AB may not.

Proof. Since A and B are symmetric, we have $A^T = A$ and $B^T = B$. By Proposition 2,

$$(A + B)^T = A^T + B^T = A + B.$$

Hence $A + B$ is symmetric.

Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix}$. Both A and B are symmetric. However,

$$AB = \begin{pmatrix} 4 & 4 \\ 1 & 2 \end{pmatrix}$$

is not symmetric.

□

5.3 The inverse of a matrix

In Section 5.1, we defined the operations of addition, subtraction and multiplication for matrices. We also emphasised that **division is not defined for matrices**. The closest we can come to defining ‘division’ for matrices is to define the **inverse** A^{-1} of a matrix A . Not all matrices, however, have inverses.

Warning. If x and y are numbers, then $x^{-1} = \frac{1}{x}$ means 1 divided by x , and $\frac{x}{y}$ means x divided by y . However, for a matrix A , we **cannot** use $\frac{I}{A}$ to represent A^{-1} . For matrices A and B , writing $\frac{A}{B}$ would be ambiguous. The reason is that matrix multiplication is not commutative, and so it is not clear whether $\frac{A}{B}$ is supposed to mean AB^{-1} or $B^{-1}A$.

Definition 1. A matrix X is said to be an **inverse** of a matrix A if both

$$AX = I \quad \text{and} \quad XA = I,$$

where I is an identity (or unit) matrix of the appropriate size.

If a matrix A has an inverse, then A is said to be an **invertible** matrix. An invertible matrix is also called a **non-singular** matrix. If a matrix A is not an invertible matrix, then it is called a **singular** matrix.

Proposition 1. All invertible matrices are square, that is, have the same number of rows as columns.

[X] *Proof.* Suppose that the matrix A has more columns than rows. Then the equation $A\mathbf{x} = \mathbf{0}$ has some non-zero solution $\mathbf{x}$. Left multiplication by the hypothetical inverse X yields $\mathbf{x} = I\mathbf{x} = (XA)\mathbf{x} = X\mathbf{0} = \mathbf{0}$, which is a contradiction. So no matrix with more columns than rows has an inverse. If the matrix A has more rows than columns and has an inverse X then X has more columns than rows and has an inverse A . As before this leads to a contradiction. $\square$

Example 1. The matrix

$$X = \begin{pmatrix} 1 & -\frac{1}{3} & -\frac{2}{3} \\ 1 & \frac{2}{3} & -\frac{2}{3} \\ 2 & \frac{2}{3} & -\frac{5}{3} \end{pmatrix}$$

is an inverse of the matrix

$$A = \begin{pmatrix} 2 & 3 & -2 \\ -1 & 1 & 0 \\ 2 & 4 & -3 \end{pmatrix},$$

since (check calculations)

$$AX = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = XA.$$

◇

Definition 2. A matrix X is said to be a **right inverse** of A if A is $r \times c$, X is $c \times r$ and $AX = I_r$. A matrix Y is said to be a **left inverse** of A if A is $r \times c$, Y is $c \times r$ and $YA = I_c$.

NOTE.

1. The conditions in the definition of an invertible matrix are very restrictive, so many square matrices are not invertible.
2. However, invertible matrices are very important as we shall begin to see below.
3. If A has left inverse then $c \leq r$. If A has a right inverse then $r \leq c$. (See proof of Proposition 1).

5.3.1 Some useful properties of inverses

Theorem 2. If the matrix A has both a left inverse Y and a right inverse X , then $Y = X$. In particular, if both Y and X are inverses of A , then $Y = X$.

That is, if A has a left and a right inverse, then A is both invertible and square.

Proof. $Y = YI_r = Y(AX) = (YA)X = I_cX = X$.

Thus, $AX = I = XA$ and so A is invertible. Hence A is a square matrix by Proposition 1. $\square$

Notation: Because the inverse of a matrix A (if one exists) is unique we can denote it by a special symbol: A^{-1} (read ‘ A inverse’).

Proposition 3. If A is an invertible matrix, then A^{-1} is also an invertible matrix and the inverse of A^{-1} is A . That is, $(A^{-1})^{-1} = A$.

Proof. If $X = A^{-1}$ exists, then $AX = I$ and $XA = I$. But these two conditions also imply that A is the inverse of X , and hence X is invertible and $X^{-1} = A$. Replacing X by A^{-1} then gives the result $(A^{-1})^{-1} = A$. $\square$

Proposition 4. If A and B are invertible matrices and the product AB exists, then AB is also an invertible matrix and $(AB)^{-1} = B^{-1}A^{-1}$.

Note the reversal of the order in the product of the inverses.

Proof. On multiplying AB from the right by $B^{-1}A^{-1}$, we have

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I.$$

Thus the matrix $X = B^{-1}A^{-1}$ satisfies $(AB)X = I$.

Similarly, on multiplying AB from the left by $B^{-1}A^{-1}$, we have

$$(B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}IB = B^{-1}B = I.$$

Thus $X = B^{-1}A^{-1}$ also satisfies $X(AB) = I$. Thus, this X is the inverse of AB , and we have $B^{-1}A^{-1} = X = (AB)^{-1}$. $\square$

This proposition can easily be extended to products with 3 or more factors to obtain, for example,

$$(ABC)^{-1} = C^{-1}(AB)^{-1} = C^{-1}B^{-1}A^{-1}.$$

These results are sometimes useful in simplifying complicated expressions.

Example 2. Assuming F , G , and H are invertible and all products exist, simplify

$$A = HG(FHG)^{-1}FG.$$

◇

SOLUTION. Replacing $(FHG)^{-1}$ by $G^{-1}H^{-1}F^{-1}$, we have

$$A = HGG^{-1}H^{-1}F^{-1}FG.$$

Then using $GG^{-1} = I$ and $F^{-1}F = I$, we have

$$A = H IH^{-1}IG = HH^{-1}G = IG = G.$$

◇

5.3.2 Calculating the inverse of a matrix

Suppose A is a 2×2 invertible matrix. Write the columns of A^{-1} as $\mathbf{x}_1, \mathbf{x}_2$ then $AA^{-1} = I$ can be written as

$$AA^{-1} = A(\mathbf{x}_1 | \mathbf{x}_2) = (\mathbf{e}_1 | \mathbf{e}_2),$$

where $\{\mathbf{e}_1, \mathbf{e}_2\}$ is the standard basis for $\mathbb{R}^2$. Then $\mathbf{x}_1$ is the solution of $A\mathbf{x} = \mathbf{e}_1$ and $\mathbf{x}_2$ is the solution of $A\mathbf{x} = \mathbf{e}_2$. We could find $\mathbf{x}_1$ by reducing $(A | \mathbf{e}_1)$ to the reduced row-echelon form and similarly for $\mathbf{x}_2$. This can be done simultaneously as shown in the following example.

Example 3. Find the inverse of $A = \begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}$.

SOLUTION.

$$(A | \mathbf{e}_1 \ \mathbf{e}_2) = \left(\begin{array}{cc|cc} 2 & 3 & 1 & 0 \\ 3 & 4 & 0 & 1 \end{array} \right) \xrightarrow{R_2 = R_2 - \frac{3}{2}R_1} \left(\begin{array}{cc|cc} 2 & 3 & 1 & 0 \\ 0 & -\frac{1}{2} & -\frac{3}{2} & 1 \end{array} \right).$$

If we further reduce the matrix to reduced row-echelon form by

$$\begin{aligned} \left(\begin{array}{cc|cc} 2 & 3 & 1 & 0 \\ 0 & -\frac{1}{2} & -\frac{3}{2} & 1 \end{array} \right) & \xrightarrow{R_2 = -2R_2} \left(\begin{array}{cc|cc} 2 & 3 & 1 & 0 \\ 0 & 1 & 3 & -2 \end{array} \right) \\ & \xrightarrow{R_1 = R_1 - 3R_2} \left(\begin{array}{cc|cc} 2 & 0 & -8 & 6 \\ 0 & 1 & 3 & -2 \end{array} \right) \\ & \xrightarrow{R_1 = \frac{1}{2}R_1} \left(\begin{array}{cc|cc} 1 & 0 & -4 & 3 \\ 0 & 1 & 3 & -2 \end{array} \right). \end{aligned}$$

Hence $\mathbf{x}_1 = \begin{pmatrix} -4 \\ 3 \end{pmatrix}$ and $\mathbf{x}_2 = \begin{pmatrix} 3 \\ -2 \end{pmatrix}$ and $A^{-1} = \begin{pmatrix} -4 & 3 \\ 3 & -2 \end{pmatrix}$. $\diamond$

In general, if A is an invertible $n \times n$ matrix and the columns of the unique inverse, A^{-1} , are $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$, then $\mathbf{x}_i$ is the unique solution of $A\mathbf{x} = \mathbf{e}_i$, where $1 \leq i \leq n$. Here $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ is the standard basis for $\mathbb{R}^n$. Conversely if any equation $A\mathbf{x} = \mathbf{e}_i$ does not have a unique solution, then A is not invertible.

This suggests a method of finding the inverse of a (invertible) matrix A :

1. Form the augmented matrix $(A \mid I)$ with n rows and $2n$ columns.
2. Use Gaussian-elimination to convert $(A \mid I)$ to row-echelon form $(U \mid C)$. Then, if all entries in the bottom row of U are zero, stop. In such circumstances A has no inverse.
3. Otherwise, use further row operations to reduce $(U \mid C)$ to reduced row-echelon form $(I \mid B)$. The right hand half of this reduced row-echelon form is the inverse.

The next proposition sums up the implications of these observations.

Proposition 5. A matrix A is invertible if and only if it can be reduced by elementary row operations to an identity matrix I and if $(A \mid I)$ can be reduced to $(I \mid B)$ then $B = A^{-1}$.

Example 4. Determine if the matrix

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 2 \\ 1 & -1 & 1 \end{pmatrix}$$

is invertible and, if it is invertible, find its inverse.

SOLUTION. We solve $AX = I$ by forming an augmented matrix with the 3 columns of I on the right, as in

$$(A \mid I) = \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 1 & 2 & 0 & 1 & 0 \\ 1 & -1 & 1 & 0 & 0 & 1 \end{array} \right).$$

As usual in solving equations, we first use Gaussian elimination to reduce the augmented matrix to row-echelon form. We obtain

$$(U \mid Y) = \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -3 & -4 & -2 & 1 & 0 \\ 0 & 0 & 2 & 1 & -1 & 1 \end{array} \right).$$

Because the bottom row of U contains a non-zero entry, A is invertible.

To obtain the inverse $X = A^{-1}$, we use further row operations to get U into reduced row-echelon form and find

$$\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{2} & \frac{5}{6} & -\frac{1}{6} \\ 0 & 1 & 0 & 0 & \frac{1}{3} & -\frac{2}{3} \\ 0 & 0 & 1 & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{array} \right).$$

We now have an identity matrix on the left, so the solution X of the equations $AX = I$ can be read off immediately as the matrix on the right of the $|$. The solution is

$$A^{-1} = X = \begin{pmatrix} -\frac{1}{2} & \frac{5}{6} & -\frac{1}{6} \\ 0 & \frac{1}{3} & -\frac{2}{3} \\ \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} \end{pmatrix}.$$

As a check, multiplication of A and the A^{-1} obtained should give I .

◇

Example 5. Determine if

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$$

is invertible and, if it is invertible, find its inverse.

SOLUTION. The augmented matrix $[A \mid I]$ is

$$(A \mid I) = \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 4 & 5 & 6 & 0 & 1 & 0 \\ 7 & 8 & 9 & 0 & 0 & 1 \end{array} \right).$$

On reducing $[A \mid I]$ to row-echelon form by Gaussian elimination, we have

$$(U \mid Y) = \left(\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & -3 & -6 & -4 & 1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 1 \end{array} \right).$$

Here, all entries in the bottom row of U are zero, so A is not invertible.

◇

5.3.3 Inverse of a 2×2 matrix

There is a simple formula of the inverse of an invertible 2×2 matrix.

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}, \quad \text{provided } ad - bc \neq 0.$$

We can easily prove this by checking that the product $\frac{1}{ad - bc} \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$ is I . This formula is useful and should be committed to memory.

Example 6. Find the inverse of $\begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}$ by the above formula.

SOLUTION. $\begin{pmatrix} 2 & 3 \\ 3 & 4 \end{pmatrix}^{-1} = \frac{1}{2 \times 4 - 3 \times 3} \begin{pmatrix} 4 & -3 \\ -3 & 2 \end{pmatrix} = \begin{pmatrix} -4 & 3 \\ 3 & -2 \end{pmatrix}.$

◇

[X] 5.3.4 Elementary row operations and matrix multiplication

In this subsection we shall see that the three elementary row operations on a matrix A in Chapter 4 can be interpreted as multiplication of A by “elementary matrices”. As applications, we can use this to prove that the algorithm to calculate the inverse of a matrix works and prove that the left inverse of a **square** matrix is two-sided in the next subsection.

Example 7. 1. If $E_1 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ and $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$ then $E_1 A = \begin{pmatrix} a & b & c \\ 3d & 3e & 3f \\ g & h & i \end{pmatrix}$ (check this), that is, the second row has been multiplied by 3.

2. If $E_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix}$ then $E_2 B = \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ m & n & o & p \\ i & j & k & l \end{pmatrix}$, that is, the 3rd and 4th rows have been swapped.

3. If $E_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -5 & 0 & 1 \end{pmatrix}$ then $E_3 A = \begin{pmatrix} a & b & c \\ d & e & f \\ g-5a & h-5b & i-5c \end{pmatrix}$, that is, row three is replaced by row three minus five times row one. $\diamond$

Proposition 6. The three elementary row operations can be effected by left multiplication by matrices. These matrices are all invertible.

Proof. Let $E_1 = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & \lambda & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}$ $E_2 = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}$ $E_3 = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}$. The entries in the blank space are zeroes.

As demonstrated by Example 7 it is easy to see that, when multiplied on the left of a matrix,

- E_1 multiplies row i by λ
- E_2 interchanges row i and row j
- E_3 adds λ times row i to row j

It is also clear that

$$E_1^{-1} = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & \lambda^{-1} & & \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}, \quad E_2^{-1} = E_2, \quad E_3^{-1} = \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & -\lambda & & 1 \\ & & & \ddots & \\ & & & & 1 \end{pmatrix}.$$

Indeed E_1^{-1} , E_2^{-1} and E_3^{-1} “undo” the operations done by E_1 , E_2 and E_3 . □

Theorem 7. *If an augmented matrix $(A \mid I)$ is reduced to a matrix of the form $(I \mid X)$ then X is a left inverse of A .*

Proof. On the reduction of $(A \mid I) \rightarrow (I \mid X)$, each step is an elementary row operation. By Proposition 6 each row operation is also left multiplication by a matrix of type E_1 , E_2 or E_3 . So there are a sequence of matrices $S_1, \dots, S_k$, each of which is of type E_1 , E_2 or E_3 such that

$$S_k S_{k-1} \cdots S_1 A = I.$$

However the same operations are performed on I yielding X , so we have

$$S_k S_{k-1} \cdots S_1 I = X$$

hence $XA = I$, that is X is a left inverse of A . □

Theorem 8. *The left inverse of A in Theorem 7 is also a right inverse of A .*

Proof. Continuing from the proof of Theorem 7, $X = S_k S_{k-1} \cdots S_1$ is a left inverse of A . By Proposition 6 $S_1, \dots, S_k$ are invertible with inverses $S_1^{-1}, \dots, S_k^{-1}$. Let $B = S_1^{-1} S_2^{-1} \cdots S_k^{-1}$. Now

$$XB = S_k S_{k-1} \cdots S_2 S_1 S_1^{-1} S_2^{-1} \cdots S_k^{-1} = I$$

and $BX = S_1^{-1} S_2^{-1} \cdots S_k^{-1} S_k S_{k-1} \cdots S_2 S_1 = I$. But $XA = I$ also, hence

$$B = BI = B(XA) = (BX)A = IA = A$$

thus B is both the left and right inverse of X as is A ! So, X actually is A^{-1} . □

Corollary 9. *The algorithm described in 5.3.2 does work. That is, the matrix calculated by the algorithm is the inverse of the given matrix.*

5.3.5 Inverses and solution of $A\mathbf{x} = \mathbf{b}$

Proposition 10. If A is a square matrix, then A is invertible if and only if $A\mathbf{x} = \mathbf{0}$ implies $\mathbf{x} = \mathbf{0}$.

Proof. Suppose A is invertible and $A\mathbf{x} = \mathbf{0}$ then $A^{-1}A\mathbf{x} = \mathbf{0}$, $I\mathbf{x} = \mathbf{0}$, $\mathbf{x} = \mathbf{0}$.

Conversely suppose $A\mathbf{x} = \mathbf{0}$ implies $\mathbf{x} = \mathbf{0}$ then the reduced echelon form of A is the identity matrix and, as discussed in 5.3.4, A^{-1} exists. $\square$

Proposition 11. Let A be an $n \times n$ square matrix. Then A is invertible if and only if the matrix equation $A\mathbf{x} = \mathbf{b}$ has a unique solution for all vectors $\mathbf{b} \in \mathbb{R}^n$. In this case, the unique solution is $\mathbf{x} = A^{-1}\mathbf{b}$.

Proof. Suppose A is invertible and $A\mathbf{x} = \mathbf{b}$ then $A^{-1}A\mathbf{x} = A^{-1}\mathbf{b}$, $I\mathbf{x} = A^{-1}\mathbf{b}$, $\mathbf{x} = A^{-1}\mathbf{b}$.

Conversely suppose $A\mathbf{x} = \mathbf{b}$ has a unique solution for all vectors $\mathbf{b}$. Then, in particular, $A\mathbf{x} = \mathbf{0}$ has a unique solution, and this is clearly $\mathbf{x} = \mathbf{0}$. Hence, by Proposition 10, A is invertible. $\square$

NOTE. In principle, this proposition can be used to obtain a solution of $A\mathbf{x} = \mathbf{b}$ by finding an inverse A^{-1} and then forming $\mathbf{x} = A^{-1}\mathbf{b}$. However, there are several serious practical problems which occur if the proposition is used in this way. Some of these problems are as follows.

1. To obtain the inverse of an $n \times n$ matrix A it is necessary to solve n equations of the form $A\mathbf{x}_j = \mathbf{e}_j$. Even with the use of the most efficient available method (the *LU*-factorisation method), the calculation of an inverse takes longer than the solution of the original equation $A\mathbf{x} = \mathbf{b}$.
2. It is very easy to forget that many matrices do not have inverses. It is then very easy to make the mistake of saying that the ‘solution’ of $A\mathbf{x} = \mathbf{b}$ is $\mathbf{x} = A^{-1}\mathbf{b}$ when, in fact, the equations might actually have no solution or might have an infinite number of solutions.
3. The Gaussian elimination and back-substitution method of solving equations always works, whereas the $\mathbf{x} = A^{-1}\mathbf{b}$ formula only works in the special case that $A\mathbf{x} = \mathbf{b}$ has a unique solution.
4. In large-scale numerical calculations, use of the inverse may produce more ‘rounding error’.

However, the propositions can be used to prove some important results.

Corollary 12. Let A be a square matrix.

1. If X is a left inverse of A , then X is the (two-sided) inverse of A . That is,

$$\text{if } XA = I, \text{ then } X = A^{-1}.$$

2. If X is a right inverse of A , then X is the (two-sided) inverse of A .

Proof. For part 1, assume $A\mathbf{x} = \mathbf{0}$. Since X is a left inverse of A ,

$$\mathbf{x} = I\mathbf{x} = (XA)\mathbf{x} = X(A\mathbf{x}) = X\mathbf{0} = \mathbf{0}.$$

Then by Proposition 10, A is invertible. Hence

$$XA = I \Rightarrow XAA^{-1} = IA^{-1} \Rightarrow X = A^{-1}.$$

We can use similar arguments to prove part 2. $\square$

5.4 Determinants

In this section we shall define the determinant function. If A is a square matrix, then the determinant of A , written $\det(A)$, is a number. Determinants arise in many areas of mathematics. As we shall see in Chapter 5, they can be used to find the volume of a parallelepiped. Determinants are very important in the theory of eigenvalues and eigenvectors, covered in Chapter 8. In later years you will need to calculate determinants to perform changes of variables in multivariable calculus.

From a theoretical point of view, the determinant is also important in determining whether a system of n linear equations in n unknowns has a solution for every right-hand-side vector, or equivalently, whether an $n \times n$ matrix A is invertible. As we shall see, A is invertible if and only if $\det(A) \neq 0$. Unfortunately, calculating $\det(A)$, even by the most efficient methods, takes the same length of time as finding the row-echelon form for A , so determinants are not used for numerical calculations of this kind. On the other hand, many general statements can be proved by using the properties of the determinant function.

Determinants are defined only for square matrices. We shall give definitions for arbitrary-size $n \times n$ matrices, but most of our examples will be restricted to determinants of 2×2 and 3×3 matrices. The case of 1×1 matrices is included for completeness. An alternative notation for $\det(A)$ is $|A|$.

5.4.1 The definition of a determinant

The determinant of a 1×1 matrix A is defined to be its sole entry. As the notation $|A|$ tends to cause confusion with the absolute value, it should be used with caution for 1×1 matrices. Thus $|(-1)| = -1$ whilst $|-1| = 1$.

There are several ways of defining determinants, each of which has advantages and disadvantages. We use a recursive definition which begins with the definition of a 2×2 determinant. Here ‘recursive’ indicates that the definition of a determinant for square matrices of a given size invokes that for matrices of a smaller size.

Definition 1. The **determinant** of a 2×2 matrix

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \quad \text{is} \quad \det(A) = a_{11}a_{22} - a_{12}a_{21}.$$

Example 1. If $A = \begin{pmatrix} 2 & -4 \\ -3 & 4 \end{pmatrix}$ then $|A| = \det(A) = 2(4) - (-4)(-3) = -4$. ◇

The definition of the determinant of a general $n \times n$ matrix can be built up recursively from the definition of the determinant of a 2×2 matrix.

We first define the concept of a minor of an entry of a matrix.

Definition 2. For a matrix A , the (row i , column j) **minor** is the determinant of the matrix obtained from A by deleting row i and column j from A .

Notation. We shall use the symbol $|A_{ij}|$ to represent the (row i , column j) minor in a matrix A .

Example 2. The (row 2, column 3) minor of the (row 2, column 3) entry in $A = \begin{pmatrix} 2 & -3 & 4 \\ 8 & -1 & -4 \\ 6 & 5 & -7 \end{pmatrix}$ is, on deleting row 2 and column 3, $|A_{23}| = \begin{vmatrix} 2 & -3 \\ 6 & 5 \end{vmatrix} = 28$. $\diamond$

Definition 3. The **determinant** of an $n \times n$ matrix A is

$$\begin{aligned} |A| &= a_{11}|A_{11}| - a_{12}|A_{12}| + a_{13}|A_{13}| - a_{14}|A_{14}| + \cdots + (-1)^{1+n}a_{1n}|A_{1n}| \\ &= \sum_{k=1}^n (-1)^{1+k} a_{1k} |A_{1k}|. \end{aligned}$$

Note that each term in the definition of the determinant $|A|$ is a product of an entry in the first row of A with its corresponding minor. The signs of the terms alternate $+$, $-$, $+$, $-$, $\dots$ starting with a $+$ on the (row 1, column 1) entry. The formula is called “expanding along the first row of the determinant.”

Example 3. Evaluate the determinant

$$\det(A) = \begin{vmatrix} 5 & 1 & 7 \\ -2 & 3 & -4 \\ 6 & -1 & 2 \end{vmatrix}.$$

SOLUTION.

$$\det(A) = 5 \begin{vmatrix} 3 & -4 \\ -1 & 2 \end{vmatrix} - 1 \begin{vmatrix} -2 & -4 \\ 6 & 2 \end{vmatrix} + 7 \begin{vmatrix} -2 & 3 \\ 6 & -1 \end{vmatrix} = 5(6 - 4) - (-4 + 24) + 7(2 - 18) = -122.$$

$\diamond$

5.4.2 Properties of determinants

In this section we present some of the basic properties of determinants. Where it is possible to do so in an efficient manner by elementary methods, we give proofs of the properties for the general $n \times n$ case. For properties where the general proof requires more advanced methods, we give proofs for the 2×2 or 3×3 cases instead.

Proposition 1. $\det(A^T) = \det(A)$

Proof. For the 3×3 case. If

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \quad \text{then} \quad A^T = \begin{pmatrix} a_{11} & a_{21} & a_{31} \\ a_{12} & a_{22} & a_{32} \\ a_{13} & a_{23} & a_{33} \end{pmatrix}.$$

By direct evaluation of $\det(A)$ and $\det(A^T)$ from Definition 3, we obtain

$$\begin{aligned}\det(A) &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} - a_{12}a_{21}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32} - a_{13}a_{22}a_{31}\end{aligned}$$

and

$$\begin{aligned}\det(A^T) &= a_{11} \begin{vmatrix} a_{22} & a_{32} \\ a_{23} & a_{33} \end{vmatrix} - a_{21} \begin{vmatrix} a_{12} & a_{32} \\ a_{13} & a_{33} \end{vmatrix} + a_{31} \begin{vmatrix} a_{12} & a_{22} \\ a_{13} & a_{23} \end{vmatrix} \\ &= a_{11}a_{22}a_{33} - a_{11}a_{32}a_{23} - a_{21}a_{12}a_{33} + a_{21}a_{32}a_{13} + a_{31}a_{12}a_{23} - a_{31}a_{22}a_{13}\end{aligned}$$

These two expressions are equal, and hence $\det(A) = \det(A^T)$. $\square$

One immediate consequence of Proposition 1 is that a determinant can also be evaluated by ‘expansion down the first column’, that is

$$|A| = a_{11}|A_{11}| - a_{21}|A_{21}| + a_{31}|A_{31}| - a_{41}|A_{41}| + \cdots + (-1)^{n+1}a_{n1}|A_{n1}| = \sum_{k=1}^n (-1)^{k+1}a_{k1}|A_{k1}|.$$

Example 4. Use expansion along the first column to evaluate the determinant of Example 3.

SOLUTION. Expanding along the first column, we have

$$\det(A) = 5 \begin{vmatrix} 3 & -4 \\ -1 & 2 \end{vmatrix} - (-2) \begin{vmatrix} 1 & 7 \\ -1 & 2 \end{vmatrix} + 6 \begin{vmatrix} 1 & 7 \\ 3 & -4 \end{vmatrix} = 5(6 - 4) + 2(2 + 7) + 6(-4 - 21) = -122.$$

$\diamond$

Proposition 2. If any two rows (or any two columns) of A are interchanged, then the sign of the determinant is reversed. More precisely if the matrix B is obtained from the matrix A by interchanging two rows (or columns), then $\det B = -\det A$.

Proof. For 2×2 case. Let

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}.$$

Then, on interchanging columns 1 and 2 of A ,

$$\det \begin{pmatrix} a_{12} & a_{11} \\ a_{22} & a_{21} \end{pmatrix} = a_{12}a_{21} - a_{11}a_{22} = -\det(A).$$

$\square$

One important application of this proposition is that a determinant can be evaluated by expanding along any row or any column, with the signs chosen from the following array.

$$\begin{pmatrix} + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ + & - & + & - & \cdots \\ - & + & - & + & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix}.$$

If we evaluate the determinant by expanding the i th row or the j th column, where $1 \leq i, j \leq n$, we have

$$|A| = \sum_{k=1}^n (-1)^{i+k} a_{ik} |A_{ik}| = \sum_{k=1}^n (-1)^{k+j} a_{kj} |A_{kj}|.$$

Example 5. Use expansion along the second row to evaluate $\det(A)$ of the previous two examples.

SOLUTION. To evaluate $\det(A)$ by expanding along the second row, we choose the signs from the

second row of $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$. So

$$\det(A) = -(-2) \begin{vmatrix} 1 & 7 \\ -1 & 2 \end{vmatrix} + 3 \begin{vmatrix} 5 & 7 \\ 6 & 2 \end{vmatrix} - (-4) \begin{vmatrix} 5 & 1 \\ 6 & -1 \end{vmatrix} = -122.$$

◇

Obviously, we should evaluate a determinant by expanding along that row or column containing the greatest number of 0's. We will soon see an even better method.

A second application of Proposition 2 is as follows.

Proposition 3. If a matrix contains a zero row or column then its determinant is zero.

Proof. Clearly, if we evaluate the determinant by expansion along the zero row or column, the value obtained will be zero. $\square$

Most of the remaining important properties of determinants centre around the question of what happens to the value of the determinant when one column (or row) of a matrix is changed, with all other columns (or rows) remaining unchanged.

Proposition 4. If a row (or column) of A is multiplied by a scalar, then the value of $\det A$ is multiplied by the same scalar. That is, if the matrix B is obtained from the matrix A by multiplying a row (or column) of A by the scalar λ , then $\det B = \lambda \det A$.

Proof. On multiplying the first row of

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}$$

by a scalar λ , we obtain

$$B = \begin{pmatrix} \lambda a_{11} & \lambda a_{12} & \cdots & \lambda a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}.$$

Then, on expanding along the first row, we have

$$\det(B) = \sum_{k=1}^n (-1)^{1+k} \lambda a_{1k} |A_{1k}| = \lambda \sum_{k=1}^n (-1)^{1+k} a_{1k} |A_{1k}| = \lambda \det(A),$$

and hence the result is proved for a multiple of the first row. Then, from Propositions 1 and 2, the result is also true for a scalar multiple of any row or column. $\square$

An immediate consequence of Propositions 2 and 4 is the following useful result.

Proposition 5. If any column of a matrix is a multiple of another column of the matrix (or any row is a multiple of another row), then the value of $\det(A)$ is zero.

Proof. The result is clearly true if the scalar multiple is zero.

Now, assume that the i th column of a matrix A is λ times the j th column. Then, if we multiply the i th column by λ^{-1} we obtain a matrix B , whose determinant, from Proposition 4, has the value $\det(B) = \lambda^{-1} \det(A)$.

This matrix B has its i th and j th columns equal. On interchanging columns i and j , we have, from Proposition 2, that the determinant of the new matrix has the value $-\det(B)$. But, interchanging equal columns does not change the matrix, so the new matrix is still B with determinant $\det(B)$. Hence, $\det(B) = -\det(B)$, and thus $\det(B) = 0$ and $\det(A) = 0$.

The proof of the proposition for rows follows immediately from Proposition 1. $\square$

Example 6.

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & -5 & 7 \\ -3 & -6 & -9 \end{vmatrix} = 0$$

as row 3 is a scalar multiple of row 1. This result can easily be checked directly by using expansion along the second row to evaluate the determinant. $\diamond$

Propositions 2 and 4 show the effect of the two elementary row operations of interchanging two rows and of multiplying a row by a scalar. The effect of the third elementary row operation of adding a multiple of one row to another row is given in the following proposition.

Proposition 6. If a multiple of one row (or column) is added to another row (or column), then the value of the determinant is not changed.

[X] *Proof.* On adding λ times row i to row 1 of

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}, \quad \text{we obtain} \quad B = \begin{pmatrix} a_{11} + \lambda a_{i1} & a_{12} + \lambda a_{i2} & \cdots & a_{1n} + \lambda a_{in} \\ \vdots & \vdots & \ddots & \vdots \\ a_{i1} & a_{i2} & \cdots & a_{in} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix}.$$

Then, on expanding along the first row, we have

$$\begin{aligned} \det(B) &= \sum_{k=1}^n (-1)^{1+k} (a_{1k} + \lambda a_{ik}) |A_{1k}| \\ &= \sum_{k=1}^n (-1)^{1+k} a_{1k} |A_{1k}| + \lambda \sum_{k=1}^n (-1)^{1+k} a_{ik} |A_{1k}| \\ &= \det(A) + \lambda \sum_{k=1}^n (-1)^{1+k} a_{ik} |A_{1k}|. \end{aligned}$$

The last sum is the determinant of the matrix which is obtained from A by replacing the first row of A by the i th row of A . Thus, the first and i th rows of this matrix are the same, and hence, from Proposition 5, its determinant is zero. Thus the second sum in the above equation is zero, and hence $\det(B) = \det(A)$. We have therefore proved the result for subtraction of scalar multiples of any row from the first row. Then, from Propositions 1 and 2, the result is true for any two rows or any two columns. $\square$

The final useful proposition that we shall give here is as follows.

Proposition 7. If A and B are square matrices such that the product AB exists, then

$$\det(AB) = \det(A)\det(B).$$

All known proofs of this result are non-trivial, so we shall not give a proof here.

Example 7. For

$$A = \begin{pmatrix} 2 & -4 \\ -3 & 5 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} -1 & -8 \\ 2 & 9 \end{pmatrix},$$

we have

$$|A| = -2, \quad |B| = 7, \quad |AB| = \begin{vmatrix} -10 & -52 \\ 13 & 69 \end{vmatrix} = -14 = |A||B|.$$

$\diamond$

5.4.3 The efficient numerical evaluation of determinants

With the exception of 2×2 and 3×3 matrices, the direct evaluation of a determinant from Definition 3 is a lengthy procedure. For example, the value of an $n \times n$ determinant is given in terms of n minors of size $(n-1) \times (n-1)$, each of which is given in terms of $n-1$ minors of size $(n-2) \times (n-2)$, etc. The evaluation ending with the evaluation of 2×2 minors. In fact, the total number of terms to be evaluated is $n!$. Even for a small number such as $n = 6$, this is already $6! = 720$ terms.

With the exception of the 2×2 and 3×3 cases, by far the most efficient method of evaluating determinants of matrices with numerical entries is to use Gaussian elimination to reduce the matrix to row-echelon form. The basic results required for this efficient method of evaluation are given in the following two propositions.

Proposition 8. If U is a square row-echelon matrix, then $\det(U)$ is equal to the product of the diagonal entries of U .

Proof. Let

$$U = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1n} \\ 0 & u_{22} & \cdots & u_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{nn} \end{pmatrix}$$

be an $n \times n$ matrix in row-echelon form. Then the proposition states that

$$\det(U) = u_{11}u_{22} \cdots u_{nn}.$$

We shall prove this result by induction. We first introduce some notation.

Let $U(j)$ be the submatrix of U which is defined by

$$U(j) = \begin{pmatrix} u_{11} & u_{12} & \cdots & u_{1j} \\ 0 & u_{22} & \cdots & u_{2j} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{jj} \end{pmatrix}.$$

Note that $U(j)$ is a square row-echelon form matrix for $1 \leq j \leq n$, with $U(1) = [u_{11}]$, $\det(U(1)) = u_{11}$ and $U(n) = U$.

We shall now prove that if the proposition is true for $\det(U(j))$, then it is also true for $\det(U(j+1))$. Now,

$$\det(U(j+1)) = \begin{vmatrix} u_{11} & u_{12} & \cdots & u_{1,j+1} \\ 0 & u_{22} & \cdots & u_{2,j+1} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{j+1,j+1} \end{vmatrix}.$$

We can evaluate $\det(U(j+1))$ by expanding along the last row, and we obtain

$$\det(U(j+1)) = (-1)^{2j} u_{j+1,j+1} \det(U(j)) = u_{j+1,j+1} \det(U(j)).$$

If we now assume that the proposition is true for $\det(U(j))$, we can replace $\det(U(j))$ by the product of its diagonal entries to obtain

$$\det(U(j+1)) = u_{11} \cdots u_{jj} u_{j+1,j+1}.$$

Hence, if the result is true for $\det(U(j))$ it is also true for $\det(U(j+1))$.

However, the result is true for $j = 1$, since in this case we have $\det(U(1)) = u_{11}$. It is therefore true for $j = 2, 3, \dots$, and the proposition is proven for all $n \geq 1$ by induction. $\square$

Example 8.

$$\begin{vmatrix} 3 & 2 & 9 & 3 & 12 \\ 0 & -4 & -5 & 6 & 2 \\ 0 & 0 & 6 & -7 & -3 \\ 0 & 0 & 0 & 5 & 20 \\ 0 & 0 & 0 & 0 & 8 \end{vmatrix} = (3)(-4)(6)(5)(8) = -2880.$$

$\diamond$

Proposition 9. If A is a square matrix and U is an equivalent row-echelon form obtained from A by Gaussian elimination using row interchanges and adding a multiple of one row to another, then $\det(A) = \epsilon \det(U)$, where $\epsilon = +1$ if an even number of row interchanges have been made, and $\epsilon = -1$ if an odd number of row interchanges have been made.

Proof. From Proposition 2, each row interchange reverses the sign of the determinant. Thus the sign is unchanged if 0, 2, 4, ... row interchanges have been made, whereas the sign is reversed if 1, 3, 5, ... row interchanges have been made.

From Proposition 6, the value of the determinant is unchanged if a multiple of one row is added to another row. Thus row addition operations do not change the value of the determinant. The proof is complete. $\square$

Propositions 8 and 9 show that the value of a determinant can be found by the highly efficient Gaussian elimination method.

Example 9. Evaluate the determinant of

$$A = \begin{pmatrix} 1 & -1 & 0 & 3 \\ 2 & -2 & 6 & -1 \\ 4 & -2 & 1 & 7 \\ 3 & 5 & -7 & 2 \end{pmatrix}.$$

SOLUTION. The first leading element for Gaussian elimination is already in the first row, so no row interchange is required. Hence, at this stage, $\epsilon = 1$. Adding multiples of the first row to subsequent rows does not change the value of the determinant. As is usual in Gaussian elimination, the multiple are chosen to make the entries below the leading element all zero. After partly reducing the original matrix we have:

$$|A| = \begin{vmatrix} 1 & -1 & 0 & 3 \\ 0 & 0 & 6 & -7 \\ 0 & 2 & 1 & -5 \\ 0 & 8 & -7 & -7 \end{vmatrix}.$$

The second leading element is in row 3. We therefore interchange rows 2 and 3 to bring the second pivot element to its required row 2 position. This interchange changes the sign of the determinant, and hence we now have $\epsilon = -1$. After interchanging rows 2 and 3 and then adding suitable multiples of the new row 2 from the new row 3 and row 4, we have

$$|A| = (-1) \begin{vmatrix} 1 & -1 & 0 & 3 \\ 0 & 2 & 1 & -5 \\ 0 & 0 & 6 & -7 \\ 0 & 0 & -11 & 13 \end{vmatrix}.$$

The third pivot element is in its correct row 3 position, so no row interchange is required. Hence ϵ remains at its previous value of -1 . After further reducing the matrix we have

$$|A| = (-1) \begin{vmatrix} 1 & -1 & 0 & 3 \\ 0 & 2 & 1 & -5 \\ 0 & 0 & 6 & -7 \\ 0 & 0 & 0 & \frac{1}{6} \end{vmatrix} = (-1)(1 \times 2 \times 6 \times \frac{1}{6}) = -2.$$

The experienced evaluator of determinants will discard entries in these matrices as the calculation progresses. This leads to the shortened calculation:

$$\begin{vmatrix} 1 & -1 & 0 & 3 \\ 2 & -2 & 6 & -1 \\ 4 & -2 & 1 & 7 \\ 3 & 5 & -7 & 2 \end{vmatrix} = \begin{vmatrix} 1 & -1 & 0 & 3 \\ 0 & 0 & 6 & -7 \\ 0 & 2 & 1 & -5 \\ 0 & 8 & -7 & -7 \end{vmatrix} = (-1) \begin{vmatrix} 2 & 1 & -5 \\ 0 & 6 & -7 \\ 0 & -11 & 13 \end{vmatrix} = -2 \begin{vmatrix} 6 & -7 \\ -11 & 13 \end{vmatrix} = -2.$$

◇

Although we can reduce any matrix to a row echelon form by the two operations—row exchange and row addition, using the row operation—multiply a row by a scalar helps, especially when we do the calculations by hand. This row operation exploits the result of Proposition 4. We illustrate the technique by the following example.

[X] **Example 10.** Factorise $\begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$.

SOLUTION.

$$\begin{aligned}
& \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix} \\
= & \begin{vmatrix} 1 & a^2 & a^3 \\ 0 & b^2 - a^2 & b^3 - a^3 \\ 0 & c^2 - a^2 & c^3 - a^3 \end{vmatrix} & \begin{array}{l} R_2 = R_2 - R_1 \\ R_3 = R_3 - R_1 \end{array} \\
= & (b-a)(c-a) \begin{vmatrix} 1 & a^2 & a^3 \\ 0 & a+b & a^2+ab+b^2 \\ 0 & a+c & a^2+ac+c^2 \end{vmatrix} & \begin{array}{l} \text{Proposition 4} \\ \text{factorise } (b-a) \text{ from } R_2 \\ \text{factorise } (c-a) \text{ from } R_3 \end{array} \\
= & (b-a)(c-a) \begin{vmatrix} 1 & a^2 & a^3 \\ 0 & a+b & a^2+ab+b^2 \\ 0 & c-b & a(c-b)+c^2-b^2 \end{vmatrix} & R_3 = R_3 - R_2 \\
= & (b-a)(c-a)(c-b) \begin{vmatrix} 1 & a^2 & a^3 \\ 0 & a+b & a^2+ab+b^2 \\ 0 & 1 & a+b+c \end{vmatrix} & \begin{array}{l} \text{Proposition 4} \\ \text{factorise } (c-b) \text{ from } R_3 \end{array} \\
= & (b-a)(c-a)(c-b)[(a+b)(a+b+c) - (a^2+ab+b^2)] & \text{expand along the first column} \\
= & (a-b)(b-c)(c-a)(ab+bc+ca)
\end{aligned}$$

◇

5.4.4 Determinants and solutions of $A\mathbf{x} = \mathbf{b}$

The propositions of Section 5.4.3 can be used to establish an important relation between the value of $\det(A)$ and the solution of $A\mathbf{x} = \mathbf{b}$ for a **square** matrix A . The basic result is as follows.

Proposition 10. Let A be an $n \times n$ matrix.

1. If $\det(A) \neq 0$, the equation $A\mathbf{x} = \mathbf{b}$ has a solution and the solution is unique for all $\mathbf{b} \in \mathbb{R}^n$.
2. If $\det(A) = 0$, the equation $A\mathbf{x} = \mathbf{b}$ either has no solution or an infinite number of solutions for a given $\mathbf{b}$.

Proof.

CASE 1. $\det(A) \neq 0$. From Propositions 8 and 9, $\det(A) \neq 0$ implies that all diagonal entries of an equivalent row-echelon matrix U for A are non-zero. Then, since A , and hence U , is a square

matrix, U has no zero rows and no non-leading columns. As there are no zero rows, the equations have a solution for all $\mathbf{b}$, and, as there are no non-leading columns, each solution is unique.

CASE 2. $\det(A) = 0$. In this case, U contains at least one zero diagonal entry, so at least one zero row and at least one non-leading column. Thus there is a non-zero solution to $A\mathbf{x} = \mathbf{0}$. The result follows from Proposition 5 of Section 4.7. $\square$

NOTE. In the case when $\det(A) \neq 0$ there is an explicit formula called *Cramer's rule*, available for the solution of $A\mathbf{x} = \mathbf{b}$. Each entry x_i of $\mathbf{x}$ is specified as the quotient of two determinants.

Example 11. Describe the type of solution for each of the following systems.

$$\begin{cases} 2x_1 + 3x_2 = b_1 \\ 4x_1 + 6x_2 = b_2 \end{cases} \quad \text{and} \quad \begin{cases} x_1 - x_2 + 3x_3 = b_1 \\ 2x_1 + 3x_2 + x_3 = b_2 \\ 3x_1 + x_2 + 4x_3 = b_3 \end{cases}.$$

SOLUTION. The determinants of the coefficient matrices of the equations are

$$|A| = \begin{vmatrix} 2 & 3 \\ 4 & 6 \end{vmatrix} = 0 \quad \text{and} \quad |B| = \begin{vmatrix} 1 & -1 & 3 \\ 2 & 3 & 1 \\ 3 & 1 & 4 \end{vmatrix} = -5 \neq 0.$$

Hence, depending on the values of b_1 and b_2 , the first system of equations either has no solution or an infinite number of solutions, whereas the second system of equations has a unique solution for all $\mathbf{b}$. $\diamond$

NOTE. With the possible exception of 2×2 and 3×3 matrices, Proposition 10 does not provide a practical method of determining the number of solutions of $A\mathbf{x} = \mathbf{b}$, because the most efficient method for evaluating $\det(A)$ is first to use Gaussian elimination to reduce A to row-echelon form U and then to multiply diagonal entries of U . However, as we have seen before, the number of solutions that an equation $A\mathbf{x} = \mathbf{b}$ has can be seen directly from the row-echelon form itself, without the need for any further calculation.

Proposition 10 leads immediately to two results which are sometimes useful.

Proposition 11. For a square matrix A , the homogeneous system of equations $A\mathbf{x} = \mathbf{0}$ has a non-zero solution if and only if $\det(A) = 0$.

Proof. $A\mathbf{x} = \mathbf{0}$ always has $\mathbf{x} = \mathbf{0}$ as a solution. From Proposition 10, if $\det(A) \neq 0$, then $\mathbf{x} = \mathbf{0}$ is the unique solution, whereas, if $\det(A) = 0$, then there are an infinite number of non-zero solutions. $\square$

Proposition 12. A square matrix A is invertible if and only if $\det(A) \neq 0$.

Proof. The result follows immediately on combining the results of Proposition 11 and of Proposition 10 of Section 5.3. $\square$

Example 12. Use determinants to check if the matrices

$$A = \begin{pmatrix} 2 & 5 \\ -4 & 1 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 1 & 3 & 4 \\ -2 & 4 & -8 \\ -3 & 1 & -12 \end{pmatrix}$$

are invertible.

SOLUTION. As $|A| = 22 \neq 0$, A is invertible. As column 3 of B is a multiple of column 1 of B , $\det(B) = 0$. Thus B has no inverse. $\diamond$

There is a very simple relationship between the determinant of a matrix and the determinant of the inverse of the matrix.

Proposition 13. If A is an invertible matrix, then $\det(A^{-1}) = \frac{1}{\det(A)}$.

Proof. If A^{-1} exists, then

$$\det(AA^{-1}) = \det(I) = 1.$$

But, from Proposition 7, the determinant of a product is the product of the determinants, and hence

$$1 = \det(AA^{-1}) = \det(A) \det(A^{-1}).$$

Finally, as A is invertible, $\det(A) \neq 0$, and hence $\det(A^{-1}) = \frac{1}{\det(A)}$. $\square$

5.5 Matrices and Maple

The Linear Algebra package becomes available on Maple only if it is ‘loaded’ with the command:

```
with(LinearAlgebra):
```

Maple treats matrices as individual entities, which can be added and subtracted (use $+$ and $-$) and multiplied by scalars (use $*$). Matrix multiplication is done with the dot.

The matrix $A = \begin{pmatrix} 11 & 12 \\ 21 & 22 \end{pmatrix}$ is entered by the command:

```
A := < < 11, 21 > | < 12, 22 > >;
```

Maple ignores irrelevant blanks! To display the matrix A you enter:

```
A;
```

In Chapter 3 we showed how to enter identity matrices. You can now try:

```
B := 3*A-2*IdentityMatrix(2);
C := B.A;
```

You can transpose a given matrix A and set B as this transpose with:

```
B := Transpose(A);
```

The determinant of a *square* matrix A can be evaluated by

```
Determinant(A);
```

The inverse of a *square* matrix A can be calculated by:

```
A^(-1);
```

though you should read Subsection 5.3.2 and then try reducing an augmented matrix.

Problems for Chapter 5

Questions marked with [R] are routine, [H] harder, [M] Maple and [X] are for MATH1141 only. You should make sure that you can do the easier questions before you tackle the more difficult questions. Questions marks with a [V] have video solutions available on Moodle.

Problems 5.1 : Matrix arithmetic and algebra

1. [R] Given the matrices

$$A = \begin{pmatrix} 2 & -3 & 4 \\ 3 & 2 & -2 \\ 1 & -1 & 3 \end{pmatrix}, \quad B = \begin{pmatrix} -2 & 1 \\ 3 & 4 \\ -1 & 5 \end{pmatrix}, \quad C = \begin{pmatrix} -3 & 2 \\ 1 & -4 \\ 6 & 2 \end{pmatrix}, \quad D = \begin{pmatrix} 2 & 3 & 1 \\ 1 & -2 & -3 \end{pmatrix}.$$

Find the following matrices if they exist, or explain why they don't exist. (I stands for an identity matrix of the appropriate size).

- | | | | | |
|---------------|------------|---------------|--------------|---------------|
| a) $3A$, | b) $-2B$, | c) $A + B$, | d) $B + C$, | e) $A + 3I$, |
| f) $B + 3I$, | g) AB , | h) BA , | i) BC , | j) CD , |
| k) A^2 , | l) B^2 , | m) $(BD)^2$. | | |

2. [H] Suppose A and B are matrices such that both AB and BA are defined.

- Show that AB and BA are both square matrices.
- If $AB = BA$, show that A and B are both square and of the same size.
- If A and B are square matrices such that $AB = BA$, show that $(A - B)(A + B) = A^2 - B^2$.
- Find two 2×2 matrices A, B for which $(A - B)(A + B) \neq A^2 - B^2$.
- Prove that $(A + B)^2 = A^2 + B^2 + 2AB$ if and only if $AB = BA$.

3. [H] Let A and B be matrices of the same size. By considering the general entries $[A]_{ij}$, $[B]_{ij}$, $[A + B]_{ij}$ and $[B + A]_{ij}$, prove the commutative laws of addition, i.e. $A + B = B + A$.

4. [H] Suppose λ is a scalar and $A, B \in M_{mn}$. Prove that $\lambda(A + B) = \lambda A + \lambda B$.

5. [H] Let A and B be two matrices such that AB is defined. By considering the general entry in both sides of the equation, show that $A(\lambda B) = \lambda AB$ where λ is any real number.

6. [R] Let

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 2 \\ 2 & -2 \\ -1 & 4 \end{pmatrix}, \quad C = \begin{pmatrix} 2 & 2 \\ 3 & -2 \\ -2 & 4 \end{pmatrix}.$$

Show that $AB = AC$ and deduce that matrices cannot in general be cancelled from products.

7. [R][V] Let

$$A = \begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix}.$$

Show that $A^2 = A + 5I$ and hence find A^6 as a linear combination of A and I .

8. [R] Let

$$N = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.$$

Find N^2 and N^3 . Show that $(I + N)(I - N + N^2) = I$.

9. [H][V] Let A and B be $n \times n$ real matrices such that $A^2 = I$, $B^2 = I$ and $(AB)^2 = I$. Prove that $AB = BA$.

10. [H] Let A be a 2×2 real matrix such that $AX = XA$ for all 2×2 real matrices X . Show that $A = \alpha I$ for some $\alpha \in \mathbb{R}$.

11. [H] Suppose

$$A = \begin{pmatrix} 1 & -2 & 3 \\ 4 & 0 & 1 \\ 3 & 2 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 7 & 0 & 3 \\ 2 & -1 & 6 \\ -1 & 0 & 5 \end{pmatrix}.$$

- Write down a column vector $\mathbf{v}$ such that $A\mathbf{v}$ is the second column of A .
 - Write down a row vector $\mathbf{v}$ such that $\mathbf{v}B$ is the third row of B .
 - Write down a column vector $\mathbf{v}$ such that $A\mathbf{v}$ is the second column of AB .
 - Write down a row vector $\mathbf{v}$ such that $\mathbf{v}B$ is the first row of AB .
12. [X]
 - Prove the associative law of matrix multiplication.
 - Suppose that $A \in M_{mn}$ and I is an $m \times m$ matrix. Prove that $IA = A$.

Problems 5.2 : The transpose of a matrix

13. [R] Find the transposes of the following matrices:

$$A = \begin{pmatrix} 1 & -2 \\ -3 & 0 \\ 4 & 5 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & -5 & 4 & 3 \\ -4 & 6 & 5 & 5 \\ 5 & 0 & 8 & 6 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 4 & 2 \\ 4 & -3 & 6 \\ 2 & 6 & 7 \end{pmatrix}.$$

14. [R] Let $\mathbf{a} = (1, 3, -2)^T$ and $\mathbf{b} = (0, 4, 2)^T$. Evaluate all of the following expressions that make sense and find those which are equal:

$$\mathbf{ab}, \quad \mathbf{a}^T\mathbf{b}, \quad \mathbf{ab}^T, \quad \mathbf{a}^T\mathbf{b}^T, \quad \mathbf{b}^T\mathbf{a}, \quad \mathbf{ba}^T.$$

15. [R][V] Suppose that A is a square matrix.

- a) Show that the matrix $B = (A + A^T)$ is symmetric.
- b) Show that the matrix $C = AA^T$ is symmetric.
- c) A matrix M with the property that $M^T = -M$ is called a *skew symmetric* matrix. Show that $D = (A - A^T)$ is a skew symmetric matrix.
- d) [H] Can you show how to write any square matrix as the sum of a symmetric and a skew symmetric matrix?
16. [H] Show by constructing an example that, in general, $A^T A \neq AA^T$, even if A is square.
17. [H] Suppose there exists a real matrix G such that $GG^T = \begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}$ where $\lambda, \mu \in \mathbb{R}$. Prove that λ and μ are non-negative. If $\lambda = 45$ and $\mu = 20$ find an example of such a matrix G with integer entries.
18. [X] Show that a matrix $A \in M_{mn}$ is a symmetric matrix if and only if:
- A is square (i.e., $m = n$), and
 - $\mathbf{x}^T A \mathbf{y} = (A \mathbf{x})^T \mathbf{y}$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

Problems 5.3 : The inverse of a matrix

19. [R][V] Find the inverses of those of the following 2×2 matrices that have inverses.
- a) $\begin{pmatrix} 2 & 7 \\ 1 & 4 \end{pmatrix}$, b) $\begin{pmatrix} -4 & 7 \\ 3 & -5 \end{pmatrix}$, c) $\begin{pmatrix} 6 & 12 \\ 3 & 6 \end{pmatrix}$, d) $\begin{pmatrix} 8 & 9 \\ 3 & 4 \end{pmatrix}$, e) $\begin{pmatrix} 0 & 1 \\ 1 & 7 \end{pmatrix}$.
20. [R] Use the matrix inversion algorithm of Section 5.3 to decide if the following matrices are invertible, and find the inverses for those which are invertible.
- $$A = \begin{pmatrix} 1 & 3 & -2 \\ 0 & -1 & 2 \\ 0 & 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 & 2 & 0 \\ 1 & 2 & 3 \\ -1 & 4 & -2 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 4 & 5 & 6 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 4 & 1 \\ 2 & 3 & 1 \\ 1 & -7 & -2 \end{pmatrix}.$$
21. [H] Write down the inverse of each of the following matrices
- a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 6 \end{pmatrix}$ b) $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 3 \\ -2 & 0 & 0 \end{pmatrix}$.
22. [R] Decide if the following matrices are invertible, and find the inverses for those that are invertible.

$$A = \begin{pmatrix} 1 & 2 & -1 & -1 \\ 1 & 2 & -2 & -2 \\ 0 & 1 & 1 & 1 \\ 1 & 4 & 0 & -1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 & 0 & 1 \\ 3 & 3 & 1 & 5 \\ 1 & 0 & 2 & 4 \\ 0 & -4 & 2 & 1 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 2 & -2 & -7 \\ 2 & 4 & 3 & 14 \\ -1 & -2 & 3 & 11 \\ 3 & 5 & 2 & 12 \end{pmatrix}.$$

23. [R] Given that A , B and C are invertible $n \times n$ matrices simplify

- a) $A(CB^2A)^{-1}C$, b) $(ABA^{-1})^6$, c) $A(A^{-1} + A)^2A^{-1}$,
 d) $A(I + (I - A) + \cdots + (I - A)^m)$.

HINT: Write the first A as $I - (I - A)$.

24. [R] a) Simplify $(B^{-1}A)^{-1}$.

b) Find $(B^{-1}A)^{-1}$ if $A^{-1} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & -1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 3 \end{pmatrix}$.

25. [H] a) Prove that $(A^T)^{-1} = (A^{-1})^T$ for any invertible matrix A .

b) If A , B , C are invertible matrices of the same size simplify

i) $A^{-1}(BA^T)^TB$, ii) $A^T(CA^T)^{-1}C^T$.

26. [R][V] Let $A = \begin{pmatrix} 1 & -1 & 1 \\ 1 & 1 & 0 \\ 3 & -2 & 2 \end{pmatrix}$.

- a) Calculate A^{-1} . b) Solve $A\mathbf{x} = \mathbf{c}$ for $\mathbf{x}$, where $\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$.

27. [H] A square matrix Q is said to be an orthogonal matrix if it has the property that $Q^TQ = I$. That is, $Q^T = Q^{-1}$. Show that the matrix

$$Q = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} & -\frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & \frac{2}{3} \end{pmatrix}$$

is orthogonal. Hence write down the solution of $Q\mathbf{x} = \mathbf{b}$ for $\mathbf{b} \in \mathbb{R}^3$.

28. [H] Show $Q = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$ is orthogonal. Show that $\mathbf{x} \in \mathbb{R}^2$ and $Q\mathbf{x}$ are equidistant from the origin. [X] Show that Q acts as a rotation on $\mathbb{R}^2$.

29. [X] A complex generalisation of Question 27 is the following. A square matrix Q is said to be a unitary matrix if it has the property that $\overline{Q}^T Q = I$, where $\overline{Q}$ is the matrix obtained from Q by taking complex conjugates of each entry of Q . Give an example of a 2×2 unitary matrix with non-real entries.

30. [X] Let Q be a square $n \times n$ orthogonal matrix, i.e., a square matrix for which $Q^TQ = I$, where I is an identity matrix. Show that $(Q\mathbf{x}) \cdot (Q\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

HINT. $\mathbf{x} \cdot \mathbf{y} = \mathbf{x}^T \mathbf{y}$.

31. [X] Let Q be a square $n \times n$ orthogonal matrix. Show that the columns of Q are a set of n orthonormal vectors in $\mathbb{R}^n$. Show that the rows of Q also form a set of n orthonormal vectors in $\mathbb{R}^n$.
32. [X] Let Q be a square $n \times n$ orthogonal matrix. Let $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$ be the n standard basis column vectors of $\mathbb{R}^n$. Show that the set of vectors $\{Q\mathbf{e}_1, Q\mathbf{e}_2, \dots, Q\mathbf{e}_n\}$ also form a set of orthonormal vectors.
33. [X] Show that the matrix

$$Q = \begin{pmatrix} \frac{1}{\sqrt{2}}i & -\frac{1}{\sqrt{2}}i & 0 \\ \frac{1}{\sqrt{2}}i & \frac{1}{\sqrt{2}}i & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

is unitary. Then use the result of Question 29 to write down the solution of $Q\mathbf{x} = \mathbf{b}$, where $\mathbf{b} = (b_1 \ b_2 \ b_3)^T$ with b_1, b_2, b_3 complex.

34. [X] a) Suppose $ab \neq 0$. Write down the inverse of $\begin{pmatrix} a & 0 \\ c & b \end{pmatrix}$.
- b) Let A, B, C be 2×2 matrices where A and B are invertible and let O be the 2×2 zero matrix. Find the inverse of the 4×4 matrix $\begin{pmatrix} A & O \\ C & B \end{pmatrix}$.

Problems 5.4 : Determinants

35. [R] Evaluate the determinants of the following 2×2 matrices and hence determine whether or not they are invertible.

a) $\begin{pmatrix} 2 & 7 \\ 1 & 4 \end{pmatrix}$, b) $\begin{pmatrix} -4 & 7 \\ 3 & -5 \end{pmatrix}$, c) $\begin{pmatrix} 5 & 2 \\ 10 & 4 \end{pmatrix}$, d) $\begin{pmatrix} 8 & 9 \\ 3 & 4 \end{pmatrix}$, e) $\begin{pmatrix} 11 & 13 \\ 12 & 14 \end{pmatrix}$.

36. [R] Evaluate the determinants for the following matrices by reducing to row echelon form.

a) $\begin{pmatrix} -1 & 1 & 2 \\ 2 & 4 & -1 \\ 0 & -1 & 1 \end{pmatrix}$, b) $\begin{pmatrix} 1 & -2 & 4 \\ 3 & 1 & -2 \\ 1 & 5 & -10 \end{pmatrix}$, c) $\begin{pmatrix} 1 & 0 & 4 \\ 3 & 1 & -2 \\ 1 & 5 & -10 \end{pmatrix}$.

37. [R] Find the determinant of the matrix $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & 7 & 1 \\ 1 & 8 & 3 & 1 \\ 1 & 1 & 1 & 4 \end{pmatrix}$.

38. [H] Suppose $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$ has determinant 5. Find

$$\begin{array}{ll} \text{a) } \det \begin{pmatrix} 3a & 3b & 3c \\ 2d & 2e & 2f \\ -g & -h & -i \end{pmatrix}, & \text{b) } \det \begin{pmatrix} a+2d & b+2e & c+2f \\ d-g & e-h & f-i \\ g & h & i \end{pmatrix}, \\ \text{c) } \det \begin{pmatrix} d & e & f \\ g & h & i \\ a & b & c \end{pmatrix}, & \text{d) } \det(7A). \end{array}$$

39. [R] Given that A is a 3×3 matrix with $\det A = -2$. Calculate:

$$\text{a) } \det A^T, \quad \text{b) } \det A^{-1}, \quad \text{c) } \det A^5.$$

40. [R] Evaluate $\det(A)$, $\det(B)$ and hence $\det(AB)$, where

$$A = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 3 & 5 \\ 3 & 4 & -2 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & -1 & 0 \\ -3 & 2 & 4 \\ 2 & 5 & 0 \end{pmatrix}.$$

41. [R] For what values of a is the matrix $\begin{pmatrix} 1 & 2 & 2 \\ 1 & 3 & 1 \\ 1 & 3 & a \end{pmatrix}$ invertible?

42. [H] Long long ago, a mathematician wrote C and C^{-1} on a piece of paper. Unfortunately insects have damaged the paper and all that is left is

$$C = \begin{bmatrix} -2 & -1 & 1 \\ 1 & 2 & -1 \\ * & * & * \end{bmatrix} \quad \text{and} \quad C^{-1} = \begin{bmatrix} * & 0 & -1 \\ 2 & * & -1 \\ 5 & 1 & * \end{bmatrix}$$

$$\text{a) Find } C^{-1}. \quad \text{b) Find } C. \quad \text{c) Find } \det C.$$

43. [H] Show that

$$\det \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1+a & 1 & 1 \\ 1 & 1 & 1+b & 1 \\ 1 & 1 & 1 & 1+c \end{pmatrix} = abc.$$

44. [H] Let U_1 and U_2 be two $n \times n$ row-echelon matrices. Prove that $\det(U_1) \det(U_2) = \det(U_1 U_2)$.

$$45. [\text{H}] \text{ Let } A = \begin{pmatrix} \alpha & 1 & -1 \\ \alpha & 2\alpha+2 & \alpha \\ \alpha-3 & \alpha-3 & \alpha-3 \end{pmatrix}.$$

- a) Factorise $\det(A)$.
b) Hence, find the values of α will there be a nonzero solution of $A\mathbf{x} = \mathbf{0}$.

46. [R] Show by constructing an example that in general $\det(A+B) \neq \det(A) + \det(B)$.

47. [R] Show by constructing an example that in general $\det(\lambda A) \neq \lambda \det(A)$.
48. [H] Use the product rule for determinants to show that a square orthogonal matrix Q (see Question 27) has a value for $\det(Q)$ of $+1$ or -1 .
49. [X] Use the product rule for determinants to show that a square unitary matrix Q (see Question 29) has $\det(Q) = e^{i\theta}$ for some angle θ .
50. [X] Let A and B be two matrices which differ only in the first column, i.e., let

$$A = (\mathbf{a}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n) \quad \text{and} \quad B = (\mathbf{b}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n),$$

where $\mathbf{a}_1$ and $\mathbf{b}_1$ are the first columns of A and B and where $\mathbf{a}_i$, $i = 2, 3, \dots, n$, are the remaining columns of both A and B . Let C be the matrix

$$C = (\mathbf{a}_1 + \mathbf{b}_1 \quad \mathbf{a}_2 \quad \cdots \quad \mathbf{a}_n)$$

obtained by replacing the first column of A (or B) by the sum of the first columns of A and B .

Show that

$$\det(C) = \det(A) + \det(B).$$

Explain why the result of this question also holds for adding two matrices which differ only in one column (not necessarily the first) or which differ only in one row.

51. [X] Prove the following relationships between “volumes” and determinants:
- a) In two dimensions, let $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$, and consider the parallelogram spanned by $\mathbf{a}$ and $\mathbf{b}$. Show that a parametric vector form for the parallelogram is

$$\mathbf{x} = \lambda_1 \mathbf{a} + \lambda_2 \mathbf{b} \quad \text{for} \quad 0 \leq \lambda_1 \leq 1, \quad 0 \leq \lambda_2 \leq 1,$$

and then show that the area of the parallelogram is equal to $|\det(A)|$, where A is the matrix with rows $\mathbf{a}$ and $\mathbf{b}$.

- b) In three dimensions, let $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$, $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ and $\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix}$, and consider the parallelepiped spanned by $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$. A parametric vector form for the equation of the parallelepiped is

$$\mathbf{x} = \lambda_1 \mathbf{a} + \lambda_2 \mathbf{b} + \lambda_3 \mathbf{c} \quad \text{for} \quad 0 \leq \lambda_1 \leq 1, \quad 0 \leq \lambda_2 \leq 1, \quad 0 \leq \lambda_3 \leq 1.$$

Show that the volume of the parallelepiped is equal to $|\det(A)|$, where A is the matrix with rows $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$.

- c) What is the one-dimensional version of these results?

52. [X] Show that

$$\det \begin{pmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{pmatrix} = (a-b)(b-c)(c-a).$$

53. [X] Show that

$$\det \begin{pmatrix} 1+x & 2 & 3 & 4 \\ 1 & 2+x & 3 & 4 \\ 1 & 2 & 3+x & 4 \\ 1 & 2 & 3 & 4+x \end{pmatrix} = x^3(x+10).$$

54. [X] Factorise the determinant

$$\det \begin{pmatrix} y+z+2x & y & z \\ x & z+x+2y & z \\ x & y & x+y+2z \end{pmatrix}.$$

55. [X] Factorise the determinant

$$\det \begin{pmatrix} z & 1 & 2 \\ 1 & z & 3 \\ 1 & 1 & z+1 \end{pmatrix}$$

and hence solve the simultaneous equations

$$zx + y = 2, \quad x + zy = 3, \quad x + y = z + 1.$$

56. [X] Suppose α, β and γ are the roots of the cubic equation $x^3 + px + q = 0$ and $s_k = \alpha^k + \beta^k + \gamma^k$. Find s_1, s_2, s_3 in terms of p and q and show that

$$\det \begin{pmatrix} s_1 & s_2 & s_3 \\ s_2 & s_3 & s_1 \\ s_3 & s_1 & s_2 \end{pmatrix} = 8p^3 + 27q^3.$$

57. [X] Let $A(x_1, y_1), B(x_2, y_2), C(x_3, y_3)$ be three points in the plane.

a) Suppose A, B and C are collinear. Show that

$$\det \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix} = 0.$$

b) Now suppose that A, B, C are not collinear. By considering the areas of some trapezia (or otherwise), show that the area of the triangle with vertices A, B, C is given by $|D|$ where

$$2D = \det \begin{pmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{pmatrix}.$$

ANSWERS TO SELECTED PROBLEMS

Chapter 1

1. a) $\mathbf{a} + \mathbf{h}$, b) $\mathbf{a} - \mathbf{h}$, c) $\mathbf{a} + \frac{1}{2}\mathbf{h}$, d) $\frac{3}{4}\mathbf{a}$, e) $\frac{3}{4}\mathbf{a} - \frac{1}{2}\mathbf{h}$.

2. a) $\mathbf{0}$, b) $2\overrightarrow{CA}$.

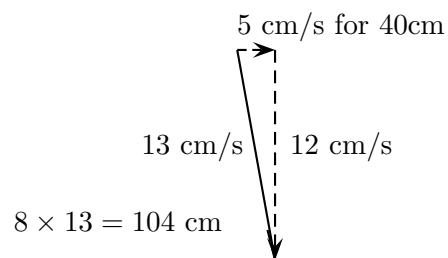
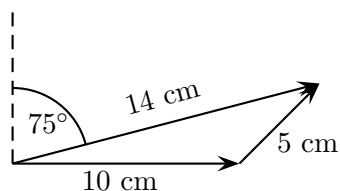
3. a) $-4\mathbf{a} + 5\mathbf{b}$, b) $(2p + 3r)\mathbf{a} + (2q - 3s)\mathbf{b}$.

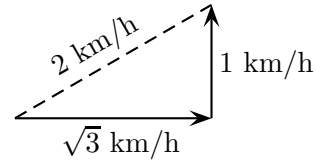
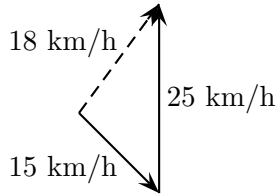
4. a) $\frac{1}{2}(\mathbf{b} + \mathbf{a})$, $\frac{1}{2}(\mathbf{b} + \mathbf{c})$

6. For part (a) and (b) we denote the vector of length 1 cm towards the east by $\mathbf{i}$ and the vector of length 1 cm towards the north by $\mathbf{j}$.

a) $\approx 14 \text{ cm N } 75^\circ \text{ E.}$
 $\approx 13.5\mathbf{i} + 3.5\mathbf{j}$

b) $\approx 104 \text{ cm S } 23^\circ \text{ E,}$
 or exactly $40\mathbf{i} - 96\mathbf{j}$



c) $\approx 18 \text{ km/h N } 36^\circ \text{ E.}$ d) The rower must row 30° upstream,
 ≈ 10.4 minutes to cross7. Approximately $28.0 \text{ km N } 51^\circ 9' \text{ E}$ from A .8. a) $\frac{1}{2}\mathbf{b} + \frac{1}{2}\mathbf{c}$, $\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{c}$, $\frac{1}{4}\mathbf{a} + \frac{3}{4}\mathbf{b}$. b) $\frac{1}{7}\mathbf{a} + \frac{3}{7}\mathbf{b} + \frac{3}{7}\mathbf{c}$.9. a) $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$, b) $\begin{pmatrix} 16 \\ 15 \\ -5 \end{pmatrix}$, c) $\begin{pmatrix} -7 \\ 2 \\ -6 \\ -1 \end{pmatrix}$, d) Not possible, e) $7\mathbf{i} - 4\mathbf{j} + 3\mathbf{k}$.10. 7.43 , $\text{N } 28^\circ \text{ E}$.16. a) not parallel, b) parallel, c) parallel.
Only in b) is $ABCD$ a parallelogram.

17. The correct sketch is:

21. $(4, 5, 0)$, $(-6, -1, 2)$, $(4, 7, 6)$ 22. $\mathbf{d} + \mathbf{e} - \mathbf{f}$, $\mathbf{d} + \mathbf{f} - \mathbf{e}$, $\mathbf{e} + \mathbf{f} - \mathbf{d}$.23. The midpoint is $(3, -1, 3)$. The point Q is $(10, -29, 31)$.24. $\mathbf{t} = \frac{1}{3}\mathbf{a} + \frac{2}{3}\mathbf{b}$ 25. $\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$

$$26. \quad 6, \frac{1}{6} \begin{pmatrix} 4 \\ -4 \\ 2 \end{pmatrix}; \quad \sqrt{14}, \frac{1}{\sqrt{14}} \begin{pmatrix} 2 \\ 1 \\ 0 \\ 3 \end{pmatrix}; \quad \sqrt{21}, \frac{1}{\sqrt{21}} \begin{pmatrix} 4 \\ 0 \\ 1 \\ -2 \\ 0 \end{pmatrix}.$$

$$27. \quad \text{a) } 15, \quad \text{b) } 12, \quad \text{c) } \sqrt{62}.$$

$$28. \quad \sqrt{35}, \sqrt{6}, \sqrt{41}.$$

$$29. \quad \text{A 4-cube has 16 vertices, say, } V = \{(a, b, c, d) \mid a, b, c, d = 0, 1\}.$$

$$30. \quad (5, 0^9)^T, (0, 5, 0^8)^T, \dots, (0^9, 5)^T. \text{ Yes, } (\alpha, \alpha, \dots, \alpha)^T \text{ where } (5 - \alpha)^2 + 9\alpha^2 = 50.$$

$$31. \quad \begin{array}{ll} \text{a) } \mathbf{x} = \begin{pmatrix} 1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 5 \end{pmatrix}, \lambda \in \mathbb{R}; & \text{b) } \mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ -3 \\ 6 \end{pmatrix}, \lambda \in \mathbb{R}; \\ \text{c) } \mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ 0 \\ 2 \end{pmatrix}, \lambda \in \mathbb{R}; & \text{d) } \mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ -1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 1 \\ 2 \\ -2 \end{pmatrix}, \lambda \in \mathbb{R}. \end{array}$$

$$32. \quad \text{Yes, it corresponds to } \lambda = 1.$$

$$33. \quad \begin{array}{ll} \text{a) } \mathbf{x} = \begin{pmatrix} 0 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \lambda \in \mathbb{R}; & \text{b) } \mathbf{x} = \begin{pmatrix} 0 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \end{pmatrix}, \lambda \in \mathbb{R}; \\ \text{c) } \mathbf{x} = \lambda \begin{pmatrix} 1 \\ -7 \end{pmatrix}, \lambda \in \mathbb{R}; & \text{d) } \mathbf{x} = \begin{pmatrix} 0 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \lambda \in \mathbb{R}; \\ \text{e) } \mathbf{x} = \begin{pmatrix} -2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}. & \end{array}$$

$$34. \quad \text{a) } \mathbf{x} = \begin{pmatrix} -4 \\ 1 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} 6 \\ 1 \\ 0 \end{pmatrix} \quad \text{or} \quad \frac{x_1 + 4}{6} = x_2 - 1, \quad x_3 = 3.$$

$$\text{b) } \mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ -3 \end{pmatrix} + \lambda \begin{pmatrix} 4 \\ -5 \\ 6 \end{pmatrix} \quad \text{or} \quad \frac{x_1 - 1}{4} = \frac{x_2 - 2}{-5} = \frac{x_3 + 3}{6}.$$

$$\text{c) } \mathbf{x} = \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ -1 \\ 2 \end{pmatrix} \quad \text{or} \quad \frac{x_1 - 1}{5} = \frac{x_2 + 1}{-1} = \frac{x_3 - 1}{2}.$$

$$\text{d) } \mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 0 \\ 3 \\ 3 \end{pmatrix} \quad \text{or} \quad x_1 = 1, \quad x_2 = x_3.$$

35. $\begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}$. a) $\mathbf{x} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ -1 \end{pmatrix}, \quad \lambda \in \mathbb{R}.$
36. a) true, b) false, c) true, d) true.
37. a) $\mathbf{x} = \mathbf{a} + \lambda(\mathbf{b} - \mathbf{a}), \quad 0 \leq \lambda \leq 1;$ b) $\mathbf{x} = \mathbf{b} + \lambda(\mathbf{a} - \mathbf{b}), \quad \lambda \geq 0;$
 c) $\mathbf{x} = \mathbf{b} + \lambda(\mathbf{a} - \mathbf{b}), \quad \lambda \geq 1;$ d) $\mathbf{x} = \mathbf{a} + \lambda(\mathbf{b} - \mathbf{a}), \quad \lambda \geq \frac{1}{2}.$
38. a) Line segment joining $(1, 3, 6)$ and $(-2, 4, 13)$.
 b) Line segment joining $(3, -3, -5, -3, -13)$ and $(-9, 27, 49, 15, 23)$.
 c) Line segment joining $(0, 4, 8, 3, -5, 4)$ and $(6, -2, 7, 2, -1, 5)$.
 d) Ray from point $(1, 4, -6, 2)$ parallel to $(3, 0, -1, 5)$.
 e) Line through $(3, 1, -4)$ parallel to $\begin{pmatrix} 6 \\ -2 \\ 7 \end{pmatrix}$ with segment from $(-9, 5, -18)$ to $(15, -3, 10)$ removed.
39. a) $\mathbf{x} = \lambda \begin{pmatrix} 3 \\ -1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 4 \\ -6 \end{pmatrix}; \quad \lambda, \mu \in \mathbb{R}.$
 b) $\mathbf{x} = \begin{pmatrix} 1 \\ 4 \\ -2 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 2 \\ 6 \end{pmatrix} + \mu \begin{pmatrix} 0 \\ -14 \\ 5 \end{pmatrix}; \quad \lambda, \mu \in \mathbb{R}.$
40. a) Plane through the origin parallel to $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} -2 \\ 3 \\ 4 \end{pmatrix}$.
 b) Line through $(3, 1, 2, 4)$ parallel to $\begin{pmatrix} -2 \\ 1 \\ 3 \\ 2 \end{pmatrix}$.
 c) Line through origin parallel to $\begin{pmatrix} 3 \\ 2 \\ 1 \\ 2 \end{pmatrix}$.
 d) Plane through $(1, 2, 3)$ parallel to $\begin{pmatrix} 4 \\ -1 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} 8 \\ 2 \\ 4 \end{pmatrix}$.
41. a) $\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix} + \lambda_2 \begin{pmatrix} -1 \\ 2 \\ -3 \end{pmatrix}$ for $\lambda_1, \lambda_2 \in \mathbb{R};$

$$\text{b) } \mathbf{x} = \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} + \lambda_1 \begin{pmatrix} -4 \\ 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ 6 \\ -6 \end{pmatrix} \text{ for } \lambda_1, \lambda_2 \in \mathbb{R};$$

$$\text{c) } \mathbf{x} = \begin{pmatrix} -2 \\ 4 \\ 1 \\ 6 \end{pmatrix} + \lambda_1 \begin{pmatrix} 5 \\ -2 \\ 5 \\ -7 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ 0 \\ -1 \\ -6 \end{pmatrix} \text{ for } \lambda_1, \lambda_2 \in \mathbb{R};$$

$$\text{d) } \mathbf{x} = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} + \lambda_1 \begin{pmatrix} -3 \\ 0 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ 4 \\ 0 \end{pmatrix} \text{ for } \lambda_1, \lambda_2 \in \mathbb{R};$$

$$\text{e) } \mathbf{x} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ \frac{6}{5} \\ 1 \end{pmatrix} \text{ for } \lambda_1, \lambda_2 \in \mathbb{R};$$

$$\text{f) } \mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} + \lambda_1 \begin{pmatrix} 4 \\ 0 \\ -4 \\ 5 \end{pmatrix} + \lambda_2 \begin{pmatrix} 7 \\ 2 \\ -3 \\ -5 \end{pmatrix} \text{ for } \lambda_1, \lambda_2 \in \mathbb{R}.$$

$$42. \quad \text{a) } \mathbf{x} = \lambda_1 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}; \lambda_1, \lambda_2 \in \mathbb{R}.$$

$$\text{b) } \mathbf{x} = \begin{pmatrix} 4 \\ 0 \\ 0 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} -4 \\ 0 \\ 3 \end{pmatrix}; \lambda_1, \lambda_2 \in \mathbb{R}.$$

$$\text{c) } \mathbf{x} = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ -6 \\ 1 \end{pmatrix}; \lambda_1, \lambda_2 \in \mathbb{R}.$$

$$\text{d) } \mathbf{x} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}; \lambda_1, \lambda_2 \in \mathbb{R}.$$

$$44. \quad \text{a) } (3, 2, 4), \quad \text{b) } (3, -4, 11).$$

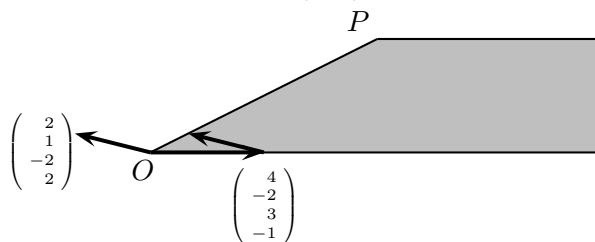
$$45. \quad \text{a) } 6x - 3y + 2z = -12, \quad \text{b) } 6x - 12y + 13z = 100.$$

$$46. \quad \text{a) } \mathbf{x} = \begin{pmatrix} 3 \\ -2 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ 3 \\ 1 \end{pmatrix} \text{ for } \lambda \in \mathbb{R}. \quad \text{b) } (-13, 22, 9).$$

$$47. \quad \text{a) } \mathbf{x} = \begin{pmatrix} 6 \\ 4 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 2 \\ -2 \end{pmatrix} \text{ for } \lambda \in \mathbb{R}. \quad \text{b) } (1, 2, 3).$$

$$48. \quad \text{a) } \text{Parallelogram with vertices } (0, 1), (1, 3), (2, 4), (3, 6).$$

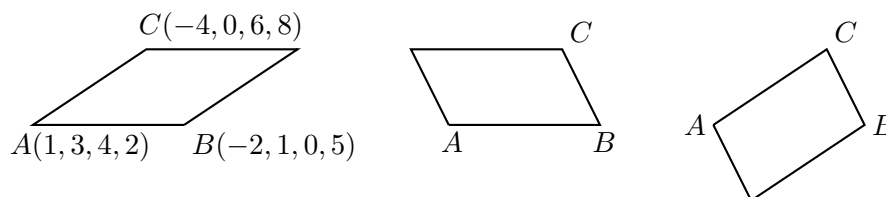
- b) Triangle with vertices $(0, 1)$, $(1, 3)$, $(3, 6)$.
- c) Parallelogram with vertices $(0, 0, 0)$, $(12, 6, -12)$, $(32, -16, 24)$, $(44, -10, 12)$.
- d) Triangle with vertices $(0, 0, 0)$, $(12, 6, -12)$, $(36, -6, 6)$.
- e) An unbounded region with vertices O and P and two of the three sides parallel to $\begin{pmatrix} 4 \\ -2 \\ 3 \\ -1 \end{pmatrix}$. At P , $\lambda_1 = \lambda_2 = 6$ and so $\overrightarrow{OP} = \begin{pmatrix} 36 \\ -6 \\ 6 \\ 6 \end{pmatrix}$.



49. a) See c).

b) $\mathbf{x} = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 2 \end{pmatrix} + \lambda_1 \begin{pmatrix} -3 \\ -2 \\ -4 \\ 3 \end{pmatrix} + \lambda_2 \begin{pmatrix} -2 \\ -1 \\ 6 \\ 3 \end{pmatrix}$ for $0 \leq \lambda_1 \leq 1$, $0 \leq \lambda_2 \leq \lambda_1$.

- c) The three parallelograms are:



The algebraic definitions are:

$$\mathbf{x} = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 2 \end{pmatrix} + \lambda_1 \begin{pmatrix} -3 \\ -2 \\ -4 \\ 3 \end{pmatrix} + \lambda_2 \begin{pmatrix} -5 \\ -3 \\ 2 \\ 6 \end{pmatrix} \text{ for } 0 \leq \lambda_1 \leq 1, 0 \leq \lambda_2 \leq 1.$$

$$\mathbf{x} = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 2 \end{pmatrix} + \lambda_1 \begin{pmatrix} -3 \\ -2 \\ -4 \\ 3 \end{pmatrix} + \lambda_2 \begin{pmatrix} -2 \\ -1 \\ 6 \\ 3 \end{pmatrix} \text{ for } 0 \leq \lambda_1 \leq 1, 0 \leq \lambda_2 \leq 1.$$

$$\mathbf{x} = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 2 \end{pmatrix} + \lambda_1 \begin{pmatrix} 2 \\ 1 \\ -6 \\ -3 \end{pmatrix} + \lambda_2 \begin{pmatrix} -5 \\ -3 \\ 2 \\ 6 \end{pmatrix} \text{ for } 0 \leq \lambda_1 \leq 1, \ 0 \leq \lambda_2 \leq 1.$$

Chapter 2

$$1. \quad \text{a) } \frac{\pi}{4} \quad \text{b) } \cos^{-1}\left(\frac{1}{10\sqrt{3}}\right) \approx 86^\circ 41', \quad \text{c) } \frac{\pi}{2}, \quad \text{d) } \cos^{-1}\left(\frac{7}{10\sqrt{13}}\right) \approx 78^\circ 48'.$$

$$2. \quad \text{a) } 0, \frac{2}{\sqrt{6}}, \frac{1}{\sqrt{3}}; \quad \text{b) } \frac{4}{3\sqrt{2}}, -\frac{5}{\sqrt{33}}, \frac{8}{\sqrt{66}}; \quad \text{c) } \frac{7}{3\sqrt{10}}, -\frac{1}{\sqrt{42}}, \frac{8}{\sqrt{105}}.$$

$$3. \quad \cos^{-1}\left(\frac{1}{3}\right) \approx 70^\circ 32'.$$

$$7. \quad \lambda_1 = \mathbf{a} \cdot \mathbf{u}_1 = \frac{1}{\sqrt{2}}, \quad \lambda_2 = \mathbf{a} \cdot \mathbf{u}_2 = -3, \quad \lambda_3 = \mathbf{a} \cdot \mathbf{u}_3 = \frac{3}{\sqrt{2}}.$$

$$8. \quad \text{a) } \begin{pmatrix} 5 \\ \frac{5}{2} \\ 1 \end{pmatrix}. \quad \text{b) } \frac{\pi}{2}. \quad \text{c) } \frac{\sqrt{66}}{2}. \quad \text{d) } \frac{1}{17} \begin{pmatrix} 80 \\ 50 \\ 22 \end{pmatrix}.$$

$$9. \quad \text{a) } \frac{1}{3} \begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix}, \quad \text{b) } \frac{3}{14} \begin{pmatrix} -1 \\ 3 \\ 0 \\ 2 \end{pmatrix}, \quad \text{c) } \begin{pmatrix} -3 \\ 3 \\ 6 \end{pmatrix}.$$

$$10. \quad \text{a) } 7, \quad \text{b) } 3, \quad \text{c) } \sqrt{6}.$$

$$11. \quad \mathbf{q} = -\mathbf{p} + 2\mathbf{a} + 2\text{proj}_{\mathbf{d}}(\mathbf{p} - \mathbf{a}).$$

$$12. \quad \text{b) } q(\lambda_0) = \mathbf{a} \cdot \mathbf{a} - \frac{(\mathbf{a} \cdot \mathbf{b})^2}{\mathbf{b} \cdot \mathbf{b}}$$

$$14. \quad \text{a) } \begin{pmatrix} 16 \\ -4 \\ -2 \end{pmatrix}, \quad \text{b) } \begin{pmatrix} -23 \\ -11 \\ 20 \end{pmatrix}, \quad \text{c) } \begin{pmatrix} -45 \\ 9 \\ -18 \end{pmatrix}.$$

$$15. \quad \begin{pmatrix} 12 \\ -8 \\ 6 \end{pmatrix}.$$

17. a) $2\sqrt{21}$, $\begin{pmatrix} 8 \\ -4 \\ 2 \end{pmatrix}$; b) $2\sqrt{2}$, $\begin{pmatrix} 0 \\ -2 \\ -2 \end{pmatrix}$.

18. a) $\sqrt{2}$; b) $\frac{15}{2}$.

19. a) $-\frac{4}{3\sqrt{2}}$; b) $\frac{1}{\sqrt{2}}$.

20. a) 2, b) $\frac{5\sqrt{2}}{6}$, c) 7.

23. a) Line through A and B is $\mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} + \lambda_1 \begin{pmatrix} -2 \\ 2 \\ 1 \end{pmatrix}$, $\lambda_1 \in \mathbb{R}$.

Line through C and D is $\mathbf{x} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix}$, $\lambda_2 \in \mathbb{R}$.

b) Shortest distance is $\frac{3}{\sqrt{17}}$.

c) Point P is $\left(-\frac{21}{17}, \frac{38}{17}, \frac{53}{17}\right)$ and Q is $\left(-\frac{30}{17}, \frac{32}{17}, \frac{47}{17}\right)$.

25. a) 14, b) 53.

27. As usual, the answers for equations of planes are not unique.

a) $\mathbf{x} = \begin{pmatrix} 3 \\ 0 \\ 0 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$; $\begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} \cdot \left(\mathbf{x} - \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}\right) = 0$;
 $x_1 - x_2 - 2x_3 = 3$.

b) $\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} + \lambda_1 \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}$; $\begin{pmatrix} -5 \\ 5 \\ -5 \end{pmatrix} \cdot \left(\mathbf{x} - \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}\right) = 0$;
 $x_1 - x_2 + x_3 = -3$.

c) $\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} + \lambda_1 \begin{pmatrix} -2 \\ -1 \\ 4 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$, $\begin{pmatrix} -7 \\ 10 \\ -1 \end{pmatrix} \cdot \left(\mathbf{x} - \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}\right) = 0$;
 $7x_1 - 10x_2 + x_3 = -15$.

d) $\mathbf{x} = \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1/2 \\ 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} -1/4 \\ 0 \\ 1 \end{pmatrix}$, $\begin{pmatrix} -4 \\ 2 \\ -1 \end{pmatrix} \cdot \left(\mathbf{x} - \begin{pmatrix} -1 \\ 0 \\ 0 \end{pmatrix}\right) = 0$;
 $4x_1 - 2x_2 + x_3 = -4$.

$$\text{e) } \mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} + \lambda_1 \begin{pmatrix} -1 \\ 2 \\ -2 \end{pmatrix} + \lambda_2 \begin{pmatrix} 14 \\ 2 \\ -6 \end{pmatrix}, \quad \begin{pmatrix} 4 \\ 17 \\ 15 \end{pmatrix} \cdot \left(\mathbf{x} - \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} \right) = 0;$$

$$4x_1 + 17x_2 + 15x_3 = 8.$$

$$28. \quad \text{a) } \mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}, \quad \text{for } \lambda, \mu \in \mathbb{R}.$$

$$\text{b) } \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix}. \quad \text{c) } x_1 + x_2 + x_3 = 7.$$

$$29. \quad \text{a) } \begin{pmatrix} 4 \\ 4 \\ 2 \end{pmatrix}, \quad \text{b) } \frac{1}{2} \begin{pmatrix} 5 \\ 2 \\ -1 \end{pmatrix}.$$

$$30. \quad \text{a) } 3, \quad \text{b) } \sqrt{6}, \quad \text{c) } \frac{13}{7}, \quad \text{d) } \frac{25}{7}.$$

$$31. \quad \text{a) } \mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} -2 \\ 1 \\ 1 \end{pmatrix}, \quad \lambda, \mu \in \mathbb{R}.$$

$$\text{b) } \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}. \quad \text{c) } \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \cdot \left(\mathbf{x} - \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} \right) = 0. \quad \text{d) } \frac{8}{\sqrt{3}}.$$

$$32. \quad \text{a) } \mathbf{c} = \text{proj}_{\mathbf{a}} \mathbf{v}, \quad \mathbf{d} = \mathbf{v} - \mathbf{c}. \quad \text{b) } \mathbf{c} = \begin{pmatrix} \frac{3}{11} \\ -\frac{1}{11} \\ -\frac{1}{11} \end{pmatrix}, \quad \mathbf{d} = \begin{pmatrix} \frac{8}{11} \\ \frac{12}{11} \\ \frac{12}{11} \end{pmatrix}.$$

Chapter 3

1.

	$x \in \mathbb{N}$	$x \in \mathbb{Z}$	$x \in \mathbb{Q}$	$x \in \mathbb{R}$
a)	-	-25	-25	-25
	3	3	3	3
	-	-3	-3	-3
	-	-	$-\frac{10}{3}$	$-\frac{10}{3}$
b)	1	1, -5	1, -5	1, -5
	5	5	$5, \frac{3}{2}$	$5, \frac{3}{2}$
	-	-	-	$\frac{1 \pm \sqrt{5}}{2}$
	-	-	-	-
c)	$3j, j \in \mathbb{N}$	$3j, j \in \mathbb{Z}$	$3j, j \in \mathbb{Z}$	$3j, j \in \mathbb{Z}$
	0	0	0	$3k\pi, k \in \mathbb{Z}$

2. No.

3. Yes. The set $\{0\}$ and the empty set $\emptyset = \{ \}$.

4. Yes.

$$5. \quad 3z = 6 + 9i, \quad z^2 = -5 + 12i, \quad z + 2w = 7i, \quad z(w + 3) = -2 + 10i, \quad \frac{z}{w} = \frac{1}{5}(4 - 7i), \\ \frac{w}{z} = \frac{1}{13}(4 + 7i).$$

$$6. \quad \text{a) } \frac{1}{5}(3 - i), \quad \text{b) } -\frac{1}{2}(1 - i).$$

$$7. \quad \text{a) } a^2 - b^2 + 2abi, \quad \text{b) } \frac{a}{a^2 + b^2} - i \frac{b}{a^2 + b^2}, \quad \text{c) } \frac{1}{(a - 1)^2 + b^2} ((a^2 - 1 + b^2) - 2ib).$$

$$8. \quad \text{a) } \frac{1}{2}(-1 \pm \sqrt{3}i), \quad \text{b) } -1 \pm \sqrt{2}i, \quad \text{c) } 3 \pm i, \quad \text{d) } \frac{1}{2}(3 \pm \sqrt{3}), \quad \text{e) } \pm i, \pm 2i.$$

10. 16

$$11. \quad \frac{8abi(a^2 - b^2)}{(a^2 + b^2)^2}$$

12.

z	$\operatorname{Re}(z)$	$\operatorname{Im}(z)$	$\bar{z}$
$-1 + i$	-1	1	$-1 - i$
$2 + 3i$	2	3	$2 - 3i$
$2 - 3i$	2	-3	$2 + 3i$
$\frac{2-i}{1+i}$	$\frac{1}{2}$	$-\frac{3}{2}$	$\frac{1+3i}{2}$
$\frac{1}{(1+i)^2}$	0	$-\frac{1}{2}$	$\frac{i}{2}$

13. $-3 + 4i, \quad \frac{11}{25} - \frac{2}{25}i.$

14. $z = 2 + 3i, \quad w = -1 + 2i.$

17. b) $z^2 - 6z + 13$

18.

z	$ z $	$\operatorname{Arg}(z)$	Polar Form
$6 + 6i$	$6\sqrt{2}$	$\frac{\pi}{4}$	$6\sqrt{2} (\cos \frac{\pi}{4} + i \sin \frac{\pi}{4})$
-4	4	π	$4(\cos \pi + i \sin \pi)$
$\sqrt{3} - i$	2	$-\frac{\pi}{6}$	$2 (\cos \frac{\pi}{6} - i \sin \frac{\pi}{6})$
$\frac{-1}{\sqrt{2}} - \frac{i}{\sqrt{2}}$	1	$-\frac{3\pi}{4}$	$\cos \frac{3\pi}{4} - i \sin \frac{3\pi}{4}$
$-7 + 3i$	$\sqrt{58}$	α	$\sqrt{58}(\cos \alpha + i \sin \alpha)$

Here $\alpha = \pi - \tan^{-1} \frac{3}{7}.$

19. $\sqrt{234}, \quad -1.$

20. $n = 4$

21. a) $\frac{3}{2}(1 + \sqrt{3}i), \quad \text{b) } \frac{3}{2}(-\sqrt{3} + i), \quad \text{c) } -\frac{3}{2}(1 + \sqrt{3}i), \quad \text{d) } \frac{3}{2}(\sqrt{3} - i),$
 e) $\frac{3}{2}(\sqrt{2 + \sqrt{2}} + i\sqrt{2 - \sqrt{2}})$ (Double angle formula used).

27. $64, \quad -(1 + \sqrt{3})i, \quad \frac{1 + \sqrt{3}}{2} + \frac{-1 + \sqrt{3}}{2}i.$

28. $\frac{7}{2}.$

29. $\pi.$

30. $\operatorname{Arg}(-1 + i) = \frac{3\pi}{4}; \operatorname{Arg}(-\sqrt{3} + i) = \frac{5\pi}{6};$
 $\operatorname{Arg}((-1 + i)(-\sqrt{3} + i)) = -\frac{5\pi}{12}; \operatorname{Arg}\left(\frac{-1 + i}{-\sqrt{3} + i}\right) = -\frac{\pi}{12}.$

$$31. \sin \frac{7\pi}{12} = \frac{1 + \sqrt{3}}{2\sqrt{2}}.$$

$$32. zw = 2\sqrt{2}e^{i\pi/12} = 2\sqrt{2} \left[\cos \left(\frac{\pi}{12} \right) + i \sin \left(\frac{\pi}{12} \right) \right]; z^9 = -512; \left(\frac{z}{w} \right)^{12} = 64e^{i\pi} = -64.$$

$$33. \quad \text{a) } 16(-\sqrt{3} + i), \quad \text{b) } -i, \quad \text{c) } -\frac{1}{2} - i\frac{\sqrt{3}}{2}.$$

$$34. \quad \text{a) } \pm(5 - 2i), \quad \text{b) } \pm(3 + 5i), \quad \text{c) } \pm(7 + 5i).$$

$$35. \quad \text{b) } \sqrt{2}e^{\frac{5\pi i}{12}} = \frac{1}{2}((\sqrt{3} - 1) + i(\sqrt{3} + 1)), \quad \text{c) } \frac{1}{\sqrt{2}}(1 + 13i).$$

$$37. \quad \text{a) } 2 + i, 1 - i; \quad \text{b) } 4 + i, 3 - 2i; \quad \text{c) } 1 - 2i, -5 + 3i.$$

$$38. e^{i\pi/7}, e^{3i\pi/7}, e^{5i\pi/7}, e^{i\pi}, e^{-i\pi/7}, e^{-3i\pi/7}, e^{-5i\pi/7}.$$

$$39. e^{in\pi/12} \quad \text{for } n = -11, -7, -3, 1, 5, 9.$$

$$40. 2e^{in\pi/15} \quad \text{for } n = -13, -7, -1, 5, 11.$$

$$41. \frac{15}{2} + i \left(\frac{3\sqrt{3}}{2} - 1 \right), \quad 3 - i, \quad \frac{15}{2} - i \left(\frac{3\sqrt{3}}{2} + 1 \right).$$

$$48. \quad \text{a) Real part} = \cos(2\theta). \text{ Imaginary part} = \sin(2\theta).$$

$$51. \quad \text{a) } \cos 6\theta = \cos^6 \theta - 15 \cos^4 \theta \sin^2 \theta + 15 \cos^2 \theta \sin^4 \theta - \sin^6 \theta$$

$$\sin 6\theta = 6 \cos^5 \theta \sin \theta - 20 \cos^3 \theta \sin^3 \theta + 6 \cos \theta \sin^5 \theta$$

$$\text{b) } \cos 6\theta = 32 \cos^6 \theta - 48 \cos^4 \theta + 18 \cos^2 \theta - 1$$

$$52. \sin 7\theta = 7 \cos^6 \theta \sin \theta - 35 \cos^4 \theta \sin^3 \theta + 21 \cos^2 \theta \sin^5 \theta - \sin^7 \theta$$

$$\cos 7\theta = \cos^7 \theta - 21 \cos^5 \theta \sin^2 \theta + 35 \cos^3 \theta \sin^4 \theta - 7 \cos \theta \sin^6 \theta.$$

$$53. \quad \text{a) } \cos \theta = \frac{1}{2}(e^{i\theta} + e^{-i\theta}).$$

$$\text{b) } \cos^6 \theta = \frac{1}{32}(\cos 6\theta + 6 \cos 4\theta + 15 \cos 2\theta + 10).$$

$$54. \quad \sin^5 \theta = \frac{1}{16}(\sin 5\theta - 5 \sin 3\theta + 10 \sin \theta)$$

$$\int \sin^5 \theta d\theta = \frac{1}{16} \left(-\frac{1}{5} \cos 5\theta + \frac{5}{3} \cos 3\theta - 10 \cos \theta \right) + C,$$

$$\cos^4 \theta = \frac{1}{8}[3 + 4 \cos(2\theta) + \cos(4\theta)]$$

$$\int \cos^4 \theta d\theta = \frac{1}{8} \left[3\theta + 2 \sin(2\theta) + \frac{1}{4} \sin(4\theta) \right] + C.$$

$$55. \quad \text{a) } \cos 5\theta = 16x^5 - 20x^3 + 5x$$

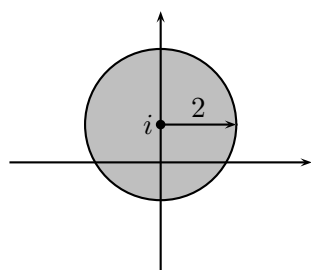
d) $-1, \cos \frac{\pi}{5}, \cos \frac{3\pi}{5}, \cos \frac{7\pi}{5}, \cos \frac{9\pi}{5}.$

56. The sum is n when k is an integer multiple of n and 0 otherwise.

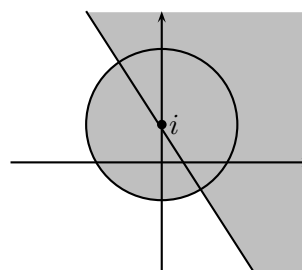
58. $\frac{\sin(\frac{1}{2}(n+1)\theta) \sin(\frac{1}{2}n\theta)}{\sin \frac{1}{2}\theta}.$

59. a) $\frac{9e^{i\theta}}{9 + e^{i2\theta}}.$

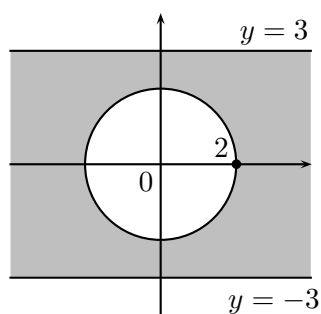
60. a) $|z - i| \leq 2$



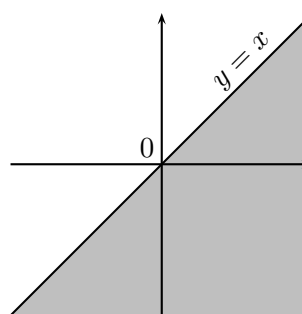
b) $|z - i| \leq 2$ or $-\frac{\pi}{3} \leq \text{Arg}(z - i) \leq \frac{2\pi}{3}$



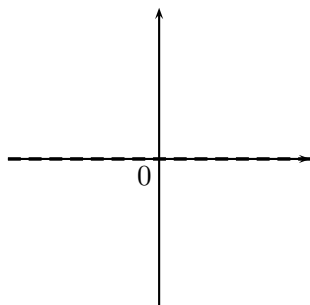
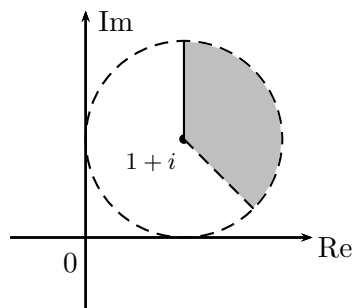
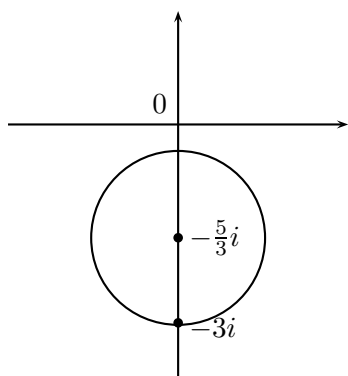
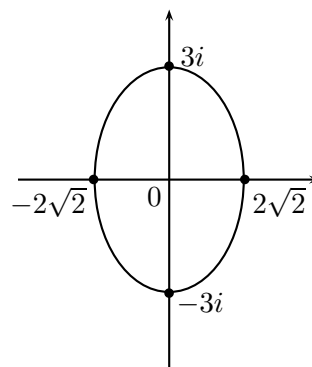
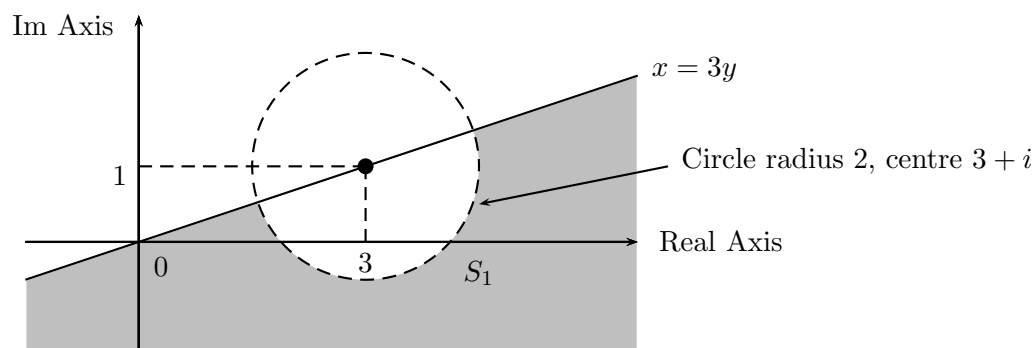
c) $|z| \geq 2$ and $|\text{Im}(z)| \leq 3$



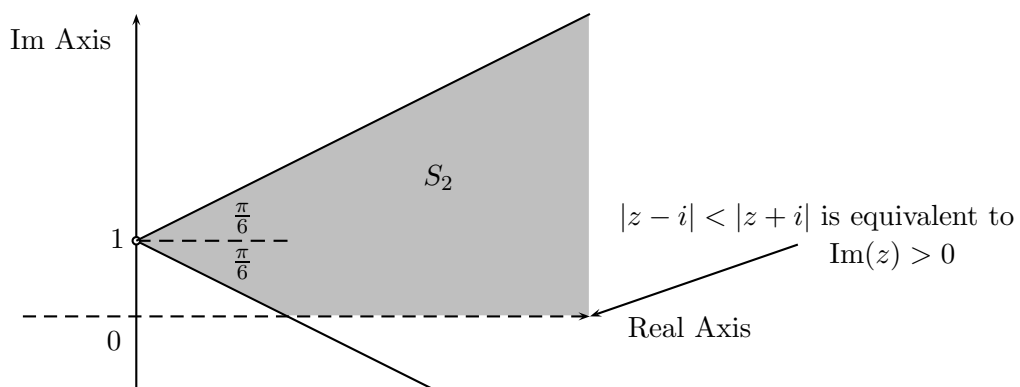
d) $y \leq x$



e) The real axis

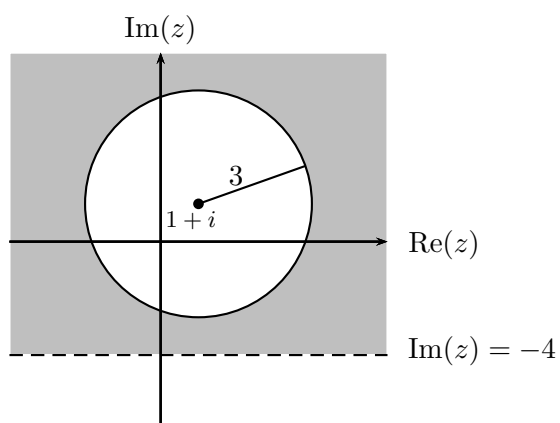
f) $|z - 1 - i| < 1$ & $-\frac{\pi}{4} < \text{Arg}(z - 1 - i) \leq \frac{\pi}{2}$ g) Circle: $x^2 + \left(y + \frac{5}{3}\right)^2 = \left(\frac{4}{3}\right)^2$ h) Ellipse: $\left(\frac{x}{2\sqrt{2}}\right)^2 + \left(\frac{y}{3}\right)^2 = 1$ 61. a) $\text{Re}(z) \geq 3 \text{Im}(z)$ and $|z - (3 + i)| > 2$ 

b) $|z - i| < |z + i|$ and $-\frac{\pi}{6} \leq \text{Arg}(z - i) \leq \frac{\pi}{6}$

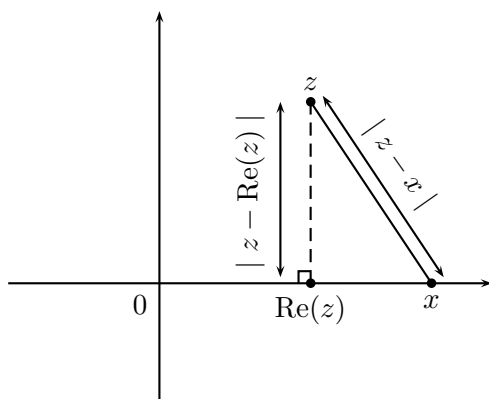


62. a) $\text{Im}(z) > -4$ and $|z - 1 - i| \geq 3$

b) Yes.

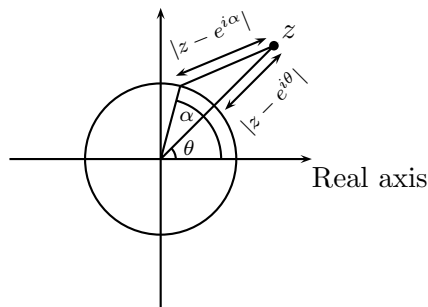
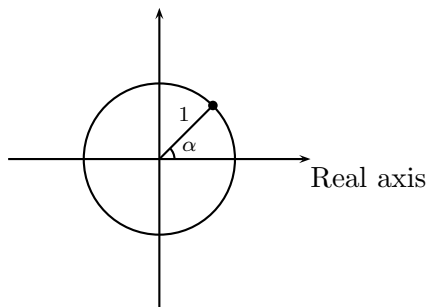


63. $|z - x| \geq |z - \text{Re}(z)|$



64. a) $w = e^{i\alpha}$, $-\pi < \alpha \leq \pi$

c) $|z - e^{i\alpha}| \geq |z - e^{i\theta}|$, $\theta = \text{Arg}(z)$



65. a) 742, b) 129, c) $1 + 9i$.

66. $p(z) = (z - 2)(2z - 5)(z + 3)$.

67. $p(z) = (z - 1)(z + 1)(z + 2)(z + 4)$.

68. a) $(z - e^{-\frac{i\pi}{10}})(z - e^{\frac{3\pi i}{10}})(z - e^{\frac{7\pi i}{10}})(z - e^{-\frac{i\pi}{2}})(z - e^{-\frac{9\pi i}{10}})$.

b) $(z - \sqrt{2}e^{\frac{i\pi}{6}})(z - \sqrt{2}e^{\frac{i\pi}{2}})(z - \sqrt{2}e^{\frac{5\pi i}{6}})(z - \sqrt{2}e^{-\frac{i\pi}{6}})(z - \sqrt{2}e^{-\frac{i\pi}{2}})(z - \sqrt{2}e^{-\frac{5\pi i}{6}})$.

69. a) $(x - 1)(x + 1)(x^2 + 1)(x^2 + \sqrt{2}x + 1)(x^2 - \sqrt{2}x + 1)$.

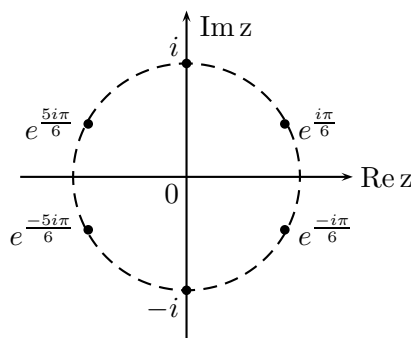
b) $(x^2 + 2)(x^2 + \sqrt{6}x + 2)(x^2 - \sqrt{6}x + 2)$.

70. $(z^2 + 2z + 2)(z^2 - 2z + 2)$

71. $(z - e^{-i\pi/8})(z - e^{i3\pi/8})(z - e^{i7\pi/8})(z - e^{-i5\pi/8})$

72. a) $e^{-\frac{5\pi i}{6}}$, $e^{-\frac{\pi i}{2}}$, $e^{-\frac{\pi i}{6}}$, $e^{\frac{\pi i}{6}}$, $e^{\frac{\pi i}{2}}$, $e^{\frac{5\pi i}{6}}$.

b) Note that the solutions are evenly spaced around the unit circle centred on 0.



c) $(z - e^{-\frac{5\pi i}{6}})(z - e^{-\frac{\pi i}{2}})(z - e^{-\frac{\pi i}{6}})(z - e^{\frac{\pi i}{6}})(z - e^{\frac{\pi i}{2}})(z - e^{\frac{5\pi i}{6}})$.

d) $(z^2 + 1)(z^2 + \sqrt{3}z + 1)(z^2 - \sqrt{3}z + 1)$.

73. a) $e^{i\pi/4}, e^{i\pi/2}, e^{i3\pi/4}, e^{-i\pi/4}, e^{-i\pi/2}, e^{-3\pi/4}$.
 b) $(z - e^{i\pi/4})(z - e^{-i\pi/4})(z - e^{i3\pi/4})(z - e^{-i3\pi/4})(z - e^{i\pi/2})(z - e^{-i\pi/2})$
 c) $(z^2 - \sqrt{2}z + 1)(z^2 + \sqrt{2}z + 1)(z^2 + 1)$.
74. a) $(z - e^{2i\pi/5})(z - e^{-2i\pi/5})(z - e^{4i\pi/5})(z - e^{-4i\pi/5})$
 b) $\left(z^2 - 2z \cos\left(\frac{2\pi}{5}\right) + 1\right)\left(z^2 - 2z \cos\left(\frac{4\pi}{5}\right) + 1\right)$
75. a) $(t + 1 - i)(t + 1 + i)(t - 2)(t + 1)(t + i)(t - i)$,
 b) $(t^2 + 2t + 2)(t - 2)(t + 1)(t^2 + 1)$.
76. $1 + i, 1 - i, \sqrt[3]{5}, \frac{\sqrt[3]{5}}{2}(-1 + i\sqrt{3}), \frac{\sqrt[3]{5}}{2}(-1 - i\sqrt{3})$.
77. a) $(z^2 + z + 1)(z^6 + z^3 + 1)$. b) $e^{\pm 2i\pi/9}, e^{\pm 4i\pi/9}, e^{\pm 8i\pi/9}$.
79. a) One of the roots is $(-2 + 2i)^{1/3} + (-2 - 2i)^{1/3}$.
80. d) $-2, 2\sqrt{2}\cos\frac{\pi}{12}, 2\sqrt{2}\cos\frac{7\pi}{12}$.
81. a) 1; b) $-1, \frac{5}{7}, \frac{1}{4}$; c) $4, \pm\frac{1}{5}$.
90. `evalc((sqrt(2)+7*I)^13);`

Chapter 4

1. a) $\left\{\frac{5}{2}\right\}, \left\{\begin{pmatrix} \frac{5}{2} \\ \lambda \end{pmatrix} : \lambda \in \mathbb{R}\right\}, \left\{\begin{pmatrix} \frac{5}{2} \\ \lambda \\ \mu \end{pmatrix} : \lambda, \mu \in \mathbb{R}\right\}$
 b) $\left\{\begin{pmatrix} 4 - 2\lambda \\ \lambda \end{pmatrix} : \lambda \in \mathbb{R}\right\}, \left\{\begin{pmatrix} 4 - 2\lambda \\ \lambda \\ \mu \end{pmatrix} : \lambda, \mu \in \mathbb{R}\right\}$
 c) $\left\{\begin{pmatrix} \lambda \\ \mu \\ 2 - 2\lambda + 3\mu \end{pmatrix} : \lambda, \mu \in \mathbb{R}\right\}$
2. a) No solution. b) Unique solution $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 8 \\ -9 \end{pmatrix}$.
 c) Infinite number of solutions on the line $\mathbf{x} = \begin{pmatrix} 5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}$.

3. For $a_{11} \neq 0$ the conditions are as follows.
- a) If $a_{11}a_{22} - a_{12}a_{21} \neq 0$, then solution is unique.
 - b) If $a_{11}a_{22} - a_{12}a_{21} = 0$ and $a_{11}b_2 - a_{21}b_1 \neq 0$, then there is no solution.
 - c) If $a_{11}a_{22} - a_{12}a_{21} = 0$ and $a_{11}b_2 - a_{21}b_1 = 0$, then there are an infinite number of solutions.

4. The general conditions are as follows.

- a) If $a_{11}a_{22} - a_{12}a_{21} \neq 0$, then solution is unique.
- b) There is no solution if $a_{11}a_{22} - a_{12}a_{21} = 0$ and either
 - i) $a_{11}b_2 - a_{21}b_1 \neq 0$, or
 - ii) $a_{12}b_2 - a_{22}b_1 \neq 0$, or
 - iii) $a_{11} = a_{12} = a_{21} = a_{22} = 0$ and b_1, b_2 are not both zero.
- c) There are an infinite number of solutions otherwise.

5. a) Solution set = $\left\{ \begin{pmatrix} 1 + \lambda \\ 2 - 2\lambda \\ \lambda \end{pmatrix} : \lambda \in \mathbb{R} \right\}$.

Planes intersect in line $\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}$.

- b) No solution. Planes are parallel.

- c) Solution set = $\left\{ \begin{pmatrix} 4 - \frac{5}{4}\lambda + \frac{1}{2}\mu \\ \lambda \\ \mu \end{pmatrix} : \lambda, \mu \in \mathbb{R} \right\}$. Equations represent the same plane.

8. a) In vector form,

$$x_1 \begin{pmatrix} 3 \\ 5 \\ -1 \end{pmatrix} + x_2 \begin{pmatrix} -3 \\ 2 \\ -1 \end{pmatrix} + x_3 \begin{pmatrix} 4 \\ -3 \\ 6 \end{pmatrix} = \begin{pmatrix} 6 \\ 7 \\ 8 \end{pmatrix}.$$

As a matrix equation and augmented matrix,

$$\begin{pmatrix} 3 & -3 & 4 \\ 5 & 2 & -3 \\ -1 & -1 & 6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 6 \\ 7 \\ 8 \end{pmatrix}; \quad (A|\mathbf{b}) = \left(\begin{array}{ccc|c} 3 & -3 & 4 & 6 \\ 5 & 2 & -3 & 7 \\ -1 & -1 & 6 & 8 \end{array} \right).$$

- b) In vector form,

$$x_1 \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 3 \\ 2 \\ 3 \end{pmatrix} + x_3 \begin{pmatrix} 7 \\ -5 \\ 6 \end{pmatrix} + x_4 \begin{pmatrix} 8 \\ -1 \\ -6 \end{pmatrix} = \begin{pmatrix} -2 \\ 7 \\ 5 \end{pmatrix}.$$

As a matrix equation and augmented matrix,

$$\begin{pmatrix} 1 & 3 & 7 & 8 \\ 3 & 2 & -5 & -1 \\ 0 & 3 & 6 & -6 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} -2 \\ 7 \\ 5 \end{pmatrix}; \quad (A|\mathbf{b}) = \left(\begin{array}{cccc|c} 1 & 3 & 7 & 8 & -2 \\ 3 & 2 & -5 & -1 & 7 \\ 0 & 3 & 6 & -6 & 5 \end{array} \right).$$

9. The system of equation is

$$\begin{array}{rrrrrr} x_1 & - & 3x_2 & & & = & 10 \\ & & 6x_2 & + & 6x_3 & = & -2 \\ -6x_1 & - & x_2 & - & 4x_3 & = & 0 \\ 7x_1 & + & 9x_2 & + & 11x_3 & = & 5 \end{array}$$

The augmented matrix form is

$$A = \left(\begin{array}{ccc|c} 1 & -3 & 0 & 10 \\ 0 & 6 & 6 & -2 \\ -6 & -1 & -4 & 0 \\ 7 & 9 & 11 & 5 \end{array} \right).$$

10. a) $R_2 = R_2 - 2R_1$, $R_3 = R_3 - 4R_1$; b) $R_1 = R_1 - R_2$, $R_2 = \frac{1}{2}R_2$.

11. $\left(\begin{array}{ccc|c} 2 & 4 & 1 & 2 \\ 9 & 14 & 7 & 7 \\ 1 & 3 & 1 & 3 \end{array} \right).$

12. All but c) and h) are in row-echelon form.

13. a) $\mathbf{x} = \begin{pmatrix} 2 \\ 3 \\ -2 \end{pmatrix}$. Point of intersection of 3 planes.

b) $\mathbf{x} = \begin{pmatrix} 2 \\ 3 \\ -2 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 0 \\ 2 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}.$

A line in $\mathbb{R}^4$ through the point $(2, 3, -2, 0)$ and parallel to $\begin{pmatrix} -1 \\ 0 \\ 2 \\ 1 \end{pmatrix}.$

14. a) $\mathbf{x} = \begin{pmatrix} 3 \\ -1 \end{pmatrix}.$ b) $\mathbf{x} = \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}.$ c) $\mathbf{x} = \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}.$

d) No solution. e) $\mathbf{x} = \begin{pmatrix} 0 \\ 5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}.$ f) No solution.

g) $\mathbf{x} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 2 \end{pmatrix}.$ h) $\mathbf{x} = \begin{pmatrix} -3 \\ 6 \\ 5 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -2 \\ -1 \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}.$

15. a) $\left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & -2 \end{array} \right).$

Solution: $\mathbf{x} = \begin{pmatrix} -1 \\ 2 \\ -2 \end{pmatrix}$, which is the position vector of a point in $\mathbb{R}^3$.

b) $\left(\begin{array}{cccc|c} 1 & 0 & 0 & -75 & -34 \\ 0 & 1 & 0 & 29 & 13 \\ 0 & 0 & 1 & 7 & 3 \end{array} \right).$

Solution: $\mathbf{x} = \begin{pmatrix} -34 \\ 13 \\ 3 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 75 \\ -29 \\ -7 \\ 1 \end{pmatrix}$, $\lambda \in \mathbb{R}$, which is a line in $\mathbb{R}^4$.

16. a) Unique solution, b) no solution, c) infinitely many solutions,
d) infinitely many solutions, e) unique solution.

17. a) $k \neq 3$, b) no such value of k , c) $k = 3$.

18. a) $\lambda = \pm 2$, b) $\lambda = 1$, c) all other values of λ .

19. a) $a \neq 0$, b) $a = 0, b \neq 0$, c) $a = b = 0$, d) $\mathbf{x} = \begin{pmatrix} 5 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} -2 \\ \frac{5}{2} \\ -1 \\ 3 \end{pmatrix}$ $\lambda \in \mathbb{R}$.

20. Perhaps, if the costs are negative or very large then you can be sure that someone is cheating.

21. No.

22. a)
$$\begin{array}{rcl} x_1 & = & 7b_1 + 5b_2 + 3b_3 \\ x_2 & = & 6b_1 + 4b_2 + 3b_3 \\ x_3 & = & 2b_1 + b_2 + b_3 \end{array}$$
 b)
$$\begin{array}{rcl} x_1 & = & \frac{3}{2}b_1 - 2b_2 - 2b_3 \\ x_2 & = & -\frac{7}{2}b_1 + 5b_2 + 4b_3 \\ x_3 & = & \frac{1}{2}b_1 - b_2 - b_3 \end{array}$$

24. a) $b_3 - \frac{1}{2}b_1 + b_2 = 0$. b) $b_1 - b_2 + b_3 = 0$ and $-2b_1 + b_2 + b_4 = 0$.

26. Yes.

27. No.

28. Yes, since $\begin{pmatrix} 1 \\ 1 \\ 4 \\ 12 \end{pmatrix} = 3 \begin{pmatrix} 3 \\ -1 \\ 4 \\ 6 \end{pmatrix} - 2 \begin{pmatrix} 4 \\ -2 \\ 4 \\ 3 \end{pmatrix}.$

29. No.

30. Yes, at $(6, 13, 11)$.

31. Yes, since $\begin{pmatrix} 5 \\ 7 \\ -1 \end{pmatrix} = 3 \begin{pmatrix} 3 \\ 5 \\ 1 \end{pmatrix} - 4 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$.

33. Meet at $(6, 9, 4)$.

34. The planes intersect at the line $\mathbf{x} = \begin{pmatrix} 6 \\ -2 \\ 3 \end{pmatrix} + \lambda \begin{pmatrix} -5 \\ 4 \\ 8 \end{pmatrix} \quad \lambda \in \mathbb{R}$.

35. Planes are not parallel as $\lambda_1 \begin{pmatrix} 2 \\ 1 \\ -2 \\ 7 \end{pmatrix} + \lambda_2 \begin{pmatrix} -3 \\ 1 \\ 5 \\ 2 \end{pmatrix} = \mu_1 \begin{pmatrix} 3 \\ -1 \\ 2 \\ 4 \end{pmatrix} + \mu_2 \begin{pmatrix} -1 \\ 4 \\ 2 \\ 6 \end{pmatrix}$
only when $\lambda_1 = \lambda_2 = \mu_1 = \mu_2 = 0$.

37. a) $\mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} \frac{1}{3} \\ \frac{2}{3} \\ 1 \end{pmatrix}, \lambda \in \mathbb{R}$. b) The planes intersect in a line.

38. $p(x) = 2x^2 - 4x + 7$

39. I am 42, my brother is 46 and my sister is 52.

40. 6 days in Bangkok, 4 each in Singapore and Kuala Lumpur.

41. 3, 1, 2.

42. a) Letting x_1 be the number of hectares of wheat, x_2 be the number of hectares of oats and x_3 be the number of hectares of barley gives the equations

$$\begin{array}{rrrrrrcl} x_1 & + & x_2 & + & x_3 & = & 12 \\ 6x_1 & + & 6x_2 & + & 2x_3 & = & 48 \\ 150x_1 & + & 100x_2 & + & 70x_3 & = & 700 \\ 72x_1 & + & 48x_2 & + & 36x_3 & = & 612 \end{array}$$

b) There is no solution.

c) The inequalities are

$$\begin{array}{rrrrrrcl} x_1 & + & x_2 & + & x_3 & \leq & 12 \\ 6x_1 & + & 6x_2 & + & 2x_3 & \leq & 48 \\ 150x_1 & + & 100x_2 & + & 70x_3 & \leq & 700 \\ 72x_1 & + & 48x_2 & + & 36x_3 & \leq & 612 \end{array}$$

and with slack variables s_1, s_2, s_3, s_4 the equations are

$$\begin{array}{ccccccccccc} x_1 & + & x_2 & + & x_3 & + & s_1 & & & & = & 12 \\ 6x_1 & + & 6x_2 & + & 2x_3 & & & + & s_2 & & = & 48 \\ 150x_1 & + & 100x_2 & + & 70x_3 & & & & & + & s_3 & = & 700 \\ 72x_1 & + & 48x_2 & + & 36x_3 & & & & & & + & s_4 & = & 612 \end{array}$$

- d) Some sensible solutions are to either plant $4\frac{2}{3}$ hectares wheat and no oats and barley, or 7 hectares oats and no wheat and barley, or 10 hectares barley and no wheat and oats. There are also an infinite number of other reasonable solutions. In each case it is the fertiliser which is restricting the planting.

44. a) Π_1 is $x + 2y - z = 2$, Π_2 is $3x + 6y - z = 12$, Π_3 is $2x + 4y - z = 7$.

b) $\mathbf{x} = \begin{pmatrix} 5 - 2t_2 \\ t_2 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \\ 3 \end{pmatrix} + t_2 \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}, \quad t_2 \in \mathbb{R}$

The intersection is a line through $(5, 0, 3)$ and parallel to $\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$.

c) $x = -2y + 5$ and $z = 3$.

45. a) $\begin{cases} x - 2y + z = a \\ 3x + 6y + 8z = b \\ 4x + 2y + 7z = c \\ 7x - 8y + 6z = d \end{cases} \quad \text{b) } d - a + 2b - 3c = 0. \quad \text{c) } \left(\frac{4}{7}, -\frac{1}{7}, \frac{1}{7} \right).$

Chapter 5

$$1. \quad a) \quad 3A = \begin{pmatrix} 6 & -9 & 12 \\ 9 & 6 & -6 \\ 3 & -3 & 9 \end{pmatrix}.$$

$$b) \quad -2B = \begin{pmatrix} 4 & -2 \\ -6 & -8 \\ 2 & -10 \end{pmatrix}.$$

c) $A + B$ is not defined.

$$d) \quad B + C = \begin{pmatrix} -5 & 3 \\ 4 & 0 \\ 5 & 7 \end{pmatrix}.$$

$$e) \quad A + 3I = \begin{pmatrix} 5 & -3 & 4 \\ 3 & 5 & -2 \\ 1 & -1 & 6 \end{pmatrix}.$$

f) $B + 3I$ is not defined.

$$g) \quad AB = \begin{pmatrix} -17 & 10 \\ 2 & 1 \\ -8 & 12 \end{pmatrix}.$$

h) BA is not defined.

i) BC is not defined.

$$j) \quad CD = \begin{pmatrix} -4 & -13 & -9 \\ -2 & 11 & 13 \\ 14 & 14 & 0 \end{pmatrix}.$$

$$k) \quad A^2 = \begin{pmatrix} -1 & -16 & 26 \\ 10 & -3 & 2 \\ 2 & -8 & 15 \end{pmatrix}.$$

l) B^2 is not defined.

$$m) \quad (BD)^2 = \begin{pmatrix} -86 & 81 & 167 \\ -47 & 38 & 85 \\ -187 & 171 & 358 \end{pmatrix}.$$

$$7. \quad 96A + 205I.$$

$$8. \quad N^2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad N^3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

$$11. \quad a) \quad \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad b) \quad (0 \ 0 \ 1), \quad c) \quad \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix}, \quad d) \quad (1 \ -2 \ 3).$$

$$13. \quad A^T = \begin{pmatrix} 1 & -3 & 4 \\ -2 & 0 & 5 \end{pmatrix}, \quad B^T = \begin{pmatrix} 2 & -4 & 5 \\ -5 & 6 & 0 \\ 4 & 5 & 8 \\ 3 & 5 & 6 \end{pmatrix}, \quad C^T = \begin{pmatrix} 1 & 4 & 2 \\ 4 & -3 & 6 \\ 2 & 6 & 7 \end{pmatrix} = C.$$

$$14. \quad \mathbf{a}^T \mathbf{b} = \mathbf{b}^T \mathbf{a} = 8, \quad \mathbf{a} \mathbf{b}^T = \begin{pmatrix} 0 & 4 & 2 \\ 0 & 12 & 6 \\ 0 & -8 & -4 \end{pmatrix}, \quad \mathbf{b} \mathbf{a}^T = \begin{pmatrix} 0 & 0 & 0 \\ 4 & 12 & -8 \\ 2 & 6 & -4 \end{pmatrix}, \quad \mathbf{a} \mathbf{b} \text{ and } \mathbf{a}^T \mathbf{b}^T \text{ are not defined.}$$

$$17. \quad \text{A possible } G = \begin{pmatrix} 3 & 6 \\ -4 & 2 \end{pmatrix}.$$

19. a) $\begin{pmatrix} 4 & -7 \\ -1 & 2 \end{pmatrix}$, b) $\begin{pmatrix} 5 & 7 \\ 3 & 4 \end{pmatrix}$, c) no inverse, d) $\frac{1}{5} \begin{pmatrix} 4 & -9 \\ -3 & 8 \end{pmatrix}$, e) $\begin{pmatrix} -7 & 1 \\ 1 & 0 \end{pmatrix}$.

20. $A^{-1} = \begin{pmatrix} 1 & 3 & -4 \\ 0 & -1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$, $B^{-1} = \begin{pmatrix} 8 & -2 & -3 \\ \frac{1}{2} & 0 & 0 \\ -3 & 1 & 1 \end{pmatrix}$, C is not invertible,

$$D^{-1} = \frac{1}{4} \begin{pmatrix} 1 & 1 & 1 \\ 5 & -3 & 1 \\ -17 & 11 & -5 \end{pmatrix}.$$

21. a) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{5} & 0 \\ 0 & 0 & \frac{1}{6} \end{pmatrix}$ b) $\begin{pmatrix} 0 & 0 & -\frac{1}{2} \\ 1 & 0 & 0 \\ 0 & \frac{1}{3} & 0 \end{pmatrix}$

22. $A^{-1} = \begin{pmatrix} 4 & -3 & -2 & 0 \\ -1 & 1 & 1 & 0 \\ 1 & -2 & -2 & 1 \\ 0 & 1 & 2 & -1 \end{pmatrix}$; $B^{-1} = \begin{pmatrix} 6 & -2 & 1 & 0 \\ 9 & -4 & 3 & -1 \\ 25 & -11 & 8 & -2 \\ -14 & 6 & -4 & 1 \end{pmatrix}$;

C^{-1} does not exist.

23. a) $(B^{-1})^2$, b) AB^6A^{-1} , c) $(A + A^{-1})^2$, d) $I - (I - A)^{m+1}$.

24. a) $A^{-1}B$. b) $\begin{pmatrix} 2 & 4 & 4 \\ 1 & -2 & 3 \\ 1 & 0 & 3 \end{pmatrix}$.

25. b) i) B^TB , ii) $C^{-1}C^T$.

26. a) $\begin{pmatrix} -2 & 0 & 1 \\ 2 & 1 & -1 \\ 5 & 1 & -2 \end{pmatrix}$. b) $\begin{pmatrix} -2c_1 + c_3 \\ 2c_1 + c_2 - c_3 \\ 5c_1 + c_2 - 2c_3 \end{pmatrix}$.

27. $\mathbf{x} = Q^T \mathbf{b}$.

29. e.g. $\begin{pmatrix} \frac{1}{\sqrt{2}}i & -\frac{1}{\sqrt{2}}i \\ \frac{1}{\sqrt{2}}i & \frac{1}{\sqrt{2}}i \end{pmatrix}$.

33. From Question 29, Q is invertible, and hence $Q\mathbf{x} = \mathbf{b}$ has the solution $\mathbf{x} = Q^{-1}\mathbf{b} = \overline{Q}^T \mathbf{b}$.

34. a) $\frac{1}{ab} \begin{pmatrix} b & 0 \\ -c & a \end{pmatrix}$. b) $\begin{pmatrix} A^{-1} & 0 \\ -B^{-1}CA^{-1} & B^{-1} \end{pmatrix}$.

35. a) 1, b) -1, c) 0, d) 5, e) -2. All are invertible except $\begin{pmatrix} 5 & 2 \\ 10 & 4 \end{pmatrix}$.

36. a) -9 , b) 0 , c) 56 .

37. -126 .

38. a) -30 , b) 5 , c) 5 , d) $5 \times 7^3 = 1715$.

39. a) -2 , b) $-\frac{1}{2}$, c) -32 .

40. $-83, -108, 8964$.

41. $a \neq 1$.

42. a) $\begin{pmatrix} 1 & 0 & -1 \\ 2 & 1 & -1 \\ 5 & 1 & -3 \end{pmatrix}$. b) $\begin{pmatrix} -2 & -1 & 1 \\ 1 & 2 & -1 \\ -3 & -1 & 1 \end{pmatrix}$. c) 1 .

45. a) $(\alpha - 3)(\alpha + 1)(\alpha + 2)$. b) $-1, -2, 3$.

46. For example, $A = B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

47. For example, $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ and $\lambda = -1$.

54. $2(x + y + z)^3$.

55. $(z - 1)(z^2 + 2z - 4)$, $x = -1$, $y = 1 \pm \sqrt{5}$, $z = -1 \pm \sqrt{5}$.

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Revision

Some important facts from MATH1131/1141

In the next couple of chapters, we shall frequently refer to some subsets of $\mathbb{R}^n$, such as lines and planes, etc. As well as the two operations addition and multiplication by a scalar, we shall also refer some other operations such as dot and cross products, etc. However, for ease of reading some definitions are restated below. When necessary, you should refer to the 1131/41 Algebra Notes.

- Suppose that $n \geq 1$. A parametric vector equation of a line in $\mathbb{R}^n$ through a point A and parallel to a non-zero vector $\mathbf{v}$ is given by

$$\mathbf{x} = \mathbf{a} + \lambda \mathbf{v}, \quad \lambda \in \mathbb{R},$$

where $\mathbf{a}$ is the position vector of A (with respect to the origin O) and $\mathbf{x}$ is the position vector of a variable point on the line.

- Suppose that $n \geq 2$. A parametric vector equation of a plane in $\mathbb{R}^n$ through a point A and parallel to two non-zero non-parallel vectors $\mathbf{u}, \mathbf{v}$ is given by

$$\mathbf{x} = \mathbf{a} + \lambda \mathbf{u} + \mu \mathbf{v}, \quad \lambda, \mu \in \mathbb{R},$$

where $\mathbf{a}$ is the position vector of A and $\mathbf{x}$ is the position vector of a variable point on the plane.

- Suppose that $\mathbf{a}$ and $\mathbf{b}$ are two vectors in $\mathbb{R}^n$, $n \geq 1$. The dot product is defined by

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + \cdots + a_n b_n.$$

The length of a vector $\mathbf{a}$ is defined to be $\sqrt{\mathbf{a} \cdot \mathbf{a}}$. (This definition is equivalent to the one given in the 1131/41 Algebra Notes.)

The vectors are said to be orthogonal if $\mathbf{a} \cdot \mathbf{b} = 0$. A set of vectors is said to be an orthonormal set if the lengths of each vectors is 1 and the vectors are mutually orthogonal.

The projection of a vector $\mathbf{a}$ on the vector $\mathbf{b}$ is

$$\text{proj}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2} \mathbf{b}.$$

- Suppose that $\mathbf{a}$ and $\mathbf{b}$ are two vectors in $\mathbb{R}^3$. An equivalent definition of cross product in determinant form is given by

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}, \quad \text{where } \mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}, \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}.$$

- The following is a point-normal form of a plane in $\mathbb{R}^3$ which passes through a point A and has a normal vector $\mathbf{n}$.

$$\mathbf{n} \cdot (\mathbf{x} - \mathbf{a}) = 0,$$

where $\mathbf{a}$ is the position vector of A and $\mathbf{x}$ is the position vector of a variable point on the plane.

Revision problems

1. [R] Let A, B, P be points in $\mathbb{R}^3$ with position vectors

$$\mathbf{a} = \begin{pmatrix} 7 \\ -2 \\ 3 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ -5 \\ 0 \end{pmatrix} \text{ and } \mathbf{p} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$

Let Q be the point on segment between A and B such that $AQ = \frac{2}{3}AB$.

- Find $\mathbf{q}$, the position vector of Q .
 - Find the parametric vector equation of the line that passes through P and Q .
2. [R] Consider the three points $A(1, 1, 1)$, $B(2, 0, 3)$ and $C(3, -1, 1)$.
- Find $\overrightarrow{AB}$ and $\overrightarrow{AC}$.
 - Find a parametric vector form of the line through A and B .
 - Find a parametric vector form of the plane through A, B and C .
 - Find $\overrightarrow{AB} \times \overrightarrow{AC}$.
 - Find a point-normal form of the plane through A, B and C .
 - Find a Cartesian equation of the plane through A, B and C .
3. [R] Given the vectors $\mathbf{p} = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$ and $\mathbf{q} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix}$, find $|\mathbf{p}|$, $|\mathbf{q}|$, $\mathbf{p} \cdot \mathbf{q}$, then the cosine of the angle between $\mathbf{p}$ and $\mathbf{q}$.
4. [R] Consider the equation
- $$\det \begin{pmatrix} x-1 & y-2 & z+1 \\ 1 & 0 & 2 \\ 2 & -1 & 0 \end{pmatrix} = 0$$
- Show that the equation represents the Cartesian equation of a plane.
 - Write the equation in point-normal form.
5. [R] For the points $P(1, 2, 0)$, $Q(1, 3, -1)$ and $R(2, 1, 1)$, find $\overrightarrow{PQ} \times \overrightarrow{PR}$ and the area of the triangle with vertices P, Q and R .

6. [R] Suppose that A is the point $(2, -1, 3)$ and Π is the plane

$$\mathbf{x} = \lambda \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -2 \\ 2 \end{pmatrix} \quad \text{for } \lambda, \mu \in \mathbb{R}.$$

- i) Find a vector $\mathbf{n}$ which is normal to Π .
- ii) Find the projection of $\overrightarrow{OA}$ on the direction $\mathbf{n}$.
- iii) Hence find the shortest distance of A from Π .

7. [R] Find the intersection (if any) of the line $\mathbf{x} = \begin{pmatrix} 0 \\ 18 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix}$ for $\mu \in \mathbb{R}$ and the plane

$$\mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 4 \end{pmatrix} + \lambda_1 \begin{pmatrix} 1 \\ 4 \\ 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} 3 \\ 1 \\ -2 \end{pmatrix} \quad \text{for } \lambda_1, \lambda_2 \in \mathbb{R}.$$

8. [R] Are the planes

$$\mathbf{x} = \begin{pmatrix} 1 \\ -4 \\ 2 \\ 3 \end{pmatrix} + \lambda_1 \begin{pmatrix} 2 \\ 1 \\ -2 \\ 7 \end{pmatrix} + \lambda_2 \begin{pmatrix} -3 \\ 1 \\ 5 \\ 2 \end{pmatrix} \quad \text{for } \lambda_1, \lambda_2 \in \mathbb{R}$$

and

$$\mathbf{x} = \begin{pmatrix} 2 \\ -4 \\ 1 \\ 3 \end{pmatrix} + \mu_1 \begin{pmatrix} 3 \\ -1 \\ 2 \\ 4 \end{pmatrix} + \mu_2 \begin{pmatrix} -1 \\ 4 \\ 2 \\ 6 \end{pmatrix} \quad \text{for } \mu_1, \mu_2 \in \mathbb{R}$$

parallel?

9. [R] Consider the following system of linear equations where k is a real number.

$$\begin{array}{rrcrcl} x_1 & + & x_2 & + & 2k x_3 & = & 3 \\ x_1 & + & 3k x_2 & + & x_3 & = & -2 \\ 2x_1 & + & 6k x_2 & + & k x_3 & = & 1 \end{array}$$

Find for which values of k the system has (I) no solutions, (II) a unique solution or (III) infinitely many solutions.

Answers to the revision problems

$$1i. \begin{pmatrix} 3 \\ -4 \\ 1 \end{pmatrix}. \quad ii. \mathbf{x} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -3 \\ -1 \end{pmatrix} \quad \text{for } \lambda \in \mathbb{R}.$$

$$2i. \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \quad \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix}. \quad ii. \mathbf{x} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \quad \text{for } \lambda \in \mathbb{R}.$$

$$iii. \mathbf{x} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 2 \\ -2 \\ 0 \end{pmatrix} \quad \text{for } \lambda, \mu \in \mathbb{R}. \quad iv. \begin{pmatrix} 4 \\ 4 \\ 0 \end{pmatrix}.$$

$$v. \begin{pmatrix} 4 \\ 4 \\ 0 \end{pmatrix} \cdot \left(\mathbf{x} - \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right) = 0.$$

$$vi. x + y - 2 = 0.$$

$$3. \sqrt{3}, \quad \sqrt{6}, \quad 4, \quad \frac{2\sqrt{2}}{3}.$$

$$4i. 2x + 4y - z = 11. \quad ii. \begin{pmatrix} 2 \\ 4 \\ -1 \end{pmatrix} \cdot \left(\begin{pmatrix} x \\ y \\ z \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ -11 \end{pmatrix} \right) = 0$$

$$5. \begin{pmatrix} 0 \\ -1 \\ -1 \end{pmatrix}, \quad \frac{\sqrt{2}}{2}.$$

$$6i. \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix}, \quad ii. -\frac{1}{9} \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix}, \quad iii. \frac{1}{3}.$$

$$7. \text{Meet at } (6, 9, 4).$$

$$8. \text{Planes are not parallel as } \lambda_1 \begin{pmatrix} 2 \\ 1 \\ -2 \\ 7 \end{pmatrix} + \lambda_2 \begin{pmatrix} -3 \\ 1 \\ 5 \\ 2 \end{pmatrix} = \mu_1 \begin{pmatrix} 3 \\ -1 \\ 2 \\ 4 \end{pmatrix} + \mu_2 \begin{pmatrix} -1 \\ 4 \\ 2 \\ 6 \end{pmatrix}$$

only when $\lambda_1 = \lambda_2 = \mu_1 = \mu_2 = 0$.

9. (I) If either $k = 2$ or $k = 1/3$, then the system will have no solutions.

(II) If $k \neq 2$ and $k \neq 1/3$ then the system will have a unique solution.

(III) There are no k values which give a system with infinitely many solutions.

Chapter 6

VECTOR SPACES

*But, keeping still the end in view
To which I hope to come,
I strove to prove the matter true
By putting everything I knew
Into an Axiom*
Lewis Carroll, Phantasmagoria.

We have studied geometric vectors and column vectors in Chapter 1 and matrices in Chapter 5. What do the following sets have in common?

- The set of geometric vectors in a three-dimensional space.
- The set of column vectors of n real components, i.e. $\mathbb{R}^n$.
- The set of $m \times n$ matrices with real entries, i.e. $M_{mn}(\mathbb{R})$.

In each of these sets, we can add two elements and we can multiply an element in the set with a scalar (in this case, a scalar is a real number) and remain inside the set we started in. We say that each set is closed under the two operations — addition and multiplication by a scalar. Such a set together with scalars and the two operations satisfy some fundamental properties. As what we have seen in Chapters 1 and 5, addition of vectors and matrices satisfies the associative and commutative laws. There is a special element $\mathbf{0}$ in each set such that $\mathbf{0} + \mathbf{v} = \mathbf{v} + \mathbf{0} = \mathbf{v}$ for all $\mathbf{v}$ in the set. For each $\mathbf{v}$ in the set, there is a negative $-\mathbf{v}$ such that $\mathbf{v} + (-\mathbf{v}) = (-\mathbf{v}) + \mathbf{v} = \mathbf{0}$. The associative law of scalar multiplication and distributive laws also hold. These are examples of **vector spaces** which we are going to study in this chapter. In a vector space, each element in the set is called a vector. The set of scalars must be a field, generally the set of real or complex numbers.

Besides the above mentioned vector spaces, there are many other examples, such as the set of polynomials with real or complex coefficients, the real functions on a given interval and the differentiable functions defined on an interval, which form a vector space. It is perhaps a remarkable fact that each of these quite different kinds of objects obey similar rules for addition and multiplication by a scalar.

In the present chapter our main objectives are to develop a general theory of vector spaces and to show how this general theory can be applied to the study of particular vector spaces. Although

all of the theorems and propositions stated in this chapter are true for **all** vector spaces, we will concentrate on giving examples and applications of the theory for the vector space $\mathbb{R}^n$. The main reason for this is that in $\mathbb{R}^n$ the theoretical results can be more easily understood, as they can often be given a geometric interpretation. Also, $\mathbb{R}^n$ is the most commonly used vector space in practical applications.

As the mathematics developed in this chapter applies to all vector spaces, it is more abstract and theoretical than that of Chapter 1, where there was always an immediate geometric picture available for all results.

Another reason for the difficulty is that an appreciable part of the language of vector spaces will be new to you. Therefore, as with any new language, **it is absolutely essential that you make a special effort to learn the definitions of any new words**. You will find that many of the fundamental vector space ideas discussed in this chapter, such as linear combination, span, linear independence, basis, dimension, and coordinate vector, are generalisations of ideas that we have already discussed in an informal geometric manner in Chapter 1 for vectors in $\mathbb{R}^n$.

In addition, you will also find that the solution of most of the problems in this chapter can be obtained by solving systems of linear equations using Gaussian Elimination developed in Chapter 4. In most cases the details are suppressed, but the reader should check them before attempting the exercises.

It is important to keep in mind in this chapter that correct **setting out** is essential, rather than just computation. When you write down a solution to a question, you must make sure it reads correctly, both logically and mathematically.

6.1 Definitions and examples of vector spaces

We start from a mathematical system which consists of the following four things.

1. A non-empty set V of elements called “vectors”.
2. A “vector-addition” rule (usually represented by $+$) for combining pairs of vectors from V . For vectors $\mathbf{v}, \mathbf{w} \in V$, the vector formed by adding $\mathbf{w}$ to $\mathbf{v}$ is denoted by $\mathbf{v} + \mathbf{w}$.
3. A field $\mathbb{F}$ of “scalars”. For example, $\mathbb{F}$ could be the rational numbers $\mathbb{Q}$ or the real numbers $\mathbb{R}$ or the complex numbers $\mathbb{C}$. There are also other important, but less common, examples of fields which can be used.
4. A “multiplication by a scalar rule” for combining a vector from V and a scalar from $\mathbb{F}$ to form a vector. If λ is a scalar and $\mathbf{v}$ is an element of V , then $\lambda * \mathbf{v}$ means the result of multiplying $\mathbf{v}$ by the scalar λ .

The system is then denoted by $(V, +, *, \mathbb{F})$. However, if we have no problem in distinguishing the product of two scalars $\lambda\mu$ and the multiplication of a vector by a scalar $\lambda * \mathbf{v}$, we shall omit the symbol $*$. Just as we write $2x$ instead of $2 \times x$, we shall write $\lambda\mathbf{v}$ instead of $\lambda * \mathbf{v}$.

We can now give a formal definition of a vector space.

Definition 1. A **vector space** V over the field $\mathbb{F}$ is a non-empty set V of vectors on which addition of vectors is defined and multiplication by a scalar is defined in such a way that the following ten fundamental properties are satisfied:

1. **Closure under Addition.** If $\mathbf{u}, \mathbf{v} \in V$, then $\mathbf{u} + \mathbf{v} \in V$.
2. **Associative Law of Addition.** If $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$, then $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
3. **Commutative Law of Addition.** If $\mathbf{u}, \mathbf{v} \in V$, then $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
4. **Existence of Zero.** There exists an element $\mathbf{0} \in V$ such that, for all $\mathbf{v} \in V$, $\mathbf{v} + \mathbf{0} = \mathbf{v}$.
5. **Existence of Negative.** For each $\mathbf{v} \in V$ there exists an element $\mathbf{w} \in V$ (usually written as $-\mathbf{v}$), such that $\mathbf{v} + \mathbf{w} = \mathbf{0}$.
6. **Closure under Multiplication by a Scalar.** If $\mathbf{v} \in V$ and $\lambda \in \mathbb{F}$, then $\lambda\mathbf{v} \in V$.
7. **Associative Law of Multiplication by a Scalar.** If $\lambda, \mu \in \mathbb{F}$ and $\mathbf{v} \in V$, then $\lambda(\mu\mathbf{v}) = (\lambda\mu)\mathbf{v}$.
8. If $\mathbf{v} \in V$ and $1 \in \mathbb{F}$ is the scalar one, then $1\mathbf{v} = \mathbf{v}$.
9. **Scalar Distributive Law.** If $\lambda, \mu \in \mathbb{F}$ and $\mathbf{v} \in V$, then $(\lambda + \mu)\mathbf{v} = \lambda\mathbf{v} + \mu\mathbf{v}$.
10. **Vector Distributive Law.** If $\lambda \in \mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in V$, then $\lambda(\mathbf{u} + \mathbf{v}) = \lambda\mathbf{u} + \lambda\mathbf{v}$.

NOTE.

1. Each of the basic rules is called an **axiom**.
2. In axiom 5, $-\mathbf{v}$ is a symbol for the negative of $\mathbf{v}$. The vector $-\mathbf{v}$ and the vector formed by multiplying $\mathbf{v}$ with the scalar -1 are not the same by definition. We shall prove that they are the same later.
3. Formally, axiom 7 says $\lambda * (\mu * \mathbf{v}) = (\lambda\mu) * \mathbf{v}$.
4. In axiom 9, the addition on the left is the addition of two scalars while the addition on the right is the addition of two vectors. They are different additions.
5. Two systems are the same only when all the four things are the same. However, we seldom discuss different vector spaces with the same set of vectors but we often discuss different sets of vectors with the same set of scalars and the same operations. When there is no confusion, we shall simply call $(V, +, *, \mathbb{F})$, the vector space V .

Example 1 (The Vector Space $\mathbb{R}^n$). The set of vectors is the set of all n -vectors of real numbers,

$$\mathbb{R}^n = \left\{ \mathbf{x} : \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \text{ for } x_1, \dots, x_n \in \mathbb{R} \right\}.$$

The set of scalars is $\mathbb{R}$.

Vector addition is defined by

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}.$$

The multiplication of a vector by a scalar $\lambda \in \mathbb{R}$ is defined by

$$\lambda \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \vdots \\ \lambda x_n \end{pmatrix}.$$

To prove that this system is a vector space it is necessary to show that all ten axioms listed in the definition are satisfied by the system.

All the axioms are general statements about arbitrary vectors and scalars. We have to prove the axioms are satisfied by any

$$\mathbf{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix}, \mathbf{v} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}, \mathbf{w} = \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix}, \quad \text{and} \quad \lambda, \mu \in \mathbb{R}.$$

1. **Closure under addition.** If $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ then $u_1 + v_1, \dots, u_n + v_n \in \mathbb{R}$ because $\mathbb{R}$ is closed under addition. Hence

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{pmatrix} \in \mathbb{R}^n.$$

2. **Associative law of addition.** If $\mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ then

$$(u_1 + v_1) + w_1 = u_1 + (v_1 + w_1), \dots, (u_n + v_n) + w_n = u_n + (v_n + w_n)$$

because addition in $\mathbb{R}$ is associative. Hence

$$\begin{aligned} (\mathbf{u} + \mathbf{v}) + \mathbf{w} &= \begin{pmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{pmatrix} + \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} = \begin{pmatrix} (u_1 + v_1) + w_1 \\ \vdots \\ (u_n + v_n) + w_n \end{pmatrix} \\ &= \begin{pmatrix} u_1 + (v_1 + w_1) \\ \vdots \\ u_n + (v_n + w_n) \end{pmatrix} = \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} + \begin{pmatrix} v_1 + w_1 \\ \vdots \\ v_n + w_n \end{pmatrix} = \mathbf{u} + (\mathbf{v} + \mathbf{w}). \end{aligned}$$

3. **Commutative law of addition.** If $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ then

$$u_1 + v_1 = v_1 + u_1, \dots, u_n + v_n = v_n + u_n$$

because addition in $\mathbb{R}$ is commutative. Hence,

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{pmatrix} = \begin{pmatrix} v_1 + u_1 \\ \vdots \\ v_n + u_n \end{pmatrix} = \mathbf{v} + \mathbf{u}.$$

4. **Existence of zero.** There is a special element

$$\mathbf{0} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \in \mathbb{R}^n$$

called the **zero vector** which has the property that

$$\mathbf{v} + \mathbf{0} = \begin{pmatrix} v_1 + 0 \\ \vdots \\ v_n + 0 \end{pmatrix} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = \mathbf{v}, \quad \text{for all } \mathbf{v} \in \mathbb{R}^n.$$

5. **Existence of Negative.** For each $\mathbf{v} \in \mathbb{R}^n$ there exists an element

$$\begin{pmatrix} -v_1 \\ \vdots \\ -v_n \end{pmatrix} \in \mathbb{R}^n,$$

the negative of $\mathbf{v}$, such that

$$\begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} + \begin{pmatrix} -v_1 \\ \vdots \\ -v_n \end{pmatrix} = \begin{pmatrix} v_1 - v_1 \\ \vdots \\ v_n - v_n \end{pmatrix} = \mathbf{0}.$$

6. **Closure under scalar multiplication.** If $\mathbf{v} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$ then $\lambda v_1, \dots, \lambda v_n \in \mathbb{R}$ because $\mathbb{R}$ is closed under multiplication. Hence $\lambda \mathbf{v} \in \mathbb{R}^n$.

7. **Associative law of multiplication by a scalar.** If $\lambda, \mu \in \mathbb{R}$ and $\mathbf{v} \in \mathbb{R}^n$ then

$$\lambda(\mu v_1) = (\lambda\mu)v_1, \dots, \lambda(\mu v_n) = (\lambda\mu)v_n$$

because multiplication in $\mathbb{R}$ is associative. Hence,

$$\lambda(\mu \mathbf{v}) = \lambda \begin{pmatrix} \mu v_1 \\ \vdots \\ \mu v_n \end{pmatrix} = \begin{pmatrix} \lambda(\mu v_1) \\ \vdots \\ \lambda(\mu v_n) \end{pmatrix} = \begin{pmatrix} (\lambda\mu)v_1 \\ \vdots \\ (\lambda\mu)v_n \end{pmatrix} = (\lambda\mu) \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = (\lambda\mu)\mathbf{v}.$$

8. If $\mathbf{v} \in \mathbb{R}^n$ then $1\mathbf{v} = \begin{pmatrix} 1v_1 \\ \vdots \\ 1v_n \end{pmatrix} = \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} = \mathbf{v}$.

9. **Scalar distributive law.** If $\lambda, \mu \in \mathbb{R}$ and $\mathbf{v} \in \mathbb{R}^n$ then

$$(\lambda + \mu)v_1 = \lambda v_1 + \mu v_1, \dots, (\lambda + \mu)v_n = \lambda v_n + \mu v_n,$$

because of the distributive law in $\mathbb{R}$. We then have

$$\begin{aligned} (\lambda + \mu) \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} &= \begin{pmatrix} (\lambda + \mu)v_1 \\ \vdots \\ (\lambda + \mu)v_n \end{pmatrix} = \begin{pmatrix} \lambda v_1 + \mu v_1 \\ \vdots \\ \lambda v_n + \mu v_n \end{pmatrix} \\ &= \begin{pmatrix} \lambda v_1 \\ \vdots \\ \lambda v_n \end{pmatrix} + \begin{pmatrix} \mu v_1 \\ \vdots \\ \mu v_n \end{pmatrix} = \lambda \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix} + \mu \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}. \end{aligned}$$

Hence $(\lambda + \mu)\mathbf{v} = \lambda\mathbf{v} + \mu\mathbf{v}$.

10. **Vector distributive law.** If $\lambda \in \mathbb{R}$ and $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ then

$$\lambda(u_1 + v_1) = \lambda u_1 + \lambda v_1, \dots, \lambda(u_n + v_n) = \lambda u_n + \lambda v_n,$$

because of the distributive law in $\mathbb{R}$. We then have

$$\lambda \begin{pmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{pmatrix} = \begin{pmatrix} \lambda(u_1 + v_1) \\ \vdots \\ \lambda(u_n + v_n) \end{pmatrix} = \begin{pmatrix} \lambda u_1 + \lambda v_1 \\ \vdots \\ \lambda u_n + \lambda v_n \end{pmatrix} = \lambda \begin{pmatrix} u_1 \\ \vdots \\ u_n \end{pmatrix} + \lambda \begin{pmatrix} v_1 \\ \vdots \\ v_n \end{pmatrix}.$$

Hence $\lambda(\mathbf{u} + \mathbf{v}) = \lambda\mathbf{u} + \lambda\mathbf{v}$.

After we have checked that all ten axioms are satisfied, we can conclude that the system is a vector space, or simply $\mathbb{R}^n$ is a vector space over $\mathbb{R}$. $\diamond$

NOTE. As special cases of $\mathbb{R}^n$, the real number line $\mathbb{R}$, the plane $\mathbb{R}^2$ and three-dimensional space $\mathbb{R}^3$ are all vector spaces over the real numbers.

Example 2 (The Vector Space $\mathbb{C}^n$). The set of vectors is the set of all column vectors with n components of complex numbers

$$\mathbb{C}^n = \left\{ \mathbf{x} : \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \text{ for } x_1, \dots, x_n \in \mathbb{C} \right\},$$

and the set of scalars is $\mathbb{C}$. Addition is defined by

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}.$$

and multiplication of a vector by a scalar $\lambda \in \mathbb{C}$ is defined by

$$\lambda \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} \lambda x_1 \\ \vdots \\ \lambda x_n \end{pmatrix}.$$

To prove that this is a vector space it is necessary to show that the ten vector space axioms are satisfied. This proof is formally identical to that for $\mathbb{R}^n$ over $\mathbb{R}$ since only basic operations are involved. $\diamond$

Example 3 (The Vector Space $M_{mn} = M_{mn}(\mathbb{R})$ of Real Matrices). M_{mn} is a vector space over $\mathbb{R}$. There is a natural and straightforward generalisation to $M_{mn}(\mathbb{F})$, where the entries come from the field $\mathbb{F}$. In the most important case of $\mathbb{F} = \mathbb{R}$ we often suppress the $\mathbb{R}$. Note that here we are thinking of matrices as vectors!

$$M_{mn} = M_{mn}(\mathbb{R}) = \left\{ A : A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}, \text{ where } a_{ij} \in \mathbb{R} \text{ for } 1 \leq i \leq m, 1 \leq j \leq n \right\}.$$

Using the notation introduced in Chapter 5, the ij th entry (i th row, j th column entry) of A is denoted by $[A]_{ij}$. We define “addition of vectors” to be matrix addition where

$$[A + B]_{ij} = [A]_{ij} + [B]_{ij}, \quad \text{for all } i, j.$$

Similarly we define the “multiplication (of the vector A) by a scalar $\lambda \in \mathbb{R}$ ” in terms of the multiplication of a matrix by a scalar. That is,

$$[\lambda A]_{ij} = \lambda [A]_{ij} \quad \text{for all } i, j$$

as in Chapter 5.

To check that M_{mn} is a vector space is routine. All of the properties are included amongst the properties developed for matrices. For example

$$A + B = B + A \quad \text{for matrices of the same size,}$$

hence the commutative law holds for the set M_{mn} regarded as a vector space. The details are left for the reader to check. $\diamond$

NOTE. $M_{mn}(\mathbb{R})$ and $M_{mn}(\mathbb{C})$ are widely used in both quantum physics and chemistry.

Example 4 (The Vector Space of Polynomials). One of the most important aspects of vector space theory is that it applies in many quite different situations. The set of all real-valued functions on $\mathbb{R}$ forms a vector space, as does the set of all continuous functions. A simpler example, perhaps, is the set $\mathbb{P}(\mathbb{R})$ of all real polynomials.

Suppose that p is the polynomial given by $p(x) = a_0 + a_1x + \cdots + a_nx^n = \sum_{k=0}^n a_kx^k$ and q is the polynomial given by $q(x) = \sum_{k=0}^m b_kx^k$ where a_k and b_k are real. Note that p is a real-valued function while $p(x)$ is the value of the function at x . (You might like to quickly read the brief review of function notation given in Appendix 6.9.)

We all know how to add and subtract these polynomials, and how to multiply a polynomial by a real number. Their sum is just the polynomial $p + q$, where the value of the function at x is

$$(p + q)(x) = p(x) + q(x) = \sum_{k=0}^{\max(n,m)} (a_k + b_k)x^k, \quad x \in \mathbb{R}.$$

(Of course we just set any missing coefficient equal to zero to do this sum.) The scalar multiple is the polynomial λp where

$$(\lambda p)(x) = \lambda(p(x)) = \sum_{k=0}^n \lambda a_k x^k, \quad x \in \mathbb{R}.$$

The proof that $\mathbb{P}(\mathbb{R})$ is a vector space over $\mathbb{R}$ is straightforward. For example, if p and q are polynomials, then $p + q$ is also a polynomial (Closure under Addition). The zero element of $\mathbb{P}(\mathbb{R})$ is just the polynomial p such that $p(x) = 0$, for all $x \in \mathbb{R}$. It is important when working in $\mathbb{P}(\mathbb{R})$ to remember that saying that two polynomials p and q are equal means that $p(x) = q(x)$ for all $x \in \mathbb{R}$, or equivalently, that the corresponding coefficients for p and q are equal. Please check the details yourself. $\diamond$

NOTE. The set $\mathbb{P}(\mathbb{F})$ of all polynomials over a field $\mathbb{F}$, with addition and multiplication by a scalar similarly defined, is also a vector space.

For any non-negative integers n , the subset of all polynomials of degrees n or less including the zero polynomial is again a vector space. That is,

$$\mathbb{P}_n(\mathbb{F}) = \{p : p \text{ is a polynomial over } \mathbb{F}, \text{ degree of } p \leq n \text{ or } p = 0\},$$

with the same addition and multiplication by a scalar as in $\mathbb{P}(\mathbb{F})$, is a vector space over $\mathbb{F}$.

As a summary, we know that the following are vectors spaces.

- $\mathbb{R}^n$ over $\mathbb{R}$, where n is a positive integer.
- $\mathbb{C}^n$ over $\mathbb{C}$, where n is a positive integer.
- $\mathbb{P}(\mathbb{F})$, $\mathbb{P}_n(\mathbb{F})$ over $\mathbb{F}$, where $\mathbb{F}$ is a field. Usually $\mathbb{F}$ is either $\mathbb{Q}$, $\mathbb{R}$ or $\mathbb{C}$.
- $M_{mn}(\mathbb{F})$ over $\mathbb{F}$, where m, n are positive integers and $\mathbb{F}$ is a field.

Furthermore, the following set, its subset of all continuous function and its subset of all differentiable functions are vector spaces over $\mathbb{R}$.

- $\mathcal{R}[X]$, the set of all possible real-valued functions with domain X .

When we refer to these vectors spaces, we assume the additions and multiplications by scalars are as defined above. However, there are systems with the same set of vectors but different operations. If we want to emphasise that the operations are the ones defined above, we shall use the terms *usual addition* and *usual multiplication by a scalar*.

We shall concentrate on the vector space $\mathbb{R}^n$. Those who want to see other examples of vector spaces can look at Section 6.8, where other vector spaces are covered in rather more depth.

We are going to end this section by an example of a system which is not a vector space. To prove a system is not a vector space, we need to prove that one of the axioms is not satisfied. To *disprove* a general statement, we only need to give a **counterexample** to illustrate that the statement is false.

Example 5. The system $(\mathbb{R}^2, +, *, \mathbb{R})$ with the usual multiplication by a scalar but the addition defined by — for any $\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$ in $\mathbb{R}^2$,

$$\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_1 + v_1 \\ 2u_2 + 2v_2 \end{pmatrix},$$

is not a vector space.

SOLUTION. We are going to give a counterexample to axiom 9. Let $\lambda = \mu = 1$ and $\mathbf{v} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$.

We have

$$(\lambda + \mu)\mathbf{v} = 2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 + 0 \\ 2 \times 1 + 2 \times 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \lambda\mathbf{v} + \mu\mathbf{v}.$$

Hence, axiom 9 is not satisfied by this system and this system is not a vector space. $\diamond$

6.2 Vector arithmetic

The axioms give a minimal set of rules needed to define a vector space. There are several other useful rules which can be proved directly from the axioms and which are therefore true in all vector spaces. In this section we shall discuss some of these properties and give examples of them for $\mathbb{R}^n$.

The first five vector space axioms apply to vector addition, and they are in fact identical to the five basic axioms of addition for integers, real numbers and complex numbers. This means that all the arithmetic properties of vector addition are identical to corresponding properties of addition of numbers. In particular, we have:

Proposition 1. In any vector space V , the following properties hold for addition.

1. **Uniqueness of Zero.** There is one and only one zero vector.
2. **Cancellation Property.** If $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ satisfy $\mathbf{u} + \mathbf{v} = \mathbf{u} + \mathbf{w}$, then $\mathbf{v} = \mathbf{w}$.
3. **Uniqueness of Negatives.** For all $\mathbf{v} \in V$, there exists only one $\mathbf{w} \in V$ such that $\mathbf{v} + \mathbf{w} = \mathbf{0}$.

Proof. For property 1, Axiom 4 ensures the existence of a zero vector in V .

Now assume that two vectors $\mathbf{0}$ and $\mathbf{0}'$ are both zero vectors in V . Then, for the reasons given in brackets, we have

$$\begin{aligned} \mathbf{0} &= \mathbf{0} + \mathbf{0}' && \text{(axiom 4 applied to the zero vector } \mathbf{0}') \\ &= \mathbf{0}' + \mathbf{0} && \text{(axiom 3)} \\ &= \mathbf{0}' && \text{(axiom 4 applied to the zero vector } \mathbf{0}) \end{aligned}$$

Hence, $\mathbf{0} = \mathbf{0}'$, and there is only one zero vector in V .

For property 2, Axiom 5 ensures the existence of the negative $-\mathbf{u}$. Hence, we have

$$\begin{aligned} (-\mathbf{u}) + (\mathbf{u} + \mathbf{v}) &= (-\mathbf{u}) + (\mathbf{u} + \mathbf{w}) && \text{(axiom 5)} \\ [(-\mathbf{u}) + \mathbf{u}] + \mathbf{v} &= [(-\mathbf{u}) + \mathbf{u}] + \mathbf{w} && \text{(axiom 2)} \\ \mathbf{0} + \mathbf{v} &= \mathbf{0} + \mathbf{w} && \text{(axiom 5)} \\ \mathbf{v} &= \mathbf{w} && \text{(axiom 4)} \end{aligned}$$

For property 3, assume $\mathbf{u}$ and $\mathbf{w}$ are both negatives of $\mathbf{v}$. By axiom 5, we have $\mathbf{v} + \mathbf{u} = \mathbf{0} = \mathbf{v} + \mathbf{w}$. By property 2, we can conclude that $\mathbf{u} = \mathbf{w}$. Hence the inverse is unique. $\square$

Example 1. i) The unique zero vector in $\mathbb{R}^n$ is $\mathbf{0} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}$.

ii) The negative of a vector is used in solving an equation such as

$$\begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} + \mathbf{v} = \begin{pmatrix} -2 \\ 5 \\ 1 \\ 7 \end{pmatrix} \quad \text{to obtain} \quad \mathbf{v} = -\begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} + \begin{pmatrix} -2 \\ 5 \\ 1 \\ 7 \end{pmatrix} = \begin{pmatrix} -3 \\ 3 \\ -2 \\ 3 \end{pmatrix}. \quad \diamond$$

A comparison of the axioms for multiplication of a vector by a scalar with the multiplication properties for fields of numbers (see Chapter 3) also shows strong similarities — the main difference being that in a field, two numbers of the same kind are being multiplied, whereas for vectors the objects being multiplied are of different kinds. As a result, some of the fundamental properties of multiplication of numbers also hold for multiplication of a vector by a scalar. In particular, we have:

Proposition 2. Suppose that V is a vector space over a field $\mathbb{F}$, $\lambda \in \mathbb{F}$, $\mathbf{v} \in V$, 0 is the zero scalar in $\mathbb{F}$ and $\mathbf{0}$ is the zero vector in V . Then the following properties hold for multiplication by a scalar:

1. **Multiplication by the zero scalar.** $0\mathbf{v} = \mathbf{0}$,
2. **Multiplication of the zero vector.** $\lambda\mathbf{0} = \mathbf{0}$.
3. **Multiplication by -1 .** $(-1)\mathbf{v} = -\mathbf{v}$ (the additive inverse of $\mathbf{v}$).
4. **Zero products.** If $\lambda\mathbf{v} = \mathbf{0}$, then either $\lambda = 0$ or $\mathbf{v} = \mathbf{0}$.
5. **Cancellation Property.** If $\lambda\mathbf{v} = \mu\mathbf{v}$ and $\mathbf{v} \neq \mathbf{0}$ then $\lambda = \mu$.

Proof. We shall prove properties 1 and 3. The readers should write out the proofs for the others as exercises. For property 1,

$$\mathbf{v} + \mathbf{0} = \mathbf{v} = 1\mathbf{v} = (1 + 0)\mathbf{v} = 1\mathbf{v} + 0\mathbf{v} = \mathbf{v} + 0\mathbf{v}.$$

You should check carefully which axioms are required. Finally, by Cancellation Property of vector addition, we have $0\mathbf{v} = \mathbf{0}$.

For property 3,

$$\mathbf{v} + (-1)\mathbf{v} = 1\mathbf{v} + (-1)\mathbf{v} = (1 + (-1))\mathbf{v} = 0\mathbf{v} = \mathbf{0}.$$

Hence by Uniqueness of Negatives $(-1)\mathbf{v} = -\mathbf{v}$. □

Example 2. The properties in the previous proposition are true for all vector spaces. In particular, for vectors in $\mathbb{R}^n$, the results can be easily proved by definitions of the operations and properties of scalars. Such as

$$\text{a) } 0 \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} = \mathbf{0},$$

$$\text{b) } (-1) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} -x_1 \\ \vdots \\ -x_n \end{pmatrix} = - \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix},$$

$$\text{c) If } \lambda \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}, \text{ then either } \lambda = 0 \text{ or } \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}. \quad \diamond$$

6.3 Subspaces

Before reading this section, you should quickly read the brief review of sets given in Appendix 6.9.

Although all examples in this section are subsets of $\mathbb{R}^n$, the definitions, theorems and corollaries apply to all vector spaces as stated.

In practice, many problems about vectors involve subsets of some vector space. For example, the points on a line in $\mathbb{R}^n$ form a subset of $\mathbb{R}^n$, the points on a plane in $\mathbb{R}^3$ form a subset of $\mathbb{R}^3$, the solutions of a system of m linear equations in n unknowns form a subset of $\mathbb{R}^n$. It is an important problem to determine the conditions under which some subset of a vector space is itself a vector space. It is convenient to begin by looking at some examples.

Example 1. The real-number line is a vector space. The question arises whether some subset of the real-number line is a vector space. For example, we might ask if some interval, for example the interval

$$S = [-5, 5] = \{x \in \mathbb{R} : -5 \leq x \leq 5\}$$

is a vector space. Geometrically, the set S represents the line segment shown in Figure 1(b).

SOLUTION. The given system is not a vector space, since it is not closed under scalar multiplication. A counterexample — $5 \in S$, but $5 + 5 = 10$ is not an element of S . A picture is given in Figure 1(b). $\diamond$

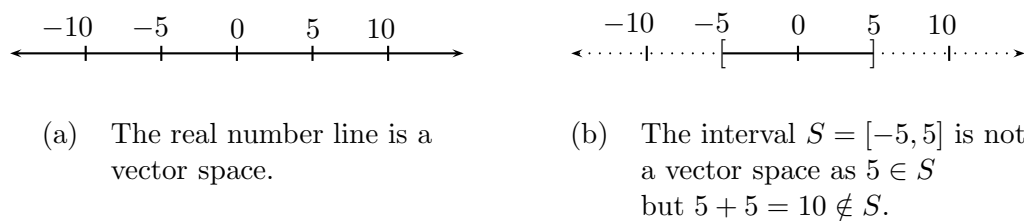


Figure 1.

Example 2. The plane $\mathbb{R}^2$ is a vector space. Show that the subset S of $\mathbb{R}^2$ given by

$$S = \left\{ \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : x_1 \geq 0 \right\}$$

is not a vector space.

SOLUTION. There are several ways to solve this problem since there are several axioms which are not satisfied. One method is to note that $\begin{pmatrix} 1 \\ 0 \end{pmatrix} \in S$, whereas $-1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix} \notin S$. Hence the set S is not closed under scalar multiplication and so S is not a vector space. $\diamond$

NOTE. Geometrically, the subset S contains the position vectors of all the points in the right half-plane as shown.

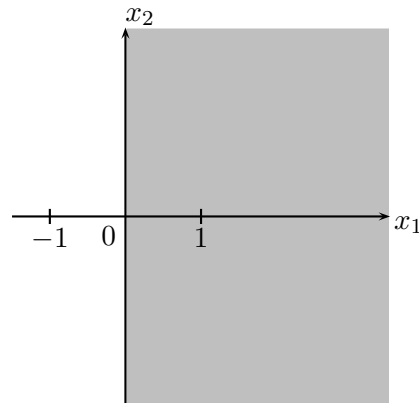


Figure 2.

Example 3. Show that the subset of $\mathbb{R}^2$ given by

$$S_1 = \left\{ \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : x_1 + x_2 = 4 \right\}$$

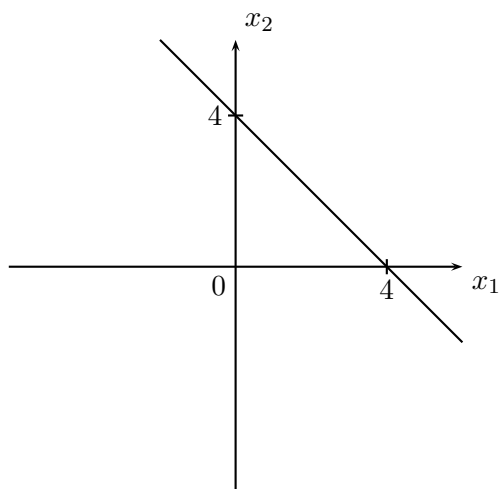
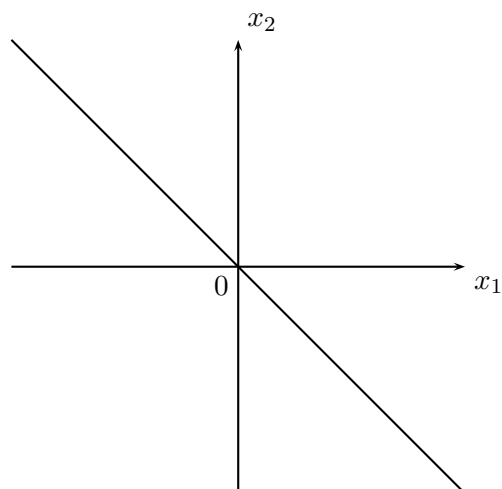
is not a vector space, whereas the subset given by

$$S_2 = \left\{ \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 : x_1 + x_2 = 0 \right\}$$

is a vector space. (Geometrically, S_1 represents a line in $\mathbb{R}^2$ which does not pass through the origin, whereas S_2 represents a line which does pass through the origin (see Figure 3).)

SOLUTION. The vector $\mathbf{v} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \notin S_1$, since $0 + 0 \neq 4$ so S_1 is not a vector space.

It is possible to show that S_2 is a vector space by the usual time-consuming and tedious process of checking that it satisfies all ten of the vector space axioms. $\diamond$

Figure 3(a): The line $x_1 + x_2 = 4$ is not a vector space.Figure 3(b): The line $x_1 + x_2 = 0$ is a vector space.

NOTE. Points in an n -dimensional space are represented by vectors in $\mathbb{R}^n$. When we say a set of vectors S in $\mathbb{R}^n$ represents a line, or simply S is a line in $\mathbb{R}^n$, we mean S is set of the position vectors of all the points on the line in the n -dimensional space.

In Examples 1, 2 and 3 we have asked whether certain subsets of the vector spaces $\mathbb{R}$ and $\mathbb{R}^2$ are themselves vector spaces. Of course the operations of the systems are the usual addition and the usual multiplication by a scalar. We have seen that it is usually fairly simple to show that a subset is not a vector space, but it will be time-consuming and tedious to show that a given subset is a vector space by checking all ten axioms. We shall now develop a simple general test for determining whether a given subset of a vector space is itself a vector space.

We first make the following definitions.

Definition 1. A subset S of a vector space V is called a **subspace** of V if S is itself a vector space over the same field of scalars as V and under the same rules for addition and multiplication by scalars.

In addition if there is at least one vector in V which is not contained in S , the subspace S is called a **proper subspace** of V .

A simple test for a subspace is given by the following theorem.

Theorem 1 (Subspace Theorem). *A subset S of a vector space V over a field $\mathbb{F}$, under the same rules for addition and multiplication by scalars, is a subspace of V if and only if*

- i) *The vector $\mathbf{0}$ in V also belongs to S .*
- ii) *S is closed under vector addition, and*
- iii) *S is closed under multiplication by scalars from $\mathbb{F}$.*

Proof. We first note that if S is a subspace then it is a vector space.

Conversely, suppose S contains the zero vector and the two closure axioms 1 and 6 are satisfied by the elements of S . Every element of S is an element of V because S is a subset of V . Furthermore, S and V are under the same operations, the vector space axioms 2, 3, 7, 8, 9, 10 are automatically satisfied by all elements of S .

Since S contains the zero vector, if $\mathbf{v} \in S$ then, $\mathbf{0} + \mathbf{v} = \mathbf{0}$ (since this is true in V and hence in S , so axiom 4 follows).

Finally, if $\mathbf{v} \in S$ then $\mathbf{v} \in V$. Hence, from part 3 of Proposition 2 of 6.2, we have $-\mathbf{v} = (-1)\mathbf{v}$. But, as S is closed under multiplication by a scalar, we have $(-1)\mathbf{v} \in S$, and hence $-\mathbf{v} \in S$. Thus, axiom 5 is satisfied for all vectors in S . The proof is complete. $\square$

If we want to check if S is a subspace of V , we should first check if the zero vector of V is in S . If the zero vector is in S we can proceed to verify the two closure axioms. Otherwise, we can draw a conclusion that S is not a subspace of V .

Example 4. Prove that the set

$$S = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} : 2x_1 - x_2 + 4x_3 = 0 \right\}$$

is a vector subspace of $\mathbb{R}^3$.

SOLUTION. As $2(0) - 0 + 4(0) = 0$, the zero vector of $\mathbb{R}^3$ is in S . We now proceed to show the two closure axioms.

For any vectors $\mathbf{u} = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$, $\mathbf{v} = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix} \in S$ and $\lambda \in \mathbb{R}$, we have

$$2u_1 - u_2 + 4u_3 = 0 \quad (1)$$

$$2v_1 - v_2 + 4v_3 = 0 \quad (2)$$

If we add (1) and (2), we have

$$(2u_1 - u_2 + 4u_3) + (2v_1 - v_2 + 4v_3) = 0 + 0.$$

Hence, we obtain

$$2(u_1 + v_1) - (u_2 + v_2) + 4(u_3 + v_3) = 0,$$

and so $\mathbf{u} + \mathbf{v} = \begin{pmatrix} u_1 + v_1 \\ u_2 + v_2 \\ u_3 + v_3 \end{pmatrix} \in S$. Thus S is closed under addition.

Now we multiply both sides of (2) by λ , we have

$$\lambda(2v_1 - v_2 + 4v_3) = \lambda \cdot 0 \quad \text{i.e.} \quad 2(\lambda v_1) - \lambda v_2 + 4(\lambda v_3) = 0.$$

Hence $\lambda \mathbf{v} = \begin{pmatrix} \lambda v_1 \\ \lambda v_2 \\ \lambda v_3 \end{pmatrix} \in S$ and so S is closed under multiplication by a scalar.

By the Subspace Theorem, the set S is a vector subspace of $\mathbb{R}^3$. ◇

Example 5. Prove that a line in $\mathbb{R}^n$ is a subspace of $\mathbb{R}^n$ if and only if it passes through the origin.

SOLUTION. Suppose that S represents a line in $\mathbb{R}^n$. If $\mathbf{0} \notin S$, then S is not a subspace. Hence, a line which does not pass through the origin is not a vector subspace.

If S is a line through the origin, we can write

$$S = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} = t\mathbf{v}, \quad t \in \mathbb{R}\},$$

where $\mathbf{v}$ is a fixed non-zero vector in $\mathbb{R}^n$.

To check if S is a subspace we check the two closure axioms.

Closure under addition. If $\mathbf{x}_1, \mathbf{x}_2 \in S$ then

$$\mathbf{x}_1 = t_1\mathbf{v} \quad \text{and} \quad \mathbf{x}_2 = t_2\mathbf{v} \quad \text{for some } t_1, t_2 \in \mathbb{R}.$$

Hence,

$$\mathbf{x}_1 + \mathbf{x}_2 = (t_1 + t_2)\mathbf{v} = t'\mathbf{v},$$

where $t' = t_1 + t_2 \in \mathbb{R}$. Thus, $\mathbf{x}_1 + \mathbf{x}_2 \in S$, and hence S is closed under addition.

Closure under multiplication by a scalar. We have

$$\mathbf{x} = t\mathbf{v} \quad \text{for some } t \in \mathbb{R},$$

and hence, if $\lambda \in \mathbb{R}$,

$$\lambda\mathbf{x} = \lambda(t\mathbf{v}) = (\lambda t)\mathbf{v} = t''\mathbf{v},$$

where $t'' = \lambda t \in \mathbb{R}$. Hence $\lambda\mathbf{x} \in S$, and thus S is closed under multiplication by a scalar.

Therefore, by the Subspace Theorem, the line S is a subspace of $\mathbb{R}^n$ if it passes through the origin. The proof is complete. $\diamond$

A similar result to that given for lines in Example 5 also holds for planes.

Example 6. Prove that a plane in $\mathbb{R}^n$ is a subspace of $\mathbb{R}^n$ if and only if it passes through the origin.

SOLUTION. If the plane does not pass through the origin, then it does not contain the zero vector, and hence is not a subspace.

If the plane passes through the origin, then it is represented by

$$S = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} = s_1\mathbf{v}_1 + s_2\mathbf{v}_2, \quad s_1, s_2 \in \mathbb{R}\},$$

where $\mathbf{v}_1$ and $\mathbf{v}_2$ are fixed, non-parallel vectors in $\mathbb{R}^n$.

We now check closure under addition and under multiplication by a scalar.

Closure under addition. If $\mathbf{x}_1, \mathbf{x}_2 \in S$, then

$$\mathbf{x}_1 = s_1\mathbf{v}_1 + s_2\mathbf{v}_2 \quad \text{and} \quad \mathbf{x}_2 = t_1\mathbf{v}_1 + t_2\mathbf{v}_2 \quad \text{for some } s_1, s_2, t_1, t_2 \in \mathbb{R}.$$

Therefore

$$\mathbf{x}_1 + \mathbf{x}_2 = (s_1 + t_1)\mathbf{v}_1 + (s_2 + t_2)\mathbf{v}_2 = s\mathbf{v}_1 + t\mathbf{v}_2,$$

where $s = s_1 + t_1$ and $t = s_2 + t_2$ are both real numbers. Thus $\mathbf{x}_1 + \mathbf{x}_2 \in S$, and hence S is closed under addition.

The proof that S is closed under multiplication by a scalar is similar and is left as an exercise. $\diamond$

In practice, some of the most important subspaces of $\mathbb{R}^n$ are connected with systems of linear equations, that is, with the matrix equation $A\mathbf{x} = \mathbf{b}$. Here is an important example of this. follows.

Example 7. Let A be an $m \times n$ matrix with real entries. Show that the subset S of $\mathbb{R}^n$ which consists of all solutions of the matrix equation $A\mathbf{x} = \mathbf{b}$ for given $\mathbf{b} \in \mathbb{R}^m$ is a subspace of $\mathbb{R}^n$ if and only if $\mathbf{b} = \mathbf{0}$.

Formally, the set of all solutions of $A\mathbf{x} = \mathbf{b}$ is given by

$$S = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{b}\}.$$

SOLUTION. We first consider the case $\mathbf{b} \neq \mathbf{0}$. Then $\mathbf{0} \in \mathbb{R}^n$ is not a solution of $A\mathbf{x} = \mathbf{b}$ as $A\mathbf{0} = \mathbf{0} \neq \mathbf{b}$, and hence S does not contain the zero vector. Thus S is not a subspace.

We next examine the case $\mathbf{b} = \mathbf{0}$. Then S is the set of solutions of $A\mathbf{x} = \mathbf{0}$. We use the Subspace Theorem to show that S is a subspace.

Closure under addition. If $\mathbf{x} \in S$ and $\mathbf{y} \in S$, then $A\mathbf{x} = \mathbf{0}$ and $A\mathbf{y} = \mathbf{0}$, and hence

$$A(\mathbf{x} + \mathbf{y}) = A\mathbf{x} + A\mathbf{y} = \mathbf{0} + \mathbf{0} = \mathbf{0}.$$

Thus $\mathbf{x} + \mathbf{y} \in S$ and S is closed under addition.

Closure under multiplication by a scalar. If $\mathbf{x} \in S$, we have $A\mathbf{x} = \mathbf{0}$, and hence for all $\lambda \in \mathbb{R}$,

$$A(\lambda\mathbf{x}) = \lambda(A\mathbf{x}) = \lambda\mathbf{0} = \mathbf{0}.$$

Thus $\lambda\mathbf{x} \in S$ and S is closed under scalar multiplication. The result is proved. $\diamond$

NOTE. Example 7 shows that the set of solutions of the matrix equation $A\mathbf{x} = \mathbf{0}$ is a subspace of $\mathbb{R}^n$. This subspace, which is of considerable practical importance, is called the **kernel** of the matrix A (see Section 7.4.1).

An important theoretical and practical problem concerning vector spaces is that of finding all their subspaces. For example, it can be shown that the **only** subspaces of $\mathbb{R}^2$ are (1) the origin, (2) lines through the origin, and (3) $\mathbb{R}^2$ itself. Similarly, for $\mathbb{R}^3$ the only subspaces (see Example 10 of Section 6.6) are (1) the origin, (2) lines through the origin, (3) planes through the origin, and (4) $\mathbb{R}^3$ itself. A listing of subspaces can be given for any vector space. However, before we can investigate this problem satisfactorily, we require further machinery. This machinery will be developed in Sections 6.4 and 6.5. In vector spaces other than $\mathbb{R}^n$ it may be difficult to get a good geometric feel for which subsets are subspaces. Nonetheless, the Subspace Theorem allows one a simple way to check whether a certain set is a subspace or not.

Example 8. Let $V = \mathbb{P}(\mathbb{R})$, the set of all real polynomials. Let $\mathbb{P}_2(\mathbb{R})$ denote the set of all real polynomials of degree less than or equal to 2. That is

$$\mathbb{P}_2(\mathbb{R}) = \{p \in \mathbb{P}(\mathbb{R}) : p(x) = a_0 + a_1x + a_2x^2 \text{ for some } a_0, a_1, a_2 \in \mathbb{R}\}.$$

Show that $\mathbb{P}_2(\mathbb{R})$ is a subspace of $\mathbb{P}(\mathbb{R})$.

SOLUTION. Clearly $\mathbb{P}_2(\mathbb{R})$ contains the zero polynomial. Suppose then that $p, q \in \mathbb{P}_2(\mathbb{R})$. Then there exist coefficients $a_0, a_1, a_2, b_0, b_1, b_2 \in \mathbb{R}$ such that

$$p(x) = a_0 + a_1x + a_2x^2, \quad q(x) = b_0 + b_1x + b_2x^2.$$

Now $(p+q)(x) = (a_0+b_0) + (a_1+b_1)x + (a_2+b_2)x^2$ which is another polynomial of degree less than or equal to 2, i.e. $p+q \in \mathbb{P}_2(\mathbb{R})$.

Suppose that $p \in \mathbb{P}_2(\mathbb{R})$ as above and that $\lambda \in \mathbb{R}$. Then $(\lambda p)(x) = (\lambda a_0) + (\lambda a_1)x + (\lambda a_2)x^2$, and so $\lambda p \in \mathbb{P}_2$.

Thus, by the Subspace Theorem, $\mathbb{P}_2(\mathbb{R})$ is a subspace of $\mathbb{P}(\mathbb{R})$. $\diamond$

Suppose that $\mathbb{P}(\mathbb{F})$ is the set of all polynomials over $\mathbb{F}$. We shall show in Section 6.8 that for any n , the set $\mathbb{P}_n(\mathbb{F})$ consisting of all polynomials over $\mathbb{F}$ of degree less than or equal to n is also a subspace of $\mathbb{P}(\mathbb{F})$.

Example 9. Let S denote the set of all real polynomials of degree exactly 3. Show that S is not a subspace of $\mathbb{P}(\mathbb{R})$.

SOLUTION. This set is not closed under either addition or scalar multiplication! For example, the polynomial p given by $p(x) = x^3$ is in S , but $0p = 0$ which does not lie in S . Also, for example, $(x^3 + x^2 + x) + (-x^3 + x + 3) \notin S$. $\diamond$

6.4 Linear combinations and spans

The two fundamental vector space operations are addition and multiplication by a scalar. By combining these two operations we arrive at the important idea of a sum of scalar multiples of vectors. This leads to the ideas of “linear combination” and “span”: a linear combination of a given set of vectors is a sum of scalar multiples of the vectors and the span of a given set of vectors is the set of **all** linear combinations of the vectors. We defined linear combinations and span of two vectors in Chapter 1. These ideas are used to develop the parametric vector forms for planes. In particular, the span of two non-parallel vectors is a plane through the origin. In this section, we generalise the ideas to a finite set of vectors.

The formal definition of ‘linear combination’ is as follows.

Definition 1. Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a finite set of vectors in a vector space V over a field $\mathbb{F}$. Then a **linear combination** of S is a sum of scalar multiples of the form

$$\lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n \quad \text{with} \quad \lambda_1, \dots, \lambda_n \in \mathbb{F}.$$

Example 1. The vector $\begin{pmatrix} 3 \\ -4 \end{pmatrix}$ is a linear combination of the vectors in the set

$$\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\} \text{ in } \mathbb{R}^2 \text{ because } \begin{pmatrix} 3 \\ -4 \end{pmatrix} = 2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + (-1) \begin{pmatrix} 2 \\ 3 \end{pmatrix} + 3 \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \quad \diamond$$

We know that a vector space (and therefore any subspace of a vector space) is closed under addition and multiplication by scalars, so we would expect that it would also be closed under the operation of forming linear combinations. This is confirmed by the following theorem. The proof of the theorem (which uses induction) is left as an exercise (Problem 36).

Proposition 1 (Closure under Linear Combinations). If S is a finite set of vectors in a vector space V , then every linear combination of S is also a vector in V .

The formal definition of ‘span’ is as follows.

Definition 2. Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a finite set of vectors in a vector space V over a field $\mathbb{F}$. Then the **span** of the set S is the set of all linear combinations of S , that is,

$$\begin{aligned} \text{span}(S) &= \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_n) \\ &= \{\mathbf{v} \in V : \mathbf{v} = \lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n \text{ for some } \lambda_1, \dots, \lambda_n \in \mathbb{F}\}. \end{aligned}$$

Example 2. The span of a single non-zero vector $\mathbf{v}$ in $\mathbb{R}^n$ is a line through the origin. In Chapter 1 we defined “the line in $\mathbb{R}^n$ spanned by $\mathbf{v}$ ” to mean the set

$$S = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} = \lambda \mathbf{v}, \text{ for some } \lambda \in \mathbb{R}\}.$$

This set is just $\text{span}(\mathbf{v})$. ◇

Example 3. If $\{\mathbf{v}, \mathbf{w}\}$ is a pair of non-zero, non-parallel vectors in $\mathbb{R}^n$ then $\text{span}(\mathbf{v}, \mathbf{w})$ is a plane containing the origin. $\diamond$

The following important theorem tells us that the span of a finite non-empty set of vectors in a vector space V is not only a subset of V , it is always a *subspace* of V .

Theorem 2 (A span is a subspace). *If S is a finite, non-empty set of vectors in a vector space V , then $\text{span}(S)$ is a subspace of V . Further, $\text{span}(S)$ is the smallest subspace containing S (in the sense that $\text{span}(S)$ is a subspace of every subspace which contains S).*

Proof. We first note that $\mathbf{0} \in S$, since we may take each scalar to be zero. Proposition 1 tells us that every linear combination of S is a vector in V , so $\text{span}(S)$ is a subset of V .

To prove that $\text{span}(S)$ is a *subspace* we will use the Subspace Theorem, so we set out to prove that $\text{span}(S)$ is closed under addition and under multiplication by scalars. Let S be the set

$$S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$$

where all $\mathbf{v}_j$ belong to V .

To show closure under addition, suppose $\mathbf{u}, \mathbf{w} \in \text{span}(S)$. Then

$$\mathbf{u} = \lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n \quad \text{for some } \lambda_1, \dots, \lambda_n \in \mathbb{F} \quad \text{and}$$

$$\mathbf{w} = \mu_1 \mathbf{v}_1 + \dots + \mu_n \mathbf{v}_n \quad \text{for some } \mu_1, \dots, \mu_n \in \mathbb{F},$$

$$\text{so } \mathbf{u} + \mathbf{w} = (\lambda_1 + \mu_1) \mathbf{v}_1 + \dots + (\lambda_n + \mu_n) \mathbf{v}_n \quad \text{with } \lambda_1 + \mu_1, \dots, \lambda_n + \mu_n \in \mathbb{F}.$$

This shows that $\mathbf{u} + \mathbf{w}$ belongs to $\text{span}(S)$, so $\text{span}(S)$ is closed under addition. To prove closure under multiplication by a scalar, suppose $\mathbf{u} \in \text{span}(S)$ and $\lambda \in \mathbb{F}$. Then

$$\begin{aligned} \lambda \mathbf{u} &= \lambda(\lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n) \\ &= (\lambda \lambda_1) \mathbf{v}_1 + \dots + (\lambda \lambda_n) \mathbf{v}_n, \end{aligned}$$

where $\lambda \lambda_1, \dots, \lambda \lambda_n \in \mathbb{F}$. This shows that $\lambda \mathbf{u}$ belongs to $\text{span}(S)$, so $\text{span}(S)$ is closed under multiplication by scalars.

We have now proved that $\text{span}(S)$ is a subspace of V . To show that it is the *smallest* subspace of V containing S , suppose W is any subspace of V containing S . Then W is itself a vector space containing S and, by what we have just proved, $\text{span}(S)$ is a subspace of W . This completes the proof by showing that $\text{span}(S)$ is a subspace of every subspace of V containing S . $\square$

Example 4. If $\mathbf{v}$ is a non-zero vector in $\mathbb{R}^3$, then the line $\text{span}(\mathbf{v})$ is a subspace of every vector space which contains $\mathbf{v}$. In particular, it is a subspace of $\mathbb{R}^3$ and of every plane through the origin parallel to $\mathbf{v}$. Further, there is no subset of the line $\text{span}(\mathbf{v})$ which both contains $\mathbf{v}$ and is a vector space. Thus, for example, a line segment cannot be a vector space. We have already seen a special case of this result in Example 1 of Section 6.3 and Figure 1. $\diamond$

We often need to find a set S in a vector space V such that $\text{span}(S)$ is the whole of V .

Definition 3. A finite set S of vectors in a vector space V is called a **spanning set** for V if $\text{span}(S) = V$ or equivalently, if every vector in V can be expressed as a linear combination of vectors in S .

Note also that we often say that “ S spans V ” instead of “ S is a spanning set for V ”.

Example 5. As shown in Chapter 1, every geometric vector $\mathbf{a}$ in three dimensions can be written as a linear combination

$$\mathbf{a} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k} \quad \text{for } a_1, a_2, a_3 \in \mathbb{R},$$

where $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ are the unit vectors along the directions of the three coordinate axes. Therefore $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is a spanning set for the vector space of all geometric vectors in three dimensions. $\diamond$

Example 6. The set $S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ -2 \end{pmatrix} \right\}$ is a spanning set of the vector space

$$\left\{ \mathbf{x} \in \mathbb{R}^3 : \mathbf{x} = \lambda \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} + \mu \begin{pmatrix} -1 \\ 2 \\ -2 \end{pmatrix}, \quad \lambda, \mu \in \mathbb{R} \right\}.$$

The set $S' = \left\{ \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}, \begin{pmatrix} -1 \\ 2 \\ -2 \end{pmatrix}, \begin{pmatrix} 0 \\ 3 \\ 1 \end{pmatrix} \right\}$ also spans the above vector space. Obviously, the third vector in S' is the sum of the other two, so $\text{span}(S) = \text{span}(S')$. Thus the third vector in S' is somewhat redundant. $\diamond$

Example 7. Let $\mathbf{v}$ be a fixed non-zero vector in $\mathbb{R}^n$. The spanning set of the vector space $\{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} = \lambda\mathbf{v}, \lambda \in \mathbb{R}\}$ is $\{\mathbf{v}\}$. $\diamond$

Example 8. Every vector $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n$ can be written as $\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$. This expresses $\mathbf{x}$ as a linear combination of the set $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$, where

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad \mathbf{e}_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}.$$

Thus $\mathbb{R}^n = \text{span}(\mathbf{e}_1, \dots, \mathbf{e}_n)$ and the set $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ spans $\mathbb{R}^n$. $\diamond$

Example 9. Let $\mathbb{P}_n$ denote the space of polynomials of degree less than or equal to n . Every polynomial $p \in \mathbb{P}_n$ can be written as a linear combination of the polynomials $\{1, x, x^2, \dots, x^n\}$, so $\mathbb{P}_n = \text{span}(1, x, x^2, \dots, x^n)$. We shall see later that there is no *finite* set of vectors whose span is all of $\mathbb{P}$ (the vector space of *all* polynomials). $\diamond$

6.4.1 Matrices and spans in $\mathbb{R}^m$

We want to have an effective way to tell whether or not a given vector in $\mathbb{R}^m$ belongs to the span of a set $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. From the definition of span, we know that $\mathbf{b}$ belongs to $\text{span}(S)$ if and only if there are $\lambda_1, \dots, \lambda_n \in \mathbb{R}$ such that

$$\mathbf{b} = \lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n.$$

This is equivalent to the condition that there is at least one solution to the vector equation

$$x_1 \mathbf{v}_1 + \dots + x_n \mathbf{v}_n = \mathbf{b},$$

where $x_1, \dots, x_n$ are the unknowns. This vector equation represents a set of simultaneous linear equations in n unknowns. Therefore the question of whether $\mathbf{b}$ belongs to $\text{span}(S)$ is a question of whether or not a particular set of linear equations has a solution. This is the sort of question which we studied in detail in MATH1131/41.

Furthermore, suppose that $\mathbf{v}_1 = \begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} a_{12} \\ \vdots \\ a_{m2} \end{pmatrix}$, $\dots$, $\mathbf{v}_n = \begin{pmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix}$, and $\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$.

If that A is the $m \times n$ matrix whose columns are the vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ then

$$\begin{aligned} A\mathbf{x} &= \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} a_{11}x_1 + \cdots + a_{1n}x_n \\ \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n \end{pmatrix} \\ &= x_1 \begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix} + \cdots + x_n \begin{pmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix} = x_1 \mathbf{v}_1 + \cdots + x_n \mathbf{v}_n. \end{aligned}$$

As a result, we have the following proposition.

Proposition 3 (Matrices, Linear Combinations and Spans). If $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a set of vectors in $\mathbb{R}^m$ and A is the $m \times n$ matrix whose columns are the vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ then

- a vector $\mathbf{b}$ in $\mathbb{R}^m$ can be expressed as a linear combination of S if and only if it can be expressed in the form $A\mathbf{x}$ for some $\mathbf{x}$ in $\mathbb{R}^n$,
- a vector $\mathbf{b}$ in $\mathbb{R}^m$ belongs to $\text{span}(S)$ if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a solution $\mathbf{x}$ in $\mathbb{R}^n$.

Example 10. For the set of three vectors

$$\mathbf{v}_1 = \begin{pmatrix} 0 \\ 5 \\ 3 \\ 6 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 5 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} -2 \\ -3 \\ -5 \\ -6 \end{pmatrix} \in \mathbb{R}^4, \quad \text{we let } A = \begin{pmatrix} 0 & 1 & -2 \\ 5 & 3 & -3 \\ 3 & 4 & -5 \\ 6 & 5 & -6 \end{pmatrix} \text{ and } \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

By expanding each side, it can easily be checked that

$$A\mathbf{x} = \begin{pmatrix} x_2 - 2x_3 \\ 5x_1 + 3x_2 - 3x_3 \\ 3x_1 + 4x_2 - 5x_3 \\ 6x_1 + 5x_2 - 6x_3 \end{pmatrix} = x_1 \begin{pmatrix} 0 \\ 5 \\ 3 \\ 6 \end{pmatrix} + x_2 \begin{pmatrix} 1 \\ 3 \\ 4 \\ 5 \end{pmatrix} + x_3 \begin{pmatrix} -2 \\ -3 \\ -5 \\ -6 \end{pmatrix}.$$

In particular,

$$A \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} = 1 \begin{pmatrix} 0 \\ 5 \\ 3 \\ 6 \end{pmatrix} + 1 \begin{pmatrix} 1 \\ 3 \\ 4 \\ 5 \end{pmatrix} + 3 \begin{pmatrix} -2 \\ -3 \\ -5 \\ -6 \end{pmatrix} = \begin{pmatrix} -5 \\ -1 \\ -8 \\ -7 \end{pmatrix}.$$

Let us denote the vector $\begin{pmatrix} -5 \\ -1 \\ -8 \\ -7 \end{pmatrix}$ by $\mathbf{b}$. Hence $\mathbf{x} = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$ is a solution of $A\mathbf{x} = \mathbf{b}$ if and only if $\mathbf{b}$ can be written as the linear combination $\mathbf{v}_1 + \mathbf{v}_2 + 3\mathbf{v}_3$. $\diamond$

Example 11. If A is an $m \times n$ matrix and $\mathbf{e}_j$ is the j th standard basis vector in $\mathbb{R}^n$ then

$$A\mathbf{e}_j = \mathbf{a}_j,$$

where $\mathbf{a}_j$ is the j th column of A . In the case of the matrix A of the last example, we find by direct matrix multiplication that

$$A\mathbf{e}_1 = \begin{pmatrix} 0 & 1 & -2 \\ 5 & 3 & -3 \\ 3 & 4 & -5 \\ 6 & 5 & -6 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 5 \\ 3 \\ 6 \end{pmatrix} = \mathbf{a}_1; \quad A \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 4 \\ 5 \end{pmatrix} = \mathbf{a}_2; \quad A \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -3 \\ -5 \\ -6 \end{pmatrix} = \mathbf{a}_3.$$

$\diamond$

When applying the results of Proposition 3, it is convenient to have a special name for the subspace of $\mathbb{R}^m$ spanned by the columns of a given $m \times n$ matrix.

Definition 4. The subspace of $\mathbb{R}^m$ spanned by the columns of an $m \times n$ matrix A is called the **column space** of A and is denoted by $\text{col}(A)$.

6.4.2 Solving problems about spans

By Proposition 3, a vector $\mathbf{b}$ in $\mathbb{R}^m$ lies in the span of a set $S = \{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ in $\mathbb{R}^m$ if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a solution, where A is the matrix with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$. The following examples show how to apply this knowledge to problems about spans in $\mathbb{R}^m$.

Example 12. Is the vector $\mathbf{b} = \begin{pmatrix} 1 \\ 4 \\ 1 \\ 2 \end{pmatrix}$ in the span of the set $S = \left\{ \begin{pmatrix} 1 \\ 3 \\ 4 \\ 2 \end{pmatrix}, \begin{pmatrix} -4 \\ -8 \\ -12 \\ 6 \end{pmatrix} \right\}$?

In geometric terms, the question is asking whether the point $(1, 4, 1, 2)$ lies on the plane through

the origin parallel to $\begin{pmatrix} 1 \\ 3 \\ 4 \\ 2 \end{pmatrix}$ and $\begin{pmatrix} -4 \\ -8 \\ -12 \\ 6 \end{pmatrix}$.

SOLUTION. Let A be the matrix whose columns are the members of S , so

$$A = \begin{pmatrix} 1 & -4 \\ 3 & -8 \\ 4 & -12 \\ 2 & 6 \end{pmatrix}.$$

As a consequence of Proposition 3, we know that $\mathbf{b}$ belongs to $\text{span}(S)$ if and only if the equation $A\mathbf{x} = \mathbf{b}$ has a solution. We form the augmented matrix $(A|\mathbf{b})$ for this system and reduce it to row-echelon form.

$$\begin{aligned} & \left(\begin{array}{cc|c} 1 & -4 & 1 \\ 3 & -8 & 4 \\ 4 & -12 & 1 \\ 2 & 6 & 2 \end{array} \right) \xrightarrow{\substack{R_2 = R_2 - 3R_1 \\ R_3 = R_3 - 4R_1 \\ R_4 = R_4 - 2R_1}} \left(\begin{array}{cc|c} 1 & -4 & 1 \\ 0 & 4 & 1 \\ 0 & 4 & -3 \\ 0 & 14 & 0 \end{array} \right) \\ & \xrightarrow{\substack{R_3 = R_3 - R_2 \\ R_4 = R_4 - \frac{7}{2}R_2}} \left(\begin{array}{cc|c} 1 & -4 & 1 \\ 0 & 4 & 1 \\ 0 & 0 & -4 \\ 0 & 0 & -\frac{7}{2} \end{array} \right) \xrightarrow{R_4 = R_4 - \frac{7}{8}R_3} \left(\begin{array}{cc|c} 1 & -4 & 1 \\ 0 & 4 & 1 \\ 0 & 0 & -4 \\ 0 & 0 & 0 \end{array} \right). \end{aligned}$$

Since the right hand column is a leading column, the system has no solution. Therefore $\mathbf{b}$ does not belong to the span of S . $\diamond$

Example 13. Find conditions which are necessary and sufficient to ensure that a vector $\mathbf{b}$ in $\mathbb{R}^3$ belongs to the span of the set $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ where $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$, $\mathbf{v}_3 = \begin{pmatrix} -1 \\ 0 \\ 5 \end{pmatrix}$.

Hence, determine if the vector $\mathbf{v} = \begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} \in \text{span}(S)$. Then give a geometric interpretation of the span.

SOLUTION. By Proposition 3, the vector $\mathbf{b}$ belongs to $\text{span}(S)$ if and only if there is a solution to the system of equations $A\mathbf{x} = \mathbf{b}$, where the three columns of A are the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$. We reduce the augmented matrix $(A|\mathbf{b})$ to row-echelon form.

$$\begin{aligned} & \left(\begin{array}{ccc|c} 1 & 1 & -1 & b_1 \\ 2 & 1 & 0 & b_2 \\ 3 & -1 & 5 & b_3 \end{array} \right) \xrightarrow{\substack{R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 3R_1}} \left(\begin{array}{ccc|c} 1 & 1 & -1 & b_1 \\ 0 & -1 & 2 & b_2 - 2b_1 \\ 0 & -4 & 8 & b_3 - 3b_1 \end{array} \right) \\ & \xrightarrow{R_3 = R_3 - 4R_2} \left(\begin{array}{ccc|c} 1 & 1 & -1 & b_1 \\ 0 & -1 & 2 & b_2 - 2b_1 \\ 0 & 0 & 0 & 5b_1 - 4b_2 + b_3 \end{array} \right). \end{aligned}$$

The system represented by this augmented matrix has a solution if and only if

$$5b_1 - 4b_2 + b_3 = 0.$$

Therefore $\mathbf{b}$ belongs to $\text{span}(S)$ if and only if this condition is satisfied.

To check that $\mathbf{v}$ is in the span or not, we substitute the components of $\mathbf{v}$ in the condition. Since $5(2) - 4(1) + 1 = 7 \neq 0$, so $\mathbf{v}$ is not in $\text{span}(S)$.

To get a geometric interpretation of this result, note that a vector $\mathbf{b}$ is in the span if and only if its components satisfy the Cartesian equation

$$5x_1 - 4x_2 + x_3 = 0$$

which is a plane through the origin with normal $\begin{pmatrix} 5 \\ -4 \\ 1 \end{pmatrix}$. Therefore $\text{span}(S)$ is this plane. $\diamond$

NOTE. As a check, note that each of the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ belongs to $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ and should satisfy the above condition. On substituting $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ for $\begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ we find $5(1) - 4(2) + 3 = 0$, so $\mathbf{v}_1$ does satisfy the above condition. You can check for yourself that $\mathbf{v}_2$ and $\mathbf{v}_3$ also satisfy this condition.

Example 14. Determine whether or not the set $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$, is a spanning set for $\mathbb{R}^3$, where $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$, $\mathbf{v}_3 = \begin{pmatrix} -1 \\ 0 \\ 5 \end{pmatrix}$ and $\mathbf{v}_4 = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$.

SOLUTION. S is a spanning set for $\mathbb{R}^3$ if and only if every vector $\mathbf{b} \in \mathbb{R}^3$ belongs to $\text{span}(S)$. By Proposition 3, every vector $\mathbf{b} \in \mathbb{R}^3$ belongs to $\text{span}(S)$ if and only if the system $A\mathbf{x} = \mathbf{b}$ has a solution for every $\mathbf{b}$ in $\mathbb{R}^3$, where A is the matrix whose columns are the members of S .

By row operations we can reduce the augmented matrix of the system to row-echelon form

$$\begin{aligned} \left(\begin{array}{cccc|c} 1 & 1 & -1 & 2 & b_1 \\ 2 & 1 & 0 & 3 & b_2 \\ 3 & -1 & 5 & 5 & b_3 \end{array} \right) & \xrightarrow[\begin{array}{l} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 3R_1 \end{array}]{\begin{array}{l} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 3R_1 \end{array}} \left(\begin{array}{cccc|c} 1 & 1 & -1 & 2 & b_1 \\ 0 & -1 & 2 & -1 & b_2 - 2b_1 \\ 0 & -4 & 8 & -1 & b_3 - 3b_1 \end{array} \right) \\ & \xrightarrow{R_3 = R_3 - 4R_2} \left(\begin{array}{cccc|c} 1 & 1 & -1 & 2 & b_1 \\ 0 & -1 & 2 & -1 & b_2 - 2b_1 \\ 0 & 0 & 0 & 3 & b_3 - 4b_2 + 5b_1 \end{array} \right). \end{aligned}$$

For every $\mathbf{b} \in \mathbb{R}^3$, the right-hand column is non-leading which means that this system has a solution. This implies that every vector $\mathbf{b} \in \mathbb{R}^3$ belongs to $\text{span}(S)$. Hence S is a spanning set for $\mathbb{R}^3$. $\diamond$

NOTE. The equations would still have a solution for all $\mathbf{b} \in \mathbb{R}^3$ if the non-leading column (column 3) were dropped from the row-echelon form matrix. This means that the vector $\mathbf{v}_3$ can be dropped from S and the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\}$ will still span $\mathbb{R}^3$. Thus, in this case,

$$\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4) = \text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4) = \mathbb{R}^3.$$

We shall see that in Section 6.5 Example 6 that in place of $\mathbf{v}_3$ we could drop either $\mathbf{v}_1$ or $\mathbf{v}_2$ from S and obtain the same span. That is

$$\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4) = \text{span}(\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4) = \text{span}(\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_4) = \mathbb{R}^3.$$

However, the removal of the vector corresponding to a non-leading column in a row-echelon form matrix gives us a simple criterion to get a subset of S which spans the same subspace $\text{span}(S)$. In general, we have the following result.

Suppose that $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is a subset of $\mathbb{R}^m$ and A , which is the matrix with the n vectors in S as columns, reduces to a row-echelon form matrix U . If the i th column of U is non-leading, then $\{\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_n\}$ spans the same set as S .

The following example shows that matrix methods can also be used to solve problems about spans in some vector spaces other than $\mathbb{R}^n$.

Example 15. Find conditions on the coefficients of $p \in \mathbb{P}_3(\mathbb{R})$ so that $p \in \text{span}(1+x, 1-x^2)$.

SOLUTION. Let $p(x) = b_0 + b_1x + b_2x^2 + b_3x^3$ be a polynomial in $\mathbb{P}_3(\mathbb{R})$. From the definition of span, we know that $p \in \text{span}(1+x, 1-x^2)$ if and only if there exist $\lambda_1, \lambda_2 \in \mathbb{R}$ such that, for all $x \in \mathbb{R}$,

$$p(x) = \lambda_1(1+x) + \lambda_2(1-x^2) = (\lambda_1 + \lambda_2) + \lambda_1x - \lambda_2x^2.$$

By comparing coefficients, we must have

$$\begin{array}{rcrcrcrcl} \lambda_1 & + & \lambda_2 & = & b_0 \\ \lambda_1 & & & = & b_1 \\ & & -\lambda_2 & = & b_2 \\ & & 0 & = & b_3. \end{array}$$

This is a system of linear equations in the variables λ_1 and λ_2 and we have to find out which choices of b_0, b_1, b_2, b_3 make it into a system which does have a solution. The augmented matrix for the system is

$$\left(\begin{array}{cc|c} 1 & 1 & b_0 \\ 1 & 0 & b_1 \\ 0 & -1 & b_2 \\ 0 & 0 & b_3 \end{array} \right).$$

This augmented matrix can be reduced to row-echelon form

$$\left(\begin{array}{cc|c} 1 & 1 & b_0 \\ 0 & 1 & b_1 - b_0 \\ 0 & 0 & b_2 - b_1 + b_0 \\ 0 & 0 & b_3 \end{array} \right).$$

This system has a solution if and only if $b_2 - b_1 + b_0 = 0$ and $b_3 = 0$, so these are the conditions under which p belongs to $\text{span}(1+x, 1-x^2)$. $\diamond$

6.5 Linear independence

Suppose that $\mathbf{v}_1, \mathbf{v}_2$ are non-zero vectors. In Chapter 1 we saw that $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$ represents a plane if $\mathbf{v}_1$ and $\mathbf{v}_2$ are not parallel to each other, but only a line if they are parallel. Similarly, if $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ are given non-zero vectors in $\mathbb{R}^3$ then $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ represents

- i) a line if the three vectors are all parallel to each other,

- ii) a plane if they are coplanar, or
- iii) the whole of $\mathbb{R}^3$ otherwise.

In this section we shall show how these results can be understood through the ideas of linear independence and linear dependence of a set of vectors.

Definition 1. Suppose that $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a subset of a vector space. The set S is a **linearly independent set** if the only values of the scalars $\lambda_1, \lambda_2, \dots, \lambda_n$ for which

$$\lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n = \mathbf{0} \quad \text{are} \quad \lambda_1 = \lambda_2 = \dots = \lambda_n = 0.$$

Definition 2. Suppose that $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a subset of a vector space. The set $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a **linearly dependent set** if it is not a linearly independent set, that is, if there exist scalars $\lambda_1, \dots, \lambda_n$, **not all zero**, such that

$$\lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n = \mathbf{0}.$$

NOTE. The linear combination

$$\lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n$$

is equal to $\mathbf{0}$ when all the scalars are zero. The essential point of the definition of linear independence is that the **only** way this linear combination is $\mathbf{0}$ is that all the scalars are zero.

Example 1. Show that the vectors $\begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$ and $\begin{pmatrix} -3 \\ -6 \\ -9 \\ 5 \end{pmatrix}$ form a linearly independent set.

SOLUTION. Applying the definition, we look for scalars λ_1, λ_2 such that

$$\lambda_1 \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} + \lambda_2 \begin{pmatrix} -3 \\ -6 \\ -9 \\ 5 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

In order to satisfy this vector equation the scalars must satisfy the four equations

$$\lambda_1 - 3\lambda_2 = 0, \quad 2\lambda_1 - 6\lambda_2 = 0, \quad 3\lambda_1 - 9\lambda_2 = 0, \quad 4\lambda_1 + 5\lambda_2 = 0.$$

Each of the first three equations is satisfied if and only if $\lambda_1 = 3\lambda_2$. By substituting this formula for λ_1 in the fourth equation we get $17\lambda_2 = 0$. Thus the only solution is $\lambda_1 = \lambda_2 = 0$ and this shows that the two given vectors form a linearly independent set. $\diamond$

The vectors in the above example are not parallel because neither is a scalar multiple of the other. This is a special case of the following geometric interpretation of linear dependence for pairs of vectors.

Example 2. Show that two non-zero vectors in $\mathbb{R}^n$ are parallel if and only if they form a linearly dependent set.

SOLUTION. We first show that two parallel vectors form a linearly dependent set. Two non-zero vectors $\{\mathbf{v}_1, \mathbf{v}_2\}$ are parallel if one is a (non-zero) scalar multiple of the other, and that is $\mathbf{v}_2 = \lambda \mathbf{v}_1$ for some $\lambda \in \mathbb{R}$. We can rewrite this equation as $\lambda \mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0}$. The coefficient of $\mathbf{v}_2$ in this expression is -1 (which is non-zero), so this equation proves that $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a linearly dependent set.

We next show that two linearly dependent non-zero vectors are parallel. If $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a linearly dependent set then there exist λ_1 and λ_2 , not both zero, such that

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 = \mathbf{0}.$$

Without loss of generality, we can assume that $\lambda_1 \neq 0$. Dividing by λ_1 and rearranging gives

$$\mathbf{v}_1 = -\frac{\lambda_2}{\lambda_1} \mathbf{v}_2.$$

This shows that $\mathbf{v}_1$ is a scalar multiple of $\mathbf{v}_2$. It also implies that $\lambda_2 \neq 0$ (otherwise $\mathbf{v}_1$ would be $\mathbf{0}$). Hence the two vectors are parallel. $\diamond$

Example 3. It is easy to verify that

$$3 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + 2 \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + (-1) \begin{pmatrix} 5 \\ 4 \\ 7 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

By Definition 2, the set $\left\{ \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 5 \\ 4 \\ 7 \end{pmatrix} \right\}$ is a linearly dependent set. $\diamond$

Note that no two of the three vectors in the previous example are parallel. The following example gives a geometric interpretation (in terms of coplanarity) for linear dependence of sets of three vectors.

Example 4. Show that three non-zero vectors in $\mathbb{R}^n$ are coplanar if and only if they form a linearly dependent set.

SOLUTION. Suppose that $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are three non-zero vectors in $\mathbb{R}^n$.

We first show that three coplanar vectors form a linearly dependent set. Suppose that $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are coplanar. We first consider the case that two of the three vectors are parallel. Without loss of generality, we can assume that $\mathbf{v}_2$ and $\mathbf{v}_3$ are parallel. By Example 2, there exist λ_2, λ_3 , not both of them are zero such that

$$\lambda_2 \mathbf{v}_2 + \lambda_3 \mathbf{v}_3 = \mathbf{0}.$$

Hence, we have

$$0\mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \lambda_3 \mathbf{v}_3 = \mathbf{0}.$$

Since, not both λ_2 and λ_3 are zero, the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ form a linearly dependent set.

Otherwise, we may assume then that $\mathbf{v}_2$ is not parallel to $\mathbf{v}_3$. Hence, $\mathbf{v}_1$ lies on the plane through the origin parallel to $\mathbf{v}_2$ and $\mathbf{v}_3$. This means that there are scalars λ_2, λ_3 such that

$$\mathbf{v}_1 = \lambda_2 \mathbf{v}_2 + \lambda_3 \mathbf{v}_3.$$

We can rearrange this to get

$$-\mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \lambda_3 \mathbf{v}_3 = \mathbf{0}.$$

At least one coefficient is non-zero (the coefficient of $\mathbf{v}_1$ is -1), so we have shown that the set is linearly dependent.

We now show that if three vectors form a linearly dependent set then they must be coplanar. If the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is linearly dependent then there exist $\lambda_1, \lambda_2, \lambda_3$, not all zero, such that

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \lambda_3 \mathbf{v}_3 = \mathbf{0}.$$

Without loss of generality we can assume that $\lambda_1 \neq 0$. Then the above equation can be rearranged as

$$\mathbf{v}_1 = -\frac{\lambda_2}{\lambda_1} \mathbf{v}_2 - \frac{\lambda_3}{\lambda_1} \mathbf{v}_3.$$

This shows that $\mathbf{v}_1$ satisfies the parametric vector form

$$\mathbf{x} = \lambda \mathbf{v}_2 + \mu \mathbf{v}_3,$$

which represents either a line or a plane through the origin. In both cases, the three vectors are coplanar. $\diamond$

6.5.1 Solving problems about linear independence

We have seen that questions about spans in $\mathbb{R}^m$ can be answered by relating them to questions about the existence of solutions for systems of linear equations. The same is true for questions about linear dependence in $\mathbb{R}^m$.

Proposition 1. If $S = \{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ is a set of vectors in $\mathbb{R}^m$ and A is the $m \times n$ matrix whose columns are the vectors $\mathbf{a}_1, \dots, \mathbf{a}_n$ then the set S is linearly dependent if and only if the system $A\mathbf{x} = \mathbf{0}$ has at least one non-zero solution $\mathbf{x} \in \mathbb{R}^n$.

Proof. As on page 20,

$$A\mathbf{x} = x_1 \mathbf{a}_1 + \dots + x_n \mathbf{a}_n \quad \text{for any vector} \quad \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n.$$

Therefore $A\mathbf{x} = \mathbf{0}$ has at least one non-zero solution $\mathbf{x} = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix}$ if and only if there are scalars $\lambda_1, \dots, \lambda_n$, not all zero, such that

$$\lambda_1 \mathbf{a}_1 + \dots + \lambda_n \mathbf{a}_n = \mathbf{0}.$$

In other words, if and only if the set $\{\mathbf{a}_1, \dots, \mathbf{a}_n\}$ is linearly dependent. $\square$

Example 5. Is the set $S = \left\{ \begin{pmatrix} 1 \\ 3 \\ 2 \\ 4 \end{pmatrix}, \begin{pmatrix} -2 \\ -1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 2 \end{pmatrix} \right\}$ a linearly independent set?

SOLUTION. Let A be the matrix whose columns are the vectors in S . By Proposition 1, the set S is linearly dependent if and only if the system $A\mathbf{x} = \mathbf{0}$ has at least one non-zero solution. We then reduce the augmented matrix $(A|\mathbf{0})$ to row-echelon form $(U|\mathbf{0})$.

$$\begin{aligned}
 \left(\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 3 & -1 & 0 & 0 \\ 2 & 0 & 1 & 0 \\ 4 & 2 & 2 & 0 \end{array} \right) & \xrightarrow{\substack{R_2 = R_2 - 3R_1 \\ R_3 = R_3 - 2R_1 \\ R_4 = R_4 - 4R_1}} \left(\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 0 & 5 & 0 & 0 \\ 0 & 4 & 1 & 0 \\ 0 & 10 & 2 & 0 \end{array} \right) \\
 & \xrightarrow{\substack{R_2 = \frac{1}{5}R_2 \\ R_4 = \frac{1}{2}R_4}} \left(\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 4 & 1 & 0 \\ 0 & 5 & 1 & 0 \end{array} \right) \\
 & \xrightarrow{\substack{R_3 = R_3 - 4R_2 \\ R_4 = R_4 - 5R_2}} \left(\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 \end{array} \right) \\
 & \xrightarrow{R_4 = R_4 - R_3} \left(\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)
 \end{aligned}$$

There are no non-leading columns in U , so the system has a unique solution, namely $\mathbf{x} = \mathbf{0}$. Since there are no non-zero solutions, the set S is linearly independent. $\diamond$

Example 6. Suppose that $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$, $\mathbf{v}_3 = \begin{pmatrix} -1 \\ 0 \\ 5 \end{pmatrix}$ and $\mathbf{v}_4 = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$.

- Prove that the set $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is a linearly dependent set.
- Find all possible ways of writing $\mathbf{0}$ as a linear combination of the vectors in S .
- Find a linearly independent subset of S with the same span as S .

SOLUTION.

- Let A be the matrix with $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ as columns. As seen in the previous example, elementary row operations do not affect the zero right-hand column. To see whether or not there is any non-zero solution for the equation $A\mathbf{x} = \mathbf{0}$, we can simply reduce the matrix A

and an equivalent row-echelon form U .

$$\begin{pmatrix} 1 & 1 & -1 & 2 \\ 2 & 1 & 0 & 3 \\ 3 & -1 & 5 & 5 \end{pmatrix} \xrightarrow[\begin{matrix} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 3R_1 \end{matrix}]{\begin{matrix} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 3R_1 \end{matrix}} \begin{pmatrix} 1 & 1 & -1 & 2 \\ 0 & -1 & 2 & -1 \\ 0 & -4 & 8 & 1 \end{pmatrix} \xrightarrow{R_3 = R_3 - 4R_2} \begin{pmatrix} 1 & 1 & -1 & 2 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

The row-echelon form matrix U has a non-leading column, the homogeneous system $A\mathbf{x} = \mathbf{0}$ has infinitely many solutions. There must be solutions other than the zero solution. Therefore the given set S is **linearly dependent**.

- b) The complete solution of $A\mathbf{x} = \mathbf{0}$ can be found by back substitution, and it is $x_4 = 0$, $x_3 = \lambda$, $x_2 = 2\lambda$, and $x_1 = -\lambda$. By substituting for x_1, x_2, x_3, x_4 in the original linear combination we get

$$\lambda(-\mathbf{v}_1 + 2\mathbf{v}_2 + \mathbf{v}_3) + 0\mathbf{v}_4 = \mathbf{0},$$

which gives all possible ways of writing $\mathbf{0}$ as a linear combination of the vectors in S .

- c) Choosing $\lambda = 1$, we have

$$-\mathbf{v}_1 + 2\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}, \quad \text{i.e.} \quad \mathbf{v}_3 = -\mathbf{v}_1 + 2\mathbf{v}_2.$$

This shows that $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4) = \text{span}(S)$. On the other hand, if we remove the third column from A , we can reduce the matrix

$$\begin{pmatrix} 1 & 1 & 2 \\ 2 & 1 & 3 \\ 3 & -1 & 5 \end{pmatrix} \quad \text{to} \quad \begin{pmatrix} 1 & 1 & 2 \\ 0 & -1 & -1 \\ 0 & 0 & 3 \end{pmatrix}.$$

Hence the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\}$ is a linearly independent subset of S with the same span as S .

Note also that $\mathbf{v}_1 = 2\mathbf{v}_2 + \mathbf{v}_3$ and $\mathbf{v}_2 = \frac{1}{2}\mathbf{v}_1 - \frac{1}{2}\mathbf{v}_3$, so both sets $\{\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ and $\{\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_4\}$ have the same span as S . It is not difficult to check that these two sets are also linearly independent. $\diamond$

The next two examples illustrate the fact that matrix methods can be used in vector spaces other than $\mathbb{R}^m$.

Example 7. Show that the set $\{1+x, 2-x\}$ is linearly independent in the vector space $\mathbb{P}(\mathbb{R})$ of all polynomials.

SOLUTION. Suppose that $\lambda_1(x+1) + \lambda_2(2-x) = 0$ for all $x \in \mathbb{R}$. Expanding the left-hand-side gives

$$(\lambda_1 + 2\lambda_2) + (\lambda_1 - \lambda_2)x = 0.$$

Comparing coefficients shows that we must have

$$\begin{aligned} \lambda_1 + 2\lambda_2 &= 0 \\ \lambda_1 - \lambda_2 &= 0. \end{aligned}$$

The augmented matrix for this system is

$$\left(\begin{array}{cc|c} 1 & 2 & 0 \\ 1 & -1 & 0 \end{array} \right),$$

which can be reduced to the row-echelon form

$$\left(\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & -3 & 0 \end{array} \right).$$

There are no non-leading variables here, so the only solution is $\lambda_1 = \lambda_2 = 0$. Therefore this set of polynomials is linearly independent. $\diamond$

Example 8. Is the set $\{1 + x, 2 - x, -1 + 2x\}$ a linearly independent subset of the vector space $\mathbb{P}(\mathbb{R})$?

SOLUTION. Suppose that $\lambda_1(x + 1) + \lambda_2(2 - x) + \lambda_3(-1 + 2x) = 0$ for all $x \in \mathbb{R}$. Expanding the left-hand-side gives

$$(\lambda_1 + 2\lambda_2 - \lambda_3) + (\lambda_1 - \lambda_2 + 2\lambda_3)x = 0.$$

Comparing coefficients shows that we must have

$$\lambda_1 + 2\lambda_2 - \lambda_3 = 0$$

$$\lambda_1 - \lambda_2 + 2\lambda_3 = 0.$$

The augmented matrix for this system is

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 1 & -1 & 2 & 0 \end{array} \right),$$

which can be reduced to the row-echelon form

$$(U|\mathbf{0}) = \left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & -3 & 3 & 0 \end{array} \right).$$

The third column in U is non-leading so there must be some solutions with λ_1, λ_2 and λ_3 not all zero. Therefore the given set of polynomials is linearly dependent. $\diamond$

NOTE. Although it is not required here, it is a good idea to find a specific nonzero solution and check that the appropriate linear combination is zero. In this example we could use back substitution to get a solution $\lambda_1 = -1, \lambda_2 = 1, \lambda_3 = 1$. Checking shows that indeed

$$-1(1 + x) + 1(2 - x) + 1(-1 + 2x) = 0$$

for all $x \in \mathbb{R}$.

6.5.2 Uniqueness and linear independence

The following theorem gives one reason why the idea of linear independence is important.

Theorem 2 (Uniqueness of Linear Combinations). *Let S be a finite, non-empty set of vectors in a vector space and let $\mathbf{v}$ be a vector which can be written as a linear combination of S . Then the values of the scalars in the linear combination for $\mathbf{v}$ are unique if and only if S is a linearly independent set.*

Proof. It turns out to be easier to prove the theorem by proving the equivalent result that:

The values of the scalars in the linear combination are non-unique if and only if S is a linearly dependent set.

Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. Suppose first that $\mathbf{v}$ has two expressions as a linear combination of S , namely

$$\mathbf{v} = \lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n \quad \text{and} \quad \mathbf{v} = \mu_1 \mathbf{v}_1 + \dots + \mu_n \mathbf{v}_n.$$

Subtracting the second equation from the first equation gives

$$(\lambda_1 - \mu_1) \mathbf{v}_1 + \dots + (\lambda_n - \mu_n) \mathbf{v}_n = \mathbf{0}. \quad (\#)$$

If the two sets of scalars are not identical then there must be at least one value of j (with $1 \leq j \leq n$) such that $\lambda_j - \mu_j \neq 0$. This means that there is at least one non-zero scalar coefficient in $(\#)$ and therefore the set S is a linearly dependent set. Conversely, if S is a linearly dependent set then there are scalars $\alpha_1, \dots, \alpha_n$, not all zero, such that

$$\alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n = \mathbf{0}.$$

If $\mathbf{v} = \lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n$ is any expression for $\mathbf{v}$ as a linear combination of S then we can get a second expression by saying

$$\begin{aligned} \mathbf{v} &= \mathbf{v} + \mathbf{0} \\ &= (\lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n) + (\alpha_1 \mathbf{v}_1 + \dots + \alpha_n \mathbf{v}_n) \\ &= (\lambda_1 + \alpha_1) \mathbf{v}_1 + \dots + (\lambda_n + \alpha_n) \mathbf{v}_n. \end{aligned}$$

The coefficients in this expression are not all the same as the coefficients in the first expression because $\alpha_1, \dots, \alpha_n$ are not all zero. Therefore the coefficients in an expression for $\mathbf{v}$ as a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_n$ are not unique. $\square$

The following is a numerical example of the result of Theorem 2.

Example 9. As shown in Example 6, the set of vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ is a linearly dependent set, where $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$, $\mathbf{v}_3 = \begin{pmatrix} -1 \\ 0 \\ 5 \end{pmatrix}$, and $\mathbf{v}_4 = \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}$. Show that $\mathbf{b} = \begin{pmatrix} 7 \\ 7 \\ -4 \end{pmatrix}$ belongs to $\text{span}(S)$ and then check that the linear combination for $\mathbf{b}$ in terms of S is not unique.

SOLUTION. We know that $\mathbf{b} \in \text{span}(S)$ if and only if the system $A\mathbf{x} = \mathbf{b}$, where A is the matrix with the vectors in S as columns. We then reduce $(A|\mathbf{b})$ to a row-echelon form $(U|\mathbf{y})$.

$$\begin{aligned} \left(\begin{array}{cccc|c} 1 & 1 & -1 & 2 & 7 \\ 2 & 1 & 0 & 3 & 7 \\ 3 & -1 & 5 & 5 & -4 \end{array} \right) & \xrightarrow[\begin{array}{l} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 3R_1 \end{array}]{\begin{array}{l} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 3R_1 \end{array}} \left(\begin{array}{cccc|c} 1 & 1 & -1 & 2 & 7 \\ 0 & -1 & 2 & -1 & -7 \\ 0 & -4 & 8 & 1 & -25 \end{array} \right) \\ & \xrightarrow{R_3 = R_3 - 4R_2} \left(\begin{array}{cccc|c} 1 & 1 & -1 & 2 & 7 \\ 0 & -1 & 2 & -1 & -7 \\ 0 & 0 & 0 & 3 & 3 \end{array} \right) \end{aligned}$$

The right-hand-side column is not a leading column, so the system does have a solution and $\mathbf{b}$ must be in $\text{span}(S)$. The third column in the row-echelon form is non-leading, so the system has infinitely many solutions and hence there are infinitely many expressions for $\mathbf{b}$ as a linear combination of S . $\diamond$

NOTE. If we want to find these expressions explicitly, we can apply back substitution to the row-echelon form and find the general solution $x_4 = 1$, $x_3 = \lambda$, $x_2 = 6 + 2\lambda$ and $x_1 = -1 - \lambda$, where λ is an arbitrary real parameter. Therefore

$$\begin{pmatrix} 7 \\ 7 \\ -4 \end{pmatrix} = (-1 - \lambda) \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + (6 + 2\lambda) \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} -1 \\ 0 \\ 5 \end{pmatrix} + \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}.$$

We found in Example 6 that if $\mathbf{v}_3$ (which is a non-leading variable in this system) is dropped then the resulting set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\}$ is linearly independent.

In this case we find that $\begin{pmatrix} 7 \\ 7 \\ -4 \end{pmatrix}$ is a *unique* linear combination of $\mathbf{v}_1$, $\mathbf{v}_2$ and $\mathbf{v}_4$. In fact, by putting $x_3 = \lambda = 0$ in the above expression we get

$$\begin{pmatrix} 7 \\ 7 \\ -4 \end{pmatrix} = -1 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + 6 \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} + 1 \begin{pmatrix} 2 \\ 3 \\ 5 \end{pmatrix}.$$

6.5.3 Spans and linear independence

Suppose that $\mathbf{v}_1$ and $\mathbf{v}_2$ are non-zero vectors in $\mathbb{R}^n$. We have seen in Chapter 1 that $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$ represents a plane if the vectors are not parallel (in other words, if they are linearly independent), whereas the span represents a line if they are parallel (in other words, linearly dependent). An equivalent way of expressing this result is to say that if $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a linearly dependent set then

$$\text{span}(\mathbf{v}_1, \mathbf{v}_2) = \text{span}(\mathbf{v}_1) = \text{span}(\mathbf{v}_2).$$

For three non-zero vectors in $\mathbb{R}^n$, we have also seen that $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ reduces to either a plane or a line when the three vectors form a linearly dependent set. In either of these cases it is possible to drop at least one vector from $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ without changing the span of the set.

In this section we show that similar results are true for the span of any number of vectors in any vector space. To do this, we need two important results. The first, which is a generalisation of the results in Examples 2 and 4, is as follows.

Theorem 3. *A set of vectors S is a linearly independent set if and only if no vector in S can be written as a linear combination of the other vectors in S , that is, if and only if no vector in S is in the span of the other vectors in S .*

NOTE. The theorem is equivalent to:

A set of vectors S is a linearly dependent set if and only if at least one vector in S is in the span of the other vectors in S .

Proof. It is easier to prove the alternative statement because in that case the method of proof can closely follow the solution given for Example 4.

If some vector $\mathbf{v}_i$ in $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is in the span of the other vectors in S then there are scalars $\mu_1, \dots, \mu_{i-1}, \mu_{i+1}, \dots, \mu_n$ such that

$$\mathbf{v}_i = \mu_1 \mathbf{v}_1 + \dots + \mu_{i-1} \mathbf{v}_{i-1} + \mu_{i+1} \mathbf{v}_{i+1} + \dots + \mu_n \mathbf{v}_n.$$

An obvious rearrangement then gives

$$\mu_1 \mathbf{v}_1 + \dots - \mathbf{v}_i + \dots + \mu_n \mathbf{v}_n = \mathbf{0}.$$

At least one coefficient in this expression (the coefficient -1 for $\mathbf{v}_i$) is non-zero, so the set is linearly dependent.

Conversely, if S is a linearly dependent set then there are scalars $\lambda_1, \dots, \lambda_n$, not all zero, such that

$$\lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n = \mathbf{0}.$$

Let i be such that $\lambda_i \neq 0$. Then we can solve for $\mathbf{v}_i$ in the preceding equation to obtain

$$\mathbf{v}_i = -\frac{1}{\lambda_i}(\lambda_1 \mathbf{v}_1 + \dots + \lambda_{i-1} \mathbf{v}_{i-1} + \lambda_{i+1} \mathbf{v}_{i+1} + \dots + \lambda_n \mathbf{v}_n),$$

This shows that $\mathbf{v}_i$ is in the span of the other vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \dots, \mathbf{v}_n\}$, so the proof is complete. $\square$

Example 10. The set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ in Example 6 is a linearly dependent set, and we found that

$$-\mathbf{v}_1 + 2\mathbf{v}_2 + \mathbf{v}_3 = \mathbf{0}.$$

We saw that $\mathbf{v}_3 = \mathbf{v}_1 - 2\mathbf{v}_2$. Hence, $\mathbf{v}_3 \in \text{span}(\mathbf{v}_1, \mathbf{v}_2)$, which implies $\mathbf{v}_3 \in \text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4)$.

Furthermore, we have $\mathbf{v}_2 \in \text{span}(\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_4)$ and $\mathbf{v}_1 \in \text{span}(\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4)$. However, $\mathbf{v}_4$ is not in the span of the other three. A geometric interpretation of this result is that $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ lie in the same plane, whereas $\mathbf{v}_4$ does not lie in this plane. $\diamond$

The second key result that we need is as follows.

If a vector is added to a set then the span of the new set is equal to the span of the original set if and only if the additional vector is in the span of the original set.

Formally, we have the following theorem.

Theorem 4. *If S is a finite subset of a vector space V and the vector $\mathbf{v}$ is in V , then $\text{span}(S \cup \{\mathbf{v}\}) = \text{span}(S)$ if and only if $\mathbf{v} \in \text{span}(S)$.*

[X] *Proof.* Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ so that $S \cup \{\mathbf{v}\} = \{\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{v}\}$.

Obviously $\mathbf{v} \in \text{span}(S \cup \{\mathbf{v}\})$, so if $\text{span}(S \cup \{\mathbf{v}\}) = \text{span}(S)$ then $\mathbf{v} \in \text{span}(S)$.

To prove the converse, we assume that $\mathbf{v} \in \text{span}(S)$ and prove $\text{span}(S) = \text{span}(S \cup \{\mathbf{v}\})$ by proving firstly that if a vector $\mathbf{u} \in \text{span}(S)$ then $\mathbf{u} \in \text{span}(S \cup \{\mathbf{v}\})$, and secondly that if a vector $\mathbf{u} \in \text{span}(S \cup \{\mathbf{v}\})$ then $\mathbf{u} \in \text{span}(S)$.

For the first proof, suppose that $\mathbf{u} \in \text{span}(S)$. Then

$$\mathbf{u} = \lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n, \quad \text{for some } \lambda_1, \dots, \lambda_n \in \mathbb{F}.$$

Then $\mathbf{u} = \lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n + 0\mathbf{v}$. Hence $\mathbf{u} \in \text{span}(S \cup \{\mathbf{v}\})$.

For the second proof, suppose that $\mathbf{u} \in \text{span}(S \cup \{\mathbf{v}\})$. Then

$$\mathbf{u} = \lambda \mathbf{v} + \lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n \quad \text{for some } \lambda, \lambda_1, \dots, \lambda_n \in \mathbb{F}.$$

But $\mathbf{v} \in \text{span}(S)$, and hence

$$\mathbf{v} = \mu_1 \mathbf{v}_1 + \dots + \mu_n \mathbf{v}_n \quad \text{for some } \mu_1, \dots, \mu_n \in \mathbb{F}.$$

On substituting this linear combination for $\mathbf{v}$ into the previous linear combination for $\mathbf{u}$, we find $\mathbf{u}$ is a linear combination of $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. Hence, $\mathbf{u} \in \text{span}(S)$ and the proof is complete. $\square$

Example 11. For the linearly dependent set of four vectors $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ of Example 6, from Example 10, we have

$$\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4) = \text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4) = \text{span}(\mathbf{v}_1, \mathbf{v}_3, \mathbf{v}_4) = \text{span}(\mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4)$$

which agrees with Theorem 3. $\diamond$

By combining Theorems 3 and 4 we get the following.

*If S is a linearly dependent set of vectors then it is possible to drop at least one vector from S to obtain a new set with the **same** span as S , whereas if S is a linearly independent set then dropping any vector from S results in a new set with a **smaller** span than $\text{span}(S)$.*

In formal terms, we have the following theorem.

Theorem 5. *Suppose that S is a finite subset of a vector space. The span of every proper subset of S is a proper subspace of $\text{span}(S)$ if and only if S is a linearly independent set.*

Example 12. For the linearly dependent set $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$ considered in Example 6, we have seen that $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4) = \text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4) = \mathbb{R}^3$, and the set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\}$ is a linearly independent set.

We have also seen in Example 14 of Section 6.4 that S spans $\mathbb{R}^3$. If we now drop any vector from the linearly independent set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\}$, we obtain a set whose span is only a plane in $\mathbb{R}^3$ and not all of $\mathbb{R}^3$. This illustrates Theorem 5. $\diamond$

For use in the next section, we need one more result involving spanning and linear independence.

Theorem 6. *If S is a finite linearly independent subset of a vector space V and $\mathbf{v}$ is in V but not in $\text{span}(S)$ then $S \cup \{\mathbf{v}\}$ is a linearly independent set.*

Proof. Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ and $S' = S \cup \{\mathbf{v}\}$. If S' is *not* linearly independent then there exist scalars $\lambda, \lambda_1, \dots, \lambda_n$, not all zero, such that

$$\lambda \mathbf{v} + \lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n = \mathbf{0}. \quad (*)$$

If $\lambda = 0$ then we must have $\lambda_i \neq 0$ for some $1 \leq i \leq n$ and this would contradict the assumption that S is linearly independent. Therefore we must have $\lambda \neq 0$ and this means that the equation $(*)$ can be rearranged to express $\mathbf{v}$ as a linear combination of the set S . This contradicts the fact $\mathbf{v}$ is not in $\text{span}(S)$, so the supposition that S' was linearly dependent must be false and hence S' is linearly independent. $\square$

6.6 Basis and dimension

In Chapter 1 we called the set of vectors $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ the standard “basis” for $\mathbb{R}^n$. We have also talked of a plane as being “two-dimensional” and ordinary space as being “three-dimensional”. The main aims of this section are to give precise definitions of the ideas of basis and dimension and to show that these ideas apply to every vector space.

Recall that S is a spanning set for a vector space V when $\text{span}(S) = V$, that is, when every vector in V can be written as a linear combination of the vectors in S . We have also shown that if S is a linearly independent set then all linear combinations formed from S are unique. Hence, if S is a **linearly independent spanning set** for V then **every** vector in V can be written as a **unique linear combination** of S .

We have also shown (Theorem 5 of Section 6.5) that a proper subset of a linearly independent set S spans a proper subspace of $\text{span}(S)$. This means, in particular, that if S is a **linearly independent spanning set** for a vector space V then **dropping any vector** from S gives a set which is **not a spanning set** for V .

Because of these two properties, linearly independent spanning sets are of fundamental importance in both the theoretical development and the practical applications of vector spaces.

6.6.1 Bases

Definition 1. *A set of vectors B in a vector space V is called a **basis** for V if:*

1. *B is a linearly independent set, and*
2. *B is a spanning set for V (that is, $\text{span}(B) = V$).*

NOTE. We exclude from our discussion the vector space consisting of only the zero vector.

Example 1. The set $\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$, where $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$, $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}$, $\dots$, $\mathbf{e}_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}$, is a linearly

independent spanning set for $\mathbb{R}^n$, so this set is a basis for $\mathbb{R}^n$. It is called the **standard basis** for $\mathbb{R}^n$. Each vector $\mathbf{a} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix}$ can be written as a **unique** linear combination $a_1\mathbf{e}_1 + \cdots + a_n\mathbf{e}_n$. $\diamond$

Example 2. Show that the set $S = \left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right\}$ is a basis for $\mathbb{R}^3$.

SOLUTION. To show that S is a spanning set for $\mathbb{R}^3$, we need to show that every vector $\mathbf{b} \in \mathbb{R}^3$ belongs to $\text{span}(S)$. This will be true if and only if the system $A\mathbf{x} = \mathbf{b}$, where

$$A = \begin{pmatrix} 2 & -1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & -1 \end{pmatrix},$$

has a solution for every right hand side $\mathbf{b} \in \mathbb{R}^3$. The augmented matrix $(A|\mathbf{b})$ can be reduced to the row-echelon form

$$\begin{aligned} \left(\begin{array}{ccc|c} 2 & -1 & 0 & b_1 \\ 1 & 0 & 1 & b_2 \\ 0 & 1 & -1 & b_3 \end{array} \right) & \xrightarrow{R_1 \leftrightarrow R_2} \left(\begin{array}{ccc|c} 1 & 0 & 1 & b_2 \\ 2 & -1 & 0 & b_1 \\ 0 & 1 & -1 & b_3 \end{array} \right) \\ & \xrightarrow{R_2 = R_2 - 2R_1} \left(\begin{array}{ccc|c} 1 & 0 & 1 & b_2 \\ 0 & -1 & -2 & b_1 - 2b_2 \\ 0 & 1 & -1 & b_3 \end{array} \right) \\ & \xrightarrow{R_3 = R_3 + R_2} \left(\begin{array}{ccc|c} 1 & 0 & 1 & b_2 \\ 0 & -1 & -2 & b_1 - 2b_2 \\ 0 & 0 & -3 & b_1 - 2b_2 + b_3 \end{array} \right). \end{aligned}$$

For all $\mathbf{b} \in \mathbb{R}^3$, the row-echelon matrix has a non-leading right hand column and hence the equation $A\mathbf{x} = \mathbf{b}$ has a solution. Therefore $\text{span}(S) = \mathbb{R}^3$.

Moreover, left side of the row-echelon matrix has *no non-leading columns*, so the only solution for a zero right hand side is $x_1 = x_2 = x_3 = 0$. This shows that S is a linearly independent set. We have now proved S is linearly independent spanning set for $\mathbb{R}^3$ and is therefore a basis for $\mathbb{R}^3$. $\diamond$

Example 3. Let $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$, $\mathbf{v}_3 = \begin{pmatrix} 2 \\ 4 \\ 2 \end{pmatrix}$, $\mathbf{v}_4 = \begin{pmatrix} 1 \\ 5 \\ 0 \end{pmatrix}$ and $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$.

Find a subset of S which is a basis for $\text{span}(S)$.

SOLUTION. From the result on page 24, we only need to reduce the matrix with the vectors in S as columns to row-echelon form, then the vectors in S corresponding to the leading columns will span the same set as S .

$$\begin{aligned} \left(\begin{array}{cccc} 1 & 2 & 2 & 1 \\ -1 & 1 & 4 & 5 \\ 2 & 3 & 2 & 0 \end{array} \right) & \xrightarrow{\substack{R_2 = R_2 + R_1 \\ R_3 = R_3 - 2R_1}} \left(\begin{array}{cccc} 1 & 2 & 2 & 1 \\ 0 & 3 & 6 & 6 \\ 0 & -1 & -2 & -2 \end{array} \right) \\ & \xrightarrow{R_3 = R_3 + \frac{1}{3}R_2} \left(\begin{array}{cccc} 1 & 2 & 2 & 1 \\ 0 & 3 & 6 & 6 \\ 0 & 0 & 0 & 0 \end{array} \right) \end{aligned}$$

The vectors corresponding to the leading columns are $\mathbf{v}_1$ and $\mathbf{v}_2$. Hence $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a spanning set for $\text{span}(S)$. Furthermore, if we remove the third and fourth columns in the matrices in the above reduction, we can see that $\{\mathbf{v}_1, \mathbf{v}_2\}$ is a linearly independent set. Therefore, this set is a basis for $\text{span}(S)$. $\diamond$

Suppose that B is a basis for a finite-dimensional vector space V . Since B is a spanning set, every vector in V can be written as a linear combination of B . Since B is also independent, the linear combination is unique. In formal terms:

Let $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for a vector space V over $\mathbb{F}$. Every vector $\mathbf{v} \in V$ can be uniquely written as

$$\mathbf{v} = \lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n, \quad \text{where } \lambda_1, \dots, \lambda_n \in \mathbb{F}.$$

Example 4. Write the vector $\mathbf{b} = \begin{pmatrix} -1 \\ 0 \\ 5 \end{pmatrix}$ as the unique linear combination of the ordered basis $\left\{ \mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \right\}$ of $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$.

SOLUTION. We first write $\mathbf{b}$ as a linear combination of $\mathbf{v}_1$ and $\mathbf{v}_2$. From Proposition 3 of Section 6.4 we know that the required scalars are the components of a solution to the system $A\mathbf{x} = \mathbf{b}$, where A is the matrix whose columns are the given vectors $\mathbf{v}_1, \mathbf{v}_2$. The augmented matrix $(A|\mathbf{b})$ can be reduced to row-echelon form.

$$\left(\begin{array}{cc|c} 1 & 1 & -1 \\ 2 & 1 & 0 \\ 3 & -1 & 5 \end{array} \right) \xrightarrow[R_3 = R_3 - 3R_1]{R_2 = R_2 - 2R_1} \left(\begin{array}{cc|c} 1 & 1 & -1 \\ 0 & -1 & 2 \\ 0 & -4 & 8 \end{array} \right) \xrightarrow{R_3 = R_3 - 4R_2} \left(\begin{array}{cc|c} 1 & 1 & -1 \\ 0 & -1 & 2 \\ 0 & 0 & 0 \end{array} \right)$$

This system has the unique solution $\mathbf{x} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$, so $\mathbf{b} = 1 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} - 2 \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$. $\diamond$

Example 5. Show that the set $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is a basis for $\mathbb{R}^3$

SOLUTION. We have seen in Section 6.4 Example 5 that $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is a spanning set. To prove that the set is linearly independent, we use the fact that $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is an orthonormal set of vectors. That is,

$$\mathbf{i} \cdot \mathbf{i} = \mathbf{j} \cdot \mathbf{j} = \mathbf{k} \cdot \mathbf{k} = 1 \quad \text{and} \quad \mathbf{i} \cdot \mathbf{j} = \mathbf{j} \cdot \mathbf{k} = \mathbf{i} \cdot \mathbf{k} = 0.$$

If we assume that

$$\mathbf{0} = a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k} \tag{\#}$$

then by taking the dot product of $(\#)$ with $\mathbf{i}$ we get

$$\begin{aligned} 0 &= \mathbf{i} \cdot \mathbf{0} = \mathbf{i} \cdot (a_1 \mathbf{i} + a_2 \mathbf{j} + a_3 \mathbf{k}) \\ &= a_1(\mathbf{i} \cdot \mathbf{i}) + a_2(\mathbf{i} \cdot \mathbf{j}) + a_3(\mathbf{i} \cdot \mathbf{k}) \\ &= a_1, \end{aligned}$$

and hence $a_1 = 0$. Similarly, by taking the dot products of (#) with $\mathbf{j}$ and $\mathbf{k}$ in turn, we find that $a_2 = 0$ and $a_3 = 0$. Therefore $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is a linearly independent set.

We have now proved that $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is a linearly independent spanning set for, and hence a basis for, $\mathbb{R}^3$. $\diamond$

The set $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ is an example of a **orthonormal basis**. An orthonormal basis is a basis whose elements are all of length 1 and are mutually orthogonal. The advantage of using an orthonormal basis is that we can write easily any vector as the unique linear combination of the basis by dot product.

Example 6. The set of vectors $B = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$, where

$$\mathbf{u}_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \quad \mathbf{u}_3 = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix},$$

is an orthonormal basis for $\mathbb{R}^3$. Write $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ as the unique linear combination of this basis.

SOLUTION. Suppose that $\mathbf{a} = x_1\mathbf{u}_1 + x_2\mathbf{u}_2 + x_3\mathbf{u}_3$. We could find x_1, x_2, x_3 in the same way as Example 4, but there is a simpler method which uses the orthonormality properties of the basis.

Given that B is orthonormal. That is

$$\mathbf{u}_1 \cdot \mathbf{u}_1 = \mathbf{u}_2 \cdot \mathbf{u}_2 = \mathbf{u}_3 \cdot \mathbf{u}_3 = 1 \quad \text{and} \quad \mathbf{u}_1 \cdot \mathbf{u}_2 = \mathbf{u}_1 \cdot \mathbf{u}_3 = \mathbf{u}_2 \cdot \mathbf{u}_3 = 0.$$

We have

$$x_1 = \mathbf{u}_1 \cdot (x_1\mathbf{u}_1 + x_2\mathbf{u}_2 + x_3\mathbf{u}_3) = \mathbf{u}_1 \cdot \mathbf{a} = \frac{1}{\sqrt{2}}a_1 - \frac{1}{\sqrt{2}}a_3,$$

and similarly

$$x_2 = \mathbf{u}_2 \cdot \mathbf{a} = \frac{1}{\sqrt{2}}a_1 + \frac{1}{\sqrt{2}}a_3 \quad \text{and} \quad x_3 = \mathbf{u}_3 \cdot \mathbf{a} = -a_2.$$

Therefore, the unique linear combination is given by

$$\mathbf{a} = \frac{1}{\sqrt{2}}(a_1 - a_3)\mathbf{u}_1 + \frac{1}{\sqrt{2}}(a_1 + a_3)\mathbf{u}_2 - a_2\mathbf{u}_3.$$

$\diamond$

Example 7. Show that the set $\{1, x, x^2, \dots, x^n\}$ is a basis for $\mathbb{P}_n(\mathbb{R})$. (This is called the **standard basis for** $\mathbb{P}_n(\mathbb{R})$.)

SOLUTION. As we have seen, $\{1, x, x^2, \dots, x^n\}$ is a spanning set for $\mathbb{P}_n(\mathbb{R})$. This set is also linearly independent because if

$$\lambda_1 1 + \lambda_2 x + \dots + \lambda_{n+1} x^n = 0$$

for all $x \in \mathbb{R}$ then $\lambda_1 = \lambda_2 = \dots = \lambda_{n+1} = 0$. This follows from the theorem that we proved in Chapter 3 which states that two polynomials agree for all values of $x \in \mathbb{R}$ if and only if their corresponding coefficients are equal. $\diamond$

6.6.2 Dimension

In this section we show that “dimension” can be defined for every vector space which is spanned by a finite set of vectors.

To do this we need two results.

Theorem 1. *The number of vectors in any **spanning** set for a vector space V is always greater than or equal to the number of vectors in any **linearly independent** set in V .*

[X] *Sketch Proof.* Suppose $I = \{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is a linearly independent set and $S = \{\mathbf{w}_1, \dots, \mathbf{w}_n\}$ is a spanning set. Then we can write

$$\begin{aligned} \mathbf{v}_1 &= a_{11} \mathbf{w}_1 + \dots + a_{1n} \mathbf{w}_n \\ &\vdots \\ \mathbf{v}_m &= a_{m1} \mathbf{w}_1 + \dots + a_{mn} \mathbf{w}_n. \end{aligned} \tag{1}$$

Suppose $m > n$ and consider the matrix $A = (a_{ij})$. A has more rows than columns, so if we reduce A to row-echelon form by the Gaussian elimination algorithm then we must end up with at least one row of zeros. Apart from a row swap to get it into the right position, this row of zeros will have been obtained from some row of our original matrix A , say row R_k , by subtracting from it multiples of other rows or linear combinations of other rows of A . Therefore there are scalars $\alpha_1, \dots, \alpha_{k-1}, \alpha_{k+1}, \dots, \alpha_m$ such that

$$R_k - (\alpha_1 R_1 + \dots + \alpha_{k-1} R_{k-1} + \alpha_{k+1} R_{k+1} + \dots + \alpha_m R_m) \tag{2}$$

is an all-zero row.

If we use the equations in (1) to express the vector

$$\mathbf{v}_k - (\alpha_1 \mathbf{v}_1 + \dots + \alpha_{k-1} \mathbf{v}_{k-1} + \alpha_{k+1} \mathbf{v}_{k+1} + \dots + \alpha_m \mathbf{v}_m) \tag{3}$$

as a linear combination of $\mathbf{w}_1, \dots, \mathbf{w}_n$ then the coefficient of each $\mathbf{w}_i$ will be the same as the i th entry in the row which is defined by (2). But we know that this row is all zeros, so the vector defined by (3) must be the zero vector. We now have a linear combination of $\mathbf{v}_1, \dots, \mathbf{v}_m$ which equals zero and at least one coefficient in this combination (the coefficient of $\mathbf{v}_k$) is non-zero. This is not compatible with the fact that the set $\{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is linearly independent. By assuming that m is greater than n we have reached a contradiction, therefore m must be less than or equal to n . $\square$

The second important theorem which guarantees the existence of a dimension for a vector space is as follows:

Theorem 2. *If a vector space V has a finite basis then every set of basis vectors for V contains the same number of vectors, that is, if $B_1 = \{\mathbf{u}_1, \dots, \mathbf{u}_m\}$ and $B_2 = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ are two bases for the same vector space V then $m = n$.*

Proof. Using the results of Theorem 1 and the fact that a basis is a linearly independent spanning set, we have

$m \geq n$, since B_1 spans V and B_2 is linearly independent, and

$n \geq m$, since B_2 spans V and B_1 is linearly independent.

Therefore $m = n$ and the proof is complete. $\square$

Since every basis for a vector space with a finite basis contains exactly the same number of vectors, the following definition makes sense for every vector space with a finite basis.

Definition 2. If V is a vector space with a finite basis, then the **dimension** of V , denoted by $\dim(V)$, is the number of vectors in any basis for V . Such a V is called a **finite dimensional vector space**.

Example 8. a) $\mathbb{R}^n$ has a basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ of n vectors, and hence $\dim(\mathbb{R}^n) = n$.

- b) The space of geometric vectors in ordinary physical space has a basis $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ of three vectors and therefore its dimension is 3.
- c) The subspace $\text{span}(S)$ in Example 3 has a basis of two vectors. The dimension of $\text{span}(S)$ is 2.
- d) We define the dimension of the vector space consisting only of the zero vector to be 0.
- e) The space $\mathbb{P}_n$ of polynomials of degree less than or equal to n has a basis $\{1, x, x^2, \dots, x^n\}$, so $\dim(\mathbb{P}_n) = n + 1$.

The following theorem summarises some useful results connecting spanning sets, linearly independent sets and dimension.

Theorem 3. Suppose that V is a finite dimensional vector space.

1. the number of vectors in any spanning set for V is greater than or equal to the dimension of V ;
2. the number of vectors in any linearly independent set in V is less than or equal to the dimension of V ;
3. if the number of vectors in a spanning set is equal to the dimension then the set is also a linearly independent set and hence a basis for V ;
4. if the number of vectors in a linearly independent set is equal to the dimension then the set is also a spanning set and hence a basis for V .

Proof. Assume that V is a vector space of dimension n .

The dimension of a vector space is equal to the number of vectors in a basis and a basis is a linearly independent spanning set. Therefore there is a linearly independent set in V which contains n vectors and there is also a spanning set for V which contains n vectors.

(1) and (2) follow from Theorem 1.

3. Assume that a spanning set S contains n vectors. Then, as no spanning set for V can contain fewer than n vectors, there is no proper subset of S which is a spanning set for V . Hence no proper subset of S has the same span as S . Thus, by Theorem 5 of Section 6.5, S is a linearly independent set, and is a basis for V .

4. Assume that $I = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a linearly independent set of n vectors in V and let $\mathbf{v}$ be any vector in V . If $\mathbf{v}$ does not belong to $\text{span}(I)$ then, by Theorem 6 of Section 6.5, the set $I \cup \{\mathbf{v}\}$ is linearly independent. This implies that V contains a linearly independent set with $n + 1 > \dim(V)$ vectors. This would contradict the result of (2), so we must have $\mathbf{v} \in \text{span}(I)$ for all $\mathbf{v} \in V$. Therefore I is a spanning set for V and hence a basis for V . $\square$

Some of the uses of Theorem 3 are illustrated in the next example.

Example 9. a) Obviously, the two vectors $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 5 \end{pmatrix}$ are non-parallel and so they are linearly independent. Hence, the set of two vectors $\left\{ \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \begin{pmatrix} 4 \\ 5 \end{pmatrix} \right\}$ is a basis for the two-dimensional space $\mathbb{R}^2$.

b) The set of 3 linearly independent vectors $\left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix} \right\}$ in Example 2 is a basis for the three-dimensional space $\mathbb{R}^3$.

c) A set of three vectors is not a spanning set for $\mathbb{R}^4$ as $\dim(\mathbb{R}^4) = 4 > 3$.

d) Any set of 10 vectors which spans $\mathbb{R}^{10}$ is a basis for $\mathbb{R}^{10}$ since $\dim(\mathbb{R}^{10}) = 10$.

e) Any linearly independent set of 325 vectors in $\mathbb{R}^{325}$ is a basis for $\mathbb{R}^{325}$ since $\dim(\mathbb{R}^{325}) = 325$.

f) A set of 1200 vectors in $\mathbb{R}^{1209}$ is not a spanning set as $\dim(\mathbb{R}^{1209}) = 1209 > 1200$. $\diamond$

Example 10. Show that the only subspaces in $\mathbb{R}^3$ are (1) the origin, (2) lines through the origin, (3) planes through the origin, and (4) $\mathbb{R}^3$ itself.

SOLUTION. By part 2 of Theorem 3, no subspace of $\mathbb{R}^3$ can have dimension greater than 3, otherwise a basis for the subspace would be a linearly independent set with more than three members. It follows that the only possible dimensions for subspaces (other than the subspace $\{\mathbf{0}\}$) are 1, 2 and 3.

A subspace of dimension 1 must be of the form $\text{span}(\mathbf{v})$, where $\mathbf{v}$ is non-zero. We know that this represents a line through the origin.

A subspace of dimension 2 must be of the form $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$, where the set $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly independent. We know that this represents a plane through the origin.

If a subspace is of dimension 3, it must be the whole of $\mathbb{R}^3$ because a basis for it will be a set of 3 linearly independent vectors in $\mathbb{R}^3$ and hence a basis for $\mathbb{R}^3$ itself. $\diamond$

6.6.3 Existence and construction of bases

In this section we examine the following two problems. Firstly, is it always possible to find a basis for a given vector space V ? Secondly, if we know that a basis does exist for a given vector space V , how can we find a basis for V ?

For the existence of a basis, we have seen in Example 3 that a finite set of vectors in $\mathbb{R}^3$ contains a subset which is a basis for $\text{span}(S)$. This is generally true for any vector space spanned by a finite set of vectors.

Theorem 4. *If S is a finite non-empty subset of a vector space then S contains a subset which is a **basis** for $\text{span}(S)$.*

*In particular, if V is any non-zero vector space which can be spanned by a **finite** set of vectors then V has a basis.*

Proof. Let S be a finite non-empty set of vectors. If S is linearly independent then it is a basis for $\text{span}(S)$ and there is nothing more to be done. If not, then (by Theorem 5 of Section 6.5) there must be a vector which can be dropped from S without changing the span. This gives a new set with fewer vectors which still spans $\text{span}(S)$. If this new set is linearly independent, we have a basis. If not, we can again remove a vector and get a smaller set which still spans the same subspace. If we continue in this way we must eventually get a set which is linearly independent and spans $\text{span}(S)$ (the process cannot continue indefinitely because the original set S had only a finite number, say n , of members and after n steps we would have no vectors left in the set).

A non-zero vector space V contains the zero vector, so it cannot be spanned by an empty set. Suppose that S is a non-empty finite spanning set for V . By the above result, there exists a subset of S which is a basis for $\text{span}(S) = V$. $\square$

This theorem shows that a spanning set can always be converted into a basis by removing some vectors from it. The next theorem proves a result about going in the opposite direction.

Theorem 5. *Suppose that V is a vector space which can be spanned by a finite set of vectors. If S is a linearly independent subset of V then there exists a basis for V which contains S as a subset. In other words, every linearly independent subset of V can be extended to a basis for V .*

Proof. Suppose S is a linearly independent set in V and that V can be spanned by a set of n vectors. If S spans V then there is nothing more to be done. If not, then there is a vector $\mathbf{v} \in V$ which is not in $\text{span}(S)$. If we add $\mathbf{v}$ to S then we get a new set $S \cup \{\mathbf{v}\}$ which is still linearly independent (by Theorem 6 of Section 6.5). If this new set spans V then we can stop. Otherwise, we can repeat the previous step and add another vector to get a larger linearly independent set. This process cannot continue beyond n steps, otherwise we would have a linearly independent set with more than n members (which is more than the number of members in a spanning set) and this would contradict Theorem 1. But the process only ends when we get a set which does span V . So eventually we must have a linearly independent spanning set or, in other words, a basis, $\square$

Note carefully that the last two theorems only apply to vector spaces which can be spanned by a finite set of vectors. Any vector space which cannot be spanned by any finite set of vectors is said to be an **infinite-dimensional** vector space. The vector space $\mathbb{P}$ of *all* polynomials is an example of an infinite-dimensional vector space (see Example 22 of Section 6.8).

In the proofs of the last two theorems we used step-by-step procedures to reduce a spanning set to a basis and to extend a linearly independent set to a basis. These procedures could be translated into algorithms for finding a basis but they would not be efficient because they involve re-testing (for linear independence or spanning) each time a vector is added or deleted. In practice, at least in $\mathbb{R}^m$, we can do all the adding or all the deleting at the same time. We form a suitable matrix, reduce it to echelon form and examine the echelon form to see which vectors we should add or delete. The details of the procedures are given in the next two theorems.

Theorem 6 (Reducing a spanning set to a basis in $\mathbb{R}^m$). *Suppose that $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is any subset of $\mathbb{R}^m$ and A is the matrix whose columns are the members of S . If U is a row-echelon form for A and S' is created from S by deleting those vectors which correspond to non-leading columns in U then S' is a basis for $\text{span}(S)$.*

Proof. Let U' be the matrix created by deleting any non-leading columns from U and let A' be created by deleting the same-numbered columns from A (so that the columns of A' are the members

of S'). The matrix U' has no non-leading columns, so the homogeneous system $A'\mathbf{y} = \mathbf{0}$ has no solutions other than the zero solution. This implies (by Proposition 1 of Section 6.5) that the set S' is linearly independent. In removing non-leading columns from U , we cannot create any new all-zero rows, so the system $A'\mathbf{y} = \mathbf{b}$ has a solution whenever $A\mathbf{x} = \mathbf{b}$ has a solution. This implies (by Proposition 3 of Section 6.4) that S' spans the same subspace as S . This completes the proof that S' is a basis for $\text{span}(S)$. $\square$

Example 11. Find a basis and the dimension for the subspace of $\mathbb{R}^4$ spanned by the set

$$S = \left\{ \begin{pmatrix} 1 \\ 1 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 4 \\ 5 \end{pmatrix}, \begin{pmatrix} -3 \\ 1 \\ -6 \\ -2 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 3 \\ 6 \end{pmatrix}, \begin{pmatrix} -2 \\ -1 \\ -4 \\ -3 \end{pmatrix} \right\}.$$

SOLUTION. As an exercise, show that the matrix A with the members of S as its columns can be reduced by elementary row operations to echelon form

$$U = \begin{pmatrix} 1 & 2 & -3 & 1 & -2 \\ 0 & 1 & 4 & 2 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

The third and fifth columns of U are non-leading, so we remove the third and fifth members from S and get

$$S' = \left\{ \begin{pmatrix} 1 \\ 1 \\ 2 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 4 \\ 5 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 3 \\ 6 \end{pmatrix} \right\}$$

as a basis for $\text{span}(S)$, which is 3 dimensional. $\diamond$

NOTE. Do not confuse the dimension of a subspace with the dimension of the vector space it lies in. In the above example, $\text{span}(S)$ is a 3 dimensional subspace of $\mathbb{R}^4$. This has nothing to do with $\mathbb{R}^3$.

Example 12. Show that the vectors $\mathbf{v}_1 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 2 \\ -1 \\ -2 \end{pmatrix}$, $\mathbf{v}_3 = \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix}$, $\mathbf{v}_4 = \begin{pmatrix} 5 \\ 4 \\ 2 \end{pmatrix}$ span $\mathbb{R}^3$

and find a basis for $\mathbb{R}^3$ which is a subset of S .

SOLUTION. Suppose that A is the matrix whose columns are the four given vectors. One way to show that S spans $\mathbb{R}^3$ is to reduce the augmented matrix $(A|\mathbf{b})$ to the echelon form as in Section 6.4.1. The augmented matrix involves a column of indeterminate $\mathbf{b}$.

By Theorem 6, we can first find a basis for $\text{span}(S)$ simply by reducing A to a row-echelon form. If the dimension of the span is 3 then the result follows.

$$\begin{pmatrix} 0 & 2 & 3 & 5 \\ 1 & -1 & 2 & 4 \\ 2 & -2 & 4 & 2 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & -1 & 2 & 4 \\ 0 & 2 & 3 & 5 \\ 2 & -2 & 4 & 2 \end{pmatrix} \xrightarrow{R_3 = R_3 - 2R_1} \begin{pmatrix} 1 & -1 & 2 & 4 \\ 0 & 2 & 3 & 5 \\ 0 & 0 & 0 & -6 \end{pmatrix}$$

The matrix U has one non-leading column, the third, so we delete the third member from the given set and find that the subset $B = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_4\}$ is a basis for $\text{span}(S)$. However B is a linearly independent set of 3 vectors in $\mathbb{R}^3$, so it is also a basis for $\mathbb{R}^3$. $\diamond$

Theorem 7 (Extending a linearly independent set to a basis in $\mathbb{R}^m$).

Suppose that $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a linearly independent subset of $\mathbb{R}^m$ and A is the matrix whose columns are the members of S followed by the members of the standard basis for $\mathbb{R}^m$. If U is a row-echelon form for A and S' is created by choosing those columns of A which correspond to leading columns in U then S' is a basis for $\mathbb{R}^m$ containing S as a subset.

Proof. Let S'' be the set of $n+m$ vectors from the columns of A . Since S'' includes all the standard basis vectors for $\mathbb{R}^m$, so $\mathbb{R}^m = \text{span}(S'')$. By Theorem 6, the set S' is a basis for $\mathbb{R}^m$.

To see that S' contains S we need to prove that the first n columns of U (which correspond to the members of S in A) are leading columns. Let B be the submatrix formed from the first n columns of A and P be the submatrix formed from the first n columns of U . Since A reduces to U , we also have B reduces to P . By Proposition 1 of Section 6.5, the linear independence of S implies $B\mathbf{x} = \mathbf{0}$ has unique solution $\mathbf{x} = \mathbf{0}$. Hence all columns of P , i.e. the first n columns of U , are leading. Hence the result follows. $\square$

Example 13. Find a basis for $\mathbb{R}^4$ containing the members of the linearly independent set

$$S = \left\{ \begin{pmatrix} 1 \\ 2 \\ 4 \\ -2 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \\ 10 \\ -5 \end{pmatrix} \right\}.$$

SOLUTION. We form the matrix A whose columns are the members of S followed by the standard basis vectors for $\mathbb{R}^4$ and reduce it to row-echelon form

$$\begin{aligned} A &= \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 2 & 5 & 0 & 1 & 0 & 0 \\ 4 & 10 & 0 & 0 & 1 & 0 \\ -2 & -5 & 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{\substack{R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 4R_1 \\ R_4 = R_4 + 2R_1}} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 & 0 \\ 0 & 2 & -4 & 0 & 1 & 0 \\ 0 & -1 & 2 & 0 & 0 & 1 \end{pmatrix} \\ &\xrightarrow{\substack{R_3 = R_3 - 2R_2 \\ R_4 = R_4 + R_2}} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 0 & -2 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \end{pmatrix} \xrightarrow{R_4 = R_4 + \frac{1}{2}R_3} \begin{pmatrix} 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 0 & -2 & 1 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2} & 1 \end{pmatrix} = U. \end{aligned}$$

The first, second, fourth and fifth columns are the leading columns in U , so we take the corresponding columns in A and get a basis

$$S' = \left\{ \begin{pmatrix} 1 \\ 2 \\ 4 \\ -2 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \\ 10 \\ -5 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

for $\mathbb{R}^4$. $\diamond$

NOTE. The procedure stated in the last theorem can also be used in situations where we have a set S which is neither linearly independent nor spanning and we want to find a basis which contains as many members of S as possible. We still form a matrix A from the members of S followed by the standard basis vectors, reduce to echelon form U and pick out from A the columns which correspond to leading columns in U . The only difference is that the first n columns are not necessarily all leading columns (because S is not necessarily linearly independent), so the new set S' does not necessarily include all the members of S .

Example 14. Find a basis for $\mathbb{R}^4$ containing as many as possible of the members of the set

$$S = \left\{ \begin{pmatrix} 1 \\ 2 \\ 4 \\ -2 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ -3 \\ 6 \end{pmatrix} \right\}.$$

SOLUTION. We form the matrix A whose columns are the members of S followed by the standard basis vectors for $\mathbb{R}^4$ and reduce it to row-echelon form.

$$\begin{pmatrix} 1 & 2 & 1 & 1 & 0 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 & 0 \\ 4 & 1 & -3 & 0 & 0 & 1 & 0 \\ -2 & 4 & 6 & 0 & 0 & 0 & 1 \end{pmatrix} \xrightarrow{\begin{matrix} R_2 = R_2 - 2R_1 \\ R_3 = R_3 - 4R_1 \\ R_4 = R_4 + 2R_1 \end{matrix}} \begin{pmatrix} 1 & 2 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 & 0 \\ 0 & -7 & -7 & -4 & 0 & 1 & 0 \\ 0 & 8 & 8 & 2 & 0 & 0 & 1 \end{pmatrix}$$

$$\xrightarrow{\begin{matrix} R_3 = R_3 + 7R_2 \\ R_4 = R_4 - 8R_2 \end{matrix}} \begin{pmatrix} 1 & 2 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 0 & -18 & 7 & 1 & 0 \\ 0 & 0 & 0 & 18 & -8 & 0 & 1 \end{pmatrix} \xrightarrow{R_4 = R_4 + R_3} \begin{pmatrix} 1 & 2 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 & 0 \\ 0 & 0 & 0 & -18 & 7 & 1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 1 & 1 \end{pmatrix}$$

The first, second, fourth and fifth columns are the leading columns in the row-echelon form matrix, so we take the corresponding columns in A and get a basis

$$S' = \left\{ \begin{pmatrix} 1 \\ 2 \\ 4 \\ -2 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \\ 1 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\}$$

for $\mathbb{R}^4$.

Note that this basis only contains two of the three vectors in the original set because S was not linearly independent (Question, what is $\dim(\text{span}(S))$?) $\diamond$

In Chapter 7 we shall need the following proposition which follows from Theorems 3 and 4.

Proposition 8. If V is a finite-dimensional space and W is a subspace of V and $\dim(W) = \dim(V)$ then $W = V$.

Proof. By Theorem 4, there exists B a basis for W . So B is a linearly independent set in V . By Theorem 3 part 4, B is also a basis for V and $V = \text{span}(B) = W$. $\square$

6.7 [X] Coordinate vectors

In Chapter 1, we have seen that the position vector $\mathbf{a}$ of a point in an n -dimensional space could be represented by a (column) coordinate vector $\begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in \mathbb{R}^n$, where the coordinates are the scalars in the linear combination which expresses $\mathbf{a}$ in terms of the standard basis vectors $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$.

As remarked in the last section, any basis B for a finite-dimensional vector space V over $\mathbb{F}$ has the property that — every vector $\mathbf{v} \in V$ can be written as a unique linear combination of B . If we specify a fixed ordering of the basis $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ then we can associate with every vector $\mathbf{v}$

the unique n -vector $\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{F}^n$, where $\mathbf{v} = x_1\mathbf{v}_1 + \dots + x_n\mathbf{v}_n$.

Note that the order of vectors in B is important because changing the order of the vectors also changes the order of the coefficients and therefore changes the n -vector corresponding to $\mathbf{v}$.

This generalise the notion of coordinates of a point in an n -dimensional real space to coordinates of a vector in any finite dimensional vector space. In this case, we can represent a vector in any vector space by a coordinate vector which is an n -vector in $\mathbb{F}^n$. Consequently, we can use all the techniques which we have learnt in the previous sections to study vector spaces by matrices over $\mathbb{F}$.

Now, we introduce the definition of coordinate vectors.

Definition 1. Let V be an n -dimensional vector space and let the ordered set of vectors $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a basis for V . If

$$\mathbf{v} = x_1\mathbf{v}_1 + \dots + x_n\mathbf{v}_n$$

then the vector

$$[\mathbf{v}]_B = \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

is called the **coordinate vector of $\mathbf{v}$ with respect to the ordered basis B** .

Example 1. With respect to the ordered basis $B = \left\{ \begin{pmatrix} 0 \\ 1 \\ 3 \\ -1 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \\ -3 \\ 1 \end{pmatrix}, \begin{pmatrix} 4 \\ -1 \\ 0 \\ 2 \end{pmatrix}, \begin{pmatrix} -6 \\ 2 \\ 1 \\ 4 \end{pmatrix} \right\}$ of $\mathbb{R}^4$,
a vector $\mathbf{v} \in \mathbb{R}^4$ has the coordinate vector $[\mathbf{v}]_B = \begin{pmatrix} 1 \\ -3 \\ 4 \\ 2 \end{pmatrix}$. Find $\mathbf{v}$.

SOLUTION.

$$\mathbf{v} = 1 \begin{pmatrix} 0 \\ 1 \\ 3 \\ -1 \end{pmatrix} - 3 \begin{pmatrix} 2 \\ 5 \\ -3 \\ 1 \end{pmatrix} + 4 \begin{pmatrix} 4 \\ -1 \\ 0 \\ 2 \end{pmatrix} + 2 \begin{pmatrix} -6 \\ 2 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} -2 \\ -14 \\ 14 \\ 12 \end{pmatrix}.$$

◇

Example 2. Find the coordinate vector for the vector $\mathbf{b} = \begin{pmatrix} -1 \\ 0 \\ 5 \end{pmatrix}$ with respect to the ordered basis $\left\{ \mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \right\}$ of $\text{span}(\mathbf{v}_1, \mathbf{v}_2)$.

SOLUTION. From the result of Example 4 in Section 6.6, the coordinate vector for $\mathbf{b}$ with respect to the ordered basis $\{\mathbf{v}_1, \mathbf{v}_2\}$ is $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$. ◇

Example 3. The set of vectors $B = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$, where

$$\mathbf{u}_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \quad \mathbf{u}_3 = \begin{pmatrix} 0 \\ -1 \\ 0 \end{pmatrix},$$

is an orthonormal basis for $\mathbb{R}^3$. Find the coordinate vector of $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ with respect to this basis.

SOLUTION. From the result of Example 6 in Section 6.6, the required coordinate vector is therefore

$$[\mathbf{a}]_B = \begin{pmatrix} \frac{1}{\sqrt{2}}(a_1 - a_3) \\ \frac{1}{\sqrt{2}}(a_1 + a_3) \\ -a_2 \end{pmatrix}.$$

◇

One of the most important aspects of coordinate vectors is that by using them we can reduce problems in any finite-dimensional real vector space to problems in $\mathbb{R}^n$, where of course we have powerful matrix techniques available.

Example 4. Find the coordinate vector of $p(x) = a_0 + a_1x + \cdots + a_nx^n \in \mathbb{P}_n(\mathbb{R})$ with respect to the ordered basis $B = \{1, x, x^2, \dots, x^n\}$.

SOLUTION. Of course there is no need to do any calculation in this case — we already know how to write p as a linear combination of elements of B . The coordinate vector of p with respect to B

$$\text{is } [p]_B = \begin{pmatrix} a_0 \\ a_1 \\ \vdots \\ a_n \end{pmatrix} \in \mathbb{R}^{n+1}.$$

◇

Things are more difficult if we are given a nonstandard basis for $\mathbb{P}_n(\mathbb{R})$.

Example 5. As a special case of Example 17 in 6.8.3, the set $\mathbb{P}_2(\mathbb{R})$ of all real polynomials which have degree two or less is a vector space. You are given an ordered basis $B = \{1 + x, x + x^2, 1 + x^2\}$ for $\mathbb{P}_2(\mathbb{R})$. Find the coordinate vector for $p(x) = 1 - x^2$ with respect to the ordered basis B .

SOLUTION. We need to find scalars $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$ such that

$$p(x) = 1 - x^2 = \alpha_1(1 + x) + \alpha_2(x + x^2) + \alpha_3(1 + x^2).$$

Expanding the right-hand-side and comparing coefficients shows that we must have

$$\begin{aligned}\alpha_1 + \alpha_3 &= 1 \\ \alpha_1 + \alpha_2 &= 0 \\ \alpha_2 + \alpha_3 &= -1.\end{aligned}$$

This reduces the problem to that of solving a system of linear equations in the unknowns α_1, α_2 and α_3 . Solving these equations by Gaussian elimination gives the unique solution $\alpha_1 = 1, \alpha_2 = -1, \alpha_3 = 0$. Therefore $[p]_B = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$. $\diamond$

Because coordinate vectors can be obtained in any finite-dimensional vector space, we can turn problems involving vectors in the vector space into problems involving coordinate vectors in $\mathbb{F}^n$. There are three important results which are fundamental to this approach.

Theorem 1. *If B is an ordered basis for a vector space V over a field $\mathbb{F}$ and $\mathbf{u}, \mathbf{v} \in V$ and $\lambda \in \mathbb{F}$, then*

- (a) $\mathbf{u} = \mathbf{v}$ if and only if $[\mathbf{u}]_B = [\mathbf{v}]_B$, that is, two vectors are equal if and only if the corresponding coordinate vectors are equal.
- (b) $[\mathbf{u} + \mathbf{v}]_B = [\mathbf{u}]_B + [\mathbf{v}]_B$, that is, the coordinate vector of the sum of two vectors is equal to the sum of the two corresponding coordinate vectors.
- (c) $[\lambda \mathbf{v}]_B = \lambda [\mathbf{v}]_B$, that is, the coordinate vector of a scalar multiple of a vector is equal to the same scalar multiple of the corresponding coordinate vector.

Proof. Equality. If $\mathbf{u}$ and $\mathbf{v}$ have the same coordinates then they are equal to the same linear combination of B and must therefore be equal to each other. Conversely, if $\mathbf{u} = \mathbf{v}$ then, because B is a basis and (by Theorem 2 of Section 6.5) no vector can have two different expressions as a linear combination of a given basis, any expressions for $\mathbf{u}$ and $\mathbf{v}$ as linear combinations of B must be the same. The coefficients in these linear combinations are the coordinates of $\mathbf{u}$ and $\mathbf{v}$, so $[\mathbf{u}]_B = [\mathbf{v}]_B$.

Addition. Let $B = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ and $[\mathbf{u}]_B = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix}$ and $[\mathbf{v}]_B = \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_n \end{pmatrix}$. By the definition of coordinate vectors, we have

$$\mathbf{u} = \lambda_1 \mathbf{b}_1 + \dots + \lambda_n \mathbf{b}_n$$

and

$$\mathbf{v} = \mu_1 \mathbf{b}_1 + \dots + \mu_n \mathbf{b}_n.$$

By adding these two equations we get

$$\mathbf{u} + \mathbf{v} = (\lambda_1 + \mu_1) \mathbf{b}_1 + \dots + (\lambda_n + \mu_n) \mathbf{b}_n$$

and this implies (by the definition of coordinate vectors) that

$$[\mathbf{u} + \mathbf{v}]_B = \begin{pmatrix} \lambda_1 + \mu_1 \\ \vdots \\ \lambda_n + \mu_n \end{pmatrix} = \begin{pmatrix} \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} + \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_n \end{pmatrix} = [\mathbf{u}]_B + [\mathbf{v}]_B.$$

Multiplication by a scalar. The proof is similar to that for addition, and is omitted. $\square$

6.8 [X] Further important examples of vector spaces

All of the material in this section is regarded as being more difficult than the material in previous sections of this chapter.

In the preceding sections we have developed a general theory of vector spaces. However, for simplicity, we have concentrated on examples which illustrate the applications of the theory to the particular vector space $\mathbb{R}^n$. In this section we shall examine three other important examples of vector spaces, namely the vector spaces of matrices, of real-valued functions, and of polynomials. We shall show how the vector space ideas of subspace, linear combination and span, linear independence, basis, dimension, and coordinate vector apply to these spaces.

Before the discussion of these vector spaces, we introduce an alternative way of proving a subset to be a subspace.

Theorem 1 (Alternative Subspace Theorem). *A subset S of a vector space V over a field $\mathbb{F}$ is a subspace of V if and only if S contains the zero vector and it satisfies the closure condition:*

$$\text{If } \mathbf{v}_1, \mathbf{v}_2 \in S, \text{ then } \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 \in S \text{ for all } \lambda_1, \lambda_2 \in \mathbb{F}. \quad (\#)$$

Proof. We prove that the closure condition (#) is satisfied if and only if both closure under addition and closure under scalar multiplication are also satisfied.

We first assume that (#) is satisfied. Then, for the special case of $\lambda_1 = \lambda_2 = 1$, condition (#) becomes

$$\mathbf{v}_1 + \mathbf{v}_2 \in S \quad \text{for all } \mathbf{v}_1, \mathbf{v}_2 \in S,$$

and hence S is closed under addition. Further, for the special case of $\lambda_2 = 0$, condition (#) becomes

$$\lambda_1 \mathbf{v}_1 \in S \quad \text{for all } \mathbf{v}_1 \in S \text{ and for all } \lambda_1 \in \mathbb{F},$$

and hence S is closed under multiplication by a scalar. Thus, if (#) is satisfied then closure under addition and closure under scalar multiplication are also satisfied.

We now assume that both closure conditions are satisfied. Then, from closure under multiplication by a scalar, we have, for all $\mathbf{v}_1, \mathbf{v}_2 \in S$ and for all $\lambda_1, \lambda_2 \in F$, that

$$\lambda_1 \mathbf{v}_1 \in S \quad \text{and} \quad \lambda_2 \mathbf{v}_2 \in S.$$

Then, on adding these scalar multiples and using closure under addition, we have that

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 \in S,$$

and hence (#) is satisfied.

Thus, (#) is satisfied if and only if closure under addition and under scalar multiplication are both satisfied. We then use the Subspace Theorem of Section 6.3 to complete the proof. $\square$

6.8.1 Vector spaces of matrices

We have seen in Example 3 in Section 6.1 that $M_{mn}(\mathbb{R})$ the set of all $m \times n$ real matrices is a vector space over $\mathbb{R}$ and $M_{mn}(\mathbb{C})$ the set of $m \times n$ complex matrices is a vector space over $\mathbb{C}$. In this section, we are going to see some examples of their subspaces, their bases and coordinate vectors with respect to these bases.

Example 1. Show that the set

$$S = \{A \in M_{22}(\mathbb{R}) : [A]_{11} = [A]_{22} = 1\}$$

is not a vector subspace of $M_{22}(\mathbb{R})$.

SOLUTION. Since the zero matrix is not in S , then vector space axiom 4 **Existence of Zero** is not satisfied. Therefore the set S is not a vector subspace. $\diamond$

Example 2. Prove that the set of $n \times n$ real symmetric matrices is a subspace of $M_{nn}(\mathbb{R})$.

SOLUTION. Recall that a square matrix A is called symmetric if $A = A^T$. Let S be the set of $n \times n$ symmetric matrices. Obvious, the zero matrix is symmetric, and so belongs to S .

Suppose that $A, B \in S$ and $\lambda, \mu \in \mathbb{R}$. By a property of transpose in Chapter 5 and the fact that A, B are symmetric, we have

$$(\lambda A + \mu B)^T = \lambda A^T + \mu B^T = \lambda A + \mu B.$$

Therefore, $\lambda A + \mu B$ is symmetric and so it is in S . Hence by the Alternative Subspace Theorem, S is a subspace. $\diamond$

Example 3. For $1 \leq i \leq m$, $1 \leq j \leq n$, let E_{ij} be the $m \times n$ matrix with all entries 0 except that the ij th entry is 1. Show that the set $B = \{E_{ij} : 1 \leq i \leq m, 1 \leq j \leq n\}$ is a basis for $M_{mn}(\mathbb{C})$.

SOLUTION. For any $A = (a_{ij}) \in M_{mn}(\mathbb{C})$, we have

$$A = a_{11}E_{11} + \cdots + a_{1n}E_{1n} + a_{21}E_{21} + \cdots + a_{mn}E_{mn} = \sum_{i=1}^m \sum_{j=1}^n a_{ij}E_{ij}.$$

Thus, S is a spanning set for $M_{mn}(\mathbb{C})$.

Suppose that $\sum_{i=1}^m \sum_{j=1}^n \lambda_{ij}E_{ij} = \mathbf{0}$, where $\mathbf{0}$ is the $m \times n$ zero matrix. Note that

$$\sum_{i=1}^m \sum_{j=1}^n \lambda_{ij}E_{ij} = \begin{pmatrix} \lambda_{11} & \cdots & \lambda_{1n} \\ \vdots & \ddots & \vdots \\ \lambda_{m1} & \cdots & \lambda_{mn} \end{pmatrix}$$

is the zero matrix. Hence $\lambda_{11} = \cdots = \lambda_{mn} = 0$ and so B is an independent set. Therefore B is a basis for $M_{mn}(\mathbb{C})$. $\diamond$

NOTE. In general, the set B is a basis called the **standard basis** for $M_{mn} = M_{mn}(\mathbb{F})$ for $\mathbb{F} = \mathbb{Q}, \mathbb{R}$ or $\mathbb{C}$. As a result the dimension of M_{mn} is mn .

Example 4. The coordinate vector of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with respect to the standard basis of

$$M_{22} \text{ is } \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix}.$$

We can also solve problems of independent sets, spanning sets and bases by Gaussian Elimination. Be careful not to mix up the augmented matrix formed from the linear combinations and the elements in M_{mn} .

Example 5. Show that set

$$\left\{ \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix} \right\}$$

is a basis for M_{22} .

SOLUTION. Note that the dimension of M_{22} is 4 and the number of vectors in the given set is also 4. To prove this set is a basis, by Theorem 3 part 4 in Section 6.6.2 we only need to show that this set is independent.

Suppose that

$$\lambda_1 \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} 1 & 1 \\ 0 & 2 \end{pmatrix} + \lambda_3 \begin{pmatrix} 1 & 0 \\ 1 & 2 \end{pmatrix} + \lambda_4 \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

By equating the corresponding entries in both sides, we have

$$\begin{array}{ccccccccc} \lambda_1 & + & \lambda_2 & + & \lambda_3 & + & 0 & = & 0 \\ \lambda_1 & + & \lambda_2 & + & 0 & + & \lambda_4 & = & 0 \\ \lambda_1 & + & 0 & + & \lambda_3 & + & \lambda_4 & = & 0 \\ 0 & + & 2\lambda_2 & + & 2\lambda_3 & + & 2\lambda_4 & = & 0 \end{array}$$

Since the right hand sides are all zeros, we can simply reduce the coefficient matrix to row-echelon form.

$$\begin{array}{c} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 2 & 2 \end{pmatrix} \xrightarrow[R_3 = R_3 - R_1]{R_2 = R_2 - R_1} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ 0 & -1 & 0 & 1 \\ 0 & 2 & 2 & 2 \end{pmatrix} \\ \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 2 & 2 & 2 \end{pmatrix} \xrightarrow{R_4 = R_4 + 2R_2 + 2R_3} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & -1 & 0 & 1 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 6 \end{pmatrix} \end{array}$$

Thus the system of equations has unique solution $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 0$. Hence the set of matrices is a basis. $\diamond$

NOTE. The four columns of the coefficient matrix are the coordinate vectors of the four matrices with respect to the standard basis.

6.8.2 Vector spaces associated with real-valued functions

Before reading this section, you might like to quickly read the brief review of function notation given in Appendix 6.9.

We know how to add two functions and how to multiply a function by a real number, so it is natural to ask whether or not these operations satisfy the axioms of a vector space. In this section we shall see that they do.

The Vector Space of Real-Valued Functions. Let X be a non-empty set. Consider the set (which we call $\mathcal{R}[X]$) of all possible real-valued functions with domain X , that is,

$$\mathcal{R}[X] = \{f : X \rightarrow \mathbb{R}\}.$$

We also let $+$ be the usual rule for adding real functions, i.e., $f + g$ is the function given by

$$(f + g)(x) = f(x) + g(x) \quad \text{for all } x \in X,$$

and we let $*$ represent the usual rule for multiplying a real function by a real number, i.e., $\lambda * f$ is the function given by

$$(\lambda * f)(x) = (\lambda f)(x) = \lambda f(x) \quad \text{for all } x \in X.$$

We then have the following result:

Proposition 2. The system $(\mathcal{R}[X], +, *, \mathbb{R})$ is a vector space over the real-number field $\mathbb{R}$.

Proof. The proof of this proposition follows the usual procedure of proving that all ten of the vector space axioms are satisfied. We give proofs of two of the axioms and leave the proofs of the other ones as exercises.

Closure under Addition. If $f, g \in \mathcal{R}[X]$, then $f(x)$ and $g(x)$ are defined and are real numbers for all $x \in X$. Then, using the usual rule for function addition, we have $(f + g)(x) = f(x) + g(x)$ for all $x \in X$.

Thus, $(f + g)(x)$ is also defined and is a real number for all $x \in X$. Hence $f + g : X \rightarrow \mathbb{R}$ and therefore $f + g \in \mathcal{R}[X]$.

Closure under Multiplication by Scalars. If $f \in \mathcal{R}[X]$ and $\lambda \in \mathbb{R}$, then, from the usual rule for multiplication of a function by a real number, we have

$$(\lambda f)(x) = \lambda f(x) \quad \text{for all } x \in X.$$

Thus $(\lambda f)(x)$ is defined and is a real number for all $x \in X$. Hence, $\lambda f : X \rightarrow \mathbb{R}$ and therefore $\lambda f \in \mathcal{R}[X]$. $\square$

In the next examples we briefly mention some subspaces of the vector space of real-valued functions. These subspaces are of importance in many areas of modern mathematics, science and engineering. The most important one, the subspace of polynomials will be discussed in the next subsection.

Example 6. Let (a, b) be an interval of $\mathbb{R}$, and let $C[(a, b)]$ be the set of all continuous real-valued functions on (a, b) . Show that $C[(a, b)]$ is a subspace of the vector space $\mathcal{R}[(a, b)]$ of all real-valued functions with domain (a, b) .

SOLUTION. We use the Alternative Subspace Theorem.

The set $C[(a, b)]$ contains the zero function, as the zero function is continuous and so is in this set.

From calculus, we know that if f and g are continuous on an interval, then $\lambda_1 f + \lambda_2 g$ is also continuous on the same interval for all real λ_1 and λ_2 . Also, $\lambda_1 f + \lambda_2 g$ is a real function, and hence $C[(a, b)]$ is a subset of $\mathcal{R}[(a, b)]$ which satisfies the condition (#) of the Alternative Subspace Theorem. Thus $C[(a, b)]$ is a subspace of $\mathcal{R}[(a, b)]$. $\diamond$

Calculus provides a very rich source of subspaces of the vector space of real-valued functions.

Example 7. Let $C^{(1)}[(a, b)]$ be the set of all real-valued functions whose first derivative exists and is continuous on an interval (a, b) of $\mathbb{R}$. Show that $C^{(1)}[(a, b)]$ is a subspace of the vector space $\mathcal{R}[(a, b)]$.

SOLUTION. We use the Alternative Subspace Theorem.

The set $C^{(1)}[(a, b)]$ contains the zero function, as the zero function has a continuous first derivative and so is in this set.

From calculus, we know that if the first derivatives of the real functions f and g exist and are continuous on an interval, then, for all $\lambda_1, \lambda_2 \in \mathbb{R}$, the function $\lambda_1 f + \lambda_2 g$ is also a real-valued function whose first derivative exists and is continuous on the same interval. Thus, $C^{(1)}[(a, b)]$ is a subset of $\mathcal{R}[(a, b)]$ which satisfies condition (#) of the Alternative Subspace Theorem, and hence it is a subspace of $\mathcal{R}[(a, b)]$.

Note that $C^{(1)}[(a, b)]$ is also a subspace of the vector space $C[(a, b)]$ of real-valued, continuous functions on (a, b) given in Example 6. Can you see why? $\diamond$

An important class of function subspace is defined by the solutions of homogeneous, linear differential equations.

Example 8. Let S be the subset of the vector space $\mathcal{R}[\mathbb{R}]$ of real-valued functions on $\mathbb{R}$ defined by

$$S = \left\{ f \in \mathcal{R}[\mathbb{R}] : \frac{d^2 f}{dx^2} - 6 \frac{df}{dx} + 5f = 0 \right\}.$$

Show that S is a subspace of $\mathcal{R}[\mathbb{R}]$.

SOLUTION. As the zero function satisfies the equation, so it belongs to S . For $f_1, f_2 \in S$ and $\lambda_1, \lambda_2 \in \mathbb{R}$, we have, on using the properties of derivatives, that

$$\begin{aligned} & \frac{d^2}{dx^2}(\lambda_1 f_1 + \lambda_2 f_2) - 6 \frac{d}{dx}(\lambda_1 f_1 + \lambda_2 f_2) + 5(\lambda_1 f_1 + \lambda_2 f_2) \\ &= \lambda_1 \left(\frac{d^2 f_1}{dx^2} - 6 \frac{df_1}{dx} + 5f_1 \right) + \lambda_2 \left(\frac{d^2 f_2}{dx^2} - 6 \frac{df_2}{dx} + 5f_2 \right) \\ &= \lambda_1 0 + \lambda_2 0 = 0. \end{aligned}$$

Hence, $\lambda_1 f_1 + \lambda_2 f_2 \in S$, and therefore S is a subspace. $\diamond$

Subspaces can also be defined by integration.

Example 9. Let S be the subset of the vector space $\mathcal{C}[-\pi, \pi]$ of all real-valued continuous functions on the interval $[-\pi, \pi]$ defined by

$$S = \left\{ f \in \mathcal{C}[-\pi, \pi] : \int_{-\pi}^{\pi} f(x)g(x) dx = 0 \right\},$$

where g is a fixed continuous function. Clearly the product $f(x)g(x)$ is integrable on $[-\pi, \pi]$, since f and g are continuous. Prove that S is a subspace of $\mathcal{C}[-\pi, \pi]$.

SOLUTION. The zero function is in S . For all $f_1, f_2 \in S$ and for all $\lambda_1, \lambda_2 \in \mathbb{R}$, we have $\lambda_1 f_1 + \lambda_2 f_2 \in \mathcal{C}[-\pi, \pi]$ and

$$\begin{aligned} \int_{-\pi}^{\pi} (\lambda_1 f_1(x) + \lambda_2 f_2(x))g(x) dx &= \lambda_1 \int_{-\pi}^{\pi} f_1(x)g(x) dx + \lambda_2 \int_{-\pi}^{\pi} f_2(x)g(x) dx \\ &= \lambda_1 0 + \lambda_2 0 = 0. \end{aligned}$$

Hence $\lambda_1 f_1 + \lambda_2 f_2 \in S$, and thus S is a subspace of $\mathcal{C}[-\pi, \pi]$. $\diamond$

Example 10. As shown in courses on differential equations, the homogeneous, linear differential equation in Example 8 has the solution

$$f(x) = \lambda_1 e^{5x} + \lambda_2 e^x \quad \text{for } \lambda_1, \lambda_2 \in \mathbb{R}.$$

Hence the set S of solutions is given by $S = \text{span}(e^{5x}, e^x)$, and thus $\{e^{5x}, e^x\}$ is a spanning set for S . $\diamond$

Example 11. Show that the set $S = \{\sin(x), \cos(x)\}$ is a linearly independent subset of the vector space $\mathcal{R}[-\pi, \pi]$ of all real-valued functions on the interval $[-\pi, \pi]$.

SOLUTION. We have to show that if

$$f(x) = \lambda_1 \sin(x) + \lambda_2 \cos(x) = 0 \quad \text{for all } x \in [-\pi, \pi]$$

then λ_1 and λ_2 are zero.

We first note that if the linear combination $f(x)$ is zero for all $x \in [-\pi, \pi]$ then $f(x)$ must also be zero for the values $x = 0$ and $x = \frac{\pi}{2}$. We then obtain

$$0 = f(0) = \lambda_2 \quad \text{and} \quad 0 = f\left(\frac{\pi}{2}\right) = \lambda_1.$$

Thus the scalars are zero, and hence the set is linearly independent. $\diamond$

Example 12. Show that the subset S_n of the vector space $\mathcal{R}[\mathbb{R}]$ defined by

$$S_n = \{\sin(kx) : k = 1, \dots, n \text{ and } x \in \mathbb{R}\}$$

is a linearly independent set.

SOLUTION. We will use a trick which is based on an extension to the vector space of real functions of the idea of orthogonality of geometric vectors and vectors in $\mathbb{R}^n$ (see Example 5 and 6 of Section 6.6).

We first prove that if k and m are integers then

$$\int_0^\pi \sin(kx) \sin(mx) dx = \begin{cases} 0 & \text{for } k \neq m \\ \frac{\pi}{2} & \text{for } k = m. \end{cases}$$

From trigonometry, we have

$$\sin(kx) \sin(mx) = \frac{1}{2} (\cos(k-m)x - \cos(k+m)x).$$

Then, for $k \neq m$,

$$\begin{aligned} \int_0^\pi \sin(kx) \sin(mx) dx &= \frac{1}{2} \int_0^\pi \cos(k-m)x dx - \frac{1}{2} \int_0^\pi \cos(k+m)x dx \\ &= \left[\frac{1}{2(k-m)} \sin(k-m)x \right]_0^\pi - \left[\frac{1}{2(k+m)} \sin(k+m)x \right]_0^\pi \\ &= 0 \end{aligned}$$

as $\sin(0) = 0$, $\sin(k-m)\pi = 0$, and $\sin(k+m)\pi = 0$ for integers k and m . We leave the proof of the result for $k = m$ as a simple exercise.

Now suppose that

$$\sum_{k=1}^n \lambda_k \sin(kx) = 0.$$

On multiplying this expression by $\sin(mx)$ and integrating from 0 to π , we have

$$0 = \sum_{k=1}^n \lambda_k \int_0^\pi \sin(kx) \sin(mx) dx = \lambda_m \frac{\pi}{2}$$

Hence, $\lambda_m = 0$ for $1 \leq m \leq n$, and thus the set S_n is linearly independent. $\diamond$

Example 13. Use the previous example, we can show that the vector space $\mathcal{R}[\mathbb{R}]$ cannot be spanned by a finite set and hence it is an infinite dimensional vector space. If it can be spanned by a finite set of m elements, by Theorem 1 in Section 6.6, all independent set has at most m elements. We can choose an integer n such that $n > m$, by the previous example, S_n will be a linearly independent set of more than m elements. Hence $\mathcal{R}[\mathbb{R}]$ cannot be spanned by a finite set. $\diamond$

6.8.3 Vector spaces associated with polynomials

Polynomials can be added, subtracted and multiplied. From the point of view of vector spaces, only addition (and subtraction) and multiplication of a polynomial by a scalar are relevant.

We will be concerned with polynomials defined over either the real or complex fields. Although it is possible to generalise all of the results in this section to the rational field $\mathbb{Q}$ and to also generalise many of the results to the finite fields $\mathbb{Z}_p$, (p a prime), we will not do so here. In the following, therefore, the field $\mathbb{F}$ should be taken as either the real numbers $\mathbb{R}$ or the complex numbers $\mathbb{C}$.

We begin by quickly reviewing the definitions of polynomial function, polynomial addition, multiplication of a polynomial by a scalar, and equality of polynomials.

Definition 1. A function $p : \mathbb{F} \rightarrow \mathbb{F}$ is called a **polynomial function** over $\mathbb{F}$ if there is a natural number $n \in \mathbb{N}$ and numbers $a_0, a_1, \dots, a_n \in \mathbb{F}$ such that

$$p(z) = a_0 + a_1z + \dots + a_nz^n = \sum_{k=0}^n a_k z^k \text{ for all } z \in \mathbb{F}.$$

For brevity we will usually refer to a polynomial function as a polynomial even though it is important in advanced mathematics courses to distinguish between the two.

Polynomials may be added and multiplied by scalars in such a way as to produce other polynomials. The formal definitions follow the usual definitions of addition and multiplication by a scalar for functions (see Appendix 6.9).

Definition 2. If p and q are polynomials over the same field $\mathbb{F}$ given by

$$p(z) = \sum_{k=0}^n a_k z^k \quad \text{and} \quad q(z) = \sum_{k=0}^m b_k z^k \quad \text{for all } z \in \mathbb{F},$$

then the **sum** polynomial is the polynomial $p + q$ given by

$$(p + q)(z) = p(z) + q(z) = \sum_{k=0}^{\max(n,m)} (a_k + b_k) z^k \quad \text{for all } z \in \mathbb{F}.$$

That is, the rule is to add corresponding coefficients. The rule for subtraction of polynomials follows immediately from the addition rule, and it is to subtract corresponding coefficients.

Definition 3. If $\lambda \in \mathbb{F}$ and p is a polynomial over $\mathbb{F}$ given by

$$p(z) = \sum_{k=0}^n a_k z^k \text{ for all } z \in \mathbb{F}$$

then the **scalar multiple** λp of p is the polynomial given by

$$(\lambda p)(z) = \lambda(p(z)) = \sum_{k=0}^n (\lambda a_k) z^k \text{ for all } z \in \mathbb{F}.$$

That is, the rule is to multiply each coefficient by the scalar.

The last main property of polynomials that we need is given by the following Uniqueness Proposition (see Chapter 3 for a proof for complex polynomials).

Proposition 3 (Uniqueness Proposition for Real and Complex Polynomials). Let p and q be polynomials over $\mathbb{F}$ given by

$$p(z) = \sum_{k=0}^n a_k z^k \quad \text{and} \quad q(z) = \sum_{k=0}^n b_k z^k \quad \text{for all } z \in \mathbb{F}.$$

Then, if the field $\mathbb{F}$ is either $\mathbb{R}$ or $\mathbb{C}$, we have that $p(z) = q(z)$ for all $z \in \mathbb{F}$ if and only if $a_k = b_k$ for all $k = 0, 1, 2, \dots, n$.

An immediate consequence of Proposition 3 is the following important result.

Corollary 4. *If $\mathbb{F}$ is the field $\mathbb{R}$ or $\mathbb{C}$, then a polynomial p over $\mathbb{F}$ has the property that $p(z) = 0$ for all $z \in \mathbb{F}$ if and only if all of its coefficients are zero.*

NOTE. that this is not true for other fields such as $\mathbb{Z}_p$, p a prime.

The unique polynomial, whose function values are all zero and which has all of its coefficients equal to zero, is called the **zero polynomial**.

The fundamental vector space associated with polynomials is defined in the following example.

Example 14 (The Vector Space of Polynomials over $\mathbb{F}$). The vector space of polynomials over a field $\mathbb{F}$ is the system $(\mathbb{P}(\mathbb{F}), +, *, \mathbb{F})$ defined as follows. The set of “vectors” is the set $\mathbb{P}(\mathbb{F})$ of all possible polynomials over the field $\mathbb{F}$, i.e.,

$$\mathbb{P}(\mathbb{F}) = \{p : p(z) = a_0 + a_1z + \cdots + a_nz^n \text{ for } n \in \mathbb{N}, a_j \in \mathbb{F}, 1 \leq j \leq n, z \in \mathbb{F}\}.$$

The rule of “vector addition” is the polynomial addition rule given in Definition 2 and the rule of multiplication is the scalar multiplication rule for polynomials given in Definition 3.

To prove that this system is a vector space it is necessary to check that all ten of the vector space axioms are satisfied. We leave the checking of the axioms as an exercise. $\diamond$

Example 15. As special cases of Example 14, we have the vector space of real polynomials $(\mathbb{P}(\mathbb{R}), +, *, \mathbb{R})$ (we have seen this in Example 4 in Section 6.1) and the vector space of complex polynomials $(\mathbb{P}(\mathbb{C}), +, *, \mathbb{C})$. $\diamond$

Notation. We will usually talk of the polynomial vector space $\mathbb{P}$ instead of the more formal $(\mathbb{P}(\mathbb{F}), +, *, \mathbb{F})$ when there can be no possibility of confusion over the field ($\mathbb{R}$ or $\mathbb{C}$) being used. When necessary, the vector space of real polynomials will be referred to as $\mathbb{P}(\mathbb{R})$ and the vector space of complex polynomials as $\mathbb{P}(\mathbb{C})$.

As shown in the following example, the field for the polynomials and the field for the scalars must be, in some sense, compatible.

Example 16. Show that the system $(\mathbb{P}(\mathbb{C}), +, *, \mathbb{R})$ of complex polynomials and real scalars is a vector space, whereas the system $(\mathbb{P}(\mathbb{R}), +, *, \mathbb{C})$ of real polynomials and complex scalars is not a vector space.

SOLUTION. It can be checked without too much difficulty that both systems satisfy nine of the ten vector space axioms. The axioms satisfied by both are the five axioms of addition, the three scalar-multiplication axioms of associativity, commutativity, multiplication by 1, and the scalar and vector distributive axioms.

The remaining axiom to check is closure under scalar multiplication.

For the system of complex polynomials and real scalars, we have that if $p \in \mathbb{P}(\mathbb{C})$ is a complex polynomial and $\lambda \in \mathbb{R}$, then λp is also a complex polynomial. The closure under scalar multiplication axiom is therefore satisfied, and hence the system of complex polynomials and real scalars is a vector space.

For the system of real polynomials and complex scalars, we note that $x \in \mathbb{P}(\mathbb{R})$, $i \in \mathbb{C}$, $ix \notin \mathbb{P}(\mathbb{R})$. The closure under scalar multiplication axiom is therefore not satisfied, and hence the system of real polynomials with complex scalars is not a vector space. $\diamond$

We now consider some of the subspaces of the polynomial vector space $\mathbb{P} = \mathbb{P}(\mathbb{F})$. The most important subspaces of $\mathbb{P}$ are the vector spaces of polynomials of degree less than or equal to n , for some n .

Example 17. Let $\mathbb{P}$ be the vector space of polynomials over $\mathbb{F}$, and let $\mathbb{P}_n$ be the subset of $\mathbb{P}$ consisting of all polynomials of degree less than or equal to some fixed integer $n > 0$, that is,

$$\mathbb{P}_n = \{p \in \mathbb{P} : \deg(p) \leq n\}.$$

Show that $\mathbb{P}_n$ is a subspace of $\mathbb{P}$.

SOLUTION. If $p, q \in \mathbb{P}_n$, then there exist coefficients $\{a_0, \dots, a_n\}$ and $\{b_0, \dots, b_n\}$ such that

$$\begin{aligned} p(z) &= a_0 + a_1z + \dots + a_nz^n \\ q(z) &= b_0 + b_1z + \dots + b_nz^n \end{aligned}$$

for all $z \in \mathbb{F}$. Then, for $\lambda p + \mu q$ with $\lambda, \mu \in \mathbb{F}$, we have

$$(\lambda p + \mu q)(z) = \lambda p(z) + \mu q(z) = (\lambda a_0 + \mu b_0) + \dots + (\lambda a_n + \mu b_n)z^n \quad \text{for all } z \in \mathbb{F}.$$

But the coefficients $\lambda a_j + \mu b_j$, $1 \leq j \leq n$, are also scalars in $\mathbb{F}$, and hence $\lambda p + \mu q \in \mathbb{P}_n$. Thus the condition in the Alternative Subspace Theorem is satisfied, and $\mathbb{P}_n$ is a subspace of $\mathbb{P}$. $\diamond$

Note that the subset of all polynomials of degree **exactly** n is not a subspace, since this subset does not contain the zero polynomial.

Subspaces of the space of all polynomials can also be formed by selecting all polynomials which have their roots at given points.

Example 18. Let $\mathbb{P}_n$ be the vector space of polynomials of degree less than or equal to n over $\mathbb{F}$. Show that the subset S of $\mathbb{P}_n$ given by

$$S = \{p \in \mathbb{P}_n : p(5) = \alpha\}$$

is a subspace of $\mathbb{P}_n$ if and only if $\alpha = 0$.

SOLUTION. We use the Alternative Subspace Theorem.

If $\alpha \neq 0$, then the zero polynomial is not in S and so S is not a subspace.

If $\alpha = 0$, then the zero polynomial is in S and so S is not empty. If $p, q \in S$ and $\lambda_1, \lambda_2 \in \mathbb{F}$, then $p(5) = 0$ and $q(5) = 0$ and

$$(\lambda_1 p + \lambda_2 q)(5) = \lambda_1 p(5) + \lambda_2 q(5) = 0.$$

Hence $\lambda_1 p + \lambda_2 q \in S$, and S is a subspace.

Therefore, S is a subspace of $\mathbb{P}_n$ if and only if $\alpha = 0$.

Note that this subspace is the set of all polynomials of degree less than or equal to n over $\mathbb{F}$ which have a root at $z = 5$. $\diamond$

As the ideas of linear combination, span and linear independence apply to all vector spaces, they apply to spaces of polynomials. The methods used in Sections 6.4 and 6.5 for $\mathbb{R}^n$ can be used to solve problems of spanning sets and independent sets.

Example 19. Does the complex polynomial p belong to $\text{span}(p_1, p_2)$, where p, p_1, p_2 are complex polynomials defined by

$$p(z) = 4 + z + 2z^2, \quad p_1(z) = 1 + z - z^2 \quad \text{and} \quad p_2(z) = 2 - z \quad \text{for all } z \in \mathbb{C}?$$

SOLUTION. $p \in \text{span}(p_1, p_2)$ if and only if p is a linear combination of p_1 and p_2 , that is, if and only if there are scalars x_1, x_2 such that

$$p(z) = x_1 p_1(z) + x_2 p_2(z) \quad \text{for all } z \in \mathbb{C}.$$

That is,

$$\begin{aligned} 4 + z + 2z^2 &= x_1(1 + z - z^2) + x_2(2 - z) \\ &= (x_1 + 2x_2) + (x_1 - x_2)z + (-x_1)z^2 \quad \text{for all } z \in \mathbb{C}. \end{aligned}$$

From the Uniqueness Proposition for Polynomials (Section 6.8.3), we know that polynomials are equal if and only if coefficients of corresponding powers of z are equal. Equating coefficients of equal powers then gives the system of linear equations

$$x_1 + 2x_2 = 4, \quad x_1 - x_2 = 1, \quad -x_1 = 2,$$

with augmented matrix

$$(A|\mathbf{b}) = \left(\begin{array}{cc|c} 1 & 2 & 4 \\ 1 & -1 & 1 \\ -1 & 0 & 2 \end{array} \right).$$

Then, p is in the span of p_1 and p_2 if and only if these equations have a solution. It is easy to see that these equations have no solution, and hence p is not in $\text{span}(p_1, p_2)$. $\diamond$

Example 20. Is the set $\{p_1, p_2, p_3\}$ of polynomials, where

$$p_1(z) = 1 + 2z - z^2; \quad p_2(z) = -3 - z + 2z^2; \quad p_3(z) = 2 + 3z + z^2,$$

a linearly independent subset of $\mathbb{P}_2$?

SOLUTION. Following the usual test for linear independence, we look for scalars x_1, x_2, x_3 such that $x_1 p_1 + x_2 p_2 + x_3 p_3 = 0$, that is, such that

$$x_1(1 + 2z - z^2) + x_2(-3 - z + 2z^2) + x_3(2 + 3z + z^2) = 0 \quad \text{for all } z \in \mathbb{C}.$$

Thus, the polynomial on the left is the zero polynomial, and hence the coefficient of each power of z is zero, that is,

$$x_1 - 3x_2 + 2x_3 = 0; \quad 2x_1 - x_2 + 3x_3 = 0; \quad -x_1 + 2x_2 + x_3 = 0.$$

This system of equations corresponds to the homogeneous system $A\mathbf{x} = \mathbf{0}$, where the matrix A and an equivalent row-echelon form U are

$$A = \begin{pmatrix} 1 & -3 & 2 \\ 2 & -1 & 3 \\ -1 & 2 & 1 \end{pmatrix}; \quad U = \begin{pmatrix} 1 & -3 & 2 \\ 0 & 5 & -1 \\ 0 & 0 & \frac{14}{5} \end{pmatrix}.$$

Then, as all columns of U are leading columns, the only solution for the scalars is $x_1 = x_2 = x_3 = 0$, and hence the set is linearly independent. $\diamond$

As we can discuss spans and independence of sets of polynomials, we also have the notion of dimensions of subspaces of polynomials. In the next example, we construct a standard basis for $\mathbb{P}_n$.

Example 21. Show that the set $\{1, z, \dots, z^n\}$ is a basis for $\mathbb{P}_n(\mathbb{F})$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$.

SOLUTION. A polynomial p of degree less than or equal to n over a field $\mathbb{F}$ is a function of the form

$$p(z) = a_0 + a_1z + \dots + a_nz^n \quad \text{with } a_j \in \mathbb{F} \text{ for } 0 \leq j \leq n \quad \text{and for } z \in \mathbb{F},$$

where any or all of the coefficients may be zero. Hence $\text{span}(1, z, \dots, z^n) = \mathbb{P}_n(\mathbb{F})$.

Furthermore, if

$$a_0 + a_1z + \dots + a_nz^n = 0 \quad \text{for all } z \in \mathbb{C}.$$

This linear combination is the zero polynomial, and hence, from the Uniqueness Proposition for Polynomials of Section 6.8.3, all of the coefficients are zero. Hence $\{1, z, \dots, z^n\}$ is independent.

Therefore, this set is a basis for $\mathbb{P}_n(\mathbb{F})$. $\diamond$

An important result which follows immediately from Example 21 is the following.

Proposition 5. The vector space $\mathbb{P}_n$ of polynomials of degree less than or equal to n has dimension $n + 1$.

As a consequence, $\mathbb{P}$ is not a finite dimensional space.

Example 22. The vector space $\mathbb{P}$ of all polynomials cannot be spanned by a finite set.

SOLUTION. We can use the same argument used for $\mathcal{R}[\mathbb{R}]$ in Example 13. However, we now use another approach.

Assume the contrary that some set S containing a finite number of polynomials is a spanning set for $\mathbb{P}$. Then, since the number of polynomials in S is finite, there must be a highest-degree polynomial in S . Let N be the degree of this polynomial. Then, no polynomial p with $\deg(p) > N$ is in $\text{span}(S)$. Hence, $\mathbb{P}$ is not spanned by any finite set of polynomials. $\diamond$

Example 23. Show that the set $S = \{2 + z, -1 + z^2, z - z^2\}$ is a basis for $\mathbb{P}_2$.

SOLUTION. If $p \in \mathbb{P}_2$, then p is given by

$$p(z) = a_0 + a_1z + a_2z^2 \quad \text{for all } z \in \mathbb{C}.$$

Then $p \in \text{span}(S)$ if there exist scalars x_1, x_2, x_3 such that

$$a_0 + a_1z + a_2z^2 = x_1(2 + z) + x_2(-1 + z^2) + x_3(z - z^2) \quad \text{for all } z \in \mathbb{C}.$$

On equating coefficients of powers of z , we obtain the system of linear equations with augmented matrix

$$(A|\mathbf{b}) = \left(\begin{array}{ccc|c} 2 & -1 & 0 & a_0 \\ 1 & 0 & 1 & a_1 \\ 0 & 1 & -1 & a_2 \end{array} \right)$$

On using Gaussian elimination, we obtain the augmented matrix

$$(U|\mathbf{y}) = \left(\begin{array}{ccc|c} 2 & -1 & 0 & a_0 \\ 0 & \frac{1}{2} & 1 & -\frac{1}{2}a_0 + a_1 \\ 0 & 0 & -3 & a_0 - 2a_1 + a_2 \end{array} \right).$$

Then, as there are no zero rows in U , there is a solution for all right hand sides. Thus, for every polynomial $p \in \mathbb{P}_2$, we have $p \in \text{span}(S)$, and hence S is a spanning set for $\mathbb{P}_2$. Further, as U has no non-leading columns, the only solution for a zero right hand side is $x_1 = x_2 = x_3 = 0$, and hence S is linearly independent. S is therefore a basis for $\mathbb{P}_2$. $\diamond$

The coordinate-vector idea applies immediately to the finite-dimensional vector space $\mathbb{P}_n$.

Example 24. For the standard basis of $\mathbb{P}_n$ consisting of powers of z , that is, $\{1, z, \dots, z^n\}$, the coordinate vector consists of the coefficients of the polynomial. For example, the polynomial $p \in \mathbb{P}_n$ defined by

$$p(z) = a_0 + a_1z + \dots + a_nz^n$$

has the coordinate vector $(a_0 \ a_1 \ \dots \ a_n)^T$ with respect to the standard basis.

NOTE. The order is important. For example, the coordinate vector of p with respect to the ordered basis $\{z^2, z, 1, z^3, \dots, z^n\}$ would be $(a_2 \ a_1 \ a_0 \ a_3 \ \dots \ a_n)^T$. $\diamond$

Example 25. Find the coordinate vector for the polynomial $p_3(z) = -1 + 5z^2$ with respect to the ordered basis $\{p_1, p_2\}$ of $\text{span}(p_1, p_2)$, where $p_1(z) = 1 + 2z + 3z^2$ and $p_2 = 1 + z - z^2$.

SOLUTION. We must find the scalars in the expression for p_3 as a linear combination of p_1 and p_2 . On writing $p_3 = x_1p_1 + x_2p_2$ and equating coefficients of equal powers of z , we get the system of equations with augmented matrix

$$(A|\mathbf{b}) = \left(\begin{array}{cc|c} 1 & 1 & -1 \\ 2 & 1 & 0 \\ 3 & -1 & 5 \end{array} \right).$$

The solution is $x_1 = 1$, $x_2 = -2$. Hence the coordinate vector of p_3 with respect to the ordered basis $\{p_1, p_2\}$ is $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$, i.e.,

$$p_3(z) = -1 + 5z^2 = 1p_1(z) - 2p_2(z) = 1(1 + 2z + 3z^2) - 2(1 + z - z^2).$$

$\diamond$

Example 26. A polynomial p has a coordinate vector $\begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix}$ with respect to the ordered basis $\{p_1 = 1, p_2 = 1 + z, p_3 = 2 - z + z^2\}$ of $\mathbb{P}_2$. Find p .

SOLUTION.

$$p = 2p_1 - p_2 + 4p_3 = 9 - 5z + 4z^2.$$

$\diamond$

6.9 A brief review of set and function notation

6.9.1 Set notation.

A set is any collection of elements. Sets are usually defined either by listing their elements or by giving a rule for selection of the elements. The elements of a set are usually enclosed in braces $\{\}$.

Example 1. $S = \{1, 4, -7\}$ is the set whose elements are 1, 4, and -7 . $\diamond$

Common notation for a set defined by a rule is shown in the following example.

Example 2. The notation

$$S = \{\mathbf{x} \in \mathbb{R}^n : x_1 \geq 0, x_3 \leq 4\}$$

is read as: the set S of vectors $\mathbf{x}$ in $\mathbb{R}^n$ such that x_1 is greater than or equal to zero and x_3 is less than or equal to 4. Note that the colon ($:$) is read as “such that” and the comma ($,$) is read as “and”. $\diamond$

Definition 1. Two sets A and B are **equal** (notation $A = B$) if every element of A is an element of B , and if every element of B is an element of A .

To prove that $A = B$ it is necessary to prove that the two conditions:

1. if $x \in A$ then $x \in B$, and
2. if $x \in B$ then $x \in A$

are both satisfied.

Definition 2. A set A is a **subset** of another set B (notation $A \subseteq B$) if every element of A is also an element of B .

To prove that $A \subseteq B$ it is necessary to prove that the condition:

if $x \in A$ then $x \in B$

is satisfied.

Definition 3. A is said to be a **proper subset** of B if A is a subset of B and at least one element of B is not an element of A .

To prove that A is a proper subset of B it is necessary to prove that the two conditions:

1. if $x \in A$ then $x \in B$, and
2. for some $x \in B$, x is not an element of A

are both satisfied.

Definition 4. The **intersection** of two sets A and B (notation: $A \cap B$) is the set of elements which are common to both sets.

That is,

$$A \cap B = \{x : x \in A \text{ and } x \in B\}.$$

Definition 5. The **union** of two sets A and B (notation: $A \cup B$) is the set of all elements which are in either or both sets.

That is,

$$A \cup B = \{x : x \in A \text{ or } x \in B\}.$$

6.9.2 Function notation

The notation $f : X \rightarrow Y$ (which is read as “ f is a function (or map) from the set X to the set Y ”) means that f is a rule which associates exactly one element $y \in Y$ to each element $x \in X$. The y associated with x is written as $y = f(x)$ and is called the “value of the function f at x ” or “the image of x under f ”. The set X is often called the **domain** of the function f and the set Y is often called the **codomain** of the function f .

Equality of Functions. Two functions $f : X \rightarrow Y$ and $g : X \rightarrow Y$ are defined to be equal if and only if $f(x) = g(x)$ for all $x \in X$.

Addition of Functions. If $f : X \rightarrow Y$ and $g : X \rightarrow Y$, and if elements of Y can be added, then the sum function $f + g$ is defined by

$$(f + g)(x) = f(x) + g(x) \text{ for all } x \in X.$$

Multiplication by a Scalar. If $f : X \rightarrow Y$ and $\lambda \in \mathbb{F}$, where $\mathbb{F}$ is a field, and if elements of Y can be multiplied by elements of $\mathbb{F}$, then the function λf is defined by

$$(\lambda f)(x) = \lambda(f(x)) \text{ for all } x \in X.$$

Multiplication of Functions. If $f : X \rightarrow Y$ and $g : X \rightarrow Y$, and if elements of Y can be multiplied, then the product function fg is defined by

$$(fg)(x) = f(x)g(x) \text{ for all } x \in X.$$

Composition of Functions. If $g : X \rightarrow W$ and $f : W \rightarrow Y$, then the composition function $f \circ g : X \rightarrow Y$ is defined by

$$(f \circ g)(x) = f(g(x)) \text{ for all } x \in X.$$

6.10 Vector spaces and MAPLE

Most of the problems in this chapter can be solved using matrix methods, which of course means that Maple can be very helpful. You might look at your session 1 notes to refresh your memory as to how Maple handles vectors and matrices. As usual, you should type

```
with(LinearAlgebra);
```

in order to load the linear algebra package.

The following commands, for example, can be used to put the vectors $\mathbf{v}_1, \mathbf{v}_2$, and $\mathbf{v}_3$ as the columns in a 3×3 matrix.

```
v1:=<1,2,3>;
v2:=<0,-1,2>;
v3:=<3,-1,3>;
A:=<v1|v2|v3>;
```

We could now test whether these three vectors are linearly independent by performing Gaussian elimination on A .

```
GaussianElimination(A);
```

In this particular example, there are no non-leading columns so the vectors are linearly independent (and hence form a basis for $\mathbb{R}^3$).

Most of the other problems concerning problems in $\mathbb{R}^n$ can be solved using similar methods. Actually, the `LinearAlgebra` package contains many ready-made commands for performing the standard calculations. For example

```
Basis({v1,v2,v3});
```

returns a subset of $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ which forms a basis for $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$.

Problems in other vector spaces can often be solved by using coordinate vectors to convert the problem to one in $\mathbb{R}^n$. Finding the coordinate vector for a vector $\mathbf{b} \in \mathbb{R}^n$ with respect to a new basis is quite simple. For example, with the ordered basis $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, the commands

```
b:=<1,1,1>;
coordvect:=LinearSolve(A,b);
```

find the coordinate vector of $\mathbf{b}$. The `firstyear` package contains commands to find the coordinate vector of a polynomial in $\mathbb{P}_n$ with respect to the standard basis $\{1, x, x^2, \dots, x^n\}$ (be careful with the capital letters).

```
with(firstyear);
p:=3*x^2-x+4;
v:=polytoVect(p,2);
Vecttopoly(v,x);
```

Problems for Chapter 6

Questions marked with [R] are routine, [H] harder, [X] extra for MATH1241, [M] Maple. You should try to solve some of the questions in Sections 6.4 to 6.7 with Maple.

Problems 6.1 : Definitions and examples of vector spaces

1. [R] Show that the set

$$S = \{\mathbf{x} \in \mathbb{R}^3 : x_1 \leq 0, \quad x_2 \geq 0\},$$

with the usual rules for addition and multiplication by a scalar in $\mathbb{R}^3$ is not a vector space by showing that at least one of the vector space axioms is not satisfied. Give a geometric interpretation of this result.

2. [R] Show that the system S with the usual rules for addition and multiplication by a scalar in $\mathbb{R}^3$, and where

$$S = \{\mathbf{x} \in \mathbb{R}^3 : 2x_1 + 3x_2^3 - 4x_3^2 = 0\},$$

is not a vector space by showing that at least one of the vector space axioms is not satisfied.

3. [R] Let $S = \left\{ \begin{pmatrix} a \\ b \\ c \end{pmatrix} \in \mathbb{R}^3 : (a-b)c = 0 \right\}$.

- Write down two non-zero elements of S .
- Show that S is not closed under vector addition.

4. [H] The set $\mathbb{C}^n$ is a vector space over $\mathbb{C}$ (see Example 2 of Section 6.1). Check that axioms 1, 2, 6, 9 are satisfied by this system.

5. [X] Let $M_{mn}(\mathbb{C})$ be the set of all $m \times n$ matrices with complex entries with addition the usual rule for addition of complex matrices, and multiplication by a scalar the usual rule for multiplication of a complex matrix by a complex scalar. Prove that the vector space $M_{mn}(\mathbb{C})$ satisfies axioms 1, 3, 6 and 10.

6. [H] Prove that the system $(\mathbb{C}^n, +, *, \mathbb{R})$ with “natural” definitions of $+$ and $*$ is a vector space, whereas the system $(\mathbb{R}^n, +, *, \mathbb{C})$ with “natural” definitions of $+$ and $*$ is not a vector space.

7. [X] Consider the system $(\mathbb{R}^2, +', *', \mathbb{R})$ in which the usual operations of “addition” and “multiplication by a scalar” are replaced by the new definitions:

$$\begin{aligned} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} +' \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} &= \begin{pmatrix} a_1 + b_1 \\ a_2 - 3b_2 \end{pmatrix} \\ \lambda *' \begin{pmatrix} a_1 \\ a_2 \end{pmatrix} &= \begin{pmatrix} 4\lambda a_1 \\ \lambda a_2 \end{pmatrix}. \end{aligned}$$

Give a list of the vector space axioms satisfied by this system, and a list of any which are not satisfied. Is this system a vector space?

Problems 6.2 : Vector arithmetic

8. [H] Prove that the following properties are true for every vector space.

a) $2\mathbf{v} = \mathbf{v} + \mathbf{v}$.

b) $n\mathbf{v} = \mathbf{v} + \cdots + \mathbf{v}$, where there are n terms on the right.

9. [H] Prove parts 2, 4 and 5 of Proposition 2 of Section 6.2.

Problems 6.3 : Subspaces

10. [R] Suppose $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$. Show that the line segment defined by

$$S = \{\mathbf{x} \in \mathbb{R}^3 : \mathbf{x} = \alpha\mathbf{v}, \quad \text{for } 0 \leq \alpha \leq 10\}$$

is not a subspace of $\mathbb{R}^3$.

11. [R] Show that the set

$$S = \{\mathbf{x} \in \mathbb{R}^3 : 2x_1 + 3x_2 - 4x_3 = 6\},$$

is not a subspace of $\mathbb{R}^3$. Give a geometric interpretation of this result.

12. [R] Let S the set

$$S = \{\mathbf{x} \in \mathbb{R}^3 : 2x_1 + 3x_2 - 4x_3 = 0\},$$

a) Find three distinct members of S .

b) Show that S is a subspace of $\mathbb{R}^3$.

c) Give a geometric interpretation of this latter result.

13. [R] Show that

$$T = \left\{ \begin{pmatrix} x \\ y \\ z \end{pmatrix} : -1 \leq x + y + z \leq 1 \right\}$$

is not a vector subspace of $\mathbb{R}^3$.

14. [R] Show that the set

$$S = \{\mathbf{x} \in \mathbb{R}^3 : 2x_1 + 3x_2 - 4x_3 = 4x_1 - 2x_2 + 3x_3 = 0\}$$

is a subspace of $\mathbb{R}^3$.

15. [R] Show that the set

$$S = \{\mathbf{x} \in \mathbb{R}^3 : 2x_1 + 3x_2 - 4x_3 = 0 \quad \text{or} \quad 4x_1 - 2x_2 + 3x_3 = 0\}$$

is not a subspace of $\mathbb{R}^3$.

16. [R] Show that the set

$$S = \{\mathbf{b} \in \mathbb{R}^2 : \mathbf{b} = A\mathbf{x} \quad \text{for some } \mathbf{x} \in \mathbb{R}^3\},$$

where

$$A = \begin{pmatrix} 2 & -3 & 1 \\ 4 & 5 & -3 \end{pmatrix},$$

is a subspace of $\mathbb{R}^2$. Explain why each column of the matrix belongs to the set S .

17. [R] For each of the following subsets of $\mathbb{R}^3$, either prove that the given subset is a subspace of $\mathbb{R}^3$ or explain why it is not a subspace.

a) $S = \left\{ \begin{pmatrix} x_1 \\ 0 \\ x_3 \end{pmatrix} \in \mathbb{R}^3 : x_1 + 2x_3 \geq 0 \right\}.$

b) $T = \left\{ \sum_{i=1}^4 \lambda_i \mathbf{v}_i : \lambda_i \in \mathbb{R}, 1 \leq i \leq 4 \right\}$, where $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ are given fixed vectors in $\mathbb{R}^3$.

c) $U = \{A\mathbf{x} : \mathbf{x} \in \mathbb{R}^5\}$, where A is a fixed 3×5 matrix.

18. [H] Suppose that $\mathbf{u} = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$. Show, by the Subspace Theorem that the set

$$S = \{\mathbf{x} \in \mathbb{R}^3 : \mathbf{x} = \lambda\mathbf{u} + \mu\mathbf{v}, \quad \text{for } \lambda, \mu \in \mathbb{R}\}$$

is a subspace of $\mathbb{R}^3$.

19. [H] Prove that the set S in Example 6 on page 15 is closed under multiplication by a scalar.

20. [H] Let $\mathbf{a}$ and $\mathbf{b}$ be two fixed non-zero vectors in $\mathbb{R}^5$. Show that

$$W = \{\mathbf{x} \in \mathbb{R}^5 : \mathbf{x} \cdot \mathbf{a} = \mathbf{x} \cdot \mathbf{b} = 0\}$$

is a subspace of $\mathbb{R}^5$.

If $\mathbf{a} = \mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$ and $\mathbf{b} = \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$, describe W .

21. [R] Show that the set $S = \{p \in \mathbb{P}_2 : p(0) = 1\}$ is NOT a subspace of $\mathbb{P}_2$.

22. [R] Show that the set

$$S = \{p \in \mathbb{P}_3 : p''(x) = 0 \quad \text{for all } x \in \mathbb{R}\}$$

is a subspace of $\mathbb{P}_3$.

23. [H] Is the set

$$S = \{p \in \mathbb{P}_3 : p'(x) + x + 1 = 0 \text{ for all } x \in \mathbb{R}\}$$

a subspace of $\mathbb{P}_3$?

24. [H] Consider the set

$$S = \{p \in \mathbb{P}_3 : (x+1)p'(x) - 3p(x) = 0 \text{ for all } x \in \mathbb{R}\}.$$

- Show that S is a subspace of $\mathbb{P}_3$, (the set of all real polynomials of degree ≤ 3).
 - Find a polynomial in S where not all the coefficients are zero.
25. [H] By constructing a counterexample, show that the union of two subspaces is not, in general, a subspace.
26. [H] Let W_1 and W_2 be two subspaces of a vector space V over the field $\mathbb{F}$. Prove that the intersection of W_1 and W_2 (i.e., the set $W_1 \cap W_2$) is a subspace of V .
27. [X] Let V be a vector space over the field $\mathbb{F}$.
- Let $\{W_k : 1 \leq k \leq m\}$ be m subspaces of V , and let W be the intersection of these m subspaces. Prove that W is a subspace of V .
 - Let S be any set of vectors in V , and let W be the intersection of all subspaces of V which contain S (that is, $\mathbf{x} \in W$ if and only if $\mathbf{x}$ lies in every subspace which contains S). Prove that W is the set of finite linear combinations of vectors from S .

Problems 6.4 : Linear combinations and spans

28. [R] Let $\mathbf{a} = \begin{pmatrix} 10 \\ 11 \\ 4 \end{pmatrix}$, $\mathbf{v}_1 = \begin{pmatrix} 2 \\ 1 \\ 4 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}$ and $\mathbf{v}_3 = \begin{pmatrix} 3 \\ 3 \\ -1 \end{pmatrix}$.

- Is $\mathbf{a} \in \text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$? If so, express $\mathbf{a}$ as a linear combination of $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$.
- Do the vectors $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ span $\mathbb{R}^3$? If not, find condition(s) on $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}$ for $\mathbf{b}$ to belong to $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ and interpret your answer geometrically.

29. [R] Repeat the preceding question using

$$\mathbf{a} = \begin{pmatrix} 9 \\ -2 \\ -4 \end{pmatrix}, \mathbf{v}_1 = \begin{pmatrix} 0 \\ 2 \\ 3 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 5 \\ -2 \\ -3 \end{pmatrix} \text{ and } \mathbf{v}_3 = \begin{pmatrix} 15 \\ -4 \\ -6 \end{pmatrix}.$$

30. [R] Repeat using $\mathbf{a} = \begin{pmatrix} 1 \\ 1 \\ -9 \\ 1 \end{pmatrix}$, $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 3 \\ 0 \\ 5 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 2 \\ 2 \\ 1 \\ 2 \end{pmatrix}$, $\mathbf{v}_3 = \begin{pmatrix} -1 \\ 0 \\ 4 \\ 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}$.

[Replace $\mathbb{R}^3$ by $\mathbb{R}^4$, of course.]

31. [R] Is the vector $\mathbf{b} = \begin{pmatrix} -2 \\ -6 \\ -4 \end{pmatrix} \in \text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4)$, where

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}, \mathbf{v}_4 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}?$$

32. [R] Is the set of vectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} -1 \\ 0 \\ 5 \end{pmatrix}$ a spanning set for $\mathbb{R}^3$?

33. [R] Does $\mathbf{v}$ belong to the column space of A , $\text{col}(A)$, where

$$\mathbf{v} = \begin{pmatrix} 2 \\ -5 \\ 19 \\ -13 \end{pmatrix} \quad \text{and} \quad A = \begin{pmatrix} 1 & 3 & -1 \\ 0 & 1 & 2 \\ 2 & -3 & -5 \\ 1 & 2 & 7 \end{pmatrix}?$$

If so, write $\mathbf{v}$ as a linear combination of the columns of A .

34. [R] Is the polynomial $p(x) = 1 + x + x^2$ in $\text{span}(1 - x + 2x^2, -1 + x^2, -2 - x + 5x^2)$?

35. [R] Is $S = \{1 + x, 1 - x^2, x + 2x^2\}$ a spanning set for $\mathbb{P}_2$?

36. [X] Prove Proposition 1 of Section 6.4.

37. [X] Use the vector space axioms to prove that we do not need to use brackets when writing down the linear combination

$$\sum_{k=1}^n \lambda_k \mathbf{v}_k = \lambda_1 \mathbf{v}_1 + \cdots + \lambda_n \mathbf{v}_n.$$

That is, prove that the result of the operations is independent of the order in which the additions are performed.

Problems 6.5 : Linear independence

38. [R] Is the set of vectors $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} -1 \\ 0 \\ 5 \end{pmatrix}$ linearly independent? Are these three vectors coplanar?

39. [R] Is the set $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} 5 \\ -4 \\ 3 \end{pmatrix}$, a linearly independent set? Are these three vectors coplanar?

40. [R] Can a set of linearly independent vectors contain a zero vector? Explain your answer.

41. [R] Given the set $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where $\mathbf{v}_1 = \begin{pmatrix} 1 \\ -3 \\ -2 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 3 \\ 2 \\ 1 \end{pmatrix}$, $\mathbf{v}_3 = \begin{pmatrix} 4 \\ -1 \\ -1 \end{pmatrix}$, do the following.
- Show that S is a linearly dependent set.
 - Show that at least one of the vectors in S can be written as a linear combination of the others, and find the corresponding linear combination.
 - Find all possible ways of writing the vector $\begin{pmatrix} 8 \\ 9 \\ 5 \end{pmatrix}$ as a linear combination of the vectors in the set.
 - Find a linearly independent subset of S with the same span as S , and then show that $\begin{pmatrix} 8 \\ 9 \\ 5 \end{pmatrix}$ can be written as a unique linear combination of this subset.
 - Give a geometric interpretation of $\text{span}(S)$.
42. [R] Repeat the previous question for the set of four vectors $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$, where
- $$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} -1 \\ 0 \\ -1 \end{pmatrix}, \mathbf{v}_4 = \begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}.$$
43. [R] Is $\{1 - x + 2x^2, -1 + x^2, -2 - x + 5x^2\}$ a linearly independent subset of $\mathbb{P}_2$? If the set is not linearly independent express one of the polynomials as a linear combination of the others.
44. [H] (For discussion). Let the set $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a linearly independent subset of a vector space V . You are standing at the origin of V and set off in the direction of $\mathbf{v}_1$. After a certain length of time, you turn and head in direction $\mathbf{v}_2$ — then in direction $\mathbf{v}_3$ and so on. Is it possible for you to return to the origin? (Note: You may walk any distance that you like along any of the directions, but you are not allowed to retrace your steps).
45. [H] What would happen in the previous question if the set S were a linearly dependent set?
46. [H] Assume that $m \leq n$ and that $S = \{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ is a set of mutually orthogonal, non-zero vectors in $\mathbb{R}^n$, that is, the dot products satisfy (see Section 5.3.1)

$$\begin{aligned} \mathbf{v}_i \cdot \mathbf{v}_j &= 0 & \text{for } i \neq j; \quad 1 \leq i, j \leq m \\ \mathbf{v}_i \cdot \mathbf{v}_i &\neq 0 & \text{for } 1 \leq i \leq m. \end{aligned}$$

Show that S is a linearly independent set.

Problems 6.6 : Basis and dimension

47. [R] Is the set $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}$, $\mathbf{v}_3 = \begin{pmatrix} 5 \\ -4 \\ 3 \end{pmatrix}$, a basis for $\mathbb{R}^3$?
48. [R] Find a basis for, and the dimension of, $W = \text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$, where
- $$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} -1 \\ 0 \\ 5 \end{pmatrix}.$$
49. [R] Without doing any calculation, explain why $\left\{ \begin{pmatrix} 1 \\ 3 \\ 0 \\ 5 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 1 \\ 3 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 4 \\ 0 \end{pmatrix} \right\}$ is not a spanning set for $\mathbb{R}^4$.
- Similarly, without doing any calculation, explain why $\left\{ \begin{pmatrix} 1 \\ 3 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -3 \\ 1 \end{pmatrix} \right\}$ is a linearly dependent set.
50. [R] Which of the following statements are true and which are false? Explain your answer.
- Any set of 6 vectors in $\mathbb{R}^5$ is linearly dependent.
 - Some sets of 6 vectors in $\mathbb{R}^5$ are linearly independent.
 - Any set of 6 vectors in $\mathbb{R}^5$ is a spanning set for $\mathbb{R}^5$.
 - Some sets of 6 vectors in $\mathbb{R}^5$ span $\mathbb{R}^5$.
 - Same as in (a) – (d), with 6 replaced by 4.
 - Any set of 5 vectors in $\mathbb{R}^5$ is a basis for $\mathbb{R}^5$.
 - Some sets of 5 vectors in $\mathbb{R}^5$ are bases for $\mathbb{R}^5$.
 - Any set of vectors which spans $\mathbb{R}^5$ is linearly independent.
 - Any set of 5 vectors which spans $\mathbb{R}^5$ is linearly independent.
 - Any 5 linearly independent vectors in $\mathbb{R}^5$ form a basis for $\mathbb{R}^5$.
51. [R] Let V be a finite dimensional real vector space, and let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a finite set of vectors in V . Suppose also that the dimension of V is ℓ . State, with brief reasons, the relationship, if any, between n and ℓ if
- S is linearly independent.
 - S is linearly dependent.
 - S spans V .
 - S is a basis for V .

52. [H] Explain why it is impossible to have a set of m mutually orthogonal, non-zero vectors in $\mathbb{R}^n$ with $m > n$.

53. [R] Consider the plane P in $\mathbb{R}^3$ whose equation is

$$x + y + z = 0.$$

- a) Prove that P is a subspace of $\mathbb{R}^3$.
- b) Find a basis for P . Give reasons for your answer.

54. [R] Find a basis for, and the dimension of, the column space of the matrix

$$A = \begin{pmatrix} 1 & 1 & -1 & -2 & 1 \\ 0 & 0 & 1 & 4 & -1 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

55. [R] Find a basis for, and the dimension of, $\text{col}(A)$, where

$$A = \begin{pmatrix} 1 & 1 & 0 & 2 & 1 \\ 0 & 0 & -1 & -2 & 2 \\ -1 & -1 & 1 & 4 & -1 \\ 1 & 1 & 0 & 4 & 2 \end{pmatrix}.$$

56. [R] Show that the columns of the matrix A given below are not a spanning set for $\mathbb{R}^4$. Then find a basis for $\mathbb{R}^4$ which contains as many of the columns of A as possible.

$$A = \begin{pmatrix} 1 & 3 & 3 & -7 & 5 \\ 2 & 6 & 5 & -8 & 1 \\ 3 & 9 & 5 & -3 & -2 \\ 4 & 12 & 5 & 2 & -5 \end{pmatrix}.$$

57. [R] Consider the set $T = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{x}\}$ where

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \\ 0 \end{pmatrix}, \mathbf{v}_2 = \begin{pmatrix} 3 \\ 6 \\ -3 \\ 0 \end{pmatrix}, \mathbf{v}_3 = \begin{pmatrix} 2 \\ 1 \\ -1 \\ 4 \end{pmatrix}, \mathbf{v}_4 = \begin{pmatrix} -1 \\ -5 \\ 2 \\ 4 \end{pmatrix}, \mathbf{x} = \begin{pmatrix} 6 \\ -3 \\ -1 \\ 20 \end{pmatrix}.$$

- a) Find a basis B for $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4, \mathbf{x})$.
- b) Explain why $\mathbf{x}$ belongs to $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4)$. Write $\mathbf{x}$ as a linear combination of B .
- c) Suppose the matrix A has columns $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$. What is the dimension of the column space of A ?

58. [R] Show that the set $\{1 - x^2 + x^3, x + 2x^2, 2 + x - x^2 + 2x^3, 2x - x^2 + x^3\}$ forms a basis for $\mathbb{P}_3$.

59. [H] Consider the set of polynomials $S = \{p_1, p_2, p_3, p_4\}$ in $\mathbb{P}_2$, where

$$p_1(z) = 1 + z - z^2, \quad p_2(z) = 2 - z, \quad p_3(z) = 5 - 4z + z^2, \quad p_4(z) = z^2.$$

Show that S is a linearly dependent spanning set for $\mathbb{P}_2$, and then find a subset of S which is a basis for $\mathbb{P}_2$.

60. [H] You are given that V is a vector space and that $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is a subset of V . Suppose that $\mathbf{w} \in \text{span}(S)$. Prove that the set

$$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{w}\}$$

is a linearly dependent set.

61. [X] Prove that the **only** subspaces of $\mathbb{R}^4$ are (i) $\{\mathbf{0}\}$, (ii) lines through the origin, (iii) planes through the origin, (iv) subspaces of the form $\text{span}(S)$, where S is any set of three linearly independent vectors in $\mathbb{R}^4$, and (v) $\mathbb{R}^4$ itself.

Problems 6.7 : [X] Coordinate vectors

62. [X] Show that the columns of the matrix

$$A = \begin{pmatrix} 1 & 2 & -1 & 1 \\ 3 & 2 & 0 & -2 \\ 0 & 1 & -1 & 1 \\ 5 & 3 & 0 & -1 \end{pmatrix}$$

are a basis for $\mathbb{R}^4$. Then find the coordinate vector of $\mathbf{v} = \begin{pmatrix} -2 \\ -6 \\ -4 \\ -2 \end{pmatrix}$ with respect to the ordered basis given by the columns of A .

63. [X] A vector $\mathbf{v} \in \mathbb{R}^4$ has the coordinate vector $\begin{pmatrix} 1 \\ 6 \\ -1 \\ 4 \end{pmatrix}$ with respect to the ordered basis formed by the columns of the matrix A of the previous question. Find $\mathbf{v}$.

64. [X] Find the vector $\mathbf{v}$ that has coordinate vector $\begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}$ with respect to the ordered basis $\left\{ \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \\ -5 \end{pmatrix}, \begin{pmatrix} 2 \\ 4 \\ 9 \end{pmatrix} \right\}$ of $\mathbb{R}^3$.

65. [X] Find the coordinates of the following vectors with respect to the given ordered bases.

a) $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ with respect to $\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ -1 \end{pmatrix} \right\}$.

b) $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ with respect to $\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -2 \\ 0 \\ -1 \end{pmatrix} \right\}$.

66. [X] With respect to the basis $B = \left\{ \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 4 \\ 6 \end{pmatrix}, \begin{pmatrix} -2 \\ 3 \\ -3 \end{pmatrix} \right\}$ of $\mathbb{R}^3$,

a) find the vector $\mathbf{v}$ with coordinate vector $[\mathbf{v}]_B = \begin{pmatrix} 3 \\ 1 \\ -3 \end{pmatrix}$;

b) find the coordinate vector of $\mathbf{w} = \begin{pmatrix} 7 \\ -3 \\ 11 \end{pmatrix}$.

67. [X] Consider the set $S = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$, where $\mathbf{v}_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}$, $\mathbf{v}_2 = \begin{pmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ 0 \end{pmatrix}$, $\mathbf{v}_3 = \begin{pmatrix} -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{2}{\sqrt{6}} \end{pmatrix}$.

Without solving systems of linear equations, do the following.

a) Show that S is an orthonormal set of vectors in $\mathbb{R}^3$.

b) Show that S is a basis for $\mathbb{R}^3$.

c) Find the coordinate vector of $\begin{pmatrix} -1 \\ 3 \\ 4 \end{pmatrix}$ with respect to the ordered basis S .

HINT. See Example 6 of Section 6.6.

68. [X] Let $S = \{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ be an orthonormal set of n vectors in $\mathbb{R}^n$. Prove that S is a basis for $\mathbb{R}^n$, and then show that the coordinate vector for any $\mathbf{v} \in \mathbb{R}^n$ is given by

$$[\mathbf{v}]_S = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad \text{where} \quad x_j = \mathbf{u}_j \cdot \mathbf{v}.$$

Problems 6.8 : [X] Further important examples of vector spaces

69. [X] Let M_{22} be the vector space of all 2×2 matrices with real entries (see Example 3 of Section 6.1). Let S be the set

$$S = \{A \in M_{22} : a_{11} + a_{22} = 5\}.$$

- a) Find three matrices in S .
- b) Is S a subspace of M_{22} ? Give a reason.

70. [X] Let T be the set

$$T = \{A \in M_{22} : a_{11} + a_{22} = 0\}.$$

- a) Find three matrices in T .
- b) Is T a subspace of M_{22} ? Give a reason.

71. [X] Show that the four matrices

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

form a basis for $M_{22}(\mathbb{R})$. This set is called the **standard basis** for $M_{22}(\mathbb{R})$.

72. [X] Show that the four matrices of the previous question also form a basis for the vector space $M_{22}(\mathbb{C})$ of all 2×2 matrices with complex entries.

HINT: Can you see why your proof of the previous question will also be valid for complex numbers?

73. [X] Show that the set of four matrices

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

form a basis for $M_{22}(\mathbb{C})$. These matrices are called the Pauli spin matrices, and they are important in quantum physics and chemistry.

74. [X] Find the coordinates of the following vectors with respect to the given ordered bases.

- a) The matrix

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

with respect to the standard basis for M_{22} given in question 71. Note that the results are the same for both real and complex numbers

- b) Repeat part (b) for the basis of Pauli spin matrices given in question 73. In this case the entries of A should be regarded as complex numbers.

75. [X] Let $R = \left\{ \begin{pmatrix} 1 & 0 \\ -2 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 3 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 5 & 1 \end{pmatrix} \right\}.$

- a) Express $\begin{pmatrix} -4 & 2 \\ -1 & -3 \end{pmatrix}$ as a linear combination of elements of R .
- b) Does R span $M_{22}(\mathbb{R})$, the space of all 2×2 matrices? Give a brief reason for your answer.

76. [X] Complete the proof of Proposition 2 of Section 6.8 that the system $(\mathcal{R}[X], +, *, \mathbb{R})$ is a vector space.
77. [X] Let $\mathcal{C}[X]$ be the set of all complex-valued functions with domain X . Show that the system $(\mathcal{C}[X], +, *, \mathbb{C})$, where $+$ and $*$ are the usual rules for addition and multiplication by a scalar of functions, satisfies vector space axioms 2, 4, 7 and 9. This system is a vector space over the complex field $\mathbb{C}$.

78. [X] Show that the set

$$S = \left\{ y \in \mathcal{R}[\mathbb{R}] : \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + 4y = 0 \right\}$$

is a subspace of the vector space $\mathcal{R}[\mathbb{R}]$ of all real-valued functions with domain $\mathbb{R}$.

79. [X] Is the set

$$S = \left\{ y \in \mathcal{R}[\mathbb{R}] : \frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} + 4y = 5 \right\}$$

a subspace of $\mathcal{R}[\mathbb{R}]$? Prove your answer.

80. [X] Let $C^{(k)}[\mathbb{R}]$ be the set of all real-valued functions with domain $\mathbb{R}$ for which the first k derivatives exist and are continuous. Prove that $C^{(k)}[\mathbb{R}]$ is a subspace of the vector space $\mathcal{R}[\mathbb{R}]$ of all real-valued functions with domain $\mathbb{R}$.
81. [X] Show that the vector spaces $C^{(k)}[\mathbb{R}]$ defined in the previous question have the property that, if $m \geq n$, then $C^{(m)}[\mathbb{R}]$ is a subspace of $C^{(n)}[\mathbb{R}]$.
82. [X] Let S be the subset of $\mathcal{R}[-\pi, \pi]$ defined by

$$S = \left\{ f \in \mathcal{R}[-\pi, \pi] : \int_{-\pi}^{\pi} \cos(x+t)f(t)dt = 0 \quad \text{for all } x \in [-\pi, \pi] \right\}.$$

Prove that S is a subspace of the vector space $\mathcal{R}[-\pi, \pi]$.

83. [X] This question generalises the results of question 46 to real-valued functions.

Let $S = \{f_1, \dots, f_n\}$ be a set of real-valued functions defined on an interval $[a, b]$ with the properties that

$$\int_a^b f_i(x)f_j(x)dx = 0 \quad \text{for } i \neq j; \quad 1 \leq i, j \leq n$$

$$\int_a^b f_i^2(x)dx \neq 0 \quad \text{for } 1 \leq i \leq n.$$

Prove that S is a linearly independent set.

NOTE. A set of functions with these properties is said to be mutually orthogonal on the interval $[a, b]$.

84. [X] Show that the set

$$S = \{p \in \mathbb{P}_n(\mathbb{R}) : 5p'(6) + 3p(6) = 0\}, \quad \text{where } p'(x) = \frac{dp}{dx},$$

is a subspace of the vector space $\mathbb{P}_n(\mathbb{R})$ of all real polynomials of degree less than or equal to n .

85. [X] Is the set

$$S = \{p \in \mathbb{P}_n(\mathbb{R}) : 5p'(6) + 3p(6) = 8\}, \quad \text{where } p'(x) = \frac{dp}{dx},$$

a subspace of $\mathbb{P}_n(\mathbb{R})$? Prove your answer.

86. [X] Let $\mathbb{P}$ be the set of all polynomials over the complex-number field $\mathbb{C}$. Show that $\mathbb{P}$ is a subspace of the vector space $\mathcal{C}[\mathbb{C}]$ of all complex-valued functions with domain $\mathbb{C}$.

87. [X] Is the polynomial $p \in \text{span}(p_1, p_2, p_3)$, where the polynomials are defined by

$$p(z) = -6 + 2z + 30z^2, \quad p_1(z) = 1 + 2z + 3z^2, \quad p_2(z) = -4 - z + 9z^2, \quad p_3(z) = -5 - z + 12z^2.$$

88. [X] Find conditions on the coefficients of the polynomial $p \in \mathbb{P}_2$ for p to be a linear combination of the three polynomials p_1, p_2, p_3 , where the polynomials are given by

$$p_1(z) = 2z + 3z^2, \quad p_2(z) = 5 - 2z - 3z^2, \quad p_3(z) = 15 - 4z - 6z^2.$$

89. [X] Are the polynomials p_1, p_2, p_3 in the previous two questions spanning sets for $\mathbb{P}_2$?

90. [X] Is the set of polynomials $S = \{p_1, p_2, p_3\}$ in $\mathbb{P}_2$, where

$$p_1(z) = 1 + z - z^2, \quad p_2(z) = 2 - z, \quad p_3(z) = 5 - 4z + z^2,$$

a linearly independent set? If not, express one of the polynomials as a linear combination of the others.

91. [X] Show that

$$\begin{aligned} p_1(z) &= -2 + 5z - 4z^2 + 15z^3 - 5z^4 + z^5, \\ p_2(z) &= 3z + 4z^2 - 3z^3 + 6z^5, \\ p_3(z) &= 2 + 3z^2 - 4z^3 + 10z^4 - 5z^5, \\ p_4(z) &= 3 + 14z^2 - 5z^3 + 6z^4 - 3z^5, \\ p_5(z) &= 3 + 8z + 17z^2 + 3z^3 + 11z^4 - z^5, \\ p_6(z) &= -3 + 11z - 7z^2 + 10z^3 - z^4 + 11z^5, \end{aligned}$$

are not a spanning set for $\mathbb{P}_5$, and then construct a basis for $\mathbb{P}_5$ containing as many of the given polynomials as possible.

HINT. Check using Maple.

92. [X] Find the coordinate vector for $p(x) = 1 + 2x + x^2$ with respect to the ordered basis $\{1 + x, 1 - x^2, x + 2x^2\}$ of $\mathbb{P}_2$.
93. [X] Find the coordinate vector of $1 + 2z + 3z^2$ with respect to the ordered basis of $\mathbb{P}_2$ given by

$$\left\{ \frac{1}{8}z(z-2), 1 - \frac{1}{4}z^2, \frac{1}{8}z(z+2) \right\}.$$

NOTE. This question and the one that follows do not require Gaussian Elimination.

94. [X] Find the coordinate vector of $a_0 + a_1z + a_2z^2$ with respect to the ordered basis of $\mathbb{P}_2$ given by

$$\left\{ \frac{1}{2}z(z-1), 1 - z^2, \frac{1}{2}z(z+1) \right\}.$$

95. [X] Let $S = \{p_1, \dots, p_n\}$ be a set of n polynomials in $\mathbb{P}_{n-1}(\mathbb{R})$ with the property that

$$\int_a^b p_i(x)p_j(x)dx = \begin{cases} 0 & \text{for } i \neq j \\ 1 & \text{for } i = j \end{cases} \quad \text{for } 1 \leq i, j \leq n.$$

A set of polynomials with this property is called an orthonormal set of polynomials on the interval $[a, b]$.

Prove that S is a basis for $\mathbb{P}_{n-1}(\mathbb{R})$, and then show that the coordinate vector for any $p \in \mathbb{P}_{n-1}(\mathbb{R})$ is given by

$$[p]_S = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad \text{where} \quad x_i = \int_a^b p_i(x)p(x)dx.$$

Chapter 7

LINEAR TRANSFORMATIONS

*“But I don’t need a Sillygism, you know,
to prove that mathematical axiom you mentioned.”
“Nor to prove that ‘all angles are equal’, I suppose?”
“Why, of course not! One takes such a simple truth as that for granted!”
Lewis Carroll, Sylvie and Bruno.*

The purpose of this chapter is to give an introduction to an extremely important class of functions called “linear transformations” or “linear maps”. Mathematical examples of linear transformations include geometric transformations such as stretching, reflection and rotation, algebraic operations such as matrix multiplication, and calculus operations such as differentiation and integration. Linear transformations are also widely used in many applications of mathematics, and objects which are often modelled (either exactly or approximately) by linear transformations are related to radio and TV sets, amplifiers and hi-fi equipment, atomic spectra, molecular vibrations, sound waves, ocean waves, oil refineries, chemical plants, profit of a company, inventory of a factory or shop, and the state of an economy.

7.1 Introduction to linear maps

Before reading this chapter you should quickly read the brief review of function notation given in Appendix 6.9.

As stated in Appendix 6.9, a function f with domain X and codomain Y (notation $f : X \rightarrow Y$) is a rule which associates exactly one element $y = f(x)$ of Y to each element $x \in X$. Note that an element x in the domain X is called an “argument” of the function and the corresponding element $y = f(x)$ in the codomain Y is usually called either “the function value of x ” or the “image of x under f ”.

Linear maps are an important special class of functions, in which both the domain and the codomain are vector spaces (that is, all arguments and values of the functions are vectors), and in which the two vector-space operations of addition and scalar multiplication are “preserved” by the function in the sense that:

Addition Condition. The function value of a sum of the two vectors is equal to the sum of the function values of the vectors.

Scalar Multiplication Condition. The function value of a scalar multiple of a vector is equal to the scalar multiple of the function value of the vector.

A more formal mathematical definition of a linear map is as follows.

Definition 1. Let V and W be two vector spaces over the same field $\mathbb{F}$. A function $T : V \rightarrow W$ is called a **linear map** or **linear transformation** if the following two conditions are satisfied.

Addition Condition. $T(\mathbf{v} + \mathbf{v}') = T(\mathbf{v}) + T(\mathbf{v}')$ for all $\mathbf{v}, \mathbf{v}' \in V$, and

Scalar Multiplication Condition. $T(\lambda \mathbf{v}) = \lambda T(\mathbf{v})$ for all $\lambda \in \mathbb{F}$ and $\mathbf{v} \in V$.

The domain and codomain of a linear map can be any vector spaces. In Section 7.5, we shall concentrate specifically on linear maps from $\mathbb{R}^n$ to $\mathbb{R}^m$. Unless otherwise stated, the following propositions and theorems are true for all linear maps.

The adjective “linear” in “linear map” suggests that the idea of a linear map arose from the geometric idea of a line. The connection between the equation of a line and a linear map is shown in Figure 1 (a) and (b) and in Example 1 below.

Example 1. Show that the function $T : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$T(x) = a_0 + a_1x \quad \text{for } x \in \mathbb{R},$$

where $a_0, a_1 \in \mathbb{R}$ are constants, is a linear map if and only if $a_0 = 0$.

SOLUTION. We check the conditions of the definition of a linear map. Firstly, the domain $\mathbb{R}$ and codomain $\mathbb{R}$ are both vector spaces as $\mathbb{R}$ is a vector space. Further, we have, for $x, x' \in \mathbb{R}$,

$$T(x + x') = a_0 + a_1(x + x'),$$

whereas,

$$T(x) + T(x') = (a_0 + a_1x) + (a_0 + a_1x') = 2a_0 + a_1(x + x').$$

Thus, the addition condition is satisfied if and only if $a_0 = 2a_0$, that is, if and only if $a_0 = 0$.

For $a_0 = 0$, we check the scalar multiplication condition, and obtain

$$T(\lambda x) = a_1(\lambda x) = \lambda(a_1x) = \lambda T(x),$$

as required.

Thus, the conditions for T to be a linear map are satisfied if and only if $a_0 = 0$. ◇

NOTE.

1. The equation $y = T(x) = a_0 + a_1x$ is the equation of a line in $\mathbb{R}^2$. Example 1 shows that the equation of a line defines a linear map if and only if the line goes through the origin.
2. The function $T(x) = a_0 + a_1x$ is a polynomial of degree 1 and is often called a linear polynomial. Example 1 shows that a “linear polynomial” is a “linear map” if and only if the constant term in the polynomial is zero.

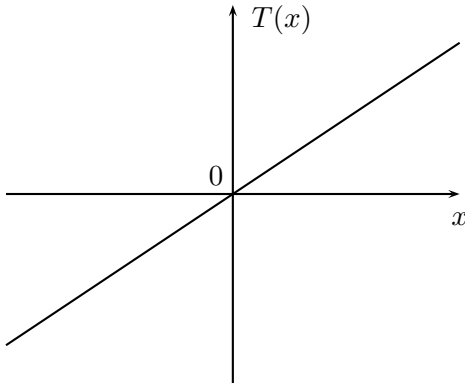


Figure 1(a). A linear map.

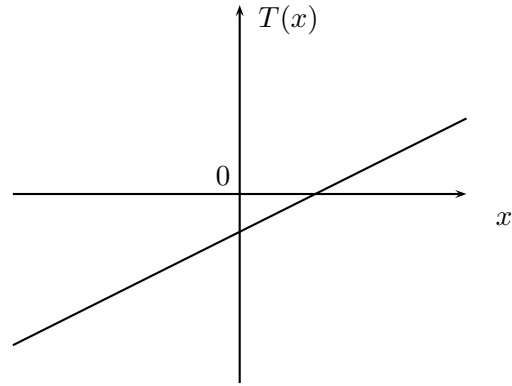


Figure 1(b). A linear polynomial which is NOT a linear map.

Example 2. Show that the function $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by

$$T(\mathbf{x}) = \begin{pmatrix} -5x_2 + 4x_3 \\ x_1 + 2x_3 \end{pmatrix} \quad \text{for } \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3,$$

is a linear map.

SOLUTION. The domain $\mathbb{R}^3$ and codomain $\mathbb{R}^2$ are both vector spaces. We next check the addition and scalar multiplication conditions.

Addition condition. For $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^3$, we have $\mathbf{x} + \mathbf{x}' = \begin{pmatrix} x_1 + x'_1 \\ x_2 + x'_2 \\ x_3 + x'_3 \end{pmatrix} \in \mathbb{R}^3$, and hence

$$\begin{aligned} T(\mathbf{x} + \mathbf{x}') &= \begin{pmatrix} -5(x_2 + x'_2) + 4(x_3 + x'_3) \\ (x_1 + x'_1) + 2(x_3 + x'_3) \end{pmatrix} \\ &= \begin{pmatrix} -5x_2 + 4x_3 \\ x_1 + 2x_3 \end{pmatrix} + \begin{pmatrix} -5x'_2 + 4x'_3 \\ x'_1 + 2x'_3 \end{pmatrix} = T(\mathbf{x}) + T(\mathbf{x}'). \end{aligned}$$

Thus the addition condition is satisfied.

Scalar multiplication condition. For $\mathbf{x} \in \mathbb{R}^3$ and $\lambda \in \mathbb{R}$, we have $\lambda\mathbf{x} = \begin{pmatrix} \lambda x_1 \\ \lambda x_2 \\ \lambda x_3 \end{pmatrix} \in \mathbb{R}^3$, and hence

$$T(\lambda\mathbf{x}) = \begin{pmatrix} -5(\lambda x_2) + 4(\lambda x_3) \\ \lambda x_1 + 2(\lambda x_3) \end{pmatrix} = \lambda \begin{pmatrix} -5x_2 + 4x_3 \\ x_1 + 2x_3 \end{pmatrix} = \lambda T(\mathbf{x}).$$

Thus, the scalar multiplication condition is also satisfied, and therefore T is a linear map. $\diamond$

We shall now summarise some useful properties that are true for all linear maps. In all of the following propositions and theorems the domain V and the codomain W are assumed to be vector spaces over the same field $\mathbb{F}$.

Here is a useful proposition regarding linear maps.

Proposition 1. If $T : V \rightarrow W$ is a linear map, then

1. $T(\mathbf{0}) = \mathbf{0}$ and
2. $T(-\mathbf{v}) = -T(\mathbf{v})$ for all $\mathbf{v} \in V$.

An informal way of stating these results is that a linear map always:

1. transforms the zero vector in the domain into the zero vector in the codomain, and
2. transforms the negative of a vector $\mathbf{v}$ in the domain into the negative of the corresponding function value $T(\mathbf{v})$ in the codomain.

Proof. (1). Since V is a vector space, we have from Proposition 2 of Section 6.2 that $0\mathbf{v} = \mathbf{0}$ for all $\mathbf{v} \in V$. Thus,

$$T(\mathbf{0}) = T(0\mathbf{v}) = 0T(\mathbf{v}) = \mathbf{0},$$

where we have first used the scalar multiplication condition of a linear map and then applied Proposition 2 of Section 6.2 to the vector $T(\mathbf{v}) \in W$.

(2). Since V is a vector space, we have from Proposition 2 of Section 6.2 that $-\mathbf{v} = (-1)\mathbf{v}$ for all $\mathbf{v} \in V$. Hence,

$$T(-\mathbf{v}) = T((-1)\mathbf{v}) = (-1)T(\mathbf{v}) = -T(\mathbf{v}),$$

where we have again used the scalar multiplication condition of a linear map, and then Proposition 2 of Section 6.2 applied to the vector $T(\mathbf{v}) \in W$. $\square$

Proposition 1 may often be used to provide a quick proof that some given function is not linear.

Example 3. Show that the function $T : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by

$$T\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 4x_1 + 3(x_2 - 6)$$

is not linear.

SOLUTION. $T\begin{pmatrix} 0 \\ 0 \end{pmatrix} = -18 \neq 0$, and hence T is not linear. $\diamond$

Example 4. Show that the function $T : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$T(x) = x^2$$

is not linear.

SOLUTION. $T(3) = 9$, but $T(6) = 36 \neq 2 \times 9$. Hence T is not linear. $\diamond$

To prove that a given map is not linear, it is easiest to provide a specific example that contravenes one of the conditions. One should also check first that the given map, takes the zero vector to the zero vector.

WARNING. The converses of the two results in Proposition 1 are **not** true in general as shown in the following example.

Example 5. The function $T(x) = x^3$ satisfies both $T(-x) = -T(x)$ and $T(0) = 0$. However, it is not a linear map by the following counterexample. For $x = 1$,

$$T(2 \times 1) = 8 \neq 2(1)^3 = 2T(1).$$

◇

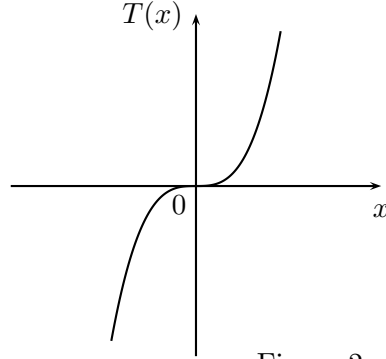


Figure 2.

The two conditions in the definition of a linear map are closely related to the two fundamental vector operations of addition and scalar multiplication. We therefore expect there to be a close relationship between linear combinations and linear maps. This relationship is given in Theorems 2 and 3 below.

Theorem 2. A function $T : V \rightarrow W$ is a linear map if and only if for all $\lambda_1, \lambda_2 \in \mathbb{F}$ and $\mathbf{v}_1, \mathbf{v}_2 \in V$

$$T(\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2) = \lambda_1 T(\mathbf{v}_1) + \lambda_2 T(\mathbf{v}_2). \quad (\#)$$

Proof. Let T be a linear function. Then,

$$\begin{aligned} T(\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2) &= T(\lambda_1 \mathbf{v}_1) + T(\lambda_2 \mathbf{v}_2) && \text{(from the addition condition)} \\ &= \lambda_1 T(\mathbf{v}_1) + \lambda_2 T(\mathbf{v}_2) && \text{(using the scalar multiplication condition twice),} \end{aligned}$$

and hence $(\#)$ is satisfied.

Conversely, suppose $(\#)$ is satisfied. Then, for $\lambda_1 = \lambda_2 = 1$, the condition the $(\#)$ becomes the addition condition, while for $\lambda_2 = 0$ the condition reduces to the scalar multiplication condition. The proof is complete. $\square$

Theorem 2 can be used to simplify the test for linearity, since it means that only one condition must be checked instead of the two separate conditions of the original definition.

Example 6. Show that the function $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$, defined by

$$T(\mathbf{x}) = \begin{pmatrix} 3x_1 - x_2 \\ 4x_2 \\ 5x_1 + 6x_2 \end{pmatrix} \quad \text{for } \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2,$$

is a linear map.

SOLUTION. For $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^2$ and $\lambda, \lambda' \in \mathbb{R}$, we have

$$\lambda \mathbf{x} + \lambda' \mathbf{x}' = \begin{pmatrix} \lambda x_1 + \lambda' x'_1 \\ \lambda x_2 + \lambda' x'_2 \end{pmatrix} \in \mathbb{R}^2,$$

and hence

$$\begin{aligned} T(\lambda \mathbf{x} + \lambda' \mathbf{x}') &= \begin{pmatrix} 3(\lambda x_1 + \lambda' x'_1) - (\lambda x_2 + \lambda' x'_2) \\ 4(\lambda x_2 + \lambda' x'_2) \\ 5(\lambda x_1 + \lambda' x'_1) + 6(\lambda x_2 + \lambda' x'_2) \end{pmatrix} \\ &= \lambda \begin{pmatrix} 3x_1 - x_2 \\ 4x_2 \\ 5x_1 + 6x_2 \end{pmatrix} + \lambda' \begin{pmatrix} 3x'_1 - x'_2 \\ 4x'_2 \\ 5x'_1 + 6x'_2 \end{pmatrix} = \lambda T(\mathbf{x}) + \lambda' T(\mathbf{x}'). \end{aligned}$$

Thus, from Theorem 2, T is a linear map. $\diamond$

A generalisation of Theorem 2 is also of considerable importance in the theory and applications of linear maps.

Theorem 3. *If T is a linear map with domain V and S is a set of vectors in V , then the function value of a linear combination of S is equal to the corresponding linear combination of the function values of S , that is, if $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ and $\lambda_1, \dots, \lambda_n$ are scalars, then*

$$T(\lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n) = \lambda_1 T(\mathbf{v}_1) + \dots + \lambda_n T(\mathbf{v}_n).$$

Proof. This, left as an exercise (see question 5), is based on an easy inductive argument.

Theorem 3 has many uses. Some examples of its use are as follows.

Example 7. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be a linear map with values

$$T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}, \quad T \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -5 \\ 6 \end{pmatrix}, \quad T \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 8 \end{pmatrix}.$$

Find the function value at $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$.

SOLUTION. We have

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + x_3 \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

From Theorem 3, the function value at $\mathbf{x}$ is

$$\begin{aligned} T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} &= x_1 T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + x_2 T \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + x_3 T \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \\ &= x_1 \begin{pmatrix} 3 \\ 7 \end{pmatrix} + x_2 \begin{pmatrix} -5 \\ 6 \end{pmatrix} + x_3 \begin{pmatrix} -2 \\ 8 \end{pmatrix} \\ &= \begin{pmatrix} 3x_1 - 5x_2 - 2x_3 \\ 7x_1 + 6x_2 + 8x_3 \end{pmatrix}. \end{aligned}$$

$\diamond$

Example 8. Show that the function $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ with function values, $T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 7 \end{pmatrix}$, $T \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -5 \\ 6 \end{pmatrix}$, $T \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ 8 \end{pmatrix}$, and $T \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -4 \\ 20 \end{pmatrix}$, is not a linear map.

SOLUTION. We have

$$\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

Hence, if T is a linear map, we have from Theorem 3 that

$$\begin{aligned} T \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} &= T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + T \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + T \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \\ &= \begin{pmatrix} 3 \\ 7 \end{pmatrix} + \begin{pmatrix} -5 \\ 6 \end{pmatrix} + \begin{pmatrix} -2 \\ 8 \end{pmatrix} = \begin{pmatrix} -4 \\ 21 \end{pmatrix}. \end{aligned}$$

But, $T \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -4 \\ 20 \end{pmatrix} \neq \begin{pmatrix} -4 \\ 21 \end{pmatrix}$, and hence T is not a linear map. $\diamond$

Examples 7 and 8 are actually special cases of the following extremely important result.

Theorem 4. For a linear map $T : V \rightarrow W$, the function values for every vector in the domain are known if and only if the function values for a basis of the domain are known.

Further, if $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ is a basis for the domain V then for all $\mathbf{v} \in V$ we have

$$T(\mathbf{v}) = x_1 T(\mathbf{v}_1) + \dots + x_n T(\mathbf{v}_n),$$

where $x_1, \dots, x_n$ are the scalars in the unique linear combination $\mathbf{v} = x_1 \mathbf{v}_1 + \dots + x_n \mathbf{v}_n$ of the basis B .

Proof. It follows from Theorem 3 that

$$T(\mathbf{v}) = x_1 T(\mathbf{v}_1) + \dots + x_n T(\mathbf{v}_n).$$

The theorem follows immediately. $\square$

7.2 Linear maps from $\mathbb{R}^n$ to $\mathbb{R}^m$ and $m \times n$ matrices

If you look at the examples of functions with domain $\mathbb{R}^n$ and codomain $\mathbb{R}^m$ given in the previous section, you will see that if $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n$, then we can write $T(\mathbf{x})$ as $A\mathbf{x}$

where A is an $m \times n$ matrix. In this section we are going to show that every matrix A represents a linear map and conversely that every linear map with domain $\mathbb{R}^n$ and codomain $\mathbb{R}^m$ can be

represented by a matrix. Because of the close relation between T and the corresponding A and we prefer to write $A\mathbf{x}$ for $T(\mathbf{x})$ instead of $\mathbf{x}A$, the vector $\mathbf{x}$ **must be a column vector**.

We begin with the following theorem.

Theorem 1. For each $m \times n$ matrix A , the function $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$, defined by

$$T_A(\mathbf{x}) = A\mathbf{x} \quad \text{for } \mathbf{x} \in \mathbb{R}^n,$$

is a linear map.

Proof. We check the addition and scalar multiplication conditions, using the properties $A(\mathbf{x} + \mathbf{x}') = A\mathbf{x} + A\mathbf{x}'$ and $A(\lambda\mathbf{x}) = \lambda A\mathbf{x}$ from Chapter 5.

Addition Condition. For all $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^n$, we have

$$T_A(\mathbf{x} + \mathbf{x}') = A(\mathbf{x} + \mathbf{x}') = A\mathbf{x} + A\mathbf{x}' = T_A(\mathbf{x}) + T_A(\mathbf{x}').$$

Scalar Multiplication Condition. For all $\lambda \in \mathbb{R}$ and $\mathbf{x} \in \mathbb{R}^n$, we have

$$T_A(\lambda\mathbf{x}) = A(\lambda\mathbf{x}) = \lambda(A\mathbf{x}) = \lambda T_A(\mathbf{x}).$$

Thus, since both the addition and scalar multiplication conditions are satisfied, $T_A(\mathbf{x}) = A\mathbf{x}$ is a linear map. □

The matrix equation $A\mathbf{x} = \mathbf{y}$ therefore has the interpretation that $\mathbf{y} = T_A(\mathbf{x}) = A\mathbf{x}$ is the function value of T_A at the point $\mathbf{x}$, or, for linear equations, the vector $\mathbf{y}$ may be regarded as the function value of the vector $\mathbf{x}$.

Example 1. Find a linear map T_A such that $T_A(\mathbf{x}) = A\mathbf{x}$ for the matrix

$$A = \begin{pmatrix} 3 & 4 \\ -1 & 0 \\ -5 & 6 \end{pmatrix}.$$

SOLUTION. Since A has 3 rows and 2 columns, the domain is $\mathbb{R}^2$, the codomain is $\mathbb{R}^3$, and the map $T_A : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is given by

$$T_A \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = A\mathbf{x} = \begin{pmatrix} 3x_1 + 4x_2 \\ -x_1 \\ -5x_1 + 6x_2 \end{pmatrix}.$$

◇

Theorem 1 and Example 1 show that a matrix can be used to define a linear map. We shall now show that every linear map with domain $\mathbb{R}^n$ and codomain $\mathbb{R}^m$ can be represented by an $m \times n$ matrix with real entries. The basic theorem which establishes this result is the following.

Theorem 2 (Matrix Representation Theorem). Let $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a linear map and let the vectors $\mathbf{e}_j$ for $1 \leq j \leq n$ be the standard basis vectors for $\mathbb{R}^n$. Then the $m \times n$ matrix A whose columns are given by

$$\mathbf{a}_j = T(\mathbf{e}_j) \quad \text{for } 1 \leq j \leq n$$

has the property that

$$T(\mathbf{x}) = A\mathbf{x} \quad \text{for all } \mathbf{x} \in \mathbb{R}^n.$$

Proof. Every vector $\mathbf{x} \in \mathbb{R}^n$ can be written as a unique linear combination of the standard basis vectors, that is,

$$\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = x_1 \mathbf{e}_1 + \cdots + x_n \mathbf{e}_n.$$

Then, from Theorem 3 of Section 7.1, we have

$$\begin{aligned} T(\mathbf{x}) &= T(x_1 \mathbf{e}_1 + \cdots + x_n \mathbf{e}_n) \\ &= x_1 T(\mathbf{e}_1) + \cdots + x_n T(\mathbf{e}_n) \\ &= x_1 \mathbf{a}_1 + \cdots + x_n \mathbf{a}_n, \end{aligned}$$

where $\mathbf{a}_j = T(\mathbf{e}_j)$. Now $\mathbf{a}_j \in \mathbb{R}^m$ for $1 \leq j \leq n$, and hence from Proposition 3 of Section 6.4 the linear combination can be rewritten in the matrix form $A\mathbf{x}$, where A is the matrix with the $\mathbf{a}_j$ as its columns. Thus, $T(\mathbf{x}) = A\mathbf{x}$ and the proof is complete. $\square$

The Representation Theorem can be used to construct a matrix for any given linear map with domain $\mathbb{R}^n$ and codomain $\mathbb{R}^m$, or more generally from $\mathbb{F}^n$ to $\mathbb{F}^m$ for any field $\mathbb{F}$.

Example 2. Find a matrix A such that $T(\mathbf{x}) = A\mathbf{x}$ for the linear map $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by

$$T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3x_1 - 5x_2 + 6x_3 \\ 5x_2 + 31x_3 \end{pmatrix}.$$

[Notice that we are using columns.]

SOLUTION. The first column of the matrix A is the vector given by

$$T(\mathbf{e}_1) = T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \end{pmatrix},$$

the second column is given by

$$T(\mathbf{e}_2) = T \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -5 \\ 5 \end{pmatrix},$$

and the third column is given by

$$T(\mathbf{e}_3) = T \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 6 \\ 31 \end{pmatrix}.$$

Thus, the matrix A is

$$A = \begin{pmatrix} 3 & -5 & 6 \\ 0 & 5 & 31 \end{pmatrix}.$$

$\diamond$

An alternative method, which is often simpler, of writing down the matrix for a given linear function is shown in the following example.

Example 3. Find a matrix A such that $T(\mathbf{x}) = A\mathbf{x}$ for the linear map $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ defined by

$$T_A \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 2x_1 - 3x_2 + 4x_3 - 5x_4 \\ -2x_1 + 3x_4 \\ x_1 - 5x_2 + 6x_3 - 8x_4 \end{pmatrix}.$$

SOLUTION. As usual for a linear map with domain $\mathbb{R}^n$ and codomain $\mathbb{R}^m$ (here $n = 4$ and $m = 3$), the components of the function value look like the left hand side of a system of linear equations. This system of equations is given by

$$T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 2x_1 & - & 3x_2 & + & 4x_3 & - & 5x_4 \\ -2x_1 & & & & & + & 3x_4 \\ x_1 & - & 5x_2 & + & 6x_3 & - & 8x_4 \end{pmatrix} = \begin{pmatrix} 2 & -3 & 4 & -5 \\ -2 & 0 & 0 & 3 \\ 1 & -5 & 6 & -8 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix}$$

Then the coefficient matrix, namely

$$A = \begin{pmatrix} 2 & -3 & 4 & -5 \\ -2 & 0 & 0 & 3 \\ 1 & -5 & 6 & -8 \end{pmatrix},$$

has the required property that $T(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^4$. ◇

In this section, we have shown that a matrix always defines a linear map and that a linear map between the vector spaces $\mathbb{R}^n$ and $\mathbb{R}^m$ can always be represented by a matrix.

This result can easily be generalised to linear maps between any two finite-dimensional vector spaces. This theorem is of fundamental importance in both the mathematical theory of linear maps and in applying the ideas of linear maps to practical problems. See Section 7.6.

7.3 Geometric examples of linear transformations

In this section we shall examine some of the geometric mappings which can be represented by linear maps and matrices. These mappings include stretching and compression, reflections, rotations, projections, and the dot and cross products with a fixed vector.

We shall begin by looking at geometric interpretations which can be given to simple types of matrices.

Example 1 (Reflection in $\mathbb{R}^2$). The simplest examples of reflections in $\mathbb{R}^2$ are reflections in one of the coordinate axes. An example of a reflection in the x_1 -axis is shown in Figure 3. Note that the reflection of the point with the position vector $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ is the point represented by the position vector $\mathbf{x}' = \begin{pmatrix} x_1 \\ -x_2 \end{pmatrix}$. This reflection can be represented by the 2×2 diagonal matrix with a negative diagonal entry given by

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

since

$$A\mathbf{x} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 \\ -x_2 \end{pmatrix} = \mathbf{x}'.$$

Note that we know that this reflection is a linear map since we have found a matrix that describes the effect of the reflection. $\diamond$

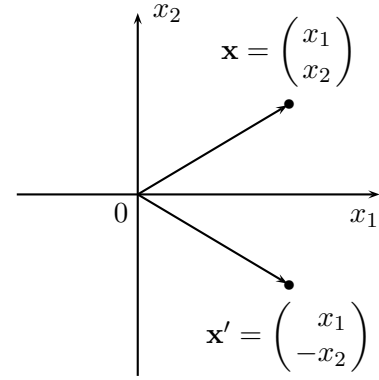


Figure 3: A reflection in the x_1 -axis.

Note that a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ will map the position vector of a point in an n -dimensional space to the position vector of a point in an m -dimensional space. The following proposition tells us a linear transformation will map a line to a line or a point.

Proposition 1. Suppose that $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear map. It maps a line in $\mathbb{R}^n$ to either a line or a point in $\mathbb{R}^m$.

Proof. In Chapter 1, a line in $\mathbb{R}^n$ through a point represented by $\mathbf{a}$ parallel to $\mathbf{v} \neq \mathbf{0}$ is defined to be the set

$$\{\mathbf{x} \in \mathbb{R}^n : \mathbf{x} = \mathbf{a} + \lambda \mathbf{v} \text{ for some } \lambda \in \mathbb{R}\}.$$

By Theorem 2 in Section 7.1, $T(\mathbf{a} + \lambda \mathbf{v}) = T(\mathbf{a}) + \lambda T(\mathbf{v})$. Hence T maps the line to the following subset of $\mathbb{R}^m$.

$$\{\mathbf{y} \in \mathbb{R}^m : \mathbf{y} = T(\mathbf{a}) + \lambda T(\mathbf{v}) \text{ for some } \lambda \in \mathbb{R}\}.$$

This set is a line when $T(\mathbf{v}) \neq \mathbf{0}$ and it contains a single vector $T(\mathbf{a})$ otherwise. $\square$

REMARK. Using similar argument, we can show that T maps a line segment with end points of position vectors $\mathbf{a}$ and $\mathbf{b}$ to a line segment with end points of position vectors $T(\mathbf{a})$ and $T(\mathbf{b})$.

Example 2 (Stretching and compression in $\mathbb{R}^2$). Let A be a 2×2 diagonal matrix with positive diagonal entries, that is, a matrix of the form

$$A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \text{ with } \lambda_1 > 0, \lambda_2 > 0.$$

Then the function value of $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ is $\mathbf{y} = T_A(\mathbf{x}) = A\mathbf{x}$, where

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} \lambda_1 x_1 \\ \lambda_2 x_2 \end{pmatrix}.$$

Thus, $b_1 = \lambda_1 x_1$ and $b_2 = \lambda_2 x_2$, and hence the effect of the matrix is simply to multiply the first component x_1 by the scalar λ_1 and the second component x_2 by the scalar λ_2 . Note that the first standard basis vector $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is transformed into $\begin{pmatrix} \lambda_1 \\ 0 \end{pmatrix} = \lambda_1 \mathbf{e}_1$, that is, its direction remains the same but it is either stretched (if $\lambda_1 > 1$) or compressed (if $\lambda_1 < 1$). Similarly, the second standard basis vector $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ is transformed into $\begin{pmatrix} 0 \\ \lambda_2 \end{pmatrix} = \lambda_2 \mathbf{e}_2$ with a resulting stretching if $\lambda_2 > 1$ or compression if $\lambda_2 < 1$.

Figure 4(a) shows a picture of a 5-point star with vertices $A(1, 5)$, $B(4, 3)$, $C(3, -1)$, $D(-1, -1)$ and $E(-2, 3)$. Suppose X is the point $(1, 3)$ on the line segment BE .

Hence the position vectors of these points are respectively

$$\begin{aligned} \mathbf{a} &= \begin{pmatrix} 1 \\ 5 \end{pmatrix}, & \mathbf{b} &= \begin{pmatrix} 4 \\ 3 \end{pmatrix}, \\ \mathbf{c} &= \begin{pmatrix} 3 \\ -1 \end{pmatrix}, & \mathbf{d} &= \begin{pmatrix} -1 \\ -1 \end{pmatrix}, \\ \mathbf{e} &= \begin{pmatrix} -2 \\ 3 \end{pmatrix} \text{ and } \mathbf{x} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}. \end{aligned}$$

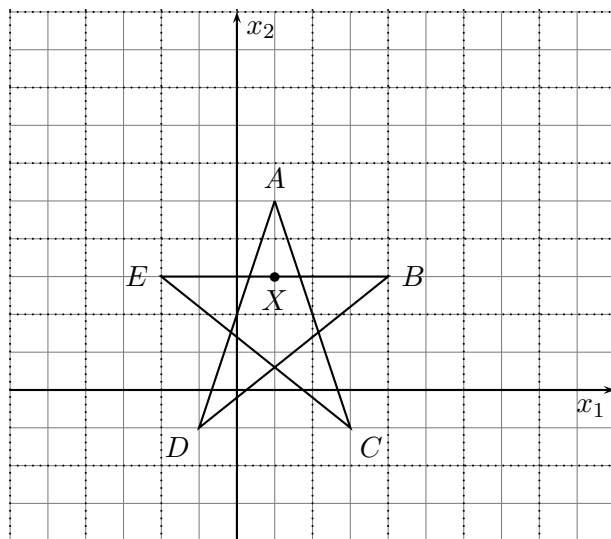


Figure 4(a): A 5-point star.

When $\lambda_1 = \lambda_2 = 2$, the matrix A is $\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$. The points in Figure 4(a) will be “transformed” to A' , B' , C' , D' , E' and X' according to

$$A\mathbf{a} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 5 \end{pmatrix} = \begin{pmatrix} 2 \\ 10 \end{pmatrix},$$

and similarly,

$$A\mathbf{b} = \begin{pmatrix} 8 \\ 6 \end{pmatrix}, \quad A\mathbf{c} = \begin{pmatrix} 6 \\ -2 \end{pmatrix}, \quad A\mathbf{d} = \begin{pmatrix} -2 \\ -2 \end{pmatrix}, \quad A\mathbf{e} = \begin{pmatrix} -4 \\ 6 \end{pmatrix} \text{ and } A\mathbf{x} = \begin{pmatrix} 2 \\ 6 \end{pmatrix}.$$

By Theorem 1 and the remark after it, the line segment AB will be transformed to $A'B'$ and so on. The star will be transformed to one shown in Figure 4(b).

When $\lambda_1 = \lambda_2 = 0.5$, the matrix A is $\begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix}$. The points in Figure 4(a) will be “transformed” according to

$$A\mathbf{a} = \begin{pmatrix} 0.5 \\ 2.5 \end{pmatrix}, A\mathbf{b} = \begin{pmatrix} 2 \\ 1.5 \end{pmatrix}, A\mathbf{c} = \begin{pmatrix} 1.5 \\ -0.5 \end{pmatrix}, A\mathbf{d} = \begin{pmatrix} -0.5 \\ -0.5 \end{pmatrix}, A\mathbf{e} = \begin{pmatrix} -1 \\ 1.5 \end{pmatrix} \quad \text{and} \quad A\mathbf{x} = \begin{pmatrix} 0.5 \\ 1.5 \end{pmatrix}.$$

The star will be transformed to one shown in Figure 4(c).

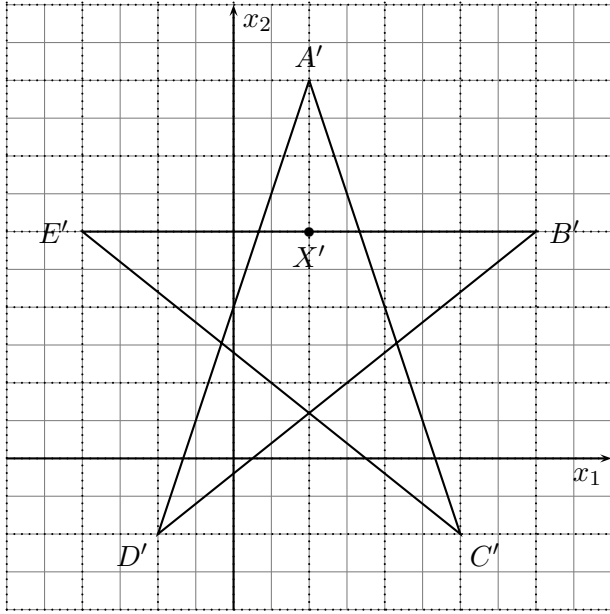


Figure 4(b): Image under $A = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$.

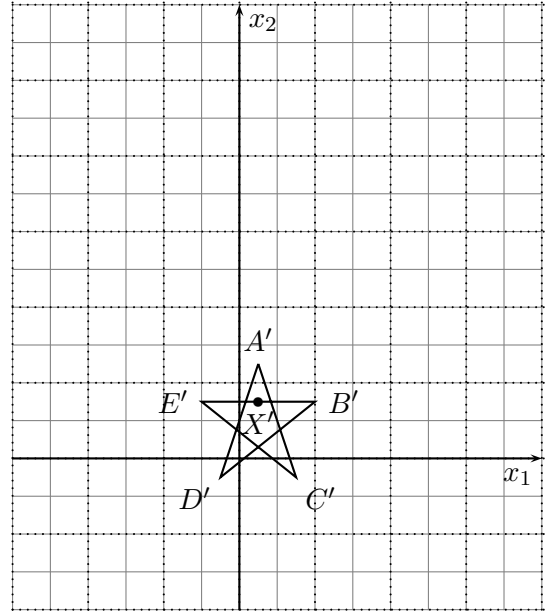


Figure 4(c): Image under $A = \begin{pmatrix} 0.5 & 0 \\ 0 & 0.5 \end{pmatrix}$.

Figure 4(d) shows the image of the 5-point star when $A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$.

The star is stretched to twice the width horizontally.

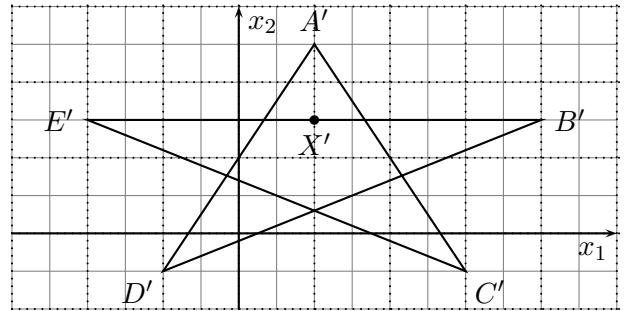


Figure 4(d): Image under $A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$.

◇

Example 3 (Rotation in a plane). Suppose that X is an arbitrary point in a plane and then X is rotated about the origin O anticlockwise by an angle α to a new position X' . Let $\mathbf{x}$ and $\mathbf{x}'$ be the position vectors of the points X and X' respectively. Show that the function $R_\alpha : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $R_\alpha(\mathbf{x}) = \mathbf{x}'$ is a linear transformation. Find the matrix A_α such that $A_\alpha \mathbf{x} = \mathbf{x}'$.

SOLUTION. We know that R_α will be linear if, for all vectors $\mathbf{a}, \mathbf{b} \in \mathbb{R}^2$ and scalar $\lambda \in \mathbb{R}$, the following two conditions are satisfied.

Addition condition	$R_\alpha(\mathbf{a} + \mathbf{b}) = R_\alpha(\mathbf{a}) + R_\alpha(\mathbf{b})$
Scalar multiplication condition	$R_\alpha(\lambda \mathbf{a}) = \lambda R_\alpha(\mathbf{a})$.

We can see the addition condition from Figure 5. The vector formed adding $\mathbf{a}$ and $\mathbf{b}$ first then rotating the sum $\mathbf{a} + \mathbf{b}$ is the same as the one formed by first rotating $\mathbf{a}$ and $\mathbf{b}$ then adding these rotated vectors.

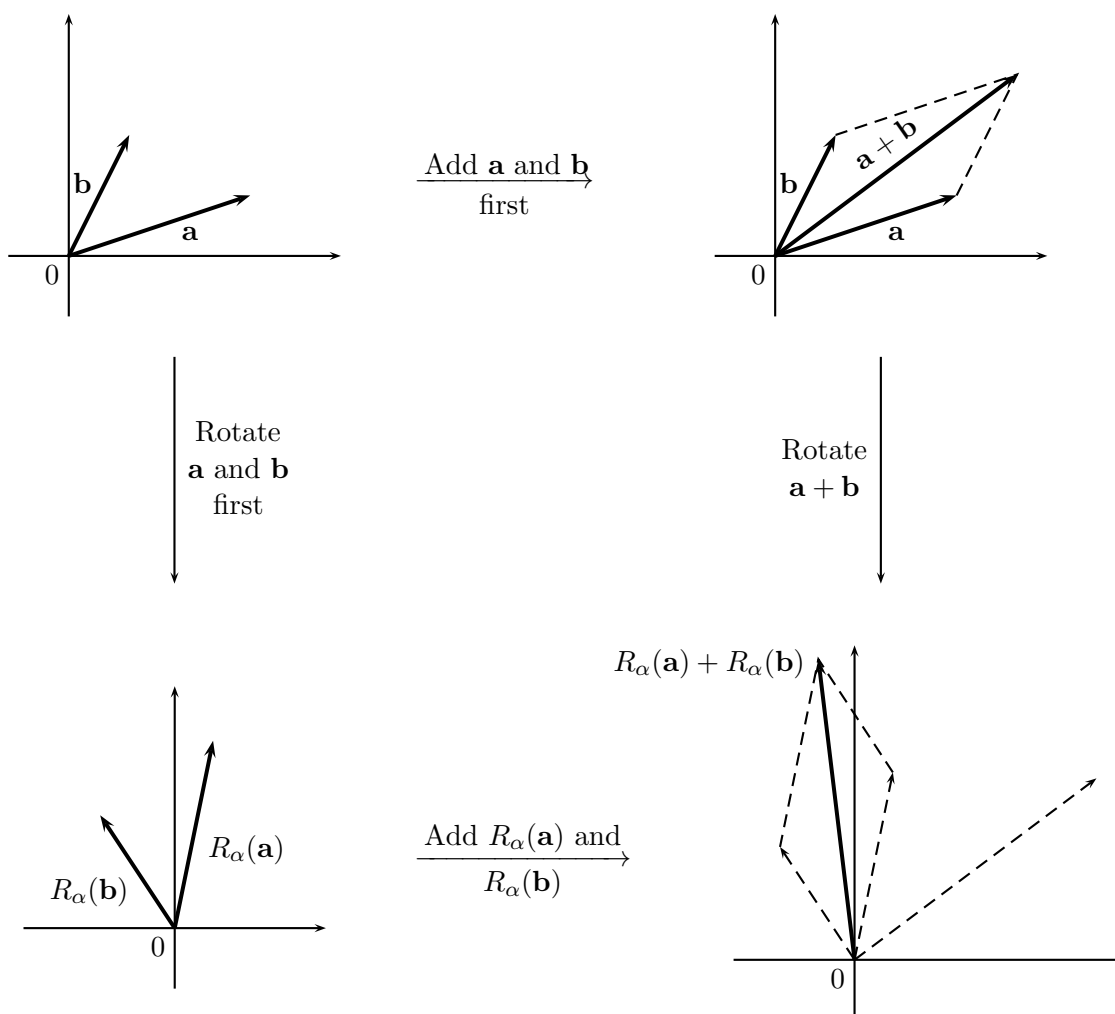


Figure 5: The geometry of the addition condition for rotations.

You should attempt to draw a picture to illustrate the scalar multiplication condition. In any case, since both the addition condition and the scalar multiplication condition hold, R_α is a linear transformation.

By Theorem 2 in Section 7.2, the columns of the matrix A_α are $R_\alpha(\mathbf{e}_1)$ and $R_\alpha(\mathbf{e}_2)$. From

Figure 6 and the fact that the length of both $R_\alpha(\mathbf{e}_1)$ and $R_\alpha(\mathbf{e}_2)$ are 1, we have

$$R_\alpha(\mathbf{e}_1) = \begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix} \quad \text{and} \quad R_\alpha(\mathbf{e}_2) = \begin{pmatrix} -\sin \alpha \\ \cos \alpha \end{pmatrix}.$$

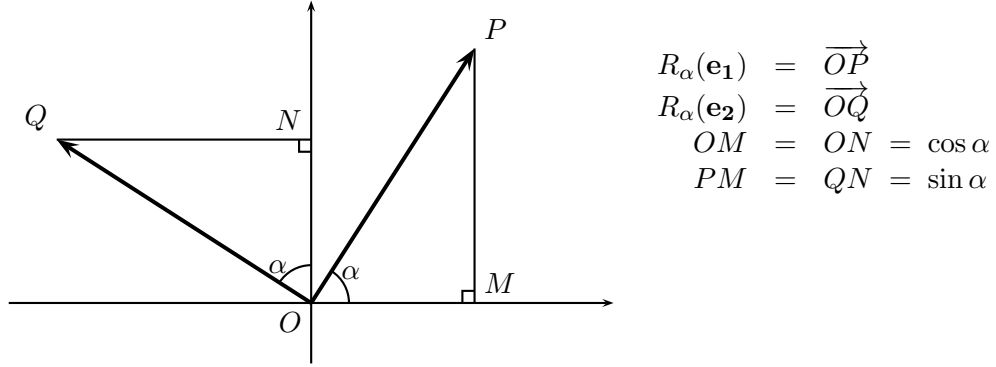


Figure 6.

The matrix

$$A_\alpha = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix},$$

is called the **rotation matrix** for angle α . ◇

Example 4 (Projections). The projection of a vector $\mathbf{x} \in \mathbb{R}^n$ on a fixed, non-zero vector $\mathbf{b} \in \mathbb{R}^n$ is given by

$$\text{proj}_{\mathbf{b}} \mathbf{x} = \frac{\mathbf{x} \cdot \mathbf{b}}{|\mathbf{b}|^2} \mathbf{b}.$$

Show that the function $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by

$$T(\mathbf{x}) = \text{proj}_{\mathbf{b}} \mathbf{x} \quad \text{for} \quad \mathbf{x} \in \mathbb{R}^n.$$

is a linear map.

SOLUTION. Clearly, the domain and codomain are vector spaces. Instead of proving that T is linear by geometric properties, we use the algebraic properties of the dot product. For all $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^n$

$$T(\mathbf{x} + \mathbf{x}') = \text{proj}_{\mathbf{b}}(\mathbf{x} + \mathbf{x}') = \frac{(\mathbf{x} + \mathbf{x}') \cdot \mathbf{b}}{|\mathbf{b}|^2} \mathbf{b} = \frac{\mathbf{x} \cdot \mathbf{b} + \mathbf{x}' \cdot \mathbf{b}}{|\mathbf{b}|^2} \mathbf{b} = T(\mathbf{x}) + T(\mathbf{x}'),$$

and hence the addition condition is satisfied.

Finally, for all $\mathbf{x} \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$,

$$T(\lambda \mathbf{x}) = \frac{(\lambda \mathbf{x}) \cdot \mathbf{b}}{|\mathbf{b}|^2} \mathbf{b} = \lambda \left(\frac{\mathbf{x} \cdot \mathbf{b}}{|\mathbf{b}|^2} \mathbf{b} \right) = \lambda T(\mathbf{x}).$$

Thus, the scalar multiplication condition is also satisfied, and therefore T is a linear map. ◇

Example 5 (Dot Product). Let $\mathbf{b}$ be a fixed vector in $\mathbb{R}^n$. Show that the function $T : \mathbb{R}^n \rightarrow \mathbb{R}$, defined by

$$T(\mathbf{x}) = \mathbf{b} \cdot \mathbf{x} \quad \text{for } \mathbf{x} \in \mathbb{R}^n,$$

is a linear map.

The proof that T is a linear map is similar to the previous example and is left as an exercise. $\diamond$

The following examples show the importance of the linear maps defined by dot product.

Example 6. From Example 5, for each $1 \leq i \leq n$ the function $P_i : \mathbb{R}^n \rightarrow \mathbb{R}$ is defined by

$$P_i(\mathbf{x}) = \mathbf{e}_i \cdot \mathbf{x} \quad \text{for } \mathbf{x} \in \mathbb{R}^n,$$

where $\mathbf{e}_i$ is the i th standard basis element. It is not difficult to see that if $\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$, the value $P_i(\mathbf{x})$ is simply x_i , the i th component of $\mathbf{x}$. $\diamond$

This example can be generalised to any basis of $\mathbb{R}^n$.

Example 7. Suppose that $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ is an orthonormal basis for $\mathbb{R}^n$ and $1 \leq i \leq n$. The function $P_i : \mathbb{R}^n \rightarrow \mathbb{R}$ is defined by

$$P_i(\mathbf{x}) = \mathbf{v}_i \cdot \mathbf{x} \quad \text{for } \mathbf{x} \in \mathbb{R}^n.$$

By the argument used in Example 6 in Section 6.6 we can prove that if $\mathbf{x} = \lambda_1 \mathbf{v}_1 + \dots + \lambda_n \mathbf{v}_n$, the value $P_i(\mathbf{x})$ is simply λ_i , the coefficient of $\mathbf{v}_i$ in the unique way of writing $\mathbf{x}$ as a linear combination of the basis vectors. $\diamond$

7.4 Subspaces associated with linear maps

There are two important subspaces associated with a linear map. These subspaces are called the **kernel** (or **null space**, which is the name Maple uses) of the linear map and the **image** (or range) of the map. Informally, the kernel is the set of zeroes of the function, and the image is the set of all function values. This is shown diagrammatically in Figure 7. You should of course not take this picture too literally — all the sets drawn as discs are vector spaces!

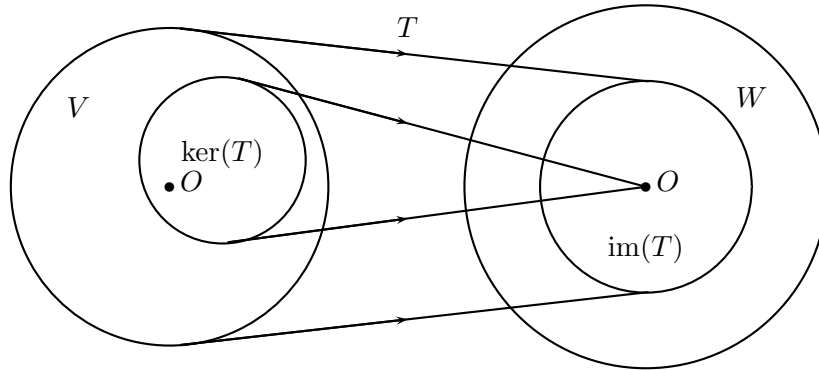


Figure 7: Kernel and image.

7.4.1 The kernel of a map

You will be familiar with the fact that one of the important properties of functions (for example of quadratics or polynomials) is the values of their zeroes. The set of zeroes of a linear map is also of importance.

Definition 1. Let $T : V \rightarrow W$ be a linear map. Then the **kernel** of T (written $\ker(T)$) is the set of all zeroes of T , that is, it is the subset of the domain V defined by

$$\ker(T) = \{\mathbf{v} \in V : T(\mathbf{v}) = \mathbf{0}\}.$$

Example 1. Showing that a vector $\mathbf{v}$ is in the kernel of a linear map T is simply a verification that $T(\mathbf{v}) = \mathbf{0}$. In particular, $\mathbf{0} \in \ker(T)$ for any linear map T , since $T(\mathbf{0}) = \mathbf{0}$. $\diamond$

Example 2 (Dot Product). In Example 5 of Section 7.3, we showed that the function $T : \mathbb{R}^n \rightarrow \mathbb{R}$ defined by $T(\mathbf{x}) = \mathbf{b} \cdot \mathbf{x}$ for $\mathbf{x} \in \mathbb{R}^n$ is a linear map. The kernel of T is

$$\ker(T) = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{b} \cdot \mathbf{x} = 0\},$$

that is, $\ker(T)$ is the set of vectors which are orthogonal to the given fixed vector $\mathbf{b}$.

For the special case that $\mathbf{x} \in \mathbb{R}^3$, the equation $\mathbf{b} \cdot \mathbf{x} = 0$ is the point-normal form of the equation of a plane in $\mathbb{R}^3$, and hence $\ker(T)$ corresponds to the points on a plane with normal $\mathbf{b}$ which passes through the origin. $\diamond$

For the important special case of a linear map $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ associated with an $m \times n$ matrix A , the kernel has a simple interpretation. For matrices, the definition of kernel becomes:

Definition 2. For an $m \times n$ matrix A , the **kernel** of A is the subset of $\mathbb{R}^n$ defined by

$$\ker(A) = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x} = \mathbf{0}\},$$

that is, it is the set of all solutions of the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

Example 3. Suppose that $A = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix}$ and $\mathbf{x} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$. Since

$$A\mathbf{x} = \begin{pmatrix} 1 & 2 \\ 3 & 6 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \times 2 + 2 \times (-1) \\ 3 \times 2 + 6 \times (-1) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix},$$

which means that $\mathbf{x} \in \ker(A)$ by definition.

To find the kernel of a matrix A , we need to find the solution set of the equation $A\mathbf{x} = \mathbf{0}$.

Example 4. Find the kernel of the matrix A , where

$$A = \begin{pmatrix} 1 & 4 & 2 & 7 \\ 3 & 6 & 0 & 15 \\ 2 & -4 & -8 & 2 \end{pmatrix}.$$

SOLUTION. The kernel is the set of all solutions of the homogeneous system of equations $A\mathbf{x} = \mathbf{0}$. An equivalent row-echelon form U for A is

$$U = \begin{pmatrix} 1 & 4 & 2 & 7 \\ 0 & -6 & -6 & -6 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

We then set parameters to the variables of the non-leading columns — $x_3 = \lambda_1$ and $x_4 = \lambda_2$. By back substitution, we obtain the solution of $A\mathbf{x} = \mathbf{0}$ as

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \lambda_1 \begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} -3 \\ -1 \\ 0 \\ 1 \end{pmatrix},$$

and hence,

$$\ker(A) = \left\{ \mathbf{x} \in \mathbb{R}^4 : \mathbf{x} = \lambda_1 \begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} -3 \\ -1 \\ 0 \\ 1 \end{pmatrix} \text{ for } \lambda_1, \lambda_2 \in \mathbb{R} \right\}.$$

In this example, the kernel can be interpreted geometrically as a plane in $\mathbb{R}^4$ through the origin parallel to $\begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} -3 \\ -1 \\ 0 \\ 1 \end{pmatrix}$. ◇

A very important property of the kernel of a linear map is given in the following theorem.

Theorem 1. *If $T : V \rightarrow W$ is a linear map, then $\ker(T)$ is a subspace of the domain V .*

Proof. We use the Subspace Theorem (Theorem 1) of Section 6.3 and prove that $\ker(T)$ is a non-empty subset of V which is closed under addition and scalar multiplication.

It is not the empty set, since $T(\mathbf{0}) = \mathbf{0}$ and so $\mathbf{0} \in \ker(T)$.

Suppose that $\mathbf{v}, \mathbf{v}' \in \ker(T)$ and $\lambda \in \mathbb{F}$. Since T is linear, it satisfies the addition and scalar multiplication conditions, so we have

$$T(\mathbf{v} + \mathbf{v}') = T(\mathbf{v}) + T(\mathbf{v}') = \mathbf{0} \quad \text{and} \quad T(\lambda\mathbf{v}) = \lambda T(\mathbf{v}) = \mathbf{0},$$

and hence both $\mathbf{v} + \mathbf{v}'$ and $\lambda\mathbf{v}$ are in $\ker(T)$. Thus, $\ker(T)$ is closed under addition and scalar multiplication, and the proof is complete. $\square$

The dimension of the kernel is important and is given a special name.

Definition 3. The **nullity** of a linear map T is the dimension of $\ker(T)$. The **nullity** of a matrix A is the dimension of $\ker(A)$.

Proposition 2. Let A be an $m \times n$ matrix with real entries and $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ the associated linear transformation. Then

$$\ker(T_A) = \ker(A)$$

Proof. $T_A(\mathbf{x}) = \mathbf{0} \Leftrightarrow A\mathbf{x} = \mathbf{0}$. $\square$

The nullity of a matrix A can be easily obtained from the properties of row-echelon forms by using the following result.

Proposition 3. For a matrix A :

$$\begin{aligned} \text{nullity}(A) &= \text{maximum number of independent vectors in the solution space of } A\mathbf{x} = \mathbf{0} \\ &= \text{number of parameters in the solution of } A\mathbf{x} = \mathbf{0} \text{ obtained by Gaussian} \\ &\quad \text{elimination and back substitution} \\ &= \text{number of non-leading columns in an equivalent row-echelon form } U \text{ for } A. \end{aligned}$$

Although a general proof of this proposition is not difficult to construct, we shall restrict ourselves to looking at an example.

Example 4 (continued). Find $\text{nullity}(A)$, and a basis for $\ker(A)$, for

$$A = \begin{pmatrix} 1 & 4 & 2 & 7 \\ 3 & 6 & 0 & 15 \\ 2 & -4 & -8 & 2 \end{pmatrix}.$$

SOLUTION. We have found that any vector $\mathbf{x} \in \ker(A)$ can be written as

$$\mathbf{x} = \lambda_1 \begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} -3 \\ -1 \\ 0 \\ 1 \end{pmatrix}.$$

This is a linear combination of the two vectors in the parametric vector form for the solution of $A\mathbf{x} = \mathbf{0}$. These two vectors are linearly independent, since if $\mathbf{x} = \mathbf{0}$ then the parameters λ_1 and λ_2 are both zero (look at the third and fourth rows of the linear combination). Thus, we obtain a basis for $\ker(A)$,

$$\left\{ \begin{pmatrix} 2 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -3 \\ -1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

and hence $\text{nullity}(A) = \dim(\ker(A)) = 2$. This illustrates Proposition 3. $\diamond$

For matrices, there is a close relationship between linear independence of the columns and the nullity of the matrix.

Proposition 4. The columns of a matrix A are linearly independent if and only if $\text{nullity}(A) = 0$.

Proof. From Proposition 1 of Section 6.5, the columns of A are linearly independent if and only if $\mathbf{x} = \mathbf{0}$ is the only solution of $A\mathbf{x} = \mathbf{0}$. That is, if and only if $\mathbf{0}$ is the only element of $\ker(A)$, in which case, $\text{nullity}(A) = \dim(\ker(A)) = 0$. $\square$

7.4.2 Image

The range or image of a function is the set of all function values (see, for example, Appendix 7.10). In this course we will usually use the term *image* rather than *range*. A formal definition of the image of a linear map is as follows.

Definition 4. Let $T : V \rightarrow W$ be a linear map. Then the **image** of T is the set of all function values of T , that is, it is the subset of the codomain W defined by

$$\text{im}(T) = \{\mathbf{w} \in W : \mathbf{w} = T(\mathbf{v}) \text{ for some } \mathbf{v} \in V\}.$$

For the special case of a linear map associated with a real $m \times n$ matrix, the definition becomes:

Definition 5. The **image** of an $m \times n$ matrix A is the subset of $\mathbb{R}^m$ defined by

$$\text{im}(A) = \{\mathbf{b} \in \mathbb{R}^m : \mathbf{b} = A\mathbf{x} \text{ for some } \mathbf{x} \in \mathbb{R}^n\}.$$

We have met this set several times before. In the language of linear equations, this set $\text{im}(A)$ is just the set of all right-hand-side vectors $\mathbf{b}$ for which the equation $A\mathbf{x} = \mathbf{b}$ has a solution, and in

vector-space language it is just the span of the columns of the matrix A , that is, the column space of A . Thus, we have

$$\begin{aligned}\text{range}(A) &= \text{im}(A) = \text{col}(A) = \text{span}(\text{columns of } A) \\ &= \{\mathbf{b} \in \mathbb{R}^m : A\mathbf{x} = \mathbf{b} \text{ has a solution}\}.\end{aligned}$$

These connections mean that any questions about the image of a matrix can be solved by the methods previously given for linear equations and spans in Chapter 6. It is useful to give an example, as it will serve to review some of the previous results on vector spaces and linear equations.

Example 4 (continued). Find conditions on a vector $\mathbf{b}$ for $\mathbf{b}$ to be in $\text{im}(A)$ where

$$A = \begin{pmatrix} 1 & 4 & 2 & 7 \\ 3 & 6 & 0 & 15 \\ 2 & -4 & -8 & 2 \end{pmatrix}.$$

SOLUTION. We look for conditions on $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \in \mathbb{R}^3$ for $A\mathbf{x} = \mathbf{b}$ to have a solution.

For hand calculations on a small system of equations, the simplest method of solution is as follows. Instead of put b_1, b_2, b_3 on the right hand side, the three right-hand columns of the following augmented matrix are coefficients of b_1, b_2, b_3 .

$$(A|\mathbf{b}) = \left(\begin{array}{cccc|ccc} 1 & 4 & 2 & 7 & 1 & 0 & 0 \\ 3 & 6 & 0 & 15 & 0 & 1 & 0 \\ 2 & -4 & -8 & 2 & 0 & 0 & 1 \end{array} \right).$$

On reduction to row-echelon form using Gaussian elimination, we find

$$(U|\mathbf{y}) = \left(\begin{array}{cccc|ccc} 1 & 4 & 2 & 7 & 1 & 0 & 0 \\ 0 & -6 & -6 & -6 & -3 & 1 & 0 \\ 0 & 0 & 0 & 0 & 4 & -2 & 1 \end{array} \right).$$

This system of equations has a solution if and only if the components of $\mathbf{b}$ satisfy

$$4b_1 - 2b_2 + b_3 = 0.$$

In this case, $\text{im}(A)$ has a geometric interpretation as a plane through the origin in $\mathbb{R}^3$ with normal $\begin{pmatrix} 4 \\ -2 \\ 1 \end{pmatrix}$. In vector-space language, $\text{im}(A)$ is a two-dimensional subspace of $\mathbb{R}^3$. $\diamond$

Note that for larger matrices it is preferable to use computer packages such as Maple to solve the equations.

An extremely important property of the image of a linear map is as follows.

Theorem 5. *Let $T : V \rightarrow W$ be a linear map between vector spaces V and W . Then $\text{im}(T)$ is a subspace of the codomain W of T .*

Proof. We use the Subspace Theorem (Theorem 1 of Section 6.3) and show that $\text{im}(T)$ is a non-empty subset of W which is closed under vector addition and scalar multiplication.

We note first that $\text{im}(T)$ is a subset of W . Since, from Proposition 1 of Section 7.1, $T(\mathbf{0}) = \mathbf{0}$, we see that $\mathbf{0} \in \text{im}(T)$.

Closure under addition. If $\mathbf{w}, \mathbf{w}' \in \text{im}(T)$, then

$$\mathbf{w} = T(\mathbf{v}) \quad \text{for some } \mathbf{v} \in V \quad \text{and} \quad \mathbf{w}' = T(\mathbf{v}') \quad \text{for some } \mathbf{v}' \in V,$$

and hence,

$$\mathbf{w} + \mathbf{w}' = T(\mathbf{v}) + T(\mathbf{v}') = T(\mathbf{v} + \mathbf{v}').$$

But, since V is a vector space, $\mathbf{v} + \mathbf{v}' \in V$, and therefore $\mathbf{w} + \mathbf{w}' \in \text{im}(T)$ and $\text{im}(T)$ is closed under addition.

Closure under scalar multiplication. If $\mathbf{w} \in \text{im}(T)$ and $\lambda \in F$, then

$$\mathbf{w} = T(\mathbf{v}) \quad \text{for some } \mathbf{v} \in V, \quad \text{and hence } \lambda\mathbf{w} = \lambda T(\mathbf{v}) = T(\lambda\mathbf{v}).$$

But, since V is a vector space, $\lambda\mathbf{v} \in V$, and therefore $\lambda\mathbf{w} \in \text{im}(T)$ and $\text{im}(T)$ is closed under scalar multiplication.

The proof is complete. □

The next result is obvious.

Proposition 6. Let A be an $m \times n$ matrix with real entries and $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ the associated linear transformation. Then

$$\text{im}(A) = \text{im}(T_A)$$

We have shown in Theorem 5 that $\text{im}(T)$ is always a subspace of the codomain of T . Thus, the fundamental vector-space properties of basis and dimension must apply to $\text{im}(T)$. The dimension of the image is very important and it has therefore been given a special name.

Definition 6. The **rank** of a linear map T is the dimension of $\text{im}(T)$. The **rank** of a matrix A is the dimension of $\text{im}(A)$.

The rank is usually regarded as one of the most important properties of a matrix, since it is the maximum number of linearly independent right-hand-side vectors for which a solution to $A\mathbf{x} = \mathbf{b}$ can be found. Some important properties of the rank of a matrix are summarised in the following proposition.

Proposition 7. For a matrix A :

$$\begin{aligned} \text{rank}(A) &= \text{maximal number of linearly independent columns of } A \\ &= \text{number of leading columns in a row-echelon form } U \text{ for } A \end{aligned}$$

Proof. From before, $\text{im}(A) = \text{col}(A) = \text{span}(\text{columns of } A)$. A basis for $\text{span}(\text{columns of } A)$ is a maximal set of linearly independent columns of A . One maximal set of linearly independent columns of A are the columns which reduce to leading columns in a row-echelon form U . Hence,

$$\begin{aligned} \text{number of leading columns} &= \text{number of linearly independent columns of } A \\ &= \text{number of vectors in basis for } \text{col}(A) \\ &= \dim(\text{col}(A)) = \dim(\text{im}(A)) = \text{rank}(A). \end{aligned}$$

□

Example 4 (continued). Find $\text{rank}(A)$, and a basis for $\text{im}(A)$, for

$$A = \begin{pmatrix} 1 & 4 & 2 & 7 \\ 3 & 6 & 0 & 15 \\ 2 & -4 & -8 & 2 \end{pmatrix}.$$

SOLUTION. Since there are two leading columns (1 and 2) in the row-echelon form

$$U = \begin{pmatrix} 1 & 4 & 2 & 7 \\ 0 & -6 & -6 & -6 \\ 0 & 0 & 0 & 0 \end{pmatrix},$$

hence $\text{rank}(A) = 2$.

A basis for $\text{im}(A)$ therefore contains two vectors. One maximal set of linearly independent columns of A is columns 1 and 2 of A , and hence a basis for $\text{im}(A)$ is $\left\{ \begin{pmatrix} 1 \\ 3 \\ 2 \end{pmatrix}, \begin{pmatrix} 4 \\ 6 \\ -4 \end{pmatrix} \right\}$. ◇

7.4.3 Rank, nullity and solutions of $A\mathbf{x} = \mathbf{b}$

Example 4 illustrates the following important fact about the rank and nullity of a matrix.

Theorem 8 (Rank-Nullity Theorem for Matrices). *For any matrix A ,*

$$\text{rank}(A) + \text{nullity}(A) = \text{number of columns of } A.$$

Proof. Let U be an equivalent row-echelon form for A obtained by the Gaussian elimination algorithm. Then, from Proposition 7, $\text{rank}(A) = \text{number of leading columns in } U$. Also, from Proposition 3, $\text{nullity}(A) = \text{number of non-leading columns in } U$. But, of course, $\text{number of leading columns} + \text{number of non-leading columns} = \text{total number of columns in } U = \text{total number of columns in } A$, and the result is proved. □

The above theorem is equivalent to the following result for linear maps between finite dimensional vector spaces.

Theorem 9 (Rank-Nullity Theorem). *Suppose V and W are finite dimensional vector spaces and $T : V \rightarrow W$ is linear. Then*

$$\text{rank}(T) + \text{nullity}(T) = \dim(V).$$

A proof of Theorem 9, using a suitably constructed basis of V , is given in Section 7.9.

For matrices, a very common use of rank and nullity is to classify the types of solution of a system of linear equations $A\mathbf{x} = \mathbf{b}$. The basic results are summarised in the following proposition.

Theorem 10. *The equation $A\mathbf{x} = \mathbf{b}$ has:*

1. *no solution if $\text{rank}(A) \neq \text{rank}([A|\mathbf{b}])$, and*
2. *at least one solution if $\text{rank}(A) = \text{rank}([A|\mathbf{b}])$. Further,*
 - i) *if $\text{nullity}(A) = 0$ the solution is unique, whereas,*
 - ii) *if $\text{nullity}(A) = \nu > 0$, then the general solution is of the form*

$$\mathbf{x} = \mathbf{x}_p + \lambda_1 \mathbf{k}_1 + \cdots + \lambda_\nu \mathbf{k}_\nu \quad \text{for } \lambda_1, \dots, \lambda_\nu \in \mathbb{R},$$

where $\mathbf{x}_p$ is any solution of $A\mathbf{x} = \mathbf{b}$, and where $\{\mathbf{k}_1, \dots, \mathbf{k}_\nu\}$ is a basis for $\ker(A)$.

Proof. Let U and $(U|\mathbf{y})$ be equivalent row-echelon forms for A and $(A|\mathbf{b})$ obtained by the Gaussian elimination algorithm.

Now, from Chapter 4, we know that $A\mathbf{x} = \mathbf{b}$ has a solution if and only if the right-hand-side column $\mathbf{y}$ is a non-leading column, and hence if and only if the numbers of leading columns in U and $(U|\mathbf{y})$ are equal. But, from Proposition 7, $\text{rank}(A) = \text{number of leading columns in } U$, and $\text{rank}([A|\mathbf{b}]) = \text{number of leading columns in } (U|\mathbf{y})$, and thus $A\mathbf{x} = \mathbf{b}$ has a solution if and only if the ranks of A and $(A|\mathbf{b})$ are equal.

The proof of parts 2(i) and 2(ii) follows immediately from the relation between solutions of a non-homogeneous system and the corresponding homogeneous system (see Chapter 4) and the fact that $\text{nullity}(A)$ is equal to the maximum number of linearly independent solutions of the homogeneous equation $A\mathbf{v} = \mathbf{0}$. $\square$

NOTE. A similar type of solution to the general solution in 2(ii) above is also obtained as the solution of a **linear** differential equation. In this differential equation case, $\mathbf{x}_p$ is called a “particular solution” and the parametric terms are called the “complementary function”. The similarity between the two types of solution is due to the fact that both the matrix and differential equation problems involve linear functions.

Example 5. Illustrate the above rules with the system of equations given by the augmented matrix

$$(A|\mathbf{b}) = \left(\begin{array}{ccccc|c} 0 & 0 & 2 & -1 & 3 & 1 \\ 1 & -2 & \frac{1}{2} & 3 & 4 & -1 \\ 3 & 2 & 1 & 4 & 6 & 0 \end{array} \right).$$

SOLUTION. Gaussian elimination gives

$$(U|\mathbf{y}) = \left(\begin{array}{ccccc|c} 1 & -2 & \frac{1}{2} & 3 & 4 & -1 \\ 0 & 8 & -\frac{1}{2} & -5 & -6 & 3 \\ 0 & 0 & 2 & -1 & 3 & 1 \end{array} \right).$$

The system has a solution as $\mathbf{y}$ is a non-leading column. The number of leading columns in U is 3, and hence $\text{rank}(A) = 3$. Similarly, $\text{rank}(A|\mathbf{b}) = 3$.

On back substitution, the parametric vector form of the solution is found to be

$$\mathbf{x} = \begin{pmatrix} -\frac{7}{16} \\ \frac{13}{32} \\ \frac{1}{2} \\ 0 \\ 0 \end{pmatrix} + \lambda_1 \begin{pmatrix} -\frac{31}{16} \\ \frac{21}{32} \\ \frac{1}{2} \\ 1 \\ 0 \end{pmatrix} + \lambda_2 \begin{pmatrix} -\frac{31}{16} \\ \frac{21}{32} \\ \frac{3}{2} \\ 0 \\ 1 \end{pmatrix} = \mathbf{x}_p + \lambda_1 \mathbf{k}_1 + \lambda_2 \mathbf{k}_2.$$

Note that

$$A \begin{pmatrix} -\frac{7}{16} \\ \frac{13}{32} \\ \frac{1}{2} \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}; \quad A \begin{pmatrix} -\frac{31}{16} \\ \frac{21}{32} \\ \frac{1}{2} \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}; \quad A \begin{pmatrix} -\frac{31}{16} \\ \frac{21}{32} \\ \frac{3}{2} \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},$$

and that the number of non-leading columns of $U = 2 = \text{nullity}(A) =$ the number of parameters in solution. This parametric vector form is the general solution of $A\mathbf{x} = \mathbf{b}$, and as expected it is the sum of a “particular solution” of $A\mathbf{x} = \mathbf{b}$ and a “complementary function” which is a linear combination of two linearly independent solutions of $A\mathbf{x} = \mathbf{0}$ (a basis for $\ker(A)$). $\diamond$

7.5 Further applications and examples of linear maps

Although the theory that we have developed so far in this chapter applies to all linear maps, most examples have been restricted to maps for which the domain is $\mathbb{R}^n$ and the codomain is $\mathbb{R}^m$. In this section we will give some examples of linear maps in which the domain and codomain are other kinds of vector spaces.

A simple, but useful, map in any vector space is the map which takes a vector to itself.

Example 1. The **identity map** $\text{id}_V : V \rightarrow V$ on a vector space V is defined by

$$\text{id}_V(\mathbf{v}) = \mathbf{v} \quad \text{for all } \mathbf{v} \in V.$$

This map is linear, since for all $\mathbf{v}, \mathbf{v}' \in V$ and all scalar λ ,

$$\begin{aligned} \text{id}_V(\mathbf{v} + \mathbf{v}') &= \mathbf{v} + \mathbf{v}' = \text{id}_V(\mathbf{v}) + \text{id}_V(\mathbf{v}') \\ \text{and } \text{id}_V(\lambda\mathbf{v}) &= \lambda\mathbf{v} = \lambda \text{id}_V(\mathbf{v}). \end{aligned}$$

$\diamond$

In Theorems 2 and 3 of Section 7.1 we showed that linear maps had the important property that they preserved linear combinations. As the next example shows, every linear combination can also be regarded as the image of a linear map.

[H] **Example 2.** Let $S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be a subset of a vector space V and let $x_1, \dots, x_n \in \mathbb{R}$. Show that the map T given by

$$T(\mathbf{x}) = x_1\mathbf{v}_1 + \dots + x_n\mathbf{v}_n \quad \text{where} \quad \mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n$$

is a linear map.

SOLUTION. The rule obviously defines a function, since $T(\mathbf{x})$ is uniquely determined for each $\mathbf{x} \in \mathbb{R}^n$.

To prove that T is linear we use Theorem 2 of Section 7.1.

Suppose $\mathbf{x}, \mathbf{x}' \in \mathbb{R}^n$ and $\lambda, \lambda' \in \mathbb{R}$. Then

$$\lambda\mathbf{x} + \lambda'\mathbf{x}' = (\lambda x_1 + \lambda'x'_1, \dots, \lambda x_n + \lambda'x'_n),$$

and hence

$$\begin{aligned} T(\lambda\mathbf{x} + \lambda'\mathbf{x}') &= (\lambda x_1 + \lambda'x'_1)\mathbf{v}_1 + \dots + (\lambda x_n + \lambda'x'_n)\mathbf{v}_n \\ &= \lambda(x_1\mathbf{v}_1 + \dots + x_n\mathbf{v}_n) + \lambda'(x'_1\mathbf{v}_1 + \dots + x'_n\mathbf{v}_n) \\ &= \lambda T(\mathbf{x}) + \lambda' T(\mathbf{x}'). \end{aligned}$$

Thus T is a linear map. ◇

This example shows that all properties of linear combinations discussed in Chapter 6 can in fact be restated in the language of linear maps.

Example 3. Let V be a vector space over the real numbers, and let $B = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be an ordered basis for V . For any $\mathbf{v} \in V$, we can write the vector uniquely as a linear combination of B ,

$$\mathbf{v} = x_1\mathbf{v}_1 + \dots + x_n\mathbf{v}_n.$$

Show that the rule $T : V \rightarrow \mathbb{R}^n$ defined by

$$T(\mathbf{v}) = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \quad \text{for} \quad \mathbf{v} \in V$$

is a linear map.

SOLUTION. Obviously the function T has domain V and codomain $\mathbb{R}^n$ which are vector spaces.

To prove that this function is a linear map, we check the addition and scalar multiplication conditions. For all $\lambda \in \mathbb{R}$ and $\mathbf{v}, \mathbf{v}' \in V$, we can write in a unique way that

$$\mathbf{v} = x_1\mathbf{v}_1 + \dots + x_n\mathbf{v}_n \quad \text{and} \quad \mathbf{v}' = x'_1\mathbf{v}_1 + \dots + x'_n\mathbf{v}_n.$$

Since

$$\mathbf{v} + \mathbf{v}' = (x_1 + x'_1)\mathbf{v}_1 + \dots + (x_n + x'_n)\mathbf{v}_n \quad \text{and} \quad \lambda\mathbf{v} = (\lambda x_1)\mathbf{v}_1 + \dots + (\lambda x_n)\mathbf{v}_n$$

we have

$$T(\mathbf{v} + \mathbf{v}') = \begin{pmatrix} x_1 + x'_1 \\ \vdots \\ x_n + x'_n \end{pmatrix} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} x'_1 \\ \vdots \\ x'_n \end{pmatrix} = T(\mathbf{v}) + T(\mathbf{v}')$$

and
$$T(\lambda \mathbf{v}) = \begin{pmatrix} \lambda x_1 \\ \vdots \\ \lambda x_n \end{pmatrix} = \lambda \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = \lambda T(\mathbf{v}),$$

and hence T is a linear map. ◇

[X] REMARK: The above example simply says that the function which maps a vector to its coordinate vector with respect to a basis is linear.

We shall now give some examples of linear maps associated with the vector spaces of polynomials and real-valued functions.

Example 4. Show that the function $T : \mathbb{C}^3 \rightarrow \mathbb{P}_2(\mathbb{C})$ defined by $T \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} = p$, where $a_0, a_1, a_2 \in \mathbb{C}$ and

$$p(z) = a_0 + a_1 + (a_2 + 3a_0)z + a_1z^2 \quad \text{for } z \in \mathbb{C},$$

is a linear map.

Before we solve this problem, note that an argument of T is a complex vector $\mathbf{a} = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} \in \mathbb{C}^3$, while the corresponding function value $p = T(\mathbf{a})$ is a complex polynomial of degree less than or equal to 2. Some function values are the polynomials given by

$$\left(T \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} \right) (z) = 1 + 2 + (3 + 3)z + 2z^2 = 3 + 6z + 2z^2,$$

$$\left(T \begin{pmatrix} -2 \\ 0 \\ i \end{pmatrix} \right) (z) = -2 + (i - 6)z, \quad \left(T \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right) (z) = 0.$$

SOLUTION. We use 7.1.2. Let $\lambda, \lambda' \in \mathbb{C}$, $\mathbf{a} = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} \in \mathbb{C}^3$, $\mathbf{a}' = \begin{pmatrix} a'_0 \\ a'_1 \\ a'_2 \end{pmatrix} \in \mathbb{C}^3$ and let

$s = T(\lambda \mathbf{a} + \lambda' \mathbf{a}')$, $p = T(\mathbf{a})$, and $q = T(\mathbf{a}')$. Then

$$s = T \begin{pmatrix} \lambda a_0 + \lambda' a'_0 \\ \lambda a_1 + \lambda' a'_1 \\ \lambda a_2 + \lambda' a'_2 \end{pmatrix},$$

and hence

$$\begin{aligned}
s(z) &= \lambda a_0 + \lambda' a'_0 + \lambda a_1 + \lambda' a'_1 + (\lambda a_2 + \lambda' a'_2 + 3(\lambda a_0 + \lambda' a'_0))z + (\lambda a_1 + \lambda' a'_1)z^2 \\
&= \lambda(a_0 + a_1 + (a_2 + 3a_0)z + a_1 z^2) + \lambda'(a'_0 + a'_1 + (a'_2 + 3a'_0)z + a'_1 z^2) \\
&= \lambda p(z) + \lambda' q(z).
\end{aligned}$$

Thus, $T(\lambda \mathbf{a} + \lambda' \mathbf{a}') = s = \lambda p + \lambda' q = \lambda T(\mathbf{a}) + \lambda' T(\mathbf{a}')$, and hence $\diamond$

As the next examples show, calculus provides many important applications of linear maps as differentiation and integration are both associated with linear maps.

Example 5 (Differentiation of polynomials). Let $\mathbb{P}_n(\mathbb{R})$ be the vector space of real polynomials of degree less than or equal to n . Show that the function $D : \mathbb{P}_n(\mathbb{R}) \rightarrow \mathbb{P}_{n-1}(\mathbb{R})$, defined by

$$D(p) = p', \quad \text{where} \quad p'(x) = \frac{dp}{dx} \quad \text{for} \quad p \in \mathbb{P}_n(\mathbb{R}) \quad \text{and} \quad x \in \mathbb{R},$$

is a linear map.

SOLUTION. Firstly, we note that if p is a polynomial of degree k then the derivative p' exists and is a polynomial of degree $k - 1$. Hence, if $p \in \mathbb{P}_n(\mathbb{R})$ then $p' \in \mathbb{P}_{n-1}(\mathbb{R})$, and thus $D : \mathbb{P}_n(\mathbb{R}) \rightarrow \mathbb{P}_{n-1}(\mathbb{R})$ is a function.

We now prove D is linear by checking the addition and scalar multiplication conditions of the definition of a linear map.

For all $p, q \in \mathbb{P}_n(\mathbb{R})$, we have from the properties of derivatives that

$$(p + q)'(x) = \frac{d}{dx}(p(x) + q(x)) = \frac{d}{dx}p(x) + \frac{d}{dx}q(x) = p'(x) + q'(x).$$

Hence, $D(p + q) = (p + q)' = p' + q' = D(p) + D(q)$, and the addition condition is satisfied.

Further, for all $p \in \mathbb{P}_n(\mathbb{R})$ and $\lambda \in \mathbb{R}$, we have that

$$\frac{d}{dx}(\lambda p(x)) = \lambda \frac{d}{dx}p(x),$$

and hence $D(\lambda p) = \lambda D(p)$ and the scalar multiplication condition is also satisfied. Thus, D is a linear map. (Note that nullity $(D) = 1$ and rank $(D) = n$.) $\diamond$

Example 6 (Integration of Polynomials). Show that the function $I : \mathbb{P}_n(\mathbb{R}) \rightarrow \mathbb{P}_{n+1}(\mathbb{R})$, defined by $I(p) = q$, where

$$q(x) = \int_0^x p(t)dt \quad \text{for} \quad p \in \mathbb{P}_n(\mathbb{R}) \quad \text{and} \quad x \in \mathbb{R},$$

is a linear map.

Before solving this problem, we give some examples of function values of I . If p is the zero polynomial we have

$$I(p) = \int_0^x 0 \, dx = 0,$$

whereas if p is the polynomial of degree 2 defined by

$$p(x) = 1 - 3x + 4x^2 \quad \text{then} \quad I(p) = \int_0^x (1 - 3t + 4t^2) dt = x - \frac{3}{2}x^2 + \frac{4}{3}x^3$$

is a polynomial of degree 3.

SOLUTION. We will prove that I is a linear map by using Theorem 2 of Section 7.1.

For ease of writing, we let $q_1 = I(p_1)$, $q_2 = I(p_2)$ and $q = I(\lambda_1 p_1 + \lambda_2 p_2)$, where $p_1, p_2 \in \mathbb{P}_n(\mathbb{R})$ and $\lambda_1, \lambda_2 \in \mathbb{R}$. Then, from the properties of integration,

$$\begin{aligned} q(x) &= \int_0^x (\lambda_1 p_1(t) + \lambda_2 p_2(t)) dt \\ &= \lambda_1 \int_0^x p_1(t) dt + \lambda_2 \int_0^x p_2(t) dt \\ &= \lambda_1 q_1(x) + \lambda_2 q_2(x). \end{aligned}$$

Thus, $I(\lambda_1 p_1 + \lambda_2 p_2) = q = \lambda_1 q_1 + \lambda_2 q_2 = \lambda_1 I(p_1) + \lambda_2 I(p_2)$, and hence I is a linear map. (Note that nullity (I) = 0 and rank(I) = $n + 1$.) $\diamond$

The next example is one of a class of so-called **integral transforms** which have many uses in mathematics, science, engineering and economics.

[X] Example 7. The Laplace transform. Let s and a be real numbers, and let V_a be the set of real-valued functions on the interval $(0, \infty)$ defined by

$$V_a = \left\{ f \in \mathcal{R}[(0, \infty)] : \int_0^\infty e^{-st} f(t) dt \text{ exists for } a < s < \infty \right\}.$$

Now, from the theory of integration, if $f, g \in V_a$ and $\lambda, \mu \in \mathbb{R}$ then

$$\int_0^\infty e^{-st} (\lambda f(t) + \mu g(t)) dt \quad \text{exists for } a < s < \infty,$$

and thus $\lambda f + \mu g \in V_a$. Hence, from the Alternative Subspace Theorem of Section 6.8, V_a is a subspace of the vector space $\mathcal{R}[(0, \infty)]$ of all real-valued functions with domain $(0, \infty)$.

We now define a function $L : V_a \rightarrow \mathcal{R}[(a, \infty)]$ with function values $L(f) = f_L$, where f_L is the function from the domain (a, ∞) to the codomain $\mathbb{R}$ defined by

$$f_L(s) = \int_0^\infty e^{-st} f(t) dt \quad \text{for } a < s < \infty.$$

f_L is called the **Laplace transform** of the function f . $\diamond$

We shall now prove that L is a linear map.

Proof. Let $f, g \in V_a$ and $\lambda, \mu \in \mathbb{R}$. Then, as noted above, the function $h = \lambda f + \mu g$ is an element of V_a , and its Laplace transform $h_L = L(\lambda f + \mu g)$ satisfies

$$\begin{aligned} h_L(s) &= \int_0^\infty e^{-st} h(t) dt \\ &= \int_0^\infty e^{-st} (\lambda f(t) + \mu g(t)) dt \\ &= \lambda \int_0^\infty e^{-st} f(t) dt + \mu \int_0^\infty e^{-st} g(t) dt \\ &= \lambda f_L(s) + \mu g_L(s). \end{aligned}$$

Thus,

$$L(\lambda f + \mu g) = h_L = \lambda f_L + \mu g_L = \lambda L(f) + \mu L(g),$$

and hence L is a linear map. $\square$

The Laplace transform is widely used in, for example, the solution of linear differential equations and the theory of dynamical systems. To understand the technique, work through question 54. It has extensive applications in electrical engineering, computer science, physics, applied and pure mathematics, and so on.

To finish this section, we shall describe some applications of linear maps in the areas of optics, chemical engineering, electrical engineering, and population dynamics.

[H] Example 8 (Optics). White light is made up of the seven colours: red, orange, yellow, green, blue, indigo, and violet. Assume that a green filter transmits 0% of red and violet, 5% of orange and indigo, 20% of yellow and blue, and 90% of the green light that falls on it. This green filter can be represented by a linear map, and the kernel and image of the map have a simple interpretation.

We let $a_1, a_2, a_3, a_4, a_5, a_6$ and a_7 be the intensities of the red, orange, yellow, green, blue, indigo and violet light respectively in the incoming light. Then the filter can be represented by the very simple linear map $T : \mathbb{R}^7 \rightarrow \mathbb{R}^7$ given by

$$T(\mathbf{a}) = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.05 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.2 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.9 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.05 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \\ a_5 \\ a_6 \\ a_7 \end{pmatrix}.$$

The kernel of the filter is the set of all possible incident light such that there is no transmitted light. The filter will transmit no light if the incoming light contains only red and violet light. Thus, a basis for the kernel of the map T which models the filter is $[1000000]^T$ (red light only) and $[0000001]^T$ (violet light only), and the nullity is 2.

The image of the filter is the transmitted light. The transmitted light can only contain orange, yellow, green, blue, and indigo, and these colours may be taken as a basis for the image. As five basic colours are transmitted, the rank is 5. Mathematically, a basis for the image of the map T is the set of 5 vectors $[0100000]^T$, $[0010000]^T$, $[0001000]^T$, $[0000100]^T$ and $[0000010]^T$. $\diamond$

[X] **Example 9** (Population Dynamics). As a simple model of the growth of a human population in a given country, we neglect males, and divide females into the six age groups of 0–14, 15–29, 30–44, 45–59, 60–74 and 75–89. It is found that, on average, 5% of the 0–14 group, 3% of the 15–29 group, 5% of the 30–44 group, 10% of the 45–59 group, 40% of the 60–74 group and 100% of the 75–89 group die in a fifteen-year period, whereas, on average, 0% of the 0–14 group, 50% of the 15–29 group, 45% of the 30–44 group, 6% of the 45–59 group and 0% of the 60–74 and 75–89 groups give birth to a female baby in a fifteen-year period.

The population at any time can be represented by a linear-transformation model as follows.

Starting at some convenient time, say January 1 1970, at which the population of the country is known, we divide time into intervals of length 15 years. Let k represent the k th of these 15-year periods, starting with $k = 0$ in the period 1970–1984. Thus, $k = 1$ represents 1985–1999, $k = 2$ represents 2000–2014, etc. We then let

$x_1(k)$	number of females of age 0–14 in interval k
$x_2(k)$	number of females of age 15–29 in interval k
$x_3(k)$	number of females of age 30–44 in interval k
$x_4(k)$	number of females of age 45–59 in interval k
$x_5(k)$	number of females of age 60–74 in interval k
$x_6(k)$	number of females of age 75–89 in interval k

Now, if we know the values of $x_j(k)$, $1 \leq j \leq 6$ in a given interval k , we can calculate the values of $x_j(k+1)$, $1 \leq j \leq 6$ in the $k+1$ th interval from the given data. For example, the females in the age group 15–29 in interval $k+1$ are the survivors of those in the age group 0–14 in interval k . Thus, $x_2(k+1) = 0.95x_1(k)$. The numbers in all groups other than the 0–15 group can be obtained in a similar fashion. In our model, the only way that females can enter the 0–15 group is to be born from mothers in the 15–29, 30–44 and 45–59 groups. Thus,

$$x_1(k+1) = 0.50x_2(k) + 0.45x_3(k) + 0.06x_4(k).$$

We therefore obtain the model

$$\mathbf{x}(k+1) = \begin{pmatrix} x_1(k+1) \\ x_2(k+1) \\ x_3(k+1) \\ x_4(k+1) \\ x_5(k+1) \\ x_6(k+1) \end{pmatrix} = \begin{pmatrix} 0 & 0.5 & 0.45 & 0.06 & 0 & 0 \\ 0.95 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.97 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.95 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.90 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.60 & 0 \end{pmatrix} \begin{pmatrix} x_1(k) \\ x_2(k) \\ x_3(k) \\ x_4(k) \\ x_5(k) \\ x_6(k) \end{pmatrix} = A\mathbf{x}(k).$$

Thus, the population vector in time interval $k+1$ is the image under the linear map whose matrix is given above of the population vector in interval k . $\diamond$

7.6 [X] Representation of linear maps by matrices

We have seen in Section 7.2 that every linear map between the vector spaces $\mathbb{R}^n$ and $\mathbb{R}^m$ can be represented by a matrix. The next theorem shows that this result can be generalised to any linear map between any finite-dimensional vector spaces.

Theorem 1 (General Matrix Representation Theorem). *Let $T : V \rightarrow W$ be a linear map from an n -dimensional vector space V to an m -dimensional vector space W , and let $B_V = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ be an ordered basis for V and $B_W = \{\mathbf{w}_1, \dots, \mathbf{w}_m\}$ be an ordered basis for W . Then, there is a unique $m \times n$ matrix A such that*

$$[T(\mathbf{v})]_{B_W} = A[\mathbf{v}]_{B_V}.$$

Further, A is the matrix whose columns are

$$\mathbf{a}_j = [T(\mathbf{v}_j)]_{B_W} \quad \text{for } 1 \leq j \leq n.$$

Proof. Let $\mathbf{w} = T(\mathbf{v}) \in W$ and

$$[\mathbf{v}]_{B_V} = \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \quad \text{i.e. } \mathbf{v} = x_1\mathbf{v}_1 + \dots + x_n\mathbf{v}_n.$$

By Theorem 3 of Section 7.1, we have

$$\mathbf{w} = T(\mathbf{v}) = T(x_1\mathbf{v}_1 + \dots + x_n\mathbf{v}_n) = x_1T(\mathbf{v}_1) + \dots + x_nT(\mathbf{v}_n).$$

Now, we have shown (Example 3 of Section 7.5) that taking coordinate vectors is a linear map, and hence, on using Theorem 3 of Section 7.1, we have

$$[\mathbf{w}]_{B_W} = [T(\mathbf{v})]_{B_W} = x_1[T(\mathbf{v}_1)]_{B_W} + \dots + x_n[T(\mathbf{v}_n)]_{B_W} = x_1\mathbf{a}_1 + \dots + x_n\mathbf{a}_n,$$

where $\mathbf{a}_j = [T(\mathbf{v}_j)]_{B_W}$. Finally, we note that $\mathbf{a}_j \in \mathbb{R}^m$ for $1 \leq j \leq n$, and hence, from Proposition 3 of Section 6.4, the linear combination can be rewritten in the matrix form $A\mathbf{x}$, and the theorem is proved. $\square$

NOTE.

1. This theorem says that if T maps $\mathbf{v}$ to $T(\mathbf{v})$ then the matrix A transforms the coordinate vector $[\mathbf{v}]_{B_V}$ for $\mathbf{v}$ with respect to an ordered basis B_V of the domain into the coordinate vector $[T(\mathbf{v})]_{B_W}$ for $T(\mathbf{v})$ with respect to an ordered basis B_W of the codomain.
2. It is important to note that the matrix A depends **only** on the map T , the ordered basis B_V and the ordered basis B_W . It does **not** depend on the particular vector $\mathbf{v}$ of V whose image is being found.

Theorem 1 provides a straightforward algorithm for finding a matrix representation.

Algorithm 1. Constructing a matrix representation for a linear map.

1. Find a basis $B_V = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ for the domain V and a basis $B_W = \{\mathbf{w}_1, \dots, \mathbf{w}_m\}$ for the codomain W .
2. Find the function values $T(\mathbf{v}_j)$, $1 \leq j \leq n$, of the domain basis vectors.
3. Find the coordinate vectors $[T(\mathbf{v}_j)]_{B_W}$ of the function values $T(\mathbf{v}_j)$ with respect to the codomain basis B_W .

4. Construct the $m \times n$ matrix A with the coordinate vectors $[T(\mathbf{v}_j)]_{B_W}$, $1 \leq j \leq n$, as its **columns**.

Example 1. Construct the matrix representation of the derivative map $D : \mathbb{P}_3(\mathbb{R}) \rightarrow \mathbb{P}_2(\mathbb{R})$, defined by

$$D(p) = p', \quad \text{where} \quad p'(x) = \frac{dp}{dx} \quad \text{for} \quad x \in \mathbb{R},$$

with respect to the standard bases of $\mathbb{P}_3(\mathbb{R})$ and $\mathbb{P}_2(\mathbb{R})$.

SOLUTION. As shown in Example 5 of Section 7.5, D is a linear map, and hence it can be represented by a matrix. We again follow algorithm 1.

For the domain $\mathbb{P}_3(\mathbb{R})$, the standard basis is $\{1, x, x^2, x^3\}$. The function values of the basis vectors are

$$D(1) = 0, \quad D(x) = 1, \quad D(x^2) = 2x, \quad D(x^3) = 3x^2.$$

The coordinate vectors of these function values with respect to the standard basis $\{1, x, x^2\}$ of the codomain are $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 2 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$ respectively. Hence the matrix is

$$A = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

As an example of the use of this matrix, we find $D(p)$ for

$$p(x) = 1 - 3x + 4x^2 + 7x^3.$$

The coordinate vector for p with respect to the standard basis $\{1, x, x^2, x^3\}$ of the domain $\mathbb{P}_3(\mathbb{R})$ is

$\begin{pmatrix} 1 \\ -3 \\ 4 \\ 7 \end{pmatrix}$. Then the coordinate vector for $D(p)$ with respect to the standard basis $\{1, x, x^2\}$ of the codomain $\mathbb{P}_2(\mathbb{R})$ is

$$A \begin{pmatrix} 1 \\ -3 \\ 4 \\ 7 \end{pmatrix} = \begin{pmatrix} -3 \\ 8 \\ 21 \end{pmatrix},$$

and hence $D(p)$ is given by

$$(D(p))(x) = -3 + 8x + 21x^2.$$

This is clearly the derivative of the polynomial p . ◇

A similar procedure to that given in Example 1 can be used to find a matrix representation for definite integration of polynomials.

In the simple case described in Example 1, it is obviously a waste of time to go through the matrix formalism. However, in more complicated examples, it is often useful to be able to use the powerful and efficient algorithms of matrix algebra to solve problems involving differentiation of polynomials.

Another example involving polynomials is as follows.

Example 2. Find the matrix with respect to standard bases in domain and codomain for the linear transformation $T : \mathbb{P}_2 \rightarrow \mathbb{C}^3$ defined by

$$T(a_0 + a_1z + a_2z^2) = \begin{pmatrix} 2a_0 + 3a_2 \\ -a_2 \\ 4a_1 + 6a_2 \end{pmatrix}.$$

SOLUTION. For the domain, the standard basis is $\{1, z, z^2\}$ and hence the images of the domain basis vectors are

$$T(1) = \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}, \quad T(z) = \begin{pmatrix} 0 \\ 0 \\ 4 \end{pmatrix}, \quad T(z^2) = \begin{pmatrix} 3 \\ -1 \\ 6 \end{pmatrix}.$$

For the codomain, the standard basis for $\mathbb{C}^3$ is $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$. The coordinate vectors for this basis are just the three image vectors given above, and hence the matrix is

$$A = \begin{pmatrix} 2 & 0 & 3 \\ 0 & 0 & -1 \\ 0 & 4 & 6 \end{pmatrix}.$$

◇

In the above examples, we have restricted the choice of bases to standard bases in domain and codomain. However, it is frequently possible to achieve great simplifications in calculations involving linear maps by using special choices of bases. Two examples of the simplification which can be achieved in this way are given below.

Example 3. A linear map $T_A : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is defined by

$$T_A(\mathbf{x}) = A\mathbf{x}, \quad \text{where} \quad A = \begin{pmatrix} 3.12 & 0.16 & -0.32 \\ 4.76 & -1.32 & -7.36 \\ 2.8 & -1.6 & -1.8 \end{pmatrix}.$$

Find the matrix which represents T_A with respect to the bases in both domain and codomain given by the columns of the matrix

$$B = \begin{pmatrix} 1 & 0 & 4 \\ -3 & 2 & 1 \\ 2 & 1 & 2 \end{pmatrix}.$$

SOLUTION. The images of the basis vectors for the domain basis are

$$T_A \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} = A \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix} = \begin{pmatrix} 2 \\ -6 \\ 4 \end{pmatrix}, \quad T_A \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -10 \\ -5 \end{pmatrix}, \quad T_A \begin{pmatrix} 4 \\ 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 12 \\ 3 \\ 6 \end{pmatrix}.$$

From algorithm 1, the columns of the matrix representing T_A with respect to the given codomain basis are just the coordinate vectors for the above images with respect to the codomain basis.

These are $\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 0 \\ -5 \\ 0 \end{pmatrix}$, and $\begin{pmatrix} 0 \\ 0 \\ 3 \end{pmatrix}$ respectively. The required matrix has these three coordinate vectors as its columns, and it is therefore

$$X = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -5 & 0 \\ 0 & 0 & 3 \end{pmatrix}.$$

◇

Note that, in this example, we have found a diagonal matrix to represent a transformation T_A . A general theory which shows the conditions under which a diagonal matrix can be found to represent a linear map will be developed in Chapter 8.

Example 4. A linear map $T_A : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ is defined by

$$T_A(\mathbf{x}) = A\mathbf{x}, \quad \text{where} \quad A = \begin{pmatrix} 1 & 4 & 2 \\ 3 & 4 & -1 \\ -2 & 0 & 5 \\ 3 & -4 & 4 \end{pmatrix}.$$

Find the matrix U which represents T_A for the standard basis in the domain $\mathbb{R}^3$ and for the basis in the codomain $\mathbb{R}^4$ which consists of the column vectors of the matrix

$$L = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 \\ -2 & -1 & 1 & 0 \\ 3 & 2 & 6 & 1 \end{pmatrix}.$$

SOLUTION. The standard basis for $\mathbb{R}^3$ is $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right\}$, and hence the images of the domain basis vectors are

$$T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = A \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ -2 \\ 3 \end{pmatrix}, \quad T \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = A \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 4 \\ 0 \\ -4 \end{pmatrix}, \quad T \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ -1 \\ 5 \\ 4 \end{pmatrix}.$$

These images are just the columns of the matrix A .

Now, following Algorithm 1, we must find the coordinate vectors for $T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$, $T \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ and $T \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ with respect to the codomain basis given by the columns of L , using the standard Gaussian elimination algorithm.

They turn out to be $\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$, $\begin{pmatrix} 4 \\ -8 \\ 0 \\ 0 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -7 \\ 2 \\ 0 \end{pmatrix}$. Thus

$$U = \begin{pmatrix} 1 & 4 & 2 \\ 0 & -8 & -7 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

Note that L is “lower triangular”, U is “upper triangular”, and that $LU = A$. $\diamond$

Both of the above examples illustrate the importance of suitable choices of bases in solving problems about linear maps. If the bases are suitably chosen, a matrix representation for a map will usually take on a very simple form from which the properties of the map can be immediately read off. We have, of course, used a special case of this approach repeatedly throughout these notes, where we have solved most problems about linear equations, vector spaces, and linear maps by constructing row-echelon form matrices from which the solutions can be immediately read off.

7.7 [X] Matrix arithmetic and linear maps

We have seen in Sections 7.2 and 7.6 that every matrix defines a linear map and that every linear map between finite-dimensional vector spaces can be represented by a matrix. We would therefore expect that properties of matrices — such as matrix addition, multiplication of a matrix by a scalar, matrix multiplication, inverse of a matrix — should also have an interpretation in terms of properties of linear maps. In this section, we shall show that matrix addition corresponds to addition of linear maps, that multiplication of a matrix by a scalar corresponds to multiplication of a linear map by a scalar, and that matrix multiplication corresponds to composition of linear maps. The relationship between inverses of linear maps and matrix inverses is discussed in Section 7.8.

Proposition 1 (Addition). Let A and B be real $m \times n$ matrices and let $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $T_B : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be the linear maps given by $T_A(\mathbf{x}) = A\mathbf{x}$ and $T_B(\mathbf{x}) = B\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$. Then the sum $T = T_A + T_B$ is the linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by $T(\mathbf{x}) = (A + B)\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$.

Proof. By definition of function addition, the sum $T = T_A + T_B$ is the linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by

$$T(\mathbf{x}) = (T_A + T_B)(\mathbf{x}) = T_A(\mathbf{x}) + T_B(\mathbf{x}) \text{ for all } \mathbf{x} \in \mathbb{R}^n.$$

Hence, on using the definition of T_A and T_B and the distributive law for matrix addition and multiplication, we have

$$T(\mathbf{x}) = A\mathbf{x} + B\mathbf{x} = (A + B)\mathbf{x} \text{ for all } \mathbf{x} \in \mathbb{R}^n.$$

□

Proposition 2 (Multiplication by a Scalar). Let A be a real $m \times n$ matrix and $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be the linear map defined by $T_A(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$. Then the scalar multiple $T = \lambda T_A$ is the linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by $T(\mathbf{x}) = (\lambda A)\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$.

Proof. By definition of multiplication of a function by a scalar, the function $T = \lambda T_A$ is given by

$$T(\mathbf{x}) = (\lambda T_A)(\mathbf{x}) = \lambda T_A(\mathbf{x}) \text{ for all } \mathbf{x} \in \mathbb{R}^n.$$

Hence, on using the definition of T_A , we have

$$T(\mathbf{x}) = \lambda(A\mathbf{x}) = (\lambda A)\mathbf{x} \text{ for all } \mathbf{x} \in \mathbb{R}^n.$$

□

Proposition 3 (Multiplication and Composition). Let A be a real $m \times n$ matrix and B be a real $n \times p$ matrix and let $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be the linear map defined by $T_A(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^n$ and let $T_B : \mathbb{R}^p \rightarrow \mathbb{R}^n$ be the linear map defined by $T_B(\mathbf{x}) = B\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^p$. Then the composite $T_A \circ T_B$ is the linear map $T : \mathbb{R}^p \rightarrow \mathbb{R}^m$ defined by $T(\mathbf{x}) = (AB)\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^p$.

Proof. The composite $T_A \circ T_B$ is the linear map $T : \mathbb{R}^p \rightarrow \mathbb{R}^m$ defined by

$$T(\mathbf{x}) = (T_A \circ T_B)(\mathbf{x}) = T_A(T_B(\mathbf{x})) \text{ for all } \mathbf{x} \in \mathbb{R}^p.$$

Hence, on substituting the function values for T_A and T_B and using the associative law of matrix multiplication, we have

$$T(\mathbf{x}) = T_A(B\mathbf{x}) = A(B\mathbf{x}) = (AB)\mathbf{x} \text{ for all } \mathbf{x} \in \mathbb{R}^p.$$

□

NOTE. B is acting first, then A .

Similar relationships hold between linear transformations on general finite-dimensional vector spaces and their corresponding matrices. We shall give just one example of these more general theorems, whose proof we shall leave to the exercises.

Proposition 4. Let U , V and W be finite-dimensional vector spaces with bases B_U , B_V and B_W respectively. Let $T : U \rightarrow V$ and $S : V \rightarrow W$. Then

- a) $S \circ T$ is a linear transformation from $U \rightarrow W$.
- b) If the matrix for T with respect to B_U and B_V is A_T , the matrix for S with respect to B_V and B_W is A_S and the matrix for $S \circ T$ with respect to B_U and B_W is A_{ST} , then $A_{ST} = A_S A_T$.

7.8 [X] One-to-one, onto and invertible linear maps and matrices

In this section we shall discuss the main results involving the ideas of one-to-one, onto and inverses for linear maps and matrices and we shall show how these results for linear maps compare with the results for general functions summarised in Appendix 7.10.

7.8.1 Linear maps

The definitions of one-to-one, onto and inverse for linear maps are virtually identical to the definitions for general functions. However, for ease of reading it is convenient to restate them here. On applying the definitions of one-to-one, onto and inverse given in Appendix 7.10 to linear maps, we obtain the following definitions.

Definition 1. A linear map $T : V \rightarrow W$ is said to be:

- a) **one-to-one** if for all $\mathbf{v}_1, \mathbf{v}_2 \in V$, $T(\mathbf{v}_1) = T(\mathbf{v}_2)$ only if $\mathbf{v}_1 = \mathbf{v}_2$.
- b) **onto** if for all $\mathbf{w} \in W$ there exists $\mathbf{v} \in V$ such that $\mathbf{w} = T(\mathbf{v})$, that is, if $\text{im}(T) = W$.

Definition 2. Let $T : V \rightarrow W$ be a linear map. Then a function $S : W \rightarrow V$ is called an **inverse** of T if it satisfies the two conditions:

- a) $S \circ T = \text{id}_V$, where id_V is the identity map on V defined by $\text{id}_V(\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v} \in V$.
- b) $T \circ S = \text{id}_W$, where id_W is the identity map on W defined by $\text{id}_W(\mathbf{w}) = \mathbf{w}$ for all $\mathbf{w} \in W$.

For linear maps, there is a simple connection between a map being one-to-one and the kernel of the map. This connection, which is not in general true for all functions, is as follows.

Proposition 1. A linear map $T : V \rightarrow W$ is one-to-one if and only if $\ker(T) = \{\mathbf{0}\}$, that is, if and only if $\text{nullity}(T) = 0$.

Proof. We first prove that if T is one-to-one then $\ker(T) = \{\mathbf{0}\}$. Now, as T is a linear map $T(\mathbf{0}) = \mathbf{0}$, and as T is one-to-one $T(\mathbf{v}) = T(\mathbf{0})$ only if $\mathbf{v} = \mathbf{0}$. Thus, $T(\mathbf{v}) = \mathbf{0}$ only if $\mathbf{v} = \mathbf{0}$, and hence $\ker(T) = \{\mathbf{0}\}$. We next prove that if $\ker(T) = \{\mathbf{0}\}$ then T is one-to-one. We let $\mathbf{v}_1, \mathbf{v}_2$ satisfy $T(\mathbf{v}_1) = T(\mathbf{v}_2)$, that is, $T(\mathbf{v}_1) - T(\mathbf{v}_2) = \mathbf{0}$. But, as T is a linear map, $T(\mathbf{v}_1) - T(\mathbf{v}_2) = T(\mathbf{v}_1 - \mathbf{v}_2)$. Then as $\ker(T) = \{\mathbf{0}\}$, we have $T(\mathbf{v}_1 - \mathbf{v}_2) = \mathbf{0}$ only if $\mathbf{v}_1 - \mathbf{v}_2 = \mathbf{0}$. Thus, $T(\mathbf{v}_1) = T(\mathbf{v}_2)$ only if $\mathbf{v}_1 = \mathbf{v}_2$, and hence T is one-to-one. $\square$

The result stated in Proposition 1 is not true in general for non-linear functions — for example, it is not true for the function of Example 1 of Appendix 7.10 which is neither one-to-one nor onto even though $f(x) = x^2 = 0$ only for $x = 0$.

Proposition 2. If the codomain W of a linear function $T : V \rightarrow W$ is finite dimensional then T is onto if and only if $\text{rank}(T) = \dim(W)$.

Proof. We know that $\text{im}(T)$ is a subspace of W and $\text{rank}(T) = \dim(\text{im}(T))$ so, by Theorem 8 of Section 6.6, $\text{rank}(T) = \dim(W)$ if and only if $\text{im}(T) = W$, that is, if and only if T is onto. $\square$

The following result is also of importance for linear maps.

Proposition 3. If the domain and codomain of a linear map $T : V \rightarrow W$ are finite-dimensional then:

- a) If T is one-to-one and onto then $\dim(V) = \dim(W)$.
- b) If $\dim(V) = \dim(W)$ and T is one-to-one then T is onto.
- c) If $\dim(V) = \dim(W)$ and T is onto then T is one-to-one.

Proof. The proofs of the three parts are based on the Rank-Nullity Theorem and Propositions 1 and 2 above.

- a) If T is one-to-one then $\text{nullity}(T) = 0$, and if T is onto then $\text{rank}(T) = \dim(W)$. Therefore, if T is one-to-one and onto, we have from the Rank-Nullity Theorem that

$$\dim(V) = \text{rank}(T) + \text{nullity}(T) = \dim(W) + 0 = \dim(W).$$

- b) If $\dim(V) = \dim(W)$ and T is one-to-one then $\text{nullity}(T) = 0$, and so

$$\text{rank}(T) = \dim(V) - \text{nullity}(T) = \dim(W),$$

and therefore T is onto.

- c) If $\dim(V) = \dim(W)$ and T is onto then $\text{rank}(T) = \dim(W)$, and so

$$\text{nullity}(T) = \dim(V) - \text{rank}(T) = \dim(V) - \dim(W) = 0,$$

and therefore T is one-to-one.

□

Proposition 4. If a linear map has an inverse then the inverse is also a linear map.

Proof. Let $T : V \rightarrow W$ be a linear map with inverse $S : W \rightarrow V$. Then, for all $\mathbf{v} \in V$ and for all $\mathbf{w} \in W$, we have $\mathbf{v} = S(\mathbf{w})$ if and only if $\mathbf{w} = T(\mathbf{v})$.

To prove S is linear, we must show that for all $\mathbf{w}_1, \mathbf{w}_2 \in W$ and all scalars $\lambda_1, \lambda_2 \in F$

$$S(\lambda_1 \mathbf{w}_1 + \lambda_2 \mathbf{w}_2) = \lambda_1 S(\mathbf{w}_1) + \lambda_2 S(\mathbf{w}_2).$$

Now if $S(\mathbf{w}_1) = \mathbf{v}_1$ and $S(\mathbf{w}_2) = \mathbf{v}_2$, we have $\mathbf{w}_1 = T(\mathbf{v}_1)$ and $\mathbf{w}_2 = T(\mathbf{v}_2)$, and hence on using the fact that T is linear we obtain

$$\lambda_1 \mathbf{w}_1 + \lambda_2 \mathbf{w}_2 = \lambda_1 T(\mathbf{v}_1) + \lambda_2 T(\mathbf{v}_2) = T(\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2).$$

Thus, by definition of S ,

$$\begin{aligned} S(\lambda_1 \mathbf{w}_1 + \lambda_2 \mathbf{w}_2) &= \lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 \\ &= \lambda_1 S(\mathbf{w}_1) + \lambda_2 S(\mathbf{w}_2). \end{aligned}$$

The proof is complete.

□

A fundamental result which summarises the main properties of inverses of linear maps is as follows.

Theorem 5. *If V and W are finite-dimensional vector spaces and if $T : V \rightarrow W$ is a linear map then the following statements are equivalent:*

1. T is invertible, that is, there exists a linear map $S : W \rightarrow V$ such that $(S \circ T)(\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v} \in V$ and such that $(T \circ S)(\mathbf{w}) = \mathbf{w}$ for all $\mathbf{w} \in W$.
2. T is one-to-one and onto.
3. $\dim(V) = \dim(W)$ and there exists $S : W \rightarrow V$ such that $(T \circ S)(\mathbf{w}) = \mathbf{w}$ for all $\mathbf{w} \in W$.
4. $\dim(V) = \dim(W)$ and there exists $S : W \rightarrow V$ such that $(S \circ T)(\mathbf{v}) = \mathbf{v}$ for all $\mathbf{v} \in V$.

Further, the map S in statements 1, 3 and 4 is the inverse of T .

NOTE. The phrase “The statements are equivalent” means that if **one** statement is true then **all** statements are true, and also that if **one** statement is false then **all** statements are false.

Proof. The equivalence of statements 1 and 2 follows immediately from Theorem 1 of Section 7.10. We shall now show that statements 1 and 3 are equivalent by first showing that statement 1 implies statement 3 and then by showing that statement 3 implies statement 1.

1 implies 3. If T is invertible, then there exists $S : W \rightarrow V$ such that $(T \circ S)(\mathbf{w}) = \mathbf{w}$ for all $\mathbf{w} \in W$. Further, as T is invertible it is also one-to-one and onto, and hence, from part (a) of Proposition 3, $\dim(V) = \dim(W)$.

3 implies 1. We first prove that T is onto. Let $\mathbf{w} \in W$. Then, as S is a function from W to V , there exists $\mathbf{v} \in V$ such that $S(\mathbf{w}) = \mathbf{v}$. Then, on first taking the function value of $\mathbf{v}$ under T and next using the property of $T \circ S$ given in statement 3, we have

$$T(\mathbf{v}) = T(S(\mathbf{w})) = (T \circ S)(\mathbf{w}) = \mathbf{w}.$$

Thus, we have shown that for all $\mathbf{w} \in W$, there exists $\mathbf{v} \in V$ such that $T(\mathbf{v}) = \mathbf{w}$, and hence T is onto. Then, from part (c) of Proposition 3, T is also one-to-one, and thus T is one-to-one and onto and therefore invertible.

To complete the proof we prove that 1 and 4 are equivalent by proving first that 1 implies 4 and then that 4 implies 1.

1 implies 4. The proof of this is virtually identical to the proof given above that 1 implies 3 and hence we omit it.

4 implies 1. We first prove T is one-to-one. Let $T(\mathbf{v}_1) = T(\mathbf{v}_2)$ for $\mathbf{v}_1, \mathbf{v}_2 \in V$. On taking the composite of both sides with S , we have

$$(S \circ T)(\mathbf{v}_1) = (S \circ T)(\mathbf{v}_2).$$

Then, using the property of $S \circ T$ given in 4, we have

$$\mathbf{v}_1 = (S \circ T)(\mathbf{v}_1) = (S \circ T)(\mathbf{v}_2) = \mathbf{v}_2.$$

Thus, $T(\mathbf{v}_1) = T(\mathbf{v}_2)$ only if $\mathbf{v}_1 = \mathbf{v}_2$, and hence T is one-to-one. Then, from part (b) of Proposition 3, T is also onto, and thus T is one-to-one and onto and therefore invertible.

The proof is complete. □

7.9 [X] Proof of the Rank-Nullity Theorem

In this section we give a proof of the general Rank-Nullity Theorem stated in Section 7.4.3.

Theorem 1 (Rank-Nullity Theorem). *If V is a finite dimensional vector space and $T : V \rightarrow W$ is linear then*

$$\text{rank}(T) + \text{nullity}(T) = \dim(V).$$

Proof. Recall that $\text{rank}(T) = \dim(\text{im}(T)) =$ number of vectors in a basis for $\text{im}(T)$, and that $\text{nullity}(T) = \dim(\ker(T)) =$ number of vectors in a basis for $\ker(T)$.

Let $\{\mathbf{w}_1, \dots, \mathbf{w}_r\}$, where $r = \text{rank}(T)$, be a basis for $\text{im}(T)$. Then, since $\mathbf{w}_j \in \text{im}(T)$, there exists an element $\mathbf{v}_j \in V$ such that $T(\mathbf{v}_j) = \mathbf{w}_j$.

Let $\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_{r+\nu}\}$, where $\nu = \text{nullity}(T)$, be a basis for $\ker(T)$.

We shall now prove that the combined set $S = \{\mathbf{v}_1, \dots, \mathbf{v}_{r+\nu}\}$ is a basis for the domain V , and hence that $r + \nu = \dim(V)$.

We first prove that S is linearly independent.

Suppose,

$$\lambda_1 \mathbf{v}_1 + \dots + \lambda_{r+\nu} \mathbf{v}_{r+\nu} = \mathbf{0}. \quad (\#)$$

Taking the image of this linear combination and using the fact that linear maps preserve linear combinations, we have

$$\lambda_1 T(\mathbf{v}_1) + \dots + \lambda_{r+\nu} T(\mathbf{v}_{r+\nu}) = T(\mathbf{0}) = \mathbf{0}.$$

Now, for $j > r$, $\mathbf{v}_j \in \ker(T)$, and hence $T(\mathbf{v}_j) = \mathbf{0}$ for $j > r$. Further, for $j \leq r$, $T(\mathbf{v}_j) = \mathbf{w}_j$. On substituting these results in the previous equation, we have

$$\lambda_1 \mathbf{w}_1 + \dots + \lambda_r \mathbf{w}_r = \mathbf{0}.$$

But the set $\{\mathbf{w}_1, \dots, \mathbf{w}_r\}$ is linearly independent (since it is a basis for $\text{im}(T)$), and thus $\lambda_j = 0$ for $j \leq r$. On substituting these values of the scalars in $(\#)$ we have

$$\lambda_{r+1} \mathbf{v}_{r+1} + \dots + \lambda_{r+\nu} \mathbf{v}_{r+\nu} = \mathbf{0}.$$

But the set $\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_{r+\nu}\}$ is linearly independent (it is a basis for $\ker(T)$), and thus $\lambda_j = 0$ for $j > r$. We have therefore proved that $(\#)$ is satisfied only if $\lambda_j = 0$ for $1 \leq j \leq r + \nu$, and hence S is linearly independent.

Now we show that S is a spanning set for V .

Since $S \subseteq V$, $\text{span}(S)$ is a subset of V . To complete the proof that $\text{span}(S) = V$ we must prove that $V \subseteq \text{span}(S)$, that is, if $\mathbf{v} \in V$ then $\mathbf{v} \in \text{span}(S)$.

Suppose $\mathbf{v} \in V$. Then $\mathbf{w} = T(\mathbf{v})$ exists and is an element of $\text{im}(T)$.

Now the set $\{\mathbf{w}_1, \dots, \mathbf{w}_r\}$ is a basis for $\text{im}(T)$, and hence there exist scalars such that

$$\mathbf{w} = \lambda_1 \mathbf{w}_1 + \dots + \lambda_r \mathbf{w}_r.$$

Using these scalars we now form the linear combination

$$\mathbf{v}_I = \lambda_1 \mathbf{v}_1 + \dots + \lambda_r \mathbf{v}_r.$$

Now, by definition of $\mathbf{v}_I$ and the $\mathbf{v}_j$ for $j \leq r$, and on using the fact that linear maps preserve linear combinations, we have $T(\mathbf{v}_I) = \mathbf{w}$. We now define

$$\mathbf{v}_R = \mathbf{v} - \mathbf{v}_I.$$

Now $\mathbf{v}_R$ satisfies

$$T(\mathbf{v}_R) = T(\mathbf{v} - \mathbf{v}_I) = T(\mathbf{v}) - T(\mathbf{v}_I) = \mathbf{w} - \mathbf{w} = \mathbf{0},$$

and hence $\mathbf{v}_R \in \ker(T)$. Then, since $\{\mathbf{v}_{r+1}, \dots, \mathbf{v}_{r+\nu}\}$ is a basis for $\ker(T)$, there exist scalars such that

$$\mathbf{v}_R = \lambda_{r+1}\mathbf{v}_{r+1} + \dots + \lambda_{r+\nu}\mathbf{v}_{r+\nu}.$$

Then, on adding the linear combinations for $\mathbf{v}_I$ and $\mathbf{v}_R$, we have

$$\mathbf{v} = \mathbf{v}_I + \mathbf{v}_R = \lambda_1\mathbf{v}_1 + \dots + \lambda_r\mathbf{v}_r + \lambda_{r+1}\mathbf{v}_{r+1} + \dots + \lambda_{r+\nu}\mathbf{v}_{r+\nu},$$

and hence $\mathbf{v} \in \text{span}(S)$.

We have therefore established that S is a linearly independent spanning set for V , and hence $\dim(V) = r + \nu = \text{rank}(T) + \text{nullity}(T)$ as asserted in the theorem. $\square$

7.10 One-to-one, onto and inverses for functions

A fundamental problem about functions is the relationship between the points in the codomain and the points in the domain. Now, we know that by the definition of function each point in the domain of a function has exactly one point in the codomain as its function value. However, for a given point in the codomain it is possible in general that either (i) it is not the function value of any point in the domain, (ii) it is the function value of exactly one point in the domain, or (iii) it is the function value of more than one point in the domain. To cover these possibilities the following definitions are introduced.

Definition 1. The **range** or **image** of a function is the set of all function values, that is, for a function $f : X \rightarrow Y$,

$$\text{im}(f) = \{y \in Y : y = f(x) \text{ for some } x \in X\}.$$

Definition 2. A function is said to be **onto** (or surjective) if the codomain is equal to the image of the function, that is, a function $f : X \rightarrow Y$ is onto if for all $y \in Y$ there exists an $x \in X$ such that $y = f(x)$.

Definition 3. A function is said to be **one-to-one** (or injective) if no point in the codomain is the function value of more than one point in the domain, that is, a function $f : X \rightarrow Y$ is one-to-one if $f(x_1) = f(x_2)$ if and only if $x_1 = x_2$.

Note that a function is onto when every point in the codomain is a function value and that it is one-to-one when each function value corresponds to exactly one point in the domain. Further, a function is one-to-one and onto if and only if every point in the codomain is the function value of exactly one point in the domain.

Example 1. The function $f : \mathbb{R} \rightarrow \mathbb{R}$, defined by

$$f(x) = x^2 \quad \text{for } x \in \mathbb{R},$$

is neither one-to-one nor onto. It is not one-to-one, since, for example, $x_1^2 = x_2^2$ is true for $x_1 = 3$ and $x_2 = -3 \neq x_1$. It is not onto, since there are some $y \in \mathbb{R}$ ($y < 0$) for which no $x \in \mathbb{R}$ exists such that $y = x^2$. $\diamond$

Example 2. The function $f : [0, \infty) \rightarrow \mathbb{R}$ defined by

$$f(x) = x^2 \quad \text{for } x \geq 0$$

is one-to-one but not onto. It is one-to-one, since if $x_1^2 = x_2^2$ (and $x_1 \geq 0$ and $x_2 \geq 0$) then $x_1 = x_2$. However, it is not onto for the reason given in Example 1. $\diamond$

Example 3. The function $f : [0, \infty) \rightarrow [0, \infty)$ defined by

$$f(x) = x^2 \quad \text{for } x \geq 0$$

is both one-to-one and onto. It is one-to-one for the reason given in Example 2. The function is also onto, since for all $y \geq 0$ there is an $x \geq 0$ such that $y = f(x) = x^2$. $\diamond$

The definition of inverse of a function is as follows.

Definition 4. Let $f : X \rightarrow Y$ be a function. Then a function $g : Y \rightarrow X$ is called an **inverse** of f if it satisfies the two conditions:

- a) $g \circ f = id_X$, where id_X is the identity function in X with the property that $id_X(x) = x$ for all $x \in X$.
- b) $f \circ g = id_Y$, where id_Y is the identity function in Y with the property that $id_Y(y) = y$ for all $y \in Y$.

An alternative way of stating (a) is that $(g \circ f)(x) = g(f(x)) = x$ for all $x \in X$, and an alternative way of stating (b) is that $(f \circ g)(y) = f(g(y)) = y$ for all $y \in Y$.

An important connection between the existence of an inverse and the properties of one-to-one and onto is given in the following proposition.

Theorem 1. A function has an inverse if and only if the function is both one-to-one and onto.

[X] *Proof.* Let $f : X \rightarrow Y$ be a function with domain X and codomain Y .

We first prove that if f has an inverse $g : Y \rightarrow X$ then f is one-to-one and onto.

To prove one-to-one, we note that if $f(x_1) = f(x_2)$ then on taking the composition with the inverse g we obtain $g(f(x_1)) = g(f(x_2))$, and hence from condition (a) of the definition of inverse

$$x_1 = g(f(x_1)) = g(f(x_2)) = x_2.$$

Thus, for all $x_1, x_2 \in X$, $f(x_1) = f(x_2)$ only if $x_1 = x_2$ and f is one-to-one.

We now prove that f is onto. From condition (b) of the definition of inverse, $y = f(g(y))$ for all $y \in Y$. But g is a function with codomain X , and hence $g(y) = x$ for $x \in X$. Thus, for all $y \in Y$ there exists an $x = g(y) \in X$ such that $y = f(x)$, and hence f is onto.

To complete the proof of the theorem, we must prove that if f is one-to-one and onto then f has an inverse function $g : Y \rightarrow X$.

Now, as f is an onto function, for each $y \in Y$ there exists $x \in X$ such that $y = f(x)$. Further, as f is one-to-one, $y_1 = f(x_1) = f(x_2) = y_2$ only if $x_1 = x_2$, and hence for each $y \in Y$ there exists a unique $x \in X$ such that $y = f(x)$. We can therefore define a function $g : Y \rightarrow X$ by the rule:

For each $y \in Y$ define $g(y) = x$ where x is the unique $x \in X$ such that $y = f(x)$.

This function g then has the property that $y = f(x) = f(g(y)) = (f \circ g)(y)$ for all $y \in Y$, and hence g satisfies condition (b) of the definition of inverse.

To complete the proof we must prove that the function g also satisfies condition (a). Now, as f is a function with domain X and codomain Y , for each $x \in X$ there exists a unique $y \in Y$ such that $y = f(x)$. But, as $y \in Y$, we have from the definition of g that $x = g(y)$. Therefore $x = g(y) = g(f(x)) = (g \circ f)(x)$ for all $x \in X$. The proof is complete. $\square$

7.11 Linear transformations and MAPLE

The main result of this chapter is the Matrix Representation Theorem. Accordingly, we can do calculations with linear transformations by getting Maple to manipulate the corresponding matrices. Usually you will have to calculate the appropriate matrix by hand, but Maple can handle some linear transformations directly. [Be aware that Maple displays vectors as rows but treats them as columns.] Consider for example the linear transformation of projecting onto a fixed vector $\mathbf{b} \in \mathbb{R}^n$ (see Section 7.3). In the following, the `LinearAlgebra` command `Norm(b,2)` calculates $\|\mathbf{b}\|$.

```
with(LinearAlgebra):
b:=<1,2,3>;
T:=a->(b.a/norm(b,2)^2).b);
a:=<1,0,-1>;
T(a);
```

We could then get Maple to find the matrix for T (with respect to the standard basis) by using

```
A:=< T(e1) | T(e2) | T(e3) >;
```

where $\mathbf{e1}, \mathbf{e2}, \mathbf{e3}$ have been previously defined as the standard basis vectors in $\mathbb{R}^3$. Indeed, you can even get Maple to check that the operator T defined above is linear.

```
x:=Vector(3,i->X[i]);  
y:=Vector(3,i->Y[i]);  
T(x+y)-T(x)-T(y);  
simplify(%);
```

This last calculation will give zero, showing that $T(\mathbf{x} + \mathbf{y}) = T(\mathbf{x}) + T(\mathbf{y})$. You should have a go at showing that T preserves scalar multiplication as well (this turns out to be just a little more complicated for Maple!). You can also check that the matrix defined above actually does give the linear transformation T by using:

```
simplify(A.x - T(x));
```

Many of the standard calculations concerning linear transformations have special commands in the **LinearAlgebra** package. In particular you should look at the procedures **NullSpace**, **ColumnSpace**, and **Rank**. As usual, the details of these commands are available using the on-line help facility. For example to find out about **NullSpace** (which calculates the kernel), enter the command **?NullSpace**.

Problems for Chapter 7

Problems 7.1 : Introduction to linear maps

1. [R] Explain why the function $S : [-1, 1] \rightarrow \mathbb{R}$ defined by $S(x) = 5x$ for $x \in [-1, 1]$ is not a linear map. Then show that the function $T : \mathbb{R} \rightarrow \mathbb{R}$ defined by $T(x) = 5x$ for $x \in \mathbb{R}$ is a linear map.

2. [R] For the following examples, determine whether T is a linear map by using Definition 1.

- a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^4$ defined by

$$T(\mathbf{x}) = \begin{pmatrix} 3x_1 - x_2 \\ 2x_1 + 4x_2 \\ -3x_1 - 3x_2 \\ x_2 \end{pmatrix} \quad \text{for } \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2.$$

- b) $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ defined by

$$T(\mathbf{x}) = \begin{pmatrix} -2x_1 + 5x_3 \\ 6x_1 - 8x_2 + 2x_4 \\ -2x_1 + 4x_2 - 3x_3 \end{pmatrix} \quad \text{for } \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in \mathbb{R}^4.$$

- c) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by

$$T(\mathbf{x}) = \begin{pmatrix} 3x_1 + 4 \\ -2x_1 + 3x_2 - x_3 \end{pmatrix} \quad \text{for } \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3.$$

- d) $T : \mathbb{R}^4 \rightarrow \mathbb{R}^3$ defined for $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in \mathbb{R}^4$ by

$$T(\mathbf{x}) = x_1 \begin{pmatrix} 1 \\ 3 \\ -2 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ -4 \\ 2 \end{pmatrix} + x_3 \begin{pmatrix} -2 \\ -3 \\ 4 \end{pmatrix} + x_4 \begin{pmatrix} -4 \\ 1 \\ 0 \end{pmatrix}.$$

- e) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ defined by

$$T(\mathbf{x}) = \begin{pmatrix} 3x_2^2 - x_3 \\ x_1 - 4x_2 \end{pmatrix} \quad \text{for } \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \in \mathbb{R}^3.$$

3. [X] Consider the complex numbers as a real vector space. Specify the “natural” domain and codomain for each of the following functions of a complex number and determine if the function is a linear function with $\mathbb{F} = \mathbb{R}$.

- a) $T(z) = \operatorname{Re}(z)$, b) $T(z) = \operatorname{Im}(z)$, c) $T(z) = |z|$,
 d) $T(z) = \operatorname{Arg}(z)$, e) $T(z) = \bar{z}$.

4. [R] Show that the sine function $T : \mathbb{R} \rightarrow \mathbb{R}$, defined by

$$T(x) = \sin(x) \quad \text{for } x \in \mathbb{R},$$

satisfies parts 1 and 2 of Proposition 1 of Section 7.1 but that it is not a linear map.

5. [H] Use proof by induction to prove Theorem 3 of Section 7.1.

6. [R] If $\{\mathbf{v}_1, \mathbf{v}_2\}$ are linearly independent in a real vector space V and $\mathbf{v}_3 = 2\mathbf{v}_1 + \mathbf{v}_2$, is there a linear map $T : W \rightarrow \mathbb{R}^2$ where $W = \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2)$ such that

$$T(\mathbf{v}_1) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, T(\mathbf{v}_2) = \begin{pmatrix} -3 \\ 2 \end{pmatrix}, T(\mathbf{v}_3) = \begin{pmatrix} -1 \\ 3 \end{pmatrix}?$$

7. [R] A linear map $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ has function values given by

$$T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}, T \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \\ 1 \\ 4 \end{pmatrix}, T \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ -5 \\ 6 \end{pmatrix}. \text{ Find } T \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} \text{ and } T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}.$$

8. [R] Show that any function $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ with function values given by $T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}$,

$$T \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -3 \\ 0 \\ 1 \\ 4 \end{pmatrix}, T \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ -5 \\ 6 \end{pmatrix}, \text{ and } T \begin{pmatrix} 1 \\ -3 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 3 \\ 1 \end{pmatrix} \text{ is not a linear map.}$$

9. [R] A linear function $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ has function values given by

$$T \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}, \quad T \begin{pmatrix} -2 \\ 1 \\ -4 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}, \quad T \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} = \begin{pmatrix} -4 \\ -2 \end{pmatrix}.$$

Write $\begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$ as a linear combination of the vectors in the basis $\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} -2 \\ 1 \\ -4 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \right\}$

of $\mathbb{R}^3$. Hence find $T \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix}$.

HINT. Use Theorem 3 of Section 7.1.

10. [H] Given that $T \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 4 \\ 1 \end{pmatrix}$, $T \begin{pmatrix} -2 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} -1 \\ 2 \end{pmatrix}$ and $T \begin{pmatrix} 1 \\ 7 \\ 13 \end{pmatrix} = \begin{pmatrix} 4 \\ -2 \end{pmatrix}$, show that T is not a linear map.

Problems 7.2 : Linear maps from $\mathbb{R}^n$ to $\mathbb{R}^m$ and $m \times n$ matrices

11. [R] For any function in question 2 which is a linear map, find a matrix A such that $T(\mathbf{x}) = A\mathbf{x}$ for $\mathbf{x}$ in the domain by using the results of the Matrix Representation Theorem of Section 7.2.
12. [R] For any function in question 2 which is a linear map, write a system of linear equations for $T(\mathbf{x})$ for $\mathbf{x}$ in the domain, and hence find a matrix A such that $T(\mathbf{x}) = A\mathbf{x}$. Check that the matrices you obtain in this question are the same as the matrices that you obtained in the previous question.

Problems 7.3 : Geometric examples of linear transformations

13. [R] For each of the following 2×2 matrices, draw a picture to show $A\mathbf{e}_1$, $A\mathbf{e}_2$, $A\mathbf{b}$, where $\mathbf{e}_1$ and $\mathbf{e}_2$ are the standard basis vectors in $\mathbb{R}^2$ and where $\mathbf{b} = 2\mathbf{e}_1 + 3\mathbf{e}_2$.

$$\text{a) } \begin{pmatrix} 2 & 0 \\ 0 & 0.7 \end{pmatrix}, \quad \text{b) } \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix}, \quad \text{c) } \begin{pmatrix} -2 & 0 \\ 0 & -3 \end{pmatrix}, \quad \text{d) } \begin{pmatrix} 6 & -2 \\ 6 & -1 \end{pmatrix}, \quad \text{e) } \begin{pmatrix} 4 & -4 \\ 3 & -4 \end{pmatrix}.$$

14. [R] Draw the image of the star in Figure 4(a) on page 90 under each of the transformations defined by the matrices in Question 13.
15. [R] Let T be the rotation in the plane $\mathbb{R}^2$ through angle $\frac{\pi}{3}$ in the anti-clockwise direction. Find the matrix which represents the linear transformation T .
16. [H] Let $\mathbf{x}$ be the position vector of a point X in $\mathbb{R}^2$, and let $\mathbf{x}'$ be the position vector of the point X' which is the reflection of X in the x_2 -axis. (That is, assume that a mirror is placed along the x_2 -axis and that X' is the reflection of X in the mirror.) Show that the function $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(\mathbf{x}) = \mathbf{x}'$ is a linear map. Find a matrix A which transforms $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ into $\mathbf{x}' = \begin{pmatrix} x'_1 \\ x'_2 \end{pmatrix}$.
17. [H] Let $\mathbf{x}$ be the position vector of a point X in $\mathbb{R}^3$ and let $\mathbf{x}'$ be the position vector of the point X' which is the reflection of X in the (x_1, x_2) -plane. Show that the function $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(\mathbf{x}) = \mathbf{x}'$ is a linear map. Find a matrix A which transforms the position vector $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ of X into the position vector $\begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \end{pmatrix}$ of X' .
18. [X] Let $\mathbf{p}$ be the position vector of a point P in $\mathbb{R}^n$ and let $\mathbf{q}$ be the position vector of the point Q which is the reflection of P in the line

$$\mathbf{x} = \lambda \mathbf{d}; \quad \lambda \in \mathbb{R}.$$

Show that the function $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by $T(\mathbf{p}) = \mathbf{q}$ is a linear map. Find a matrix

A which transforms $\mathbf{p} = \begin{pmatrix} p_1 \\ \vdots \\ p_n \end{pmatrix}$ into $\mathbf{q} = \begin{pmatrix} q_1 \\ \vdots \\ q_n \end{pmatrix}$.

19. [R] Let $\mathbf{b}$ be a fixed vector in $\mathbb{R}^3$. Is the function $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$T(\mathbf{x}) = \mathbf{b} \times \mathbf{x} \quad \text{for } \mathbf{x} \in \mathbb{R}^3,$$

where $\mathbf{b} \times \mathbf{x}$ is the cross product, a linear map? Prove your answer. Find a matrix A

which transforms the vector $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$ into its function value $T(\mathbf{x}) = \begin{pmatrix} x'_1 \\ x'_2 \\ x'_3 \end{pmatrix}$.

20. [H] Suppose that $\mathbf{b} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$. Prove that the projection function $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ of vectors onto $\mathbf{b}$, which is defined by

$$T(\mathbf{a}) = \text{proj}_{\mathbf{b}}(\mathbf{a}) = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{b}|^2} \mathbf{b} \quad \text{for } \mathbf{a} \in \mathbb{R}^3,$$

is a linear map. Find a matrix A which transforms the vector $\mathbf{a}$ into its projection $T(\mathbf{a})$.

21. [H] If $\mathbf{a}$ is regarded as a fixed non-zero vector, is the function $S : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by

$$S(\mathbf{b}) = \begin{cases} \text{proj}_{\mathbf{b}} \mathbf{a} & \text{for } \mathbf{b} \in \mathbb{R}^n \setminus \{\mathbf{0}\} \\ \mathbf{0} & \mathbf{b} = \mathbf{0} \end{cases}$$

a linear map? Prove your answer.

22. [X] Let A_ϕ , A_θ and $A_{\phi+\theta}$ be the matrices for rotations in the plane by angles ϕ , θ and $\phi + \theta$ respectively (see Example 3 of Section 7.3). Prove that

$$A_\theta A_\phi = A_{\phi+\theta}.$$

What is this saying geometrically?

23. [X] Let $B = \{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ be an ordered orthonormal basis (Cartesian coordinate system) for a three-dimensional geometric vector space. Let $\mathbf{a}$ be a three-dimensional geometric vector and let $\mathbf{a}' = R_\alpha(\mathbf{a})$ be the vector obtained by rotating $\mathbf{a}$ anticlockwise by an angle α

about an axis parallel to $\mathbf{j}$. If the coordinate vectors of $\mathbf{a}$ and $\mathbf{a}'$ are $[\mathbf{a}]_B = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$ and

$[\mathbf{a}']_B = \begin{pmatrix} a'_1 \\ a'_2 \\ a'_3 \end{pmatrix}$, find the rule $R_\alpha \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} a'_1 \\ a'_2 \\ a'_3 \end{pmatrix}$ and show that it defines a linear map from $\mathbb{R}^3$ to $\mathbb{R}^3$. Also find the matrix A such that $\mathbf{a}' = A\mathbf{a}$.

Problems 7.4 : Subspaces associated with linear maps

24. [R] Show that the set $\left\{ \begin{pmatrix} \lambda \\ -2\lambda \\ \lambda \end{pmatrix} : \lambda \in \mathbb{R} \right\}$ is the kernel of the matrix $\begin{pmatrix} 3 & 1 & -1 \\ 8 & 3 & -2 \end{pmatrix}$.

25. [R] Find the kernel and the nullity of each of the following matrices.

$$\begin{aligned} \text{a) } A &= \begin{pmatrix} 2 & -1 & 3 \\ 1 & -2 & 3 \\ 4 & 1 & -1 \end{pmatrix}, & \text{b) } B &= \begin{pmatrix} 0 & 5 & 15 \\ 2 & -2 & -4 \\ 3 & -3 & -6 \end{pmatrix}, \\ \text{c) } C &= \begin{pmatrix} 1 & 2 & -1 & 1 \\ 3 & 2 & 0 & -2 \\ 0 & 1 & -1 & 1 \end{pmatrix}. \end{aligned}$$

Where possible, give a geometric interpretation of the kernels.

26. [R] Find a basis for the kernel, and the nullity, of each of the following matrices.

$$\text{a) } D = \begin{pmatrix} 1 & 2 & 5 & 0 \\ 1 & -1 & -4 & 0 \\ -1 & 0 & 1 & 1 \end{pmatrix}, \quad \text{b) } E = \begin{pmatrix} 1 & 1 & -1 \\ 2 & -1 & 0 \\ 5 & -4 & 1 \\ 0 & 0 & 1 \end{pmatrix}.$$

27. [H] Let $W = \left\{ (x_1 \ x_2 \ x_3 \ x_4)^T : x_1 + x_2 + x_3 + x_4 = x_1 + 2x_2 + 3x_3 + 4x_4 = 0 \right\}$. Find a matrix A such that $W = \ker A$.

28. [R] Find $\ker(T)$ and $\text{nullity}(T)$ for the linear functions of question 2.

29. [R] Find $\ker(T)$ and $\text{nullity}(T)$ for the linear functions of questions 16 through 20. Give a geometric interpretation of the kernels.

30. [H] Suppose that $\mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$.

- Prove that the mapping $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, given by $T(\mathbf{x}) = \mathbf{b} \times \mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^3$, is a linear mapping.
- Find the dimension of the kernel of this mapping.

31. [R] For each given vector $\mathbf{b}$ and matrix A , determine if $\mathbf{b} \in \text{im}(A)$.

$$\begin{aligned} \text{a) } \mathbf{b} &= \begin{pmatrix} 11 \\ 10 \\ 4 \end{pmatrix}, & A &= \begin{pmatrix} 1 & -2 & 3 \\ 2 & -1 & 3 \\ 4 & 1 & -1 \end{pmatrix}. \\ \text{b) } \mathbf{b} &= \begin{pmatrix} 9 \\ -2 \\ -4 \end{pmatrix}, & A &= \begin{pmatrix} 0 & 5 & 15 \\ 2 & -2 & -4 \\ 3 & -3 & -6 \end{pmatrix}. \end{aligned}$$

$$\text{c) } \mathbf{b} = \begin{pmatrix} -2 \\ -6 \\ -4 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 2 & -1 & 1 \\ 3 & 2 & 0 & -2 \\ 0 & 1 & -1 & 1 \end{pmatrix}.$$

32. [R] Find conditions on b_1, b_2, b_3 for the vector $\mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \in \mathbb{R}^3$ for $\mathbf{b}$ to belong to $\text{im}(A)$ for the matrices in the preceding question.

33. [R] Find a basis for the image, and the rank, of each of the matrices in questions 25 and 26.

34. [R] By comparing the answers to questions 25, 26 and 33, verify the conclusion of the Rank-Nullity Theorem.

35. [R] Find a basis for the image, and the rank, of each of the matrices

$$A = \begin{pmatrix} 1 & 1 & -1 & -2 & 1 \\ 0 & 0 & 1 & 4 & -1 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 1 & 0 & 2 & 1 \\ 0 & 0 & -1 & -2 & 2 \\ -1 & -1 & 1 & 4 & -1 \\ 1 & 1 & 0 & 4 & 2 \end{pmatrix}.$$

36. [R] Find a basis for $\mathbb{R}^3$ which contains a basis of $\text{im}(C)$, where

$$C = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & -4 & 6 & -2 \\ -1 & 2 & -3 & 1 \end{pmatrix}.$$

37. [R] Find a basis for $\mathbb{R}^4$ which contains a basis of $\text{im}(D)$, where

$$D = \begin{pmatrix} 1 & 3 & 3 & -1 & 7 \\ 2 & 6 & 3 & 1 & 8 \\ 3 & 9 & 3 & 4 & 7 \\ 4 & 12 & 0 & 8 & 4 \end{pmatrix}.$$

38. [R] A linear map $T : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ has the property that

$$T \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad T \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 4 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad T \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -1 \\ 3 \\ 0 \\ 0 \end{pmatrix}, \quad T \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ -3 \\ 0 \\ 1 \end{pmatrix}.$$

- Write down the matrix representation of T with respect to the standard basis (in both domain and co-domain).
- Find a basis for the image of T and find the rank of T .
- State the dimension of the kernel of T .

- d) Does the vector $\begin{pmatrix} 8 \\ -3 \\ 1 \\ 2 \end{pmatrix}$ belong to the image of T ? Give reasons.
39. [H] Let $A \in M_{nn}(\mathbb{R})$. Show that the following statements are equivalent, that is, show that if any statement is true then all are true, whereas if any statement is false then all are false.
- For all $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$, $A\mathbf{x} = A\mathbf{y}$ if and only if $\mathbf{x} = \mathbf{y}$.
 - $\ker(A) = \{\mathbf{0}\}$.
 - $\text{nullity}(A) = 0$.
 - $\text{rank}(A) = n$.
 - $\text{im}(A) = \mathbb{R}^n$.
 - The columns of A form a basis for $\mathbb{R}^n$.
40. [H] Let $A \in M_{mn}(\mathbb{R})$, and let $\{\mathbf{e}_j : 1 \leq j \leq m\}$ be the set of m standard basis vectors of $\mathbb{R}^m$. If A is of rank r , explain why at most r of the m equations $A\mathbf{x}_j = \mathbf{e}_j$ can have solutions.
41. [H] Let A and $\mathbf{e}_j$ be as in the previous question. If $\text{nullity}(A) = \nu$, explain why at least $m - n + \nu$ of the m equations $A\mathbf{x}_j = \mathbf{e}_j$ **do not** have solutions.
42. [H] Let $A \in M_{nn}(\mathbb{R})$, $\text{rank}(A) = n$, and $\mathbf{e}_j$ be the standard basis vectors for $\mathbb{R}^n$. Prove that each of the n equations $A\mathbf{x}_j = \mathbf{e}_j$, $1 \leq j \leq n$, has a unique solution.
43. [X] Let $T : V \rightarrow V$ be a linear map and assume that $\dim(V) = n$. Show that the following statements are equivalent.
- $T(\mathbf{v}) = T(\mathbf{w})$ if and only if $\mathbf{v} = \mathbf{w}$ for all $\mathbf{v}, \mathbf{w} \in V$.
 - $\ker(T) = \{\mathbf{0}\}$.
 - $\text{nullity}(T) = 0$.
 - $\text{rank}(T) = n$.
 - $\text{im}(T) = V$.

Problems 7.5 : Further applications and examples of linear maps

44. [X] Show that the function $T : \mathbb{R}^4 \rightarrow M_{22}(\mathbb{R})$ defined by

$$T(\mathbf{a}) = \begin{pmatrix} a_1 & a_2 \\ a_3 & a_4 \end{pmatrix} \quad \text{for } \mathbf{a} = (a_1, a_2, a_3, a_4) \in \mathbb{R}^4$$

is a linear map.

45. [X] Show that the function $T : \mathbb{R}^4 \rightarrow M_{22}(\mathbb{R})$ defined by

$$T(a_1, a_2, a_3, a_4) = \begin{pmatrix} 3a_1 - 2a_4 & a_4 + 2a_3 \\ -5a_2 + 3a_3 & a_1 \end{pmatrix}$$

is a linear map.

46. [X] Is the function $T : M_{23}(\mathbb{R}) \rightarrow \mathbb{R}^6$ defined by

$$T(A) = (a_{11} \ a_{12} \ a_{13} \ a_{21} \ a_{22} \ a_{23})^T \quad \text{for} \quad A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} \in M_{23}(\mathbb{R})$$

a linear map?

47. [R] Show that the function $T : \mathbb{P}_2 \rightarrow \mathbb{C}^3$ defined by

$$T(a_0 + a_1z + a_2z^2) = \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix}$$

is a linear map.

[X] Note that T maps a polynomial in $\mathbb{P}_2$ into its coordinate vector with respect to the standard basis $\{1, z, z^2\}$.

48. [H] Show that the function $T : \mathbb{C}^4 \rightarrow \mathbb{P}_4$ defined by $T(\mathbf{a}) = p$ for $\mathbf{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{pmatrix} \in \mathbb{C}^4$, where

$$p(z) = (a_1 - 3a_2) + (2a_3 - 3a_4)z + a_2z^3 + (3a_1 - a_2 + 2a_3 + 4a_4)z^4 \quad \text{for all } z \in \mathbb{C},$$

is a linear map.

49. [R] Show that the function $T : \mathbb{P}_3(\mathbb{R}) \rightarrow \mathbb{P}_3(\mathbb{R})$ defined by

$$T(p) = 4p' + 3p, \quad \text{where} \quad p'(x) = \frac{dp}{dx},$$

is a linear map.

50. [R] Is the function $T : \mathbb{P}_3(\mathbb{R}) \rightarrow \mathbb{P}_3(\mathbb{R})$ defined by $T(p) = q$, where

$$q(x) = 4xp'(x) - 8p(x) \quad \text{for } x \in \mathbb{R},$$

a linear map? Prove your answer.

51. [H] Show that the function $T : \mathbb{P}_3(\mathbb{R}) \rightarrow \mathbb{P}_4(\mathbb{R})$ defined by $T(p) = q$, where

$$q(x) = \int_0^x p(t)dt \quad \text{for } x \in \mathbb{R},$$

is a linear map.

52. [X] Let V be the subset of the vector space $\mathcal{R}[\mathbb{R}]$ of all real-valued functions on $\mathbb{R}$ defined by

$$V = \left\{ f \in \mathcal{R}[\mathbb{R}] : \int_0^x f(t)dt \text{ exists for all } x \in \mathbb{R} \right\}.$$

Show that V is a subspace of $\mathcal{R}[\mathbb{R}]$, and then show that the rule $T : V \rightarrow \mathcal{R}[\mathbb{R}]$ defined by $T(f) = g$, where

$$g(x) = \int_0^x f(t)dt \quad \text{for } f \in V \quad \text{and } x \in \mathbb{R},$$

is a linear map.

53. [X] A function $S : \mathbb{R} \rightarrow \mathbb{Z}$ is defined by $S(x) = y$, where y is the integer obtained on rounding x to the nearest integer. Is S a linear map? A function $T : \mathbb{R} \rightarrow \mathbb{R}$ is defined by $T(x) = y$, where y is the integer obtained on rounding x to the nearest integer. Is T a linear map?
54. [X] Let y be a real-valued function with domain $\mathbb{R}$ such that y and its first two derivatives y' and y'' exist, and such that the Laplace transforms (see Example 7 of Section 7.5) of y , y' and y'' also exist on the interval $(0, \infty)$. Given that $y(0) = 1$ and $y'(0) = 2$ and that y satisfies the differential equation

$$y''(x) + 4y'(x) + 3y = e^{-3x},$$

find an explicit formula for the Laplace transform $y_L(s)$ of y in terms of s .

HINT. Take the Laplace transform of the differential equation and use integration by parts to find formulae for the Laplace transforms of y' and y'' in terms of $y_L(s)$.

55. [H] Consider the mapping $T : \mathbb{P}_3(\mathbb{R}) \rightarrow \mathbb{R}^2$ defined by

$$T(p(x)) = \begin{pmatrix} a-b \\ c-d \end{pmatrix} \quad \text{where } p(x) = a + bx + cx^2 + dx^3.$$

- a) Prove that T is linear.
 b) Show that $p(x) = 3x^3 + 3x^2 - 2x - 2$ is in the kernel of T .

56. [H] Consider the function $T : \mathbb{R}^4 \rightarrow \mathbb{P}_1$ defined by

$$T \begin{pmatrix} a \\ b \\ c \\ d \end{pmatrix} = (a - 2b) + (c + d)x.$$

a) Find $T \begin{pmatrix} 1 \\ -3 \\ 2 \\ -4 \end{pmatrix}$.

- b) Show T is a linear transformation.
 c) Write down a non-zero vector in $\mathbb{R}^4$ which lies in $\ker(T)$.

57. [H] A linear map $T : \mathbb{C}^3 \rightarrow \mathbb{P}_3$ has function values given by

$$T \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 1 + (2+i)z - 3z^3, \quad T \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = (4-3i)z + z^2, \quad T \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} = -2 \quad \text{for } z \in \mathbb{C}.$$

Find $T \begin{pmatrix} i \\ 2 \\ -1 \end{pmatrix}$ and $T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$.

58. [X] Let $\mathbb{P}_n$ be the real vector space of polynomials of degree less than or equal to n , and take its standard basis to be

$$\{1, x, x^2, \dots, x^n\}.$$

For $p(x) \in \mathbb{P}_3$, let $(T(p))(x) \in \mathbb{P}_4$ be defined by

$$(T(p))(x) = \int_0^x p(t)dt.$$

- Show that T is a linear transformation from $\mathbb{P}_3$ to $\mathbb{P}_4$.
 - Calculate the matrix A of this linear transformation with respect to the standard bases of $\mathbb{P}_3$ and $\mathbb{P}_4$.
 - Find a basis for the image of T , $\text{im}(T)$.
 - Find a basis for the kernel of T , $\text{ker}(T)$.
59. [R] A car manufacturer produces a station wagon, a four-wheel drive, a hatchback and a sedan model. Each model is made from steel, plastics, rubber and glass, and it also requires a number of hours of labour to produce. The requirements per car of these inputs for each model are as shown in the following table.

	steel (tonnes)	plastics (tonnes)	rubber (tonnes)	glass (tonnes)	labour (hours)
station wagon	1	0.5	0.1	0.2	1
4-wheel drive	1.5	0.6	0.2	0.15	1.5
hatchback	0.8	0.7	0.2	0.2	1.1
sedan	0.9	0.6	0.25	0.3	0.9

Construct a matrix which can be used to express the factory *input* as a linear function of the factory *output*.

Problems 7.6 : [X] Representation of linear maps by matrices

60. [R] Let $\text{id}_{\mathbb{R}^2}$ be the identity map for $\mathbb{R}^2$. Find a matrix representation of this map with respect to standard bases in domain and codomain.
61. [X] Let $\text{id}_{\mathbb{R}^2}$ be the identity map for $\mathbb{R}^2$. Find a matrix representation of this map with respect to the domain basis $\left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \end{pmatrix} \right\}$ and the codomain basis $\left\{ \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \end{pmatrix} \right\}$.
62. [X] A linear mapping $G : \mathbb{P}_2 \rightarrow \mathbb{P}_2$ has matrix representation

$$A = \begin{pmatrix} 1 & 2 & -1 \\ -1 & 1 & 0 \\ 3 & 4 & 1 \end{pmatrix}$$

with respect to the standard basis $\{1, x, x^2\}$ in both domain and co-domain. Find $G(p)$, where $p(x) = -3 + x + 5x^2$.

63. [X] For each of the linear maps in questions 48 to 51, find a matrix which represents the linear map for standard bases in the domain and codomain.
64. [X] Using your results of the previous question or otherwise, find the kernel, nullity, image and rank of the linear maps in questions 48 to 51.
65. [X] For the linear map $T : \mathbb{P}_n(\mathbb{R}) \rightarrow \mathbb{P}_n(\mathbb{R})$ defined by $T(p) = q$, where

$$q(x) = x^2 \frac{d^2 p}{dx^2} - 3x \frac{dp}{dx} + 3p(x) \quad \text{for } x \in \mathbb{R},$$

find a matrix which represents T with respect to standard bases in domain and codomain. Hence, or otherwise, find the kernel and nullity of T .

66. [X] Let V be a vector space and let $B = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be an orthonormal basis for V . Let $\mathbf{a} \in V$ be a vector whose coordinate vector with respect to B is $\begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix}$. Let $\begin{pmatrix} a'_1 \\ a'_2 \\ a'_3 \end{pmatrix}$ be the coordinate vector of $\mathbf{a}$ with respect to the basis $B' = \{\mathbf{u}'_1, \mathbf{u}'_2, \mathbf{u}'_3\}$ given by

$$\begin{aligned} \mathbf{u}'_1 &= \frac{1}{\sqrt{2}}\mathbf{u}_1 + \frac{1}{\sqrt{2}}\mathbf{u}_3, \\ \mathbf{u}'_2 &= -\frac{1}{\sqrt{2}}\mathbf{u}_1 + \frac{1}{\sqrt{2}}\mathbf{u}_3, \\ \mathbf{u}'_3 &= -\mathbf{u}_2 \end{aligned}$$

Show that B' is an orthonormal basis, then show that the rule $T \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} a'_1 \\ a'_2 \\ a'_3 \end{pmatrix}$ is a linear map from $\mathbb{R}^3$ to $\mathbb{R}^3$, and find a matrix representation for this function with respect to standard bases for $\mathbb{R}^3$ in domain and codomain. Finally, show that the matrix you have constructed is a matrix representation of the identity map $\text{id}_V : V \rightarrow V$ with respect to the basis B in domain and B' in codomain.

Problems 7.7 : [X] Matrix arithmetic and linear maps

67. [X] Let $T : V \rightarrow W$ and $S : V \rightarrow W$ be linear maps. Let B_V be a basis for V and B_W be a basis for W and let A and B be the matrices representing T and S with respect to the bases B_V and B_W . Prove that $A + B$ is the matrix representing the sum function $T + S$ with respect to the bases B_V and B_W .
68. [X] Let $T : U \rightarrow V$ and $S : V \rightarrow W$ be linear maps. Let B_U , B_V and B_W be bases for U , V and W respectively. Let A be the matrix representing T with respect to bases B_U and B_V and let B be the matrix representing S with respect to bases B_V and B_W . Prove that the matrix product BA is the matrix which represents the composition function $S \circ T : U \rightarrow W$ with respect to the bases B_U and B_W .

Problems 7.8 : [X] One-to-one, onto and invertible linear maps and matrices

69. [X] Let $V = C[\mathbb{R}]$, the vector space of all continuous real-valued functions on $\mathbb{R}$. Let

$$B = \{e^x, (x-1)e^x, (x-1)(x-2)e^x\} \subseteq V$$

- a) Prove that B is linearly independent.
- b) Let $W = \text{span}(B)$, and let $D : W \rightarrow W$ denote the linear transformation

$$D(f) = f',$$

where f' is the derivative of f .

- i) Find the matrix for D with respect to the ordered basis B of W .
- ii) Find the matrix for the linear transformation $T = D \circ D$.
- iii) Hence or otherwise, prove that for every $g \in W$, there exists $f \in W$ such that

$$f'' = g.$$

70. [X] For a field $\mathbb{F}$, define $T : M_{22}(\mathbb{F}) \rightarrow \mathbb{F}^3$ by

$$T\left(\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}\right) = \begin{pmatrix} a_{22} \\ a_{12} - a_{21} \\ 3a_{11} + a_{12} \end{pmatrix}.$$

- a) Show that T is a linear transformation.
- b) Find the kernel of T and the nullity of T .
- c) Find the rank of T .
- d) Is T one-to-one (injective)? Give a brief reason for your answer.
- e) Find the matrix of T with respect to the standard bases of $M_{22}(\mathbb{F})$ and $\mathbb{F}^3$.

Problems 7.11 : Linear transformations and MAPLE

71. [M] Consider the following MAPLE output

```
> with(LinearAlgebra):
> A:=<<2,4,-2,4>|<-3,-6,3,-6>|<1,2,-2,1>|<-1,-3,2,-2>|<1,-1,-1,-1>>;
```

$$A := \begin{bmatrix} 2 & -3 & 1 & -1 & 1 \\ 4 & -6 & 2 & -3 & -1 \\ -2 & 3 & -2 & 2 & -1 \\ 4 & -6 & 1 & -2 & -1 \end{bmatrix}$$

```
> b:=<a1,a2,a3,a4>;
```

$$b := \begin{bmatrix} a1 \\ a2 \\ a3 \\ a4 \end{bmatrix}$$

> GaussianElimination(<A|b>);

$$\begin{bmatrix} 2 & -3 & 1 & -1 & 1 & a1 \\ 0 & 0 & -1 & 1 & 0 & a3 + a1 \\ 0 & 0 & 0 & -1 & -3 & a2 - 2 a1 \\ 0 & 0 & 0 & 0 & 0 & a4 - a1 - a3 - a2 \end{bmatrix}$$

- a) Find a basis for $\text{col}(A)$, the column space of A .
- b) What is the dimension of $\text{col}(A)$?
- c) Under what conditions does (a_1, a_2, a_3, a_4) belong to $\text{col}(A)$.
- d) Find a basis for the kernel, or null space, of A .
- e) What are the values of the rank and nullity of A ?

Chapter 8

EIGENVALUES AND EIGENVECTORS

... she set to work very carefully, nibbling first at one and then at the other, and growing sometimes taller, and sometimes shorter,...
Lewis Carroll, Alice in Wonderland.

Eigenvalues and eigenvectors are of great theoretical and practical importance. Some practical applications of eigenvalues and eigenvectors include the following.

1. Oscillations. For example, vibrating strings, organ pipes, wing flutter on an aircraft, vibrations of buildings and bridges, etc.
2. Quantum Physics and Chemistry. Structure of atoms, molecules, nuclei, solids etc.
3. Electronics and Electrical Engineering. Microwave oscillators, amplifiers, signal transmission, communications networks, etc.
4. Economics. Stability of economic systems, dynamic econometric models, Leontief input-output models, inventory models, stock market models, etc.
5. Biological and Ecological Systems. Solution of population models, stability of ecological systems etc.

In this chapter we shall only be able to give a brief introduction to this extremely important topic. A general theory of eigenvalues and eigenvectors and some applications of them is given in the second year mathematics courses.

8.1 Definitions and examples

We are concerned with linear maps in which the domain and the codomain are the same vector space, that is, with linear maps of the form $T : V \rightarrow V$. The fundamental questions asked are:

1. Given a map T , are there vectors $\mathbf{v} \in V$ which are related in a very simple way to their images $T(\mathbf{v}) \in V$?

2. [X] Is there a choice of basis for V such that the matrix representing T for this basis takes on a very simple form?

The answer to both of these questions is yes.

For question 1, we look for vectors for which $T(\mathbf{v})$ is a multiple of $\mathbf{v}$. Formally, we have

Definition 1. Let $T : V \rightarrow V$ be a linear map. Then if a scalar λ and **non-zero** vector $\mathbf{v} \in V$ satisfy

$$T(\mathbf{v}) = \lambda \mathbf{v},$$

then λ is called an **eigenvalue** of T and $\mathbf{v}$ is called an **eigenvector** of T for the eigenvalue λ .

NOTE. An eigenvector is non-zero, but zero can be an eigenvalue.

Example 1. For infinitely differentiable real-valued functions f , the derivative

$$D(f) = f', \quad \text{where} \quad f'(x) = \frac{df}{dx} \quad \text{for} \quad x \in \mathbb{R}$$

defines a linear map D . The exponential function satisfies

$$D(e^{\lambda x}) = \lambda e^{\lambda x},$$

and hence $e^{\lambda x}$ is an eigenvector of D with eigenvalue λ . It should be noted that the great importance of exponential functions in calculus is due to the fact that they are the only functions f where f' is a multiple of f . $\diamond$

Calculus and its applications provides a very rich source of eigenvalue and eigenvector problems. However, we are mainly concerned in this course with algebraic problems involving linear maps between finite-dimensional vector spaces. These linear maps can always be represented by matrices, and hence we will be concerned in the remainder of this chapter with eigenvalues and eigenvectors of matrices.

When dealing with eigenvalues and eigenvectors of matrices we will be forced to use complex numbers for our scalar field. The fundamental reason for this is that the eigenvalues of a matrix are actually zeroes of some polynomial and, as we have seen in Chapter 3, we can only be certain of finding zeroes when the polynomials are complex polynomials. Thus, the “natural” field of scalars for eigenvalues and eigenvectors is the set of complex numbers $\mathbb{C}$, and the “natural” vector spaces are the complex vector spaces $\mathbb{C}^n$ (see Example 2 of Section 6.1).

For the special case of a matrix, Definition 1 becomes:

Definition 2. Let $A \in M_{nn}(\mathbb{C})$ be a square matrix. Then if a scalar $\lambda \in \mathbb{C}$ and non-zero vector $\mathbf{v} \in \mathbb{C}^n$ satisfy

$$A\mathbf{v} = \lambda \mathbf{v},$$

then λ is called an **eigenvalue** of A and $\mathbf{v}$ is called an **eigenvector** of A for the eigenvalue λ .

Example 2. For the diagonal 2×2 matrix,

$$A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix},$$

the standard basis vectors $\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$ and $\mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ satisfy

$$\begin{aligned} A \begin{pmatrix} 1 \\ 0 \end{pmatrix} &= \begin{pmatrix} \lambda_1 \\ 0 \end{pmatrix} = \lambda_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \\ A \begin{pmatrix} 0 \\ 1 \end{pmatrix} &= \begin{pmatrix} 0 \\ \lambda_2 \end{pmatrix} = \lambda_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \end{aligned}$$

Thus, $\mathbf{e}_1$ is an eigenvector of A with eigenvalue λ_1 and $\mathbf{e}_2$ is an eigenvector of A with eigenvalue λ_2 . A picture of this result is shown in Figure 1 for the special case of $\lambda_1 = 3$ and $\lambda_2 = -2$. $\diamond$

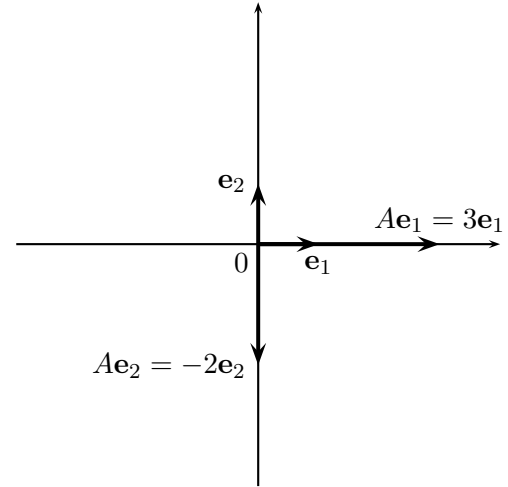


Figure 1: The eigenvectors of the diagonal matrix $\begin{pmatrix} 3 & 0 \\ 0 & -2 \end{pmatrix}$.

Example 3. For the matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 20 & -24 & 9 \end{pmatrix} \quad \text{and the vector} \quad \mathbf{v} = \begin{pmatrix} 1 \\ 5 \\ 25 \end{pmatrix}$$

by matrix multiplication, we have $A\mathbf{v} = 5\mathbf{v}$, and hence $\mathbf{v}$ is an eigenvector of A for the eigenvalue $\lambda = 5$. $\diamond$

8.1.1 Some fundamental results

The fundamental theoretical results for eigenvalues and eigenvectors draw on results given in previous chapters on linear equations, polynomials, vector spaces, linear maps, and determinants.

The following theorem is extremely important.

Theorem 1. A scalar λ is an eigenvalue of a square matrix A if and only if $\det(A - \lambda I) = 0$, and then $\mathbf{v}$ is an eigenvector of A for the eigenvalue λ if and only if $\mathbf{v}$ is a non-zero solution of the homogeneous equation $(A - \lambda I)\mathbf{v} = \mathbf{0}$, i.e., if and only if $\mathbf{v} \in \ker(A - \lambda I)$ and $\mathbf{v} \neq \mathbf{0}$.

Proof. From Definition 2, A is a square matrix, and an eigenvalue λ and corresponding eigenvector $\mathbf{v}$ of A satisfy the equation

$$A\mathbf{v} = \lambda\mathbf{v}, \quad \text{where } \mathbf{v} \neq \mathbf{0}.$$

This equation can be rearranged in the form

$$\mathbf{0} = A\mathbf{v} - \lambda\mathbf{v} = A\mathbf{v} - \lambda I\mathbf{v} = (A - \lambda I)\mathbf{v},$$

where I is an identity matrix of the same size as A .

Now, $A - \lambda I$ is a square matrix, and hence (by a proposition in Chapter 5) the equation

$$(A - \lambda I)\mathbf{v} = \mathbf{0}$$

can have a non-zero solution if and only if $\det(A - \lambda I) = 0$. Thus, λ is an eigenvalue if and only if $\det(A - \lambda I) = 0$ and the first part of the theorem is proved.

Then, if λ is an eigenvalue, $\mathbf{v}$ is an eigenvector if and only if it is a non-zero solution of the above homogeneous equation, that is, if and only if $\mathbf{v} \in \ker(A - \lambda I)$ and $\mathbf{v} \neq \mathbf{0}$. The proof is complete. $\square$

NOTE. The set of all eigenvectors of A for eigenvalue λ is therefore equal to $\ker(A - \lambda I)$ with $\mathbf{0}$ removed. Also, there are infinitely many eigenvectors corresponding to a single eigenvalue.

A second fundamental result for the theory of eigenvalues is the following.

Theorem 2. *If A is an $n \times n$ matrix and $\lambda \in \mathbb{C}$, then $\det(A - \lambda I)$ is a complex polynomial of degree n in λ .*

This theorem can be proved in a straightforward, but tedious, fashion by direct expansion of the determinant $\det(A - \lambda I)$. For example, for $n = 3$, we have

$$A - \lambda I = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{pmatrix}.$$

Then, by direct evaluation of $\det(A - \lambda I)$ by expansion along the first column, we obtain

$$\begin{aligned} \det(A - \lambda I) &= (a_{11} - \lambda)((a_{22} - \lambda)(a_{33} - \lambda) - a_{32}a_{23}) \\ &\quad - a_{21}(a_{12}(a_{33} - \lambda) - a_{32}a_{13}) + a_{31}(a_{12}a_{23} - (a_{22} - \lambda)a_{13}) \\ &= -\lambda^3 + \text{terms containing } \lambda^2, \lambda \text{ and constants.} \end{aligned}$$

Hence, for $n = 3$, $\det(A - \lambda I)$ is a complex polynomial of degree 3 as stated in Proposition 2.

Definition 3. *For a square matrix A , the polynomial $p(\lambda) = \det(A - \lambda I)$ is called the **characteristic polynomial** for the matrix A .*

Example 4. For the 3×3 matrix,

$$A = \begin{pmatrix} 1 & -1 & 2 \\ 3 & -4 & -1 \\ 5 & 1 & 2 \end{pmatrix},$$

the characteristic polynomial is the cubic

$$p(\lambda) = \begin{vmatrix} 1 - \lambda & -1 & 2 \\ 3 & -4 - \lambda & -1 \\ 5 & 1 & 2 - \lambda \end{vmatrix} = -\lambda^3 - \lambda^2 + 16\lambda + 50.$$

$\diamond$

We can now apply the theory of roots of complex polynomials developed in Chapter 3 to obtain the following fundamental result.

Theorem 3. *An $n \times n$ matrix A has exactly n eigenvalues in $\mathbb{C}$ (counted according to their multiplicities). These eigenvalues are the zeroes of the characteristic polynomial $p(\lambda) = \det(A - \lambda I)$.*

Proof. From Proposition 2, the characteristic polynomial $p(\lambda) = \det(A - \lambda I)$ is a polynomial of degree n over the complex field. Thus, from the Factorisation Theorem of Chapter 3, p has exactly n zeroes (counted according to their multiplicities) which from Theorem 1 are the eigenvalues of A . $\square$

Example 5. For the matrix A of Example 4, the roots of the cubic characteristic polynomial are (to 4-figure accuracy) 4.688 , $-2.844 + 1.605i$, $-2.844 - 1.605i$, and these are the three eigenvalues of A . $\diamond$

NOTE.

1. The equation $p(\lambda) = 0$ is called the **characteristic equation** for A .
2. Theorem 3 is of fundamental theoretical importance, as it proves the existence of eigenvalues of a matrix. However, with the exception of 2×2 and specially constructed larger matrices, modern methods of finding eigenvalues of matrices do not make use of this theorem. These efficient modern methods are currently available in standard matrix software packages such as MAPLE, MATLAB.

8.1.2 Calculation of eigenvalues and eigenvectors

As stated above, Theorem 3 provides a practical method for finding eigenvalues of 2×2 or specially constructed larger matrices. The corresponding eigenvectors can then be obtained from Theorem 1. Some examples of the calculation of eigenvalues and eigenvectors for simple matrices are as follows:

Example 6. For a diagonal matrix the diagonal entries are eigenvalues and the standard basis vectors are eigenvectors. For example, if

$$A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

then

$$\det(A - \lambda I) = \det \left(\begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \right) = \begin{vmatrix} \lambda_1 - \lambda & 0 \\ 0 & \lambda_2 - \lambda \end{vmatrix} = (\lambda_1 - \lambda)(\lambda_2 - \lambda) = 0$$

has the solutions $\lambda = \lambda_1$ and $\lambda = \lambda_2$.

Then for the eigenvector corresponding to $\lambda = \lambda_1$, we solve

$$(A - \lambda_1 I)\mathbf{v} = \begin{pmatrix} 0 & 0 \\ 0 & \lambda_2 - \lambda_1 \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \mathbf{0}.$$

It is obvious that one of the solutions is $\mathbf{v} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \mathbf{e}_1$.

Similarly, $\mathbf{e}_2$ is a solution of the homogeneous equation $(A - \lambda_2 I)\mathbf{v} = \mathbf{0}$, and hence $\mathbf{e}_2$ is one of the eigenvectors corresponding to the eigenvalue λ_2 . $\diamond$

Example 7. Find eigenvalues and eigenvectors of

$$A = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}.$$

SOLUTION. The first step is to find the eigenvalues from the characteristic polynomial.

We have

$$p(\lambda) = \det(A - \lambda I) = \begin{vmatrix} 2 - \lambda & 2 \\ 2 & 2 - \lambda \end{vmatrix} = \lambda^2 - 4\lambda.$$

Note that $A - \lambda I$ is obtained from A by subtracting λ from each diagonal element of A , and that the characteristic polynomial is a quadratic.

The roots of the characteristic equation are

$$\lambda_1 = 0 \quad \text{and} \quad \lambda_2 = 4.$$

Note that, as asserted in Theorem 3, there are two eigenvalues for the 2×2 matrix A .

The next step is to find an eigenvector for each eigenvalue by finding $\ker(A - \lambda I)$, first for $\lambda = 0$, and then for $\lambda = 4$.

For eigenvalue $\lambda_1 = 0$, the eigenvectors are the non-zero vectors in $\ker(A)$. By row reduction,

$$\left(\begin{array}{cc|c} 2 & 2 & 0 \\ 2 & 2 & 0 \end{array} \right) \xrightarrow{R_2 = R_2 - R_1} \left(\begin{array}{cc|c} 2 & 2 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

and then, back substitution gives $\ker(A) = \text{span}(\mathbf{v}_1)$ where $\mathbf{v}_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$. The set of eigenvectors corresponding to the eigenvalue 0 is then

$$\left\{ t \begin{pmatrix} -1 \\ 1 \end{pmatrix} : t \neq 0 \right\}.$$

For $\lambda_2 = 4$, the required eigenvectors are $\ker(A - 4I)$ (with $\mathbf{0}$ deleted) where

$$A - 4I = \begin{pmatrix} -2 & 2 \\ 2 & -2 \end{pmatrix}.$$

Solving $(A - 4I)\mathbf{v} = \mathbf{0}$ in the same way, we find that $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is a basis for $\ker(A - 4I)$ and the set of eigenvectors corresponding to the eigenvalue 4 is then

$$\left\{ t \begin{pmatrix} 1 \\ 1 \end{pmatrix} : t \neq 0 \right\}.$$

[Note that the scalar field is assumed to be $\mathbb{C}$, so $t \in \mathbb{C}$.]

◇

Example 8. Find eigenvalues and eigenvectors of

$$A = \begin{pmatrix} 2 & 1 \\ -1 & 4 \end{pmatrix}.$$

SOLUTION. The eigenvalues are solutions of the characteristic equation

$$\begin{vmatrix} 2-\lambda & 1 \\ -1 & 4-\lambda \end{vmatrix} = (2-\lambda)(4-\lambda) + 1 = (\lambda-3)^2 = 0.$$

Hence, there is one eigenvalue $\lambda = 3$ with multiplicity 2.

Eigenvectors. The eigenvectors are vectors $\mathbf{v} \neq \mathbf{0} \in \ker(A - 3I)$, where

$$A - 3I = \begin{pmatrix} -1 & 1 \\ -1 & 1 \end{pmatrix}.$$

On solving $(A - 3I)\mathbf{v} = \mathbf{0}$, we find that the only solution is $\mathbf{v} = t \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ for $t \in \mathbb{C}$. A matrix with fewer linearly independent eigenvectors than columns, as in this example, is called a **defective matrix** (poor thing). $\diamond$

As the next example shows, it is also possible to have a 2×2 matrix A with one eigenvalue (with multiplicity 2) and two linearly independent eigenvectors.

Example 9. The matrix

$$A = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$$

has eigenvalue $\lambda = 3$ (with multiplicity 2) and $\ker(A - 3I)$ is $\text{span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$. $\diamond$

Example 10. Find all eigenvalues and eigenvectors of the matrix

$$A = \begin{pmatrix} 1 & 2 \\ -2 & 1 \end{pmatrix}.$$

SOLUTION. As usual the eigenvalues are solutions of the characteristic equation $\det(A - \lambda I) = 0$, that is, of

$$\begin{vmatrix} 1-\lambda & 2 \\ -2 & 1-\lambda \end{vmatrix} = \lambda^2 - 2\lambda + 5 = 0.$$

In this case the roots of the quadratic are the complex numbers

$$\lambda_1 = 1 + 2i \quad \text{and} \quad \lambda_2 = 1 - 2i.$$

The eigenvectors for $\lambda_1 = 1 + 2i$,

$$A - (1 + 2i)I = \begin{pmatrix} -2i & 2 \\ -2 & -2i \end{pmatrix}.$$

An equivalent row-echelon form is

$$U = \begin{pmatrix} -2i & 2 \\ 0 & 0 \end{pmatrix},$$

and the eigenvectors are $\mathbf{v} = t \begin{pmatrix} -i \\ 1 \end{pmatrix}$ with $t \in \mathbb{C}$, $t \neq 0$.

For $\lambda_2 = 1 - 2i$ the eigenvectors are $\mathbf{v} = t \begin{pmatrix} i \\ 1 \end{pmatrix}$ with $t \in \mathbb{C}$, $t \neq 0$. $\diamond$

The characteristic polynomial of a real 2×2 matrix has real coefficients, so has two real roots, one real root with multiplicity 2, or a pair of distinct conjugate complex roots, so the matrix has two real eigenvalues, one real eigenvalue with multiplicity 2, or two distinct conjugate complex eigenvalues. Examples 7 – 10 above show all these possibilities.

If A is a complex 2×2 matrix, its characteristic polynomial has complex coefficients, and either two distinct complex roots or one complex root with multiplicity 2, and the matrix has two eigenvalues or one eigenvalue with multiplicity 2.

For each eigenvalue the space spanned by its corresponding eigenvector(s) is called the **eigenspace** for that eigenvalue. Thus, when we write down *the* eigenvectors for a given eigenvalue, we are really recording the basis vectors for the corresponding eigenspace.

8.2 Eigenvectors, bases, and diagonalisation

In the examples of the preceding section, we have seen that, with one exception (Example 8), we obtain two linearly independent eigenvectors for a 2×2 matrix. Since a 2×2 matrix A represents a linear map whose domain is $\mathbb{C}^2$, these two eigenvectors form a basis for the domain. The matrix of Example 8 has one independent eigenvector, and it does not form a basis for the domain.

These results can be generalised to matrices of arbitrary size.

Theorem 1. *If an $n \times n$ matrix has n distinct eigenvalues then it has n linearly independent eigenvectors.*

[X] *Proof.* Let the set of n distinct eigenvalues of the $n \times n$ matrix A be $\{\lambda_1, \dots, \lambda_n\}$ and let $\mathbf{v}_i$ be a corresponding eigenvectors for λ_i , $1 \leq i \leq n$. We shall now prove that

$$S = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}.$$

is linearly independent.

Suppose

$$\mu_1 \mathbf{v}_1 + \dots + \mu_n \mathbf{v}_n = \mathbf{0}. \quad (\#)$$

We show that $\mu_1 = 0$. In similar fashion, $\mu_2 = \dots = \mu_n = 0$, so $\mathbf{v}_1, \dots, \mathbf{v}_n$ are linearly independent. Apply the matrix $(A - \lambda_2 I)(A - \lambda_3 I) \dots (A - \lambda_n I)$ to both sides of #, then we have If $j \neq 1$,

$$\begin{aligned} (A - \lambda_2 I) \dots (A - \lambda_n I) \mathbf{v}_j &= (\lambda_j - \lambda_2)(\lambda_j - \lambda_3) \dots (\lambda_j - \lambda_n) \mathbf{v}_j = \mathbf{0} \quad \text{if } j \neq 1, \\ (A - \lambda_2 I) \dots (A - \lambda_n I) \mathbf{v}_1 &= (\lambda_1 - \lambda_2)(\lambda_1 - \lambda_3) \dots (\lambda_1 - \lambda_n) \mathbf{v}_1 \neq \mathbf{0}. \end{aligned}$$

So $\mu_1(\lambda_1 - \lambda_2) \dots (\lambda_1 - \lambda_n) \mathbf{v}_1 = \mathbf{0}$, and that is $\mu_1 = 0$. $\square$

NOTE. Even if the $n \times n$ matrix does not have n distinct eigenvalues, it may have n linearly independent eigenvectors.

In Examples 2 and 6 of Section 8.1, we have seen that it is very easy to write down eigenvalues and eigenvectors of diagonal matrices. The next theorem shows that it is sometimes possible to find an equivalent diagonal matrix for a given matrix.

Theorem 2. *If an $n \times n$ matrix A has n linearly independent eigenvectors, then there exists an invertible matrix M and a diagonal matrix D such that*

$$M^{-1}AM = D.$$

Further, the diagonal elements of D are the eigenvalues of A and the columns of M are the eigenvectors of A , with the j th column of M being the eigenvector corresponding to the j th element of the diagonal of D .

Conversely if $M^{-1}AM = D$ with D diagonal then the columns of M are n linearly independent eigenvectors of A .

[X] *Proof.* Let the n linearly independent eigenvectors of A be $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$. We now form the matrix M with these vectors as its columns, i.e.,

$$M = (\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_n).$$

Then, from the usual rules of matrix multiplication, we have

$$AM = (A\mathbf{v}_1 \quad A\mathbf{v}_2 \quad \cdots \quad A\mathbf{v}_n),$$

and from $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$ we have

$$AM = (\lambda_1 \mathbf{v}_1 \quad \lambda_2 \mathbf{v}_2 \quad \cdots \quad \lambda_n \mathbf{v}_n).$$

Following the usual rules of matrix multiplication, we can rewrite this equation in the matrix form

$$AM = (\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_n) \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} = MD,$$

where

$$D = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}$$

is the diagonal matrix of eigenvalues. Thus, $AM = MD$. Further, since the columns of M are a basis for $\mathbb{C}^n$, the equation $M\mathbf{x} = \mathbf{b}$ has a unique solution for all $\mathbf{b} \in \mathbb{C}^n$, and hence M is invertible. Then, on multiplying the equation $AM = MD$ on the left by M^{-1} , we have $M^{-1}AM = D$.

Conversely if $M^{-1}AM = D$ then $AM = MD$ and M is invertible. Let

$$M = (\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_n) \quad \text{and} \quad D = \begin{pmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{pmatrix}$$

then from the first columns of the matrix products on the two sides of $AM = MD$, we have $A\mathbf{v}_1 = \lambda_1 \mathbf{v}_1$. Similarly $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$, $1 \leq i \leq n$.

Finally the columns of an invertible matrix are linearly independent. $\square$

Definition 1. A square matrix A is said to be a **diagonalisable matrix** if there exists an invertible matrix M and diagonal matrix D such that $M^{-1}AM = D$.

Example 1. Show that the matrix

$$A = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}$$

is diagonalisable and find an invertible matrix M and diagonal matrix D such that $M^{-1}AM = D$.

SOLUTION. We first find the eigenvalues and eigenvectors of A in the usual way. The eigenvalues are $\lambda_1 = 5, \lambda_2 = 1$ and corresponding eigenvectors are $\mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. Clearly, $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent. (Theorem 1 guarantees this, since $\lambda_1 \neq \lambda_2$.) Thus we may apply Theorem 2, letting D be a diagonal matrix with the eigenvalues as its diagonal elements, and M be the matrix with corresponding eigenvectors as its columns. For example,

$$D = \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix} \quad \text{and} \quad M = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

are the required diagonal matrix D and a suitable invertible matrix M . ◇

NOTE.

1. The results can be checked by direct multiplication of $M^{-1}AM$. In the above example, we readily obtain

$$M^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix},$$

and then

$$M^{-1}AM = M^{-1} \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 5 & 1 \\ 5 & -1 \end{pmatrix} = \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix} = D.$$

2. The choice of D and M is not unique. For example, we could reverse the order of the eigenvalues and set

$$D = \begin{pmatrix} 1 & 0 \\ 0 & 5 \end{pmatrix}, \quad M = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}.$$

Also non-zero multiples of eigenvectors are eigenvectors, so multiplying any column of M by a non-zero scalar would produce another valid diagonalising matrix.

8.3 Applications of eigenvalues and eigenvectors

Some important practical applications have already been noted at the beginning of this chapter. Many of these applications arise from the study of dynamical systems. A dynamical system is essentially any system which changes in time. Some examples of such systems include an electrical power network, a bridge oscillating in a wind, the population of a city or country, an ant colony, a forest, the Australian economy, an atom, an atomic nucleus.

8.3.1 Powers of A

A typical problem in, for example, the study of dynamical systems is to find A^k for positive integers k . There are two results which enable us to easily solve this problem.

Proposition 1. Let D be the diagonal matrix

$$D = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}.$$

Then, for $k \geq 1$,

$$D^k = \begin{pmatrix} \lambda_1^k & 0 & \cdots & 0 \\ 0 & \lambda_2^k & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^k \end{pmatrix}.$$

Proof. The proof is by induction.

The result is obviously true for $k = 1$.

Now assume that the result is true for $k = m$. Then, on multiplying out,

$$\begin{aligned} D^{m+1} &= DD^m = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix} \begin{pmatrix} \lambda_1^m & 0 & \cdots & 0 \\ 0 & \lambda_2^m & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^m \end{pmatrix} \\ &= \begin{pmatrix} \lambda_1^{m+1} & 0 & \cdots & 0 \\ 0 & \lambda_2^{m+1} & & 0 \\ \vdots & & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^{m+1} \end{pmatrix}. \end{aligned}$$

Hence, if the result is true for m it is also true for $m + 1$. But, we have already seen that it is true for $m = 1$, and hence it is true for all positive integers k . $\square$

The second result that we need is as follows:

Proposition 2. If A is diagonalisable, that is, if there exists an invertible matrix M and diagonal matrix D such that $M^{-1}AM = D$, then

$$A^k = MD^kM^{-1} \quad \text{for integer } k \geq 1.$$

Proof. The proof is by induction.

On multiplying $M^{-1}AM = D$ on the left by M and on the right by M^{-1} , we obtain

$$A = MDM^{-1},$$

and hence the statement of the proposition is true for $k = 1$.

Now suppose the statement of the proposition is true for $k = m$. Then

$$A^{m+1} = AA^m = MDM^{-1}MD^mM^{-1} = MDD^mM^{-1} = MD^{m+1}M^{-1},$$

and hence the statement of the proposition is also true for $m + 1$. Thus, the statement of the proposition is true for all positive integers k . $\square$

Example 1. Find A^k for

$$A = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}.$$

SOLUTION. The first step is to check that A is diagonalisable, and, if it is, to find the matrix M of eigenvectors and diagonal matrix D of eigenvalues such that $A = MDM^{-1}$. From Example 1 of Section 8.2, suitable matrices are:

$$D = \begin{pmatrix} 5 & 0 \\ 0 & 1 \end{pmatrix}; \quad M = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}; \quad M^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}.$$

Then,

$$\begin{aligned} A^k &= MD^kM^{-1} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 5^k & 0 \\ 0 & 1^k \end{pmatrix} M^{-1} \\ &= \begin{pmatrix} 5^k & 1 \\ 5^k & -1 \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2}(5^k + 1) & \frac{1}{2}(5^k - 1) \\ \frac{1}{2}(5^k - 1) & \frac{1}{2}(5^k + 1) \end{pmatrix}. \end{aligned}$$

As a check on this solution, note that we obtain I if we substitute $k = 0$, and A if we substitute $k = 1$. $\diamond$

NOTE. [X] Given a diagonalisable matrix A , we can give meaning to its exponential, using the power series expansion of e^x . We substitute A into the expansion

$$e^x = 1 + x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \dots$$

replacing 1 by I .

Since A is diagonalisable, we can write $A = PDP^{-1}$, with D diagonal, and a simple calculation shows that

$$I + PDP^{-1} + \frac{1}{2!}(PDP^{-1})^2 + \frac{1}{3!}(PDP^{-1})^3 + \dots = P(I + D + \frac{1}{2!}D^2 + \frac{1}{3!}D^3 + \dots)P^{-1}.$$

We define this to be the exponential of the matrix.

In the case of a 2×2 matrix with distinct eigenvalues λ_1, λ_2 , by adding the entries in the matrix, we have

$$e^A = P \begin{pmatrix} 1 + \lambda_1 + \frac{1}{2!}\lambda_1^2 + \frac{1}{3!}\lambda_1^3 + \dots & 0 \\ 0 & 1 + \lambda_2 + \frac{1}{2!}\lambda_2^2 + \frac{1}{3!}\lambda_2^3 + \dots \end{pmatrix} P^{-1}$$

$$= P \begin{pmatrix} e^{\lambda_1} & 0 \\ 0 & e^{\lambda_2} \end{pmatrix} P^{-1}.$$

In a similar way, one can define the sine and cosine (etc) of a matrix.

8.3.2 Solution of first-order linear differential equations

A typical problem in many applications is to find the solution of a pair of first-order linear differential equations with constant coefficients of the form

$$\begin{aligned} \frac{dy_1}{dt} &= a_{11}y_1 + a_{12}y_2 \\ \frac{dy_2}{dt} &= a_{21}y_1 + a_{22}y_2, \end{aligned}$$

with initial conditions $y_1(0)$ and $y_2(0)$ given.

If t represents time, this pair of equations represents a simple “continuous-time dynamical system.” For example, in a model of a population, $y_1(t)$ might be the number of females at time t and $y_2(t)$ the number of males at time t . The system of equations then describes how the numbers of females and males change with time.

One method of solution of this system is as follows. We first write the equations in matrix form, with

$$\mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}, \quad A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix},$$

and obtain

$$\frac{d\mathbf{y}}{dt} = A\mathbf{y}, \quad \text{with} \quad \mathbf{y}(0) = \begin{pmatrix} y_1(0) \\ y_2(0) \end{pmatrix}.$$

In this matrix form it is clear that there is no real restriction on the number of components of the vector $\mathbf{y}$. Equations of this type are important in the study of dynamical systems, where they are given the special name of **state-space equations**. The vector $\mathbf{y}$ in the equation is then called the **state vector** and t represents time.

This type of equation is a generalisation of the one-dimensional, first-order, linear differential equation with constant coefficients that you have met in calculus, and which is of the form

$$\frac{dy}{dt} = ay; \quad y(0) = y_0 = \text{constant}.$$

This equation has a solution

$$y(t) = y_0 e^{at}.$$

It is therefore plausible to guess that the n -dimensional equation will have a similar exponential type of solution. We therefore guess an exponential solution of the form:

$$\mathbf{y} = \mathbf{u}(t) = \mathbf{v}e^{\lambda t},$$

where λ is a constant scalar and $\mathbf{v}$ is a constant vector. On substituting this guess or “trial solution” into the matrix equation we obtain

$$\frac{d\mathbf{y}}{dt} = \lambda \mathbf{v}e^{\lambda t} = A\mathbf{y} = A\mathbf{v}e^{\lambda t},$$

which can be rearranged to give

$$e^{\lambda t}(A\mathbf{v} - \lambda\mathbf{v}) = \mathbf{0}.$$

Now, $e^{\lambda t} \neq 0$ for all t , and hence our guess is actually a solution only if $(A - \lambda I)\mathbf{v} = \mathbf{0}$. We therefore arrive at the result:

Proposition 3. $\mathbf{y}(t) = \mathbf{v}e^{\lambda t}$ is a solution of

$$\frac{d\mathbf{y}}{dt} = A\mathbf{y}$$

if and only if λ is an eigenvalue of A and $\mathbf{v}$ is an eigenvector for the eigenvalue λ .

Example 2. Find solutions of

$$\frac{d\mathbf{y}}{dt} = A\mathbf{y} \quad \text{where} \quad A = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}.$$

SOLUTION. We first find the eigenvalues and eigenvectors of A . We have obtained these previously, and they are:

$$\lambda_1 = 5, \quad \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

and

$$\lambda_2 = 1, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Hence, two solutions of the equation are

$$\mathbf{u}_1(t) = e^{5t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{and} \quad \mathbf{u}_2(t) = e^t \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

◇

The next point to notice is that the linearity of the differential equation leads to the following proposition.

Proposition 4. If $\mathbf{u}_1(t)$ and $\mathbf{u}_2(t)$ are two solutions of the equation

$$\frac{d\mathbf{y}}{dt} = A\mathbf{y},$$

then any linear combination of $\mathbf{u}_1$ and $\mathbf{u}_2$ is also a solution.

Proof. Let

$$\mathbf{y}(t) = \alpha_1 \mathbf{u}_1(t) + \alpha_2 \mathbf{u}_2(t),$$

where α_1 and α_2 are scalars. Then

$$\begin{aligned} \frac{d}{dt}(\alpha_1 \mathbf{u}_1(t) + \alpha_2 \mathbf{u}_2(t)) &= \alpha_1 \frac{d\mathbf{u}_1}{dt} + \alpha_2 \frac{d\mathbf{u}_2}{dt} \\ &= \alpha_1 A\mathbf{u}_1 + \alpha_2 A\mathbf{u}_2 \\ &= A(\alpha_1 \mathbf{u}_1 + \alpha_2 \mathbf{u}_2), \end{aligned}$$

and the result is proved. □

Example 2 (continued). In our example, we therefore have that

$$\mathbf{y}(t) = \alpha_1 e^{5t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \alpha_2 e^t \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

is a solution of the linear differential equation.

Although we have not proved it, the above solution is the general solution of the original differential equation, that is, every solution of the differential equation is of the above form.

Now, since there are two unknown scalars in the general solution, two extra conditions must be specified in order to completely determine the solution. Typical conditions are that the value of $\mathbf{y}(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix}$ is given at some t , for example, at $t = 0$. $\diamond$

Example 2 (continued). Find the solution of

$$\frac{d\mathbf{y}}{dt} = A\mathbf{y} \quad \text{for} \quad A = \begin{pmatrix} 3 & 2 \\ 2 & 3 \end{pmatrix}, \quad \text{given that} \quad \mathbf{y}(0) = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

SOLUTION. On substituting $t = 0$ into our general solution of the differential equation, and equating $\mathbf{y}(0)$ to the given vector, we obtain

$$\mathbf{y}(0) = \alpha_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}.$$

We can now obtain α_1 and α_2 by solving this pair of linear equations in the usual way. We find $\alpha_1 = -\frac{1}{2}$ and $\alpha_2 = \frac{3}{2}$, and hence the solution of the differential equation is

$$\mathbf{y}(t) = -\frac{1}{2}e^{5t} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{3}{2}e^t \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

$\diamond$

One reason for considering these linear first-order systems of differential equations is that every linear differential equation can be written as a system of linear first-order differential equations. We will illustrate the method with an example.

Example 3. Convert the second-order differential equation

$$\frac{d^2 y}{dt^2} + 4\frac{dy}{dt} - 5y = 0$$

to a system of first-order differential equations.

SOLUTION. First define new variables by

$$y_1 = y \quad \text{and} \quad y_2 = \frac{dy_1}{dt} = \frac{dy}{dt}.$$

Then, on differentiating y_2 and using the differential equation, we find

$$\frac{dy_2}{dt} = \frac{d^2 y}{dt^2} = 5y - 4\frac{dy}{dt} = 5y_1 - 4y_2.$$

The original second-order equation is therefore equivalent to the pair of first-order equations

$$\begin{aligned}\frac{dy_1}{dt} &= y_2 \\ \frac{dy_2}{dt} &= 5y_1 - 4y_2.\end{aligned}$$

This pair of equations can then be rewritten in matrix form as

$$\frac{d\mathbf{y}}{dt} = A\mathbf{y}, \quad \text{where} \quad A = \begin{pmatrix} 0 & 1 \\ 5 & -4 \end{pmatrix}.$$

◇

It is useful to compare the matrix method of solution of a second-order linear differential equation with the method of solution usually taught in calculus courses. The final results obtained by the two methods are, of course, the same.

Example 4 (Matrix method). Guess a solution of form $\mathbf{y}(t) = \mathbf{u}(t) = e^{\lambda t}\mathbf{v}$ and substitute in the differential equation. Then, $\mathbf{u}$ is a solution if λ and $\mathbf{v}$ satisfy the eigenvector equation $A\mathbf{v} = \lambda\mathbf{v}$.

The eigenvalues are solutions of the characteristic equation $\det(A - \lambda I) = 0$, that is, of

$$\det(A - \lambda I) = \begin{vmatrix} -\lambda & 1 \\ 5 & -4 - \lambda \end{vmatrix} = \lambda^2 + 4\lambda - 5 = 0.$$

The roots of the quadratic give the eigenvalues $\lambda_1 = -5$ and $\lambda_2 = 1$. A solution to $(A + 5I)\mathbf{v}_1 = 0$ is the eigenvector $\mathbf{v}_1 = \begin{pmatrix} -1 \\ 5 \end{pmatrix}$, and a solution to $(A - I)\mathbf{v}_2 = 0$ is the eigenvector $\mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$. The general solution is therefore

$$\mathbf{y}(t) = \alpha_1 e^{-5t} \begin{pmatrix} -1 \\ 5 \end{pmatrix} + \alpha_2 e^t \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

Since $y(t) = y_1(t)$, the solution for $y(t)$ in the original second-order equation is

$$y(t) = y_1(t) = -\alpha_1 e^{-5t} + \alpha_2 e^t.$$

◇

Example 5 (Calculus method). We first guess a solution $y(t) = e^{\lambda t}$, and substitute in the original second-order differential equation to obtain the so-called characteristic equation

$$\lambda^2 + 4\lambda - 5 = 0.$$

Note that this characteristic equation is identical to the characteristic equation $\det(A - \lambda I) = 0$ of the matrix method. See question 24 of the problems for a generalisation of this result.

The roots of the quadratic are $\lambda_1 = -5$ and $\lambda_2 = 1$, and hence the general solution is

$$y(t) = \alpha_1 e^{-5t} + \alpha_2 e^t,$$

which is identical to the solution from the matrix method.

◇

In the above example, it is clear that the calculus method gives a much quicker solution than the matrix method. However, the matrix method has the great advantage that it works for a much larger class of differential equations than does the calculus method. One reason that the matrix method works for a larger class of differential equations is that any single higher-order differential equation can be easily converted into a system of first-order equations, but it is extremely difficult to convert a given system of first-order equations into a single higher-order differential equation. It should also be pointed out that the matrix method described above will not work if the matrix A is not diagonalisable. However, an extension of the matrix method which uses “Jordan forms” can be developed to handle this case.

Example 6. The atoms in a laser can exist in two states, an “excited state” and a “ground state”. The laser is initially pumped so that it has 80% of its atoms in the excited state and the remaining 20% in the ground state. When the laser is operating, 70% of the excited atoms decay to the ground state per second, whereas 40% of the ground state atoms are pumped up to the excited state per second.

Find the percentage of atoms in each state at a time t seconds after the laser starts to operate.

SOLUTION. Let

$$\begin{aligned}x_1(t) &= \% \text{ of atoms in excited state at time } t \\x_2(t) &= \% \text{ of atoms in ground state at time } t.\end{aligned}$$

During operation the laser is described by the pair of differential equations

$$\begin{aligned}\frac{dx_1}{dt} &= -70x_1(t) + 40x_2(t) \\ \frac{dx_2}{dt} &= 70x_1(t) - 40x_2(t)\end{aligned}$$

That is, in matrix form,

$$\frac{d\mathbf{x}}{dt} = A\mathbf{x}(t), \quad \text{where} \quad A = \begin{pmatrix} -70 & 40 \\ 70 & -40 \end{pmatrix}.$$

The eigenvalues of A are $\lambda_1 = 0$ and $\lambda_2 = -1.1$, and corresponding eigenvectors are $\mathbf{v}_1 = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$. The general solution is therefore

$$\mathbf{x}(t) = \alpha_1 \begin{pmatrix} 4 \\ 7 \end{pmatrix} + \alpha_2 e^{-1.1t} \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

The initial condition of the laser is given as $\mathbf{x}(0) = \begin{pmatrix} 80 \\ 20 \end{pmatrix}$, and hence the values of α_1 and α_2 can be determined from the equations

$$\mathbf{x}(0) = \begin{pmatrix} 80 \\ 20 \end{pmatrix} = \alpha_1 \begin{pmatrix} 4 \\ 7 \end{pmatrix} + \alpha_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix},$$

for which the solution is $\alpha_1 = 9\frac{1}{11}$ and $\alpha_2 = -43\frac{7}{11}$.

Thus, the complete solution is

$$\begin{aligned}x_1(t) &= \frac{1}{11} (400 + 440e^{-1.1t}) \\x_2(t) &= \frac{1}{11} (700 - 440e^{-1.1t})\end{aligned}$$

Note that, as $t \rightarrow \infty$, $e^{-1.1t} \rightarrow 0$, and hence the laser settles into a “steady-state” operation in which there are $400/11 = 36\frac{4}{11}\%$ of the atoms in the excited state and $700/11 = 63\frac{7}{11}\%$ of the atoms in the ground state. The “steady-state” solution for large t is a scalar multiple of the eigenvector $\mathbf{v}_1 = \begin{pmatrix} 4 \\ 7 \end{pmatrix}$ corresponding to the eigenvalue $\lambda_1 = 0$. $\diamond$

8.3.3 [X] Markov chains

Matrices are very useful in studying many discrete-time dynamical systems. Dynamical systems are ones where the state of the system at stage $k + 1$ depends solely on the state at stage k .

Example 7. In a certain experiment, a psychologist was testing the learning abilities of rats by getting them to run a maze. The experimenter started with 100 rats, none of which had previously run the maze. She then set each of the 100 rats in turn at the maze and noted whether it successfully ran the maze. She then repeated the process several more times.

She found that, on average, 10% of the rats which failed at one attempt were successful on their next attempt, whereas 95% of the rats which were successful at one attempt were also successful at their next attempt. (These numbers are meant for illustration only. They are not taken from actual experimental data).

For this experiment, calculate the approximate number of rats which successfully run the maze on the 3rd run, the 20th run and the 50th run.

SOLUTION. Let

$$\begin{aligned}x_1(k) &= \text{number of rats successfully completing the maze at the } k\text{th run} \\x_2(k) &= \text{number of rats failing the maze at the } k\text{th run.}\end{aligned}$$

Then, in the $(k + 1)$ th run, we have

$$\begin{aligned}x_1(k + 1) &= 0.95x_1(k) + 0.10x_2(k) \\x_2(k + 1) &= 0.05x_1(k) + 0.90x_2(k),\end{aligned}$$

which can be written in matrix form as

$$\mathbf{x}(k + 1) = A\mathbf{x}(k), \quad \text{where} \quad A = \begin{pmatrix} 0.95 & 0.10 \\ 0.05 & 0.90 \end{pmatrix}.$$

We note that the unique solution of the equation is

$$\mathbf{x}(k) = A^k \mathbf{x}(0),$$

as can easily be checked by direct substitution in $\mathbf{x}(k + 1) = A\mathbf{x}(k)$.

In our problem, $\mathbf{x}(0) = \begin{pmatrix} 0 \\ 100 \end{pmatrix}$, since at the beginning there were 100 rats, none of which had successfully run the maze.

We now calculate A^k to complete the solution. The eigenvalues of A are $\lambda_1 = 1$ and $\lambda_2 = 0.85$, and corresponding eigenvectors are $\mathbf{v}_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ and $\mathbf{v}_2 = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$. Thus, A is diagonalisable and suitable choices for M , D and M^{-1} are

$$M = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 1 & 0 \\ 0 & 0.85 \end{pmatrix}, \quad \text{and} \quad M^{-1} = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{pmatrix}.$$

Thus,

$$\begin{aligned} \mathbf{x}(k) &= MD^k M^{-1} \mathbf{x}(0) = \begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1^k & 0 \\ 0 & (0.85)^k \end{pmatrix} \begin{pmatrix} \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{pmatrix} \begin{pmatrix} 0 \\ 100 \end{pmatrix} \\ &= \frac{100}{3} \begin{pmatrix} 2(1 - (0.85)^k) \\ (1 + 2(0.85)^k) \end{pmatrix}. \end{aligned}$$

Note: As a check, if we substitute $k = 0$ in this expression, we obtain $\mathbf{x}(0) = \begin{pmatrix} 0 \\ 100 \end{pmatrix}$. Further, for $k = 1$, we have

$$\mathbf{x}(1) = \frac{100}{3} \begin{pmatrix} 2(1 - 0.85) \\ (1 + 2(0.85)) \end{pmatrix} = \begin{pmatrix} 10 \\ 90 \end{pmatrix},$$

which equals

$$A\mathbf{x}(0) = \begin{pmatrix} 0.95 & 0.10 \\ 0.05 & 0.90 \end{pmatrix} \begin{pmatrix} 0 \\ 100 \end{pmatrix} = \begin{pmatrix} 10 \\ 90 \end{pmatrix}.$$

Then, for $k = 3$ the solution is $\mathbf{x}(3) = (25.72, 74.28)$, and hence approximately 26 rats will successfully complete the maze on the third run. For $k = 20$, the solution is $\mathbf{x}(20) = \begin{pmatrix} 64.08 \\ 35.92 \end{pmatrix}$, and hence approximately 64 rats will successfully complete the maze on the 20th run. For $k = 50$, the solution is $\mathbf{x}(50) = \begin{pmatrix} 66.65 \\ 33.35 \end{pmatrix}$, and hence approximately 67 rats will successfully complete the maze on the 50th run. Note that, for large values of k , $\mathbf{x}(k)$ is approximately equal to $\begin{pmatrix} 66\frac{2}{3} \\ 33\frac{1}{3} \end{pmatrix}$, which corresponds to approximately 67 rats successfully completing the maze on a given run. Note that this solution for large values of k is a scalar multiple of the eigenvector $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$ corresponding to the eigenvalue $\lambda_1 = 1$, which is the eigenvalue of A with largest magnitude. $\diamond$

Systems such as the one in Example 7 are called Markov chains. In these systems the objects or individuals can be in one of a certain number of states and the system is modelled by the matrix equation

$$\mathbf{x}(k+1) = A\mathbf{x}(k)$$

where $\mathbf{x}(k) = \begin{pmatrix} x_1(k) \\ \vdots \\ x_n(k) \end{pmatrix}$ gives the number of individuals in each of the n states at time k . The $n \times n$ matrix A has the property that all its entries are non-negative, and, for $j = 1, \dots, n$, $\sum_{i=1}^n a_{ij} = 1$. In other words, each column sums to 1. The number a_{ij} is the probability that an individual changes from state j to state i . Usually what we are interested in is finding the long term behaviour of such a system. That is, how A^k behaves as $k \rightarrow \infty$. It turns out that the behaviour exhibited in Example 7 is typical of these systems. For any such matrix, $\lambda = 1$ is an eigenvalue, and indeed is the eigenvalue of largest magnitude. In almost all cases, $A^k \mathbf{x}(0)$ converges to a multiple of the eigenvector corresponding to $\lambda = 1$. The limit vector in these cases depends only on the number of individuals involved, and not on the initial distribution of the individuals into the particular states. We conclude this section by proving that $\lambda = 1$ is always an eigenvalue of such matrix, i.e. the columns sums are all one.

Lemma 5. *If λ is an eigenvalue of A , then λ is also a eigenvalue of A^T .*

Proof. Question 13 in the problems for this chapter. □

Theorem 6. *Suppose that A is $n \times n$ matrix and that the sum of each of the columns of A is 1. Then A has 1 as an eigenvalue.*

Proof. The hypothesis on $A = (a_{ij})$ is that

$$\begin{aligned} a_{11} + a_{21} + \cdots + a_{n1} &= 1 \\ a_{12} + a_{22} + \cdots + a_{n2} &= 1 \\ \vdots & \\ a_{1n} + a_{2n} + \cdots + a_{nn} &= 1 \end{aligned}$$

or equivalently,

$$A^T \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}$$

Thus, 1 is an eigenvalue of A^T , and by the preceding lemma, is thus an eigenvalue of A . (Note that, in general, $(1 \ 1 \ \cdots)^T$ will not be its eigenvector.) □

8.4 Eigenvalues and MAPLE

The `LinearAlgebra` package in Maple has procedures for doing all the calculations described in this section. The command

```
with(LinearAlgebra):
```

loads the `LinearAlgebra` commands. If A is a square matrix then

```
Determinant(A-t);
```

produces the characteristic polynomial for A . Actually, Maple has a command which will also do this directly. A slight complication here is that

```
CharacteristicPolynomial(A,t);
```

gives the polynomial $\det(tI - A)$. This of course has the same roots as the polynomial we use. You can (and should at least once!) use `solve` or `fsolve` to find the roots of this equation, or you can use the `eigenvalues` command directly. You can then use `kernel` to find the eigenvectors.

```
evals:=Eigenvalues(A);
NullSpace(A-evals[1]);
```

This will give you a set containing a basis for the eigenspace. Use `op` to strip off the braces if necessary.

You can also get Maple to do all this for you at once, and more. The command

```
EV:=Eigenvectors(A);
```

returns a sequence with two elements. The first is a Vector with the eigenvalues as entries and the second is a Matrix whose columns are the eigenvectors in the same order. This matrix is thus a diagonalising matrix for A , if one exists. Thus if you then do

```
EV[2]^(-1).A.EV[2]
```

you will get a diagonal matrix — the same matrix that

```
DiagonalMatrix(EV[1]);
```

would give.

Problems for Chapter 8

Problems 8.1 : Definitions and examples

1. [R] Let

$$A = \begin{pmatrix} 3 & 0 \\ 0 & -4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} -3 & 0 \\ 0 & 0 \end{pmatrix},$$

and let $\mathbf{e}_1$ and $\mathbf{e}_2$ be the standard basis vectors for $\mathbb{R}^2$.

- a) Write down the eigenvalues and eigenvectors of A , B and C .
 - b) Draw a sketch of $\mathbf{e}_1$, $A\mathbf{e}_1$, $\mathbf{e}_2$, $A\mathbf{e}_2$. Then, for some vector $\mathbf{x}$ which is not parallel to either $\mathbf{e}_1$ or $\mathbf{e}_2$ draw a sketch of $\mathbf{x}$ and $A\mathbf{x}$.
 - c) Repeat part (b) for the matrix B . Comment on any differences you observe between the results for A and B .
 - d) Repeat part (b) for the matrix C . Again comment on any differences you observe between the results for A and C .
 - e) For $\mathbf{x} \neq \mathbf{0}$, prove algebraically that $A\mathbf{x}$ is parallel to $\mathbf{x}$ if and only if $\mathbf{x}$ is parallel to either $\mathbf{e}_1$ or $\mathbf{e}_2$, that $B\mathbf{x}$ is parallel to $\mathbf{x}$ for all $\mathbf{x}$ and that $C\mathbf{x}$ is parallel to $\mathbf{e}_1$ for all $\mathbf{x}$.
2. [R] Show that the vector $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ is an eigenvector of the matrix $\begin{pmatrix} 5 & -3 \\ 2 & 0 \end{pmatrix}$ and find the corresponding eigenvalue.
3. [X] Let A be a fixed 3×3 matrix and define a linear map $T : M_{33} \rightarrow M_{33}$ by $T(X) = AX$. If λ is a real eigenvalue of T corresponding to an invertible eigenvector X , find λ in terms of $\det(A)$.
4. [H] Let T be the linear map which reflects vectors in $\mathbb{R}^2$ about the line $y = x$.
- a) Explain why $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 1 \\ -1 \end{pmatrix}$ are eigenvectors of T and give their corresponding eigenvalues.
 - b) Find the matrix A such that $T\mathbf{x} = A\mathbf{x}$ for all $\mathbf{x} \in \mathbb{R}^2$.
5. [R] Find the eigenvalues and eigenvectors for
- a) $A = \begin{pmatrix} 6 & -2 \\ 6 & -1 \end{pmatrix}$, b) $A = \begin{pmatrix} -5 & 2 \\ -6 & 3 \end{pmatrix}$.
6. [X] For each of the matrices in the preceding question find two independent eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$. On one diagram sketch the lines

$$\begin{aligned} \ell_1 &= \{\mathbf{x} : \mathbf{x} = \mu\mathbf{v}_1, \mu \in \mathbb{R}\} \\ \ell_2 &= \{\mathbf{x} : \mathbf{x} = \mu\mathbf{v}_2, \mu \in \mathbb{R}\} \end{aligned}$$

and the parallelogram

$$P = \{\mathbf{x} : \mathbf{x} = \mu_1 \mathbf{v}_1 + \mu_2 \mathbf{v}_2, \quad \text{for } 0 \leq \mu_1 \leq 1, 0 \leq \mu_2 \leq 1\}.$$

Then identify and sketch (on a separate diagram)

$$\{\mathbf{y} : \mathbf{y} = A\mathbf{x}, \mathbf{x} \in \ell_1\}$$

$$\{\mathbf{y} : \mathbf{y} = A\mathbf{x}, \mathbf{x} \in \ell_2\}$$

$$\{\mathbf{y} : \mathbf{y} = A\mathbf{x}, \mathbf{x} \in P\}.$$

Describe the linear mapping $T\mathbf{x} = A\mathbf{x}$ geometrically.

7. [R] Find the eigenvalues and eigenvectors of the following matrices. In each case, note if the eigenvalues are real, occur in complex conjugate pairs, or are general complex numbers. Also note if the eigenvectors form a basis for $\mathbb{C}^2$.

$$\begin{array}{lll} \text{a)} \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}, & \text{b)} \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}, & \text{c)} \begin{pmatrix} 3 & 5 \\ 0 & -6 \end{pmatrix}, \\ \text{d)} \begin{pmatrix} 0 & -2 \\ 1 & 2 \end{pmatrix}, & \text{e)} \begin{pmatrix} 4 & 2i \\ 2i & 6 \end{pmatrix}, & \text{f)} \begin{pmatrix} 4 & -2i \\ 2i & 6 \end{pmatrix}. \end{array}$$

8. [H] Show that the eigenvalues of a square row-echelon form matrix U are equal to the diagonal elements of the matrix. (A square row-echelon form matrix is also called an upper triangular matrix).
9. [R] Find the eigenvalues and eigenvectors of the row-echelon matrix

$$U = \begin{pmatrix} 2 & -4 & 1 & 3 \\ 0 & -2 & 1 & -3 \\ 0 & 0 & 3 & 3 \\ 0 & 0 & 0 & 5 \end{pmatrix}.$$

10. [R] Find the eigenvalues and eigenvectors of the following matrices.

$$\text{a)} \quad A = \begin{pmatrix} 1 & 3 & 0 \\ 2 & 2 & 0 \\ 0 & 0 & 6 \end{pmatrix}, \quad \text{b)} \quad B = \begin{pmatrix} 3 & 0 & 0 \\ 0 & -4 & -1 \\ 0 & 6 & 3 \end{pmatrix}.$$

Problems 8.2 : Eigenvectors, bases, and diagonalisation

11. [R] For each of the matrices in Questions 7, 9 and 10, decide if the matrix is diagonalisable, and if it is find an invertible matrix M and a diagonal matrix D such that $D = M^{-1}AM$.
12. [H] Show that if λ is an eigenvalue of A then λ is also an eigenvalue of the matrix $A' = B^{-1}AB$, where B is any invertible matrix. Also show that if $\mathbf{v}$ is an eigenvector of A for eigenvalue λ then $B^{-1}\mathbf{v}$ is an eigenvector of A' for eigenvalue λ .

13. [H] Show that if λ is an eigenvalue of A then λ is also an eigenvalue of A^T .

HINT: Use the characteristic equation and the properties of determinants.

14. [X] Let A be an $n \times n$ matrix. Let $T_A : \mathbb{C}^n \rightarrow \mathbb{C}^n$ be the linear transformation defined by

$$T_A(\mathbf{x}) = A\mathbf{x} \quad \text{for } \mathbf{x} \in \mathbb{C}^n.$$

Let the columns of an $n \times n$ matrix B be an ordered basis for $\mathbb{C}^n$. Show that the matrix representing T_A with respect to the basis formed by the columns of B is $B^{-1}AB$.

HINT. The method used in Example 3 of Section 7.6 might be helpful.

NOTE. Modern methods of finding eigenvalues search for a change of basis which makes $A' = B^{-1}AB$ into an upper triangular matrix. As shown in question 4 the eigenvalues are then the diagonal elements of the upper triangular matrix. The actual algorithms for finding the change of basis are complicated.

15. [X] Let $T : V \rightarrow V$ be linear. Show that if B is any basis for V and A is the matrix representing T with respect to the basis B in both domain and codomain of T then the eigenvalues of T and A are the same. What is the relation between the eigenvectors of T and A ?

Problems 8.3 : Applications of eigenvalues and eigenvectors

16. [R] Let $A = \begin{pmatrix} 0 & 6 \\ 1 & -1 \end{pmatrix}$. Diagonalise A and hence find A^5 .

17. [R] Let $A = \begin{pmatrix} 0 & 3 \\ 8 & 2 \end{pmatrix}$.

- a) Find the eigenvalues and eigenvectors of A .
- b) Find matrices P and D such that

$$A = PDP^{-1}.$$

- c) Write down an expression for A^n in terms of P and D . Hence evaluate $A^n P$.

18. [R] A first-order linear difference equation (often called a first-order linear recurrence relation) is an equation of the form

$$\mathbf{x}_{k+1} = A\mathbf{x}_k, \quad \text{where } k = 0, 1, 2, \dots,$$

and where A is a fixed matrix.

The solution of this equation is

$$\mathbf{x}_k = A^k \mathbf{x}_0,$$

as you can check by direct substitution. For the diagonalisable matrices of questions 7 a,c, find A^k and hence evaluate $\mathbf{x}_k$.

19. [R] For each of the diagonalisable matrices of questions 5, 9 and 10 find general solutions of the differential equations

$$\frac{d\mathbf{y}}{dt} = A\mathbf{y}.$$

20. [R] a) Find the eigenvalues and eigenvectors of $\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$.
 b) Hence solve the system of differential equations:

$$\begin{cases} \frac{dx_1}{dt} = 2x_1 + 3x_2, \\ \frac{dx_2}{dt} = x_1 + 4x_2. \end{cases}$$

21. [R] Solve the following systems of differential equations, given that $x(0) = y(0) = 100$.

$$\text{a) } \begin{cases} \frac{dx}{dt} = 5x - 8y \\ \frac{dy}{dt} = x - y \end{cases} \quad \text{b) } \begin{cases} \frac{dx}{dt} = 3x - 15y \\ \frac{dy}{dt} = x - 5y \end{cases}$$

22. [R] Solve the following second-order linear differential equations with constant coefficients by the “calculus method” and by the matrix method and compare your answers.

a) $5\frac{d^2y}{dt^2} - 6\frac{dy}{dt} + y = 0.$

b) $\frac{d^2y}{dt^2} - 16y = 0.$

23. [R] What happens if you try to solve the second-order equation

$$\frac{d^2y}{dt^2} + 4\frac{dy}{dt} + 4y = 0$$

by the matrix method?

24. [X] Consider the second-order linear differential equation

$$a\frac{d^2y}{dt^2} + b\frac{dy}{dt} + cy = 0,$$

where $a, b, c \in \mathbb{R}$ and $a \neq 0$.

- a) Assume that the solutions to the characteristic equation $a\lambda^2 + b\lambda + c = 0$ for this second-order differential equation are distinct. By making the substitutions $y_1 = y$ and $y_2 = \frac{dy_1}{dt}$, convert the differential equation into a system of first-order linear differential equations

$$\frac{d\mathbf{y}}{dt} = A\mathbf{y}, \quad \text{where } \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}.$$

- b) Using matrix methods, show that the general solution of this system is

$$\mathbf{y} = \alpha_1 e^{\lambda_1 t} \begin{pmatrix} 1 \\ \lambda_1 \end{pmatrix} + \alpha_2 e^{\lambda_2 t} \begin{pmatrix} 1 \\ \lambda_2 \end{pmatrix}.$$

Compare this solution with that obtained using the usual “calculus method” of solving the original second-order linear differential equation.

25. [X] A radioactive isotope A decays at the rate of 2% per century into a second radioactive isotope B , which in turn decays at a rate of 1% per century into a stable isotope C .

- a) Find a system of linear differential equations to describe the decay process. If we start with pure A , what are the proportions of A , B , and C after 500 years, after 1000 years, and after 1000000 years?

First solve this problem using matrix methods, and then try to solve the problem directly by solving the original two differential equations in the right order.

- b) Explain how the problem would be different if the rates of decay of A and B were both 2% per century.

26. [X] There are 3 mathematics lecturers A , B and C who are teaching parallel streams in algebra to a total of 900 students. At the first lecture equal numbers go to each lecture group. After each lecture a certain percentage of the students in each group decide to stay with the same lecturer while the remaining percentage divide evenly among the other two lecturers for the next lecture. If 98% of A 's students stay with A each time, 96% of B 's students stay with B and 94% of C 's students stay with C , find the numbers of students in each group in the 12th lecture and in the 24th lecture. Make the assumption that no students stop attending lectures.

HINT. Set up a model as a difference equation of the type given in Question 18. You may use MAPLE to find all eigenvalues and eigenvectors. Alternatively, if you wish to solve the problem by hand calculations, you will need to know that one of the eigenvalues is 1.

NOTE. This problem is an example of a Markov chain process. Markov chain processes are important in many areas of mathematics and its applications, such as statistics, psychology, finance, economics, operations research, queueing theory, inventory theory, diffusion processes, theory of epidemics etc.

27. [X] Repeat the previous question on the following assumptions. After each lecture, 1% of each group stop attending lectures altogether, and the remaining percentage either stay with the same lecturer or divide equally among the other two lecturers for the next lecture. If 97% of A 's students stay with A each time, 95% of B 's students stay with B and 93% of C 's students stay with C , find the numbers of students in each group in the 12th lecture and in the 24th lecture. Also find the total number of students attending lectures in the 12th lecture and in the 24th lecture.
28. [X] Consider a modified version of the population dynamics model of Example 9 of Section 7.5, in which all females are assumed to die at age 74 instead of at age 89, as in the model given. Use eigenvalues and eigenvectors to solve this modified model, given that there are

one million females in each age group at January 1, 1970. What happens to the population for large values of k ?

NOTE. You will need to use Maple to find the eigenvalues and eigenvectors of the matrix A , and you may also use Maple, if you wish, to carry out all other matrix manipulations required to solve the problem.

29. [X] Let A be a 2×2 matrix with the property that all its entries are non-negative and both its columns sum to 1. Show that $\lambda_1 = 1$ is always an eigenvalue for A , and that if λ_2 is another eigenvalue of A then $-1 \leq \lambda_2 \leq 1$.

Problems 8.4 : Eigenvalues and MAPLE

30. [M] Show that the matrix of the original population dynamics model of Example 9 of Section 7.5 is not diagonalisable.

HINT. Use Maple to find the eigenvalues, and then show that the eigenvalue $\lambda = 0$ has multiplicity 2 and that $\dim(\ker(A)) = 1$, i.e., a basis for the kernel of $A - 0I$ consists of one vector.

NOTE. The original population dynamics model can be solved by a generalisation of the eigenvalue-eigenvector methods which makes use of “Jordan forms”, and is covered in our second year linear algebra subjects.

31. [M] Using the following Maple session,

```
> with(LinearAlgebra):
> M:=<<6|2|2>,<-2|8|4>,<0|1|7>>;
```

$$M := \begin{bmatrix} 6 & 2 & 2 \\ -2 & 8 & 4 \\ 0 & 1 & 7 \end{bmatrix}$$

```
> I3:=IdentityMatrix(3);
```

$$I3 := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

```
> p:=Determinant(M-t*I3);
```

$$p := 336 - 146t + 21t^2 - t^3$$

```
> solve(p,t);
```

$$6, 7, 8$$

```
> NullSpace(M-6*I3);
```

$$\left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$$

```
> NullSpace(M-7*I3);
```

$$\left\{ \begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix} \right\}$$

> NullSpace(M-8*I3);

$$\left\{ \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \right\}$$

- a) state the eigenvalues and corresponding eigenvectors for the matrix M ,
- b) find a matrix A such that $A^{-1}MA$ is a diagonal matrix D and write down D ,
- c) calculate A^{-1} and hence find an explicit formula for M^k where k is a positive integer.

Chapter 9

INTRODUCTION TO PROBABILITY AND STATISTICS

*“What IS the use of repeating all that stuff?”
the Mock Turtle interrupted . . .
Lewis Carroll, Alice in Wonderland.*

This chapter introduces mathematical probability, random variables, and probability distributions. The concepts, methods, and applications are required in statistics courses that include MATH2801/MATH2901 – (Higher) Theory of Statistics, a core subject for the mathematics and statistics majors, and MATH2089/MATH2099, which are compulsory courses for many second year engineering students.

Statistics is the science of turning raw data into reliable information on which decisions can be made, given randomness or variation in the original data. As a science, it aims to uncover patterns in observations that can be described by mathematical or heuristic models. It is also concerned with formulating and testing various hypotheses about the context from which the data are drawn.

For instance in order to predict voting patterns in an election, opinions are sought from voters by carrying out opinion polls. It might not be possible to obtain the views of all voters, so the preferences of a relatively small sample of voters are obtained. As a result, the opinion poll can only provide an *estimate* of the true proportion of voters who favour a particular political party. It is important to quantify how accurate this estimate can be expected to be.

How many voters must be polled in order for this estimate to be reasonably accurate? How big is the measurement error? Statistical science has the answers to these frequently asked and important questions. These answers have had immediate utility. It traditionally cost time and resources to poll voters, so it mattered whether 1,000 voters formed an adequate sample or whether 10,000 voters were required.

This simple but typical example illustrates nicely the three essential aspects of statistical science: data production, data analysis, and statistical inference.

Data production

How many sample units should be taken, how should they be selected, and what data should be measured on each unit? An important part of data production is controlling the measurement error that invariably arises.

Data Analysis

In order to be presented transparently, data must be organised into easily understood forms, often as graphics, tables, and summary “statistics”, such as sample means or sample proportions. For opinion polling, a simple summary of the sample proportions provide the information sought.

Statistical Inference

This is the process of drawing valid conclusions about a whole population based on information obtained from a part of the population. An essential ingredient here is a **random sample** from the population. Given the data obtained from a random sample of voters, what can one infer about the general voting patterns?

The transition from population to random sample is one instance in which the notion of probability becomes important. To create a random sample, we must know the probability that a given member of the population will be selected in the sample. Conversely, probability allows us to model and forecast real-world behaviour in terms of random processes. In these ways, probability theory and statistics play important roles in countless contexts, such as clinical trials, weather forecasting, finance, or traffic control, to name just a few.

First, let us recollect some background set theory and notation.

9.1 Some Preliminary Set Theory

A probability model consists of two components:

1. A set of possible outcomes;
2. The probability of each outcome or set of outcomes.

In this section, we present basic set theory as background material for the first of these components. The second component will be addressed in Section 9.2.

Definition 1. A *set* is a collection of objects. These objects are called **elements**.

We write $x \in A$ to express that x is an element of a set A . If x is not an element of A , then we write $x \notin A$.

Example 1. The set $A = \{1, 2, 3\}$ has elements 1, 2, and 3. Thus, $1 \in A$ but $4 \notin A$, say. $\diamond$

The above definition is circular and imprecise. For instance, it is vulnerable to Russell’s Paradox (briefly discussed in MATH1081 Discrete Mathematics). One could improve the definition by insisting that each set must have the property that each conceivable element is either completely in the set or completely outside of the set, but not both. However, one must improve the definition further in order to guard it from contradiction, and this is in fact difficult. Fortunately for our purposes, the above naive definition suffices.

Definition 2.

- A set A is a **subset** of a set B (written $A \subseteq B$) if and only if each element of A is also an element of B ; that is, if $x \in A$, then $x \in B$.
- The **power set** $\mathcal{P}(A)$ of A is set of all subsets of A .
- The **universal set** S is the set that denotes all objects of given interest.
- The **empty set** $\emptyset$ (or $\{\}$) is the set with no elements.

Example 2. The set $A = \{1, 2, 3\}$ has eight subsets. For instance, $\{2, 3\} \subseteq A$. The power set of A is the set of these eight subsets, namely

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}. \quad \diamond$$

Example 3. For problems in 3-dimensional vector geometry, the universal set is usually $S = \mathbb{R}^3$. Points, lines, and planes are then subsets of S . $\diamond$

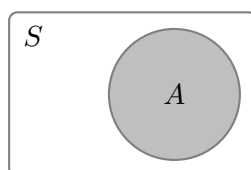
Definition 3. A set S is **countable** if its elements can be listed as a sequence.

More formally, S is countable if and only if there is a one-to-one function from S to $\mathbb{N}$.

Example 4.

- Every finite set is countable.
- The integers are countable since we can list them as follows: $0, 1, -1, 2, -2, \dots$
- The rationals are countable. (**Challenge:** can you list them as a sequence?)
- The reals are not countable; this can be shown by a simple and elegant proof known as Cantor's Diagonal Argument.

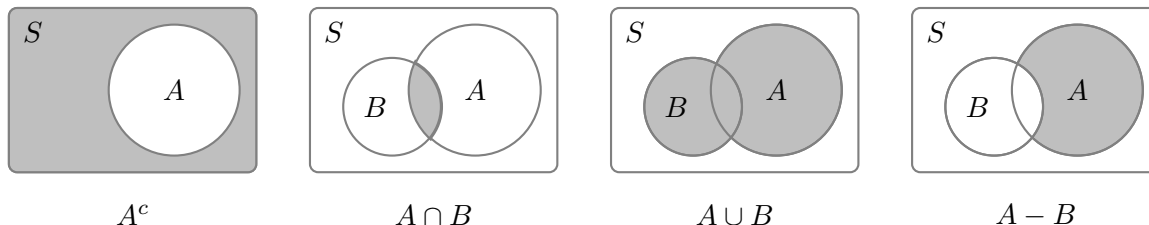
Sets are often visualised by a **Venn diagram** as regions in the plane. For instance, here is a Venn diagram of a universal set S containing a set A :



Definition 4. For all subsets $A, B \subseteq S$, define the following **set operations**:

- **complement** of A : $A^c = \{x \in S : x \notin A\}$
- **intersection** of A and B : $A \cap B = \{x \in S : x \in A \text{ and } x \in B\}$
- **union** of A and B : $A \cup B = \{x \in S : x \in A \text{ or } x \in B\}$
- **difference**: $A - B = \{x \in S : x \in A \text{ but } x \notin B\} = A \cap B^c$

Following mathematical convention, “or” in the union definition means “one or the other or both”. Venn diagrams describing the above set operations are given below.



Example 5. Let S be all students enrolled in MATH1231, let A be those who are 20 years or older, and let B be those who are engineering students. Then

A^c are MATH1231 students at most 19 years old

$A \cap B$ are MATH1231 students studying engineering and who are 20 years or older

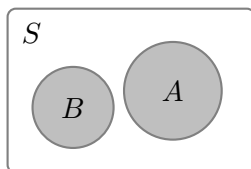
$A \cup B$ are MATH1231 students who study engineering or are 20 years or older

$A - B$ are MATH1231 students who do not study engineering but who are 20 years or older. $\diamond$

Definition 5. Sets A and B are **disjoint** (or **mutually exclusive**) if and only if

$$A \cap B = \emptyset$$

A Venn diagram showing two disjoint sets A and B is given below:



$$A \cap B = \emptyset$$

Example 6. Let S be the set of people in Australia, let A be the set of people enrolled in MATH1231, and let B be the set of people aged under 10. Then A and B are disjoint. $\diamond$

Definition 6. Disjoint subsets $A_1, \dots, A_k$ **partition** a set B if and only if

$$A_1 \cup \dots \cup A_k = B$$

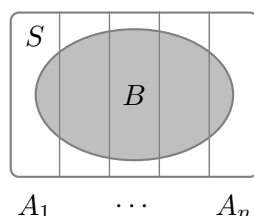
Note that A and A^c partition the universal set S for each subset A of S .

Example 7. The sets $A_1 = \{1, 3\}$, $A_2 = \{2\}$, $A_3 = \{4, 5\}$ partition the set $B = \{1, 2, 3, 4, 5\}$. $\diamond$

The following simple result will often be used in the rest of the chapter, sometimes implicitly.

Lemma 1. *If $A_1, \dots, A_n$ partition S and B is a subset of S , then $A_1 \cap B, \dots, A_n \cap B$ partition B .*

This result is illustrated below.

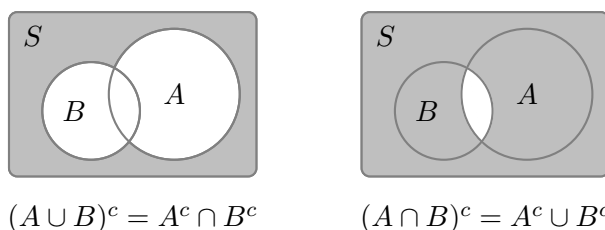


There are many laws governing set operations. Here are just a few:

$$\begin{array}{ll} \textbf{Distributive Laws} & A \cap (B \cup C) = (A \cap B) \cup (A \cap C) \\ & A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \end{array}$$

$$\begin{array}{ll} \textbf{De Morgan's Laws} & (A \cup B)^c = A^c \cap B^c \\ & (A \cap B)^c = A^c \cup B^c \end{array}$$

These laws can be proved by logical arguments or by sketching the Venn diagrams for the left-hand and right-hand sides of the identities. Venn diagrams for De Morgan's Laws are given here:



Definition 7. *If A is a set, then $|A|$ is the **number of elements** in A .*

Note that if A and B are disjoint, then

$$|A \cup B| = |A| + |B|$$

The Inclusion-Exclusion Principle. $|A \cup B| = |A| + |B| - |A \cap B|$

This result is clear once a Venn diagram is drawn.

The Inclusion-Exclusion Principle may be extended to any finite number of sets. For instance,

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

Note that for any subset A of S , we have $S = A \cup A^c$ and so $|A^c| = |S| - |A|$. Hence, for example, $|(A \cup B)^c| = |S| - |A \cup B|$. The following example makes use of this idea.

Example 8. Of 20 music students, 7 play guitar, 8 play piano, and 3 play both guitar and piano. How many play neither guitar nor piano?

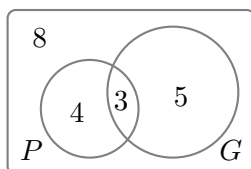
SOLUTION. Let S be the set of all the music students; let G be the set of students who play guitar; and let P be the set of students who play piano. By the information given,

$$|S| = 20 \quad |G| = 7 \quad |P| = 8 \quad |P \cap G| = 3.$$

By the Inclusion-Exclusion Principle, the number of students who play neither piano nor guitar is

$$\begin{aligned} |(G \cup P)^c| &= |S| - |G \cup P| \\ &= |S| - (|G| + |P| - |G \cap P|) = 20 - (7 + 8 - 3) = 8. \end{aligned}$$

Alternatively, we can draw a Venn diagram of the problem and deduce the answer by filling in the number of elements in each region:



◇

9.2 Probability

The notion of luck is ancient and has often been seen as an inherent quality that individuals or objects might possess and whose nature is determined by Fate, whims of the Gods, karmic justice, mana-like association with other instances of luck, and many other mechanisms. By associating with lucky individuals or objects, by acting righteously, or by appealing to the Gods, one might improve one's luck during one's present life. Gambling is the competitive realisation of this belief in influencing one's luck, and it too is ancient. Good gamblers have appeared throughout history, and many prominent and talented mathematicians have focused much of their work on gambling problems and strategies, particularly in the 16-18th centuries. However, most of this work addressed specific problems and was stunted by incorrect intuitions and by an unfortunate focus on ratios and odds. This focus is still present in gambling today, where odds are given, rather than percentages.

Apart from a few important exceptions, it was only relatively recently, in the first half of the 20th century, that the notion of luck was treated rigorously and systematically by mathematicians. Of note, A. Kolmogorov put forth a set of axioms in 1933 that provided a solid framework for dealing mathematically with the notion of luck, or in mathematical terms: **probability**.

9.2.1 Sample Space and Probability Axioms

In order to develop a framework for probability, we will first think of any given situation that leads randomly to a set of outcomes as an **experiment**. Thus, the roll of a die is seen as an experiment, as is the Melbourne Cup; countless other such experiments abound, including financial markets, the weather, election outcomes, or what grade you might get for this course.

Definition 1. A **sample space** of an experiment is a set of all possible outcomes.

Outcomes are also called **sample points**.

Example 1. Tossing a coin may be seen as an experiment. An appropriate sample space is the set $S = \{H, T\}$ where H (“head”) and T (“tail”) are the two possible outcomes.

Example 2. Tossing a coin 3 times can be seen as another experiment. If the object of the experiment is to determine the resulting coin-flip sequence, then an appropriate sample space is

$$S_1 = \{HHH, HHT, HTH, THH, HTT, THT, TTH, TTT\}.$$

On the other hand, if the object of the experiment were to determine the number of resulting heads, then an appropriate sample space is

$$S_2 = \{0, 1, 2, 3\}.$$

Thus, the experiment and its sample space depends on the type of data that we wish to observe.

It is often useful to consider sets of outcomes, particularly if the number of outcomes is large. This leads to the next definition.

Definition 2. An **event** is a subset of a sample space.

Note that the set of all events in a sample space S is exactly the power set $\mathcal{P}(S)$. Note also that the empty set $\emptyset$ and the whole space S are events.

Example 3. Toss a coin 3 times and consider the event A that we toss 2 heads. This is the subset of the sample space S_1 of Example 2 given by

$$A = \{HHT, HTH, THH\}.$$

Note that each of the outcomes in A forms an event by itself: $\{HHT\}$, $\{HTH\}$, $\{THH\}$.

In each of the above examples, each possible outcome has equal probability. This is not generally true, so we must define probability in full generality.

Definition 3. A **probability** P on a sample space S is any real function on $\mathcal{P}(S)$ that satisfies the following conditions:

- (a) $0 \leq P(A) \leq 1$ for all $A \subseteq S$;
- (b) $P(\emptyset) = 0$;
- (c) $P(S) = 1$;
- (d) If A and B are disjoint, then $P(A \cup B) = P(A) + P(B)$.

Example 4. Toss a coin and observe whether H or T is tossed. The appropriate sample space is the set $S = \{H, T\}$. Define the probability P on S as follows, for each event $A \subseteq S$:

$$P(A) = \frac{|A|}{2}.$$

Then $P(\{H\})$ is the probability of tossing H , namely $P(\{H\}) = \frac{1}{2}|\{H\}| = \frac{1}{2}$. Similarly, $P(\{T\}) = \frac{1}{2}$. Note that the probability of tossing neither H nor T is $P(\emptyset) = 0$, and that the probability of tossing either H or T is $P(S) = 1$.

This probability is exactly the probability that one would usually think of when tossing a coin. However, there are many other possible probabilities. For instance, let p be some real number between 0 and 1, and define the function Q on S by

$$Q(\emptyset) = 0 \quad Q(\{H\}) = p \quad Q(\{T\}) = 1 - p \quad Q(S) = 1.$$

It is easy to verify that Q is a probability on S . To find a physical interpretation of this probability, one could think of a coin that is twisted or bent, so that the probability of tossing H is not necessarily the same as that of tossing T . Bear in mind, however, that a probability is a mathematical object that need not always model a real-world phenomenon.

Example 5. Toss a coin 3 times and observe the resulting (ordered) sequence of H and T , as in Example 2 above. Let S be the natural sample space consisting of all 8 such sequences. The appropriate probability P is then given as follows, for each event $A \subseteq S$:

$$P(A) = \frac{|A|}{8}.$$

For instance, consider the event A that we toss 2 heads. The probability of this happening is

$$P(A) = \frac{|A|}{8} = \frac{|\{HHT, HTH, THH\}|}{8} = \frac{3}{8}.$$

Now consider the event $B = \{HHH\}$ that we toss 3 heads. Since $|B| = 1$, we see that $P(B) = \frac{1}{8}$.

The probability of tossing at least 2 heads is $P(A \cup B)$ which, since A and B are disjoint, equals

$$P(A \cup B) = P(A) + P(B) = \frac{3}{8} + \frac{1}{8} = \frac{1}{2}.$$

Example 6. Roll a die and observe the resulting number. An appropriate sample space is then $S = \{1, \dots, 6\}$. The appropriate probability P is given as follows, for each event $A \subseteq S$:

$$P(A) = \frac{|A|}{6}.$$

For example, consider the event A that we roll an even number. The probability of this occurring is

$$P(A) = \frac{|A|}{6} = \frac{|\{2, 4, 6\}|}{6} = \frac{3}{6} = \frac{1}{2}.$$

Theorem 1. Let P be a probability on a sample space S , and let A be an event in S .

1. If S is finite (or countable), then $P(A) = \sum_{a \in A} P(\{a\})$.

2. If S is finite and $P(\{a\})$ is constant for all outcomes $a \in S$, then $P(A) = \frac{|A|}{|S|}$.
3. If S is finite (or countable), then $\sum_{a \in S} P(\{a\}) = 1$.

Note that if S is finite, then $P(A)$ may be seen as *size*, or *ratio*, of A compared to S . In general, $P(A)$ may be seen as a *measure* of how large A is compared to S . Outcomes whose probabilities are all equal are often referred to as “equally likely”.

Proof. The finite case of statement 1 follows by induction using the additive condition (d) in the definition of a probability. We will ignore the general case of statement 1 but note that it is often given as an axiom for probabilities. Statement 3 follows immediately from statement 1 and by noting that $P(S) = 1$. Let us now prove the statement 2, so suppose that S is finite and that $P(\{a\})$ is equal to the constant p for all outcomes $a \in S$. By statement 3,

$$1 = \sum_{a \in S} P(\{a\}) = \sum_{a \in S} p = |S|p,$$

so $p = \frac{1}{|S|}$. By statement 1,

$$P(A) = \sum_{a \in A} P(\{a\}) = \sum_{a \in A} \frac{1}{|S|} = \frac{|A|}{|S|}. \quad \diamond$$

Example 7. The natural probabilities P in Examples 4–6 may each be expressed as

$$P(A) = \frac{|A|}{|S|}$$

where A is any event in S . This reflects the fact that each outcome is equally likely.

Example 8. Pick a ball at random from a bag containing 3 red balls and 7 blue balls. If each ball has the same chance as being picked as any other ball, then the chance of picking a red ball is $\frac{3}{10}$.

Let us express this in mathematical terms. Let S be the sample space consisting of all 10 balls. Next, let A be the event that a red ball is chosen; A is then the set containing the three red balls. Since the probability of each outcome is the same, the probability of picking a red ball is

$$P(A) = \frac{|A|}{|S|} = \frac{3}{10},$$

as expected. $\diamond$

The definition of probability only states what is required of a probability; it does not help us decide upon an appropriate probability for a given experiment. This sort of decision is called “allocating the probabilities” and is generally based on one of the following three methods:

Method 1. Allocate the probabilities on the basis of any inherent symmetry in the situation. This is what is applied in games of chance, as illustrated by the die-rolling or coin-tossing examples that we have seen. It is how you calculated with probability at school by using counting, permutations, and combinations with equally likely outcomes.

It is also used to allocate probabilities in the following sort of experiment. A “wheel of fortune” wheel is spun. The probability that it points to some region which subtends an angle θ is $\frac{\theta}{360}$. This is an example of an experiment with a sample space that is not countable, let alone finite.

Method 2. Allocate the probabilities on the basis of experience or large amounts of data.

This is what actuaries regularly do when creating life tables. The probability of a typical Australian male aged 60 living to age 65 is 95.4%, based on past history of lots of males aged 60 years.

Method 3. Guess or use intuition.

This is common in society but should be avoided by non-experts when dealing with serious issues. Indeed, the gambling industry, insurances, supermarket prices, and so on, and even much of politics and lawmaking, all rely on the common person's inability to properly understand probabilities, particularly when it comes to odds of winning or risk of injury. Actuaries and mathematicians educate themselves to avoid falling for common misconceptions; however, even these experts should not rely heavily on their intuition of probability.

9.2.2 Rules for Probabilities

Theorem 2. Let A and B be events of a sample space S .

1. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ (**Addition Rule**)
2. $P(A^c) = 1 - P(A)$
3. If $A \subseteq B$, then $P(A) \leq P(B)$.

Proof.

1. This result is connected to the Inclusion-Exclusion Principle.
2. The sets A and A^c partition S , so $P(A) + P(A^c) = P(A \cup A^c) = P(S) = 1$.
3. The sets A and $B - A$ partition B , so $P(A) \leq P(A) + P(B - A) = P(B)$. □

Example 9. What is the probability that at least two of n people share the same birthday?

SOLUTION. Ignoring leap years, let Y be the set of the 365 days of the year. An experiment could here be to discover the n birthdays, and an associated sample space is

$$S_n = \{(b_1, \dots, b_n) : b_1, \dots, b_n \in Y\}.$$

We wish to calculate $P(A_n)$ for the event $A_n \subseteq S_n$ that at least two of the n people share the same birthday. Assume that the probability of each person being born on a given date does not depend on the person or on the date. The outcomes of the sample space then have constant probability, namely $\frac{1}{|S_n|} = \frac{1}{365^n}$. Therefore, $P(A_n) = 1 - P(A_n^c) = 1 - \frac{|A_n^c|}{|S_n|}$. Now, A_n^c is the event that none of the n birthdays are the same. Therefore,

$$P(A_n) = 1 - P(A_n^c) = 1 - \frac{|A_n^c|}{|S_n|} = 1 - \frac{365 \times 364 \times \dots \times (365 - n + 1)}{365^n}.$$

It is thus slightly more likely than not that at least two of 23 people have the same birthday ($P(A_{23}) = 50.7\%$), and it is highly likely that at least two of 57 people share the same birthday ($P(A_{57}) = 99.01\%$). Of course, there will always be at least two people with the same birthday whenever there are more people than days in the year, and this is expressed by the probability $P(A_n) = 1$ for $n > 365$. ◇

Example 10. In some town, 80% of the population has comprehensive car policies, 60% has house cover, and 10% has neither. What percent has both covers?

SOLUTION. Let A be the event “a person has comprehensive car cover” and let B be the event “a person has house cover”. For any random person,

$$P(A) = 0.8, \quad P(B) = 0.6, \quad \text{and} \quad P(A^c \cap B^c) = 0.1.$$

Hence, $P(A \cup B) = 1 - P((A \cup B)^c) = 1 - P(A^c \cap B^c) = 1 - 0.1 = 0.9$. Therefore,

$$P(A \cap B) = P(A) + P(B) - P(A \cup B) = 0.8 + 0.6 - 0.9 = 0.5.$$

In other words, 50% of the population has both covers. $\diamond$

9.2.3 Conditional Probabilities

We now consider what happens if we restrict the sample space from S to some event in S .

Example 11. In Example 10 above, 80% of people have comprehensive car cover. However, of those people who have house cover, the percentage who also have comprehensive car cover is

$$\frac{50}{60} = 0.833 \quad \text{or} \quad 83.3\%.$$

Thus when we restrict our sample space to those having house cover, the percentage of those having comprehensive cover changes. We say that the conditional probability of a person having comprehensive cover **given** that they have house cover is 0.833. $\diamond$

Definition 4. The **conditional probability** of A **given** B is denoted and defined by

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad \text{provided that} \quad P(B) \neq 0.$$

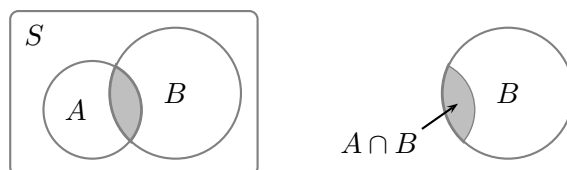
Lemma 3. For any fixed event B , the function $P(A|B)$ is a probability on S .

Proof. Check that the probability conditions are satisfied for $P(A|B)$. $\square$

Since $P(S) = 1$, we can write, for each event A of S ,

$$P(A) = \frac{P(A \cap S)}{P(S)} = P(A|S).$$

Just as $P(A)$ can be seen as a measure of A compared to S , $P(A|B) = \frac{P(A \cap B)}{P(B)}$ can be seen as a measure of A (or the part of A that is contained in B) compared to B . This is illustrated by the following Venn diagrams:



Example 12. We roll a die and let A and B be the events that we roll a six and that we roll an even number, respectively. Then $P(A) = \frac{1}{6}$ and $P(B) = \frac{3}{6} = \frac{1}{2}$. Since in this case $A \cap B = A$,

$$P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{P(A)}{P(B)} = \frac{(\frac{1}{6})}{(\frac{1}{2})} = \frac{1}{3}.$$

In other words, given that we rolled an even number, the probability of having rolled a six is $\frac{1}{3}$. This is as one would expect since there are 3 even rolls (2,4,6) of which one is six.

In contrast, the probability of rolling an even number, given that we rolled a six, is

$$P(B|A) = \frac{P(B \cap A)}{P(A)} = \frac{P(A)}{P(A)} = 1. \quad \diamond$$

Example 13. Consider a bag containing 3 red balls and 3 blue balls. First draw one ball from the bag and then draw another. Let R_i be the events that a red ball is chosen on the i th draw, where $i = 1, 2$, and define $B_1 (= R_1^c)$ and $B_2 (= R_2^c)$ similarly for the blue balls. The probability of first drawing a red ball or a blue is the same, namely $P(R_1) = P(B_1) = \frac{3}{6} = \frac{1}{2}$.

Now, suppose that we first draw a red ball. The bag then contains 2 red balls and 3 blue balls, so the probability of drawing a red ball on the second draw is $P(R_2|R_1) = \frac{2}{5}$. We can also calculate this probability from the definition of conditional probability. Of the $\frac{6 \times 5}{2} = 15$ ways of choosing two of the six balls without order, three of ways give us two red balls. Therefore, we see that $P(R_1 \cap R_2) = \frac{3}{15} = \frac{1}{5}$, so

$$P(R_2|R_1) = \frac{P(R_1 \cap R_2)}{P(R_1)} = \frac{(\frac{1}{5})}{(\frac{1}{2})} = \frac{2}{5}.$$

In contrast, $P(R_2) = P(B_2) = \frac{1}{2}$ since there are equally many red and blue balls to begin with.

Rearranging the terms in the definition of conditional probability yields the following identities.

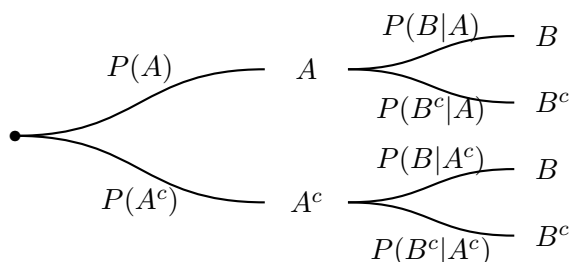
Multiplication Rule $P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$

Example 14. Consider the bag of red and blue balls in Example 13. We saw that the probability of drawing a red ball on the second draw, given that we first drew a red ball, is $P(R_2|R_1) = \frac{2}{5}$. Similarly, it is easy to see that $P(R_2|B_1) = \frac{3}{5}$. To calculate $P(R_2)$ without using the symmetry argument in Example 13, first note that R_2 is partitioned by $R_2 \cap R_1$ and $R_2 \cap B_1$: either a red ball is first drawn, followed by another red ball, or a blue ball and then a red ball are drawn. Hence by the Multiplication Rule,

$$\begin{aligned} P(R_2) &= P((R_2 \cap R_1) \cup (R_2 \cap B_1)) = P(R_2 \cap R_1) + P(R_2 \cap B_1) \\ &= P(R_2|R_1)P(R_1) + P(R_2|B_1)P(B_1) = \frac{2}{5} \times \frac{1}{2} + \frac{3}{5} \times \frac{1}{2} = \frac{1}{2}, \end{aligned}$$

as expected. $\diamond$

Conditional probabilities are implicitly used whenever a tree diagram is drawn. Thus in a typical two stage experiment we have the following tree diagram



Example 15. Consider 3 urns containing red and blue balls:

- Urn 1 contains 10 balls, of which 3 are red and 7 are blue;
- Urn 2 contains 20 balls, of which 4 are red and 16 are blue; and
- Urn 3 contains 10 balls, of which 0 are red and 10 are blue.

First, an urn is chosen at random; then a ball is chosen from it at random.

- What is the probability that a red ball is chosen from urn 2?
- What is the probability of choosing a red ball?
- If a red ball were chosen, what is then the probability that it came from Urn 2?

SOLUTION. Assume that we are equally likely to choose any urn and, given an urn, are equally likely to choose any ball in it. Let U_1 , U_2 , and U_3 denote the event of choosing Urn 1, 2, and 3, respectively, and let R and B denote the event of then choosing a red or blue ball, respectively.

Then $P(U_1) = P(U_2) = P(U_3) = \frac{1}{3}$ and

$$\begin{array}{lll} P(R|U_1) = \frac{3}{10} & P(R|U_2) = \frac{4}{20} = \frac{1}{5} & P(R|U_3) = 0 \\ P(B|U_1) = \frac{7}{10} & P(B|U_2) = \frac{16}{20} = \frac{4}{5} & P(B|U_3) = 1. \end{array}$$

- Therefore, $P(R \cap U_2) = P(R|U_2)P(U_2) = \frac{1}{5} \times \frac{1}{3} = \frac{1}{15}$.
- Similarly, $P(R \cap U_1) = \frac{3}{10} \times \frac{1}{3} = \frac{1}{10}$ and $P(R \cap U_3) = \frac{0}{10} \times \frac{1}{3} = 0$.

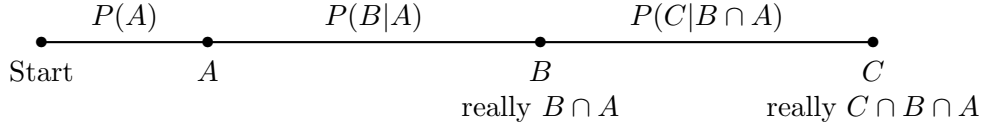
Now, $R = (R \cap U_1) \cup (R \cap U_2) \cup (R \cap U_3)$ and the three terms are disjoint, so

$$P(R) = P(R \cap U_1) + P(R \cap U_2) + P(R \cap U_3) = \frac{1}{10} + \frac{1}{15} + 0 = \frac{1}{6}.$$

- Finally, $P(U_2|R) = \frac{P(U_2 \cap R)}{P(R)} = \frac{(\frac{1}{15})}{(\frac{1}{6})} = \frac{2}{5}$.

◇

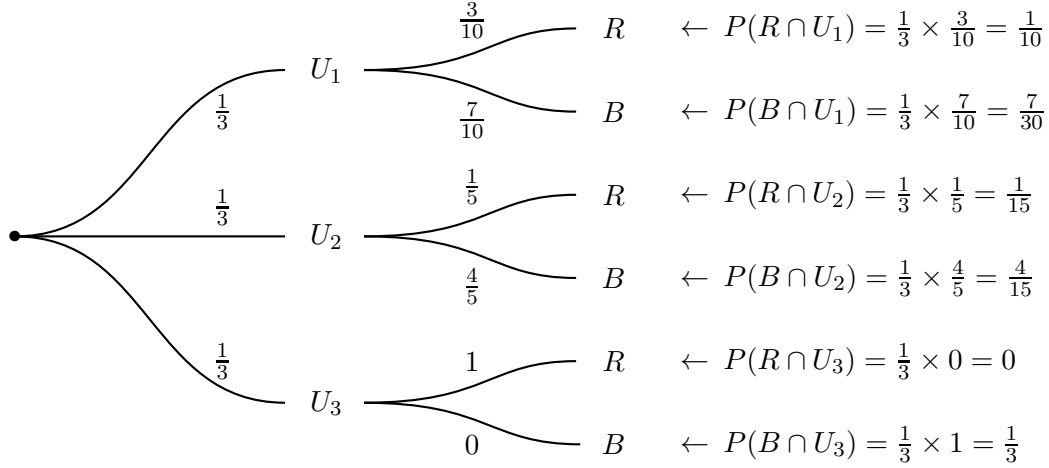
The above example shows one instance of a multi-stage experiment. The conditional probabilities and the multiplication rule for these sort of experiments can be illustrated and applied using **tree diagrams**. Such diagrams illustrate all possible outcomes of the multi-stage experiment as well as the way in which one is able to arrive at those outcomes via the various stages. The branches carry the conditional probability of the right hand node given that you are at the left node. The probability of getting from one node to a node to its right is obtained by multiplying the probability on the connecting branches. A typical sequence of branches is of the form



Here,

$$P(C \cap B \cap A) = P(C|B \cap A)P(B \cap A) = P(C|B \cap A)P(B|A)P(A).$$

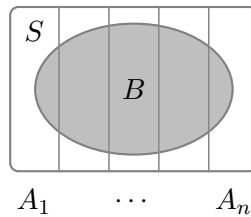
Example 16. The tree diagram for the multi-stage experiment given in Example 15 is as follows:



Since sequences of branches represent disjoint events,

$$P(R) = P(R \cap U_1) + P(R \cap U_2) + P(R \cap U_3) = \frac{1}{10} + \frac{1}{15} + 0 = \frac{1}{6}. \quad \diamond$$

Tree diagrams are very useful for visualising and calculating problems involving small numbers of conditional probabilities. However, tree diagrams are infeasible when these numbers are large or only implicitly given. We now derive a mathematical rule that enables us to deal with such cases. In particular, suppose that the n events $A_1, \dots, A_n$ partition the sample space S :



Since the sets $A_1 \cap B, \dots, A_n \cap B$ partition the event B , we see that

$$P(B) = P(A_1 \cap B) + \dots + P(A_n \cap B).$$

Applying the Multiplication Rule to each of these n terms yields the following rule.

Total Probability Rule

If $A_1, \dots, A_n$ partition S and B is an event, then
$$P(B) = \sum_{i=1}^n P(B|A_i)P(A_i).$$

Note that we have already used this rule implicitly, for instance to calculate $P(R_2)$ in Example 13, and to calculate $P(R)$ in part (b) of Example 15.

Although the Total Probability Rule is a very simple and almost obvious result, it is very useful. Furthermore, it implies Bayes' Rule, which is non-trivial and which often offers surprising results.

Bayes' Rule

If $A_1, \dots, A_n$ partition S and B is an event, then $P(A_j|B) = \frac{P(B|A_j)P(A_j)}{\sum_{i=1}^n P(B|A_i)P(A_i)}$.

Proof. Apply the Total Probability Rule to the identity $P(A_j|B) = \frac{P(A_j \cap B)}{P(B)}$. □

Example 17. Consider the urns and the ball of Example 15 above, and suppose that we drew a blue ball from one of the urns. What is then the probability $P(U_1|B)$ that we drew it from Urn 1?

SOLUTION. Now, U_1, U_2, U_3 partition S since we must choose exactly one urn. Thus by Bayes' Rule,

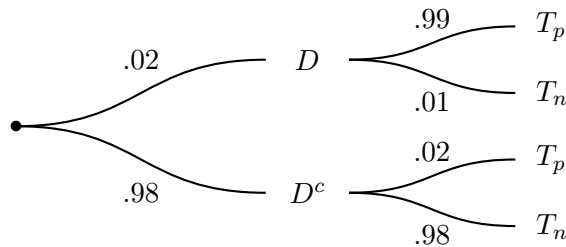
$$P(U_1|B) = \frac{P(B|U_1)P(U_1)}{P(B|U_1)P(U_1) + P(B|U_2)P(U_2) + P(B|U_3)P(U_3)} = \frac{\frac{7}{10} \times \frac{1}{3}}{\frac{7}{10} \times \frac{1}{3} + \frac{4}{5} \times \frac{1}{3} + 1 \times \frac{4}{5}} = \frac{7}{25} \quad \diamond$$

This example illustrates how Bayes' Rule allows reverse-inference. To some, this can seem counter-intuitive and has even caused contention and controversy. Nevertheless, Bayes' Rule remains a very useful statistical tool that is widely used in medical trials, court cases, and elsewhere.

Example 18. A certain diagnostic test for a disease X indicates with 99% accuracy that a person has X when that person actually has it. Similarly, the test indicates with 98% accuracy that someone does not have X when they do not in fact have it. In medical terms, the test is "positive" if it indicates that a person has the disease and is "negative" otherwise. Suppose that 2% of the population has the disease.

Find the probability of a false positive, namely that a person without X still tests positive.

SOLUTION. One might guess that this probability would be very small since the test seems so accurate. Let us calculate whether or not this is true. Thus, let D be the event that the person has disease X , let T_p be the event that the test shows positive, and set $T_n = T_p^c$. The tree diagram is



so by Bayes' Rule,

$$P(D^c|T_p) = \frac{P(T_p|D^c)P(D^c)}{P(T_p|D^c)P(D^c) + P(T_p|D)P(D)} = \frac{.02 \times .98}{.02 \times .98 + .99 \times .02} = 0.497.$$

Thus, almost 50% positives are false, which might seem outrageously inaccurate. Fortunately, the probability of a false negative, that is, someone with X testing negative, is almost negligible:

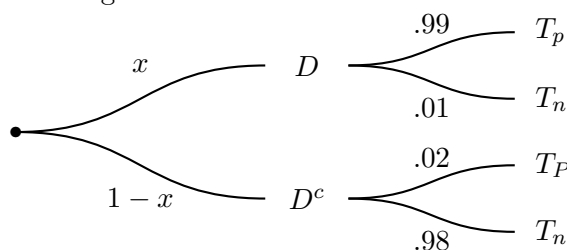
$$P(D|T_n) = \frac{P(T_n|D)P(D)}{P(T_n|D)P(D) + P(T_n|D^c)P(D^c)} = \frac{.01 \times .02}{.01 \times .02 + .98 \times .98} = .000208.$$

Thus, if someone is tested for disease X and the test shows that they do not have X , then they almost certainly do not have it. On the other hand, if the test is positive, then they might possibly have X , and more accurate (and presumably more expensive or time-consuming) tests can then be made to determine whether they do in fact have X . $\diamond$

Example 19. We modify Example 18 slightly by supposing that fraction x of population has disease X and that 5% test positive when a large random sample is tested.

What percentage of the population has disease X ?

SOLUTION. Here, the tree diagram is



By the Total Probability Rule,

$$0.05 = P(T_p) = P(T_p|D)P(D) + P(T_p|D^c)P(D^c) = .99x + .02(1 - x) = .97x + .02.$$

Therefore, $x = \frac{0.03}{0.97} \approx .031 = 3.1\%$ of the population has the disease. $\diamond$

9.2.4 Statistical Independence

Intuitively, two events A and B are mutually independent if one does not influence the probability of the other. This can be expressed as $P(A|B) = P(A)$ and $P(B|A) = P(B)$. That is, the probability of A does not depend on whether or not B is given, and the same is true for the probability of B . Since $P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$, we can express this independence quite elegantly:

Definition 5. Events A and B are **(statistically) independent** if and only if

$$P(A \cap B) = P(A)P(B)$$

Note that in contrast to conditional probability, this definition allows all probabilities, including 0.

To visualise statistical independence, let A and B be independent events with $P(B) \neq 0$. Then, since $P(S) = 1$,

$$\frac{P(A)}{P(S)} = \frac{P(A \cap B)}{P(B)}.$$

Thus, the probability measure of A is just as great in comparison to the whole sample space S as is $A \cap B$, the part of A in B , when compared to B . The following Venn diagram illustrates this:

S	B
A	$A \cap B$

Note that independence and disjointness are not the same concept, as might be supposed. Indeed, these two concepts are almost opposite in nature: non-empty events A and A^c are disjoint but are strongly dependent, for if A occurs, then A^c cannot (independently) occur, and vice versa.

Example 20. We roll a die and let A and B be the events that we roll a six and that we roll an even number, respectively. Then $P(A) = \frac{1}{6}$ and $P(B) = \frac{3}{6} = \frac{1}{2}$. Since in this case $A \cap B = A$,

$$P(A \cap B) = P(A) = \frac{1}{6} \quad \text{and} \quad P(A)P(B) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12},$$

so $P(A \cap B) \neq P(A)P(B)$. Therefore, A and B are not independent; that is, they are dependent. This is as one would expect since if A occurs, then B must necessarily occur.

Now, define A' to be the event that we roll either a five or a six. Then since $A' \cap B = A$,

$$P(A') = \frac{2}{6} = \frac{1}{3} \quad \text{and} \quad P(A' \cap B) = P(A) = \frac{1}{6},$$

so $P(A')P(B) = \frac{1}{3} \times \frac{1}{2} = \frac{1}{6} = P(A' \cap B)$. Therefore, A' and B are independent. In other words, the probability of rolling an even number is the same, namely $\frac{1}{2}$, whether or not one of the numbers five and six is rolled, and the converse is equally true. $\diamond$

Example 21. Roll a dice twice and, for each $i = 1, \dots, 6$, let X_i and Y_i denote the events that we get i on the first roll and second roll, respectively. Under usual conditions, we may assume that the first and second throws have no influence on each other. Therefore, X_i and Y_j are independent for all i, j , so

$$P(X_i \cap Y_j) = P(X_i)P(Y_j) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}.$$

This identity allows us to calculate more complicated events, such as the event S_4 that the sum of the two rolls is 4. Since S_4 is partitioned by $X_1 \cap Y_3$, $X_2 \cap Y_2$, $X_3 \cap Y_1$, we see that

$$\begin{aligned} P(S_4) &= P((X_1 \cap Y_3) \cup (X_2 \cap Y_2) \cup (X_3 \cap Y_1)) \\ &= P(X_1 \cap Y_3) + P(X_2 \cap Y_2) + P(X_3 \cap Y_1) = \frac{1}{36} + \frac{1}{36} + \frac{1}{36} = \frac{1}{12}. \end{aligned}$$

This result could also have been obtained by viewing this experiment as having a sample space of the 36 equally likely outcomes

$$S = \{(1, 1), (1, 2), \dots, (6, 6)\}$$

where, for instance, $(3, 5)$ denotes that we first rolled three and then five. Then

$$P(S_4) = \frac{|\{(1, 3), (2, 2), (3, 1)\}|}{|S|} = \frac{3}{36} = \frac{1}{12},$$

as before. $\diamond$

We now consider the statistical independence of any finite number of events.

Definition 6. Events $A_1, \dots, A_n$ are **mutually independent** if and only if, for any $A_{i_1}, \dots, A_{i_k}$ of these,

$$P(A_{i_1} \cap \dots \cap A_{i_k}) = P(A_{i_1}) \times \dots \times P(A_{i_k}).$$

Example 22. Events A, B, C are mutually independent if and only if these four identities all hold:

$$\begin{aligned} P(A \cap B) &= P(A)P(B) \\ P(A \cap C) &= P(A)P(C) \\ P(B \cap C) &= P(B)P(C) \\ P(A \cap B \cap C) &= P(A)P(B)P(C) \end{aligned} \quad \diamond$$

In general for n events to be mutually independent, they must satisfy non-trivial $2^n - n - 1$ identities such as the ones above, one for each subset of $\{1, \dots, n\}$ with at least two elements. None of these many identities imply any of the others in general, so we cannot make do with a smaller set of identities. This is illustrated by the following example.

Example 23. Draw a ball from a bag containing four balls marked 0 to 3. For $i = 0, \dots, 3$, let A_i be the event that ball i is drawn, and let $B_i = A_0 \cup A_i$ be the event that ball 0 or ball i is drawn. Then for all distinct $i, j = 1, 2, 3$,

$$P(B_i) = P(A_0 \cup A_i) = \frac{2}{4} = \frac{1}{2} \quad \text{and} \quad P(B_i \cap B_j) = P(A_0) = \frac{1}{4}.$$

Hence, $P(B_i \cap B_j) = P(B_i)P(B_j)$, so B_i and B_j are independent. In contrast,

$$P(B_1 \cap B_2 \cap B_3) = P(A_0) = \frac{1}{4} \neq \frac{1}{8} = P(B_1)P(B_2)P(B_3),$$

so B_1, B_2, B_3 are not mutually independent. $\diamond$

If events A and B are independent, then A and B^c are also independent:

$$P(A \cap B^c) = P(A) - P(A \cap B) = P(A) - P(A)P(B) = P(A)(1 - P(B)) = P(A)P(B^c).$$

By modifying these calculations slightly and using induction, we can prove the more general result:

Theorem 4. If events $A_1, \dots, A_n$ are mutually independent and B_i is either A_i or A_i^c for each $i = 1, \dots, n$, then $B_1, \dots, B_n$ are also mutually independent.

Suppose that events A, B , and C are mutually independent. Then by Theorem 4,

$$\begin{aligned} P(A \cap (B \cup C)) &= P(A \cap ((B - C) \cup (B \cap C) \cup (C - B))) \\ &= P(A \cap B \cap C^c) + P(A \cap B \cap C) + P(A \cap C \cap B^c) \\ &= P(A)P(B \cap C^c) + P(A)P(B \cap C) + P(A)P(C \cap B^c) \\ &= P(A)(P(B - C) + P(B \cap C) + P(C - B)) \\ &= P(A)P(B \cup C). \end{aligned}$$

We see that A and $B \cup C$ are also independent. By generalising the above calculations, one may prove the following result.

Theorem 5. *If events $A_{1,1}, \dots, A_{1,n_1}, A_{2,1}, \dots, A_{m,n_m}$ are mutually independent and for each $i = 1, \dots, m$, the event B_i is obtained from $A_{i,1}, \dots, A_{i,n_i}$ by taking unions, intersections, and complements, then $B_1, \dots, B_n$ are also mutually independent.*

Example 24 (A reliability example).

A 3-engine plane has a central engine and two wing engines. The plane will crash if the central engine and at least one of the wing engines fail. On any given flight, the central engine fails with probability 0.005, and each wing engine fails with probability 0.008. Assuming that the three engines fail mutually independently, find the probability that the plane will crash during a flight.

SOLUTION. Let A be the event that the port engine fails, let B be the event that the starboard engine fails, and C be the event that the central engine fails. Then $P(A) = P(B) = 0.008$ and $P(C) = 0.005$. Let D denote the event that the plane crashes and note that $D = C \cap (A \cup B)$. Since A , B , and C are mutually independent, C and $A \cup B$ are independent by Theorem 5. Therefore,

$$\begin{aligned} P(D) &= P(C \cap (A \cup B)) = P(C)P(A \cup B) = P(C)[P(A) + P(B) - P(A \cap B)] \\ &= P(C)[P(A) + P(B) - P(A)P(B)] \\ &= 0.005[0.008 + 0.008 - 0.008 \times 0.008] = 0.00007968. \end{aligned}$$

(Note: There are other ways to do this problem.)

Under our assumptions, the plane will crash on a given flight with probability slightly less than eighty in one million. These are dangerous assumptions, however, since it is highly optimistic to hope that the engines will fail independently of each other. For instance, if engine failure is caused by volcanic ash, then all three engines will be at risk of failure, and there are countless other factors, like shared electric wiring, that might similarly introduce dependence. In view of this, the real probabilities might be considerably higher than stated. $\diamond$

9.3 Random Variables

It can often be useful to label the outcomes of an experiment by numbers. This often makes event notation more flexible, and it allows us to perform arithmetic on the outcomes.

Definition 1. A **random variable** is a real function defined on a sample space.

Example 1. Toss a coin and let $S = \{H, T\}$ be the associated sample space.

Two random variables X and Y on S are given as follows, for each outcome $s \in S$:

$$X(s) = \begin{cases} 1 & \text{if } s = H \\ 0 & \text{if } s = T \end{cases} \quad Y(s) = \begin{cases} 1 & \text{if } s = H \\ -1 & \text{if } s = T \end{cases}$$

Example 2. Roll a die and let $S = \{1, \dots, 6\}$ be the associated sample space.

Two random variables X and Y on S are given as follows, for each outcome $s \in S$:

$$X(s) = s \quad Y(s) = \begin{cases} -1 & \text{if } s \text{ is odd} \\ 1 & \text{if } s \text{ is even} \end{cases}$$

Definition 2. For a random variable X on some sample space S , define for all subsets $A \subseteq S$ and real numbers $r \in \mathbb{R}$,

- $\{X \in A\} = \{s \in S : X(s) \in A\}$
- $\{X = r\} = \{s \in S : X(s) = r\}$
- $\{X \leq r\} = \{s \in S : X(s) \leq r\}$
- ... and so on.

We suppress the curly brackets when expressing the probability of these events. For instance, $P(\{X = r\})$ is written as $P(X = r)$.

Example 3. Roll a die and let X and Y be the random variables defined in Example 2. Then

$$P(X > 4) = P(\{5, 6\}) = \frac{1}{3} \quad P(Y = 1) = P(\{2, 4, 6\}) = \frac{1}{2} \quad P(Y = \pi) = P(\emptyset) = 0.$$

Example 4. Toss a coin three times and let the random variable X count the number of heads tossed. Then $X(S) = \{0, 1, 2, 3\}$ and

$$\begin{aligned} P(X = 0) &= P(\{TTT\}) = \frac{1}{8} \\ P(X = 1) &= P(\{HTT, THT, TTH\}) = \frac{3}{8} \\ P(X = 2) &= P(\{HHT, HTH, THH\}) = \frac{3}{8} \\ P(X = 3) &= P(\{HHH\}) = \frac{1}{8}. \end{aligned}$$

Definition 3. The **cumulative distribution function** of a random variable X is given by

$$F_X(x) = P(X \leq x) \quad \text{for } x \in \mathbb{R}.$$

We often refer to $F_X(x)$ as just $F(x)$. Note that F is non-decreasing and that if $a \leq b$, then $P(a < X \leq b) = F(b) - F(a)$ and

$$0 = \lim_{x \rightarrow -\infty} F(x) \leq F(a) \leq F(b) \leq \lim_{x \rightarrow \infty} F(x) = 1.$$

Example 5. Toss a coin three times and let random variable X be the number of heads tossed, as in Example 4. Then

$$F\left(\frac{3}{2}\right) = P\left(X \leq \frac{3}{2}\right) = P(X = 0) + P(X = 1) = \frac{1}{8} + \frac{3}{8} = \frac{1}{2}.$$

9.3.1 Discrete Random Variables

The *image* of a function is the set of its function values.

Definition 4. A random variable X is **discrete** if its image is countable.

The random variables in Examples 1–5 are each discrete since their images are finite and thus countable. We shall for now only consider discrete random variables but will consider certain non-discrete random variables in Section 9.5, namely those that are *continuous*.

Definition 5. The **probability distribution** of a discrete random variable X is some description of all the probabilities of all events associated with X .

We sometimes write the probabilities as $p_k = P(X = x_k)$.

Note that for a discrete random variable X , the cumulative distribution function $F(x)$ is

$$F(x) = \sum_{k \leq x} p_k.$$

Thus, in practice to show that $\{p_k\}_{k \geq 0}$ is a probability distribution, we need to show that:

- (i) $p_k \geq 0$ and
- (ii) $\sum_k p_k = 1$.

Example 6. Roll a die and define random variables X and Y as in Example 2. The probability distributions of X and Y can for instance be represented as

$$P(X = x) = \begin{cases} \frac{1}{6} & \text{if } x \in \{1, \dots, 6\} \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad \begin{array}{|c|c|c|} \hline y_k & -1 & 1 \\ \hline p_k = P(Y = y_k) & \frac{1}{2} & \frac{1}{2} \\ \hline \end{array}$$

Clearly in each case $p_k \geq 0$ and for the random variable X , we have $\sum_k p_k = 6 \times \frac{1}{6} = 1$, while

for the random variable Y , $\sum_k p_k = \frac{1}{2} + \frac{1}{2} = 1$. Hence, these are probability distributions for the random variables X and Y respectively. ◇

Example 7. Roll a die twice and let $S = \{(i, j) : i, j = 1, \dots, 6\}$ be the associated sample space. Let X be the random variable defined by $X(i, j) = i + j$, that is, X is the sum of the numbers showing. The probability distribution of X is

x_k	2	3	4	5	6	7	8	9	10	11	12
p_k	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Clearly, $p_k \geq 0$. Also,

$$\sum_k p_k = \frac{1}{36}(1 + 2 + 3 + 4 + 5 + 6 + 5 + 4 + 3 + 2 + 1) = 1.$$

Hence p_k is a probability distribution. $\diamond$

Note that, in the above example, we did not actually need to specify the sample space; it would suffice to define X to be the sum of rolls. Indeed, we often specify the probability distribution without defining the sample space or even, at times, a random variable with that distribution.

Example 8. The probability distribution of discrete random variable X is given as follows:

x_k	0	1	2	4	7
$p_k = P(X = x_k)$	0.2	$0.3c$	0.2	c^2	0.5

(a) Find the value of c .

(b) Find $P(X \geq 4)$.

SOLUTION.

(a) $\sum p_k = c^2 + 0.3c + 0.9 = 1$, or $c^2 + 0.3c - 0.1 = 0$.

Solving this gives $c = -0.5$ or $c = 0.2$. Since $0.3c = p_1 \geq 0$, we conclude that $c = 0.2$.

(b) By (a), the probability distribution is

x_k	0	1	2	4	7
p_k	0.2	0.06	0.2	0.04	0.5

Hence, $P(X \geq 4) = P(X = 4) + P(X = 7) = 0.04 + 0.5 = 0.54$. $\diamond$

9.3.2 The Mean and Variance of a Discrete Random Variable

As we have seen in the above examples, the use of random variables can simplify the description of events and their probabilities. We will now see how they also enable us to perform arithmetic on outcomes. In particular, we can calculate the weighted averages of outcome values; this is the expected value, or mean. We can also measure the average of the squares of the distances from the mean to the outcome values; this is called the variance.

Definition 6. The **expected value** (or **mean**) of a discrete random variable X with probability distribution p_k

$$E(X) = \sum_{\text{all } k} x_k p_k.$$

The expected value $E(X)$ is often denoted by μ or μ_X .

Example 9. Toss a coin three times and let X count the number of heads tossed as in Example 4. The probability distribution of X is

x_k	0	1	2	3
p_k	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

so the expected value of X is

$$E(X) = \sum_k p_k x_k = 0 \times \frac{1}{8} + 1 \times \frac{3}{8} + 2 \times \frac{3}{8} + 3 \times \frac{1}{8} = \frac{3}{2}.$$

This agrees with our intuition: on average, half of the throws will be heads.

Example 10. Roll a die twice and let the random variable X be the sum of the rolls. Since a die roll i is as likely as the roll $7 - i$, we see that X and $14 - X$ have the same probability distribution, so the expected value of X is $E(X) = \frac{14}{2} = 7$. Let us check this using the definition of $E(X)$ and the probabilities $P(X = x)$ given in Example 10:

$$E(X) = 2 \times \frac{1}{36} + \cdots + 7 \times \frac{6}{36} + \cdots + 12 \times \frac{1}{36} = 7.$$

◇

Theorem 1. Let X be a discrete random variable with probability distribution $p_k = P(X = x_k)$. Then for any real function $g(x)$, the expected value of $Y = g(X)$ is

$$E(Y) = E(g(X)) = \sum_k g(x_k) p_k.$$

[X] *Proof.* Let $\{y_j\} = \{g(x_k)\}$ be the set of function values of Y , and note that

$$P(Y = y_j) = P(\{s \in S : g(X(s)) = y_j\}) = P\left(\bigcup_{k: g(x_k)=y_j} \{s \in S : X(s) = x_k\}\right) = \sum_{k: g(x_k)=y_j} P(X = x_k).$$

By changing order of summation, we therefore see that

$$\begin{aligned} E(Y) &= \sum_j y_j P(Y = y_j) = \sum_j y_j \sum_{k: g(x_k)=y_j} P(X = x_k) = \sum_j \sum_{k: g(x_k)=y_j} g(x_k) p_k \\ &= \sum_k \sum_{j: y_j=g(x_k)} g(x_k) p_k \\ &= \sum_k g(x_k) p_k. \end{aligned}$$

The final equality is valid because the second sum only sums over a single element j . □

Example 11. Toss a coin three times and let X be the number of heads tossed, as in Examples 4 and 9. By Theorem 1, the expected value of X^2 is

$$E(X^2) = \sum_k p_k x_k^2 = 0^2 \times \frac{1}{8} + 1^2 \times \frac{3}{8} + 2^2 \times \frac{3}{8} + 3^2 \times \frac{1}{8} = 3.$$

◇

The expected value of a random variable X describes where the values of X are centred. We can also measure how widely the values of X spread, namely by the average distance (squared) between the values and the mean.

Definition 7. The **variance** of a discrete random variable X is

$$\text{Var}(X) = E((X - E(X))^2).$$

The **standard deviation** of X is $SD(X) = \sqrt{\text{Var}(X)}$.

The standard deviation is often denoted by σ or σ_X , and the variance is often written as σ^2 or σ_X^2 .

Theorem 2. $\text{Var}(X) = E(X^2) - (E(X))^2$.

This formula is often useful in hand calculations.

Proof. Write μ for $E(X)$. By the definition of the variance we have,

$$\begin{aligned} \text{Var}(X) &= \sum_k (x_k - \mu)^2 p_k \\ &= \sum_k (x_k^2 - 2x_k\mu + \mu^2) p_k \\ &= \sum_k x_k^2 p_k - 2\mu \sum_k x_k p_k + \mu^2 \sum_k p_k = E(X^2) - 2\mu^2 + \mu^2 = E(X^2) - (E(X))^2. \quad \square \end{aligned}$$

Example 12. Toss a coin three times and let X be the number of heads tossed. We saw in Examples 9 and 11 that $E(X) = \frac{3}{2}$ and $E(X^2) = 3$. Thus by Theorem 2, the variance of X is

$$\text{Var}(X) = E(X^2) - (E(X))^2 = 3 - \left(\frac{3}{2}\right)^2 = \frac{3}{4}. \quad \diamond$$

Example 13. Consider a random variable X with probability distribution given below:

x_k	0	1	2	4	7
p_k	0.2	0.06	0.2	0.04	0.5

The expected values of X and X^2 are

$$\begin{aligned} E(X) &= \sum x_k p_k = 0 \times 0.2 + 1 \times 0.06 + 2 \times 0.2 + 4 \times 0.04 + 7 \times 0.5 = 4.12 \\ E(X^2) &= \sum x_k^2 p_k = 0^2 \times 0.2 + 1^2 \times 0.06 + 2^2 \times 0.2 + 4^2 \times 0.04 + 7^2 \times 0.5 = 26.0 \end{aligned}$$

and the variance of X is

$$\text{Var}(X) = E(X^2) - (E(X))^2 = 26 - (4.12)^2 = 9.256.$$

Thus, the average root mean square distance of the values of X to the mean $E(X)$ is roughly $\sqrt{9.256} \approx 3$. $\diamond$

There is generally no easily-described relationship between $\text{Var}(Y)$ and $\text{Var}(X)$ when $Y = g(X)$. However, if $Y = aX + B$ is a linear function of X , then we have the following simple identities:

Theorem 3. *If a and b are constants, then*

$$\begin{aligned} E(aX + b) &= aE(X) + b \\ \text{Var}(aX + b) &= a^2 \text{Var}(X) \\ \text{SD}(aX + b) &= |a| \text{SD}(X). \end{aligned}$$

Proof. Write μ for $E(X)$, we have

$$E(aX + b) = \sum (ax_k + b)p_k = a \sum x_k p_k + b \sum p_k = aE(X) + b.$$

By Theorem 2 and the above identity,

$$\begin{aligned} \text{Var}(aX + b) &= E(aX + b - E(aX + b))^2 = E(aX - aE(X))^2 \\ &= a^2 E(X - E(X))^2 = a^2 \text{Var}(X). \end{aligned}$$

The third statement follows by definition from the second. $\square$

The variance $\sigma^2 = \text{Var}(X)$ of a random variable X gives a measure of the average square distance from the expected value $E(X)$ to the values of X .

9.4 Special Distributions

Often, statistical models incorporate specific classes of probability distributions whose expected value and variance are known. In this section, we shall consider two such classes, namely the Binomial distributions and the Geometric distributions. These both involve **Bernoulli trials** and **Bernoulli processes**. A Bernoulli trial is an experiment with two outcomes, often “success” and “failure”, or Y(es) and N(o), or $\{1, 0\}$, where $P(Y)$ and $P(N)$ are denoted by p and $q = 1 - p$, respectively. A Bernoulli process is an experiment composed of a sequence of identical and mutually independent Bernoulli trials. More particularly, the events A_i , denoting the success of the i th trial, are mutually independent. We have already seen examples of Bernoulli processes in previous sections, such as tossing a coin repeatedly and considering head-outcomes ($p = \frac{1}{2}$); rolling a die multiple times to obtain sixes ($p = \frac{1}{6}$); or asking each of several people whether it is their birthday ($p = \frac{1}{365}$).

Example 1. Tossing a coin three times is a Bernoulli process with $n = 3$ identical trials that each result in either H or T , with probabilities $p = q = \frac{1}{2}$. The trials are mutually independent since the coin tosses do not influence each other. Let us formally verify this claim. Let A_1 , A_2 , and A_3 to be the events that H is tossed on the 1st, 2nd, and 3rd toss, respectively. Then

$$P(A_1) = P(\{HTT, HTH, HHT, HHH\}) = \frac{4}{8} = \frac{1}{2}.$$

Similarly, $P(A_2) = P(A_3) = \frac{1}{2}$. Therefore,

$$P(A_1 \cap A_2) = P(\{HHT, HHH\}) = \frac{2}{8} = \frac{1}{4} = P(A_1)P(A_2),$$

and, similarly, $P(A_2 \cap A_3) = P(A_2)P(A_3)$ and $P(A_1 \cap A_3) = P(A_1)P(A_3)$. Finally,

$$P(A_1 \cap A_2 \cap A_3) = P(\{HHH\}) = \frac{1}{8} = P(A_1)P(A_2)P(A_3).$$

We see that the events A_1, A_2, A_3 are indeed mutually independent. $\diamond$

Throughout the remainder of this section, let p be a real number with $0 < p < 1$ and let $q = 1 - p$.

9.4.1 The Binomial Distribution

Recall that the expression $\binom{n}{k}$ denotes the number of ways to select k objects from n distinct objects, with order unimportant, and is given by

$$\frac{n!}{k!(n-k)!}.$$

Definition 1. The **Binomial distribution** $B(n, p)$ for $n \in \mathbb{N}$ is the function

$$B(n, p, k) = \binom{n}{k} p^k (1-p)^{n-k} \quad \text{where } k = 0, 1, \dots, n.$$

Note that $B(n, p, k)$ is a probability distribution. To see this, we can use the Binomial Theorem:

$$\sum_k B(n, p, k) = \sum_{k=0}^n B(n, p, k) = \sum_{k=0}^n \binom{n}{k} p^k q^{n-k} = (p+q)^n = 1^n = 1.$$

Since $B(n, p, k)$ is nonnegative, we conclude that $0 \leq B(n, p, k) \leq 1$ for all k .

Theorem 1. If X is the random variable that counts the successes of some Bernoulli process with n trials having success probability p , then X has the binomial distribution $B(n, p)$.

We write $X \sim B(n, p)$ to denote that X is a random variable with this distribution.

Proof. The variable X can assume values $k = 0, 1, \dots, n$ so we must calculate $p_k = P(X = k)$ for these values. Suppose that the first k trials each results in Y(es) and the rest each results in N(o):

$$\underbrace{Y \cdots Y}_k \underbrace{N \cdots N}_{n-k}$$

The trials are independent, so this outcome has probability $p^k(1-p)^{n-k}$. In general, there are

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

ways for precisely n trials with k Y's to occur. Therefore,

$$p_k = P(X = k) = \binom{n}{k} p^k (1-p)^{n-k} = B(n, p, k).$$

□

Example 2. Toss a coin $n = 3$ times and let X be the random variable counting the number of resulting heads (H). The tosses are identical and mutually independent with probability $p = \frac{1}{2}$ of resulting in H. Thus, Theorem 1 implies that $X \sim B(3, \frac{1}{2})$. This tells us everything about the probabilities of X ; for instance, $P(X = 2) = \binom{3}{2} \left(\frac{1}{2}\right)^2 \left(\frac{1}{2}\right)^{3-2} = \frac{3}{8}$. ◇

Probabilities such as $P(X \geq t)$, $P(X > t)$, and $P(|X - E(X)| > t)$ are each referred to as a **tail probability**.

Example 3. Roll a die $n = 12$ times and let X be the number of resulting sixes. The rolls are identical and mutually independent with probability $p = \frac{1}{6}$ of resulting in a six, so by Theorem 1, $X \sim B(12, \frac{1}{6})$. Thus for instance, we can calculate the following tail probability:

$$\begin{aligned} P(X > 9) &= P(X = 10) + P(X = 11) + P(X = 12) \\ &= \binom{12}{10} \left(\frac{1}{6}\right)^{10} \left(\frac{5}{6}\right)^2 + \binom{12}{11} \left(\frac{1}{6}\right)^{11} \left(\frac{5}{6}\right) + \left(\frac{1}{6}\right)^{12} \approx 7.86 \times 10^{-7}. \end{aligned}$$

We see that the likelihood of rolling more than nine sixes is less than one in a million. $\diamond$

Example 4. Ask $n = 40$ people whether today is their birthday and let X count the Yes-answers. Assume that these questions form identical and mutually independent trials with probability $p = \frac{1}{365}$ of resulting in a Yes-answer. By Theorem 1, $X \sim B(40, \frac{1}{365})$. The likelihood of today being the birthday of at least one of these people is

$$P(X \geq 1) = 1 - P(X = 0) = 1 - \left(\frac{364}{365}\right)^{40} \approx 10.4\%.$$

$\diamond$

Theorem 2. If X is a random variable and $X \sim B(n, p)$, then

- $E(X) = np$;
- $\text{Var}(X) = npq = np(1 - p)$.

Proof. First note that for $k \geq 1$,

$$k \binom{n}{k} = k \frac{n!}{k!(n-k)!} = n \frac{(n-1)!}{(k-1)!(n-1-(k-1))!} = n \binom{n-1}{k-1}.$$

Hence,

$$\begin{aligned} E(X) &= \sum_{k=0}^n k p_k = \sum_{k=0}^n k \binom{n}{k} p^k q^{n-k} = \sum_{k=1}^n k \binom{n}{k} p^k q^{n-k} \\ &= n \sum_{k=1}^n \binom{n-1}{k-1} p^k q^{n-k} \\ &= n \sum_{j=0}^{n-1} \binom{n-1}{j} p^{j+1} q^{n-1-j} \\ &= np \sum_{j=0}^{n-1} \binom{n-1}{j} p^j q^{n-1-j} = np \sum_{j=0}^{n-1} B(n-1, p, j) = np. \end{aligned}$$

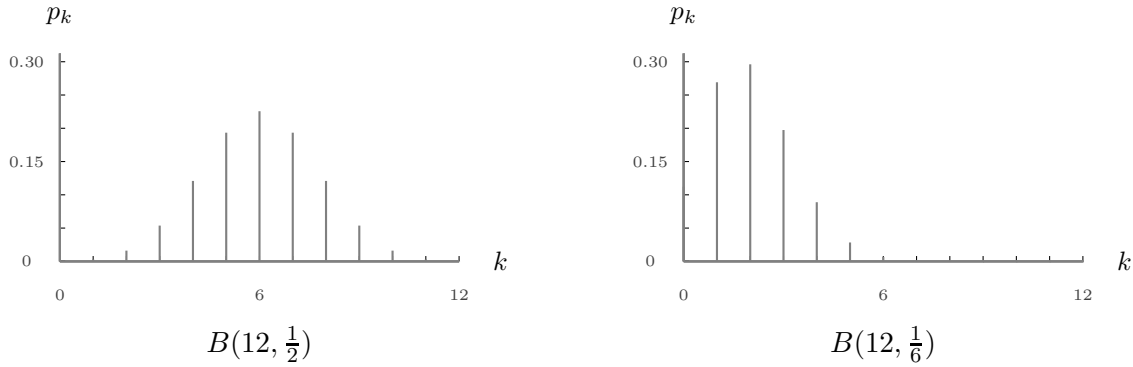
See Problem 36 for the second half of the proof. $\square$

Example 5. Toss a coin $n = 3$ times and let X count the number of ensuing heads. By Example 1, $X \sim B(3, \frac{1}{2})$, so by Theorem 2, the expected number of resulting heads is $E(X) = np = \frac{3}{2}$ and the average square distance between the number of heads and $E(X)$ is $\text{Var}(X) = npq = \frac{3}{4}$.

If we now toss the coin $n = 12$ times, then $X \sim B(12, \frac{1}{2})$, so the expected number of resulting heads is $E(X) = np = \frac{12}{2} = 6$, and $\text{Var}(X) = npq = 3$. $\diamond$

Example 6. If we roll a die $n = 12$ times, then one might intuitively expect to roll $12 \times \frac{1}{6} = 2$ sixes on average. This is also what we find by the following calculations. If X is the random variable counting the number of sixes rolled, then by Example 3, $X \sim B(12, \frac{1}{6})$. Thus by Theorem 2, the expected number of resulting sixes is $E(X) = np = 12 \times \frac{1}{6} = 2$, as we expected. $\diamond$

The distributions $B(12, \frac{1}{2})$ and $B(12, \frac{1}{6})$ appearing in Example 2 and 6 are illustrated below. Note that the function values $B(12, \frac{1}{2}, k)$ are centered symmetrically around $E(X) = 6$ and spread out gradually to the extremities $k = 0, 12$. In contrast, the function values $B(12, \frac{1}{6}, k)$ are clustered asymmetrically about the expected value $E(X) = 2$ and taper rapidly off, so that $B(12, \frac{1}{6}, k)$ is nearly zero for $k \geq 7$. Thus, it is extremely highly unlikely that we would roll at least seven sixes when rolling a die twelve times.



9.4.2 Geometric Distribution

Definition 2. The **Geometric distribution** $G(p)$ is the function

$$G(p, k) = (1 - p)^{k-1}p = q^{k-1}p \quad \text{where } k = 1, 2, \dots$$

Note that $G(p, k)$ is a probability distribution since $0 \leq G(p, k) \leq 1$ for all k and since

$$\sum_k G(p, k) = \sum_{k=1}^{\infty} G(p, k) = \sum_{k=1}^n q^{k-1}p = p \sum_{k=0}^n q^k = p \frac{1}{1-q} = p \frac{1}{p} = 1.$$

Theorem 3. Consider an infinite Bernoulli process of trials each of which has success probability p . If the random variable X is the number of trials conducted until success occurs for the first time, then X has the geometric distribution $G(p)$.

We write $X \sim G(p)$ to denote that X has this distribution. Note that it is theoretically possible for a success never to happen ($X = \infty$); however, this has zero probability. We therefore omit the all-failure outcome from the sample space so that X is a well-defined finite number.

Proof. The variable X can assume values $k = 1, 2, \dots$, so we must find $p_k = P(X = k)$ for these values. The event $\{X = k\}$ consists of the outcome in which the first $k - 1$ trials each result in $N(o)$ and the k th trial results in $Y(es)$:

$$\underbrace{N \cdots N}_{k-1} Y$$

The trials are independent, so this outcome has probability

$$p_k = P(X = k) = (1 - p)^{k-1}p = G(p, k).$$

□

Example 7. Toss a coin until H (ead) is tossed and let X count the number of these tosses. The tosses are identical and mutually independent with probability $p = \frac{1}{2}$ of resulting in H . Theorem 3 then implies that $X \sim G(\frac{1}{2})$. Thus, the likelihood of having to toss the coin seven times before tossing H is

$$P(X = 7) = \left(1 - \frac{1}{2}\right)^{7-1} \frac{1}{2} = \frac{1}{2^7} \approx 0.8\%.$$

◇

Tail probabilities are very easily expressed for geometrically distributed random variables:

Theorem 4. If $X \sim G(p)$ and n is a positive integer, then $P(X > n) = (1 - p)^n = q^n$.

Proof.
$$P(X > n) = \sum_{k=n+1}^{\infty} P(X = k) = \sum_{k=n+1}^{\infty} q^{k-1}p = q^n \sum_{k=1}^{\infty} q^{k-1}p = q^n \sum_{k=1}^{\infty} G(p, k) = q^n.$$

□

Theorem 4 gives us a simple expression for the cumulative distribution function $F(x)$ of X :

Corollary 5. If $X \sim G(p)$, then the cumulative distribution function F is given by $F(x) = P(X \leq x) = 1 - (1 - p)^{\lfloor x \rfloor} = 1 - q^{\lfloor x \rfloor}$ for $x \in \mathbb{R}$.

Note that $\lfloor x \rfloor$ denotes the largest integer less or equal to x .

Example 8. Roll a die until six is rolled, and let X count the number of these rolls. The rolls are identical and mutually independent with probability $p = \frac{1}{6}$ of resulting in six. Theorem 3 implies that $X \sim G(\frac{1}{6})$. By Theorem 4, the likelihood of rolling a six within at most four rolls is

$$F(4) = P(X \leq 4) = 1 - P(X > 4) = 1 - \left(1 - \frac{1}{6}\right)^4 \approx 52\%,$$

or close to half. Similarly, $F(6) = P(X \leq 6) \approx \frac{2}{3}$ and $P(X > 7) \approx 28\%$.

◇

Theorem 6. If X is a random variable and $X \sim G(p)$, then

- $E(X) = \frac{1}{p}$;
- $\text{Var}(X) = \frac{1-p}{p^2}$.

Proof. First note that for $x \neq 0$, using Power Series results from Calculus,

$$\sum_{k=0}^{\infty} kx^{k-1} = \frac{d}{dx} \sum_{k=0}^{\infty} x^k = \frac{d}{dx} \frac{1}{1-x} = \frac{1}{(1-x)^2}.$$

Hence,

$$E(X) = \sum_{k=1}^{\infty} kq^{k-1}p = \frac{p}{(1-q)^2} = \frac{p}{p^2} = \frac{1}{p}.$$

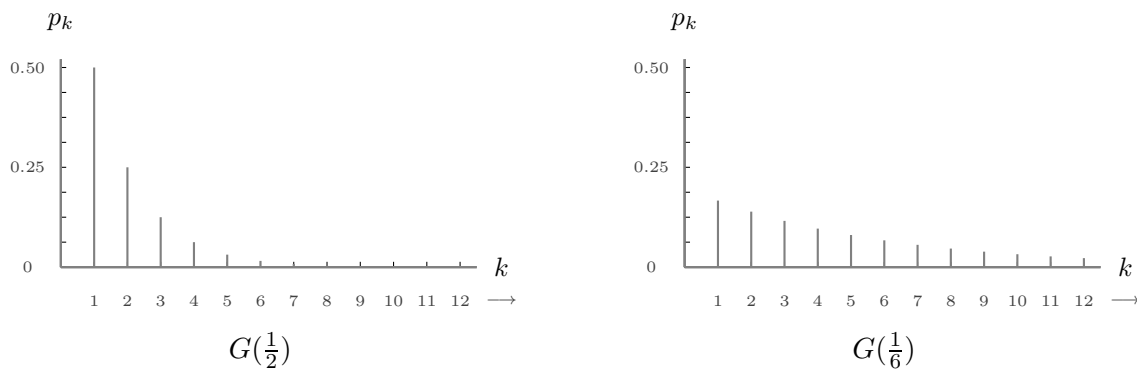
The second part of the proof is left as an exercise.

□

Example 9. Toss a coin until $H(\text{ead})$ is tossed and let X count the number of these tosses. As seen in Example 7, $X \sim G(\frac{1}{2})$. Thus, we must expect to on average have to toss the die $E(X) = \frac{1}{p} = (\frac{1}{2})^{-1} = 2$ times in order to toss a head; this is presumably what most would expect intuitively. Note that this is a relatively accurate estimate of the average since the average squared distance from $E(X) = 2$ to the infinitely many values of X is only $\text{Var}(X) = \frac{1-p}{p^2} = (1-\frac{1}{2})/(\frac{1}{2})^2 = 2$. Indeed by Theorem 4, the likelihood that we must toss the coin more than three times to get a head is $P(X > 3) = (\frac{1}{2})^3 = \frac{1}{8} = 12.5\%$, which is relatively small. $\diamond$

Example 10. Roll a die until six is rolled, and let X count the number of these rolls as in Example 7. In that example, we saw that $X \sim G(\frac{1}{6})$, so we must expect on average to have to toss the die $E(X) = \frac{1}{p} = (\frac{1}{6})^{-1} = 6$ times in order to toss a head. As in the coin-tossing example above, this expected value is what one might guess intuitively. However, in contrast to that example, the present expected value $E(X) = 6$ is not a particularly precise estimate of an average tossing count. In particular, the average squared distance from $E(X) = 2$ to the infinitely many values of X is $\text{Var}(X) = \frac{1-p}{p^2} = (1-\frac{1}{6})/(\frac{1}{6})^2 = 30$. Indeed by Example 7, the likelihood of requiring at most four rolls in order to roll a six is (slightly) more than half, and the likelihood of requiring at least eight rolls is more than a quarter. $\diamond$

The distributions $G(\frac{1}{2})$ and $G(\frac{1}{6})$ appearing in Example 8 and 10 are illustrated below. Note that the function values $G(\frac{1}{2}, k)$ are large to begin with but very quickly decrease, which is reflected by the small expected value and variance that both equal $E(X) = \text{Var}(X) = 2$. In contrast, the function values $G(\frac{1}{6}, k)$ are small to begin with but only gradually decrease. This is indicated by the expected value $E(X) = 6$ and the large variance $\text{Var}(X) = 30$. Thus, it is very likely that we would toss a head after only a few tosses of a coin; whereas it would not be possible to form such a good estimate about the number of die-rolls required to roll a six.



9.4.3 Sign Tests

Often, we have a sample of data consisting of independent observations of some quantity of interest, and it might be of interest to see whether the observed values differ systematically from some fixed and pre-determined value.

Example 11. Crop research shows that a new variety of corn yields, in bushels per acre, for 15 plots of land:

138.0	139.1	113.0	132.5	140.7
109.7	118.0	134.8	109.6	127.3
115.6	130.4	130.2	117.7	105.5

A variety of corn currently used yields 110 bushels per acre. We want to know whether the new variety improves on the existing one – that is, are the above values centred around a true value for yield of 110, or are they systematically different from the value 110?

To answer this question, one may use a “sign test” approach as follows:

1. Count the number of observations that are strictly greater than the target value (“+”).
2. Count the total number of observations that are either strictly greater (“+”) or strictly smaller (“-”) than the target value.
3. Calculate the tail probability that measures how often one would expect to observe as many increases (“+”) as were observed, if there were equal probability of “+” and “-”.

Using this approach, we can now determine whether the new variety of corn has a higher yield than the current variety.

1. The yield was strictly greater (“+”) than 110 bushels/acre in 12 plots.
2. The yield was either strictly greater (“+”) than or strictly smaller (“-”) than 110 bushels/acre in all 15 plots.
3. Assuming that probabilities of greater-than (“+”) yield probabilities for each plot are identical and mutually independent, we can model these yields binomially. In particular, let X be the random variable that counts the yields exceeding the average yield of 110 bushels per acre; then $X \sim B(15, \frac{1}{2})$. The probability $p = \frac{1}{2}$ is set under the assumption that smaller-than (“-”) yields are as likely as greater-than (“+”) yields and that no equal-to yields occur; this assumes that the new crop has the same yield as the old one. The tail probability that 12 or more plots have a greater-than (“+”) yield is then

$$P(X \geq 12) = \sum_{k=12}^{15} \binom{15}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{15-k} = 1.76\%.$$

Under the assumption that the average yield for the new variety has not improved, it is quite unlikely (less than 2%) that we would have observed 12 of the 15 yields above the old average yield. We therefore conclude that the new variety has improved yield.

In this course, we will say that if the tail probability is less than 5% then we will regard this as significant.

9.5 Continuous random variables

In the previous sections, we considered discrete random variables. These can assume countably many values assigned to the outcomes of an experiment. Although there may be infinitely many of these values, this is not sufficient to model many real-life experiments in which outcomes may be assigned any real values from some interval, such as the height or weight of individuals, or the half-life of a radioactive isotope. We will therefore now consider a type of non-discrete random variable,

called a **continuous** variable X . In contrast to the discrete random variables, these cannot be defined by probabilities $P(X = x)$ of single values x , since these probabilities each equal 0. Instead, we will define continuous random variables in terms of the cumulative distribution function

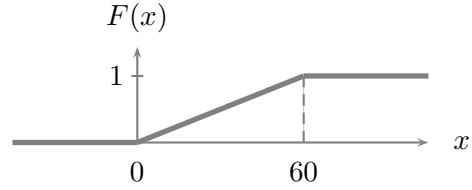
$$F(x) = F_X(x) = P(X \leq x) \quad \text{for } x \in \mathbb{R}.$$

Definition 1. Random variable X is **continuous** if and only if $F_X(x)$ is continuous.

Strictly speaking, $F_X(x)$ must actually be *piecewise differentiable*, which means that $F_X(x)$ is differentiable except for at most countably many points. However, the above definition is good enough for our present purposes.

Example 1. At a random point during the day, we take note of the time, ignoring the date and number of hours. This gives us a real number (of minutes) that we can represent by a random variable X with function values lying in the interval $[0, 60)$. If x is one of these values, then, assuming that any time is as likely as another, our intuition tells us that $F(x) = P(X \leq x) = P(0 \leq X \leq x)$ is the size of the interval $[0, x]$ compared the size of $[0, 60)$; measured in lengths, this is $\frac{x}{60}$. We therefore find that $F(x)$ is the function

$$F(x) = P(X \leq x) = \begin{cases} 0 & \text{if } x \leq 0 \\ \frac{x}{60} & \text{if } 0 < x \leq 60 \\ 1 & \text{if } x > 60 \end{cases}$$



This is a continuous function, so X is a continuous random variable. $\diamond$

For discrete random variables X , the cumulative distribution function $F(x)$ is a sum over probability distribution values $p_k = P(X = x_k)$. For continuous random variables, $F(x)$ is an integral over continuous function analogues of the discrete probability distributions. These analogues are given in the following definition.

Definition 2. The **probability density function** $f(x)$ of a continuous random variable X is defined by

$$f(x) = f_X(x) = \frac{d}{dx}F(x), \quad x \in \mathbb{R}$$

if $F(x)$ is differentiable, and $\lim_{x \rightarrow a^-} \frac{d}{dx}F(x)$ if $F(x)$ is not differentiable at $x = a$.

Since $F(x)$ is non-decreasing and $\lim_{x \rightarrow \infty} F(x) = 1$, the probability density function satisfies

$$f(x) \geq 0 \quad \text{for all } x \quad \text{and} \quad \int_{-\infty}^{\infty} f(x) dx = 1.$$

Theorem 1. $F(x) = \int_{-\infty}^x f(t) dt$.

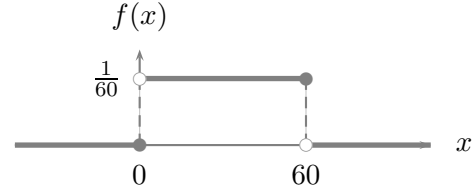
Proof. This follows from the Fundamental Theorem of Calculus since $\lim_{x \rightarrow -\infty} F(x) = 0$. $\square$

Note that if $a \leq b$, then

$$P(a \leq X \leq b) = P(a < X \leq b) = F(b) - F(a) = \int_a^b f(x)dx.$$

Example 2. At a random point during the day, we take note of the time as in Example 1, and let X again be the continuous random variable X that tell us how far past the hour the time is. The density function $f(x)$ is calculated by differentiating the cumulative distribution function $F(x)$ found in Example 1:

$$f(x) = \frac{d}{dx}F(x) = \begin{cases} \frac{1}{60} & , 0 < x \leq 60 \\ 0 & , \text{otherwise.} \end{cases}$$



or, more compactly written,

$$f(x) = \frac{1}{60} \quad \text{for } x \in (0, 60].$$

Here, we differentiated $F(x)$ from the left at the points $x = 0, 60$.

The probability that we noted the time between a quarter past and half past the hour is

$$P(15 \leq X \leq 30) = \int_{15}^{30} f(x)dx = \int_{15}^{30} \frac{1}{60} dx = \frac{1}{4},$$

as we would intuitively expect. ◇

9.5.1 The mean and variance of a continuous random variable

The mean of continuous random variable is obtained (using the notion of Riemann Sums) by replacing sums by integrals, and probability distributions by probability density functions:

Definition 3. The **expected value** (or **mean**) of a continuous random variable X with probability density function $f(x)$ is defined to be

$$\mu = E(X) = \int_{-\infty}^{\infty} xf(x)dx.$$

Here, and in the following, we assume that all improper integrals converge.

The following theorem is the continuous analogue of Theorem 1 in Section 9.3.2.

Theorem 2. If X is a continuous random variable with density function $f(x)$, and $g(x)$ is a real function, then the expected value of $Y = g(X)$ is

$$E(Y) = E(g(X)) = \int_{-\infty}^{\infty} g(x)f(x)dx.$$

The variance of a continuous random variable is defined exactly as for discrete random variables:

Definition 4. The **variance** of a continuous random variable X is

$$\text{Var}(X) = E((X - E(X))^2) = E(X^2) - (E(X))^2.$$

The standard deviation of X is $\sigma = \text{SD}(X) = \sqrt{\text{Var}(X)}$.

Note that by Theorem 2,

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x) dx.$$

Example 3. At a random point during the day, we take note of the time as in Examples 1 and 2, and let X again be the continuous random variable giving the number of minutes past the hour. In Example 2, we found the density function $f(x)$ to be $f(x) = \frac{1}{60}$ for $x \in (0, 60]$, so

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx = \int_{-\infty}^0 0 dx + \int_0^{60} \frac{x}{60} dx + \int_{60}^{\infty} 0 dx = 0 + \frac{1}{60} \left[\frac{1}{2}x^2 \right]_0^{60} = 30,$$

as we would expect. Similarly by Theorem 2,

$$E(X^2) = \int_{-\infty}^{\infty} x^2 f(x)dx = \int_0^{60} x^2 \frac{1}{60} dx = \frac{1}{60} \left[\frac{1}{3}x^3 \right]_0^{60} = 1200,$$

so $\text{Var}(X) = 1200 - 30^2 = 300$ and $\text{SD}(X) = \sqrt{300} \approx 17.3$. ◇

The mean and variance have the same properties under linear scaling as in the discrete case.

Theorem 3. If a and b are constants, then

$$\begin{aligned} E(aX + b) &= aE(X) + b \\ \text{Var}(aX + b) &= a^2 \text{Var}(X) \\ \text{SD}(aX + b) &= |a| \text{SD}(X). \end{aligned}$$

An immediate consequence of these properties is

Theorem 4. If $E(X) = \mu$ and $\text{Var}(X) = \sigma^2$, and $Z = \frac{X - \mu}{\sigma}$, then $E(Z) = 0$ and $\text{Var}(Z) = 1$.

The random variable $Z = \frac{X - \mu}{\sigma}$ is referred to as the **standardised** random variable obtained from X . Note that this theorem holds for discrete and continuous random variables alike.

Proof. By Theorem 3,

$$\begin{aligned} E(Z) &= E\left(\frac{X - \mu}{\sigma}\right) = \frac{1}{\sigma}E(X) - \frac{\mu}{\sigma} = \frac{\mu}{\sigma} - \frac{\mu}{\sigma} = 0; \\ \text{Var}(Z) &= \text{Var}\left(\frac{X - \mu}{\sigma}\right) = \frac{1}{\sigma^2} \text{Var}(X) = \frac{\sigma^2}{\sigma^2} = 1. \end{aligned}$$

□

9.6 Special Continuous Distributions

In this section, we consider two well-known continuous probability distributions, namely the normal and exponential distributions. It turns out that these are limiting cases of the discrete probability distributions that we have already seen, namely the binomial and geometric distributions, respectively, but we will not prove this here.

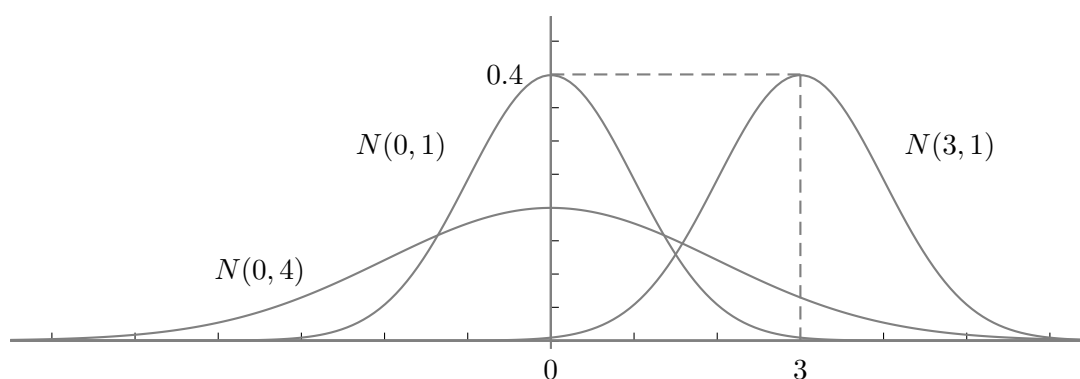
9.6.1 The Normal Distribution

A widely used probability distribution in statistics is the **normal** or **Gaussian** distribution.

Definition 1. A continuous random variable X has **normal distribution** $N(\mu, \sigma^2)$ if it has probability density

$$\phi(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \quad \text{where} \quad -\infty < x < \infty.$$

We write $X \sim N(\mu, \sigma^2)$ to denote that X has the normal distribution $N(\mu, \sigma^2)$. The normal probability density is bell-shaped, symmetric about the value $x = \mu$, and narrower for smaller σ . The probability densities for $N(0, 1)$, $N(3, 1)$, and $N(0, 4)$ are illustrated below.



The distribution $N(0, 1)$ is called the **standard normal distribution**.

The mean and variance of a random variable $X \sim N(\mu, \sigma^2)$ are simply μ and σ^2 :

Theorem 1. If X is a continuous random variable and $X \sim N(\mu, \sigma^2)$, then

- $E(X) = \mu$
- $\text{Var}(X) = \sigma^2$.

Proof. These are left as an exercise. □

Theorem 2. If $X \sim N(\mu, \sigma^2)$, then $\frac{X - \mu}{\sigma} \sim N(0, 1)$.

Proof. This (almost) follows from Theorem 1 above and Theorem 4 from the previous section, but a proof is required that the new random variable is actually normal. □

Note that when we standardise a normal random variable, the resulting distribution is also normal.

To find a probability involves evaluating the integral of the density function which is very hard, since this function does not have an elementary primitive. Thus, if X is normally distributed with mean μ and standard deviation σ ,

$$P(X \leq x) = F_X(x) = \int_{-\infty}^x \frac{1}{\sqrt{2\pi}\sigma^2} e^{-\frac{1}{2}\left(\frac{t-\mu}{\sigma}\right)^2} dt.$$

To evaluate this integral, we convert to the standard normal distribution $Z \sim N(0, 1)$, using the change of variable $Z = \frac{X - \mu}{\sigma}$ outlined above. This gives

$$P(Z \leq z) = F_Z(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}t^2} dt.$$

The value of this integral for various z has been tabulated numerically and is available either via a calculator or the table given on the following page. This table gives the values of this integral for z in the range -3 to 3 . For z less than -3 , the value is essentially zero, while for z greater than 3 , the value is essentially 1.

Example 1. Suppose X is normally distributed with mean $\mu = 20$ and standard deviation $\sigma = 3$. Find $P(X \leq 24)$.

SOLUTION. We change to the standard normal distribution, so

$$P(X \leq 24) = P\left(\frac{X - \mu}{\sigma} \leq \frac{24 - 20}{3}\right) \approx P(Z \leq 1.33) \approx 0.9082,$$

from the tables. ◇

Example 2. Suppose that the weekly wages of secretaries are normally distributed with mean \$800 and standard deviation \$50. What is the probability of a secretary having a weekly wage higher than \$900 and how many secretaries out of a group of 2000 randomly selected secretaries would you expect to have a weekly wage greater than \$900?

SOLUTION. Let X denote the weekly wage of a secretary. Then the mean of X is $\mu = 800$ and the standard deviation of X is $\sigma = 50$. Since $X \sim N(\mu = 800, \sigma^2 = 50^2)$,

$$Z = \frac{X - 800}{50} \sim N(0, 1) \quad \text{and} \quad X = 900 \quad \text{when} \quad Z = \frac{900 - 800}{50} = 2.$$

Thus,

$$P(X > 900) = P(Z > 2) = 1 - P(Z \leq 2) = 1 - 0.9772 = 0.0228.$$

Therefore, in a group of 2000 secretaries we would expect $.0228 \times 2000 = 45.6$ (i.e., about 46) of them to have a weekly wage in excess of \$900. ◇

In the above example, we used the fact that $P(Z \geq a) = 1 - P(Z \leq a)$; this is true because $P(Z = a) = 0$ since Z is continuous. Note also that for $a \leq b$, $P(a \leq Z \leq b) = P(Z \leq b) - P(Z \leq a)$.

Standard normal probabilities $P(Z \leq z)$										
z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
-2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
-2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
-2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
-2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
-2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
-2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
-2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
-2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
-2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
-2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
-1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
-1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
-1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
-1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
-1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
-1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
-1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
-1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
-1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
-1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
-0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
-0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
-0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
-0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
-0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
-0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
-0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
-0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
-0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
-0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986

The normal distribution is used, among other things, to approximate the binomial distribution $B(n, p)$ when n grows large. Before the advent of powerful computers, calculations involving many $B(n, p)$ terms was very laborious or even impossible, so it was easier to approximate such calculations by ones involving integration of the probability density of the normal distribution $N(\mu, \sigma^2)$ with the same mean ($\mu = np$) and variance ($\sigma^2 = np(1 - p)$) as the binomial distribution. These integrals can be evaluated by transforming to the standard normal distribution as we did above and using tables.

These days, computers can calculate most probabilities involving binomial sums; however, the normal distribution is occasionally still used instead.

Example 3. Toss a coin $n = 10$ times and let X be the random variable counting the number of resulting heads. As we have seen previously, $X \sim B(10, \frac{1}{2})$. The probability of tossing at most four heads is thus

$$P(X \leq 4) = F_X(4) = \sum_{k=0}^4 \binom{10}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{10-k} = \frac{193}{512} \approx 37.7\%.$$

We could also have approximated this probability by calculating $F_Y(4)$ of a continuous random variable Y with $Y \sim N(\mu, \sigma^2)$ where $\mu = E(X) = 5$ and $\sigma^2 = \text{Var}(X) = \frac{5}{2}$. To get an even better approximation, we can calculate $F_Y(4.5)$ since $P(X \leq 4) = P(X < 5)$ and 4.5 lies in the middle of the interval from 4 to 5:

$$P(X \leq 4) = F_X(4) \approx P(Y \leq 4.5) \approx P(Z \leq -0.32) \approx 37.45\%,$$

which differs from the true value by only 0.25%.

Now toss a coin $n = 50$ times and let X again be the random variable counting the number of resulting heads. Then $X \sim B(50, \frac{1}{2})$. The probability of tossing at most 23 heads is thus

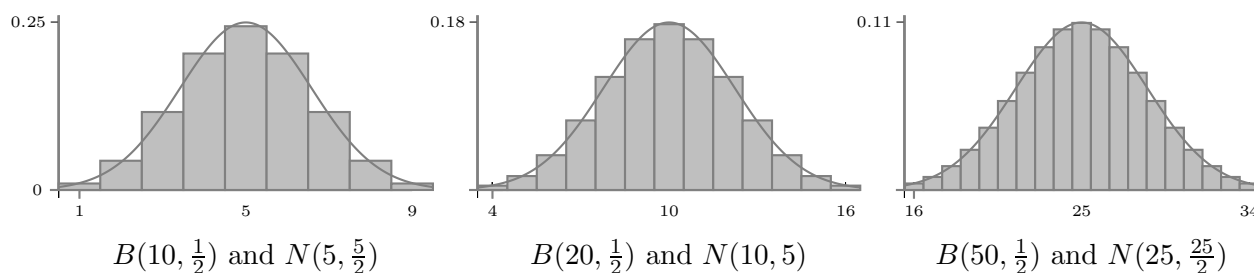
$$P(X \leq 23) = F_X(23) = \sum_{k=0}^{23} \binom{50}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{50-k} \approx 33.59\%.$$

The above calculation is obtained at once when using Maple but would be very time-consuming to calculate by hand (however, can you find a simple short-cut to perform this particular calculation with far less effort?). We could also have approximated this probability by calculating $F_Y(23.5)$ of a continuous random variable Y with $Y \sim N(\mu, \sigma^2)$ where $\mu = E(X) = 25$ and $\sigma^2 = \text{Var}(X) = \frac{25}{2}$:

$$P(X \leq 23) \approx P(Y \leq 23.5) \approx P(Z \leq -0.42) \approx 33.72\%,$$

which only differs from the true value by 0.13%. ◇

The three figures below show how binomial distributions may be approximated by normal distributions. The first and third figures illustrate $B(10, \frac{1}{2})$ and $N(5, \frac{5}{2})$, and $B(50, \frac{1}{2})$ and $N(25, \frac{25}{2})$, respectively, that appeared in the above example. Note that the first and second coordinates of the three figures are differently scaled and truncated.



More generally, normal distributions are used to model experiments involving large numbers of identical and independent trials that have several possible outcomes. Typical examples include the final-grade distributions of the high-school graduates in a particular country in a given year; or distributions of height, or of weight, or of IQ-test results, of the citizens of a country, and so on.

Example 4. A six-sided die, which is believed to be biased, is rolled 720 times and shows a ‘6’ 100 times.

- a. Write down the formula for the tail probability of getting 100 or less 6’s in 720 rolls of a fair die.
- b. Using the normal approximation to the binomial distribution, calculate the probability in (i), giving your answer to 3 decimal places.

Solution: a. Let X be the number of sixes in 720 rolls of a fair die.

Then X is binomial with $n = 720, p = \frac{1}{6}$.

Hence

$$P(X \leq 100) = \sum_{k=0}^{100} \binom{720}{k} \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{720-k}.$$

- b. Now X can be approximated by the continuous random variable $Y \sim N(\mu, \sigma^2)$, with

$$\mu = E(X) = np = 720 \times \frac{1}{6} = 120,$$

$$\sigma^2 = \text{Var}(X) = np(1-p) = 720 \times \frac{1}{6} \times \frac{5}{6} = 100.$$

Then

$$P(X \leq 100) \approx P(Y \leq 100.5) = P\left(Z \leq \frac{100.5 - 120}{10}\right) = P(Z \leq -1.95) = 0.026,$$

where $Z = \frac{Y-\mu}{\sigma} \sim N(0, 1)$.

Since $0.026 = 2.6\% < 5\%$, the tail probability is significantly low and so there is good evidence that the die is biased.

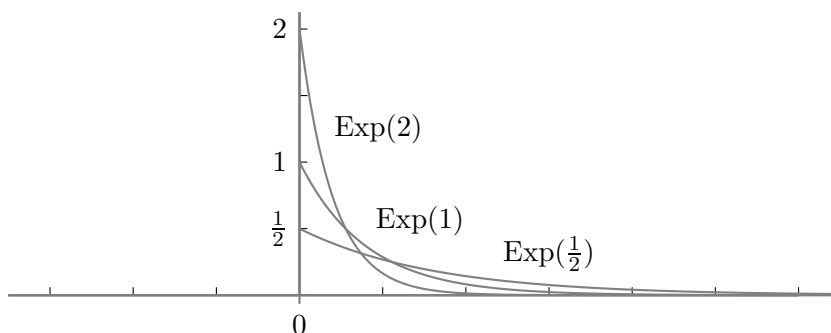
9.6.2 [X] The Exponential Distribution

In the previous subsection, we saw that the binomial distribution can be approximated by the normal distribution. Similarly, the geometric distribution has an analogous continuous probability distribution, namely the **exponential distribution**.

Definition 2. A continuous random variable T has **exponential distribution** $\text{Exp}(\lambda)$ if it has probability density

$$f(t) = \begin{cases} \lambda e^{-\lambda t} & \text{if } t \geq 0 \\ 0 & \text{if } t < 0. \end{cases}$$

We write $T \sim \text{Exp}(\lambda)$ to denote that T has the exponential distribution $\text{Exp}(\lambda)$. The probability densities for $\text{Exp}(\frac{1}{2})$, $\text{Exp}(1)$, and $\text{Exp}(2)$ are illustrated below.



The mean and variance associated with the exponential distribution are given as follows:

Theorem 3. If T is a continuous random variable and $T \sim \text{Exp}(\lambda)$, then

- $E(T) = \frac{1}{\lambda}$
- $\text{Var}(T) = \frac{1}{\lambda^2}$.

Proof. Using integration by parts,

$$\begin{aligned} E(T) &= \int_{-\infty}^{\infty} t f(t) dt = \int_{-\infty}^0 0 dt + \int_0^{\infty} t \lambda e^{-\lambda t} dt \\ &= 0 + [-te^{-\lambda t}]_0^{\infty} + \int_0^{\infty} e^{-\lambda t} dt = 0 - 0 + \frac{1}{\lambda} [-e^{-\lambda t}]_0^{\infty} = \frac{1}{\lambda} (0 - (-e^0)) = \frac{1}{\lambda}. \end{aligned}$$

The variance $\text{Var}(T)$ is calculated similarly. □

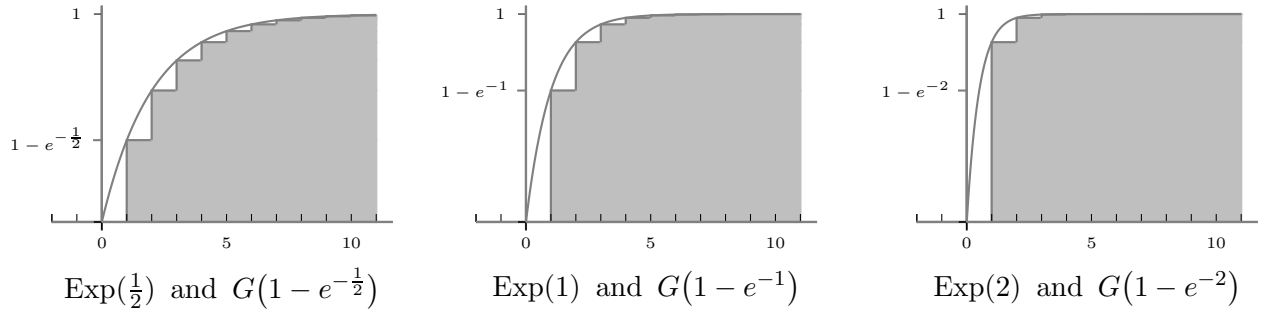
The cumulative distribution function $F_T(t)$ of an exponentially distributed random variable $T \sim \text{Exp}(\lambda)$ is easily expressed:

$$F_T(t) = P(T \leq t) = \int_{-\infty}^t f(x) dx = \begin{cases} 1 - e^{-\lambda t} & , t \geq 0 \\ 0 & , t < 0. \end{cases}$$

If we set $p = 1 - e^{-\lambda}$ and let $n \in \mathbb{Z}$ be an integer, then

$$F_T(n) = \begin{cases} 1 - (1 - p)^n & , n \geq 0 \\ 0 & , n < 0. \end{cases}$$

By Corollary 5 of Theorem 9.4.2, this is the value $F_X(n)$ of the cumulative distribution function of a discrete random variable X that is geometrically distributed with parameter $p = 1 - e^{-\lambda}$. In other words, the exponential distribution $\text{Exp}(\lambda)$ is approximated by the geometric distribution $G(p)$:



Conversely, the geometric distribution $G(p)$ is interpolated by the exponential distribution $\text{Exp}(\lambda)$ where $\lambda = \ln \frac{1}{1-p}$.

Example 5. An insurance company has collected data on one of its insurance policies, and it turns out that, on average, $p = 0.0502$ of these policies are claimed each year. For one of these policies, find the

- probability that the first claim occurs within the first six years;
- probability that the first claim occurs within the first 6.5 years;
- probability that the first claim occurs during the first half of the sixth year;
- expected number of years until the first claim occurs.

SOLUTION. We assume that claims occur independently of each other and with equal probability $p = 0.0502$. If claims only occurred at the end of each year, then we could model the behaviour of the first occurring claim by a discrete random variable $X \sim G(p)$ that counted the number of years until that first claim occurred. However, claims might occur at any positive time from the policy's inception, so the discrete model does not suffice; instead, let T be continuous random variable that gives the time until the first claim. Then $T \sim \text{Exp}(\lambda)$ where $\lambda = \ln \frac{1}{1-0.0502} \approx 0.0515$.

- The probability that the first claim occurs within the first six years is

$$P(X \leq 6) = F_X(6) = 1 - (1 - p)^6 = 1 - (1 - 0.0502)^6 \approx 26.58\%.$$

We could also have calculated this probability as follows:

$$P(T \leq 6) = F_T(6) = 1 - e^{-\lambda \times 6} = 1 - e^{-0.0515 \times 6} \approx 26.58\%.$$

- The probability that the first claim occurs within the first 6.5 years is

$$P(T \leq 6.5) = F_T(6.5) = 1 - e^{-\lambda \times 6.5} = 1 - e^{-0.0515 \times 6.5} \approx 28.45\%.$$

- (c) The probability that the first claim occurs during the first half of the sixth year is

$$\begin{aligned} P(5 < T \leq 5.5) &= P(T \leq 5.5) - P(T \leq 5) = F_T(5.5) - F_T(5) \\ &= (1 - e^{-\lambda \times 5.5}) - (1 - e^{-\lambda \times 5}) \\ &= e^{-0.0515 \times 5} - e^{-0.0515 \times 5.5} \\ &\approx 1.965\%. \end{aligned}$$

A second way to calculate this probability is as follows:

$$P(5 < T \leq 5.5) = \int_5^{5.5} \lambda e^{-\lambda t} dt = [-e^{-\lambda t}]_5^{5.5} = e^{-0.0515 \times 5} - e^{-0.0515 \times 5.5} \approx 1.965\%.$$

- (d) The expected number of years until the first claim occurs is $E(T) = \frac{1}{\lambda} = \frac{1}{0.0515} \approx 19.42$. This is roughly approximated by $E(X) = \frac{1}{p} = \frac{1}{0.0502} \approx 19.53$.

These values are what we might intuitively estimate: “just under $\frac{1}{0.05} = 20$ ”.

◇

9.6.3 Useful Web Applets to Illustrate Probability Reasoning

Web applets for long-run frequency illustrations:

<http://www.shodor.org/interactivate/activities/AdjustableSpinner/>

<http://socr.stat.ucla.edu/Applets.dir/DiceApplet.html>

An applet to illustrate conditional probability and independence:

<http://www.stat.berkeley.edu/users/stark/Java/Html/Venn.htm>

An applet for Bayes' Rule:

<http://www.stat.berkeley.edu/users/stark/Java/Html/Venn.htm>

An applet for the Birthday Problem:

<http://www.mste.uiuc.edu/reese/birthday/>

9.7 Probability and MAPLE

Most of the problems in this chapter reduce to sums of various probability expressions. Maple's ability to do summation symbolically is especially useful here, as is its ability to sum infinite series. Even for finite sums, Maple can simplify the calculations significantly. For example, we can implement the binomial distribution function $B(n, p, k) = \binom{n}{k} p^k (1-p)^{n-k}$ as follows

```
B := proc(n,p,k) binomial(n,k)*p^k*(1-p)^(n-k) end proc;
```

Maple allows us to check that the function values of $B(n, p, k)$ sum to 1 (even though we have not specified what n and p are!):

```
simplify(sum(B(n,p,k), k = 0..n));
```

Indeed, Maple allows us to perform many exact calculations on discrete distributions. For example, it is somewhat time-consuming and awkward to calculate by hand and calculator the probability that 1000 coin tosses result in between 510 and 530 heads. However, this calculation is very quick and easy to conduct with Maple:


```
n := 1000;  
p := 1/2;  
sum(B(n,p,k), k = 510..530);  
evalf(%);
```

Problems for Chapter 9

Problems 9.1 : Some Preliminary Set Theory

1. [R] Let $A = \{a, c, d, e\}$ and $B = \{d, e, f\}$.
Suppose that the universal set is $S = \{a, b, c, d, e, f\}$.
Write down the following sets.

a) $A - B$,	b) $B - A$,	c) $A \cup A^c$,	d) $B \cap B^c$,
e) $A \cup B^c$,	f) $A^c \cap B$,	g) $(A \cup B)^c$,	h) $A^c \cap B^c$.
2. [R] A survey was carried out in a new development area to gain data on home-delivered newspapers. 110 homes were selected at random and the occupants asked whether they had the daily paper or the weekend paper home delivered. 74 received the daily paper, 58 received the weekend paper and 10 received no paper at all. How many homes visited in this survey received both the daily and weekend papers.
3. [R] Of the students taking PHYS1121 and MATH1131 in a hypothetical year, 90% passed MATH1131, 85% passed PHYS1121 and 6% passed neither. What percentage passed both? What percentage of those who passed PHYS1121 also passed MATH1131?
4. [R] A brewery brews one type of beer which is marketed under three different brands. In a survey of 150 first year students, 58 drink at least brand A, 49 drink at least brand B and 57 drink at least brand C. 14 drink brand A and brand C, 13 drink brand A and brand B and 17 drink both brand B and brand C. 4 students drink all three brands. How many students drink none of these three brands?
5. [R] Suppose A , B and C represent three events. Using unions, intersections and complements, find expressions representing the events
 - a) only A occurs,
 - b) at least one event occurs,
 - c) at least two events occur,
 - d) exactly one event occurs,
 - e) exactly two events occur.

Problems 9.2 : Probability

6. [R] Two fair dice are thrown.
 - a) What is the probability that the sum of the two numbers obtained is 6?
 - b) What is the probability that both dice show the same number?
 - c) What is the probability that at least one of the dice shows an even number?

7. [R] Suppose that 30% of computer users use a Macintosh, 50% use a Microsoft Windows PC and that 20% use Linux. Also suppose that 60% of the Macintosh users have succumbed to a computer virus, 80% of the Windows PC users get the virus and 10% of the Linux users get the virus. A computer user is selected at random and it is found that her computer was infected with the virus. What is the probability that she is a Windows PC user?
8. [R] Employment data at a large company reveal that 72% of the workers are married, that 44% are university graduates and that half of the university graduates are married. What is the probability that a randomly chosen worker
- is neither married nor a university graduate?
 - is married but not a university graduate?
 - is married or is a university graduate?
9. [R] On the basis of the health records of a particular group of people, an insurance company accepted 60% of the group for a 10 year life policy. Ten years later it examined the survival rates for the whole group and found that 80% of those accepted for the policy had survived the 10 years, while 50% of those rejected had survived the 10 years. What percentage of the group did not survive 10 years? If a person did survive 10 years, what is the probability that they had been refused cover?
10. [R] Urn 1 contains 2 red balls and 3 black balls. Urn 2 contains 4 red balls and 5 black balls.
- If an urn is randomly selected and a ball drawn at random from it, what is the probability that the ball is red?
 - Suppose a ball is drawn at random from Urn 1 and placed into Urn 2. If a ball is then drawn at random from the 10 balls in Urn 2, what is the probability that it is red?
 - In the previous part, given that the ball drawn from Urn 2 is red, what is the probability that the ball transferred from Urn 1 was black?
11. [R] Down's syndrome is a disorder that affects 1 in 270 babies born to mothers aged 35 or over. A new blood test for the condition has a sensitivity (*i.e.* the probability of a positive test result given the Down's syndrome is present) of 89%. The specificity (*i.e.* the probability of a negative test result given that Down's syndrome is absent) of the new test is 75%.
- What proportion of women over age 35 would test positive on this new blood test?
 - A mother over age 35 receives a positive test result. What is the chance that Down's syndrome is actually present?
 - A mother over age 35 receives a negative test result. What is the chance that Down's syndrome is actually present?
12. [R] The following is a table of the annual promotion probabilities at a particular workplace, broken down by gender.

	Promoted	Not promoted	Total
Male	0.17	0.68	0.85
Female	0.03	0.12	0.15

Is there gender bias in promotion?

13. [R] A system has n independent components and each fail with probability p . Calculate the probability that the system will fail when
- the components are in parallel, so the system fails only when all of the components fail;
 - the components are in series, so the system fails if any one of the components fail;
14. [X] Tom and Bob play a game by each tossing a fair coin. The game consists of tossing the two coins together, until for the first time either two heads appear when Tom wins, or two tails appear when Bob wins.
- Show that the probability that Tom wins at or before the n^{th} toss is $\frac{1}{2} - \frac{1}{2^{n+1}}$.
 - Show that the probability that the game is decided at or before the n^{th} toss is $1 - \frac{1}{2^n}$.
15. [X] Extend the Multiplication Rule of section 9.2.3 to 3 events A_1, A_2, A_3 and show that

$$P(A_1 \cap A_2 \cap A_3) = P(A_3|A_1 \cap A_2)P(A_2|A_1)P(A_1).$$

The same pattern applies to higher numbers of events. Write this down.

This law is particularly useful when we have a sequence of dependent trials. To gain entry to a selective high school students must pass 3 tests. 20% fail the first test and are excluded. Of the 80% who pass the first, 30% fail the second and are excluded. Of those who pass the second, 60% pass the third.

What proportion of students pass the first two tests? Use the multiplicative law to answer this question.

What proportion of students gain entry to the selective high school?

What proportion pass the first two tests, but fail the third?

16. [X] Use the additive law of probability to establish, using mathematical induction, Boole's Law:

$$P(A_1 \cup A_2 \cup \cdots \cup A_n) \leq P(A_1) + P(A_2) + \cdots + P(A_n)$$

17. [X] Establish, using mathematical induction, Bonferoni's inequality:

$$P(A_1 \cap A_2 \cap \cdots \cap A_n) \geq 1 - [P(A_1^c) + P(A_2^c) + \cdots + P(A_n^c)]$$

Problems 9.3 : Random Variables

18. [R] Show that each of the following sequences p_k satisfies $p_k \geq 0$ and $\sum_{k=0}^{\infty} p_k = 1$. Note that in the distributions below $p_k = P(X = k)$ where X is the random variable under consideration.

a) **Uniform Distribution.**

$$p_k = \frac{1}{n} \quad \text{for } 1 \leq k \leq n$$

and 0 otherwise. Here, n is a fixed positive integer.

b) **Binomial Distribution.**

$$p_k = B(n, p, k) = \binom{n}{k} p^k (1-p)^{n-k} \quad \text{for } 0 \leq k \leq n,$$

and 0 otherwise. Here, p is a constant with $0 < p < 1$.

c) **Geometric Distribution.**

$$p_k = G(p, k) = (1-p)^{k-1} p \quad \text{for } 1 \leq k < \infty,$$

where p is a constant with $0 < p < 1$.

d) **Poisson Distribution.**

$$p_k = e^{-\lambda} \frac{\lambda^k}{k!} \quad \text{for } 0 \leq k < \infty,$$

where $\lambda > 0$ is a constant. To solve this question you will need the Maclaurin series for e^λ .

19. [R] A box contains four red and two black balls. Two balls are drawn from the box. Let X be the number of red balls obtained. Find the probability distribution for X .
20. [R] A busy switchboard receives 150 calls an hour on average. Assume that the probability, p_k , of getting k calls in a given minute is

$$p_k = e^{-\lambda} \frac{\lambda^k}{k!}$$

where λ the average number of calls per minute. (This is called a *Poisson* distribution.)

- a) Find the probability of getting exactly 3 calls in a given minute.
- b) Find the probability of getting at least 2 calls in a given minute.
21. [R] In a biased lottery with tickets numbered 1 to 50, the probability that ticket number n wins is

$$p_n = \frac{n}{1275} \quad \text{for } n = 1, 2, 3, \dots, 50.$$

What is the probability that the winning ticket bears a number less than or equal to 25?

22. [H] Let X be a random variable with probability distribution

$$P(X = k) = \frac{c}{k!}, \quad \text{for } k = 0, 1, 2, \dots$$

- Determine the value of c .
 - Calculate $P(X = 2)$.
 - Calculate $P(X < 2)$.
 - Calculate $P(X \geq 4)$.
23. [X] In a biochemical experiment, n organisms are placed in a nutrient medium, and the number of organisms X which survive for a given period is recorded. The probability distribution of X is assumed to be given by

$$P(X = k) = \frac{2(k+1)}{(n+1)(n+2)} \quad \text{for } 0 \leq k \leq n,$$

and 0 otherwise.

- Check that $\sum_{k=0}^n P(X = k) = 1$.
 - Calculate the probability that at most a proportion α of the organisms survive, and deduce that for large n this is approximately α^2 .
 - Find the smallest value of n for which the probability of there being at least one survivor among the n organisms is at least 0.95.
24. [H] A genetic experiment on cell division can give rise to at most $2n$ cells. The probability distribution of the number of cells X recorded is

$$P(X = k) = \frac{\theta^k(1-\theta)}{1-\theta^{2n+1}} \quad \text{for } 0 \leq k \leq 2n,$$

where θ is a constant with $0 < \theta < 1$.

What are the probabilities that

- an odd number of cells is recorded,
 - at most n cells are recorded?
25. [R] Let X be a discrete random variable with the following probability distribution

$$\begin{array}{cccccc} k : & 0 & 1 & 2 & 3 & 4 \\ P(X = k) = p_k : & 0.1 & 2c & 0.2 & 0.1 & 4c \end{array}$$

- Find the value of c .
 - Calculate $E(X)$ and $\text{Var}(X)$.
 - Let $Y = 1 - 4X$. Calculate $E(Y)$ and $\text{Var}(Y)$.
26. [R] Find the mean and variance for the uniform distribution in Question 18(a).

27. [X] Let X be a random variable with the Poisson probability distribution given in Question 18(d). Find $E((1 + X)^{-1})$.
28. [X] Assuming that one can differentiate a power series term by term, one obtains from the formula

$$\sum_{k=0}^{\infty} x^k = \frac{1}{1-x}, \quad |x| < 1$$

the formulas

$$\sum_{k=1}^{\infty} kx^{k-1} = \frac{1}{(1-x)^2}; \quad \sum_{k=2}^{\infty} k(k-1)x^{k-2} = \frac{2}{(1-x)^3}, \quad |x| < 1.$$

(You will see that this is justified in your MATH1231/41 Calculus lectures).

From these formulas, show that

$$\sum_{k=0}^{\infty} kx^k = \frac{x}{(1-x)^2}; \quad \sum_{k=0}^{\infty} k^2x^k = \frac{x(x+1)}{(1-x)^3}, \quad |x| < 1.$$

and hence calculate the mean and the variance for geometric distribution in Question 18(c).

Problems 9.4 : Special Distributions

29. [R] A coin is tossed 50 times. What is the probability of it coming down heads exactly 25 times?
30. [R] A test paper contains 8 multiple choice questions, each with 4 potential answers to choose from. A correct answers gains 1 mark, a wrong answer 0 marks and 4 is the pass mark. If a student simply guesses, what is probability that she will pass?
31. [R] The probability of dying from a particular disease is 0.3. 10 people in a hospital are suffering from the disease. Find the probability that at least 8 survive.
32. [R] How many times must a coin be tossed until the probability of getting 2 or more heads exceeds 0.99? (You need to try different n values after an initial guess.)
33. [X] For the $B(n, p)$ distribution, by considering $\frac{p_k}{p_{k-1}}$, show that p_k is largest when $k = \lfloor (n+1)p \rfloor$. This k is called the “mode” of the distribution.
34. [R] Consider the game of “rock, scissors, paper” in which two players instantaneously choose one of “rock”, “scissors” or “paper”. If both players pick the same item, they play again; if the two players make different choices one of them wins (rock beats scissors, scissors beats paper and paper beats rock).

Let X be the number of times the game is played until someone wins. Find the probability distribution for X when each player chooses randomly from rock, scissors or paper.

35. [H] A die is rolled repeatedly.
- Find the probability that the third time a ‘6’ shows is on the 20th roll.
 - [X] Generalise to the k th time on the n th roll. This is an example of the “negative binomial” distribution, which generalises the geometric distribution.

36. [X] a) Show that for non-negative integers k, m, n ,

$$\binom{k}{m} \binom{n}{k} = \binom{n}{m} \binom{n-m}{k-m}$$

- b) Show that

$$\sum_{k=m}^n \binom{k}{m} \binom{n}{k} p^k (1-p)^{n-k} = \binom{n}{m} p^m.$$

- c) By considering the cases $m = 1$ and $m = 2$ in the preceding formula, prove the variance formula for the Binomial distribution, as stated in Theorem 2 of Section 9.4.
37. [R] A certain type of car is known to sustain damage 25% of the time in 15 km/hr crash tests. A modified bumper design has been proposed in an effort to decrease this percentage. In a trial, cars with the new type of bumper were damaged in one of 15 test crashed.
- What distribution could be used to model the number cars damaged in trials, if the new bumpers perform no better than the old ones?
 - Calculate a tail probability that measures how unusual it would be to observe as few as one damaged car, under the assumptions of part (a).
 - Does the trial indicate that the new bumpers protect cars from damage better than the old bumpers?

38. [R] An extensive study was conducted in the northern hemisphere of butterfly distributions, comparing where butterfly species are presently found with where they were found a century ago (Parmesan *et al* 1999, Nature 399, 579-583). It was hypothesised that due to climate change, more species would have shifted northwards in distribution than shifted southwards.

It was found that of the 23 butterfly species whose distribution had shifted, 22 had shifted northwards in distribution, and 1 had shifted southwards in distribution.

- What distribution could be used to model the number of butterfly species moving northward, if they are just as likely to move north as south (that is, if there is no influence of climate change on butterfly species)?
- Calculate a tail probability that measures how unusual it would be to observe as many as 22 butterfly species moving northwards, under the assumptions of part (a).
- Do these data provide evidence that climate change has affected the distribution of butterfly species?

39. [R] Sydney's dam levels recently reached historic lows, which may in part be due to lower than average rainfall over recent years. The following data are total annual rainfall in Sydney for eight recent calendar years.

Year	2000	2001	2002	2003	2004	2005	2006	2007
Annual rainfall	812.6	1359.0	860.0	1207.6	995.2	816.0	994.0	1499.2

Historic rainfall levels prior to the year 2000 was 1302.2mm per year. We will use a sign test approach to see how much evidence there is that the recent Sydney rainfall is less than historic levels (that is, less than 1302.2mm).

It is reasonable to assume that annual rainfall is independent across years.

- In how many years was the annual rainfall less than 1302.2mm?
 - What distribution could be used to model the number of years in which rainfall was less than 1302.2mm rather than being greater than this value, if both outcomes were equally likely?
 - Calculate a tail probability that measures how unusual it would be to observe as many years with a total rainfall less than 1302.2mm as was observed, if annual rainfall was just as likely to be greater than 1302.2 as less than 1302.2.
 - Do you think there is evidence that Sydney rainfall has decreased in recent years? (That is, that the values are systematically smaller than 1302.2mm?)
40. [R] Do ravens intentionally fly towards gunshot sounds (to scavenge on the carcass they expect to find)? Crow White addressed this question by counting raven numbers at a location, firing a gun, then counting raven numbers 10 minutes later (Ecology 2005, 86:1057-1060). He did this in 12 locations. Results:

location	1	2	3	4	5	6	7	8	9	10	11	12
before	1	0	1	0	0	0	0	5	1	1	2	0
after	2	3	2	2	1	1	2	2	4	2	0	3

We would like to find out if there is evidence that ravens fly towards the location of gunshots.

- In how many locations was there an increase in number of ravens, after the gunshot?
- In how many locations was there a change in number of ravens after the gunshot?
- What distribution could be used to model the number of locations in which there was an increase in the number of ravens rather than a decrease, if both outcomes were equally likely?
- Calculate a tail probability that measures how unusual it would be to observe as many locations with an increase in number of ravens as was observed, if increases and decreases were equally likely.
- Do you think there is evidence that the ravens fly towards gunshot sounds? (That is, was there a systematic increase in the number of ravens present after the gunshot sound?)

Problems 9.5

41. [R] Verify that the following functions f are probability densities, that is, that

$$f(x) \geq 0 \quad \text{and} \quad \int_{-\infty}^{\infty} f(x) dx = 1.$$

Also sketch the graph of each function.

- a) **Uniform Distribution (Assume $a < b$).**

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise.} \end{cases}$$

- b) **Pareto Distribution.** For $k > 0$,

$$f(x) = \begin{cases} \frac{k}{x^{k+1}} & \text{for } x \geq 1 \\ 0 & \text{otherwise.} \end{cases}$$

- c) **Gamma Distribution f_n .** For $n > 0$,

$$f_n(x) = \begin{cases} \frac{1}{n!} x^n e^{-x} & \text{for } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

NOTE. The formula $\int_0^{\infty} x^n e^{-x} dx = n!$ will be useful. You can prove it using integration by parts.

- d) **Laplace Distribution.**

$$f(x) = \frac{1}{2} e^{-|x|} \quad \text{for } -\infty < x < \infty.$$

42. [R] Calculate the means and variances for the distributions in the preceding question. (Note that for the Pareto distribution the mean is only defined if $k > 1$ and the variance is only defined if $k > 2$.)

43. [R] The probability density of the **Cauchy Distribution** is given by

$$f(x) = \frac{\alpha}{1+x^2} \quad \text{for } -\infty < x < \infty.$$

- a) Find the value of α .

- b) If X has a Cauchy distribution, find a number c such that

$$P(X \leq c) = 0.25.$$

- c) [H] What can be said about $E(X)$ and $\text{Var}(X)$ for the Cauchy distribution?

44. [R] X has the probability density function

$$f(x) = \begin{cases} \frac{1}{8}x & \text{for } 3 \leq x \leq 5 \\ 0 & \text{otherwise.} \end{cases}$$

Calculate

- a) $P(X \geq 4)$, b) $P(X \leq 4)$, c) $P(3.2 \leq X \leq 4.1)$, d) $E(X)$.

45. [R] Y has the probability density function

$$f(y) = \begin{cases} \frac{c}{y} & \text{for } 10 \leq y \leq 100 \\ 0 & \text{otherwise.} \end{cases}$$

- a) Determine the value of the constant c .
 b) Obtain the cumulative distribution function of Y .
 c) Find b such that $P(Y \leq b) = 0.50$.
 d) Find the expected value of $E(Y)$.

46. [R] Let F be the function defined by

$$F(x) = \begin{cases} 0 & \text{for } x < 2 \\ \frac{1}{4}(x-2) & \text{for } 2 \leq x < 3 \\ \frac{1}{4} + \frac{3}{8}(x-3) & \text{for } 3 \leq x < 5 \\ 1 & \text{for } x \geq 5. \end{cases}$$

- a) Sketch the graph of F .
 b) Find a probability density function f which would have F as its cumulative distribution function. Sketch the graph of f .
 c) Find $E(X)$ for this probability density function.

47. [H] Suppose X has a probability density given by

$$f(x) = \begin{cases} \alpha e^{-\alpha x} & \text{for } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$$

Find the mean and variance of $Y = 2X + 3$.

Problems 9.6

48. [R] Suppose Z is the standard normal random variable. Use the standard normal tables to write down
- a) $P(Z \leq 1.23)$
 - b) $P(Z \leq -2.3)$
 - c) $P(Z \geq 0.36)$
 - d) $P(Z \geq -1.24)$
 - e) $P(1 \leq Z \leq 2)$
 - f) $P(-0.5 \leq Z \leq 0.5)$
49. [R] Suppose X is normally distributed with mean μ and standard deviation σ . Use the standard normal tables to find:
- a) $P(X \leq 12)$ with $\mu = 10, \sigma = 2$
 - b) $P(X \geq 53)$ with $\mu = 50, \sigma = 4.5$
 - c) $P(X \leq 47)$ with $\mu = 50, \sigma = 4.5$
 - d) $P(X \geq 32)$ with $\mu = 36, \sigma = 8$
 - e) $P(21 \leq X \leq 24)$ with $\mu = 20, \sigma = 3$
 - f) $P(15 \leq X \leq 20)$ with $\mu = 18, \sigma = 1.5$
50. [R] Suppose X is normally distributed with mean μ and standard deviation σ . Use the standard normal tables to find the value of c such that:
- a) $P(X \leq c) = 0.8238$ with $\mu = 0, \sigma = 1$
 - b) $P(X \geq c) = 0.0495$ with $\mu = 0, \sigma = 1$
 - c) $P(X \leq c) = 0.2514$ with $\mu = 50, \sigma = 4.5$
 - d) $P(X \geq c) = 0.6915$ with $\mu = 36, \sigma = 8$
51. [R] The mean heights of men in a certain country is estimated to be 1.69m with a standard deviation of 0.06m. We assume that the heights are normally distributed.
- a) Find the approximate probability that a man chosen at random from this country has a height between 1.60m and 1.74m.
 - b) If 400 men are chosen from this country, how many would you expect on average to have heights greater than 1.74?
52. [R] The length of life of a particular make of T.V. is approximately normally distributed, with a mean of 3.1 years and a standard deviation of 1.2 years. If this type of T.V. is guaranteed for one year, what fraction of the T.V.s sold will require replacement under the guarantee?

53. [R] Experience has shown that the I.Q. scores of University students are normally distributed with a mean of 112 and a standard deviation of 8. Calculate the percentage of students who will have an I.Q. score
- a) higher than 130 b) lower than 100 c) between 105 and 125.
54. [R] The lengths of studs turned out by a certain automatic machine are normally distributed with a mean of 3.220 cm and a standard deviation of 0.003cm. If the acceptable length of a stud is between 3.226 and 3.212 cm, determine to one decimal place the percentage rejected as under size and over size respectively.
55. [R] An unbiased die is tossed 600 times. Use the normal approximation to the binomial to find the approximate probability that a 6 appears more than 120 times.
56. [R] Olof Jonsson was a controversial psychic whose psychic abilities were tested in an experiment. In the experiment a computer showed 4 cards to the subject and (randomly) picked one of them, and the subject was to guess which card they thought the computer had picked. This process was repeated 288 times, and Olof managed to pick the right card 88 times.
- a) What distribution could be used to model the number of cards Olof correctly selects, if he is not psychic (and so can do no better than random guessing)?
- b) Write down an expression for a tail probability that measures how unusual it is to correctly pick as many cards as Olof did.
- c) Use the normal approximation to the binomial to find this probability, under the assumption that Olof is not psychic.
- d) Is this evidence that Olof was psychic?
57. [X] If X is a normal random variable with mean μ and variance σ^2 , show that

$$E(|X - \mu|) = \sqrt{\frac{2}{\pi}} \sigma.$$

58. [X] Find $E(X)$ and $\text{Var}(X)$ for the random variable X with probability density function proportional to
- $$e^{-x^2+x}.$$
59. [X] Let T be a continuous random variable and $T \sim \text{Exp}(\lambda)$; that is T has an exponential distribution with parameter λ . Prove that

$$\text{Var}(T) = \frac{1}{\lambda^2}.$$

60. [X] Suppose a continuous random variable T has an exponential distribution with parameter λ and

$$P(T \leq t) = p, \quad 0 < p < 1.$$

- a) Find t in terms of p .
- b) Hence find the median m . In other words, find m such that $P(T \leq m) = 0.5$.

61. [X] (Memoryless property) A continuous random variable T has an exponential distribution and $T \sim \text{Exp}(\lambda)$. Prove that

$$P(T \geq t + t_0 | T \geq t_0),$$

is independent of t_0 for positive t and t_0 .

62. [X] Suppose the time, in minutes, required by any particular student to complete a certain two-hour examination has the exponential distribution for which the mean is 90 minutes. The examination starts at 10:00 am.

- a) What is the probability that a student completes the examination before 11:00 am?
- b) What is the probability that a student completes the examination between 11:00 am and 11:30 am?
- c) What is the probability that a student could not complete the examination within 2 hours?
- d) What is the median time for completing the examination?

63. [X] During a whale watch season in Sydney, the time, measured in hours from the moment that a cruise enters the whale watch area, to spot the first whale can be modelled by the exponential distribution with parameter $\lambda = 0.4$. A person joins a Sydney whale watch tour during the season. After entering the area,

- a) what is the probability that no whale can be spotted in the first hour?
- b) what is the probability that the time to spot the first whale exceeds the mean by one standard deviation?
- c) The organiser claims a 90 % success rate of finding whales in a trip. How long should the cruise stay in the whale watching area to achieve that?

64. [X] A system consists of 3 independent components connected in series. Hence the system fails when at least one of the components fails. Suppose the lengths of life of the components, measured in hours, have the exponential distribution $\text{Exp}(0.001)$, $\text{Exp}(0.003)$, $\text{Exp}(0.004)$. Find the probability that the system can last at least 100 hours.

65. [X] A system consists of n independent components connected in series. The lengths of life of the components, in hours, have the exponential distributions $\text{Exp}(\lambda_i)$, $1 \leq i \leq n$. Let T be the random variable that gives the time until the system fails.

- a) Find $P(T \leq t)$ and hence write down the cumulative distribution function of the random variable T .
- b) Find the probability density function of T .
- c) Name the probability distribution of T , and find the expected value and variance of T .

ANSWERS TO SELECTED PROBLEMS

Chapter 6

3. a) $\begin{pmatrix} 1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 4 \end{pmatrix}.$
7. Axioms 1, 4, 5, 6, 10 are satisfied, others are not. It is not a vector space.
8. a) $2\mathbf{v} = (1+1)\mathbf{v} = 1\mathbf{v} + 1\mathbf{v} = \mathbf{v} + \mathbf{v}.$ Identify the axioms that have been used.
b) Use induction.
9. For part 5: If $\lambda\mathbf{v} = \mu\mathbf{v}$ then $\lambda\mathbf{v} - \mu\mathbf{v} = \mathbf{0},$ so $(\lambda - \mu)\mathbf{v} = \mathbf{0}.$ But $\mathbf{v} \neq \mathbf{0}$ so, by part 4 of the proposition, $\lambda - \mu = 0.$ So $\lambda = \mu.$
11. The position vectors of the points on a plane which does not pass through the origin do not form a vector subspace.
12. a) For example, $\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$ are in S
c) The position vectors of the points on a plane which passes through the origin form a vector subspace.
16. Column 1 belongs to S as $A \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \\ 4 \end{pmatrix}.$ Similar arguments apply for columns 2 and 3.
17. a) No. b) Yes. c) Yes.
20. $W = \left\{ \begin{pmatrix} 0 \\ 0 \\ \alpha \\ \beta \\ \gamma \end{pmatrix} : \alpha, \beta, \gamma \in \mathbb{R} \right\},$ a copy of $\mathbb{R}^3.$

23. No, S is not a subspace because the zero polynomial is not in it.
24. b) $x^3 + 3x^2 + 3x + 1$.
25. HINT: Suppose that $S_1 = \left\{ \begin{pmatrix} x \\ 0 \end{pmatrix} : x \in \mathbb{R} \right\}$ and $S_2 = \left\{ \begin{pmatrix} 0 \\ y \end{pmatrix} : y \in \mathbb{R} \right\}$. Show that S_1 and S_2 are subspaces of $\mathbb{R}^2$ but $S_1 \cup S_2$ is not.
28. a) Yes, $\mathbf{a} = 2\mathbf{v}_1 - 3\mathbf{v}_2 + \mathbf{v}_3$. b) Yes.
29. a) No. b) No, $3b_2 - 2b_3 = 0$. $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ is a plane in $\mathbb{R}^3$.
30. a) Yes, $\mathbf{a} = \mathbf{v}_1 - \mathbf{v}_2 - 2\mathbf{v}_3$
 b) No, $\mathbf{b} \in \text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ if and only if $b_1 - 2b_2 + b_4 = 0$.
 The span is a 3-dimensional subspace in $\mathbb{R}^4$.
31. Yes, $\mathbf{b} = -10\mathbf{v}_1 + 12\mathbf{v}_2 + 16\mathbf{v}_3$.
32. No, $\text{span}(\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$ is plane in $\mathbb{R}^3$ given by $5b_1 - 4b_2 + b_3 = 0$.
33. Yes. $\mathbf{v} = 3\mathbf{v}_1 - \mathbf{v}_2 - 2\mathbf{v}_3$, where $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are the columns of A .
34. No.
35. Yes.
38. Linearly dependent, coplanar.
39. Linearly independent, not coplanar.
40. Set containing $\mathbf{0}$ is linearly dependent as coefficient of zero vector can be varied to make linear combination non-unique.
41. b) $\mathbf{v}_3 = \mathbf{v}_1 + \mathbf{v}_2$.
 e) $\text{span}(S)$ is plane through origin parallel to $\mathbf{v}_1$ and $\mathbf{v}_2$.
42. b) $\mathbf{v}_4 = -2\mathbf{v}_1 + 2\mathbf{v}_2 + \mathbf{v}_3$ e) $\text{span}(S) = \mathbb{R}^3$.
43. No. $-2 - x + 5x^2 = (1 - x + 2x^2) + 3(-1 + x^2)$.
44. No, it is impossible to return to origin.
45. Yes, it is possible to return to origin.
47. Yes, it is a basis.

48. A basis for W is $\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ -1 \end{pmatrix} \right\}$. $\text{Dim}(W) = 2$.

50. a) True. b) False. c) False. d) True. e) False, True, False, False.
 f) False. g) True. h) False. i) True. j) True.

51. a) $n \leq \ell$ b) No relation. c) $n \geq \ell$. d) $n = \ell$.

53. b) $\left\{ \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \right\}$.

54. A basis for $\text{col}(A)$ is $\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -2 \\ 4 \\ 2 \\ 0 \end{pmatrix} \right\}$. $\text{Dim}(\text{col}(A)) = 3$.

55. A basis for $\text{col}(A)$ is $\left\{ \begin{pmatrix} 1 \\ 0 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ -2 \\ 4 \\ 4 \end{pmatrix} \right\}$. $\text{Dim}(\text{col}(A)) = 3$.

56. A basis is $\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 5 \\ 5 \\ 5 \end{pmatrix}, \begin{pmatrix} -7 \\ -8 \\ -3 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} \right\}$.

57. a) $B = \{\mathbf{v}_1, \mathbf{v}_3\}$. b) $\mathbf{x} = -4\mathbf{v}_1 + 5\mathbf{v}_3$, c) $\dim(\text{col}(A)) = 2$.

59. Bases are $\{p_1, p_2, p_4\}$ or $\{p_1, p_3, p_4\}$ or $\{p_2, p_3, p_4\}$. $\{p_1, p_2, p_3\}$ is not a basis; why not?

62. Coordinate vector is $\begin{pmatrix} -2 \\ 4 \\ 12 \\ 4 \end{pmatrix}$.

63. $\mathbf{v} = \begin{pmatrix} 18 \\ 7 \\ 11 \\ 19 \end{pmatrix}$.

64. $\mathbf{v} = \begin{pmatrix} 1 \\ 1 \\ 10 \end{pmatrix}$.

65. a) $\begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}$. b) $\begin{pmatrix} -a_1 - a_2 + 2a_3 \\ a_2 \\ -a_1 + a_3 \end{pmatrix}$.

66. a) $\begin{pmatrix} 12 \\ -8 \\ 21 \end{pmatrix}$; b) $\begin{pmatrix} -2 \\ 1 \\ -3 \end{pmatrix}$.

67. c) Coordinates are $\lambda_1 = \begin{pmatrix} -1 \\ 3 \\ 4 \end{pmatrix} \cdot \mathbf{v}_1 = -2\sqrt{2}$, $\lambda_2 = \begin{pmatrix} -1 \\ 3 \\ 4 \end{pmatrix} \cdot \mathbf{v}_2 = 2\sqrt{3}$,
 $\lambda_3 = \begin{pmatrix} -1 \\ 3 \\ 4 \end{pmatrix} \cdot \mathbf{v}_3 = \sqrt{6}$. Coordinate vector is $\begin{pmatrix} -2\sqrt{2} \\ 2\sqrt{3} \\ \sqrt{6} \end{pmatrix}$.

69. a) For example, $\begin{pmatrix} 1 & 0 \\ 0 & 4 \end{pmatrix}$, $\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $\begin{pmatrix} -1 & 0 \\ 0 & 6 \end{pmatrix}$ are in S .
 b) S is not a subspace because the 2×2 zero matrix is not in it.

70. a) For example, $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$, $\begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$, $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ are in S .
 b) Use the Subspace Theorem to prove that S is a subspace.

74. a) $\begin{pmatrix} a_{11} \\ a_{12} \\ a_{21} \\ a_{22} \end{pmatrix}$. b) $\begin{pmatrix} \frac{1}{2}(a_{11} + a_{22}) \\ \frac{1}{2}(a_{12} + a_{21}) \\ \frac{1}{2i}(-a_{12} + a_{21}) \\ \frac{1}{2}(a_{11} - a_{22}) \end{pmatrix}$.

75. a) $\begin{pmatrix} -4 & 2 \\ -1 & -3 \end{pmatrix} = -4 \begin{pmatrix} 1 & 0 \\ -2 & 0 \end{pmatrix} + 2 \begin{pmatrix} 0 & 1 \\ 3 & 0 \end{pmatrix} - 3 \begin{pmatrix} 0 & 0 \\ 5 & 1 \end{pmatrix}$. b) No.

79. Not a subspace.

85. Not a subspace.

87. No.

88. Let $p \in \mathbb{P}_2$ be $p(z) = a_0 + a_1z + a_2z^2$. Then the condition is $-\frac{3}{2}a_1 + a_2 = 0$. (An equivalent condition is $p'(\frac{-1}{3}) = 0$.)

89. No for question 87. No for question 88.

90. No. $p_3 = -p_1 + 3p_2$.

91. A basis is $\{p_1, p_2, p_3, p_4, 1, z\}$.

$$92. \begin{pmatrix} 2 \\ -1 \\ 0 \end{pmatrix}.$$

$$93. \begin{pmatrix} 9 \\ 1 \\ 17 \end{pmatrix}.$$

$$94. \begin{pmatrix} a_0 - a_1 + a_2 \\ a_0 \\ a_0 + a_1 + a_2 \end{pmatrix}.$$

Chapter 7

1. S is not a linear map as the domain $[-1, 1]$ is not a vector space.
2. a) Linear. b) Linear. c) Not Linear. d) Linear. e) Not Linear.
3. a) Domain $\mathbb{C}$, codomain $\mathbb{R}$, linear. b) Domain $\mathbb{C}$, codomain $\mathbb{R}$, linear.
c) Domain $\mathbb{C}$, codomain $\mathbb{R}_+ = \{x \in \mathbb{R} : x \geq 0\}$, not linear.
d) Domain $\mathbb{C} - \{0\}$, codomain $(-\pi, \pi]$, not linear. e) Domain $\mathbb{C}$, codomain $\mathbb{C}$, linear.
6. No.

$$7. T \begin{pmatrix} 2 \\ -1 \\ 4 \end{pmatrix} = \begin{pmatrix} 21 \\ 4 \\ -15 \\ 28 \end{pmatrix} \quad \text{and} \quad T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} x_1 - 3x_2 + 4x_3 \\ 2x_1 \\ 3x_1 + x_2 - 5x_3 \\ 4x_1 + 4x_2 + 6x_3 \end{pmatrix}.$$

$$9. \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \begin{pmatrix} -2 \\ 1 \\ -4 \end{pmatrix} + \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \text{ and so } T \begin{pmatrix} 0 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}.$$

$$11. \quad \text{a) } \begin{pmatrix} 3 & -1 \\ 2 & 4 \\ -3 & -3 \\ 0 & 1 \end{pmatrix}. \quad \text{b) } \begin{pmatrix} -2 & 0 & 5 & 0 \\ 6 & -8 & 0 & 2 \\ -2 & 4 & -3 & 0 \end{pmatrix}. \quad \text{d) } \begin{pmatrix} 1 & 0 & -2 & -4 \\ 3 & -4 & -3 & 1 \\ -2 & 2 & 4 & 0 \end{pmatrix}.$$

12. Same as for 11.
13. a) $A\mathbf{e}_1 = 2\mathbf{e}_1$, $A\mathbf{e}_2 = 0.7\mathbf{e}_2$, $A\mathbf{b} = 4\mathbf{e}_1 + 2.1\mathbf{e}_2$. ($\mathbf{e}_1$ is stretched to twice its length, $\mathbf{e}_2$ is compressed to 0.7 of its length and $\mathbf{b}$ is stretched and rotated.)
d) Notice that $A\mathbf{b} = 3\mathbf{b}$. This means $\mathbf{b}$ is stretched to three times its length.

- e) Notice that $A\mathbf{b} = -2\mathbf{b}$. This means the direction of $\mathbf{b}$ is reversed and it is stretched to twice its length.

15. $\frac{1}{2} \begin{pmatrix} 1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix}.$

16. $\mathbf{x}' = T(\mathbf{x}) = \begin{pmatrix} -x_1 \\ x_2 \end{pmatrix}, \quad A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$

17. $\mathbf{x}' = T(\mathbf{x}) = \begin{pmatrix} x_1 \\ x_2 \\ -x_3 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$

18. $\mathbf{q} = A\mathbf{p}$, where diagonal entries of A are $a_{ii} = -1 + 2d_i^2/|\mathbf{d}|^2$ and off-diagonal entries of A are $a_{ij} = 2d_i d_j / |\mathbf{d}|^2$.

19. T is linear. For $T(\mathbf{x}) = A\mathbf{x}$, the matrix is

$$A = \begin{pmatrix} 0 & -b_3 & b_2 \\ b_3 & 0 & -b_1 \\ -b_2 & b_1 & 0 \end{pmatrix}.$$

20. $T(\mathbf{a}) = A\mathbf{a}$, where $A = \frac{1}{5} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 0 & 0 \\ 2 & 0 & 4 \end{pmatrix}.$

21. S is not linear.

23. $\begin{pmatrix} a'_1 \\ a'_2 \\ a'_3 \end{pmatrix} = R_\alpha \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} a_1 \cos \alpha + a_3 \sin \alpha \\ a_2 \\ -a_1 \sin \alpha + a_3 \cos \alpha \end{pmatrix}.$

25. a) $\ker(A) = \{\mathbf{0}\}$, $\text{nullity}(A) = 0$, no basis.

b) Kernel: $\left\{ \lambda \begin{pmatrix} -1 \\ -3 \\ 1 \end{pmatrix} : \lambda \in \mathbb{R} \right\}, \text{ nullity}(B) = 1.$

c) Kernel: $\left\{ \lambda \begin{pmatrix} 2 \\ -2 \\ -1 \\ 1 \end{pmatrix} : \lambda \in \mathbb{R} \right\}, \text{ nullity}(C) = 1.$

26. a) $\left\{ \begin{pmatrix} 1 \\ -3 \\ 1 \\ 0 \end{pmatrix} \right\}, \text{ nullity}(D) = 1. \quad \text{b) } \{\}, \text{ nullity}(E) = 0.$

27. For example $A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 3 & 4 \end{pmatrix}$.

28. a) $\ker(A) = \{\mathbf{0}\}$, $\text{nullity}(A) = 0$.

b) $\ker(A) = \left\{ \lambda \begin{pmatrix} 5 \\ 4 \\ 2 \\ 1 \end{pmatrix} : \lambda \in \mathbb{R} \right\}$, $\text{nullity}(A) = 1$.

d) $\ker(A) = \left\{ \lambda \begin{pmatrix} 6 \\ 4 \\ 1 \\ 1 \end{pmatrix} : \lambda \in \mathbb{R} \right\}$, $\text{nullity}(A) = 1$.

29. For questions 16, 17 and 18, $\ker(T) = \{\mathbf{0}\}$ and $\text{nullity}(T) = 0$.

For question 19, $\ker(T) = \{\mathbf{x} \in \mathbb{R}^3 : \mathbf{x} = \lambda \mathbf{b} \text{ for } \lambda \in \mathbb{R}\}$. $\text{Nullity}(T) = 1$. Kernel is set of all vectors parallel to $\mathbf{b}$.

For question 20, $\ker(T) = \{\mathbf{x} \in \mathbb{R}^n : \mathbf{b} \cdot \mathbf{x} = 0\}$. $\text{Nullity}(T) = n - 1$ (Why?). Kernel is set of all vectors orthogonal to $\mathbf{b}$.

30. b) 1

31. a) $\mathbf{b} \in \text{im}(A)$ as $A \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} = \begin{pmatrix} 11 \\ 10 \\ 4 \end{pmatrix}$.

b) $\mathbf{b}$ is not in $\text{im}(A)$ as $A\mathbf{x} = \begin{pmatrix} 9 \\ -2 \\ -4 \end{pmatrix}$ has no solution.

c) $\mathbf{b} \in \text{im}(A)$, since, for example, $\mathbf{x} = \begin{pmatrix} -10 \\ 12 \\ 16 \\ 0 \end{pmatrix}$ is a solution of $A\mathbf{x} = \mathbf{b}$.

32. a) No conditions, $\text{im}(A) = \mathbb{R}^3$. b) $3b_2 - 2b_3 = 0$.

c) No conditions, $\text{im}(A) = \mathbb{R}^3$.

33. $\text{rank}(A) = 3$. Columns 1,2,3 of A form a basis for $\text{im}(A)$.

$\text{rank}(B) = 2$. Columns 1,2 of B form a basis for $\text{im}(B)$.

$\text{rank}(C) = 3$. Columns 1,2,3 of C form a basis for $\text{im}(C)$.

$\text{rank}(D) = 3$. Columns 1,2,4 of D form a basis for $\text{im}(D)$.

$\text{rank}(E) = 3$. Columns 1,2,3 of E form a basis for $\text{im}(E)$.

35. $\text{rank}(A) = 3$. Columns 1,3,4 of A form a basis for $\text{im}(A)$.

$\text{rank}(B) = 3$. Columns 1,3,4 of B form a basis for $\text{im}(B)$.

36. One possible answer is $\left\{ \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}.$

37. One possible answer is $\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix}, \begin{pmatrix} 3 \\ 3 \\ 3 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \\ 4 \\ 8 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} \right\}.$

38. a) $\begin{pmatrix} 3 & 4 & -1 & 0 \\ 0 & 0 & 3 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$ b) $\left\{ \begin{pmatrix} 3 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -1 \\ 3 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ -3 \\ 0 \\ 1 \end{pmatrix} \right\},$ 3. c) 1. d) No.

46. T is linear.

50. T is a linear function.

53. S is not linear as $\mathbb{Z}$ is not a vector space.

T is not linear as, for example, $T(1.5) + T(1.5) = 2 + 2 = 4$ while $T(1.5 + 1.5) = T(3) = 3$.

54. $y_L(s) = \frac{s^2 + 9s + 19}{(s + 3)^2(s + 1)}.$

56. a) $7 - 2x$ c) $\begin{pmatrix} 2 \\ 1 \\ 1 \\ -1 \end{pmatrix}.$

57. $T \begin{pmatrix} i \\ 2 \\ -1 \end{pmatrix} = (2 + i) + (7 - 4i)z + 2z^2 - 3iz^3,$

$T \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = (x_1 - 2x_3) + [(2 + i)x_1 + (4 - 3i)x_2]z + x_2z^2 - 3x_1z^3.$

58. b) $\begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & \frac{1}{4} \end{pmatrix}.$ c) $\{x, x^2, x^3, x^4\}.$ d) The empty set.

59. Let input vector be $\mathbf{b} = (b_1 \ b_2 \ b_3 \ b_4 \ b_5)^T$, where b_1, b_2, b_3, b_4 and b_5 are the amounts of steel, plastics, rubber, glass and labour used respectively. Let the output vector be $\mathbf{x} = (x_1 \ x_2 \ x_3 \ x_4)^T$, where x_1, x_2, x_3, x_4 are the numbers of station wagons, 4-wheel drives, hatchbacks and sedans made. Then the factory is represented by the linear map

$T_A : \mathbb{R}^4 \rightarrow \mathbb{R}^5$, where $T_A(\mathbf{x}) = \mathbf{b} = A\mathbf{x}$ with

$$A = \begin{pmatrix} 1 & 1.5 & 0.8 & 0.9 \\ 0.5 & 0.6 & 0.7 & 0.6 \\ 0.1 & 0.2 & 0.2 & 0.25 \\ 0.2 & 0.15 & 0.3 & 0.3 \\ 1 & 1.5 & 1.1 & 0.9 \end{pmatrix}.$$

60. The matrix is $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$.

61. The matrix is $\begin{pmatrix} -1 & -3 \\ 2 & 4 \end{pmatrix}$.

62. $-6 + 4x$

63. For 48, $\begin{pmatrix} 1 & -3 & 0 & 0 \\ 0 & 0 & 2 & -3 \\ 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 3 & -1 & 2 & 4 \end{pmatrix}.$

For 49, $\begin{pmatrix} 3 & 4 & 0 & 0 \\ 0 & 3 & 8 & 0 \\ 0 & 0 & 3 & 12 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$

For 50, $\begin{pmatrix} -8 & 0 & 0 & 0 \\ 0 & -4 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix};$

For 51, $\begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{3} & 0 \\ 0 & 0 & 0 & \frac{1}{4} \end{pmatrix}.$

64. 48: $\text{im}(T) = \{p \in \mathbb{P}_4(\mathbb{R}) : p(z) = \lambda_0 + \lambda_1 z + \lambda_2 z^3 + \lambda_3 z^4 \text{ for } \lambda_0, \lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}\},$
(note z^2 is not in $\text{im}(T)$) $\text{rank} = 4$, $\ker(T) = \{0\}$, $\text{nullity}(T) = 0$.

49: $\text{im}(T) = \mathbb{P}_3(\mathbb{R})$, $\text{rank}(T) = 4$, $\ker(T) = \{0\}$, $\text{nullity}(T) = 0$.

50: $\text{im}(T) = \{p \in \mathbb{P}_3(\mathbb{R}) : p(x) = \lambda_0 + \lambda_1 x + \lambda_2 x^3 \text{ for } \lambda_0, \lambda_1, \lambda_2 \in \mathbb{R}\}$, $\text{rank}(T) = 3$,
 $\ker(T) = \{p \in \mathbb{P}_3(\mathbb{R}) : p(x) = \lambda x^2 \text{ for } \lambda \in \mathbb{R}\}$, $\text{nullity}(T) = 1$.

51: $\text{im}(T) = \{p \in \mathbb{P}_4(\mathbb{R}) : p(x) = \lambda_1 x + \lambda_2 x^2 + \lambda_3 x^3 + \lambda_4 x^4 \text{ for } \lambda_1, \lambda_2, \lambda_3, \lambda_4 \in \mathbb{R}\},$
 $\text{rank}(T) = 4$, $\ker(T) = \{0\}$, $\text{nullity}(T) = 0$.

65. The matrix A is diagonal with diagonal elements $a_{kk} = (k-1)(k-2) - 3(k-1) + 3$ for $1 \leq k \leq n+1$. The kernel is $\alpha_1 x + \alpha_2 x^3$ and the nullity is 2. Note that the kernel is the solution of the homogeneous differential equation.

66. The matrix is $\begin{pmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & -1 & 0 \end{pmatrix}.$

69. b) i) $\begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$. ii) $\begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{pmatrix}$.

70. b) $\lambda \begin{pmatrix} 1 & -3 \\ -3 & 0 \end{pmatrix}, \lambda \in \mathbb{F};$ 1. c) 3. d) No. e) $\begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 3 & 1 & 0 & 0 \end{pmatrix}$.

71. a) $\left\{ \begin{pmatrix} 2 \\ 4 \\ -2 \\ 4 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ -2 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -3 \\ 2 \\ -2 \end{pmatrix} \right\}$. b) 3. c) $a_4 - a_1 - a_3 - a_2 = 0$.

d) $\left\{ \begin{pmatrix} \frac{3}{2} \\ 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -\frac{1}{2} \\ 0 \\ -3 \\ -3 \\ 1 \end{pmatrix} \right\}$. e) 3, 2.

Chapter 8

1. a) In each case, the eigenvalues are the diagonal entries and the respective eigenvectors are $t\mathbf{e}_1$ and $t\mathbf{e}_2$ ($t \neq 0$).

For interpretations of (b), (c) and (d), see part (e) of question.

2. Eigenvalue is 2.

3. $\lambda = (\det A)^{1/3}$.

4. b) $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

5. a) $\lambda = 2$ with eigenvectors $\left\{ t \begin{pmatrix} 1 \\ 2 \end{pmatrix} : t \neq 0 \right\}$ and $\lambda = 3$ with eigenvectors $\left\{ t \begin{pmatrix} 2 \\ 3 \end{pmatrix} : t \neq 0 \right\}$,

- b) $\lambda = -3$ with eigenvectors $\left\{ t \begin{pmatrix} 1 \\ 1 \end{pmatrix} : t \neq 0 \right\}$ and $\lambda = 1$ with eigenvectors $\left\{ t \begin{pmatrix} 1 \\ 3 \end{pmatrix} : t \neq 0 \right\}$.

7. a) $\lambda = 3$ with eigenvectors $\left\{ t \begin{pmatrix} 1 \\ 1 \end{pmatrix} : t \neq 0 \right\}$ and
 $\lambda = -1$ with eigenvectors $\left\{ t \begin{pmatrix} -1 \\ 1 \end{pmatrix} : t \neq 0 \right\}$.

- b) Only one eigenvalue $\lambda = 2$ with multiplicity 2 and eigenvectors $\left\{ t \begin{pmatrix} 1 \\ 0 \end{pmatrix} : t \neq 0 \right\}$.

- c) $\lambda = 3$ with eigenvectors $\left\{ t \begin{pmatrix} 1 \\ 0 \end{pmatrix} : t \neq 0 \right\}$ and
 $\lambda = -6$ with eigenvectors $\left\{ t \begin{pmatrix} -5 \\ 9 \end{pmatrix} : t \neq 0 \right\}$.
- d) $\lambda = 1 \pm i$ with eigenvectors $\left\{ t \begin{pmatrix} -1 \pm i \\ 1 \end{pmatrix} : t \neq 0 \right\}$.
- e) $\lambda = 5 \pm i\sqrt{3}$ with eigenvectors $\left\{ t \begin{pmatrix} \pm \frac{\sqrt{3}}{2} + \frac{1}{2}i \\ 1 \end{pmatrix} : t \neq 0 \right\}$.
- f) $\lambda = 5 \pm \sqrt{5}$ with eigenvectors $\left\{ t \begin{pmatrix} \frac{1}{2}(1 \mp \sqrt{5})i \\ 1 \end{pmatrix} : t \neq 0 \right\}$.

9. The eigenvalues are the diagonal entries, 2, -2, 3, 5. Corresponding eigenvectors are

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ 5 \\ 0 \end{pmatrix}, \quad \mathbf{v}_4 = \begin{pmatrix} 25 \\ -3 \\ 21 \\ 14 \end{pmatrix}.$$

10. a) $-1, 4, 6; \begin{pmatrix} -3 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$. b) $2, -3, 3; \begin{pmatrix} 0 \\ -1 \\ 6 \end{pmatrix}, \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$.

11. In each of the following answers, the diagonal entries in D and the columns in M may be rearranged in the same way and the answer is still correct. Also, any column in M may be multiplied by a scalar and the new M is still correct.

For Question 7:

- a) $D = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}, \quad M = \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}$.
- b) The matrix is not diagonalisable.
- c) $D = \begin{pmatrix} 3 & 0 \\ 0 & -6 \end{pmatrix}, \quad M = \begin{pmatrix} 1 & -5 \\ 0 & 9 \end{pmatrix}$.
- d) $D = \begin{pmatrix} 1+i & 0 \\ 0 & 1-i \end{pmatrix}, \quad M = \begin{pmatrix} 1+i & 1-i \\ 1 & 1 \end{pmatrix}$.
- e) $D = \begin{pmatrix} 5+i\sqrt{3} & 0 \\ 0 & 5-i\sqrt{3} \end{pmatrix}, \quad M = \begin{pmatrix} \frac{\sqrt{3}}{2} + \frac{1}{2}i & -\frac{\sqrt{3}}{2} + \frac{1}{2}i \\ 1 & 1 \end{pmatrix}$.
- f) $D = \begin{pmatrix} 5+\sqrt{5} & 0 \\ 0 & 5-\sqrt{5} \end{pmatrix}, \quad M = \begin{pmatrix} \frac{1}{2}(1-\sqrt{5})i & \frac{1}{2}(1+\sqrt{5})i \\ 1 & 1 \end{pmatrix}$.

For Question 9:

$$D = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & -2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 5 \end{pmatrix}, \quad M = \begin{pmatrix} 1 & 1 & 1 & 25 \\ 0 & 1 & 1 & -3 \\ 0 & 0 & 5 & 21 \\ 0 & 0 & 0 & 14 \end{pmatrix}.$$

For Question 10:

$$\text{a) } D = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 6 \end{pmatrix}, \quad M = \begin{pmatrix} -3 & 1 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

$$\text{b) } D = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 3 \end{pmatrix}, \quad M = \begin{pmatrix} 0 & 0 & 1 \\ -1 & -1 & 0 \\ 6 & 1 & 0 \end{pmatrix}.$$

15. If $\mathbf{v}$ is an eigenvector of T then the coordinate vector $[\mathbf{v}]_B$ of $\mathbf{v}$ with respect to the basis B is the corresponding eigenvector for the matrix A .

$$16. A^5 = \begin{pmatrix} -78 & 330 \\ 55 & -133 \end{pmatrix}.$$

$$17. \quad \text{a) } 6, \quad t \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad t \neq 0, \quad t \in \mathbb{R}; \quad -4, \quad \begin{pmatrix} -3 \\ 4 \end{pmatrix}, \quad t \neq 0, \quad t \in \mathbb{R}.$$

$$\text{b) } P = \begin{pmatrix} 1 & -3 \\ 2 & 4 \end{pmatrix}, \quad D = \begin{pmatrix} 6 & 0 \\ 0 & -4 \end{pmatrix}. \quad \text{c) } \begin{pmatrix} 6^n & (-3)(-4)^n \\ 2 \times 6^n & 4(-4)^n \end{pmatrix}.$$

18. When A is diagonalisable, $A^k = MD^kM^{-1}$ and $\mathbf{x}_k = A^k\mathbf{x}_0$. As a check, if you put $k = 0$ in the answers below you should get $A^0 = I$, whereas if you put $k = 1$ you should get $A^1 = A$.

For Question 7:

$$\text{a) } A^k = \frac{1}{2} \begin{pmatrix} 3^k + (-1)^k & 3^k + (-1)^{k+1} \\ 3^k + (-1)^{k+1} & 3^k + (-1)^k \end{pmatrix}.$$

$$\text{c) } A^k = \frac{1}{9} \begin{pmatrix} 3^{k+2} & 5(3^k - (-6)^k) \\ 0 & 9(-6)^k \end{pmatrix}.$$

19. $\alpha_1, \alpha_2, \alpha_3$ and α_4 are arbitrary real numbers.

For Question 5:

$$\text{a) } \mathbf{y}(t) = \alpha_1 e^{3t} \begin{pmatrix} 2 \\ 3 \end{pmatrix} + \alpha_2 e^{2t} \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

$$\text{b) } \mathbf{y}(t) = \alpha_1 e^t \begin{pmatrix} 1 \\ 3 \end{pmatrix} + \alpha_2 e^{-3t} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

For Question 9:

$$\mathbf{y}(t) = \alpha_1 e^{2t} \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} + \alpha_2 e^{-2t} \begin{pmatrix} 1 \\ 1 \\ 0 \\ 0 \end{pmatrix} + \alpha_3 e^{3t} \begin{pmatrix} 1 \\ 1 \\ 5 \\ 0 \end{pmatrix} + \alpha_4 e^{5t} \begin{pmatrix} 25 \\ -3 \\ 21 \\ 14 \end{pmatrix}.$$

For Question 10:

$$\text{a) } \mathbf{y}(t) = \alpha_1 e^{-t} \begin{pmatrix} -\frac{3}{2} \\ 1 \\ 0 \end{pmatrix} + \alpha_2 e^{4t} \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + \alpha_3 e^{6t} \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}.$$

$$\text{b) } \mathbf{y}(t) = \alpha_1 e^{2t} \begin{pmatrix} 0 \\ -\frac{1}{6} \\ 1 \end{pmatrix} + \alpha_2 e^{-3t} \begin{pmatrix} 0 \\ -1 \\ 1 \end{pmatrix} + \alpha_3 e^{3t} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}.$$

$$20. \quad \text{a) } 1, \quad \lambda \begin{pmatrix} 3 \\ -1 \end{pmatrix}, \lambda \neq 0; \quad 5, \quad \lambda \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \lambda \neq 0. \quad \text{b) } \begin{cases} x_1 = 3\alpha e^t + \beta e^{5t}, \\ x_2 = -\alpha e^t + \beta e^{5t}. \end{cases}$$

$$21. \quad \text{a) } x = 300e^t - 200e^{3t}, y = 150e^t - 50e^{3t}. \quad \text{b) } x = -500 + 600e^{-2t}, y = -100 + 200e^{-2t}.$$

22. The solutions by the two methods are:

$$\text{a) } \mathbf{y}(t) = \begin{pmatrix} y(t) \\ \dot{y}(t) \end{pmatrix} = \alpha_1 e^t \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \alpha_2 e^{t/5} \begin{pmatrix} 1 \\ \frac{1}{5} \end{pmatrix}, \quad (\text{matrix method})$$

$$y(t) = \alpha_1 e^t + \alpha_2 e^{t/5}. \quad (\text{calculus method})$$

$$\text{b) } \mathbf{y}(t) = \begin{pmatrix} y(t) \\ \dot{y}(t) \end{pmatrix} = \alpha_1 e^{4t} \begin{pmatrix} 1 \\ 4 \end{pmatrix} + \alpha_2 e^{-4t} \begin{pmatrix} 1 \\ -4 \end{pmatrix}, \quad (\text{matrix method})$$

$$y(t) = \alpha_1 e^{4t} + \alpha_2 e^{-4t}. \quad (\text{calculus method})$$

23. The matrix method given in notes is not applicable as the matrix is not diagonalisable.

$$24. \quad \text{a) } A = \begin{pmatrix} 0 & 1 \\ -\frac{c}{a} & -\frac{b}{a} \end{pmatrix}.$$

25. a) 500 years — 0.9048 : 0.0928 : 0.0024
 1000 years — 0.8187 : 0.1722 : 0.0091
 1000000 years — 1.4×10^{-87} : 7×10^{-44} : 1
 b) The associated matrix is not diagonalisable.

$$26. \quad \text{In the 12th: } \begin{pmatrix} 0.98 & 0.02 & 0.03 \\ 0.01 & 0.96 & 0.03 \\ 0.01 & 0.02 & 0.94 \end{pmatrix}^{11} \begin{pmatrix} 300 \\ 300 \\ 300 \end{pmatrix} \approx \begin{pmatrix} 378 \\ 293 \\ 229 \end{pmatrix},$$

$$\text{In the 24th: } \begin{pmatrix} 0.98 & 0.02 & 0.03 \\ 0.01 & 0.96 & 0.03 \\ 0.01 & 0.02 & 0.94 \end{pmatrix}^{23} \begin{pmatrix} 300 \\ 300 \\ 300 \end{pmatrix} \approx \begin{pmatrix} 426 \\ 280 \\ 194 \end{pmatrix}.$$

$$27. \quad \text{In the 12th: } \begin{pmatrix} 339 \\ 262 \\ 205 \end{pmatrix} \text{ total} = 806; \quad \text{In the 24th: } \begin{pmatrix} 339 \\ 222 \\ 154 \end{pmatrix} \text{ total} = 715.$$

28. The population settles to the proportions $1.156 : 1.124 : 1.116 : 1.086 : 1$ but eventually dies out.

31. a) $6, \quad t \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}; \quad 7, \quad t \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}; \quad 8, \quad t \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}, \quad t \in \mathbb{R}.$

b) $A = \begin{pmatrix} 1 & 2 & 2 \\ -1 & 0 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad D = \begin{pmatrix} 6 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & 8 \end{pmatrix}.$

c) $A^{-1} = \begin{pmatrix} -1 & 0 & 2 \\ 2 & -1 & -3 \\ -1 & 1 & 2 \end{pmatrix},$
 $M^k = \begin{pmatrix} -6^k + 4 \times 7^k - 2 \times 8^k & -2 \times 7^k + 2 \times 8^k & 2 \times 6^k - 6 \times 7^k + 4 \times 8^k \\ 6^k - 8^k & 8^k & -2 \times 6^k + 2 \times 8^k \\ -6^k + 2 \times 7^k - 8^k & -7^k + 8^k & 2 \times 6^k - 3 \times 7^k + 2 \times 8^k \end{pmatrix}$

Chapter 9

1. a) $\{a, c\},$ b) $\{f\},$ c) $S,$ d) $\emptyset,$
e) $\{a, b, c, d, e\},$ f) $\{f\},$ g) $\{b\},$ h) $\{b\}.$
2. 32.
3. 81%, 95.3%.
4. 26.
5. a) $A \cap B^c \cap C^c,$ b) $A \cup B \cup C,$ c) $(A \cap B) \cup (A \cap C) \cup (B \cap C),$
d) $(A \cap B^c \cap C^c) \cup (A^c \cap B \cap C^c) \cup (A^c \cap B^c \cap C),$
e) $(A^c \cap B \cap C) \cup (A \cap B^c \cap C) \cup (A \cap B \cap C^c).$
6. a) $\frac{5}{36}.$ b) $\frac{1}{6}.$ c) $\frac{3}{4}.$
7. $\frac{2}{3}.$
8. a) $\frac{3}{50},$ b) $\frac{1}{2},$ c) $\frac{47}{50}.$
9. 32%, $\frac{5}{17}.$
10. a) $\frac{19}{45},$ b) $\frac{11}{25},$ c) $\frac{6}{11}.$

11. a) 25.24%, b) 0.0131, c) 0.000545.

12. No.

13. a) p^n ; b) $1 - (1 - p)^n$.

15. $P(A_1 \cap A_2 \cap \cdots \cap A_n) = P(A_n | A_1 \cap \cdots \cap A_{n-1})P(A_{n-1} | A_1 \cap \cdots \cap A_{n-2}) \cdots P(A_2 | A_1)P(A_1)$,
56%, 33.6%, 22.4%.

19.

x	0	1	2
$P(X = x)$	$\frac{1}{15}$	$\frac{8}{15}$	$\frac{2}{5}$

20. a) 0.214, b) 0.713.

21. $\frac{13}{51}$.

22. a) $c = \frac{1}{e}$. b) $P(X = 2) = \frac{1}{2e}$. c) $P(X < 2) = \frac{2}{e}$. d) $P(X \geq 4) = 1 - \frac{8}{3e}$.

23. b) $\frac{(\lfloor \alpha n \rfloor)^2 + 3\lfloor \alpha n \rfloor + 2}{n^2 + 3n + 2}$, c) $n = 5$.

24. a) $\frac{\theta(1 - \theta^{2n})}{(1 - \theta^{2n+1})(1 + \theta)}$, b) $\frac{1 - \theta^{n+1}}{1 - \theta^{2n+1}}$.

25. a) $c = 0.1$, b) $E(X) = 2.5$, $\text{Var}(X) = 2.05$, c) $E(Y) = -9$, $\text{Var}(Y) = 32.8$.

26. $\mu = \frac{n+1}{2}$; $\sigma^2 = \frac{n^2-1}{12}$.

28. $\mu = \frac{\alpha}{1-\alpha}$; $\sigma^2 = \frac{\alpha}{(1-\alpha)^2}$.

29. 0.1123.

30. $70p^4q^4 + 56p^5q^3 + 28p^6q^2 + 8p^7q + p^8$ where $p = \frac{1}{4}$, $q = \frac{3}{4}$. This evaluates to $\frac{7459}{65536} \simeq 0.1138$.

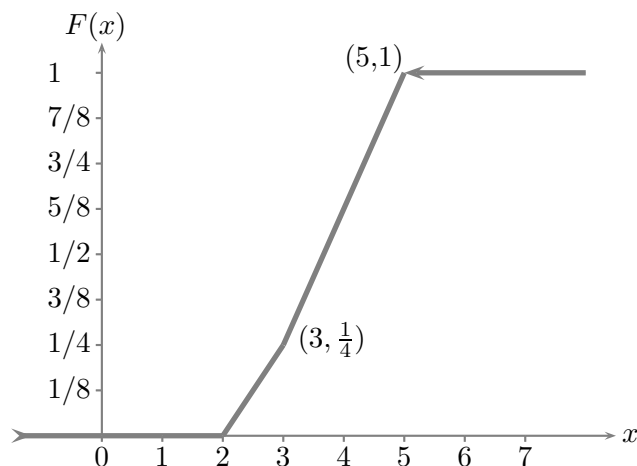
31. 0.383.

32. 11.

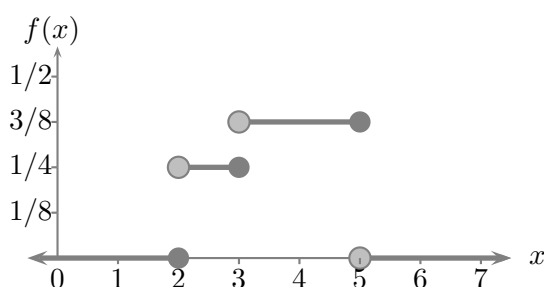
34. $P(X = k) = \frac{2}{3^k}$ for $k = 0, 1, 2, 3, \dots$

35. a) $\binom{19}{2} \left(\frac{1}{6}\right)^3 \left(\frac{5}{6}\right)^{17}$. b) $\binom{n-1}{k-1} \left(\frac{1}{6}\right)^k \left(\frac{5}{6}\right)^{n-k}$.

37. a) $B(15, 0.25)$. b) 0.08018. c) No.
38. a) $B(23, 0.5)$. b) 2.86×10^{-6} . c) Yes.
39. a) 6. b) $B(8, 0.5)$ c) 0.1445. d) No.
40. a) 10. b) 12. c) $B(12, 0.5)$. d) 0.01929. e) Yes.
42. a) $\mu = \frac{1}{2}(a+b)$; $\sigma^2 = \frac{1}{12}(a-b)^2$.
- b) $\mu = \frac{k}{k-1}$ for $k > 1$; $\sigma^2 = \frac{k}{(k-1)^2(k-2)}$ for $k > 2$.
- c) $\mu = n+1$, $\sigma^2 = n+1$.
- d) $\mu = 0$, $\sigma^2 = 2$.
43. a) $\alpha = \frac{1}{\pi}$. b) $c = -1$. c) neither exists.
44. a) $\frac{9}{16}$, b) $\frac{7}{16}$, c) 0.4106, d) $\frac{49}{12}$.
45. a) $\frac{1}{\log 10}$. b) $F(y) = \begin{cases} 0 & y < 10 \\ \frac{\log(y/10)}{\log 10} & 10 \leq y \leq 100 \\ 1 & y > 100 \end{cases}$.
- c) $10^{3/2} \approx 31.62$. d) $\frac{90}{\log 10} \approx 39.09$.
46. a) The graph of $F(x)$ is ...



b) The function $f(x) = \begin{cases} 0 & x \leq 2 \\ \frac{1}{4} & 2 < x \leq 3 \\ \frac{3}{8} & 3 < x \leq 5 \\ 0 & x > 5 \end{cases}$ will do. Its graph is shown below.



c) $E(X) = 3\frac{5}{8}$.

47. a) $E(Y) = \frac{2}{\alpha} + 3$; $\text{Var}(Y) = \frac{4}{\alpha^2}$.

48. a) 0.8907. b) 0.0107. c) 0.3594 d) 0.8925. e) 0.1359 f) 0.3830

49. a) 0.8413. b) 0.2514. c) 0.2514 d) 0.6915. e) 0.2789 f) 0.8854

50. a) 0.93. b) -1.65. c) 47 d) 32

51. a) 0.7299. b) 81

52. 0.0401.

53. a) 1.2%, b) 6.7%, c) 75.9%.

54. 2.3% over, 0.4% under.

55. 0.0122

56. a) Binomial(288, 0.25). b) $\sum_{k=88}^{288} \binom{288}{k} \left(\frac{1}{4}\right)^k \left(\frac{3}{4}\right)^{288-k}$. c) 0.017. d) Yes.

58. $E(X) = \text{Var}(X) = \frac{1}{2}$.

60. a) $-\frac{1}{\lambda} \log(1-p)$ b) $\frac{1}{\lambda} \log 2$.

62. a) 0.487 b) 0.146 c) 0.264 d) 62.4 min.

63. a) 0.6703 b) 0.1353 c) more than 5.76 hours.

64. 0.4493

65. Let $\lambda = \lambda_1 + \cdots + \lambda_n$

a) $P(T \leq t) = 1 - e^{-\lambda t}, \quad \begin{cases} 1 - e^{-\lambda t}, & t > 0 \\ 0, & t \leq 0. \end{cases}$

b) $\begin{cases} \lambda e^{-\lambda t}, & t > 0 \\ 0, & t \leq 0. \end{cases}$

c) The exponential distribution $\text{Exp}(\lambda)$, mean $= \frac{1}{\lambda}$, variance $= \frac{1}{\lambda^2}$.

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